

Krishna's

TEXT BOOK on

Differential Equations & Integral Transforms



(For B.A. and B.Sc. IInd year students of All Colleges affiliated to universities in Uttar Pradesh)

As per U.P. UNIFIED Syllabus

(w.e.f. 2012-2013)



By

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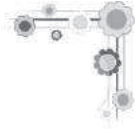
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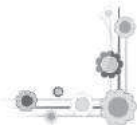
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Jai Shri Radhey Shyam

Dedicated
to
Lord
Krishna

Authors & Publishers



Preface

This book on **DIFFERENTIAL EQUATIONS & INTEGRAL TRANSFORMS** has been specially written according to the latest **Unified Syllabus** to meet the requirements of the **B.A. and B.Sc. Part-II Students** of all Universities in Uttar Pradesh.

The subject matter has been discussed in such a simple way that the students will find no difficulty to understand it. The proofs of various theorems and examples have been given with minute details. Each chapter of this book contains complete theory and a fairly large number of solved examples. Sufficient problems have also been selected from various university examination papers. At the end of each chapter an exercise containing objective questions has been given.

We have tried our best to keep the book free from misprints. The authors shall be grateful to the readers who point out errors and omissions which, inspite of all care, might have been there.

The authors, in general, hope that the present book will be warmly received by the students and teachers. We shall **indeed** be very thankful to our colleagues for their recommending this book to their students.

The authors wish to express their thanks to **Mr. S.K. Rastogi**, *Managing Director*, **Mr. Sugam Rastogi**, *Executive Director*, **Mrs. Kanupriya Rastogi** *Director* and **entire team of KRISHNA Prakashan Media (P) Ltd., Meerut** for bringing out this book in the present form.

The authors will feel amply rewarded if the book serves the purpose for which it is meant. Suggestions for the improvement of the book are always welcome.

Preface to the Revised Edition

The authors feel great pleasure in presenting the thoroughly revised edition of the book **DIFFERENTIAL EQUATIONS & INTEGRAL TRANSFORMS** and wish to record thanks to the teachers and students for their warm reception to the previous edition.

The present edition has been specially designed, made up-to-date and well organised in a systematic order according to the latest syllabus.

The authors have always endeavoured to keep the text update in the best interests of the students community- a gesture which the authors hope would be appreciated by the students and teachers alike.

Suggestions for the improvement of the book will be thankfully received.

— *Authors*

Syllabus

Differential Equations & Integral Transforms

U.P. UNIFIED (*w.e.f.* 2012-13)

B.A./B.Sc. Paper-II

M.M. : 33 / 65

Differential Equations

Unit-1: Formation of a differential equation (D.E.), Degree, order and solution of a D.E., Equations of first order and first degree : Separation of variables method, Solution of homogeneous equations, linear equations and exact equations, Linear differential equations with constant coefficients, Homogeneous linear differential equations,

Unit-2: Differential equations of the first order but not of the first degree, Clairaut's equations and singular solutions, Orthogonal trajectories, Simultaneous linear differential equations with constant coefficients, Linear differential equations of the second order (including the method of variation of parameters),

Unit-3: Series solutions of second order differential equations, Legendre and Bessel functions (P_n and J_n only) and their properties. Order, degree and formation of partial differential equations, Partial differential equations of the first order, Lagrange's equations, Charpit's general method, Linear partial differential equations with constant coefficients.

Unit-4(i): Partial differential equations of the second order, Monge's method.

Integral Transforms

Unit-4(ii): The concept of transform, Integral transforms and kernel, Linearity property of transforms, Laplace transform, Inverse Laplace transform, Convolution theorem, Applications of Laplace transform to solve ordinary differential equations.

Unit-5: Fourier transforms (finite and infinite), Fourier integral, Applications of Fourier transform to boundary value problems, Fourier series.

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SECTION

A



DIFFERENTIAL EQUATIONS

Chapters

A large, light gray arrow pointing from left to right, positioned below the 'Chapters' header and above the list of chapters.

1. Differential Equations of First Order
And First Degree

2. Differential Equations of The First
Order But Not of The First Degree

3. Orthogonal Trajectories

4. Linear Differential Equations With
Constant Coefficients

5. Homogeneous Linear Differential
Equations



6.

Ordinary Simultaneous Differential Equations

7.

Linear Equations of Second Order With Variable Coefficients

8.

Partial Differential Equations of The First Order

9.

Linear Partial Differential Equation of Second and Higher Order with Constant Coefficients

10.

Partial Differential Equations of Second Order with Variable Coefficients

11.

Monge's Method

12.

Series Solutions of Differential Equations

13.

Legendre's Functions

14.

Bessel's Functions

Chapter

1



Differential Equations of First Order and First Degree

1.1 Definitions

A **differential equation** is an equation containing the dependent and independent variables and different derivatives of the dependent variables w.r.t. one or more independent variables.

The **order of a differential equation** is the order of the highest derivative (or differential coefficient) occurring in the equation.

(Lucknow 2007; Meerut 09B, 10B; Bundelkhand 10)

The **degree of a differential equation** is the degree of the highest derivative (or diff. coeff.) which occurs in it, after the differential equation has been rationalized (*i.e.*, made free from radicals and fractions so far as derivatives are concerned). A differential equation is called **ordinary**, if the unknown function depends on only one argument (independent variable).

(Lucknow 2007; Meerut 09B, 10B; Bundelkhand 10)

A differential equation is said to be **partial** if there are *two or more* independent variables.

A differential equation is said to be **linear** if the dependent variable, say, ' y ' and all its derivatives occur in the first degree, otherwise it is **non-linear**.

A function $y = f(x)$ is called a **Solution** (or the primitive) of a differential equation if, when substituted into the equation, it reduces the equation to an identity and the process of finding all the solutions is called integrating (or solving) the differential equation.

General solution:

(Lucknow 2007; Gorakhpur 09)

A solution of a differential equation, containing independent arbitrary constants equal in number to the order of the differential equation is called its *general solution*.

Particular solution:

(Lucknow 2007)

A solution obtained by giving particular values to the arbitrary constants in the general solution is called a *particular solution* or *particular integral*.

Arbitrary Constants: The solution of a differential equation may contain as many arbitrary constants as is the order of the differential equation *i.e.*, the solution of an n th order differential equation may contain n arbitrary constants.

Illustrative Examples

Example 1: Find the differential equation of the family of curves $y = Ae^x + (B/e^x)$ for different values of A and B .

Solution: We have $y = Ae^x + Be^{-x}$.

...(1)

To obtain the required differential equation the constants A and B are to be eliminated with the help of the given equation (1) and the two equations obtained by differentiating (1) once and twice. Thus differentiating (1), we get

$$\frac{dy}{dx} = Ae^x - Be^{-x}. \quad \dots(2)$$

Now differentiating (2), we get

$$\frac{d^2y}{dx^2} = Ae^x + Be^{-x}. \quad \dots(3)$$

Eliminating A and B between (1), (2) and (3), we obtain $\frac{d^2y}{dx^2} = y$, which is the required differential equation.

Example 2: Find the differential equation of all circles of radius a , or By the elimination of the constants h and k , find the differential equation of which $(x - h)^2 + (y - k)^2 = a^2$, is a solution.

Solution: The equation of all circles of radius a is given by

$$(x - h)^2 + (y - k)^2 = a^2, \quad \dots(1)$$

h and k being parameters (*i.e.*, arbitrary constants).

Differentiating (1), we get

$$2(x-h) + 2(y-k) \frac{dy}{dx} = 0. \quad \dots(2)$$

Differentiating (2), we get

$$1 + (y-k) \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0. \quad \dots(3)$$

From (2) and (3), we obtain

$$(x-h) = -(y-k) (dy/dx) \text{ and } (y-k) = -\frac{[1 + (dy/dx)^2]}{d^2y/dx^2}.$$

Substituting these values in (1) and simplifying, we obtain

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = a^2 \left(\frac{d^2y}{dx^2}\right)^2,$$

which is the required differential equation.

Comprehensive Exercise 1

1. Find the differential equation of the family of curves

$$y = Ae^{2x} + Be^{-2x},$$

for different values of A and B .

(Bundelkhand 2003)

2. Find the differential equation corresponding to

$$y = ae^{2x} + be^{-3x} + ce^x,$$

where a, b, c are arbitrary constants.

3. Find the differential equation of the family of curves

$$y = e^x (A \cos x + B \sin x).$$

where A and B are arbitrary constants.

4. By eliminating the constants a and b obtain the differential equation of which $xy = ae^x + be^{-x} + x^2$ is a solution.

(Purvanchal 2014)

5. Find the differential equation corresponding to the family of curves $y = c(x-c)^2$, where c is an arbitrary constant.

6. Show that $Ax^2 + By^2 = 1$ is the solution of

$$x \left[y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 \right] - y \frac{dy}{dx} = 0.$$

7. Show that $v = (a/r) + B$ is a solution of

$$\frac{d^2v}{dr^2} + \frac{2}{r} \frac{dv}{dr} = 0.$$

Answers 1

1. $\frac{d^2 y}{dx^2} = 4y$

2. $\frac{d^3 y}{dx^3} - 7 \frac{dy}{dx} + 6y = 0$

3. $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + 2y = 0$

4. $x \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} - xy + x^2 - 2 = 0$

5. $8y^2 = 4xy \frac{dy}{dx} - \left(\frac{dy}{dx}\right)^3$

1.2 Differential Equations of First Order and First Degree

The differential equations of first order and of first degree can always be written in the form

$$M + N (dy/dx) = 0 \quad \text{or} \quad M dx + N dy = 0,$$

where M and N are some functions of x and y or are constants.

Here the differential equation being of 1st order, its general solution (or primitive) will contain only one arbitrary constant. All differential equations of the first order cannot be always solved. However they can be solved by suitable methods if they belong to any one of the following standard forms :

- Variables separable,
- Homogeneous equations,
- Equations of the form $\frac{dy}{dx} = \frac{ax + by + c}{a_1x + b_1y + c_1}$,
- Linear equations,
- Equations reducible to the linear form,
- Exact differential equations,
- Equations reducible to exact form,
- Equations which by suitable change of variable can be transformed to any of the above forms.

1.2.1 Variables Separable

If a differential equation of the first order and of the first degree is of the form

$$f_1(x) dx = f_2(y) dy, \quad \dots(1)$$

where $f_1(x)$ is a function of x only and $f_2(y)$ is a function of y only, then we say that the variables are separable in the differential equation.

In such equations it is possible to get dx and all the terms involving x on one side and dy along with all the terms involving y on the other side.

To solve such a differential equation *integrate the two sides* and add an arbitrary constant of integration to any one of the two sides.

Thus, integrating both the sides of (1), we get its solution as

$$\int f_1(x) dx = \int f_2(y) dy + c,$$

where c is an arbitrary constant. The arbitrary constant can be chosen in any form suitable for the answer, i.e., we can replace it by $\log c$, $\tan^{-1} c$, $\sin c$, e^c , etc.

Illustrative Examples

Example 3: Solve $(1 + x^2) dy = (1 + y^2) dx$

(Meerut 2003)

Solution: The given equation can be written as $dy/(1 + y^2) = dx/(1 + x^2)$, in which the variables have been separated.

$\therefore$ integrating, we get $\tan^{-1} y = \tan^{-1} x + \tan^{-1} c$, where c is an arbitrary constant

or $\tan^{-1} y - \tan^{-1} x = \tan^{-1} c$

or $\tan^{-1} \frac{y - x}{1 + yx} = \tan^{-1} c$, or $\frac{y - x}{1 + yx} = c$

or $(y - x) = c(1 + yx)$, which is the required solution.

Example 4: Solve $(1 + e^x) y dy = (y + 1) e^x dx$.

Solution: Here $\frac{y}{y+1} dy = \frac{e^x}{1+e^x} dx$, (the variables being separated)

or $\left[1 - \frac{1}{y+1}\right] dy = \frac{e^x}{1+e^x} dx$.

$\therefore$ integrating, we get

$$y - \log(y + 1) = \log(1 + e^x) + \log c, \text{ (} c \text{ being an arbitrary constant)}$$

or $y = \log[c(y + 1)(1 + e^x)]$

or $c(y + 1)(1 + e^x) = e^y$, which is the required solution.

Example 5: Solve $dy/dx = (x + y)^2$.

(Avadh 2011)

Solution: Here the variables are not separable but some suitable substitution will reduce the differential equation to a form in which the variables are separable. Here we put $x + y = v$.

Differentiating both sides w.r.t., 'x', we have

$$1 + \frac{dy}{dx} = \frac{dv}{dx} \quad \text{or} \quad \frac{dy}{dx} = \frac{dv}{dx} - 1.$$

By these substitutions the given equation reduces to

$$\frac{dv}{dx} - 1 = v^2, \quad \text{or} \quad \frac{dv}{dx} = v^2 + 1, \quad \text{or} \quad \frac{dv}{v^2 + 1} = dx.$$

Thus the variables being separated and so integrating, we get

$$\int \frac{dv}{v^2 + 1} = \int dx + c, \quad \text{or} \quad \tan^{-1} v = x + c$$

or $\tan^{-1} (x + y) = x + c, \quad [\because v = x + y]$

or $x + y = \tan (x + c)$ as the required solution where c is an arbitrary constant.

Example 6: Solve $dy/dx = \sin (x + y) + \cos (x + y)$.

Solution: Let $x + y = v$. Then differentiating, we have

$$1 + \frac{dy}{dx} = \frac{dv}{dx}, \quad \text{or} \quad \frac{dy}{dx} = \frac{dv}{dx} - 1.$$

Substituting these values in the given equation, we get

$$\frac{dv}{dx} - 1 = \sin v + \cos v, \quad \text{or} \quad \frac{dv}{dx} = 1 + \sin v + \cos v$$

or $\frac{dv}{(1 + \cos v) + \sin v} = dx$, separating the variables

or $\frac{dv}{2 \cos^2 \frac{1}{2} v + 2 \sin \frac{1}{2} v \cos \frac{1}{2} v} = dx \quad \text{or} \quad \frac{\frac{1}{2} \sec^2 \frac{1}{2} v}{1 + \tan \frac{1}{2} v} dv = dx.$

Now integrating, we get $\log (1 + \tan \frac{1}{2} v) = x + c$

or $\log \{1 + \tan \frac{1}{2} (x + y)\} = x + c,$

which is the required solution containing an arbitrary constant c .

Comprehensive Exercise 2

Solve the following differential equations :

- (i) $(1 + x) y dx + (1 - y) x dy = 0.$

(ii) $(1 - x^2) (1 - y) dx = xy (1 + y) dy.$
- (i) $x^2 (y + 1) dx + y^2 (x - 1) dy = 0.$

(ii) $\left(y - x \frac{dy}{dx}\right) = a \left(y^2 + \frac{dy}{dx}\right).$ (Purvanchal 2006; Avadh 10; Gorakhpur 08, 11)
- (i) $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0.$ (Agra 2006; Meerut 09B)

(ii) $dy/dx = e^{x-y} + x^2 e^{-y}.$ (Agra 2005; Avadh 07; Meerut 09; Purvanchal 10, 11; Rohilkhand 10; Bundelkhand 04)
- (i) $(e^y + 1) \cos x dx + e^y \sin x dy = 0.$ (Rohilkhand 2010)

(ii) $3e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0.$ (Meerut 2008; Bundelkhand 04)

5. (i) $(ds/dx) + x^2 = x^2 e^{3s}$.
 (ii) $(dy/dx) \tan y = \sin(x+y) + \sin(x-y)$.
6. (i) $\log(dy/dx) = ax + by$.
 (ii) $xy(dy/dx) = (1+y^2)(1+x+x^2)/(1+x^2)$.
7. (i) If $dy/dx = e^{x+y}$ and it is given that for $x=1, y=1$, find y when $x=-1$.
 (ii) $(x-y)^2 (dy/dx) = a^2$.
8. (i) $\cos(x+y) dy = dx$. (Lucknow 2005; Gorakhpur 09; Purvanchal 10; Avadh 14)
 (ii) $\cos(x+y) dx = dy$. (Gorakhpur 2005)
9. (i) $\sin^{-1}(dy/dx) = x + y$.
 (ii) $(dy/dx) + 1 = e^{x+y}$.
10. (i) $dy/dx = (4x + y + 1)^2$. (Lucknow 2006)
 (ii) $\left(\frac{x+y-a}{x+y-b} \right) \frac{dy}{dx} = \left(\frac{x+y+a}{x+y+b} \right)$.
11. $(y-x)(dy/dx) = a^2$. (Meerut 2006B)

Answers 2

1. (i) $xy = ce^{y-x}$
 (ii) $\log\{x(1-y)^2\} = \frac{1}{2}x^2 - \frac{1}{2}y^2 - 2y + c$
2. (i) $\frac{1}{2}(x^2 + y^2) + (x-y) + \log\{c(x-1)(y+1)\} = 0$
 (ii) $y = c(a+x)(1-ay)$
3. (i) $\tan x \tan y = c$ (ii) $e^y = e^x + \frac{1}{3}x^3 + c$
4. (i) $\sin x (e^y + 1) = c$ (ii) $\tan y = c(1 - e^x)^3$
5. (i) $(e^{3s} - 1) = c_1 e^{(3s+x^3)}$, where $c_1 = e^{3c}$ (ii) $\sec y = c - 2 \cos x$
6. (i) $-(1/b)e^{-by} = (1/a)e^{ax} + c$
 (ii) $\frac{1}{2} \log(1+y^2) = \log x + \tan^{-1} x + \log c$
7. (i) $y = -1$ (ii) $y + c = \frac{a}{2} \log \left\{ \frac{x-y-a}{x-y+a} \right\}$
8. (i) $y = c + \tan \frac{1}{2}(x+y)$ (ii) $\tan \frac{1}{2}(x+y) = x + c$
9. (i) $[-2/(x+c)] = 1 + \tan \frac{1}{2}(x+y)$ (ii) $(x+c)e^{(x+y)} + 1 = 0$

10. (i) $4x + y + 1 = 2 \tan (2x + k)$
 (ii) $(b - a) \log \{(x + y)^2 - ab\} = 2(x - y + c)$
11. $y + a^2 \log (a^2 + x - y) = c$

1.2.2 Homogeneous Equations

Definition: A differential equation of the form $\frac{dy}{dx} = \frac{f_1(x, y)}{f_2(x, y)}$, where $f_1(x, y)$ and $f_2(x, y)$ are homogeneous functions of x and y of the same degree, is called a *homogeneous equations*. (Bundelkhand 2005)

To solve such an equation, put

$$y = vx ; \text{ whence } \frac{dy}{dx} = v + x \frac{dv}{dx}.$$

These substitutions transform the given equation into an equation of the form

$$v + x \frac{dv}{dx} = f(v) \quad \text{i.e.,} \quad x \frac{dv}{dx} = f(v) - v.$$

The variables are now separable. Separating the variables and integrating, we get

$$\int \frac{dv}{f(v) - v} = \log x + c, \text{ where } c \text{ is an arbitrary constant.}$$

Now replacing v by (y / x) after integration, we get the required solution.

Working Rule. To get the solution of a homogeneous differential equation proceed as follows :

- (i) Put $y = vx$ and $\frac{dy}{dx} = v + x \frac{dv}{dx}$.
- (ii) The equation thus obtained will be of the form in which the variables are separable. Separate the variables and integrate.
- (iii) After integration replace v by y/x and get the required solution.

Illustrative Examples

Example 7: Solve $(x^2 - y^2) dx + 2xy dy = 0$.

(Gorakhpur 2008)

Solution: The given equation can be written as

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy} \quad \dots(1)$$

This is a homogeneous differential equation as each term in the N' and D' of R.H.S. is of the second degree in x and y .

$\therefore$ putting $y = vx$ and consequently $\frac{dy}{dx} = v + x \frac{dv}{dx}$ in (1), we get

$$v + x \frac{dv}{dx} = \frac{v^2 x^2 - x^2}{2vx^2} = \frac{v^2 - 1}{2v}$$

$$\text{or } x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v = \frac{v^2 - 1 - 2v^2}{2v} = \frac{-1 - v^2}{2v}$$

$$\text{or } \frac{2v dv}{v^2 + 1} = -\frac{dx}{x}, \quad \text{in which the variables have been separated.}$$

$\therefore$ integrating, we get

$$\log (v^2 + 1) = -\log x + \log c$$

$$\text{or } \log \{(v^2 + 1) \cdot x\} = \log c \quad \text{or } x(v^2 + 1) = c$$

$$\text{or } x \{(y/x)^2 + 1\} = c \quad [\text{putting } v = y/x]$$

$$\text{or } (y^2 + x^2) = cx, \text{ which is the required solution.}$$

Example 8: Solve $y^2 dx + (xy + x^2) dy = 0$.

(Avadh 2010)

Solution: The given equation can be written as

$$\frac{dy}{dx} = -\frac{y^2}{xy + x^2}, \text{ which is homogeneous.}$$

$\therefore$ putting $y = vx$, this equation reduces to

$$\left(v + x \frac{dv}{dx}\right) = -\frac{v^2 x^2}{x \cdot vx + x^2} = -\frac{v^2}{v + 1}$$

$$\text{or } x \frac{dv}{dx} = -\frac{v^2}{v + 1} - v = -\frac{v^2 + v^2 + v}{v + 1} = -\frac{2v^2 + v}{v + 1},$$

in which the variables are separable.

$$\therefore -\frac{dx}{x} = \frac{(v + 1) dv}{(2v^2 + v)} = \frac{(v + 1)}{v(2v + 1)} dv.$$

Integrating, we get

$$-\log x + \log c = \int \frac{v + 1}{v(2v + 1)} dv, \quad \text{where } c \text{ is an arbitrary constant.}$$

$$\text{Now let } \frac{v + 1}{v(2v + 1)} = \frac{A}{v} + \frac{B}{2v + 1}. \text{ Then } A = 1, B = -1.$$

$$\therefore \int \frac{v + 1}{v(2v + 1)} dv = \int \left[\frac{1}{v} - \frac{1}{2v + 1} \right] dv = \log v - \frac{1}{2} \log (2v + 1).$$

Hence the solution is

$$-\log x + \log c = \log v - \frac{1}{2} \log (2v + 1)$$

$$\text{or } \frac{1}{2} \log (2v + 1) = \log v + \log x - \log c$$

$$\text{or } \log (2v + 1)^{1/2} = \log (vx/c)$$

$$\text{or } \{2(y/x) + 1\}^{1/2} = y/c \quad [\because v = y/x]$$

$$\text{or } (2y + x)/x = y^2/c^2 \quad \text{or } c^2(2y + x) = xy^2.$$

Example 9: Solve $x^2 y \, dx - (x^3 + y^3) \, dy = 0$.

(Gorakhpur 2007; Agra 08)

Solution: The given equation can be written as

$$\frac{dy}{dx} = \frac{x^2 y}{x^3 + y^3}. \quad \dots(1)$$

This is a homogeneous equation as each term in the N^r and D^r on the R.H.S. is of the same degree (3rd degree).

$\therefore$ putting $y = vx$ and $\frac{dy}{dx} = v + x \frac{dv}{dx}$ in (1), we get

$$v + x \frac{dv}{dx} = \frac{x^2 \cdot vx}{x^3 + v^3 x^3} = \frac{v}{1 + v^3}$$

$$\text{or} \quad x \frac{dv}{dx} = \frac{v}{1 + v^3} - v = -\frac{v^4}{1 + v^3} \quad \text{or} \quad \frac{dx}{x} = -\frac{(1 + v^3) \, dv}{v^4}$$

$$\text{or} \quad \frac{dx}{x} = -\frac{1}{v^4} \, dv - \frac{1}{v} \, dv.$$

$\therefore$ integrating, we get

$$\log x + \log c = 1/(3v^3) - \log v$$

$$\text{or} \quad \log (cxv) = 1/(3v^3)$$

$$\text{or} \quad 3 \log (cxv) = 1/v^3 \quad \text{or} \quad \log (cxv)^3 = 1/v^3$$

$$\text{or} \quad c^3 x^3 v^3 = e^{1/v^3} \quad \text{or} \quad c^3 y^3 = e^{x^3/y^3} \quad [\because v = y/x]$$

$$\text{or} \quad y^3 = ke^{x^3/y^3}, \text{ which is the required solution.}$$

Example 10: Solve $(x^3 - 3xy^2) \, dx = (y^3 - 3x^2 y) \, dy$.

(Meerut 2007)

Solution: The given equation can be written as

$$\frac{dy}{dx} = \frac{x^3 - 3xy^2}{y^3 - 3x^2 y}, \quad \dots(1)$$

which is homogeneous.

$\therefore$ putting $y = vx$ and $\frac{dy}{dx} = v + x \frac{dv}{dx}$ in (1), we get

$$v + x \frac{dv}{dx} = \frac{x^3 - 3x \cdot v^2 x^2}{v^3 x^3 - 3x^2 \cdot vx} = \frac{1 - 3v^2}{v^3 - 3v}$$

$$\text{or} \quad x \frac{dv}{dx} = \frac{1 - 3v^2}{v^3 - 3v} - v = \frac{1 - 3v^2 - v^4 + 3v^2}{v^3 - 3v} = \frac{1 - v^4}{v^3 - 3v}$$

$$\text{or} \quad \frac{dx}{x} = \frac{v^3 - 3v}{1 - v^4} \, dv = \left[\frac{1}{2(v+1)} + \frac{1}{2(v-1)} - \frac{2v}{v^2+1} \right] \, dv,$$

(by partial fractions)

$\therefore$ integrating, we get

$$\log x + \log c = \frac{1}{2} \log (v+1) + \frac{1}{2} \log (v-1) - \log (v^2+1)$$

$$\begin{aligned}
 \text{or} \quad & \log (cx) = \log \left[\{(v+1)^{1/2} (v-1)^{1/2}\} / (v^2 + 1) \right] \\
 \text{or} \quad & cx (v^2 + 1) = (v^2 - 1)^{1/2} \quad \text{or} \quad c^2 x^2 (v^2 + 1)^2 = v^2 - 1 \\
 \text{or} \quad & c^2 x^2 \{(y^2/x^2) + 1\}^2 = \{(y^2/x^2) - 1\}, \quad [\because v = y/x] \\
 \text{or} \quad & c^2 (y^2 + x^2)^2 = (y^2 - x^2), \text{ which is the required solution.}
 \end{aligned}$$

Example 11: Solve $x \left(\cos \frac{y}{x} \right) (y dx + x dy) = y \left(\sin \frac{y}{x} \right) (x dy - y dx)$.

Solution: The given equation can be written as

$$\begin{aligned}
 & x \left(\cos \frac{y}{x} \right) \left(y + x \frac{dy}{dx} \right) = y \left(\sin \frac{y}{x} \right) \left(x \frac{dy}{dx} - y \right) \\
 \text{or} \quad & x \frac{dy}{dx} \left(x \cos \frac{y}{x} - y \sin \frac{y}{x} \right) = -y^2 \sin \frac{y}{x} - xy \cos \frac{y}{x} \\
 \text{or} \quad & \frac{dy}{dx} = \frac{y \{ y \sin (y/x) + x \cos (y/x) \}}{x \{ y \sin (y/x) - x \cos (y/x) \}} \quad \dots(1)
 \end{aligned}$$

Now putting $y = vx$ and $\frac{dy}{dx} = v + x \frac{dv}{dx}$ in (1), we get

$$\begin{aligned}
 v + x \frac{dv}{dx} &= \frac{vx (vx \sin v + x \cos v)}{x (vx \sin v - x \cos v)} \\
 \text{or} \quad v + x \frac{dv}{dx} &= \frac{v (v \sin v + \cos v)}{(v \sin v - \cos v)} \\
 \text{or} \quad x \frac{dv}{dx} &= \frac{v^2 \sin v + v \cos v - v^2 \sin v + v \cos v}{v \sin v - \cos v}, \text{ variables separable} \\
 \text{or} \quad & - \left(\frac{-v \sin v + \cos v}{v \cos v} \right) dv = \frac{2 dx}{x}, \text{ separating the variables.}
 \end{aligned}$$

$\therefore$ integrating, we get

$$\begin{aligned}
 & -\log (v \cos v) = 2 \log x + \log c \\
 \text{or} \quad & \log \{1/(v \cos v)\} = \log (cx^2) \quad \text{or} \quad cx^2 = 1/(v \cos v) \\
 \text{or} \quad & cx^2 v \cos v = 1 \quad \text{or} \quad cx^2 \cdot (y/x) \cos (y/x) = 1 \\
 \text{or} \quad & cxy \cos (y/x) = 1, \text{ which is the required solution.}
 \end{aligned}$$

Comprehensive Exercise 3

Solve the following differential equations :

- $x + y (dy/dx) = 2y$. (Gorakhpur 2005)
- $y - x (dy/dx) = x + y (dy/dx)$. (Gorakhpur 2010)
- $(x^2 + y^2) dx - 2xy dy = 0$. (Avadh 2006; Bundelkhand 03)

4. $(x^2 + y^2) (dy/dx) = xy$. (Kanpur 2002)
5. $dy/dx = y^2/(xy - x^2)$.
6. $x^2 dy + y(x + y) dx = 0$.
7. $x(x - y) dy = y(x + y) dx$.
8. $x \frac{dy}{dx} + \frac{y^2}{x} = y$.
9. $x dy - y dx = \sqrt{x^2 + y^2} dx$. (Meerut 2008; Lucknow 10)
10. $(1 + e^{x/y}) dx + e^{x/y} \{1 - (x/y)\} dy = 0$. (Meerut 2006)
11. $\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$.
12. $x \frac{dy}{dx} = y - x \cos^2 \frac{y}{x}$.
13. $x \left(\frac{dy}{dx} \right) = y + x e^{y/x}$.
14. $x \frac{dy}{dx} \cos \left(\frac{y}{x} \right) = y \cos \left(\frac{y}{x} \right) - x$.

Answers 3

1. $\log(y - x) = c + x/(y - x)$
2. $\frac{1}{2} \log(x^2 + y^2) + \tan^{-1}(y/x) = \log c$
3. $x^2 - y^2 = cx$
4. $cy = e^{x^2/(2y^2)}$
5. $y = ke^{y/x}$
6. $(y + 2x) = c^2 x^2 y$
7. $c^2 xy = e^{-x/y}$
8. $cx = e^{x/y}$
9. $y + \sqrt{(y^2 + x^2)} = cx^2$
10. $x + ye^{x/y} = c$
11. $\sin(y/x) = cx$
12. $\tan(y/x) = \log(c/x)$
13. $e^{-y/x} + \log x = c$
14. $\sin\left(\frac{y}{x}\right) + \log x = c$

1.2.3 Equations Reducible To Homogeneous Form

A differential equation of the form

$$\frac{dy}{dx} = \frac{ax + by + c}{a_1x + b_1y + c_1}, \text{ where } \frac{a}{a_1} \neq \frac{b}{b_1}$$

can be reduced to homogeneous form by taking new variables X and Y such that $x = X + h$ and $y = Y + k$, where h and k are constants to be so chosen as to make the given equation homogeneous. With the above substitutions we get $dx = dX$ and $dy = dY$, so that $dy/dx = dY/dX$. Hence the given equation becomes

$$\frac{dY}{dX} = \frac{a(X + h) + b(Y + k) + c}{a_1(X + h) + b_1(Y + k) + c_1} = \frac{aX + bY + (ah + bk + c)}{a_1X + b_1Y + (a_1h + b_1k + c_1)}.$$

Now choose h and k such that

$$ah + bk + c = 0, \quad \text{and} \quad a_1h + b_1k + c_1 = 0.$$

Then the differential equation becomes

$$\frac{dY}{dX} = \frac{aX + bY}{a_1X + b_1Y}, \text{ which is homogeneous.}$$

Now this equation can be solved as in 1.2.2 substituting $Y = vX$. Finally by replacing X by $(x - h)$ and Y by $(y - k)$ we shall get the solution in original variables x and y .

If, however, $a/a_1 = b/b_1 = m$ (say) then the differential equation becomes of the form

$$\frac{dy}{dx} = \frac{m(a_1x + b_1y) + c}{a_1x + b_1y + c_1}.$$

To solve such a differential equation put $v = a_1x + b_1y$, get rid of y and then the transformed equation will be such that the variables are separable.

Illustrative Examples

Example 12: Solve $\frac{dy}{dx} = \frac{y - x + 1}{y + x + 5}$.

Solution: Here $\frac{a}{a_1} \neq \frac{b}{b_1}$ i.e., the coefficients of x and y in the numerator and denominator of the expression for (dy/dx) are not proportional. Such equations can be reduced to homogeneous form by taking new variables X and Y such that $x = X + h$ and $y = Y + k$, where h and k are constants to be taken at our choice. With these substitutions the given equation reduces to

$$\frac{dY}{dX} = \frac{Y + k - X - h + 1}{Y + k + X + h + 5} = \frac{Y - X + (k - h + 1)}{Y + X + (k + h + 5)} \quad \dots(1)$$

Choose h and k such that

$$k - h + 1 = 0 \quad \text{and} \quad k + h + 5 = 0.$$

Solving these, we get $k = -3, h = -2$.

With these values of h and k the equation (1) becomes

$$\frac{dY}{dX} = \frac{Y - X}{Y + X} \quad \dots(2)$$

Now (2) is a homogeneous equation, so putting $Y = vX$, we have

$$v + X \frac{dv}{dX} = \frac{v - 1}{v + 1}, \quad \text{or} \quad X \frac{dv}{dX} = \frac{v - 1}{v + 1} - v = \frac{-1 - v^2}{v + 1}$$

$$\text{or} \quad \frac{v + 1}{v^2 + 1} dv = -\frac{dX}{X} \quad \text{or} \quad \frac{v}{v^2 + 1} dv + \frac{dv}{v^2 + 1} = -\frac{dX}{X}$$

$$\text{or} \quad \frac{2v}{v^2 + 1} dv + \frac{2}{v^2 + 1} dv = -\frac{2}{X} dX.$$

Integrating, we have

$$\log(v^2 + 1) + 2 \tan^{-1} v = -2 \log X + c$$

$$\text{or} \quad \log \{(v^2 + 1) X^2\} = -2 \tan^{-1} v + c$$

$$\text{or} \quad \log (Y^2 + X^2) = -2 \tan^{-1} (Y/X) + c, \quad [\because v = Y/X]$$

$$\text{or} \quad \log [(y+3)^2 + (x+2)^2] + 2 \tan^{-1} \{(y+3)/(x+2)\} = c, \\ [\because Y = y - k = y + 3, X = x - h = x + 2]$$

which is the required solution.

Example 13: Solve $\frac{dy}{dx} = \frac{2x + 2y - 2}{3x + y - 5}$.

(Lucknow 2005)

Solution: Here $(a/a_1) \neq (b/b_1)$, therefore putting $x = X + h$ and $y = Y + k$ the given equation reduces to

$$\frac{dY}{dX} = \frac{2(X+h) + 2(Y+k) - 2}{3(X+h) + (Y+k) - 5} = \frac{2X + 2Y + (2h + 2k - 2)}{3X + Y + (3h + k - 5)} \quad \dots(1)$$

Now choose h and k such that

$$2h + 2k - 2 = 0 \quad \text{and} \quad 3h + k - 5 = 0.$$

Solving these, we get $h = 2$ and $k = -1$.

With these values of h and k , the equation (1) becomes

$$\frac{dY}{dX} = \frac{2X + 2Y}{3X + Y}.$$

This is a homogeneous equation so putting $Y = vX$, we get

$$v + X \frac{dv}{dX} = \frac{2X + 2vX}{3X + vX} = \frac{2 + 2v}{3 + v}$$

$$\text{or} \quad X \frac{dv}{dX} = \frac{2 + 2v}{3 + v} - v = \frac{2 + 2v - 3v - v^2}{3 + v} = \frac{2 - v - v^2}{3 + v}$$

$$\text{or} \quad -\frac{dX}{X} = \frac{3 + v}{v^2 + v - 2} dv = \frac{3 + v}{(v+2)(v-1)} dv \\ = \left[\frac{4}{3(v-1)} - \frac{1}{3(v+2)} \right] dv, \text{ by resolving into partial fractions}$$

$$\text{or} \quad 3 \frac{dX}{X} = \left[\frac{1}{v+2} - \frac{4}{v-1} \right] dv.$$

Integrating, we get

$$3 \log X = \log (v+2) - 4 \log (v-1) + \log c$$

$$\text{or} \quad \log \{X^3 (v-1)^4\} = \log \{c (v+2)\}$$

$$\text{or} \quad X^3 [(Y/X) - 1]^4 = c \{(Y/X) + 2\}, \quad [\because v = Y/X]$$

$$\text{or} \quad (Y - X)^4 = c (Y + 2X)$$

$$\text{or} \quad \{(y+1) - (x-2)\}^4 = c \{(y+1) + 2(x-2)\}$$

$$\text{or} \quad (y - x + 3)^4 = c (2x + y - 3), \text{ which is the required solution.}$$

Example 14: Solve $(2x + y + 1) dx + (4x + 2y - 1) dy = 0$. (Lucknow 2011; Kanpur 12)

Solution: We have $\frac{dy}{dx} = -\frac{2x + y + 1}{4x + 2y - 1}$.

Here $\frac{a}{a_1} = \frac{b}{b_1}$.

$\therefore$ let $2x + y = v$ so that $2 + (dy/dx) = dv/dx$.

With these substitutions the given equation reduces to

$$\frac{dv}{dx} - 2 = -\frac{v+1}{2v-1},$$

or
$$\frac{dv}{dx} = 2 - \frac{v+1}{2v-1} = \frac{3v-3}{2v-1}$$

or
$$3 dx = \frac{2v-1}{v-1} dv = \frac{2(v-1)+1}{v-1} dv = \left(2 + \frac{1}{v-1}\right) dv.$$

$\therefore$ integrating, we have

$$3x + c = 2v + \log(v-1)$$

or
$$3x + c = 2(2x + y) + \log(2x + y - 1), \quad [\because v = 2x + y]$$

or
$$x + 2y + \log(2x + y - 1) = c, \text{ which is the required solution.}$$

Comprehensive Exercise 4

Solve the following differential equations :

- $(2x + y - 3) dy = (x + 2y - 3) dx$. (Bundelkhand 2010; Lucknow 09; Rohilkhand 11)
- $dy/dx = (2x - y + 1)/(x + 2y - 3)$.
- $(2x + 3y - 5)(dy/dx) + (3x + 2y - 5) = 0$.
- $(x - y - 2) dx + (x - 2y - 3) dy = 0$.
- $(x + y - 1) dy = (x + y) dx$.
- $(x + y)(dx - dy) = dx + dy$. (Gorakhpur 2008)
- $(x - y - 2) dx = (2x - 2y - 3) dy$. (Kashi 2011)
- $\frac{dy}{dx} = \frac{x - y + 3}{2x - 2y + 5}$.
- $(2x + 4y + 1) dy = (2y + x + 1) dx$.
- $(2x - 2y + 5) dy - (x - y + 3) dx = 0$.

Answers 4

- $(x + y - 2) = c^2 (x - y)^3$
- $(5y - 7)^2 + (5x - 1)(5y - 7) - (5x - 1)^2 = c$
- $3x^2 + 4xy + 3y^2 - 10x - 10y = c$

4. $\log \{c^2 (2Y^2 - X^2)\} = (1/\sqrt{2}) \log \{(Y/\sqrt{2} - X)/(Y/\sqrt{2} + X)\}$,
 where $X = x - 1$ and $Y = y + 1$.
5. $2(y - x) - \log(2x + 2y - 1) = c$ 6. $(y - x) + \log(x + y) = c$
7. $\log(x - y - 1) = x - 2y - c$ 8. $x - 2y + \log(x - y + 2) = c$
9. $4(2y - x) - \log(4x + 8y + 3) = c$
10. $x - 2y + \log(x - y + 2) = c$

1.2.4. Linear Differential Equations

Definition: A differential equation is said to be linear when the dependent variable y and all its derivatives occur in the first degree only and are not multiplied together.

An equation of the form

$$\frac{dy}{dx} + Py = Q,$$

where P and Q are functions of x only is called a linear differential equation of the first order with y as the dependent variable. To solve such an equation, multiply both the sides by $e^{\int P dx}$. The equation then becomes

$$e^{\int P dx} (dy/dx) + Pye^{\int P dx} = Qe^{\int P dx}$$

or
$$\frac{d}{dx} \{ye^{\int P dx}\} = Qe^{\int P dx}.$$

Integrating both sides w.r.t., ' x ', we get

$$ye^{\int P dx} = \int Qe^{\int P dx} dx + c,$$

which is the solution of the differential equation.

Note 1: The factor $e^{\int P dx}$, on multiplying by which the L.H.S. of the differential equation becomes a differential coefficient of some function of x and y , is called an **integrating factor** (I.F.) of the differential equation. (Agra 2007)

Note 2: Sometimes a given differential equation becomes linear if we take y as the independent variable and x as the dependent variable.

Then it is of the form $\frac{dx}{dy} + Px = Q$,

where P and Q are some functions of y only. The integrating factor (I.F.) in this case is $e^{\int P dy}$ and the solution is

$$x \cdot e^{\int P dy} = \int Q \cdot e^{\int P dy} dy + c.$$

Working Rule :

(i) Write the given equation in the form

$$\frac{dy}{dx} + Py = Q \quad \text{or} \quad \frac{dx}{dy} + Px = Q \text{ as the case may be.}$$

(ii) Find the integrating factor $e^{\int P dx}$ or $e^{\int P dy}$.

(iii) The solution of the differential equation is either

$$y \cdot (\text{I.F.}) = \int \{Q \cdot (\text{I.F.})\} dx + c$$

or $x \cdot (\text{I.F.}) = \int \{Q \cdot (\text{I.F.})\} dy + c$ as the case may be.

Illustrative Examples

Example 15: Solve $\frac{dy}{dx} + 2xy = e^{-x^2}$.

(Meerut 2001, 03, 10B)

Solution: The given differential equation is $\frac{dy}{dx} + 2xy = e^{-x^2}$, which is linear with y as the dependent variable.

Here $P = 2x$, and $Q = e^{-x^2}$.

We have $\int P dx = \int 2x dx = 2 \cdot \frac{1}{2} x^2 = x^2$.

Therefore I.F. = $e^{\int P dx} = e^{x^2}$.

Hence the solution is $y \cdot (\text{I.F.}) = \int \{Q \cdot (\text{I.F.})\} dx + c$, where c is an arbitrary constant

$$\text{or } y \cdot e^{x^2} = \int (e^{-x^2} \cdot e^{x^2}) dx + c$$

$$\text{or } y \cdot e^{x^2} = \int dx + c \quad \text{or } ye^{x^2} = x + c.$$

Example 16: Solve $(x^2 - 1)(dy/dx) + 2xy = 1$.

(Meerut 2010)

Solution: The given equation can be written as

$$\frac{dy}{dx} + \frac{2x}{x^2 - 1} y = \frac{1}{x^2 - 1} \quad \left[\text{From } \frac{dy}{dx} + Py = Q \right]$$

Here $P = \frac{2x}{x^2 - 1}$, and so $\int P dx = \int \frac{2x dx}{x^2 - 1} = \log(x^2 - 1)$.

$\therefore$ integrating factor (I.F.) = $e^{\int P dx} = e^{\log(x^2 - 1)} = (x^2 - 1)$.

$\therefore$ the solution is $y \cdot (x^2 - 1) = \int \frac{1}{(x^2 - 1)} \cdot (x^2 - 1) dx + c$, $\left[\because Q = \frac{1}{(x^2 - 1)} \right]$

$$\text{or } y(x^2 - 1) = \int dx + c = x + c.$$

Hence $y(x^2 - 1) = x + c$ is the required solution.

Example 17: Solve $x(dy/dx) + 2y = x^2 \log x$.

Solution: The given equation can be written as

$$\frac{dy}{dx} + \frac{2}{x} y = x \log x, \quad \left(\text{from } \frac{dy}{dx} + Py = Q \right)$$

Here $P = (2/x)$ and so $\int P \, dx = \int (2/x) \, dx = 2 \log x = \log x^2$.

$$\therefore \text{I.F.} = e^{\int P \, dx} = e^{\log x^2} = x^2.$$

$\therefore$ the solution is

$$y \text{ (I.F.)} = \int Q \cdot (\text{I.F.}) \, dx + c$$

$$\text{i.e.,} \quad yx^2 = \int (x \log x) x^2 \, dx + c, \quad [\because Q = x \log x]$$

$$\text{or} \quad yx^2 = \int x^3 \log x \, dx + c$$

$$\text{or} \quad yx^2 = (\log x) \cdot \frac{x^4}{4} - \int \frac{1}{x} \cdot \frac{x^4}{4} \, dx + c, \quad [\text{integrating by parts}]$$

$$\text{or} \quad yx^2 = \frac{1}{4} x^4 \log x - \frac{1}{4} \cdot \left(\frac{1}{4} x^4\right) + c$$

Hence $4yx^2 = x^4 \log x - \frac{1}{4} x^4 + k$ is the required solution.

Example 18: Solve $\cos^2 x (dy/dx) + y = \tan x$. (Meerut 2003, 13; Avadh 05)

Solution: The given equation can be written as $\frac{dy}{dx} + \sec^2 x \cdot y = \tan x \sec^2 x$, which is linear with y as the dependent variable.

Here $P = \sec^2 x$, and $Q = \tan x \sec^2 x$.

$$\text{We have } \int P \, dx = \int \sec^2 x \, dx = \tan x.$$

$$\therefore \text{I.F.} = e^{\int P \, dx} = e^{\tan x}.$$

$$\therefore \text{the solution is } y \cdot e^{\tan x} = \int \tan x \sec^2 x \cdot e^{\tan x} \, dx + c$$

$$\text{or} \quad ye^{\tan x} = \int t e^t \, dt + c, \text{ putting } \tan x = t \text{ and } \sec^2 x \, dx = dt$$

$$= [t e^t - \int e^t \, dt] + c, \text{ integrating by parts}$$

$$= t e^t - e^t + c = \tan x e^{\tan x} - e^{\tan x} + c$$

$$\text{or} \quad ye^{\tan x} = e^{\tan x} (\tan x - 1) + c.$$

Example 19: Solve $(1 + y^2) \, dx = (\tan^{-1} y - x) \, dy$.

(Agra 2005; Avadh 07; Rohilkhand 10; Bundelkhand 03, 04; Meerut 13B)

Solution: The given equation can be written as

$$\frac{dx}{dy} = \frac{\tan^{-1} y - x}{1 + y^2} \quad \text{or} \quad \frac{dx}{dy} + \frac{1}{1 + y^2} \cdot x = \frac{\tan^{-1} y}{1 + y^2}.$$

This is linear equation with y as independent variable and x as dependent variable, Here $P = 1/(1 + y^2)$.

$$\therefore \int P \, dy = \int \{1/(1+y^2)\} \, dy = \tan^{-1} y.$$

$$\text{Hence I.F.} = e^{\int P \, dy} = e^{\tan^{-1} y}.$$

$$\text{Now the solution is } x \cdot (\text{I.F.}) = \int \{Q \cdot (\text{I.F.})\} \, dy + c$$

$$\begin{aligned} \text{i.e., } x e^{\tan^{-1} y} &= \int \frac{\tan^{-1} y}{1+y^2} \cdot e^{\tan^{-1} y} \, dy + c, & \left(\because Q = \frac{\tan^{-1} y}{1+y^2} \right) \\ &= \int t e^t \, dt + c, \text{ putting } \tan^{-1} y = t \text{ and } dy/(1+y^2) = dt \\ &= [t e^t - \int 1 \cdot e^t \, dt] + c, \text{ integrating by parts} \\ &= t e^t - e^t + c = (t-1) e^t + c \end{aligned}$$

$$\text{or } x \cdot e^{\tan^{-1} y} = (\tan^{-1} y - 1) \cdot e^{\tan^{-1} y} + c.$$

Example 20: Solve $(dy/dx) + 2y \tan x = \sin x$, given that $y = 0$ when $x = \pi/3$.

Solution: We have $\frac{dy}{dx} + 2 \tan x \cdot y = \sin x$, which is linear with y as dependent variable.

$$\text{Here } P = 2 \tan x \text{ and so } \int P \, dx = \int 2 \tan x \, dx = 2 \log \sec x = \log \sec^2 x.$$

$$\therefore \text{I.F.} = e^{\int P \, dx} = e^{\log \sec^2 x} = \sec^2 x.$$

$$\therefore \text{the solution is } y \cdot (\text{I.F.}) = \int [Q \cdot (\text{I.F.})] \, dx + c$$

$$\text{i.e., } y \sec^2 x = \int \sin x \cdot \sec^2 x \, dx + c = \int \tan x \sec x \, dx + c$$

$$\text{or } y \sec^2 x = \sec x + c. \quad \dots(1)$$

$$\text{Now it is given that } y = 0 \text{ at } x = \frac{1}{3} \pi.$$

Therefore, from (1)

$$0 \cdot \sec^2 \frac{1}{3} \pi = \sec \frac{1}{3} \pi + c \quad \text{or} \quad c = -\sec \frac{1}{3} \pi = -2.$$

Hence from (1), $y \sec^2 x = \sec x - 2$ is the required solution.

Comprehensive Exercise 5

Solve the following differential equations :

$$1. (dy/dx) + (1/x)y = x^n. \quad 2. x^2 (dy/dx) + y = 1.$$

$$3. x (dy/dx) + y = x^2 + 3x + 2$$

(Avadh 2010)

$$4. (\sin x) (dy/dx) + 3y = \cos x.$$

(Meerut 2004)

5. $(x^2 + 1)(dy/dx) + 2xy = 4x^2$. (Agra 2008)
6. $(1 + x^2)(dy/dx) + 2xy = \cos x$. (Meerut 2009)
7. $x(x^2 + 1)(dy/dx) = y(1 - x^2) + x^3 \log x$. (Meerut 2007B)
8. $(dy/dx) + (2y/x) = \sin x$.
9. $(dy/dx) + (y/x) = \sin x^2$
10. $(y \sin x - 1)dx + \cos x dy = 0$. (Bundelkhand 2008)
11. $\sin 2x (dy/dx) - y = \tan x$. (Agra 2007)
12. $(1 + x)(dy/dx) - xy = 1 - x$.
13. $(x + 2y^3)dy = y dx$. (Rohilkhand 2006; Avadh 11)
14. $(x + y + 1)(dy/dx) = 1$. (Gorakhpur 2006; Rohilkhand 08)
15. $(1 + y^2)dx + (x - e^{\tan^{-1}y})dy = 0$. (Gorakhpur 2007; Lucknow 09; Purvanchal 11; Bundelkhand 04)

Answers 5

1. $xy = x^{n+2} / (n+2) + c$
2. $y = 1 + ce^{1/x}$
3. $6xy = 2x^3 + 9x^2 + 12x + c$
4. $y \tan^3 (\frac{1}{2}x) = 2 \tan (\frac{1}{2}x) - x - \frac{1}{3} \tan^3 (\frac{1}{2}x) + c$
5. $y(1 + x^2) = \frac{4}{3}x^3 + c$
6. $y(1 + x^2) = \sin x + c$
7. $y(x^2 + 1) = cx + \frac{1}{2}x^3 \log x - \frac{1}{4}x^3$
8. $yx^2 = -x^2 \cos x + 2x \sin x + 2 \cos x + c$
9. $yx = -\frac{1}{2} \cos x^2 + c$
10. $y \sec x = \tan x + c$
11. $y = \tan x + c \sqrt{(\tan x)}$
12. $y(1 + x) = x + ce^x$
13. $x = y^3 + cy$
14. $x = ce^y - y - 2$
15. $xe^{\tan^{-1}y} = \frac{1}{2}e^{2 \tan^{-1}y} + c$

1.2.5. Equations Reducible to the Linear Form

Some equations can be reduced to the linear form by making suitable substitutions, and hence can be solved easily.

(a) Bernoulli's Equation:

(Gorakhpur 2008)

A differential equation of the form

$$\frac{dy}{dx} + Py = Qy^n,$$

where P and Q are functions of x alone is called Bernoulli's equation.

To solve this equation dividing both sides by y^n , we get

$$y^{-n} \frac{dy}{dx} + P y^{-n+1} = Q$$

[Note that the R.H.S. is now a function of x alone.]

Now putting $y^{-n+1} = v$, so that $(1-n) y^{-n} \frac{dy}{dx} = \frac{dv}{dx}$, the equation (1) transforms to

$$\frac{1}{1-n} \frac{dv}{dx} + P v = Q$$

or
$$\frac{dv}{dx} + (1-n) P \cdot v = (1-n) Q,$$

which is a linear differential equation with v as dependent variable and can be solved by the method discussed in 1.2.4.

(b) Equations of the form

$$\frac{dy}{dx} + P \phi(y) = Q f(y),$$

where P and Q are functions of x alone.

To reduce such an equation to the linear form, dividing both sides by $f(y)$ and thus getting rid of $f(y)$ from the R.H.S., we have

$$\frac{1}{f(y)} \cdot \frac{dy}{dx} + P \cdot \frac{\phi(y)}{f(y)} = Q.$$

Now we try the substitution $\frac{\phi(y)}{f(y)} = v$.

If
$$\frac{dv}{dx} = \frac{d}{dx} \left\{ \frac{\phi(y)}{f(y)} \right\} = K \cdot \frac{1}{f(y)} \frac{dy}{dx},$$

where K is some constant, the equation reduces to the form $\frac{1}{K} \frac{dv}{dx} + P v = Q$, or

$\frac{dv}{dx} + K P v = K Q$, which is a linear differential equation, with v as the dependent variable.

Illustrative Examples

Example 21: Solve $\frac{dy}{dx} + \frac{1}{x} y = x^2 y^6$.

Solution: The given equation on dividing out by y^6 , becomes

$$\frac{1}{y^6} \frac{dy}{dx} + \frac{1}{x} \cdot \frac{1}{y^5} = x^2. \quad \dots(1)$$

Put $\frac{1}{y^5} = v$, so that

$$-\frac{5}{y^6} \frac{dy}{dx} = \frac{dv}{dx} \quad \text{or} \quad \frac{1}{y^6} \frac{dy}{dx} = -\frac{1}{5} \frac{dv}{dx}.$$

With these substitutions the equation (1) becomes

$$-\frac{1}{5} \frac{dv}{dx} + \frac{1}{x} \cdot v = x^2 \quad \text{or} \quad \frac{dv}{dx} - \frac{5}{x} v = -5x^2.$$

This is a linear equation with v as the dependent variable. Here $P = -5/x$ and $Q = -5x^2$.

We have

$$\int P dx = \int (-5/x) dx = -5 \log x = \log x^{-5} = \log (1/x^5).$$

$$\therefore \text{I.F.} = e^{\int P dx} = e^{\log (1/x^5)} = 1/x^5.$$

$\therefore$ the solution is

$$v \cdot (\text{I.F.}) = \int \{Q \cdot (\text{I.F.})\} dx + c$$

$$\text{i.e.,} \quad v \cdot (1/x^5) = \int -5x^2 \cdot (1/x^5) dx + c = -5 \int x^{-3} dx + c$$

$$\text{i.e.,} \quad (1/y^5) \cdot (1/x^5) = \frac{5}{2} x^{-2} + c. \quad [\text{Note that } v = 1/y^5]$$

Hence $1/(x^5 y^5) = \frac{5}{2} \cdot (1/x^2) + c$ is the required solution.

Example 22: Solve $\frac{dy}{dx} + \left(\frac{x}{1-x^2}\right)y = x \sqrt{y}$.

Solution: Dividing both sides of the given equation by $2y^{1/2}$, we have

$$\frac{1}{2\sqrt{y}} \frac{dy}{dx} + \frac{1}{2} \left(\frac{x}{1-x^2} \right) \sqrt{y} = \frac{1}{2} x. \quad \dots(1)$$

Now put $y^{1/2} = v$, so that $\frac{1}{2} y^{-1/2} (dy/dx) = (dv/dx)$.

With these substitutions the equation (1) becomes

$$\frac{dv}{dx} + \frac{1}{2} \left(\frac{x}{1-x^2} \right) \cdot v = \frac{1}{2} x,$$

which is linear with v as the dependent variable.

Here $P = \frac{1}{2} \{x/(1-x^2)\}$ and $Q = \frac{1}{2} x$.

$$\text{We have } \int P dx = \int \frac{1}{2} \frac{x}{1-x^2} dx = -\frac{1}{4} \int \frac{-2x}{1-x^2} dx$$

$$= -\frac{1}{4} \log (1-x^2) = \log (1-x^2)^{-1/4} = \log \{1/(1-x^2)^{1/4}\}.$$

$$\therefore \text{I.F.} = e^{\int P dx} = e^{\log \{1/(1-x^2)^{1/4}\}} = 1/(1-x^2)^{1/4}.$$

$\therefore$ the solution is

$$v/(1-x^2)^{1/4} = \frac{1}{2} \int \{x/(1-x^2)^{1/4}\} dx + c$$

$$\text{or} \quad v/(1-x^2)^{1/4} = -\frac{1}{4} \int t^{-1/4} dt, \quad \text{putting } (1-x^2) = t \text{ so that } -2x dx = dt$$

$$\text{or} \quad \sqrt[4]{y/(1-x^2)^{1/4}} = -\frac{1}{4} \cdot \frac{4}{3} t^{3/4} + c, \quad [\because v = \sqrt[4]{y}]$$

$$\text{or} \quad \sqrt[4]{y/(1-x^2)^{1/4}} = -\frac{1}{3} (1-x^2)^{3/4} + c. \quad [\because t = (1-x^2)]$$

Example 23: Solve $x (dy/dx) + y = y^2 \log x$.

(Bundelkhand 2006; Gorakhpur 10; Purvanchal 07, 08, 10; Kanpur 06)

Solution: The given equation can be written as

$$\frac{dy}{dx} + \frac{1}{x} y = \frac{y^2}{x} \log x, \text{ dividing throughout by } x$$

$$\text{or} \quad \frac{1}{y^2} \frac{dy}{dx} + \frac{1}{x} \cdot \frac{1}{y} = \frac{\log x}{x}, \quad \dots(1)$$

dividing throughout by y^2 .

Now put $1/y = v$, so that $(-1/y^2)(dy/dx) = (dv/dx)$.

With these substitutions equation (1) becomes

$$-\frac{dv}{dx} + \frac{1}{x} v = \frac{\log x}{x}$$

$$\text{or} \quad \frac{dv}{dx} - \frac{1}{x} \cdot v = -\frac{\log x}{x}.$$

This is linear with v as the dependent variable.

Here $P = -1/x$ and $Q = -(\log x)/x$.

$$\therefore \text{I.F.} = e^{\int P dx} = e^{\int (-1/x) dx} = e^{-\log x} = e^{\log(1/x)} = (1/x).$$

Hence the solution is

$$v(1/x) = \int \{-(\log x)/x\} \cdot (1/x) dx + c$$

$$\text{or} \quad v/x = -\int (1/x^2) \log x dx + c$$

$$= -[(-1/x) \log x - \int (1/x)(-1/x) dx] + c,$$

integrating by parts taking $1/x^2$ as the second function

$$= (1/x) \log x - \int (1/x^2) dx + c = (1/x) \log x + (1/x) + c$$

$$\text{or} \quad 1/(xy) = (1/x)(1 + \log x) + c, \quad [\because v = 1/y]$$

$$\text{or} \quad 1 = y(1 + \log x) + cxy.$$

Example 24: Solve $\frac{dy}{dx} + \frac{y}{x} \log y = \frac{y}{x^2} (\log y)^2$.

Solution: Dividing both sides of the given equation by $y(\log y)^2$, we have

$$\frac{1}{y(\log y)^2} \frac{dy}{dx} + \frac{1}{x} \cdot \frac{1}{\log y} = \frac{1}{x^2} \quad \dots(1)$$

Now put $1/\log y = v$, so that

$$-\{1/(\log y)^2\} \cdot (1/y) \cdot (dy/dx) = dv/dx.$$

With these substitutions the equation (1) becomes

$$-\frac{dv}{dx} + \frac{v}{x} = +\frac{1}{x^2} \quad \text{or} \quad \frac{dv}{dx} - \frac{1}{x} \cdot v = -\frac{1}{x^2}.$$

This is linear with v as the dependent variable.

Here $P = -1/x$ and $Q = -1/x^2$,

$$\therefore \text{I.F.} = e^{\int P dx} = e^{-\int (1/x) dx} = e^{-\log x} = e^{\log(1/x)} = 1/x.$$

$$\therefore \text{the solution is } v/x = \int (-1/x^2) \cdot (1/x) dx + c$$

$$\text{or} \quad (1/\log y)(1/x) = -\int x^{-3} dx + c, \quad [\because v = 1/\log y]$$

$$\text{or} \quad \frac{1}{x \log y} = \frac{1}{2x^2} + c.$$

Example 25: Solve $\frac{dy}{dx} + \frac{1}{x} = \frac{e^y}{x^2}$.

Solution: On dividing out by e^y , the given equation becomes

$$e^{-y} \frac{dy}{dx} + e^{-y} \cdot \frac{1}{x} = \frac{1}{x^2}. \quad \dots(1)$$

Now put $e^{-y} = v$, so that $-e^{-y} (dy/dx) = (dv/dx)$.

With these substitutions equation (1) becomes

$$-\frac{dv}{dx} + \frac{1}{x} \cdot v = \frac{1}{x^2} \quad \text{or} \quad \frac{dv}{dx} - \frac{1}{x} \cdot v = -\frac{1}{x^2}.$$

This is linear with v as the dependent variable.

Here $P = -1/x$ and $Q = -1/x^2$.

$$\therefore \text{I.F.} = e^{\int P dx} = e^{\int -(1/x) dx} = e^{-\log x} = e^{\log(1/x)} = 1/x.$$

$$\therefore \text{the solution is } v \cdot (\text{I.F.}) = \int \{Q \cdot (\text{I.F.})\} dx + c$$

$$\text{i.e.,} \quad v \cdot \frac{1}{x} = -\int \frac{1}{x^2} \cdot \frac{1}{x} dx + c = \frac{1}{2x^2} + c$$

$$\text{or} \quad e^{-y} \cdot (1/x) = 1/(2x^2) + c, \quad (\because v = e^{-y})$$

$$\text{or} \quad e^{-y}(1/x) = (1 + 2cx^2)/(2x^2), \quad \text{or} \quad 2x = (2cx^2 + 1) e^{-y}.$$

Comprehensive Exercise 6

Solve the following differential equations :

1. $(dy/dx) + (y/x) = y^2$.

(Bundelkhand 2007)

2. $(dy/dx) + \{y/(x-1)\} = xy^{1/3}$.

(Meerut 2005B)

3. $(x^3 y^2 + xy) dx = dy$. (Kanpur 2002)
4. $dy/dx = x^3 y^3 - xy$. (Gorakhpur 2005; Meerut 13B)
5. $3(dy/dx) + 2y/(x+1) = x^3/y^2$.
6. $y(2xy + e^x) dx - e^x dy = 0$. (Kanpur 2001)
7. $(xy^2 - e^{1/x^3}) dx - x^2 y dy = 0$. (Lucknow 2007)
8. $dy/dx = e^{x-y}(e^x - e^y)$.
9. $(dy/dx) - y \tan x = -y^2 \sec x$. (Meerut 2009B)
10. $\cos x dy = (\sin x - y) y dx$. (Rohilkhand 2007)
11. $(dy/dx) + x \sin 2y = x^3 \cos^2 y$. (Gorakhpur 2007, 11; Purvanchal 08; Avadh 14)
12. $(dy/dx) + y \cot x = y^2 \sin^2 x$.
13. $2(dy/dx) - y \sec x = y^3 \tan x$.
14. $(1-x^2) \frac{dy}{dx} + xy = xy^2$. (Purvanchal 2006)
15. $x \frac{dy}{dx} + y \log y = xy e^x$. (Bundelkhand 2005)
16. Solve $xy^2 (dy/dx) - 2y^3 = 2x^3$, given that $y = 1$ when $x = 1$.
17. Solve $\frac{dy}{dx} + y \frac{d\phi}{dx} = \phi(x) \frac{d\phi}{dx}$, where ϕ is some function of x only.

Answers 6

1. $xy(c - \log x) = 1$
2. $y^{2/3}(x-1)^{2/3} = \frac{2}{5}x(x-1)^{5/2} - \frac{3}{20}(x-1)^{8/3} + c$
3. $1 = 2y(1 - \frac{1}{2}x^2) - cy e^{-\frac{1}{2}x^2}$
4. $(1/y^2) = ce^{x^2} + x^2 + 1$
5. $y^3(x+1)^2 = \frac{1}{6}x^6 + \frac{2}{5}x^5 + \frac{1}{4}x^4 + c$
6. $e^x + y(c + x^2) = 0$
7. $3y^2 = 2x^2 e^{1/x^3} + 3cx^2$
8. $e^y = e^x - 1 + ce^{-(e^x)}$
9. $\sec x = y(\tan x - c)$
10. $\sec x = y(\tan x - c)$
11. $2 \tan y = (x^2 - 1) + 2ce^{-x^2}$
12. $(1/y) = (\cos x - c) \sin x$
13. $-(\sec x + \tan x) = y^2(\sec x + \tan x - x + c)$
14. $1/y^{n-1} = 2 \sin x - \{2/(1-n)\} + ce^{(n-1)\sin x}$
15. $x \log y = e^x(x-1) + c$
16. $y^3 = -2x^3 + 3x^6$
17. $ye^\phi = \int \phi e^\phi d\phi + c$

1.2.6. Exact Differential Equations

(Lucknow 2006)

A differential equation is said to be exact if it can be obtained from its primitive (solution) directly by differentiation, without involving any subsequent process of multiplication, elimination, etc. Thus the differential equation of the form $M dx + N dy = 0$, where M and N are some functions of x and y , is **exact** if it can be obtained directly by differentiating an equation of the form $u = c$, where u is some function of x and y and c is an arbitrary constant.

Theorem: The necessary and sufficient condition for the ordinary differential equation $M dx + N dy = 0$ to be exact is that

(Lucknow 2006, 09)

$$\partial M / \partial y = \partial N / \partial x .$$

The condition is necessary: Suppose the differential equation

$$M dx + N dy = 0 \quad \dots(1)$$

is exact. Let the primitive of (1) be $u = c$, where u is some function of x and y and c is an arbitrary constant.

Since u is a function of x and y , therefore from partial differentiation, we have

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy .$$

$$\text{Now } u = c \Rightarrow du = 0 \Rightarrow \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = 0 . \quad \dots(2)$$

Since the equation (1) is exact, therefore equation (2) must be identical with (1), therefore, we have

$$\frac{\partial u}{\partial x} = M \quad \dots(3)$$

$$\text{and } \frac{\partial u}{\partial y} = N . \quad \dots(4)$$

Differentiating (3) and (4) partially with respect to y and x respectively, we obtain

$$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial M}{\partial y} , \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial N}{\partial x} .$$

$$\text{But } \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y} . \text{ Therefore } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} .$$

Hence the condition is necessary.

The condition is sufficient: We have to show that if $\partial M / \partial y = \partial N / \partial x$, then $M dx + N dy = 0$ must be an exact differential equation. Let $\int M dx = P$, then

$$\frac{\partial P}{\partial x} = M, \text{ so that } \frac{\partial^2 P}{\partial y \partial x} = \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} , \text{ by hypothesis.}$$

$$\text{Now } \frac{\partial N}{\partial x} = \frac{\partial^2 P}{\partial x \partial y} \Rightarrow \frac{\partial N}{\partial x} = \frac{\partial^2 P}{\partial x \partial y} \Rightarrow \frac{\partial N}{\partial x} = \frac{\partial}{\partial x} \left(\frac{\partial P}{\partial y} \right)$$

$$\Rightarrow N = \frac{\partial P}{\partial y} + f(y), \text{ where } f(y) \text{ is some function of } y \text{ alone.}$$

Now putting $M = \frac{\partial P}{\partial x}$ and $N = \frac{\partial P}{\partial y} + f(y)$ in $M dx + N dy$, we have

$$\begin{aligned} M dx + N dy &= \frac{\partial P}{\partial x} dx + \left\{ \frac{\partial P}{\partial y} + f(y) \right\} dy \\ &= \frac{\partial P}{\partial x} dx + \frac{\partial P}{\partial y} dy + f(y) dy = dP + f(y) dy \\ &= d[P + F(y)], \text{ where } d[F(y)] = f(y) dy \\ &= \text{an exact differential of some function of } x \text{ and } y. \end{aligned}$$

Therefore the differential equation $M dx + N dy = 0$ is exact and hence the condition is sufficient.

Working rule for solving an exact differential equation: (Meerut 2008)

To solve a differential equation of the form $M dx + N dy = 0$, first ascertain with the help of the condition $\partial M / \partial y = \partial N / \partial x$ whether the equation is exact or not. If the equation is exact, then

- (i) Integrate M with respect to x treating y as constant.
- (ii) Integrate w.r.t. y only those terms of N which do not contain x .
- (iii) Equate the sum of these two integrals [found in (i) and (ii)] to an arbitrary constant and thus get the required solution.

Thus if the differential equation $M dx + N dy = 0$ is exact, its solution is

$$\int M dx \quad + \quad \int N dy \quad = c$$

treating y as constant take only those terms in
 N which do not contain x .

Illustrative Examples

Example 26: Solve $(ax + hy + g) dx + (hx + by + f) dy = 0$. (Meerut 2003)

Solution: Here $M = ax + hy + g$ and $N = hx + by + f$.

We have $\frac{\partial M}{\partial y} = h$ and $\frac{\partial N}{\partial x} = h$. Thus we observe that $\partial M / \partial y = \partial N / \partial x$ and therefore the given differential equation is exact. Hence its solution is

$$\int M dx \quad + \quad \int N dy \quad = c$$

treating y as a constant taking only those terms in
 N which do not contain x .

i.e.,

$$\int (ax + hy + g) dx \quad + \quad \int (by + f) dy = c$$

treating y as a constant

$$\text{i.e.,} \quad \frac{1}{2} ax^2 + hxy + gx + \frac{1}{2} by^2 + fy = c$$

$$\text{i.e.,} \quad ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 ,$$

where c is an arbitrary constant. We have replaced $-2c$ by c .

Example 27: Solve $(4x + 3y + 1) dx + (3x + 2y + 1) dy = 0$.

(Purvanchal 2009; Meerut 11)

Solution: Here $M = 4x + 3y + 1$ and $N = 3x + 2y + 1$.

We have $\frac{\partial M}{\partial y} = 3$ and $\frac{\partial N}{\partial x} = 3$ i.e., $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, and hence the given equation is exact.

Integrating M , i.e., $4x + 3y + 1$ with respect to x , treating y as a constant, we obtain
 $4 \cdot \frac{1}{2} x^2 + 3xy + x$ i.e., $2x^2 + 3xy + x$.

Again, the only terms in N which do not contain x are $2y + 1$.

Integrating $(2y + 1)$ with respect to y , we obtain

$$2 \cdot \frac{1}{2} y^2 + y \text{ i.e., } y^2 + y .$$

Hence the solution of the given differential equation is

$$2x^2 + 3xy + x + y^2 + y = c, \text{ where } c \text{ is an arbitrary constant}$$

$$\text{i.e.,} \quad 2x^2 + 3xy + y^2 + x + y = c .$$

Example 28: Solve $(x^2 - ay) dx - (ax - y^2) dy = 0$.

Solution: Here $M = x^2 - ay$ and $N = -(ax - y^2)$.

We have

$$\frac{\partial M}{\partial y} = -a \text{ and } \frac{\partial N}{\partial x} = -a \text{ i.e., } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

and hence the given equation is exact.

Now $\int M dx$ (regarding y as a constant)

$$= \int (x^2 - ay) dx = \frac{1}{3} x^3 - ayx \quad \dots(1)$$

and $\int N dy$ (taking in N only those terms which do not contain x)

$$= \int y^2 dy = \frac{1}{3} y^3 \quad \dots(2)$$

$\therefore$ the required solution is,

$$(1) + (2) = c \text{ (an arbitrary constant)}$$

$$\text{or} \quad \frac{1}{3} x^3 - ayx + \frac{1}{3} y^3 = c ,$$

$$\text{or} \quad x^3 + y^3 - 3axy = 3c .$$

Example 29: Solve $x dx + y dy + \frac{x dy - y dx}{x^2 + y^2} = 0$.

(Bundelkhand 2007)

Solution: The given equation can be written as

$$\left[x - \frac{y}{x^2 + y^2} \right] dx + \left[y + \frac{x}{x^2 + y^2} \right] dy = 0.$$

Here $M = x - \frac{y}{x^2 + y^2}$ and $N = y + \frac{x}{x^2 + y^2}$.

We have

$$\frac{\partial M}{\partial y} = 0 - \frac{1 \cdot (x^2 + y^2) - y \cdot 2y}{(x^2 + y^2)^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2},$$

and $\frac{\partial N}{\partial x} = 0 + \frac{1 \cdot (x^2 + y^2) - x \cdot 2x}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}.$

Thus $\partial M / \partial y = \partial N / \partial x$ and hence the given equation is exact. Therefore its solution is

$$\int \left[x - \frac{y}{x^2 + y^2} \right] dx + \int y dy = c$$

regarding y as a constant

taking only those terms in N which do not contain x

or $\frac{x^2}{2} - y \cdot \frac{1}{y} \tan^{-1} \frac{x}{y} + \frac{y^2}{2} = c$

or $x^2 - 2 \tan^{-1} (x/y) + y^2 = 2c = k.$

Comprehensive Exercise 7

Solve the following differential equations :

1. $dy/dx = (2x - y)/(x + 2y - 5).$

2. $(2ax + by) y dx + (ax + 2by) x dy = 0.$

3. $(1 + 4xy + 2y^2) dx + (1 + 4xy + 2x^2) dy = 0.$

(Bundelkhand 2004)

4. $x dx + y dy = a^2 \frac{x dy - y dx}{x^2 + y^2}.$

(Agra 2006; Meerut 08; Gorakhpur 11; Bundelkhand 11)

5. $[1 + e^{x/y}] dx + e^{x/y} [1 - (x/y)] dy = 0.$

(Meerut 2004B, 06)

6. $(e^y + 1) \cos x dx + e^y \sin x dy = 0.$

(Agra 2006)

7. $(\sin x \cos y + e^{2x}) dx + (\cos x \sin y + \tan y) dy = 0.$

8. $(y \sin 2x) dx - (1 + y^2 + \cos^2 x) dy = 0.$

9. $[y \{1 + (1/x)\} + \cos y] dx + [x + \log x - x \sin y] dy = 0.$

Answers 7

1. $x^2 - y^2 - xy + 5y = c$.
2. $ayx^2 + bxy^2 = c$
3. $x + 2x^2y + 2xy^2 + y = c$
4. $x^2 + y^2 + 2a^2 \tan^{-1}(x/y) = c$
5. $x + y e^{x/y} = c$
6. $(e^{-y} + 1) \sin x = c$
7. $\frac{1}{2} e^{2x} - \cos x \cos y + \log \sec y = c$
8. $y \cos 2x + 2y + \frac{2}{3} y^3 = c$
9. $y(x + \log x) + x \cos y = c$

1.3 Integrating Factors

(Lucknow 07)

A differential equation which is not exact can sometimes be made exact by multiplying by some suitable function of x and y . Such a function is called an **integrating factor** (I.F.) of the equation.

Methods used to find the integrating factors:

Method 1: Integrating factor found by inspection: Sometimes integrating factors are found by inspection.

The students should **remember** the following exact differentials. These help us in finding the integrating factors.

- (i) $d(xy) = x dy + y dx$,
- (ii) $d\left(\frac{x}{y}\right) = \frac{y dx - x dy}{y^2}$,
- (iii) $d\left(\frac{y}{x}\right) = \frac{x dy - y dx}{x^2}$,
- (iv) $d\left(\frac{x^2}{y}\right) = \frac{2yx dx - x^2 dy}{y^2}$,
- (v) $d\left(\frac{y^2}{x}\right) = \frac{2xy dy - y^2 dx}{x^2}$,
- (vi) $d\left(\tan^{-1} \frac{x}{y}\right) = \frac{y dx - x dy}{x^2 + y^2}$,
- (vii) $d\left(\tan^{-1} \frac{y}{x}\right) = \frac{x dy - y dx}{x^2 + y^2}$,
- (viii) $d\left(\log \frac{x}{y}\right) = \frac{y dx - x dy}{xy}$,
- (ix) $d\left(\log \frac{y}{x}\right) = \frac{x dy - y dx}{xy}$,
- (x) $d\left(\frac{e^x}{y}\right) = \frac{ye^x dx - e^x dy}{y^2}$,
- (xi) $d\{\log \sqrt{(x^2 + y^2)}\} = \frac{x dx + y dy}{x^2 + y^2}$.

Illustrative Examples

Example 30: Solve $(1 + xy) y dx + (1 - xy) x dy = 0$.

Solution: The given equation can be written as

$$(y dx + x dy) + xy^2 dx - x^2 y dy = 0,$$

or $d(yx) + xy^2 dx - x^2 y dy = 0$. [from (i) of article (1.3)]

Dividing out by $x^2 y^2$, we have

$$\frac{d(yx)}{x^2 y^2} + \frac{1}{x} dx - \frac{1}{y} dy = 0 . \quad \dots(1)$$

Now putting $xy = v$ in (1), we have

$$(1/v^2) du + (1/x) dx - (1/y) dy = 0 .$$

Now integrating each term, we get

$$-(1/v) + \log x - \log y = c, \text{ where } c \text{ is a constant}$$

or $-(1/xy) + \log x - \log y = c, \quad (\because v = xy)$

or $\log(x/y) = c + (1/xy)$, is the required solution.

Example 31: Solve $y dx - x dy + (1 + x^2) dx + x^2 \sin y dy = 0$. (Lucknow 2005)

Solution: Dividing each term by x^2 , the given equation becomes

$$\frac{y dx - x dy}{x^2} + \left(\frac{1}{x^2} + 1 \right) dx + \sin y dy = 0$$

or $-\frac{x dy - y dx}{x^2} + \left\{ \frac{1}{x^2} + 1 \right\} dx + \sin y dy = 0$

or $-d\left\{ \frac{y}{x} \right\} + \left\{ \frac{1}{x^2} + 1 \right\} dx + \sin y dy = 0$, [Refer article 1.3, result (iii)]

Integrating each term, we have

$$-\frac{y}{x} - \frac{1}{x} + x - \cos y = -c, \quad \text{or} \quad \frac{y}{x} + \frac{1}{x} - x + \cos y = c.$$

Comprehensive Exercise 8

Solve the following differential equations :

1. $x dy - y dx = xy^2 dx$.

2. $x dy - y dx + 2x^3 dx = 0$.

3. $(y^2 e^x + 2xy) dx - x^2 dy = 0$.

(Gorakhpur 2008)

4. $y (axy + e^x) dx - e^x dy = 0$.

5. $y (2x^2 y + e^x) dx - (e^x + y^3) dy = 0$.

(Gorakhpur 2009; Avadh 09)

6. $y \sin 2x dx = (1 + y^2 + \cos^2 x) dy$.

7. $x dy - y dx = (x^2 + y^2) dx$.

8. $x dx + y dy + (x^2 + y^2) dy = 0$.

9. $(x^2 + y^2 + a^2) y dy + (x^2 + y^2 - a^2) x dx = 0$.

(Rohilkhand 2006, 07)

Answers 8

1. $yx^2 + 2x = 2cy$.

2. $y + x^3 = cx$.

3. $e^x + (x^2/y) = c$.

4. $\frac{1}{2}ax^2 + (e^x/y) = c$.

5. $\frac{2}{3}x^3 - \frac{1}{2}y^2 + (e^x/y) = c$.

6. $-y \cos^2 x = y + \frac{1}{3}y^3 + c$.

7. $\tan^{-1}(y/x) = x + c$.

8. $x^2 + y^2 = e^{c-2y}$.

9. $(x^2 + y^2)^2 + 2a^2(y^2 - x^2) = c$.

Method 2: If the equation $M dx + N dy = 0$ is of the form

$$[f(x, y)] y dx + [F(x, y)] x dy = 0,$$

and $Mx - Ny \neq 0$, then

$$1/(Mx - Ny) \text{ is an integrating factor.}$$

Note: If $Mx - Ny = 0$, then $Mx = Ny$ or $M/y = N/x$

i.e., the differential equation $M dx + N dy = 0$ reduces to $y dx + x dy = 0$ whose solution is $xy = c$.

Example 32: Solve $(x^3 y^3 + x^2 y^2 + xy + 1) y dx + (x^3 y^3 - x^2 y^2 - xy + 1) x dy = 0$.

Solution: Here

$$\begin{aligned} Mx - Ny &= (x^3 y^3 + x^2 y^2 + xy + 1) y \cdot x - (x^3 y^3 - x^2 y^2 - xy + 1) xy \\ &= 2(x^3 y^3 + x^2 y^2) = 2x^2 y^2 (xy + 1). \end{aligned}$$

$$\therefore \text{I.F.} = 1 / \{2x^2 y^2 (xy + 1)\}.$$

The given equation may be written as

$$[x^2 y^2 (xy + 1) + (xy + 1)] y dx + [(x^3 y^3 + 1) - xy (xy + 1)] x dy = 0.$$

Now multiplying by I.F. this becomes

$$\frac{[x^2 y^2 (xy + 1) + (xy + 1)] y dx}{2x^2 y^2 (xy + 1)} + \frac{[(xy + 1) \{(x^2 y^2 - xy + 1) - xy\}] x dy}{2x^2 y^2 (xy + 1)} = 0$$

$$\text{or} \quad \left[\frac{x^2 y^2 + 1}{x^2 y^2} \right] y dx + \left[\frac{(x^2 y^2 - 2xy + 1)}{x^2 y^2} \right] x dy = 0$$

$$\text{or} \quad (y dx + x dy) + \frac{y dx + x dy}{x^2 y^2} - \frac{2x^2 y}{x^2 y} dy = 0$$

$$\text{or} \quad d(xy) + \frac{d(xy)}{x^2 y^2} - \frac{2}{y} dy = 0.$$

Now integrating each term, we get

$$xy + (-1/xy) - 2 \log y = c.$$

Comprehensive Exercise 9

Solve the following differential equations :

1. $(xy^2 + 2x^2y^3) dx + (x^2y - x^3y^2) dy = 0$.
2. $(x^2y^2 + xy + 1) y dx + (x^2y^2 - xy + 1) x dy = 0$.
3. $(xy \sin xy + \cos xy) y dx + (xy \sin xy - \cos xy) x dy = 0$.

Answers 9

1. $x^2 = cy e^{1/(xy)}$
2. $xy + \log x - \log y - 1/(xy) = c$
3. $x \sec(xy) = cy$

Method 3: When $Mx + Ny \neq 0$ and the equation is homogeneous, then the integrating factor of $M dx + N dy = 0$, is $1/(Mx + Ny)$.

Example 33: Solve $x^2 y dx - (x^3 + y^3) dy = 0$.

Solution: Here $Mx + Ny = x^3 y - x^3 y - y^4 = -y^4 \neq 0$

and the equation is homogeneous.

$\therefore$ I.F. $= 1/(Mx + Ny) = -1/y^4$.

Multiplying the given equation by the I.F., we get

$$-\frac{x^2}{y^3} dx + \frac{x^3 + y^3}{y^4} dy = 0, \quad \text{or} \quad \frac{dy}{y} = \frac{x^2}{y^3} dx - \frac{x^3}{y^4} dy$$

or
$$\frac{dy}{y} = \frac{1}{3} d\left(\frac{x^3}{y^3}\right).$$

$\therefore$ integrating each term, we get

$$\log y = \frac{1}{3} \frac{x^3}{y^3} + \log c, \quad \text{or} \quad y = ce^{x^3/3y^3}.$$

Method 4: If $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)$ is a function of x alone, say $f(x)$, then the integrating factor for $M dx + N dy = 0$ is $e^{\int f(x) dx}$.

Example 34: Solve $(y + \frac{1}{3}y^3 + \frac{1}{2}x^2) dx + \frac{1}{4}(x + xy^2) dy = 0$.

(Lucknow 2007)

Solution: Here $M = y + \frac{1}{3}y^3 + \frac{1}{2}x^2$ and $N = \frac{1}{4}(x + xy^2)$.

$\therefore \quad \frac{\partial M}{\partial y} = 1 + y^2 \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{1}{4}(1 + y^2).$

$$\begin{aligned}\therefore \quad \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) &= \frac{4}{x(1+y^2)} \{ (1+y^2) - \frac{1}{4}(1+y^2) \} \\ &= \frac{4}{x(1+y^2)} \cdot \frac{3}{4} (1+y^2) = \frac{3}{x}, \text{ which is a function of } x \text{ alone, say } f(x).\end{aligned}$$

$$\therefore \quad \text{I.F.} = e^{\int f(x) dx} = e^{\int (3/x) dx} = e^{3 \log x} = e^{\log x^3} = x^3.$$

Multiplying the given diff. equation by the I.F. x^3 , we get

$$(x^3 y + \frac{1}{3} x^3 y^3 + \frac{1}{2} x^5) dx + \frac{1}{4} (x^4 + x^4 y^2) dy = 0,$$

which is an exact diff. equation and its solution is

$$\int M dx \text{ (treating } y \text{ as a constant)} + \int N dy \text{ (taking in } N \text{ terms without } x) = c$$

$$\text{i.e.,} \quad \frac{x^4 y}{4} + \frac{x^4 y^3}{12} + \frac{x^6}{12} = c,$$

$$\text{or} \quad 3x^4 y + x^4 y^3 + x^6 = 12c = k.$$

Method 5: If $\frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$ is a function of y alone, say $f(y)$, then the integrating factor for $M dx + N dy = 0$ is $e^{\int f(y) dy}$.

Example 35: Solve $(xy^3 + y) dx + 2(x^2 y^2 + x + y^4) dy = 0$.

(Lucknow 2006)

Solution: Here $M = xy^3 + y$ and $N = 2(x^2 y^2 + x + y^4)$.

$$\therefore \quad \frac{\partial M}{\partial y} = 3xy^2 + 1 \text{ and } \frac{\partial N}{\partial x} = 2(2xy^2 + 1).$$

$$\begin{aligned}\therefore \quad \frac{1}{M} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) &= \frac{1}{xy^3 + y} \{ 4xy^2 + 2 - 3xy^2 - 1 \} = \frac{1}{y(xy^2 + 1)} (xy^2 + 1) \\ &= \frac{1}{y}, \text{ which is a function of } y \text{ alone, say } f(y).\end{aligned}$$

$$\therefore \quad \text{I.F.} = e^{\int f(y) dy} = e^{\int (1/y) dy} = e^{\log y} = y.$$

Multiplying the given equation with the I.F. y , we have

$$(xy^4 + y^2) dx + 2(x^2 y^3 + xy + y^5) dy = 0,$$

which is an exact differential equation and its solution is

$$\begin{aligned}&\int M dx \text{ (treating } y \text{ as a constant)} \\ &+ \int N dy \text{ (taking in } N \text{ only those terms which do not contain } x) \\ &= c, \text{ (a constant)}\end{aligned}$$

$$\text{i.e.,} \quad \int (xy^4 + y^2) dx \text{ (} y \text{ constant)} + \int 2y^5 dy = c$$

$$\text{or} \quad \frac{1}{2} x^2 y^4 + xy^2 + \frac{1}{3} y^6 = c.$$

Comprehensive Exercise 10

Solve the following differential equations :

1. $(x^2 y - 2xy^2) dx - (x^3 - 3x^2 y) dy = 0$. (Lucknow 2011)
2. $(x^2 + y^2) dx - 2xy dy = 0$. (Gorakhpur 2008)
3. $(3x^2 y^4 + 2xy) dx + (2x^3 y^3 - x^2) dy = 0$.
4. $(xy^2 - x^2) dx + (3x^2 y^2 + x^2 y - 2x^3 + y^2) dy = 0$.

Answers 10

1. $(x/y) - 2 \log x + 3 \log y = c$
2. $x^2 - y^2 = cx$
3. $x^3 y^2 + (x^2 / y) = c$
4. $e^{6y} [x^2 (\frac{1}{2} y^2 - \frac{1}{3} x) + (\frac{1}{6} y^2 - \frac{1}{18} y + \frac{1}{108})] = 0$

Method 6: If the equation is of the form

$$x^a y^b (my dx + nx dy) + x^c y^d (py dx + qx dy) = 0, \quad \dots (A)$$

where a, b, m, n, c, d, p, q are constants, then the *integrating factor* is $x^h y^k$, where h, k can be obtained by applying the condition that after multiplication by $x^h y^k$, the equation (A) must become exact.

Example 36: Solve $(y^2 + 2x^2 y) dx + (2x^3 - xy) dy = 0$.

Solution: The given equation can be written as

$$y (y dx - x dy) + 2x^2 (y dx + x dy) = 0.$$

This equation is of the form (A), as mentioned above. So let the possible integrating factor be $x^h y^k$.

Multiplying the given equation by the proposed I.F. $x^h y^k$, we have

$$(x^h y^{k+2} + 2x^{h+2} y^{k+1}) dx + (2x^{h+3} y^k - x^{h+1} y^{k+1}) dy = 0. \quad \dots (1)$$

Now $(\partial M / \partial y) = (k+2) x^h y^{k+1} + 2(k+1) x^{h+2} y^k$

and $(\partial N / \partial x) = 2(h+3) x^{h+2} y^k - (h+1) x^h y^{k+1}.$

If the equation (1) is exact, then

$$(\partial M / \partial y) = (\partial N / \partial x).$$

$\therefore$ equating the coefficients of $x^h y^{k+1}$ and $x^{h+2} y^k$ on both sides, we get

$$k + 2 = -h - 1 \quad \text{i.e.,} \quad h + k + 3 = 0$$

and $2k + 2 = 2h + 6 \quad \text{i.e.,} \quad h - k + 2 = 0 .$

Solving these, we get $h = -5/2$ and $k = -1/2$.

$\therefore$ the integrating factor is $x^h y^k = x^{-5/2} y^{-1/2}$.

With these values of h and k the equation (1) becomes

$$(x^{-5/2} y^{3/2} + 2x^{-1/2} y^{1/2}) dx + (2x^{1/2} y^{-1/2} - x^{-3/2} y^{1/2}) dy = 0 .$$

This differential equation is of the form $M dx + N dy = 0$ and is exact as can be seen by verifying the condition $\partial M / \partial y = \partial N / \partial x$.

Hence its solution is

$$\begin{aligned} & \int M dx \text{ (treating } y \text{ as a constant)} \\ & + \int N dy \text{ (taking in } N \text{ only those terms which do not contain } x) \\ & = c, \text{ (a constant)} \end{aligned}$$

i.e., $-\frac{2}{3} x^{-3/2} y^{3/2} + 4x^{1/2} y^{1/2} + 0 = c .$

Hence $-\frac{2}{3} x^{-3/2} y^{3/2} + 4x^{1/2} y^{1/2} = c$ is the required solution.

(Gorakhpur 2007)

Comprehensive Exercise 11

Solve the following differential equations :

- $(2y dx + 3x dy) + 2xy (3y dx + 4x dy) = 0 .$
- $(3x + 2y^2) y dx + 2x (2x + 3y^2) dy = 0 .$
- $x (3y dx + 2x dy) + 8y^4 (y dx + 3x dy) = 0 .$

(Lucknow 2010)

Answers 11

- $x^2 y^3 + 2x^3 y^4 = c$
- $x^3 y^4 + x^2 y^6 = c$
- $x^3 y^2 + 4y^6 x^2 = c$

1.4 Change of Variables

In some cases a suitable substitution (change of variables) reduces a given differential equation to one or the other of the forms already discussed and hence the equation can be solved.

Illustrative Examples

Example 37: Solve $\frac{x dx + y dy}{x dy - y dx} = \sqrt{\frac{a^2 - x^2 - y^2}{x^2 + y^2}}$.

(Rohilkhand 2005; Purvanchal 09; Avadh 09; Bundelkhand 03)

Solution: Put $x = r \cos \theta$ and $y = r \sin \theta$. Then

$$x^2 + y^2 = r^2, \quad \dots(1)$$

$$\text{and} \quad y/x = \tan \theta. \quad \dots(2)$$

Differentiating (1), we get

$$2x dx + 2y dy = 2r dr \quad \text{or} \quad x dx + y dy = r dr.$$

Differentiating (2), we get

$$\frac{x dy - y dx}{x^2} = \sec^2 \theta d\theta,$$

$$\text{or} \quad x dy - y dx = x^2 \sec^2 \theta d\theta = r^2 \cos^2 \theta \sec^2 \theta d\theta = r^2 d\theta.$$

Substituting these values in the given differential equation, we get

$$\frac{r dr}{r^2 d\theta} = \sqrt{\frac{a^2 - r^2}{r^2}} = \frac{\sqrt{a^2 - r^2}}{r}.$$

$$\therefore \quad \frac{dr}{d\theta} = \sqrt{a^2 - r^2} \quad \text{or} \quad \frac{dr}{\sqrt{a^2 - r^2}} = d\theta, \text{ [separating the variables]}$$

Integrating, we get

$$\sin^{-1}(r/a) = \theta + c, \quad \text{or} \quad r = a \sin(\theta + c)$$

$$\text{or} \quad \sqrt{x^2 + y^2} = a \sin\{\tan^{-1}(y/x) + c\}.$$

$$[\because r = \sqrt{x^2 + y^2} \text{ and } \theta = \tan^{-1}(y/x)]$$

1.5 Geometrical Problems

Remember the following formulae of the differential calculus:

$$\text{Length of the tangent} = y \sqrt{1 + (dx/dy)^2}.$$

$$\text{Length of the subtangent} = y (dx/dy) = y / (dy/dx).$$

$$\text{Length of the normal} = y \sqrt{1 + (dy/dx)^2}.$$

$$\text{Length of the subnormal} = y (dy/dx).$$

Intercept made by the tangent on the x -axis is $x - y (dx/dy)$, and on the y -axis is $y - x (dy/dx)$.

Angle ϕ between the radius vector and the tangent is given by

$$\tan \phi = r (d\theta/dr), \quad \text{or} \quad \cot \phi = (1/r) (dr/d\theta).$$

Length of the polar subtangent = $r^2 (d\theta/dr)$.

Length of the polar subnormal = $(dr/d\theta)$.

Radius of curvature $\rho = [1 + (dy/dx)^2]^{3/2} / (d^2y/dx^2)$.

Illustrative Examples

Example 38: Find the curves in which the polar subnormal is of constant length.

Solution: Let the constant length of the polar subnormal be a .

But the length of the polar subnormal = $dr/d\theta$.

$\therefore$ as given in the question, $dr/d\theta = a$.

This is the differential equation of the required curves.

To solve, put it into the form $dr = a d\theta$, [separating the variables]

$\therefore$ integrating, we get $r = a\theta + c$, where c is an arbitrary constant.

This is the polar equation of the required family of curves.

Example 39: Find the curves for which the sum of the reciprocals of the radius vector and the polar subtangent is constant.

Solution: The polar subtangent = $r^2 (d\theta/dr)$.

$\therefore$ as given in the question, $\frac{1}{r} + \frac{1}{r^2} \frac{dr}{d\theta} = \lambda$, where λ is a constant.

This is the differential equation of the required curves. This equation may be written as

$$\frac{dr}{d\theta} = r^2 \left(\lambda - \frac{1}{r} \right) = r(\lambda r - 1)$$

$$\text{or} \quad d\theta = \frac{dr}{r(\lambda r - 1)} = \left(\frac{\lambda}{\lambda r - 1} - \frac{1}{r} \right) dr, \quad [\text{separating the variables}]$$

$\therefore$ integrating, we have

$$\theta + c = \log(\lambda r - 1) - \log r, \text{ where } c \text{ is an arbitrary constant}$$

$$\text{or} \quad \theta + c = \log \{(\lambda r - 1)/r\}, \quad \text{or} \quad (\lambda r - 1)/r = e^{\theta + c}.$$

Hence $\lambda r - 1 = re^{\theta + c}$ is the equation of the required family of curves.

Example 40: Find the curves in which the cartesian subtangent varies as the abscissa.

Solution: The cartesian subtangent is $y/(dy/dx)$.

$\therefore$ as given in the question $y/(dy/dx) = \lambda x$, where λ is a constant

or $\lambda dx = \lambda x dy$, or $(dx/x) = \lambda (dy/y)$, separating the variables.

$\therefore$ integrating, we have $\log x = \lambda \log y + \log c$, where c is an arbitrary constant

Hence $\log x = \log (y^\lambda \cdot c)$ or $x = cy^\lambda$ is the required family of curves.

Example 41: Show that the curve in which the angle between the tangent and the radius vector at every point is half of the vectorial angle is a cardioid.

Solution: As given, $\phi = \frac{1}{2}\theta$, where θ is the vectorial angle.

$$\therefore \quad \tan \phi = \tan \frac{1}{2}\theta. \text{ But } \tan \phi = r (d\theta/dr).$$

$$\therefore \quad r (d\theta/dr) = \tan \frac{1}{2}\theta, \quad \text{or} \quad (dr/r) = (\cot \frac{1}{2}\theta) d\theta,$$

in which the variables have been separated.

$$\therefore \quad \text{integrating, we have } \log r = 2 \log \left(\sin \frac{1}{2}\theta \right) + \log c,$$

where c is an arbitrary constant

$$\text{or} \quad \log r = \log \left(\sin^2 \frac{1}{2}\theta \right) + \log c, \quad \text{or} \quad \log (r/c) = \log \left(\sin^2 \frac{1}{2}\theta \right)$$

$$\text{or} \quad (r/c) = \sin^2 \frac{1}{2}\theta = \frac{1}{2}(1 - \cos \theta), \quad \text{or} \quad r = \frac{1}{2}c(1 - \cos \theta)$$

$$\text{or} \quad r = a(1 - \cos \theta), \text{ where } a = \frac{1}{2}c. \text{ This is a cardioid.}$$

Example 42: Find the curve for which the tangent at each point makes a constant angle α with the radius vector.

Solution: As given $\phi = \alpha$ (a constant)

$$\text{or} \quad \tan \phi = \tan \alpha. \text{ But } \tan \phi = r (d\theta/dr).$$

$$\therefore \quad r (d\theta/dr) = \tan \alpha, \quad \text{or} \quad (dr/r) = (\cot \alpha)d\theta.$$

Now the variables have been separated. Therefore integrating, we have

$$\log r = \theta \cot \alpha + \log c, \text{ where } c \text{ is an arbitrary constant}$$

$$\text{or} \quad \log r - \log c = \theta \cot \alpha,$$

$$\text{or} \quad \log (r/c) = \theta \cot \alpha$$

$$\text{or} \quad r/c = e^{\theta \cot \alpha}.$$

Hence $r = ce^{\theta \cot \alpha}$ is the required curve.

Example 43: Show that the curve in which the slope of the tangent at any point equals the ratio of the abscissa to the ordinate of the point is a rectangular hyperbola.

Solution: The slope of the tangent at any point $(x, y) = \tan \psi = dy/dx$.

$$\therefore \quad \text{as given, } \frac{dy}{dx} = \frac{\text{abscissa}}{\text{ordinate}} = \frac{x}{y}$$

$$\text{or} \quad y dy = x dx, \text{ in which the variables have been separated.}$$

$$\therefore \quad \text{integrating, we have } \frac{1}{2}y^2 = \frac{1}{2}x^2 + c, \text{ where } c \text{ is an arbitrary constant.}$$

Hence $y^2 - x^2 = 2c = a^2$ (say) is the required curve which is the equation of a rectangular hyperbola.

Example 44: Find the family of curves whose tangent forms an angle $\pi/4$ with the hyperbola $xy = c$.

Solution: Let m_1 and m_2 be the gradients of the tangents of the required family of curves and the given hyperbola respectively. Since the angle between these tangents is given to be $\frac{1}{4}\pi$, therefore

$$\tan(\pi/4) = \frac{m_1 - m_2}{1 + m_1 m_2}, \quad \text{or} \quad 1 = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$\text{or} \quad 1 + m_1 m_2 = m_1 - m_2 \quad \text{or} \quad m_1 = (1 + m_2) / (1 - m_2). \quad \dots(1)$$

Now $m_1 =$ gradient of the tangent of the required family of curves $= dy/dx$

and $m_2 = (dy/dx)$ for hyperbola $xy = c$.

$$\therefore m_2 = \frac{d}{dx}\left(\frac{c}{x}\right) = -\frac{c}{x^2}, \quad \left[\because y = \frac{c}{x}\right]$$

Hence from (1), we have

$$\frac{dy}{dx} = \frac{1 + (-c/x^2)}{1 - (-c/x^2)} = \frac{x^2 - c}{x^2 + c} = \left(1 - \frac{2c}{x^2 + c}\right) \text{ or } dy = \left(1 - \frac{2c}{x^2 + c}\right) dx,$$

in which the variables have been separated.

$$\therefore \text{integrating, we have } y = x - \frac{2c}{\sqrt{c}} \tan^{-1}\left(\frac{x}{\sqrt{c}}\right) + c_1,$$

where c_1 is an arbitrary constant. This is the required family of curves.

Comprehensive Exercise 12

1. By the substitution $y^2 = v - x$ reduce the equation $y^3 (dy/dx) + x + y^2 = 0$ to the homogeneous form and hence solve the equation.
2. Find the curve in which the polar subtangent is constant.
3. Find the equation of the family of curves for which the sum of the reciprocals of the radius vector and the polar subnormal is constant.
4. Find the curve in which the cartesian sub-normal is equal to the abscissa.
5. Find the equation of the curve for which the cartesian subtangent varies as the reciprocal of the square of the abscissa.
6. Show that the parabola is the only curve in which the sub-normal is constant.
7. Show that the curve for which the normal at every point passes through a fixed point is a circle.
8. Find the equation of the curve in which the perpendicular from the origin on any tangent is equal to the abscissa of the point of contact.

9. The normal PN to a curve meets the x -axis in N . If the distance of N from the origin is twice the abscissa of P , prove that the curve is a rectangular hyperbola.
10. The tangent of any point P of a curve meets the x -axis in Q . If Q is on the positive side of the origin O and $OP = OQ$, show that the family of curves having this property are parabolas whose common axis is the x -axis.

Answers 12

1. $\tan^{-1} \{(y^2 + x)/x\} - \frac{1}{2} \log \{x^2 + (y^2 + x)^2\} = \log c$
2. $r(\theta + c) + a = 0$
3. $\theta = \lambda r - \log r + c$
4. $y^2 - x^2 = c^2$
5. $\lambda \log y = \frac{1}{3} x^3 + c$
6. $y^2 = 2\lambda \{x + (c/2\lambda)\}$
8. $y^2 + x^2 = cx$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. The order of the differential equation

$$\frac{d^4 y}{dx^4} - 3 \left(\frac{d^3 y}{dx^3} \right)^2 + 4 \frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0 \text{ is}$$

(Rohilkhand 2005)

- (a) 3
 - (b) 6
 - (c) 4
 - (d) 2
2. The integrating factor of the differential equation $\cos^2 x \frac{dy}{dx} + y = \tan x$ is
- (a) $\sec^2 x$
 - (b) $e^{\tan x}$
 - (c) $\cos^2 x$
 - (d) $\tan x$
3. The integrating factor of the differential equation $\frac{dy}{dx} + y \cot x = 2 \cos x$ is
- (a) $\sin x$
 - (b) $\log \sin x$
 - (c) $\cot x$
 - (d) $\cos x$
4. The differential equation $M dx + N dy = 0$, where M and N are functions of x and y , is exact if
- (a) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
 - (b) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$
 - (c) $M + N = 0$
 - (d) $M = N$
- (Avadh 2005)

5. The complete solution of the differential equation

$$(x^2 - ay) dx - (ax - y^2) dy = 0 \text{ is}$$

- (a) $x^3 + y^3 + 3axy = c$ (b) $x^3 + y^3 - 3axy = 4$
(c) $x^3 + y^3 - 3axy = c$ (d) $x^3 + y^3 - 3axy = 0$

6. The integrating factor of differential equation $\frac{dx}{dy} + P(y)x = Q(y)$ is
(Bundelkhand 2007)

- (a) $e^{\int P dy}$ (b) $e^{\int P dx}$
(c) $e^{\int Q dx}$ (d) $e^{\int -P dx}$

7. The differential equation $(x^2 + y^2 + a^2) y \frac{dy}{dx} + x(x^2 + y^2 - a^2) = 0$ is

(Rohilkhand 2007)

- (a) variables separable (b) linear
(c) Homogeneous (d) Exact

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. The order of the differential equation

$$\frac{d^3 y}{dx^3} - 2 \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 2y = 0 \text{ is } \dots\dots$$

2. The degree of the differential equation $\left(\frac{dy}{dx}\right)^3 - 4xy \frac{dy}{dx} + 8y^2 = 0$ is

3. The order and degree of the differential equation $\left(\frac{d^2 y}{dx^2}\right)^2 - \left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = 0$ is
..... and
(Rohilkhand 2006)

4. A differential equation of the form $\frac{dy}{dx} = \frac{f_1(x, y)}{f_2(x, y)}$, where $f_1(x, y)$ and $f_2(x, y)$ are homogeneous functions of x and y of the same degree, is called a

5. To solve the differential equation $x^2 y dx - (x^3 + y^3) dy = 0$, we put.....

6. To solve the differential equation $\frac{dy}{dx} = \frac{2x + y + 1}{4x + 2y - 1}$, we put

7. A differential equation is said to be when the dependent variable y and all its derivatives occur in the first degree only and are not multiplied together.

8. The integrating factor of the differential equation $\frac{dy}{dx} + Py = Q$, where P and Q are functions of x only, is

9. The integrating factor of the differential equation $(x^2 - 1) \frac{dy}{dx} + 2xy = 1$ is

10. The integrating factor of the differential equation $\frac{dy}{dx} + 2y \tan x = \sin x$ is

11. The ordinary differential equation $M dx + N dy = 0$, where M and N are functions of x and y , is exact if and only if
(Meerut 2003; Bundelkhand 08; Rohilkhand 10)
12. A differential equation which is not exact can sometimes be made exact by multiplying by some suitable function of x and y . Such a function is called an of the equation.
13. The differential equation $\frac{dx}{dy} + \frac{1}{1+y^2} \cdot x = \frac{\tan^{-1} y}{1+y^2}$ is linear with as dependent variable.
14. If the differential equation $M dx + N dy = 0$ is homogeneous and if $Mx + Ny \neq 0$, then the integrating factor of this differential equation is
(Bundelkhand 2007)
15. The differential equation $(x + y) dy = a^2 dx$ is linear with as independent variable.
16. A solution of differential equation of three order has constants.
(Agra 2007)

True or False

Write 'T' for true and 'F' for false statement.

1. The differential equation $y^2 dx + (xy + x^2) dy = 0$ is homogeneous.
2. The order of the differential equation $\frac{d^3 y}{dx^3} - 7 \left(\frac{dy}{dx} \right)^2 + 6y = 0$ is 2.
3. The degree of the differential equation $\frac{d^2 y}{dx^2} + 4 \left(\frac{dy}{dx} \right)^2 - 5y = 0$ is 2.
4. The differential equation $x^2 \frac{dy}{dx} + y = 1$ is linear with y as dependent variable.
5. The integrating factor of the differential equation $\frac{dy}{dx} + y \sec x = \tan x$ is $(\sec x + \tan x)$.
6. The differential equation $(x + y + 1) \frac{dy}{dx} = 1$ is linear with y as dependent variable.
7. The differential equation $(4x + 3y + 1) dx + (3x + 2y + 1) dy = 0$ is exact.
8. The integrating factor of the differential equation $\frac{dy}{dx} + \frac{y}{x} = \sin x^2$ is $\log x$.
9. The differential equation $\frac{dy}{dx} + \frac{1}{x} y = x^2 y^6$ is linear with y as dependent variable.
10. The differential equation $x \frac{dy}{dx} + y = x^2 y^4$ can be made linear by putting $\frac{1}{y^3} = v$.

11. A differential equation of the form $\frac{dy}{dx} + Py = Qy^n$,

where P and Q are functions of x alone, is called Bernoulli's equation.

(Meerut 2003)

12. If the differential equation $M dx + N dy = 0$ is exact, its solution is

(Meerut 2003)

$$\int M dx + \int N dy = c$$

treating y as constant taking only those terms in N which do not contain x

Answers

Multiple Choice Questions

- | | | | | |
|--------|--------|--------|--------|--------|
| 1. (c) | 2. (b) | 3. (a) | 4. (a) | 5. (c) |
| 6. (a) | 7. (d) | | | |

Fill in the Blank(s)

- | | | | |
|--------------|-------------------------|---|-------------------------|
| 1. 3 | 2. 3 | 3. 2, 2 | 4. homogeneous equation |
| 5. $y = vx$ | 6. $2x + y = v$ | 7. linear | 8. $e^{\int P dx}$ |
| 9. $x^2 - 1$ | 10. $\sec^2 x$ | 11. $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ | 12. integrating factor |
| 13. x | 14. $\frac{1}{Mx + Ny}$ | 15. y | 16. 3 |

True or False

- | | | | | |
|-------|-------|------|------|-------|
| 1. T | 2. F | 3. F | 4. T | 5. T |
| 6. F | 7. T | 8. F | 9. F | 10. T |
| 11. T | 12. T | | | |



Chapter

2



Differential Equations of The First Order But Not of The First Degree

2.1 Introduction

In the present chapter we shall discuss the solutions of differential equations which are of the first order but are of degree higher than one. Such differential equations will contain only the first differential coefficient dy/dx but it will occur in a degree higher than one. It is usual to denote dy/dx by p . The general form of such a differential equation is then

$$p^n + A_1 p^{n-1} + A_2 p^{n-2} + \dots + A_{n-1} p + A_n = 0,$$

where $A_1, A_2, \dots, A_n$ are some functions of x and y .

Now we shall consider the various methods of solving the differential equations of the above type.

2.2 Equations Solvable for p

Suppose a differential equation of first order and of degree n can be solved for *p i.e.*, it can be resolved into n linear factors in p of the type

$$\{p - f_1(x, y)\} \{p - f_2(x, y)\} \dots \{p - f_n(x, y)\} = 0.$$

We can equate each factor to zero and the resulting differential equations of the first order and first degree can be solved. Let their solutions be

$$\phi_1(x, y, c_1) = 0, \phi_2(x, y, c_2) = 0, \dots, \phi_n(x, y, c_n) = 0,$$

where $c_1, c_2, \dots, c_n$ are arbitrary constants. There is no loss of generality if we replace the arbitrary constants $c_1, c_2, \dots, c_n$ by a single arbitrary constant c because in any of the above n solutions c is free to take any real value. Thus the n solutions of the given differential equation are

$$\phi_1(x, y, c) = 0, \phi_2(x, y, c) = 0, \dots, \phi_n(x, y, c) = 0.$$

Combining the above equations, we get a single composite solution as

$$\phi_1(x, y, c) \phi_2(x, y, c) \dots \phi_n(x, y, c) = 0.$$

Illustrative Examples

Example 1: Solve $p^2 - 7p + 12 = 0$.

(Bundelkhand 2008; Kanpur 04)

Solution: Resolving into linear factors, the given differential equation can be written as $(p - 3)(p - 4) = 0$.

Its component equations are $p = 3, p = 4$.

Solving the differential equation $p = 3$ i.e., $dy/dx = 3$, we get $y = 3x + c$. Also the solution of the differential equation $p = 4$ is $y = 4x + c$.

So the solutions of the given differential equation are $y = 3x + c, y = 4x + c$.

The single combined solution is

$$(y - 3x - c)(y - 4x - c) = 0.$$

Example 2: Solve $p^2 + 2py \cot x = y^2$.

(Gorakhpur 2005; Rohilkhand 09; Kanpur 08; Bundelkhand 04)

Solution: The given differential equation is

$$p^2 + 2py \cot x - y^2 = 0.$$

Solving for p , we get

$$\begin{aligned} p = \frac{dy}{dx} &= \frac{-2y \cot x \pm \sqrt{(4y^2 \cot^2 x + 4y^2)}}{2} \\ &= -y \cot x \pm y \operatorname{cosec} x = y(-\cot x \pm \operatorname{cosec} x). \end{aligned}$$

Thus the component equations are

$$\frac{dy}{dx} = y(-\cot x + \operatorname{cosec} x), \quad \dots(1)$$

$$\text{and} \quad \frac{dy}{dx} = -y(\cot x + \operatorname{cosec} x). \quad \dots(2)$$

In each of the above differential equations, the variables are separable.

From (1), separating the variables, we have

$$\frac{dy}{y} = (-\cot x + \operatorname{cosec} x) dx.$$

Integrating, we get

$$\log y - \log c = -\log \sin x + \log \tan \frac{1}{2} x$$

$$\begin{aligned} \text{or} \quad \log \left(\frac{y}{c} \right) &= \log \left\{ \frac{\tan \frac{1}{2} x}{\sin x} \right\} = \log \left\{ \frac{\sin \frac{1}{2} x / \cos \frac{1}{2} x}{2 \sin \frac{1}{2} x \cos \frac{1}{2} x} \right\} \\ &= \log \left\{ \frac{1}{2 \cos^2 \frac{1}{2} x} \right\} = \log \left\{ \frac{1}{1 + \cos x} \right\}. \end{aligned}$$

$$\therefore y/c = 1/(1 + \cos x) \quad \text{or} \quad y = c/(1 + \cos x). \quad \dots(3)$$

From (2), separating the variables, we have

$$\frac{dy}{y} = -(\cot x + \operatorname{cosec} x) dx.$$

Integrating, we get

$$\log y - \log c = -(\log \sin x + \log \tan \frac{1}{2} x)$$

$$\begin{aligned} \text{or} \quad \log (y/c) &= -\log \{ (\sin x) (\tan \frac{1}{2} x) \} \\ &= -\log \{ 2 \sin \frac{1}{2} x \cos \frac{1}{2} x \cdot (\sin \frac{1}{2} x / \cos \frac{1}{2} x) \} = -\log (2 \sin^2 \frac{1}{2} x) \\ &= -\log (1 - \cos x) = \log (1 - \cos x)^{-1} = \log \{ 1/(1 - \cos x) \}. \\ \therefore y/c &= 1/(1 - \cos x) \quad \text{or} \quad y = c/(1 - \cos x). \quad \dots(4) \end{aligned}$$

Thus the solutions of the given differential equation are given by (3) and (4). The single combined solution is

$$\left(y - \frac{c}{1 + \cos x} \right) \left(y - \frac{c}{1 - \cos x} \right) = 0.$$

Example 3: Solve $(p - xy)(p - x^2)(p - y^2) = 0$.

Solution: Equating each factor to zero, the component equations are

$$p - xy = 0, \quad \dots(1) \quad p - x^2 = 0, \quad \dots(2)$$

$$\text{and} \quad p - y^2 = 0. \quad \dots(3)$$

From (1), $p = xy$ or $dy/dx = xy$ or $(1/y) dy = x dx$.

Integrating, we get

$$\log y = \frac{1}{2} x^2 + \log c \quad \text{or} \quad \log (y/c) = \frac{1}{2} x^2$$

$$\text{or} \quad y/c = e^{\frac{1}{2} x^2} \quad \text{or} \quad y = ce^{\frac{1}{2} x^2}. \quad \dots(4)$$

From (2), $p = x^2$ or $dy/dx = x^2$ or $dy = x^2 dx$.

Integrating, we get

$$y = \frac{1}{3}x^3 + \frac{1}{3}c \quad \text{or} \quad 3y - x^3 = c. \quad \dots(5)$$

$$\text{From (3), } p = y^2 \quad \text{or} \quad dy/dx = y^2 \quad \text{or} \quad (1/y^2) dy = dx.$$

Integrating, we have

$$-(1/y) = x + c \quad \text{or} \quad xy + cy + 1 = 0. \quad \dots(6)$$

Now (4), (5) and (6) are the solutions of the given differential equation. The single combined solution is

$$\left(y - ce^{\frac{1}{2}x^2} \right) (3y - x^3 - c) (xy + cy + 1) = 0.$$

Example 4: Solve $p(p - y) = x(x + y)$.

Solution: The given differential equation can be written as

$$p^2 - py - (x^2 + xy) = 0.$$

Solving for p , we get

$$p = \frac{dy}{dx} = \frac{y \pm \sqrt{(y^2 + 4x^2 - 4xy)}}{2} = \frac{y \pm \sqrt{(y + 2x)^2}}{2} = \frac{y \pm (y + 2x)}{2}.$$

Thus the component equations are

$$dy/dx = \{y + (y + 2x)\}/2 = y + x, \quad \dots(1)$$

$$\text{and} \quad dy/dx = (y - y - 2x)/2 = -x. \quad \dots(2)$$

From (1), we have $(dy/dx) - y = x$, which is a linear differential equation with y as the dependent variable.

Integrating factor is

$$e^{\int -dx} = e^{-x}.$$

Therefore its solution is

$$ye^{-x} = \int xe^{-x} dx + c$$

$$\text{or} \quad ye^{-x} = -xe^{-x} - \int (-e^{-x}) dx + c, \text{ integrating by parts}$$

$$\text{or} \quad ye^{-x} = -xe^{-x} - e^{-x} + c$$

$$\text{or} \quad ye^{-x} = -e^{-x}(x + 1) + c$$

$$\text{or} \quad y = -(x + 1) + ce^x \quad \text{or} \quad y + x + 1 - ce^x = 0. \quad \dots(3)$$

From (2), we have $dy = -x dx$.

Integrating, we get

$$y = -\frac{1}{2}x^2 + \frac{1}{2}c \quad \text{or} \quad 2y + x^2 - c = 0. \quad \dots(4)$$

Thus (3) and (4) are the required solutions of the differential equation. The single combined solution is

$$(y + x + 1 - ce^x)(2y + x^2 - c) = 0.$$

Example 5: Solve $\left(1 - y^2 + \frac{y^4}{x^2}\right) p^2 - 2 \frac{y}{x} p + \frac{y^2}{x^2} = 0$.

Solution: Multiplying throughout by x^2 , the given differential equation becomes

$$(x^2 - x^2 y^2 + y^4) p^2 - 2xyp + y^2 = 0$$

$$\text{or } x^2 p^2 - 2xyp + y^2 = p^2 (x^2 y^2 - y^4)$$

$$\text{or } (xp - y)^2 = p^2 y^2 (x^2 - y^2)$$

$$\therefore xp - y = \pm py \sqrt{(x^2 - y^2)} \quad \text{or} \quad p[x \pm y \sqrt{(x^2 - y^2)}] = y$$

$$\text{or } \frac{1}{p} = \frac{dx}{dy} = \frac{x \pm y \sqrt{(x^2 - y^2)}}{y}$$

Putting $x = vy$ so that $(dx/dy) = v + y (dv/dy)$, we have

$$v + y \frac{dv}{dy} = \frac{vy \pm \sqrt{(v^2 y^2 - y^2)}}{y} = v \pm \sqrt{(v^2 - 1)}$$

$$\text{or } y \frac{dv}{dy} = \pm \sqrt{(v^2 - 1)}$$

$$\text{or } \frac{dv}{dy} = \pm \sqrt{(v^2 - 1)}$$

$$\text{or } dy = \pm [1/\sqrt{(v^2 - 1)}] dv, \text{ separating the variables.}$$

Integrating, we get $y + c = \pm \cosh^{-1} v$

$$\text{or } y + c = \pm \cosh^{-1} (x/y)$$

$$\text{or } \cosh^{-1} (x/y) = \pm (y + c)$$

$$\text{or } x/y = \cosh \{ \pm (y + c) \}.$$

Since $\cosh(-x) = \cosh x$, therefore, we have

$$x = y \cosh (y + c) \text{ and } x = y \cosh (y - c), \text{ as the required solutions.}$$

The single combined solution is $[x - y \cosh (y + c)]^2 = 0$.

Comprehensive Exercise 1

Solve the following differential equations :

- $p^2 - 5p + 6 = 0$. (Rohilkhand 2006; Kanpur 02)
- $y^2 + xyp - x^2 p^2 = 0$. (Meerut 2007B)
- $xyp^2 - (x^2 + y^2)p + xy = 0$. (Gorakhpur 2011)
- $xyp^2 + p(3x^2 - 2y^2) - 6xp = 0$.
- $x(dy/dx)^2 + (y - x)(dy/dx) - y = 0$. (Meerut 2006B; Bundelkhand 09, 10)

6. $y (dy/dx)^2 + (x - y)(dy/dx) - x = 0.$

(Meerut 2005B)

Answers 1

1. $(y - 2x - c)(y - 3x - c) = 0$
2. $(y^2 - cx^{1+\sqrt{5}})(y^2 - cx^{1-\sqrt{5}}) = 0$
3. $(x^2 - y^2 - c)(y - cx) = 0$
4. $(y - cx^2)(y^2 + 3x^2 - c) = 0$
5. $(y - x - c)(xy - c) = 0$
6. $(y - x - c)(x^2 + y^2 - c^2) = 0$

2.3 Equations Solvable for y

Suppose the given differential equation is solvable for y . Then it can be put in the form

$$y = f(x, p). \quad \dots(1)$$

Differentiating (1) w.r.t. x and denoting dy/dx by p , we obtain

$$p = \phi\left(x, p, \frac{dp}{dx}\right), \quad \dots(2)$$

which is a differential equation in two variables x and p . Suppose it is possible to solve the differential equation (2). Let its solution be

$$F(x, p, c) = 0, \quad \dots(3)$$

where c is the arbitrary constant.

Eliminating p between (1) and (3), we get the required solution of (1) in the form

$$\psi(x, y, c) = 0.$$

If it is not easily practicable to eliminate p between (1) and (3), we may solve (1) and (3) to get x and y in terms of p and c in the form

$$x = f_1(p, c), \quad y = f_2(p, c), \quad \dots(4)$$

which give us the required solution of (1) in the form of parametric equations, the parameter being p .

Special case: Equations that do not contain x : In this case the equation has the form $f(y, p) = 0$. If it is solvable for p , it will give

$$p = \phi(y) \quad \text{i.e.,} \quad dy/dx = \phi(y),$$

which can be easily solved by separating the variables.

If it is solvable for y , it will give $y = \psi(p)$, which can be solved by the method just explained in article 2.3.

Illustrative Examples

Example 6: Solve $y + px = p^2 x^4$.

(Rohilkhand 2006; Avadh 08; Agra 08; Purvanchal 11; Lucknow 11)

Solution: Solving for y , the given differential equation can be written as

$$y = -px + p^2 x^4. \quad \dots(1)$$

Differentiating (1) w.r.t. x and denoting dy/dx by p , we get

$$p = -p - x \frac{dp}{dx} + 2p x^4 \frac{dp}{dx} + 4p^2 x^3$$

$$\text{or} \quad 2p - 4p^2 x^3 + x \frac{dp}{dx} (1 - 2x^3 p) = 0$$

$$\text{or} \quad 2p(1 - 2x^3 p) + x \frac{dp}{dx} (1 - 2x^3 p) = 0$$

$$\text{or} \quad (1 - 2x^3 p) \left(2p + x \frac{dp}{dx} \right) = 0.$$

$$\therefore \quad 2p + x \frac{dp}{dx} = 0, \quad \dots(2)$$

$$\text{and} \quad 1 - 2x^3 p = 0.$$

From (2), $dp/dx = -2p/x$ or $dp/p = -(2/x) dx$.

Integrating, we get $\log p = -2 \log x + \log c$

$$\text{or} \quad \log px^2 = \log c \quad \text{or} \quad px^2 = c \quad \text{or} \quad p = c/x^2.$$

Substituting this value of p in (1), we get

$$y = -x(c/x^2) + (c^2/x^4).x^4$$

$$\text{or} \quad y = -(c/x) + c^2 \quad \text{or} \quad xy = c^2 x - c, \text{ as the required solution of (1).}$$

Note: If we eliminate p between (1) and (3), we get

$$y = -(1/2x^3)x + (1/4x^6).x^4 = -(1/2x^2) + (1/4x^2) = -1/4x^2,$$

which is also a solution of (1) because it satisfies (1). This solution itself does not contain any arbitrary constant. Also it cannot be obtained from the general solution $xy = c^2 x - c$ by giving any particular value to c . Such a solution is called the **singular solution** and we shall discuss it in details later on.

Example 7: Solve $y = 2px - p^2$.

(Meerut 2001, 04B; Kanpur 07)

Solution: The given differential equation is

$$y = 2px - p^2. \quad \dots(1)$$

Differentiating (1) w.r.t. x and denoting dy/dx by p , we get

$$p = 2p + 2x \frac{dp}{dx} - 2p \frac{dp}{dx}$$

$$\text{or} \quad p + 2 \frac{dp}{dx} (x - p) = 0 \quad \text{or} \quad p \frac{dx}{dp} + 2x = 2p$$

$$\text{or} \quad \frac{dx}{dp} + \frac{2}{p} x = 2, \quad \dots(2)$$

which is a linear differential equation with x as dependent variable and p as the independent variable.

$$\therefore \text{ integrating factor} = e^{\int (2/p) dp} = e^{2 \log p} = e^{\log p^2} = p^2.$$

$$\therefore \text{ the solution of (2) is } xp^2 = \int 2p^2 dp + c = \frac{2}{3} p^3 + c$$

$$\text{or} \quad x = \frac{2}{3} p + cp^{-2} \quad \dots(3)$$

Here it is not easily practicable to eliminate p between (1) and (3). So putting the value of x from (3) in (1), we get $y = 2p \left(\frac{2}{3} p + cp^{-2} \right) - p^2$

$$\text{or} \quad y = \frac{1}{3} p^2 + 2cp^{-1}. \quad \dots(4)$$

The equations (3) and (4), which express x and y in terms of a parameter p , constitute the required solution of (1).

Example 8: Solve $y - x = x \frac{dy}{dx} + \left(\frac{dy}{dx} \right)^2$. (Meerut 2008)

Solution: Denoting dy/dx by p , the given differential equation can be written as

$$y - x = xp + p^2.$$

$$\text{Solving it for } y, \text{ we get } y = x(1 + p) + p^2. \quad \dots(1)$$

Differentiating (1) w.r.t. x and writing p for dy/dx , we get

$$p = 1 + p + x \frac{dp}{dx} + 2p \frac{dp}{dx}$$

$$\text{or} \quad 1 + x \frac{dp}{dx} + 2p \frac{dp}{dx} = 0$$

$$\text{or} \quad \frac{dx}{dp} + x = -2p, \quad \dots(2)$$

which is a linear differential equation.

$$\text{Here the I.F.} = e^{\int 1 dp} = e^p.$$

$\therefore$ the solution of (2) is

$$x e^p = \int -2p e^p dp + c$$

$$\text{or} \quad x e^p = -2(p e^p - e^p) + c, \text{ integrating by parts}$$

$$\text{or} \quad x e^p = -2e^p(p - 1) + c$$

$$\text{or} \quad x = -2(p - 1) + ce^{-p}. \quad \dots(3)$$

Substituting this value of x in (1), we get

$$y = (1 + p) \{-2(p - 1) + ce^{-p}\} + p^2$$

$$\text{or } y = c(1 + p)e^{-p} + 2 - p^2. \quad \dots(4)$$

The equations (3) and (4), which express x and y in terms of a parameter p , constitute the required solution of (1).

Example 9: Solve $y = a\sqrt{1 + p^2}$.

Solution: The given differential equation is

$$y = a\sqrt{1 + p^2}. \quad \dots(1)$$

Differentiating (1) w.r.t. x and writing p for dy/dx , we get

$$p = a \cdot \frac{1}{2}(1 + p^2)^{-1/2} \cdot 2p(dp/dx) \quad \text{or} \quad \frac{a dp}{\sqrt{1 + p^2}} = dx.$$

Integrating, we get

$$x = a \log \{p + \sqrt{1 + p^2}\} + c. \quad \dots(2)$$

Now from (1),

$$y^2 = a^2 + a^2 p^2 \quad \text{or} \quad a^2 p^2 = y^2 - a^2 \quad \text{or} \quad p = \sqrt{(y^2 - a^2)/a}.$$

Putting $\sqrt{1 + p^2} = y/a$ and $p = \sqrt{(y^2 - a^2)/a}$ in (2), we get

$$x = a \log \left\{ \frac{\sqrt{(y^2 - a^2)}}{a} + \frac{y}{a} \right\} + c$$

$$\text{or } x = a \log \{y + \sqrt{(y^2 - a^2)}\} - a \log a + c$$

$$\text{or } x = a \log \{y + \sqrt{(y^2 - a^2)}\} + c,$$

writing c for $-a \log a + c$ because c is an arbitrary constant.

Hence the required solution is

$$x = \log \{y + \sqrt{(y^2 - a^2)}\} + c.$$

Comprehensive Exercise 2

Solve the following differential equations :

- $y = 2px + p^4 x^2.$ (Meerut 2006; Avadh 09)
- $4p^3 + 3xp = y.$
- $x^2 + p^2 x = yp.$
- $y - 2xp = f(xp^2).$ (Meerut 2007)
- $y = x\{p + \sqrt{1 + p^2}\}.$ (Kanpur 2009; Rohilkhand 09)
- $y = \sin p - p \cos p.$ (Gorakhpur 2006)
- By differentiating with respect to x the equation $p^3 + xp^2 = y$, obtain its general solution in the form $x = f(p)$, $y = \phi(p).$

Answers 2

1. $(y - c^2)^2 = 4cx$
2. $x = -\frac{12}{7}p^2 + cp^{-3/2}, y = -\frac{3}{7}p^3 + 3cp^{-1/2}$
3. $x = -\frac{1}{3}p^3 + cp^{1/2}, y = (-\frac{1}{3}p^2 + cp^{1/2})^2 / p + p^2(-\frac{1}{3}p^2 + cp^{1/2})$
4. $y = 2c \sqrt{x} + f(c^2)$
5. $x^3 + y^2 - 2xc = 0$
6. $x = c - \cos p, y = \sin p - \cos p$
7. $x = \frac{\frac{1}{2}(2c + 3p^2 - 2p^3)}{(p-1)^2}, y = \frac{2cp^2 + 2p^3 - p^4}{2(p-1)^2}$

2.4 Equations Solvable for x

Suppose the given differential equation is solvable for x . Then it can be put in the form

$$x = f(y, p). \quad \dots(1)$$

Differentiating (1) w.r.t. y and writing $1/p$ for dx/dy , we get

$$\frac{1}{p} = \phi\left(y, p, \frac{dp}{dy}\right), \quad \dots(2)$$

which is a differential equation in two variables y and p . Suppose it is possible to solve the differential equation (2). Let its solution be

$$F(y, p, c) = 0, \quad \dots(3)$$

where c is the arbitrary constant.

Eliminating p between (1) and (3), we get the required solution of (1) in the form

$$\psi(x, y, c) = 0.$$

In case it is not easily practicable to eliminate p between (1) and (3), we may solve (1) and (3) to get x and y in terms of p and c in the form

$$x = f_1(p, c), y = f_2(p, c), \quad \dots(4)$$

which give us the required solution of (1) in the form of parametric equations, the parameter being p .

Special Case: Equations that do not contain y :

In this case the equation has the form $f(x, p) = 0$.

If it is solvable for p , it will give

$$p = \phi(x) \quad \text{i.e.,} \quad dy/dx = \phi(x), \text{ which can be easily integrated.}$$

If it is solvable for x , it will give $x = \psi(p)$, which can be solved by the method just explained in article 2.4.

Illustrative Examples

Example 10: Solve $y = 2px + y^2 p^3$. (Purvanchal 2007, 10; Lucknow 10; Avadh 07)

Solution: Solving for x , the given differential equation can be written as

$$2px = y - y^2 p^3$$

$$\text{or} \quad x = \frac{y}{2p} - \frac{y^2 p^2}{2}. \quad \dots(1)$$

Differentiating (1) w.r.t. y and writing $1/p$ for dx/dy , we obtain

$$\frac{1}{p} = \frac{1}{2p} - \frac{y}{2p^2} \frac{dp}{dy} - \frac{2yp^2}{2} - \frac{y^2}{2} \cdot 2p \frac{dp}{dy}$$

$$\text{or} \quad \frac{1}{2p} + yp^2 + \left(\frac{y}{2p^2} + py^2 \right) \frac{dp}{dy} = 0$$

$$\text{or} \quad p \left(\frac{1}{2p^2} + py \right) + y \left(\frac{1}{2p^2} + py \right) \frac{dp}{dy} = 0$$

$$\text{or} \quad \left(\frac{1}{2p^2} + py \right) \left(p + y \frac{dp}{dy} \right) = 0.$$

$$\therefore \quad p + y \frac{dp}{dy} = 0 \quad \text{or} \quad \frac{1}{2p^2} + py = 0.$$

The equation $\frac{1}{2p^2} + py = 0$ will give us the singular solution of (1).

$$\text{From } p + y \frac{dp}{dy} = 0, \text{ we have } \frac{dp}{dy} = -\frac{p}{y}$$

$$\text{or} \quad (1/p)dp = -(1/y)dy.$$

Integrating, we get

$$\log p = -\log y + \log c \quad \text{or} \quad \log py = \log c$$

$$\text{or} \quad py = c \quad \text{or} \quad p = c/y.$$

Substituting this value of p in the given differential equation, we get

$$y = 2x \cdot (c/y) + y^2 (c^3/y^3) \quad \text{or} \quad y = 2cx/y + c^3/y$$

$$\text{or} \quad y^2 = 2cx + c^3, \text{ which is the required solution.}$$

Example 11: Solve the differential equation

$$p^3 - 4xyp + 8y^2 = 0.$$

Solution: Solving the given differential equation for x , we get

$$4xyp = p^3 + 8y^2 \quad \text{or} \quad x = \frac{p^2}{4y} + \frac{2y}{p}. \quad \dots(1)$$

Differentiating (1) w.r.t. y and writing $1/p$ for dx/dy , we get

$$\frac{1}{p} = -\frac{p^2}{4y^2} + \frac{2p}{4y} \frac{dp}{dy} + \frac{2}{p} - \frac{2y}{p^2} \frac{dp}{dy}$$

$$\text{or} \quad \frac{dp}{dy} \left(\frac{p}{2y} - \frac{2y}{p^2} \right) = \frac{p^2}{4y^2} - \frac{1}{p}$$

$$\text{or} \quad \frac{dp}{dy} \left(\frac{p}{2y} - \frac{2y}{p^2} \right) = \frac{p}{2y} \left(\frac{p}{2y} - \frac{2y}{p^2} \right)$$

$$\text{or} \quad \frac{dp}{dy} = \frac{p}{2y}, \quad \begin{array}{l} \text{cancelling the common factor on either side which} \\ \text{corresponds to the singular solution of (1)} \end{array}$$

$$\text{or} \quad (2/p) dp = (1/y) dy, \text{ separating the variables.}$$

Integrating, we get $2 \log p = \log y + \log c$

$$\text{or} \quad \log p^2 = \log cy \quad \text{or} \quad p^2 = cy. \quad \dots(2)$$

Now we shall eliminate p between the given differential equation and the equation (2).

From the given differential equation, we have

$$8y^2 = p(4xy - p^2)$$

$$\text{or} \quad 8y^2 = (cy)^{1/2}(4xy - cy), \text{ substituting for } p \text{ from (2)}$$

$$\text{or} \quad 8y^2 = c^{1/2} y^{3/2} (4x - c) \quad \text{or} \quad 8y^{1/2} = c^{1/2} (4x - c)$$

$$\text{or} \quad 64y = c(4x - c)^2 \quad \text{or} \quad 64y = 16c(x - \frac{1}{4}c)^2,$$

$$\text{or} \quad y = \frac{1}{4}c(x - \frac{1}{4}c)^2 \quad \text{or} \quad y = c(x - c), \text{ writing } c \text{ for } \frac{1}{4}c.$$

Hence the required solution is $y = c(x - c)^2$.

Example 12: Solve $ap^2 + py - x = 0$.

(Rohilkhand 2005; Meerut 13B)

Solution: Solving the given differential equation for x , we get

$$x = py + ap^2. \quad \dots(1)$$

Differentiating (1) w.r.t. y and writing $1/p$ for dx/dy , we get

$$\frac{1}{p} = p + y \frac{dp}{dy} + 2ap \frac{dp}{dy} \quad \text{or} \quad \frac{1 - p^2}{p} = y \frac{dp}{dy} + 2ap \frac{dp}{dy}$$

$$\text{or} \quad \frac{1 - p^2}{p} \frac{dy}{dp} - y = 2ap, \text{ multiplying both sides by } dy / dp$$

$$\text{or} \quad \frac{dy}{dp} + \frac{p}{p^2 - 1} y = -\frac{2ap^2}{p^2 - 1}, \quad \dots(2)$$

which is a linear differential equation.

Here the I.F. = $e^{\int \{p/(p^2-1)\} dp} = e^{\frac{1}{2} \log (p^2-1)} = (p^2-1)^{1/2}$.

$\therefore$ the solution of (2) is

$$\begin{aligned} y(p^2-1)^{1/2} &= \int \frac{-2ap^2}{p^2-1} (p^2-1)^{1/2} dp + c \\ &= -2a \int \frac{(p^2-1)+1}{\sqrt{(p^2-1)}} dp + c \\ &= -2a \int \left[\sqrt{(p^2-1)} + \frac{1}{\sqrt{(p^2-1)}} \right] dp + c \\ &= -2a \left[\frac{1}{2} p \sqrt{(p^2-1)} - \frac{1}{2} \cosh^{-1} p + \cosh^{-1} p \right] + c \\ &= -ap \sqrt{(p^2-1)} - a \cosh^{-1} p + c \end{aligned}$$

or
$$y = \frac{c - a \cosh^{-1} p}{\sqrt{(p^2-1)}} - ap. \quad \dots(3)$$

Substituting this value of y in (1), we get

$$x = p \left(\frac{c - a \cosh^{-1} p}{\sqrt{(p^2-1)}} - ap \right) + ap^2$$

or
$$x = \frac{p(c - a \cosh^{-1} p)}{\sqrt{(p^2-1)}}. \quad \dots(4)$$

The equations (3) and (4) constitute the parametric equations of the required solution.

Comprehensive Exercise 3

Solve the following differential equations :

- $y^2 \log y = xyp + p^2$. (Agra 2007; Purvanchal 08)
- $p^2 y + 2px - y = 0$. (Meerut 2005; Purvanchal 06; Lucknow 08)
- $4(xp^2 + yp) = y^4$.
- $(2x - b)p = y - ayp^2$.
- $xp^3 = a + bp$.
- $p = \tan \{x - p/(1 + p^2)\}$.
- $x + p/\sqrt{(1 + p^2)} = a$.

Answers 3

- $\log y = cx + c^2$
- $y^2 - 2cx + c^2 = 0$
- $y = 4c(xyc + 1)$
- $ac^2 + (2x - b)c - y^2 = 0$

5. $x = (a/p^3) + (b/p^2), y = (3a/2p^2) + (2b/p) + c$
6. $x = p/(1 + p^2) + \tan^{-1} p, y = c - 1/(1 + p^2)$
7. $(x - a)^2 + (y + c)^2 = 1$

2.5 Clairaut's Equation

(Meerut 2004; Lucknow 08)

(a) *The differential equation*

$$y = px + f(p), \quad \dots(1)$$

is known as Clairaut's equation. Here $f(p)$ is some function of p only. This differential equation is very important and the students should note its form carefully. To solve this differential equation, we shall obviously adopt the method explained in article 2.3. Thus differentiating (1) w.r.t. x and writing p for dy/dx , we get

$$p = p + x \frac{dp}{dx} + f'(p) \frac{dp}{dx}$$

$$\text{or} \quad \frac{dp}{dx} \{x + f'(p)\} = 0.$$

$$\therefore \quad \frac{dp}{dx} = 0. \quad \dots(2)$$

$$\text{or} \quad x + f'(p) = 0. \quad \dots(3)$$

$$\text{From (2), we have } p = \text{constant} = c, \text{ (say).} \quad \dots(4)$$

Eliminating p between (1) and (4), we obtain

$$y = cx + f(c), \quad \dots(5)$$

which is the required general solution of (1).

If we eliminate p between (1) and (3), we get the **singular solution** of (1).

Remember: To obtain the general solution of a differential equation in Clairaut's form simply replace p by c .

(b) *To solve the differential equation*

$$y = xf_1(p) + f_2(p). \quad \dots(1)$$

This differential equation is not in Clairaut's form. However it can be solved by the method we adopted in solving Clairaut's equation. Thus differentiating (1) w.r.t. x and writing p for dy/dx , we get

$$p = f_1(p) + x f_1'(p) \frac{dp}{dx} + f_2'(p) \frac{dp}{dx}$$

$$\text{or} \quad p - f_1(p) = x f_1'(p) \frac{dp}{dx} + f_2'(p) \frac{dp}{dx}$$

$$\text{or} \quad [p - f_1(p)] \frac{dx}{dp} - x f_1'(p) = f_2'(p)$$

$$\text{or} \quad \frac{dx}{dp} + \frac{f_1'(p)}{f_1(p) - p} \cdot x = \frac{f_2'(p)}{p - f_1(p)} \quad \dots(2)$$

which is a linear differential equation with x as the dependent variable and p as the independent variable.

Let the solution of (2) be

$$\phi(x, p, c) = 0. \quad \dots(3)$$

Then eliminating p between (1) and (3), we get the required solution.

Illustrative Examples

Example 13: Solve $y = x(dy/dx) + (dy/dx)^2$. (Meerut 2011)

Solution: Denoting dy/dx by p , the given differential equation is $y = px + p^2$, which is in Clairaut's form. So replacing p by the arbitrary constant c , the required solution is $y = cx + c^2$.

Example 14: Solve $p = \log(px - y)$. (Gorakhpur 2007; Lucknow 09)

Solution: The given differential equation can be written as $px - y = e^p$ or $y = px - e^p$, which is in Clairaut's form. So replacing p by the arbitrary constant c in the given differential equation, the required solution is $c = \log(cx - y)$.

Example 15: Solve $\sin px \cos y = \cos px \sin y + p$. (Meerut 2003; Rohilkhand 06, 08, 10; Lucknow 07)

Solution: The given differential equation is

$$\sin px \cos y - \cos px \sin y = p$$

$$\text{or} \quad \sin(px - y) = p$$

$$\text{or} \quad px - y = \sin^{-1} p$$

$$\text{or} \quad y = px - \sin^{-1} p, \text{ which is in Clairaut's form.}$$

So changing p to the arbitrary constant c , the required solution is

$$y = cx - \sin^{-1} c.$$

Example 16: Solve $p^2(x^2 - a^2) - 2pxy + y^2 - b^2 = 0$.

Solution: The given differential equation can be written as

$$p^2 x^2 - 2pxy + y^2 = p^2 a^2 + b^2 \quad \text{or} \quad (y - px)^2 = b^2 + a^2 p^2$$

$$\text{or} \quad y = px \pm \sqrt{(b^2 + a^2 p^2)},$$

each of which is in Clairaut's form. Hence the required solution is

$$(y - cx)^2 = b^2 + a^2 c^2.$$

Example 17: Solve $y = 2px + p^n$.

Solution: The given differential equation is

$$y = 2px + p^n. \quad \dots(1)$$

Differentiating (1) w.r.t. x and writing p for dy/dx , we get

$$p = 2p + 2x (dp/dx) + np^{n-1} (dp/dx)$$

$$\text{or } p + 2x (dp/dx) = -np^{n-1} (dp/dx)$$

$$\text{or } p (dx/dp) + 2x = -np^{n-1}, \text{ multiplying both sides by } dx/dp$$

$$\text{or } \frac{dx}{dp} + \frac{2}{p} x = -np^{n-2}, \quad \dots(2)$$

which is a linear differential equation.

$$\text{Hence the I.F.} = e^{\int (2/p) dp} = e^{2 \log p} = e^{\log p^2} = p^2.$$

$$\therefore \text{ the solution of (2) is } xp^2 = - \int n p^{n-2} p^2 dp + c$$

$$\text{or } xp^2 = -n \int p^n dp + c = -n p^{n+1} / (n+1) + c$$

$$\text{or } x = cp^{-2} - \{n/(n+1)\} p^{n-1}. \quad \dots(3)$$

Substituting this value of x in (1), we get

$$y = 2p[cp^{-2} - \{n/(n+1)\} p^{n-1}] + p^n$$

$$\text{or } y = 2cp^{-1} + p^n - \frac{2n}{n+1} p^n = 2cp^{-1} - \frac{n-1}{n+1} p^n. \quad \dots(4)$$

The equations (3) and (4), which express x and y in terms of a parameter p , constitute the required solution.

2.6 Equations Reducible to Clairaut's Form

Some differential equations by suitable change of variables may be reduced to Clairaut's form.

Example 18: Use the transformation $x^2 = u$, $y^2 = v$ to solve

$$(px - y)(py + x) = h^2 p. \quad (\text{Agra 2006; Lucknow 08, 10})$$

Solution: The given differential equation can be written as

$$p^2 xy + p(x^2 - y^2 - h^2) - xy = 0. \quad \dots(1)$$

Putting $x^2 = u$ and $y^2 = v$, we have $2x dx = du$ and $2y dy = dv$.

$$\therefore \frac{y}{x} \frac{dy}{dx} = \frac{dv}{du} \quad \text{or} \quad \frac{dy}{dx} = \frac{x}{y} \frac{dv}{du} \quad \text{or} \quad p = \frac{x}{y} \frac{dv}{du}.$$

Putting this value of p in the differential equation (1), we get

$$xy \frac{x^2}{y^2} \left(\frac{dv}{du} \right)^2 + \frac{x}{y} \frac{dv}{du} (x^2 - y^2 - h^2) - xy = 0$$

$$\text{or } \frac{x}{y} \left[x^2 \left(\frac{dv}{du} \right)^2 + (x^2 - y^2 - h^2) \frac{dv}{du} - y^2 \right] = 0$$

$$\text{or } x^2 (dv/du)^2 + (x^2 - y^2 - h^2) (dv/du) - y^2 = 0$$

$$7. (y - cx)^2 = 4/c^2$$

$$8. y = cx - \cos^{-1} c$$

$$9. x = \frac{1}{3} p^2 + cy^{-1}, y = -xp \log p + \frac{1}{9} (2 + 3 \log p) p^3$$

$$10. y^2 = cx^2 + c^2$$

$$11. e^y = ce^x + c^3$$

2.7 Geometrical Meaning of a Differential Equation of the First Order

(Avadh 2010)

Let $f(x, y, p) = 0$, ...(1)

be a differential equation of the first order, where $p = dy/dx$.

Suppose the general solution of (1) is $\phi(x, y, c) = 0$, ...(2)

where c is the arbitrary constant.

For each real value of c the equation (2) represents a curve. Thus (2) is the equation of a family of curves; c being the parameter for the family. Hence *every differential equation of the first order represents a family of curves.*

Let us take any point (x_1, y_1) in the plane. If we substitute the coordinates of this point in (1) and (2) and solve the resulting equations in p and c , the values of c so obtained are the values of the parameter for the curves of the family (2) which pass through (x_1, y_1) and the values of p are the slopes of the tangents to these curves at the point (x_1, y_1) . Naturally the degree of c in (2) must be equal to the degree of p in (1).

2.8 Singular Solutions

(Gorakhpur 2009)

Sometimes a differential equation (for degree higher than one) possesses a solution which does not contain any arbitrary constant and which cannot be derived from the general solution of the differential equation by giving a particular value to the arbitrary constant. Such a solution is called a **singular solution** and it is generally not included in the general solution of the differential equation.

The singular solution of a differential equation is given by the envelope of family of curves represented by that differential equation.

Whenever the envelope of the family of curves

$$\phi(x, y, c) = 0, \quad \text{...(1)}$$

represented by the general solution of the differential equation

$$f(x, y, p) = 0 \quad \text{...(2)}$$

exists, the equation of the envelope is the singular solution of the differential equation (2).

Suppose that the family of curves (1) possesses an envelope. For any point $P(x, y)$ on the envelope, there exists a curve of the family (1), say $\phi(x, y, c) = 0$, which touches the envelope at (x, y) . The values of $x, y, dy/dx$ for the curve at P satisfy the differential equation (2). But the values of $x, y, dy/dx$ at P for the envelope are the same as for the curve. Hence the values of $x, y, dy/dx$ at each point of the envelope satisfy the differential equation (2). Consequently the envelope of (1) is also a solution of (2). This solution does not contain any arbitrary constant and in general, cannot be obtained from (1) by giving any particular value to the arbitrary constant c . Hence this envelope is the singular solution of (2).

2.9 Determination of Singular Solution with the Help of c -Discriminant and p -Discriminant Relations

The discriminant: Let $F(c) = 0$ be an algebraic equation. The discriminant of this equation is the simplest function of the coefficients the vanishing of which represents the condition that the equation should have two equal roots. Thus the discriminant of the quadratic $Ac^2 + Bc + C = 0$ is $B^2 - 4AC$. The equation

$B^2 - 4AC = 0$ is called the *discriminant relation*.

Let the given differential equation be

$$f(x, y, p) = 0 \quad \dots(1)$$

and let its general solution be

$$\phi(x, y, c) = 0 \quad \dots(2)$$

The envelope of the family of curves (2) is *contained* in the locus obtained on eliminating c , between (2) and

$$\frac{\partial \phi(x, y, c)}{\partial c} = 0 \quad \dots(3)$$

Let the c -eliminant between (2) and (3) be

$$\psi(x, y) = 0 \quad \dots(4)$$

Since the equation (4) may sometimes represent other loci besides the envelope, therefore only that part of the locus (4) is the singular solution which also satisfies the given differential equation (1). Therefore if the equation (4) fails to satisfy (1), we should resolve it into others that are simpler. Then we should try for each part whether it satisfies the differential equation (1) or not. Only those parts will constitute the singular solution which satisfy (1).

From our knowledge of the theory of equations we know that the c -eliminant between (2) and (3) gives the condition for the equation (2) in c to have two equal roots. Therefore we shall call this c -eliminant the **c -discriminant relation**. The singular solution is contained in this c -discriminant relation. From the geometrical point of view the c -discriminant relation (4) is the locus of the points (x, y) such that two of the

curves of the family (2) through (x, y) coincide. Now if two curves of the family coincide, their tangents must also coincide and so the corresponding values of p i.e., (dy/dx) given by (1) must also coincide. The condition for the equation (1) in p to have two equal roots is obtained on eliminating p between (1) and

$$\frac{\partial f(x, y, p)}{\partial p} = 0. \quad \dots(5)$$

The p -eliminant between (1) and (5) is called the **p -discriminant relation** and the envelope and hence the singular solution is also contained in it.

Note 1: From the above discussion it is clear that if p occurs only in the first degree in the differential equation, there will be no singular solution.

Note 2: A differential equation which possesses a singular solution is not considered completely solved until the singular solution also has been found. Therefore while solving a differential equation of the first order but higher degree we must find the general solution and also discuss the singular solution.

2.10 Working Rule for Finding the Singular Solution

Let the given differential equation be $f(x, y, p) = 0$(1)

Find the general solution of (1) and let it be $\phi(x, y, c) = 0$(2)

Find the **c -discriminant relation** by eliminating c between

$$\phi(x, y, c) = 0 \text{ and } \frac{\partial \phi(x, y, c)}{\partial c} = 0.$$

Also find the **p -discriminant relation** by eliminating p between

$$f(x, y, p) = 0 \text{ and } \frac{\partial f(x, y, p)}{\partial p} = 0.$$

The locus common to both the c -discriminant and the p -discriminant relations gives us the singular solution of (1) provided it satisfies (1).

However, we can find only one of these two discriminant relations and only that part of the locus contained in it is the singular solution which also satisfies the given differential equation (1).

2.11 The Singular Solution of Clairaut's Equation

We know that the general solution of the Clairaut's equation

$$y = px + f(p) \quad \dots(1)$$

is $y = cx + f(c) \quad \dots(2)$

Differentiating (2) partially w.r.t. c , we get $0 = x + f'(c)$(3)

The singular solution, which is the envelope of (2), is obtained by eliminating c between (2) and (3).

Now differentiating (1) partially w.r.t. p , we get $0 = x + f'(p)$ (4)

The equation (1) and (4) differ from the equations (2) and (3) only in having p instead of c . Therefore the c -eliminant between (2) and (3) and the p -eliminant between (1) and (4) are identical and either of them gives us the singular solution. **Hence the singular solution of the Clairaut's equation (1) is obtained by eliminating p between (1) and (4).**

Note: The equation (4) is the same as the equation (3) of article 2.5 (a).

Illustrative Examples

Example 19: Find the general and singular solutions of

$$y^2 - 2pxy + p^2(x^2 - 1) = m^2. \quad (\text{Bundelkhand 2010})$$

Solution: The given differential equation can be written as

$$(y - px)^2 = p^2 + m^2$$

or $y - px = \pm \sqrt{p^2 + m^2}$ or $y = px \pm \sqrt{p^2 + m^2}$,

either of which is in Clairaut's form. Hence the general solution is obtained on replacing p by the arbitrary constant c in the given differential equation. Thus the general solution is

$$(y - cx)^2 = c^2 + m^2 \quad \text{or} \quad c^2(x^2 - 1) - 2xyc + y^2 - m^2 = 0. \quad \dots(1)$$

Now from the given differential equation and from the general solution (1) both the p and the c discriminant relations are

$$4x^2y^2 - 4(x^2 - 1)(y^2 - m^2) = 0$$

or $x^2m^2 + y^2 - m^2 = 0 \quad \text{or} \quad y^2 + m^2x^2 = m^2,$

which is the singular solution.

Example 20: Find the general and singular solution of the differential equation.

$$(xp - y)^2 = p^2 - 1. \quad (\text{Bundelkhand 2008; Lucknow 06})$$

Solution: From the given differential equation, we have

$$xp - y = \pm \sqrt{p^2 - 1} \quad \text{or} \quad y = px \pm \sqrt{p^2 - 1}$$

either of which is in Clairaut's form. Hence the general solution is obtained on replacing p by the arbitrary constant c in the given differential equation.

Thus the general solution is

$$(xc - y)^2 = c^2 - 1 \quad \text{or} \quad c^2(x^2 - 1) - 2xyc + y^2 + 1 = 0. \quad \dots(1)$$

The envelope of the family of curves (1) is the singular solution.

From (1) the c -discriminant relation is

$$4x^2y^2 - 4(x^2 - 1)(y^2 + 1) = 0$$

or $-x^2 + y^2 + 1 = 0 \quad \text{or} \quad x^2 - y^2 = 1,$

which is obviously also the p -discriminant relation. Since the c -discriminant relation contains only one locus, therefore it gives us the envelope of the family of curves (1).

Hence the singular solution is $x^2 - y^2 = 1$.

Example 21: Find the general and singular solution of

$$9p^2 (2 - y)^2 = 4 (3 - y).$$

Solution: Solving the given differential equation for p , we have

$$p = \frac{dy}{dx} = \pm \frac{2}{3} \frac{\sqrt{3 - y}}{2 - y}$$

or
$$dx = \pm \frac{3}{2} \frac{2 - y}{\sqrt{3 - y}} dy, \text{ separating the variables.}$$

Integrating, we have

$$\begin{aligned} x + c &= \pm \frac{3}{2} \int \frac{2 - y}{\sqrt{3 - y}} dy \\ &= \pm \frac{3}{2} \int \frac{2 - (3 - t^2)}{t} (-2t) dt, \end{aligned}$$

putting $3 - y = t^2$ so that $-dy = 2t dt$

$$= \pm 3 \int (t^2 - 1) dt = \pm 3 \left(\frac{1}{3} t^3 - t \right) = \pm t (t^2 - 3)$$

$$= \pm \sqrt{3 - y} (-y), \quad [\because 3 - y = t^2]$$

$$\therefore (x + c)^2 = y^2 (3 - y), \quad \dots(1)$$

which is the general solution of the given differential equation.

Differentiating (1) partially w.r.t. c , we get

$$2(x + c) = 0 \quad \text{i.e., } x + c = 0. \quad \dots(2)$$

Eliminating c between (1) and (2), we get the c -discriminant relation as

$$y^2 (3 - y) = 0.$$

Now $y = 0$ gives $dy/dx = p = 0$. Putting $y = 0$ and $p = 0$ in the given differential equation, we see that these values of y and p do not satisfy it. Therefore $y = 0$ is not a solution.

Again $3 - y = 0$ gives $y = 3$ and so $dy/dx = p = 0$. Putting $y = 3$ and $p = 0$ in the given differential equation, we see that these values of y and p satisfy it. Therefore $y = 3$ is the singular solution.

Example 22: Reduce the differential equation $(px - y)(x - py) = 2p$ to Clairaut's form by substituting $x^2 = u$ and $y^2 = v$ and find its complete primitive and the singular solution.

Solution: We put $u = x^2$ and $v = y^2$, so that

$$du/dx = 2x \quad \text{and} \quad dv/dx = 2y (dy/dx) = 2py.$$

$$\therefore \frac{dv}{du} = \frac{dv/dx}{du/dx} = \frac{2py}{2x} = \frac{py}{x}$$

or $P = py/x$, where $P = dv/du$ or $p = xP/y$.

Substituting this value of P in the given differential equation, we get

$$\left(\frac{x^2 P}{x} - y \right) (x - xP) = 2x \frac{P}{y} \quad \text{or} \quad x(x^2 P - y^2)(1 - P) = 2xP$$

$$\text{or} \quad (x^2 P - y^2)(1 - P) = 2P \quad \text{or} \quad (uP - v)(1 - P) = 2P$$

$$\text{or} \quad uP - v = 2P/(1 - P) \quad \text{or} \quad v = uP - 2P/(1 - P),$$

which is in Clairaut's form.

Hence its general solution is $v = uc - 2c/(1 - c)$

$$\text{or} \quad y^2 = x^2 c - 2c/(1 - c) \quad \text{or} \quad y^2(1 - c) = x^2 c(1 - c) - 2c$$

$$\text{or} \quad c^2 x^2 - c(x^2 + y^2 - 2) + y^2 = 0. \quad \dots(1)$$

Now (1) is a quadratic in c . So the c -discriminant relation is

$$(x^2 + y^2 - 2)^2 - 4x^2 y^2 = 0$$

$$\text{or} \quad (x^2 - 2xy + y^2 - 2)(x^2 + 2xy + y^2 - 2) = 0$$

$$\text{or} \quad \{(x - y)^2 - 2\} \{(x + y)^2 - 2\} = 0$$

$$\text{or} \quad (x - y - \sqrt{2})(x - y + \sqrt{2})(x + y - \sqrt{2})(x + y + \sqrt{2}) = 0.$$

All the four equations $x - y - \sqrt{2} = 0$, $x - y + \sqrt{2} = 0$, $x + y - \sqrt{2} = 0$ and $x + y + \sqrt{2} = 0$ satisfy the given differential equation and are therefore the singular solutions.

Comprehensive Exercise 5

- Find the complete primitive and singular solution of $y = px + \sqrt{(b^2 + a^2 p^2)}$. Interpret your results geometrically. (Avadh 2010)
- Find the general and singular solutions of $y = px + a/p$.
- Solve and examine for singular solution the differential equation $x^3 p^2 + x^2 yp + a^3 = 0$. (Gorakhpur 2007)
- Find the general and singular solution of $27y - 8p^3 = 0$.
- Solve and examine for singular solution the equation $xp^2 - (x - a)^2 = 0$.
- Find the general and singular solution of
 - $(y - px)^2 + a^2 p = 0$.
 - $3xy = 2px^2 - 2p^2$.
 - $p^2 + y^2 = 1$.
- Examine $y^2(1 + p^2) = r^2$ for singular solution. (Bundelkhand 2009)

8. Reduce the equation $x^2 p^2 + py(2x + y) + y^2 = 0$, where $p = (dy / dx)$ to Clairaut's form by putting $u = x$ and $v = xy$ and find its complete primitive and also its singular solution. (Gorakhpur 2010)
9. Solve the differential equation $(px^2 + y^2)(px + y) = (p + 1)^2$ by reducing it to Clairaut's form and find its singular solution. (Purvanchal 2009)
10. Reduce the differential equation $y = 2px + y^2 p^3$ to Clairaut's form by putting $y^2 = v$ and hence find its general and singular solutions.
11. Solve and test for singular solutions $p^3 - 4pxy + 8y^2 = 0$.

Answers 5

1. $y = cx + \sqrt{(b^2 + a^2 c^2)}; x^2 / a^2 + y^2 / b^2 = 1$
2. $y = cx + a/c; y^2 = 4ax$
3. $c^2 + cxy + a^3 x = 0; x = 0, xy^2 - 4a^3 = 0$
4. $y^2 = (x + c)^3, y = 0$
5. $9(y + c)^2 = 4x(x - 3a)^2, x = 0$
6. (a) $(y - cx)^2 + a^2 c = 0; xy = a^2 / 4$ (b) $(2y + 2c)^2 = 4cx^3; x^3 - 6y = 0$
(c) $y = \cos(x + c); y = \pm 1$
7. $y^2 + (x + c)^2 = r^2; y = \pm r$
8. $xy = cy + c^2, y = 0, y + 4x = 0$
9. $c^2(x + y) - cxy - 1 = 0; x^2 y^2 + 4(x + y) = 0$
10. $y^2 = cx + c^3 / 8; 27y^4 + 32x^3 = 0$
11. $y = c(x - c)^2; y = 0, y = \frac{4}{27} x^3$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. Solution of the differential equation $p^2 - 8p + 15 = 0$ is
 (a) $p = 5, p = 3$ (b) $(y - 5x - c)(y - 3x - c) = 0$
 (c) $(y + 5x)(y + 3x + c) = 0$ (d) None of these

2. Solution of the equation $y^2 \log y = xyp + p^2$ is
 - (a) $\log y = cx + x^2$
 - (b) $\log y = cx^2 + e^x$
 - (c) $\log x = cy + y^2$
 - (d) None of these.
3. Solution of the equation $y = px + \log p$ is
 - (a) $y = e^x + c$
 - (b) $y = cx + \log c$
 - (c) $y = \log cx$
 - (d) $x = e^y + c$.
4. Which of the following equations is Clairaut's equation ?
 - (a) $x = py + f(p)$
 - (b) $y = px + f(c)$
 - (c) $y = px + f(p)$
 - (d) $y = x^2 + c$

Fill in The Blanks

Fill in the blanks "....." so that the following statements are complete and correct.

1. We usually denote $\frac{dy}{dx}$ by the letter
2. The differential equation $y = px + f(p)$, is known as
(Meerut 2003; Avadh 2005)
3. $p^n + A_1 p^{n-1} + A_2 p^{n-2} + \dots + A_{n-1} p + A_n = 0$,
where $A_1, A_2, \dots, A_n$ are some functions of x and y , is a differential equation of order and degree.
4. Solution of the equation $p^2 - 7p + 12 = 0$ is
5. Solution of the equation $\left(\frac{dy}{dx}\right)^2 - ax^3 = 0$ is
6. Solution of the differential equation $y = px + e^p$ is
7. Solution of the equation $y = x\left(\frac{dy}{dx}\right) + \left(\frac{dy}{dx}\right)^3$ is
8. Solution of the equation $\cos y \cos px + \sin y \sin px = p$ is
9. Solution of the equation $(x-a)p^2 + (x+y)p - y = 0$ is

True or False

Write 'T' for true and 'F' for false statement.

1. Every differential equation of the first order represents a family of curves.
2. The singular solution of a differential equation is given by the envelope of the family of curves represented by that differential equation.
3. Only those differential equations possess a singular solution in which p (or dy/dx) occurs only in the first degree.

Answers

Multiple Choice Questions

1. (b)
2. (a)
3. (b)
4. (c)

Fill in the Blank(s)

1. p
2. Clairaut's equation
3. first, n th
4. $(y - 3x - c)(y - 4x - c) = 0$
5. $25(y + c)^2 = 4ax^5$
6. $y = cx + e^c$
7. $y = cx + c^3$
8. $y = cx - \cos^{-1} c$
9. $(x - a)c^2 + (x - y)c - y = 0$

True or False

1. T
2. T
3. F



Chapter

3



Orthogonal Trajectories

3.1 Trajectory

(Lucknow 2005; Gorakhpur 06; Bundelkhand 08; Meerut 13, 13B)

Definition: A trajectory of a given system of curves is defined to be a curve which cuts all the members of the family according to a given law. Here we propose to find the equation of the trajectories (a family of curves) each member of which cuts each member of a given family of curves at a constant angle. If the angle is a right angle, the trajectories are called **orthogonal trajectories**, when it is other than a right angle they are called to be **oblique trajectories**.

3.2 Trajectories

Cartesian Co-ordinates: To find the trajectories which cut every member of a given family of curves at a constant angle.

Let the equation of the given family of curves be

$$f(x, y, c) = 0, \quad \dots(1)$$

c being the arbitrary parameter. Let the required trajectories cut the given curves (1) at a constant angle α .

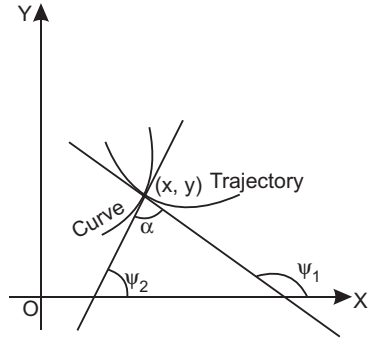
Differentiating (1) with respect to x , we have

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} = 0. \quad \dots(2)$$

Eliminating c between (1) and (2), we get the differential equation of the given family of curves (1). Let it be

$$\phi \left(x, y, \frac{dy}{dx} \right) = 0. \quad \dots(3)$$

Let (x, y) be the coordinates of a point of intersection of a member of the given curves (1) and a member of the required trajectories. If the tangent to the trajectory at this point makes an angle ψ_1 with x -axis and if the tangent to the given curve at this point makes an angle ψ_2 with x -axis, we have



$$dy/dx \text{ given by (3)} = \tan \psi_2 = \tan (\psi_1 - \alpha) = \frac{\tan \psi_1 - \tan \alpha}{1 + \tan \psi_1 \tan \alpha}.$$

Putting this value of (dy/dx) in (3), we get

$$\phi \left(x, y, \frac{\tan \psi_1 - \tan \alpha}{1 + \tan \psi_1 \tan \alpha} \right) = 0. \quad \dots(4)$$

Now dy/dx for the trajectory is $\tan \psi_1$. So replacing $\tan \psi_1$ in the equation (4) by dy/dx , we get the differential equation of the required trajectories as

$$\phi \left(x, y, \frac{dy/dx - \tan \alpha}{1 + (dy/dx) \tan \alpha} \right) = 0.$$

Solving this differential equation, we shall obtain the equation of the required trajectories.

Orthogonal trajectories: If α is a right angle, we have

$$\tan \psi_1 \times \tan \psi_2 = -1.$$

$$\therefore \frac{dy}{dx} \text{ given by (3)} = \tan \psi_2 = -\frac{1}{\tan \psi_1}.$$

Putting this value of dy/dx in (3) of the last article, we get

$$\phi \left(x, y, -\frac{1}{\tan \psi_1} \right) = 0. \quad \dots(5)$$

Now (dy/dx) for the trajectory is $\tan \psi_1$. So replacing $\tan \psi_1$ in the equation (5) by dy/dx , we get differential equation of the orthogonal trajectories of (1) as

$$\phi \left(x, y, -\frac{dx}{dy} \right) = 0.$$

Solving this differential equation, we shall get the equation of the orthogonal trajectories of the given family (1).

Remember: To obtain the differential equation of the orthogonal trajectories, we have to **write** $-dx/dy$ for dy/dx in the differential equation of the original family of curves.

3.3 Orthogonal Trajectories

Polar Co-ordinates: Let the equation of the given family of curves be

$$f(r, \theta, c) = 0, \quad \dots(1)$$

c being the parameter i.e., c can take any real value. Differentiating (1) w.r.t. θ and then eliminating c , let the differential equation of the given family (1) be

$$\phi\left(r, \theta, \frac{dr}{d\theta}\right) = 0. \quad \dots(2)$$

Now if at a point of intersection $P(r, \theta)$, the tangents to the trajectory and the given curve make angles ϕ_1 and ϕ_2 with the common radius vector OP of that point, then

$$\tan 90^\circ = \tan(\phi_1 - \phi_2) = \frac{\tan \phi_1 - \tan \phi_2}{1 + \tan \phi_1 \tan \phi_2}$$

and therefore $\tan \phi_1 \tan \phi_2 = -1$.

Now $r \frac{d\theta}{dr}$ given by (2)

$$= \tan \phi_2 = -\frac{1}{\tan \phi_1}.$$

$\therefore dr/d\theta$ given by (2) $= -r \tan \phi_1$.

$\therefore$ putting this value of $dr/d\theta$ in (2), we have

$$\phi(r, \theta, -r \tan \phi_1) = 0$$

But $\tan \phi_1 = r \frac{d\theta}{dr}$ for the trajectory.

$\therefore$ the differential equation of the required orthogonal trajectories is

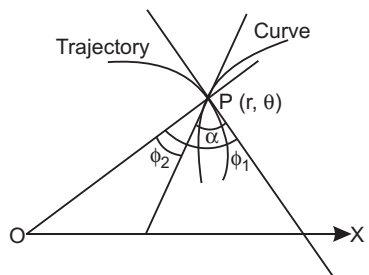
$$\phi\left(r, \theta, -r^2 \frac{d\theta}{dr}\right) = 0.$$

Solving this we shall get the required equation of the orthogonal trajectories.

Remember: The differential equation of the orthogonal trajectories is obtained from the differential equation (in polar co-ordinates) of a given family of curves by writing

$$-r^2 \frac{d\theta}{dr} \quad \text{for} \quad \frac{dr}{d\theta} \quad \text{or} \quad -r \frac{d\theta}{dr} \quad \text{for} \quad \frac{1}{r} \frac{dr}{d\theta}$$

$$\text{or} \quad -\frac{1}{r} \frac{dr}{d\theta} \quad \text{for} \quad r \frac{d\theta}{dr}.$$



Illustrative Examples

Example 1: Find the orthogonal trajectories of the family of curves $y = ax^n$. (Meerut 2011)

Solution: The equation of the given family of curves is

$$y = ax^n, \quad a \text{ being the parameter} \quad \dots(1)$$

Differentiating (1) with respect to x , we get

$$dy/dx = a \cdot nx^{n-1}. \quad \dots(2)$$

Dividing (2) by (1) to eliminate a between (1) and (2), we have

$$\frac{dy/dx}{y} = \frac{anx^{n-1}}{ax^n} = \frac{n}{x}$$

$$\text{or} \quad x(dy/dx) = ny, \quad \dots(3)$$

which is the differential equation of the given family of curves. Now to obtain the differential equation of the orthogonal trajectories, write $-dx/dy$ for dy/dx in (3).

Hence the orthogonal trajectories are given by the differential equation

$$x(-dx/dy) = ny. \quad \dots(4)$$

Separating the variables, we have $-x dx = ny dy$.

Integrating, we have $-\frac{1}{2}x^2 = n\frac{1}{2}y^2 + c$, where c is constant of integration.

Thus $x^2 + ny^2 = 2c$ or $x^2 + ny^2 = c^2$ is the required family of orthogonal trajectories.

Example 2: Find the orthogonal trajectories of the family of parabolas $y^2 = 4ax$, where ' a ' is the variable parameter. (Meerut 2013B)

Solution: Differentiating the given equation $y^2 = 4ax$, \dots(1)

with respect to x , we get

$$2y(dy/dx) = 4a. \quad \dots(2)$$

Eliminating a between (1) and (2), we get

$$2x(dy/dx) = y, \quad \dots(3)$$

which is the differential equation of the given family of parabolas.

So to obtain the differential equation of the orthogonal trajectories, write $-dx/dy$ for dy/dx in (3). Thus the differential equation of the required family of orthogonal trajectories is

$$2x(-dx/dy) = y \quad \text{or} \quad 2x dx = -y dy.$$

Integrating, we get

$$x^2 = -\frac{1}{2}y^2 + c^2 \quad \text{or} \quad 2x^2 + y^2 = k^2,$$

which is the required family of orthogonal trajectories.

Example 3: Find the orthogonal trajectories of the family of rectangular hyperbolas $xy = c^2$.

(Meerut 2003; Rohilkhand 06, 07; Kanpur 07, 12; Bundelkhand 09; Avadh 10, 11)

Solution: The equation of the given family of curves is

$$xy = c^2, \quad c \text{ being the parameter.} \quad \dots(1)$$

Differentiating (1) w.r.t. x , we get

$$x(dy/dx) + y = 0, \quad \dots(2)$$

which is independent of the parameter c . Hence the equation (2) is the differential equation of the given family of curves.

So to obtain the differential equation of the orthogonal trajectories, write $-dx/dy$ for dy/dx in (2).

Thus the differential equation of the required family of orthogonal trajectories is $x(-dx/dy) + y = 0$

or $x dx = y dy$, in which the variables are separated.

$\therefore$ Integrating, $\frac{1}{2} x^2 = \frac{1}{2} y^2 + c$, where c is constant of integration

or $x^2 - y^2 = 2c = k^2$ (say) is the required family of orthogonal trajectories.

Example 4: Find the orthogonal trajectories of the family of parabolas $y^2 = 4a(x + a)$, where 'a' is the parameter. (Meerut 2006; Purvanchal 06; Lucknow 06; Avadh 07, 09)

Solution: The equation of the given family of parabolas is

$$y^2 = 4a(x + a), \quad \dots(1)$$

a being the parameter.

Differentiating (1) w.r.t. x , we get

$$2y(dy/dx) = 4a. \quad \dots(2)$$

Now eliminating a between (1) and (2), we get

$$y^2 = 2y(dy/dx)[x + \frac{1}{2}y(dy/dx)]$$

$$\text{or } y = 2x(dy/dx) + y(dy/dx)^2, \quad \dots(3)$$

which is the differential equation of the given family of parabolas. So to obtain the differential equation of the orthogonal trajectories, write $-dx/dy$ for dy/dx in (3).

Thus the differential equation of the required family of orthogonal trajectories is

$$y = 2x(-dx/dy) + y(-dx/dy)^2$$

$$\text{or } y(dy/dx)^2 + 2x(dy/dx) = y. \quad \dots(4)$$

Now we observe that the differential equation (4) of the orthogonal trajectories is the same as the differential equation (3) of the given family of parabolas. Therefore the given family of parabolas (1) is self-orthogonal i.e., the orthogonal trajectories of the system belong to the system itself. Hence the equation of the orthogonal trajectories of (1) is $y^2 = 4c(x + c)$, c being the parameter.

Example 5: Find the orthogonal trajectory of the family of circles $x^2 + y^2 = 2ax$, a being the parameter. (Meerut 2005, 10)

$$\text{Solution: Differentiating } x^2 + y^2 = 2ax \quad \dots(1)$$

with respect to x , we get

$$2x + 2y(dy/dx) - 2a = 0$$

$$\text{or } x + y(dy/dx) - a = 0. \quad \dots(2)$$

Eliminating 'a' between (1) and (2), we get

$$x^2 + y^2 = 2x\{x + y(dy/dx)\} \text{ or } x^2 + 2xy(dy/dx) - y^2 = 0, \quad \dots(3)$$

which is the differential equation of the given family of circles. So to obtain the differential equation of the orthogonal trajectories, write $-dx/dy$ for dy/dx in (3).

Thus the differential equation of the required family of orthogonal trajectories is

$$x^2 + 2xy(-dx/dy) - y^2 = 0$$

$$\text{or} \quad dx/dy = (x^2 - y^2)/2xy. \quad \dots(4)$$

This is a homogeneous differential equation. To solve it putting $x = vy$, so that

$$dx/dy = v + y(dv/dy), \text{ in (4), we get}$$

$$v + y(dv/dy) = y^2(v^2 - 1)/(2vy^2)$$

$$\text{or} \quad y \frac{dv}{dy} = \frac{v^2 - 1}{2v} - v = -\frac{1 + v^2}{2v}, \quad \text{or} \quad \frac{2v dv}{1 + v^2} = -\frac{dy}{y}.$$

Integrating, we have

$$\log(1 + v^2) = -\log y + \log b \quad \text{or} \quad 1 + v^2 = b/y$$

$$\text{or} \quad 1 + (x^2/y^2) = b/y, \text{ since } v = x/y.$$

Hence $x^2 + y^2 = by$ is the equation of the required orthogonal trajectories.

Example 6: Find the orthogonal trajectories of the family of curves

$$\{x^2/(a^2 + \lambda)\} + \{y^2/(b^2 + \lambda)\} = 1,$$

where λ is the parameter.

(Meerut 2007B, 09B; Lucknow 10; Rohilkhand 10;
Gorakhpur 11; Purvanchal 07, 09, 11)

Solution: Differentiating the given equation with respect to x , we get

$$2x/(a^2 + \lambda) + \{2y/(b^2 + \lambda)\} \cdot (dy/dx) = 0$$

$$\text{or} \quad x(b^2 + \lambda) + y(dy/dx)(a^2 + \lambda) = 0$$

$$\text{or} \quad \lambda \{x + y(dy/dx)\} = -\{b^2x + a^2y(dy/dx)\}.$$

$$\therefore \quad \lambda = -\{b^2x + a^2y(dy/dx)\} \div \{x + y(dy/dx)\}.$$

$$\text{Thus} \quad a^2 + \lambda = (a^2 - b^2)x / \{x + y(dy/dx)\}$$

$$\text{and} \quad b^2 + \lambda = -(a^2 - b^2)y / \{x + y(dy/dx)\}.$$

Substituting these values of $(a^2 + \lambda)$ and $(b^2 + \lambda)$ in the given equation, we get the differential equation of the given family of curves as

$$\frac{x^2 \{x + y(dy/dx)\}}{(a^2 - b^2)x} - \frac{y^2 \{x + y(dy/dx)\}}{(a^2 - b^2)y(dy/dx)} = 1$$

$$\text{or} \quad x^2 - y^2 + xy \left(\frac{dy}{dx} - \frac{1}{dy/dx} \right) = a^2 - b^2. \quad \dots(1)$$

Hence putting $-dx/dy$ for dy/dx in (1), the differential equation of the orthogonal trajectories is

$$x^2 - y^2 + xy \left(-\frac{dx}{dy} + \frac{dy}{dx} \right) = a^2 - b^2$$

or
$$x^2 - y^2 + xy \left(\frac{dy}{dx} - \frac{1}{dy/dx} \right) = a^2 - b^2,$$

which is the same as (1). Therefore solving it we shall get

$$x^2 / (a^2 + \mu) + y^2 (b^2 + \mu) = 1,$$

μ being the parameter of the orthogonal trajectories. In other words the system of given confocal conics $x^2 / (a^2 + \lambda) + y^2 / (b^2 + \lambda) = 1$ is self-orthogonal.

Example 7: Find the orthogonal trajectories of the cardioids $r = a(1 - \cos \theta)$, a being the parameter. (Meerut 2006B; Gorakhpur 05; Lucknow 11; Kashi 11)

Solution: The given family of curves is $r = a(1 - \cos \theta)$, ... (1)

the parameter being a . Differentiating (1) w.r.t. θ , we get

$$dr/d\theta = a \sin \theta. \quad \dots (2)$$

Now we shall eliminate a between (1) and (2). So dividing (1) by (2), we get

$$r (d\theta/dr) = (1 - \cos \theta) / \sin \theta, \quad \dots (3)$$

which is the differential equation of the given family of curves (1).

To obtain the differential equation of the orthogonal trajectories, write $-(1/r) (dr/d\theta)$ for $r (d\theta/dr)$ in (3). Thus we get

$$-\frac{1}{r} \frac{dr}{d\theta} = \frac{1 - \cos \theta}{\sin \theta} = \frac{2 \sin^2 \frac{1}{2} \theta}{2 \sin \frac{1}{2} \theta \cos \frac{1}{2} \theta} = \tan \frac{1}{2} \theta,$$

which is the differential equation of the orthogonal trajectories. To solve it, separating the variables, we get

$$\frac{dr}{r} = -\tan \frac{1}{2} \theta d\theta = \frac{-\sin \frac{1}{2} \theta}{\cos \frac{1}{2} \theta} d\theta = 2 \cdot \frac{-\frac{1}{2} \sin \frac{1}{2} \theta}{\cos \frac{1}{2} \theta} d\theta.$$

Integrating, we have

$$\log r = 2 \log \cos \frac{1}{2} \theta + \log c$$

or
$$\log (r/c) = \log \cos^2 \frac{1}{2} \theta \quad \text{or} \quad r/c = \cos^2 \frac{1}{2} \theta = \frac{1}{2} (1 + \cos \theta)$$

or
$$r = \frac{1}{2} c (1 + \cos \theta) \quad \text{or} \quad r = k (1 + \cos \theta),$$

which is the required family of orthogonal trajectories.

Example 8: Find the orthogonal trajectories of the system of curves $r^n = a^n \cos n\theta$, a being the parameter. (Meerut 2007, 10B; Bundelkhand 05; Gorakhpur 06; Lucknow 08; Kanpur 08)

Solution: The given family of curves is $r^n = a^n \cos n\theta$, ... (1)

the parameter being a . Taking logarithm of both sides of (1), we get

$$n \log r = n \log a + \log \cos n\theta.$$

Now differentiating both sides w.r.t. θ , we get

$$\frac{n}{r} \frac{dr}{d\theta} = 0 + \frac{-n \sin n\theta}{\cos n\theta} \quad \text{or} \quad \frac{1}{r} \frac{dr}{d\theta} = -\tan n\theta \quad \dots(2)$$

which is the differential equation of the family of curves (1).

Now putting $-r \frac{d\theta}{dr}$ for $\frac{1}{r} \frac{dr}{d\theta}$ in (2), the differential equation of the orthogonal trajectories is

$$-r \frac{d\theta}{dr} = -\tan n\theta \quad \text{or} \quad \frac{dr}{r} = \cot n\theta \, d\theta.$$

Integrating, we get $\log r = (1/n) \log \sin n\theta + \log c$

$$\text{or} \quad n \log (r/c) = \log \sin n\theta$$

$$\text{or} \quad \log (r/c)^n = \log \sin n\theta$$

$$\text{or} \quad r^n = c^n \sin n\theta,$$

which is the required family of orthogonal trajectories, the parameter being c .

Comprehensive Exercise 1

1. Find the equation of the family of curves orthogonal to the family $y = ax^3$.
(Avadh 2005; Rohilkhand 09; Kanpur 11)
2. Find the orthogonal trajectories of the semi-cubical parabolas $ay^2 = x^3$, where a is the variable parameter.
(Gorakhpur 2007; Kashi 13)
3. Find the orthogonal trajectories of the family of circles $x^2 + y^2 = a^2$, where ' a ' is the parameter.
(Bundelkhand 2008; Kanpur 10)
4. Find the orthogonal trajectories of the family of curves $x^{2/3} + y^{2/3} = a^{2/3}$, where a is parameter.
5. Find the equation of the family of curves that is orthogonal to $ax^2 + y^2 = 1$.
6. Find the orthogonal trajectories of the system of curves $(dy/dx)^2 = a/x$.
(Bundelkhand 2007, 10; Agra 08; Purvanchal 14)
7. Find the differential equation of the family of curves given by the equation $x^2 - y^2 + 2\lambda xy = 1$, where λ is a parameter.

Obtain the differential equation of its orthogonal trajectories and solve it.

8. Show that the orthogonal trajectories of the family of conics $y^2 - x^2 + 4xy - 2cx = 0$ consist of a family of cubics, with the common asymptote $x + y = 0$.
(Meerut 2009)
9. Find the orthogonal trajectories of the family of circles $x^2 + y^2 + 2fy + 1 = 0$, where f is the parameter.
(Meerut 2004; Gorakhpur 08, 11)

10. Find the orthogonal trajectories of the family of curves

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 + \lambda} = 1, \lambda \text{ being parameter.}$$

(Lucknow 2005; Meerut 05B; Purvanchal 14)

11. Find the orthogonal trajectories of the family of coaxial circles $x^2 + y^2 + 2gx + c = 0$, where g is a parameter and c is a constant. (Agra 2007)

12. Find the orthogonal trajectories of

(i) $r\theta = a$,

(ii) $r = a\theta$.

(Rohilkhand 2008)

13. Find the orthogonal trajectories of the family of cardioids $r = a(1 + \cos \theta)$.

(Rohilkhand 2006; Lucknow 06; Purvanchal 08, 10; Gorakhpur 10)

14. Find the orthogonal trajectories of $r = e^{a\theta}$.

15. Find the orthogonal trajectories of $r^n \sin n\theta = a^n$.

(Meerut 2001, 04B)

Answers 1

1. $x^2 + 3y^2 = c^2$
2. $x^2 + \frac{3}{2}y^2 = c^2$
3. $x = cy$
4. $x^{4/3} - y^{4/3} = c^{4/3}$
5. $y^2 = ce^{(x^2 + y^2)}$
6. $(x^{3/2} + c^{3/2})^2 = \frac{9}{4}ay^2$
7. $\frac{1}{2}(x^2 + y^2)^2 + y^2 - x^2 = c^2$
8. $y^3 + 3x^2y + 4x^3 = k^3$
9. $y^2 + x^2 - 1 = cx$
10. $x^2 + y^2 - 2a^2 \log x = c$
11. $x^2 + y^2 + 2fy - c = 0$, f being parameter.
12. (i) $r^2 = c^2 \cdot e^{\theta^2}$
- (ii) $r = ce^{-\frac{1}{2}\theta^2}$
13. $r = c(1 - \cos \theta)$
14. $\theta^2 + (\log r)^2 = c^2$
15. $r^n \cos n\theta = c^n$

Objective Type Questions

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. The differential equation of the orthogonal trajectories is obtained from the differential equation of the given family of curves by writing for dy/dx .
2. The differential equation of the orthogonal trajectories is obtained from the differential equation (in polar coordinates) of the given family of curves by writing for $\frac{dr}{d\theta}$.

3. Orthogonal trajectories of the system of curves $r^n = a^n \cos n\theta$, a being the parameter, is

True or False

Write 'T' for true and 'F' for false statement.

1. A family of curves is said to be self orthogonal if the differential equation of the orthogonal trajectories is the same as the differential equation of the given family of curves.
2. Family of curves $\{x^2 / (a^2 + \lambda)\} + \{y^2 / (b^2 + \lambda)\} = 1$, where λ is the parameter, is self orthogonal.

Answers

Fill in the Blank(s)

1. $-\frac{dx}{dy}$
2. $-r^2 \frac{d\theta}{dr}$
3. $r^n = c^n \sin n\theta$

True or False

1. T
2. T



Chapter

4



Linear Differential Equations With Constant Coefficients

4.1 Definitions

A linear differential equation is an equation in which the dependent variable and its derivatives appear only in the first degree.

A linear differential equation of order n of the form

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = Q, \quad \dots(1)$$

where $a_1, a_2, \dots, a_{n-1}, a_n$ are constants and Q is any function of x is called a **linear differential equation with constant coefficients**.

For convenience, the operators $\frac{d}{dx}, \frac{d^2}{dx^2}, \frac{d^3}{dx^3}, \dots, \frac{d^n}{dx^n}$ are also denoted by $D, D^2, D^3, \dots, D^n$ respectively.

Thus the equation (1) can also be written as

$$D^n y + a_1 D^{n-1} y + \dots + a_{n-1} D y + a_n y = Q$$

or $[D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n] y = Q. \quad \dots(2)$

If $y = f(x)$ is the general solution of

$$[D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n] y = 0, \quad \dots(3)$$

and $y = \phi(x)$ is any particular solution of the equation (2) not containing any arbitrary constant, then

$$y = f(x) + \phi(x),$$

is the general solution of (2).

Thus the method of solving a linear equation is divided into two parts :

First, we find the general solution of the equation (3).

It is called the *complementary function* (C.F.). It must contain as many arbitrary constants as is the order of the given differential equation.

Next, we find a solution of (2) which does not contain an arbitrary constant. This is called the *particular integral* (P.I.).

If we add (C.F.) and (P.I.), we get the general solution of (2). Thus the general solution of (2) is

$$y = \text{C. F.} + \text{P. I.} \quad (\text{Gorakhpur 2005})$$

4.2 Determination of Complementary Function (C.F.)

Consider a linear n th order differential equation with constant coefficients of the form $f(D)y = 0$ i.e.,

$$[D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_n] y = 0. \quad \dots(1)$$

This is equivalent to

$$[(D - m_1)(D - m_2) \dots (D - m_n)] y = 0. \quad \dots(2)$$

The solution of any one of the equations

$$(D - m_1)y = 0, (D - m_2)y = 0, \dots, (D - m_n)y = 0 \quad \dots(3)$$

is also a solution of (2) and we know that the general solution of $(D - m_1)y = 0$ is

$$y = Ae^{m_1 x}.$$

Hence we can assume that a solution of the equation (2) is of the form $y = e^{mx}$. Then, substituting e^{mx} for y in (1), so that $Dy = me^{mx}$, $D^2 y = m^2 e^{mx}$, ..., $D^n y = m^n e^{mx}$, we get

$$e^{mx} (m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n) = 0$$

or $m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n = 0$, because $e^{mx} \neq 0$.

Hence e^{mx} will be a solution of (1) if m has the value obtained from the equation $m^n + a_1 m^{n-1} + \dots + a_n = 0$(4)

The equation (4) is called the **auxiliary equation (A.E.)** and is obtained by putting $D = m$ in $f(D) = 0$.

It will give in general n roots, say, $m_1, m_2, m_3, \dots, m_n$.

Case I: If all the roots of the Auxiliary equation (A.E.) are distinct:

If the roots $m_1, m_2, m_3, \dots, m_n$ are all distinct, then $e^{m_1 x}, e^{m_2 x}, \dots, e^{m_n x}$ are all distinct and linearly independent. So the general solution of (1) in this case is

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \dots + c_n e^{m_n x}. \quad \dots(5)$$

Case II: Auxiliary equation having equal roots :

If two roots are equal say $m_1 = m_2$, then the solution (5) becomes

$$y = c_1 e^{m_1 x} + c_2 e^{m_1 x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$$

or
$$y = (c_1 + c_2) e^{m_1 x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}.$$

Now $(c_1 + c_2)$ can be replaced by single constant say c .

Therefore this solution has only $(n - 1)$ arbitrary constants and so it is not the general solution.

To obtain the general solution, consider the differential equation $(D - m_1)^2 y = 0$ in which the two roots are equal.

This can be written as $(D - m_1) [(D - m_1) y] = 0$(6)

Now putting $(D - m_1) y = v$, we get $(D - m_1) v = 0$,

or $dv/dx = m_1 v$ or $dv/v = m_1 dx$, (variables being separated).

$\therefore$ integrating,

$$\log v = m_1 x + \log c_1 \quad \text{or} \quad \log (v/c_1) = m_1 x \quad \text{or} \quad v = c_1 e^{m_1 x}.$$

Thus $(D - m_1) y = v = c_1 e^{m_1 x}$

and putting it in (6), we get $(D - m_1) y = c_1 e^{m_1 x}$

or $Dy - m_1 y = c_1 e^{m_1 x}$ or $\frac{dy}{dx} - m_1 y = c_1 e^{m_1 x}$. $\left[\because D \equiv \frac{d}{dx} \right]$

This is a linear equation of the first order. Hence the I.F. = $e^{\int -m_1 dx} = e^{-m_1 x}$.

$\therefore$ The solution of this equation is

$$\begin{aligned} y \cdot e^{-m_1 x} &= \int c_1 e^{m_1 x} \cdot e^{-m_1 x} dx + c_2 \\ &= c_1 \int dx + c_2 = c_1 x + c_2 \end{aligned}$$

or $y = (c_1 x + c_2) e^{m_1 x}$ or $y = (c_2 + c_1 x) e^{m_1 x}$.

Hence the general solution of $f(D) y = 0$ in this case is

$$y = (c_1 + c_2 x) e^{m_1 x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}.$$

Similarly if three roots of the auxiliary equation are equal say, $m_1 = m_2 = m_3$, the general solution of $f(D) y = 0$ will be

$$y = (c_1 + c_2 x + c_3 x^2) e^{m_1 x} + c_4 e^{m_4 x} + \dots + c_n e^{m_n x}$$

and so on.

Case III: Auxiliary equation having complex roots:

(Meerut 2003)

Let the two roots of the auxiliary equation be complex, say $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, (where $i = \sqrt{-1}$).

The solution corresponding to these two roots will be

$$\begin{aligned} y &= c_1 e^{(\alpha + i\beta)x} + c_2 e^{(\alpha - i\beta)x} = c_1 e^{\alpha x} \cdot e^{i\beta x} + c_2 e^{\alpha x} \cdot e^{-i\beta x} \\ &= c_1 e^{\alpha x} (\cos \beta x + i \sin \beta x) + c_2 e^{\alpha x} (\cos \beta x - i \sin \beta x) \\ &= (c_1 + c_2) e^{\alpha x} \cos \beta x + i(c_1 - c_2) e^{\alpha x} \sin \beta x \end{aligned}$$

or $y = e^{\alpha x} [A_1 \cos \beta x + A_2 \sin \beta x]$, where $A_1 = c_1 + c_2$ and $A_2 = i(c_1 - c_2)$

or $y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$, changing the constants.

If the imaginary roots are repeated, say $\alpha + i\beta$ and $\alpha - i\beta$ occur twice then the solution will be

$$y = e^{\alpha x} [(c_1 + c_2 x) \cos \beta x + (c_3 + c_4 x) \sin \beta x],$$

and so on.

Note 1: The expression $e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$ can also be written as

$$c_1 e^{\alpha x} \sin (\beta x + c_2) \quad \text{or} \quad c_1 e^{\alpha x} \cos (\beta x + c_2).$$

Note 2: If a pair of the roots of the auxiliary equation are irrational i.e., they are $\alpha \pm \sqrt{\beta}$, where β is positive, then the corresponding term in the C.F. will be

$$e^{\alpha x} (c_1 \cosh \sqrt{\beta} x + c_2 \sinh \sqrt{\beta} x),$$

or $c_1 e^{\alpha x} \sinh (\sqrt{\beta} x + c_2) \quad \text{or} \quad c_1 e^{\alpha x} \cosh (\sqrt{\beta} x + c_2).$

If these irrational roots are repeated twice, then the corresponding portion of the solution will be

$$e^{\alpha x} \{(c_1 + c_2 x) \cosh \sqrt{\beta} x + (c_3 + c_4 x) \sinh \sqrt{\beta} x\}.$$

Illustrative Examples

Example 1: Solve $\frac{d^2 y}{dx^2} - 7 \frac{dy}{dx} + 12y = 0$.

Solution: The given differential equation is $(D^2 - 7D + 12)y = 0$.

$\therefore$ the auxiliary equation is $m^2 - 7m + 12 = 0$

or $(m - 3)(m - 4) = 0 \quad \therefore \quad m = 3, 4.$

Hence the solution is

$$y = c_1 e^{3x} + c_2 e^{4x}.$$

Example 2: Solve $(D^3 + 6D^2 + 11D + 6)y = 0$.

(Meerut 2010)

Solution: The auxiliary equation is

$$m^3 + 6m^2 + 11m + 6 = 0$$

or $(m + 1)(m^2 + 5m + 6) = 0$

or $(m + 1)(m + 2)(m + 3) = 0.$

$$\therefore m = -1, -2, -3.$$

Hence the solution is

$$y = c_1 e^{-x} + c_2 e^{-2x} + c_3 e^{-3x}.$$

Example 3: Solve $\frac{d^2 x}{dt^2} - 3 \frac{dx}{dt} + 2x = 0$, given that when $t = 0$, $x = 0$ and $dx/dt = 0$.

(Gorakhpur 2008)

Solution: The auxiliary equation is $m^2 - 3m + 2 = 0$

$$\text{or } (m-1)(m-2) = 0 \quad \text{or } m = 1, 2.$$

$$\text{Hence the solution is } x = c_1 e^t + c_2 e^{2t}, \quad \dots(1)$$

where c_1 and c_2 are arbitrary constants.

Now $x = 0$ when $t = 0$;

$$\therefore 0 = c_1 + c_2. \quad \dots(2)$$

Also $dx/dt = c_1 e^t + 2c_2 e^{2t}$, and $dx/dt = 0$ when $t = 0$.

$$\therefore 0 = c_1 + 2c_2. \quad \dots(3)$$

Solving (2) and (3), we get $c_1 = 0, c_2 = 0$.

Now putting the values of c_1 and c_2 in (1), we get the required solution as $x = 0$.

Example 4: Solve $(D^3 - 3D + 2)y = 0$.

Solution: The auxiliary equation is $m^3 - 3m + 2 = 0$

$$\text{or } (m-1)(m^2 + m - 2) = 0 \quad \text{or } (m-1)\{(m-1)(m+2)\} = 0.$$

$$\therefore m = 1, 1, -2.$$

Hence the solution is

$$y = (c_1 + c_2 x) e^x + c_3 e^{-2x}.$$

Example 5: Solve $\frac{d^3 y}{dx^3} - 8y = 0$.

(Meerut 2010B)

Solution: The auxiliary equation is $m^3 - 8 = 0$

$$\text{or } (m-2)(m^2 + 2m + 4) = 0 \quad \text{i.e., } m - 2 = 0 \quad \text{and } m^2 + 2m + 4 = 0.$$

$$\therefore m = 2 \quad \text{and } m = -1 \pm i\sqrt{3}.$$

Hence the solution is

$$y = e^{-x} (c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x) + c_3 e^{2x}.$$

Example 6: Solve $\frac{d^4 y}{dx^4} - 2 \frac{d^3 y}{dx^3} - 2 \frac{dy}{dx} - y = 0$.

Solution: The auxiliary equation is

$$m^4 - 2m^3 - 2m - 1 = 0$$

$$\text{or } (m^4 - 1) - 2m(m^2 + 1) = 0$$

$$\text{or } (m^2 + 1)(m^2 - 1) - 2m(m^2 + 1) = 0$$

$$\text{or} \quad (m^2 + 1)(m^2 - 2m - 1) = 0 \quad \text{i.e., } m^2 = -1 \quad \text{or} \quad m^2 - 2m - 1 = 0.$$

$$\therefore \quad m = 0 \pm i, 1 \pm \sqrt{2}.$$

Hence the solution is

$$y = e^{0x}(c_1 \cos x + c_2 \sin x) + e^x(c_3 \cosh \sqrt{2}x + c_4 \sinh \sqrt{2}x)$$

$$\text{or} \quad y = c_1 \cos x + c_2 \sin x + e^x \{c_3 \cosh \sqrt{2}x + c_4 \sinh \sqrt{2}x\}.$$

Example 7: Solve $(D^4 + k^4)y = 0$.

(Meerut 2003, 04B, 05B, 06)

Solution: The auxiliary equation is $m^4 + k^4 = 0$

$$\text{or} \quad (m^2 + k^2)^2 - 2k^2m^2 = 0$$

$$\text{or} \quad (m^2 + k^2)^2 - (\sqrt{2}km)^2 = 0$$

$$\text{or} \quad (m^2 + k^2 - \sqrt{2}km)(m^2 + k^2 + \sqrt{2}km) = 0$$

$$\text{or} \quad m^2 - \sqrt{2}km + k^2 = 0 \text{ and } m^2 + \sqrt{2}km + k^2 = 0$$

$$\text{or} \quad m = \frac{\sqrt{2}k \pm \sqrt{(2k^2 - 4k^2)}}{2} \text{ and } m = \frac{-\sqrt{2}k \pm \sqrt{(2k^2 - 4k^2)}}{2}$$

$$\text{or} \quad m = \frac{k}{\sqrt{2}} \pm i \frac{k}{\sqrt{2}} \text{ and } -\frac{k}{\sqrt{2}} \pm i \frac{k}{\sqrt{2}}.$$

Hence the solution is

$$y = e^{kx/\sqrt{2}} \{c_1 \cos(kx/\sqrt{2}) + c_2 \sin(kx/\sqrt{2})\} \\ + e^{-kx/\sqrt{2}} \{c_3 \cos(kx/\sqrt{2}) + c_4 \sin(kx/\sqrt{2})\}.$$

Comprehensive Exercise 1

Solve the following differential equations :

$$1. \frac{d^2 y}{dx^2} + (a + b) \frac{dy}{dx} + aby = 0. \quad 2. \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} - 4y = 0.$$

$$3. \frac{d^3 y}{dx^3} + 6 \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} - 10y = 0. \quad 4. \frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = 0.$$

$$5. (D^3 - 4D^2 + 5D - 2)y = 0.$$

$$6. \frac{d^4 y}{dx^4} + 2 \frac{d^3 y}{dx^3} - 3 \frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = 0.$$

$$7. (i) (D^4 + 1)y = 0. \quad (\text{Agra 2006}) \quad (ii) \frac{d^4 y}{dx^4} - k^4 y = 0.$$

$$8. (D^4 + 8D^2 + 16)y = 0.$$

9. $(D^2 \pm \mu^2) y = 0$.

(Avadh 05, 14)

10. Solve $(d^2 y/dx^2) + y = 0$, given $y = 2$ for $x = 0$ and $y = -2$ for $x = \frac{1}{2}\pi$.

Answers 1

1. $y = c_1 e^{-ax} + c_2 e^{-bx}$

2. $y = c_1 e^{-x} + c_2 e^{4x}$

3. $y = c_1 e^x + c_2 e^{-2x} + c_3 e^{-5x}$

4. $y = (c_1 + c_2 x) e^{2x}$

5. $y = (c_1 + c_2 x) e^x + c_3 e^{2x}$

6. $y = (c_1 + c_2 x) e^x + (c_3 + c_4 x) e^{-2x}$

7. (i) $y = e^{x/\sqrt{2}} \{c_1 \cos(x/\sqrt{2}) + c_2 \sin(x/\sqrt{2})\}$
 $+ e^{-x/\sqrt{2}} \{c_3 \cos(x/\sqrt{2}) + c_4 \sin(x/\sqrt{2})\}.$

(ii) $y = c_1 e^{kx} + c_2 e^{-kx} + c_3 \cos kx + c_4 \sin kx$

8. $y = (c_1 + c_2 x) \cos 2x + (c_3 + c_4 x) \sin 2x$

9. $y = c_1 \cos \mu x + c_2 \sin \mu x$ and $y = c_3 e^{\mu x} + c_4 e^{-\mu x}$

10. $y = 2\sqrt{2} \cos\left(x + \frac{1}{4}\pi\right)$

4.3 The Particular Integral (P.I.)

As already shown in 4.1 the complete solution of

$$(D^n + a_1 D^{n-1} + a_2 D^{n-2} + \dots + a_{n-1} D + a_n) y = Q \quad \dots(1)$$

or $F(D) y = Q$ is $y = \text{C.F.} + \text{P.I.}$,

where the C.F. consists of the general solution of the differential equation

$$F(D) y = 0.$$

In article 4.2 we have discussed different methods of finding the complementary function by taking the differential equation as $F(D) y = 0$. Methods of finding the particular integral will be discussed now.

The particular integral of the differential equation $F(D) y = Q$ is $\frac{1}{F(D)} Q$. It is

obviously a function of x which when operated by $F(D)$ gives Q .

Now as $F(D) \frac{1}{F(D)} Q = Q$, therefore $\frac{1}{F(D)}$ can be regarded as the inverse operator of

$F(D)$. Similarly D and $1/D$ are inverse operations. If D stands for differentiation then $1/D$ will stand for integration.

Hence the particular integral of the equation

$$F(D)y = Q \text{ will be } \frac{1}{F(D)} Q$$

because it satisfies the given equation.

4.4 Particular Integral in Some Special Cases

Case I: To find P.I. when Q is of the form e^{ax} , where a is any constant and

$$F(a) \neq 0.$$

By simple differentiation we know that $D(e^{ax}) = ae^{ax}$;

$$D^2(e^{ax}) = a^2 e^{ax}; D^3(e^{ax}) = a^3 e^{ax}, \dots, D^n(e^{ax}) = a^n e^{ax}.$$

It suggests that $F(D)e^{ax} = F(a)e^{ax}$(1)

Let $F(a) \neq 0$.

Operating on both sides of (1) with $1/F(D)$, we have

$$\frac{1}{F(D)} F(D)e^{ax} = \frac{1}{F(D)} \{F(a)e^{ax}\}$$

or $e^{ax} = F(a) \cdot \frac{1}{F(D)} e^{ax}$, because $F(a)$ is a constant

or $\frac{1}{F(a)} e^{ax} = \frac{1}{F(D)} e^{ax}$, because $F(a) \neq 0$

$\therefore$ P.I. = $\frac{1}{F(D)} e^{ax} = \frac{1}{F(a)} e^{ax}$, provided $F(a) \neq 0$.

Working Rule : If P.I. = $\{1/F(D)\} e^{ax}$, then put a for D in $F(D)$ and we get the P.I., provided $F(a) \neq 0$.

Examples on case I:

Example 9: Solve $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^{5x}$.

(Avadh 2008; Purvanchal 11; Kanpur 12)

Solution: The given equation can be written as

$$(D^2 - 3D + 2)y = e^{5x}, \text{ where } d/dx \equiv D.$$

Here $F(D) = D^2 - 3D + 2$ and $Q = e^{5x}$.

The auxiliary equation is $m^2 - 3m + 2 = 0$

or $(m-1)(m-2) = 0$;

$\therefore m = 1, 2$.

$\therefore$ The C.F. = $c_1 e^x + c_2 e^{2x}$, the roots of the A.E. being distinct.

and P.I. = $\frac{1}{F(D)} e^{ax} = \frac{1}{D^2 - 3D + 2} e^{5x} = \frac{e^{5x}}{5^2 - 3 \cdot 5 + 2} = \frac{e^{5x}}{12}$.

[We have put 5 for D in $F(D)$, because here $a = 5$]

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

or
$$y = c_1 e^x + c_2 e^{2x} + \frac{1}{12} e^{5x}.$$

Example 10: Solve $\frac{d^2 y}{dx^2} + 31 \frac{dy}{dx} + 240 y = 272 e^{-x}$.

(Meerut 2010B, 11)

Solution: The auxiliary equation is

$$m^2 + 31m + 240 = 0 \quad \text{or} \quad (m + 15)(m + 16) = 0;$$

$\therefore m = -15, -16.$

$\therefore \text{C.F.} = c_1 e^{-15x} + c_2 e^{-16x}.$

and
$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + 31D + 240} (272e^{-x}) = 272 \cdot \frac{1}{D^2 + 31D + 240} e^{-x} \\ &= 272 \cdot \frac{1}{(-1)^2 + 31(-1) + 240} e^{-x} = \frac{272}{210} e^{-x} = \frac{136}{105} e^{-x}. \end{aligned}$$

Hence the general solution is $y = (\text{C.F.}) + (\text{P.I.})$

i.e.,
$$y = c_1 e^{-15x} + c_2 e^{-16x} + \frac{136}{105} e^{-x}.$$

Example 11: Obtain the complete solution of the differential equation

$$\frac{d^2 y}{dx^2} - 7 \frac{dy}{dx} + 6y = e^{2x},$$

and determine the constant so that $y = 0$ when $x = 0$.

Solution: The given equation is $[D^2 - 7D + 6] y = e^{2x}.$

$\therefore$ auxiliary equation is $m^2 - 7m + 6 = 0$

or $(m - 1)(m - 6) = 0 \quad \text{or} \quad m = 1, 6.$

$\therefore \text{C.F.} = c_1 e^x + c_2 e^{6x}.$

and
$$\text{P.I.} = \frac{1}{D^2 - 7D + 6} e^{2x} = \frac{1}{2^2 - 7 \cdot 2 + 6} e^{2x} = -\frac{e^{2x}}{4}.$$

$\therefore$ the general solution is $y = (\text{C.F.}) + (\text{P.I.})$

i.e.,
$$y = c_1 e^x + c_2 e^{6x} - \frac{1}{4} e^{2x}. \quad \dots(1)$$

Now when $y = 0, x = 0$;

$\therefore 0 = c_1 + c_2 - \frac{1}{4}, \text{ from (1)}$

so that
$$c_2 = \left(\frac{1}{4} - c_1 \right).$$

Hence from (1), we have

$$y = c_1 e^x + \left(\frac{1}{4} - c_1 \right) e^{6x} - \frac{1}{4} e^{2x} = c_1 (e^x - e^{6x}) + \frac{1}{4} (e^{6x} - e^{2x})$$

or
$$y = c_1 (e^x - e^{6x}) + \frac{1}{4} e^{2x} (e^{4x} - 1) \text{ is the required solution.}$$

Comprehensive Exercise 2

Solve the following differential equations :

1. $\frac{d^2 y}{dx^2} - 2k \frac{dy}{dx} + k^2 y = e^x.$

2. $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y = 2e^{2x}.$

(Bundelkhand 2001)

3. $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = e^x.$

4. $[D^2 + D + 1] y = e^{-x}.$

5. $\frac{d^3 y}{dx^3} + 6 \frac{d^2 y}{dx^2} + 11 \frac{dy}{dx} + 6y = e^{2x}.$

Answers 2

1. $y = (c_1 + c_2 x) e^{kx} + e^x / (1 - k)^2$

2. $y = (c_1 + c_2 x) e^{-x} + \frac{2}{9} e^{2x}$

3. $y = e^{-x/2} \{c_1 \cos(x \frac{1}{2} \sqrt{3}) + c_2 \sin(x \frac{1}{2} \sqrt{3})\} + \frac{1}{3} e^x$

4. $y = c_1 e^{-x/2} \cos\{x(\sqrt{3}/2) + c_2\} + e^{-x}$

5. $y = c_1 e^{-x} + c_2 e^{-2x} + c_3 e^{-3x} + \frac{1}{60} e^{2x}$

Case II: To find P.I. when Q is of the form $\sin ax$ or $\cos ax$ and $F(-a^2) \neq 0$.

By simple differentiation we know that

$$D(\sin ax) = a \cos ax; D^2(\sin ax) = -a^2 \sin ax;$$

$$D^3(\sin ax) = -a^3 \cos ax; D^4(\sin ax) = (-a^2)^2 \sin ax,$$

$$\dots\dots, (D^2)^n \sin ax = (-a^2)^n \sin ax.$$

It suggests that

$$F(D^2) \sin ax = F(-a^2) \sin ax. \quad \dots(1)$$

Let $F(-a^2) \neq 0$.

Now operating both sides of (1) with $1/F(D^2)$, we get

$$\frac{1}{F(D^2)} F(D^2) \sin ax = \frac{1}{F(D^2)} \{F(-a^2) \sin ax\}$$

or $\sin ax = F(-a^2) \cdot \frac{1}{F(D^2)} \sin ax.$

Thus $\frac{1}{F(D^2)} \sin ax = \frac{1}{F(-a^2)} \sin ax$, provided $F(-a^2) \neq 0$.

Similarly $\frac{1}{F(D^2)} \cos ax = \frac{1}{F(-a^2)} \cos ax$, provided $F(-a^2) \neq 0$.

Working Rule: If P.I. = $\{1 / F(D)\} \sin ax$ or $\cos ax$, put $-a^2$ for D^2 , $-a^2 D$ for D^3 , $(-a^2)^2$ i.e., a^4 for D^4 , $a^4 D$ for D^5 , $-a^6$ for D^6 etc. in $F(D)$ and calculate the P.I.

Note: Linear factors in D of the form $(pD \pm q)$ appearing in the denominator are removed by first multiplying the Nr. and Deno. by the conjugate factors $(pD \mp q)$ and then putting $-a^2$ for D^2 in the denominator. The operation left in the numerator can be easily worked out because the operator D stands for differentiation with respect to x .

Examples on Case II:

Example 12: Solve $\frac{d^2 y}{dx^2} - \frac{dy}{dx} - 2y = \sin 2x$.

(Meerut 2008, 10)

Solution: The given equation is $[D^2 - D - 2] y = \sin 2x$.

$\therefore$ auxiliary equation is $m^2 - m - 2 = 0$

or $(m + 1)(m - 2) = 0$; $\therefore m = -1, 2$.

$\therefore$ C.F. = $c_1 e^{-x} + c_2 e^{2x}$.

And P.I. = $\frac{1}{D^2 - D - 2} \sin 2x = \frac{1}{-4 - D - 2} \sin 2x$,

putting -2^2 i.e., -4 for D^2

$$= -\frac{1}{D + 6} \sin 2x = -\frac{(D - 6)}{(D + 6)(D - 6)} \sin 2x = -\frac{(D - 6)}{D^2 - 36} \sin 2x$$

$$= -\frac{(D - 6)}{-4 - 36} \sin 2x, \text{ putting } -4 \text{ for } D^2$$

$$= \frac{1}{40} (D - 6) \sin 2x = \frac{1}{40} \{D(\sin 2x) - 6 \sin 2x\}$$

$$= \frac{1}{40} (2 \cos 2x - 6 \sin 2x) = \frac{1}{20} \cos 2x - \frac{3}{20} \sin 2x.$$

$\therefore$ the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

or $y = c_1 e^{-x} + c_2 e^{2x} + \frac{1}{20} \cos 2x - \frac{3}{20} \sin 2x$.

Example 13: Solve $(D^2 - 5D + 6) y = \sin 3x$.

Solution: Here the auxiliary equation is $m^2 - 5m + 6 = 0$.

or $(m - 2)(m - 3) = 0$; $\therefore m = 2, 3$.

$\therefore$ C.F. = $c_1 e^{2x} + c_2 e^{3x}$.

$$\begin{aligned}
 \text{And } P.I. &= \frac{1}{D^2 - 5D + 6} \sin 3x = \frac{1}{-3^2 - 5D + 6} \sin 3x = \frac{1}{-9 - 5D + 6} \sin 3x \\
 &= \frac{-1}{5D + 3} \sin 3x = \frac{-(5D - 3)}{(5D + 3)(5D - 3)} \sin 3x = \frac{-(5D - 3)}{25D^2 - 9} \sin 3x \\
 &= \frac{-(5D - 3)}{25 \cdot (-3^2) - 9} \sin 3x = \frac{1}{234} \{5D (\sin 3x) - 3 \sin 3x\} \\
 &= \frac{1}{234} (5 \cdot 3 \cos 3x - 3 \sin 3x) = \frac{1}{78} (5 \cos 3x - \sin 3x).
 \end{aligned}$$

$\therefore$ the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = c_1 e^{2x} + c_2 e^{3x} + (1/78) (5 \cos 3x - \sin 3x).$$

Example 14: Solve $\frac{d^2 y}{dx^2} + 9y = \cos 2x + \sin 2x$.

Solution: The auxiliary equation is

$$m^2 + 9 = 0, \quad \therefore m = \pm 3i.$$

$$\therefore \text{C.F.} = c_1 \cos 3x + c_2 \sin 3x \quad \text{or} \quad = c_1 \cos (3x + c_2)$$

$$\begin{aligned}
 \text{And } P.I. &= \frac{1}{(D^2 + 9)} (\cos 2x + \sin 2x) = \frac{1}{D^2 + 9} \cos 2x + \frac{1}{D^2 + 9} \sin 2x \\
 &= \frac{\cos 2x}{-2^2 + 9} + \frac{\sin 2x}{-2^2 + 9} = \frac{\cos 2x}{5} + \frac{\sin 2x}{5} = \frac{1}{5} (\cos 2x + \sin 2x).
 \end{aligned}$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = c_1 \cos (3x + c_2) + \frac{1}{5} (\cos 2x + \sin 2x).$$

Comprehensive Exercise 3

Solve the following differential equations :

1. $(d^2 y/dx^2) + 9y = \cos 4x$.
2. (i) $(D^2 - 3D + 2)y = \sin 3x$. (ii) $(D^2 - 2D + 5)y = \sin 3x$.
3. (i) $(D^2 - 3D + 2)y = \cos 3x$
(ii) $(D^3 + D^2 - D - 1)y = \cos 2x$ (Avadh 2008; Kanpur 06)
4. $\frac{d^2 y}{dx^2} - 8 \frac{dy}{dx} + 9y = 40 \sin 5x$.
5. $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + y = a \sin 2x$. (Meerut 2006B)
6. $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 10y + 37 \sin 2x = 0$. (Lucknow 2005; Gorakhpur 07)

Answers 3

1. $y = c_1 \cos 3x + c_2 \sin 3x - \frac{1}{7} \cos 4x$
2. (i) $y = c_1 e^x + c_2 e^{2x} + \frac{1}{130} (9 \cos 3x - 7 \sin 3x)$
 (ii) $y = e^{2x} (c_1 \cos 2x + c_2 \sin 2x) + (1/26) (3 \cos 3x - 2 \sin 3x)$
3. (i) $y = c_1 e^x + c_2 e^{2x} - \frac{1}{130} (7 \cos 3x + 9 \sin 3x)$
 (ii) $y = (c_1 + c_2 x) e^{-x} + c_3 e^x - (1/25) (2 \sin 2x + \cos 2x)$
4. $y = c_1 e^{4x} \cosh(x\sqrt{7} + c_2) + \frac{5}{29} (5 \cos 5x - 2 \sin 5x)$
5. $y = c_1 e^{2x} \cosh(\sqrt{3}x + c_2) + (1/73) a (8 \cos 2x - 3 \sin 2x)$
6. $y = e^{-x} (c_1 \cos 3x + c_2 \sin 3x) + \frac{37}{26} (2 \cos 2x - 3 \sin 2x)$

Case III: To find P.I. when Q is of the form x^m , where m is a positive integer.

Consider first $\{1/(D - a)\} x^m$. We have

$$\begin{aligned}
 \frac{1}{(D - a)} x^m &= -\frac{1}{(a - D)} x^m = -\frac{1}{a \{1 - (D/a)\}} x^m \\
 &= -\frac{1}{a} \left(1 - \frac{D}{a}\right)^{-1} x^m \\
 &= -\frac{1}{a} \left(1 + \frac{D}{a} + \frac{D^2}{a^2} + \dots\right) x^m, \text{ expanding by the binomial theorem} \\
 &= -\frac{1}{a} \left\{x^m + \frac{1}{a} m x^{m-1} + \frac{1}{a^2} m(m-1) x^{m-2} + \dots\right\}.
 \end{aligned}$$

Here we observe that in the expansion by the binomial theorem, the terms of the expansion beyond the m th power of D need not be written since $D^{m+1} x^m = 0$, $D^{m+2} x^m = 0$, etc.

Working Rule: In order to evaluate $\{1/F(D)\} x^m$, bring out common the lowest degree term in D from $F(D)$ so that remaining factor in the denominator is of the form $[1 + f(D)]$ or $[1 - f(D)]$ which is taken in the numerator with a negative index. Next we expand $[1 \pm f(D)]^{-1}$ in powers of D by the binomial theorem and operate upon x^m with the expansion obtained.

This expansion should be done upto the term D^m , since $D^{m+1} x^m = 0$ and all higher differential coefficients of x^m are zero. The whole process will be clear from the following examples.

The following binomial expansions should be remembered well.

$$(i) \quad (1-x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \dots$$

$$(ii) \quad (1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots$$

$$(iii) \quad (1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$(iv) \quad (1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$$

Examples on Case III:

Example 15: Solve $(d^2 y/dx^2) - 4y = x^2$.

(Lucknow 2010)

Solution: The auxiliary equation is $m^2 - 4 = 0$ or $m = \pm 2$.

$$\therefore \quad \text{C.F.} = c_1 e^{2x} + c_2 e^{-2x}.$$

$$\begin{aligned} \text{And} \quad \text{P.I.} &= \frac{1}{D^2 - 4} x^2 = \frac{1}{-4 \left[1 - \frac{1}{4} D^2\right]} x^2 = -\frac{1}{4} \left[1 - \frac{1}{4} D^2\right]^{-1} x^2 \\ &= -\frac{1}{4} \left[1 + \frac{1}{4} D^2 + \dots\right] x^2, \end{aligned}$$

expanding by binomial theorem upto the terms containing D^2

$$\begin{aligned} &= -\frac{1}{4} \left[x^2 + \frac{1}{4} D^2 (x^2) \right], \text{ because all the remaining terms vanish} \\ &= -\frac{1}{4} \left[x^2 + \frac{1}{4} \cdot 2 \right] = -\frac{1}{4} \left[x^2 + \frac{1}{2} \right]. \end{aligned}$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or} \quad y = c_1 e^{2x} + c_2 e^{-2x} - \frac{1}{4} \left(x^2 + \frac{1}{2} \right).$$

Example 16: Solve $\frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} = 1 + x^2$.

(Gorakhpur 2006; Avadh 06)

Solution: The auxiliary equation is $m^3 - m^2 - 6m = 0$

$$\text{or} \quad m(m+2)(m-3) = 0. \quad \therefore \quad m = 0, -2, 3.$$

$$\therefore \quad \text{C.F.} = c_1 e^{0x} + c_2 e^{-2x} + c_3 e^{3x} = c_1 + c_2 e^{-2x} + c_3 e^{3x}.$$

$$\begin{aligned} \text{And} \quad \text{P.I.} &= \frac{1}{D^4 - D^2 - 6D} (1 + x^2) = \frac{1}{-6D \left[1 + \frac{1}{6} D - \frac{1}{6} D^2\right]} (1 + x^2) \\ &= -\frac{1}{6D} \left[1 - \frac{1}{6} (-D + D^2)\right]^{-1} (1 + x^2) \\ &= -\frac{1}{6D} \left[1 + \frac{1}{6} (-D + D^2) + \frac{1}{36} (-D + D^2)^2 + \dots\right] (1 + x^2), \end{aligned}$$

the terms in the expansion being needed only upto D^2

$$\begin{aligned} &= -\frac{1}{6D} \left[1 - \frac{1}{6} D + \frac{1}{6} D^2 + \frac{1}{36} D^2 + \dots\right] (1 + x^2) \\ &= -\frac{1}{6D} \left[1 - \frac{1}{6} D + \frac{7}{36} D^2 + \dots\right] (1 + x^2) \end{aligned}$$

$$= -\frac{1}{6D} \left[(1+x^2) - \frac{1}{6} D(1+x^2) + \frac{7}{36} D^2(1+x^2) \right],$$

because all other terms vanish

$$\begin{aligned} &= -\frac{1}{6D} \left(1+x^2 - \frac{1}{3}x + \frac{7}{18} \right) = -\frac{1}{6D} \left(\frac{25}{18} - \frac{1}{3}x + x^2 \right) \\ &= -\frac{1}{6} \int \left(\frac{25}{18} - \frac{1}{3}x + x^2 \right) dx, \quad \left[\because 1/D \equiv \int dx \right] \\ &= -\frac{1}{6} \left[\frac{25}{18}x - \frac{1}{6}x^2 + \frac{1}{3}x^3 \right]. \end{aligned}$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

or $y = c_1 + c_2 e^{-2x} + c_3 e^{3x} - \frac{1}{6} \left[\frac{25}{18}x - \frac{1}{6}x^2 + \frac{1}{3}x^3 \right].$

Example 17: Solve $(D^3 + 8)y = x^4 + 2x + 1$.

Solution: The auxiliary equation is $m^3 + 8 = 0$

or $(m+2)(m^2 - 2m + 4) = 0 \quad \therefore m = -2, 1 \pm i\sqrt{3}$

Hence C.F. = $c_1 e^{-2x} + e^x \{c_2 \cos(x\sqrt{3}) + c_3 \sin(x\sqrt{3})\}$.

And
$$\begin{aligned} \text{P.I.} &= \frac{1}{D^3 + 8} (x^4 + 2x + 1) = \frac{1}{8 \left[1 + \frac{1}{8} D^3 \right]} (x^4 + 2x + 1) \\ &= \frac{1}{8} [1 + \frac{1}{8} D^3]^{-1} (x^4 + 2x + 1) \\ &= \frac{1}{8} [1 - \frac{1}{8} D^3 + \dots] (x^4 + 2x + 1), \end{aligned}$$

the other terms in the expansion being of no need

$$\begin{aligned} &= \frac{1}{8} [(x^4 + 2x + 1)] - \frac{1}{8} D^3 (x^4 + 2x + 1) \\ &= \frac{1}{8} [x^4 + 2x + 1 - 3x] = \frac{1}{8} (x^4 - x + 1). \end{aligned}$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

or $y = c_1 e^{-2x} + e^x \{c_2 \cos(x\sqrt{3}) + c_3 \sin(x\sqrt{3})\} + \frac{1}{8} (x^4 - x + 1).$

Example 18: Solve $(D^2 + D - 2)y = x + \sin x$.

Solution: The auxiliary equation is $m^2 + m - 2 = 0$

or $(m-1)(m+2) = 0.$

$\therefore m = 1, -2. \quad \therefore \text{C.F.} = c_1 e^x + c_2 e^{-2x}.$

And
$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + D - 2} (x + \sin x) = \frac{1}{(D^2 + D - 2)} x + \frac{1}{D^2 + D - 2} \sin x \\ &= \frac{1}{-2 \left[1 - \frac{1}{2} D - \frac{1}{2} D^2 \right]} x + \frac{1}{-1^2 + D - 2} \sin x \end{aligned}$$

$$\begin{aligned}
&= -\frac{1}{2} \left[1 - \left(\frac{1}{2} D + \frac{1}{2} D^2 \right) \right]^{-1} x + \frac{(D+3)}{(D-3)(D+3)} \sin x \\
&= -\frac{1}{2} \left[1 + \frac{1}{2} D + \dots \right] x + \frac{(D+3)}{D^2-9} \sin x = -\frac{1}{2} \left(x + \frac{1}{2} \right) + \frac{D+3}{-1-9} \sin x \\
&= -\frac{1}{2} \left(x + \frac{1}{2} \right) - \frac{1}{10} \{ D(\sin x) + 3 \sin x \} . \\
&= -\frac{1}{2} x - \frac{1}{4} - \frac{1}{10} \{ \cos x + 3 \sin x \} .
\end{aligned}$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

or
$$y = c_1 e^x + c_2 e^{-2x} - \frac{1}{2} x - \frac{1}{4} - \frac{1}{10} \cos x - \frac{3}{10} \sin x.$$

Example 19: Solve $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = x^2 + e^x + \cos 2x$.

(Meerut 2001)

Solution: The auxiliary equation is $m^2 - 4m + 4 = 0$.

$\therefore m = 2, 2.$

Hence C.F. = $(c_1 + c_2 x) e^{2x}$,

And
$$\text{P.I.} = \frac{1}{(D^2 - 4D + 4)} (x^2 + e^x + \cos 2x) = P_1 + P_2 + P_3, (\text{say}),$$

where
$$\begin{aligned}
P_1 &= \frac{1}{D^2 - 4D + 4} x^2 = \frac{1}{(D-2)^2} x^2 = \frac{1}{4} \left[1 - \frac{1}{2} D \right]^{-2} x^2 \\
&= \frac{1}{4} \left[1 + D + \frac{3}{4} D^2 + \dots \right] x^2 = \frac{1}{4} \left[x^2 + 2x + \frac{3}{2} \right],
\end{aligned}$$

$$P_2 = \frac{1}{(D^2 - 4D + 4)} e^x = \frac{e^x}{1^2 - 4 \cdot 1 + 4} = \frac{e^x}{1} = e^x,$$

and
$$P_3 = \frac{1}{(D^2 - 4D + 4)} \cos 2x = \frac{1}{-2^2 - 4D + 4} \cos 2x = -\frac{1}{4D} \cos 2x$$

$$= -\frac{1}{4} \int \cos 2x \, dx = -\frac{1}{4} \cdot \frac{1}{2} \sin 2x = -\frac{1}{8} \sin 2x.$$

(Rohilkhand 2008)

Hence the complete solution is $y = \text{C.F.} + \text{P.I.}$

or
$$y = \text{C.F.} + P_1 + P_2 + P_3$$

or
$$y = (c_1 + c_2 x) e^{2x} + \frac{1}{4} (x^2 + 2x + \frac{3}{2}) + e^x - \frac{1}{8} \sin 2x.$$

Comprehensive Exercise 4

1. $(D^2 + D - 6)y = x$.

(Bundelkhand 2007)

2. $(d^3 y / dx^3) + 3(d^2 y / dx^2) + 2(dy / dx) = x^2$.

(Bundelkhand 2001)

$$3. \frac{d^3 y}{dx^3} - 4 \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} - 2 = 0.$$

$$4. (D^2 + 2D + 1) y = 2x + x^2.$$

$$5. (D^2 + 3D + 2) y = x^2.$$

(Bundelkhand 2001)

$$6. (D^4 + D^2 + 16) y = 16x^2 + 256.$$

(Lucknow 2005)

$$7. (D^2 - 5D + 6) y = x + \sin 3x.$$

(Kanpur 2007)

$$8. (D^2 - 4D + 3) y = e^{-x} + 5.$$

$$9. (d^2 y / dx^2) - 4y = \sin^2 x.$$

(Gorakhpur 2007)

$$10. (D^2 - 2D + 3) y = \cos x + x^2.$$

(Meerut 2009B; Bundelkhand 11)

$$11. \frac{d^3 y}{dx^3} + 2 \frac{d^2 y}{dx^2} + \frac{dy}{dx} = e^{2x} + x^2 + x.$$

(Meerut 2009; Gorakhpur 10)

Answers 4

$$1. y = c_1 e^{2x} + c_2 e^{-3x} - \frac{1}{36} (6x + 1)$$

$$2. y = c_1 + c_2 e^{-x} + c_3 e^{-2x} + \frac{1}{12} x (2x^2 - 9x + 21)$$

$$3. y = c_1 + c_2 e^{2x} \cos (x + c_3) + \frac{2}{5} x$$

$$4. y = (c_1 + c_2 x) e^{-x} + (x^2 - 2x + 2)$$

$$5. y = c_1 e^{-x} + c_2 e^{-2x} + \frac{1}{2} x^2 - \frac{3}{2} x + \frac{7}{4}.$$

$$6. y = c_1 e^{-(1/2)\sqrt{7}x} \sin\left(\frac{3}{2}x + c_2\right) + c_3 e^{(1/2)\sqrt{7}x} \sin\left(\frac{3}{2}x + c_4\right) + x^2 + \frac{127}{8}$$

$$7. y = c_1 e^{2x} + c_2 e^{3x} + \frac{1}{6} \left(x + \frac{5}{6}\right) - \frac{1}{234} (3 \sin 3x - 15 \cos 3x)$$

$$8. y = c_1 e^x + c_2 e^{3x} + \frac{1}{8} e^{-x} + \frac{5}{3}$$

$$9. y = c_1 e^{2x} + c_2 e^{-2x} - \frac{1}{8} + \frac{1}{16} \cos 2x$$

$$10. y = e^x [c_1 \cos \sqrt{2}x + c_2 \sin \sqrt{2}x] + \frac{1}{3} x^2 + \frac{4}{9} x + \frac{2}{27} + \frac{1}{4} (\cos x - \sin x)$$

$$11. y = c_1 + (c_2 + c_3 x) e^{-x} + \frac{1}{18} e^{2x} + \frac{1}{3} x^3 - \frac{3}{2} x^2 + 4x$$

4.5 P.I. when $Q = e^{ax} V$, where V is any Function of x

By successive differentiation we notice that

$$\begin{aligned} D(e^{ax}V) &= e^{ax} D(V) + Vae^{ax} = e^{ax} (D+a)V, \\ D^2(e^{ax}V) &= D\{e^{ax} (D+a)V\} \\ &= e^{ax} D(D+a)V + ae^{ax} (D+a)V = e^{ax} (D+a)^2 V. \end{aligned}$$

Similarly, $D^3(e^{ax}V) = e^{ax} (D+a)^3 V, \dots,$

$$D^n(e^{ax}V) = e^{ax} (D+a)^n V.$$

$$\therefore F(D)(e^{ax}V) = e^{ax} F(D+a)V. \quad \dots(1)$$

The result (1) is true for any function V of x . Taking $\frac{1}{F(D+a)}V$ in place of V in (1), we

have

$$F(D)\left\{e^{ax} \frac{1}{F(D+a)}V\right\} = e^{ax} F(D+a)\left\{\frac{1}{F(D+a)}V\right\}$$

$$i.e., \quad F(D)\left\{e^{ax} \frac{1}{F(D+a)}V\right\} = e^{ax}V. \quad \dots(2)$$

Operating by $\frac{1}{F(D)}$ on both sides of (2), we get

$$e^{ax} \frac{1}{F(D+a)}V = \frac{1}{F(D)}(e^{ax}V).$$

$$\text{Thus,} \quad \frac{1}{F(D)}(e^{ax}V) = e^{ax} \frac{1}{F(D+a)}V.$$

Working Rule: Replace D by $(D+a)$ and take out e^{ax} before the operator $1/F(D)$. Then determine $\{1/F(D+a)\}V$ by the methods discussed in 4.4.

This method also enables us to find $\{1/F(D)\}e^{ax}$ when $F(a)$ is zero. We shall discuss it later on in article 4.6.

Illustrative Examples

Example 20: Solve $(D^2 - 2D + 1)y = x^2 e^{3x}$.

(Agra 2005; Avadh 09)

Solution: The auxiliary equation is

$$m^2 - 2m + 1 = 0$$

$$\text{or} \quad (m-1)^2 = 0.$$

$$\therefore m = 1, 1.$$

$$\therefore \text{C.F.} = (c_1 + c_2 x)e^x.$$

$$\begin{aligned}\text{And } \text{P.I.} &= \frac{1}{D^2 - 2D + 1} x^2 e^{3x} = \frac{1}{(D-1)^2} e^{3x} x^2 \\ &= e^{3x} \frac{1}{\{(D+3)-1\}^2} x^2,\end{aligned}$$

putting $D+3$ for D and bringing e^{3x} before the operator

$$\begin{aligned}&= e^{3x} \frac{1}{(D+2)^2} x^2 = e^{3x} \frac{1}{4\left(1+\frac{1}{2}D\right)^2} x^2 = \frac{1}{4} e^{3x} \left(1+\frac{1}{2}D\right)^{-2} x^2 \\ &= \frac{1}{4} e^{3x} \left[1 - 2 \cdot \frac{1}{2} D + 3 \cdot \frac{1}{4} D^2 + \dots\right] x^2,\end{aligned}$$

expanding by binomial theorem

$$\begin{aligned}&= \frac{1}{4} e^{3x} \left[1 - D + \frac{3}{4} D^2 + \dots\right] x^2 \\ &= \frac{1}{4} e^{3x} \left\{x^2 - 2x + \frac{3}{4} \cdot 2\right\} = \frac{1}{4} e^{3x} \left\{x^2 - 2x + \frac{3}{2}\right\}.\end{aligned}$$

Hence the required solution is $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = (c_1 + c_2 x) e^x + \frac{1}{4} e^{3x} \left\{x^2 - 2x + \frac{3}{2}\right\}.$$

Example 21: Solve $(D^2 - 2D + 5) y = e^{2x} \sin x$.

Solution: The auxiliary equation is $m^2 - 2m + 5 = 0$.

$$\therefore m = \{2 \pm \sqrt{4 - 20}\} / 2 = 1 \pm 2i.$$

Hence C.F. = $e^x (c_1 \cos 2x + c_2 \sin 2x)$.

$$\begin{aligned}\text{And } \text{P.I.} &= \frac{1}{D^2 - 2D + 5} e^{2x} \sin x \\ &= e^{2x} \frac{1}{(D+2)^2 - 2(D+2) + 5} \sin x = e^{2x} \frac{1}{D^2 + 2D + 5} \sin x \\ &= e^{2x} \frac{1}{(-1^2 + 2D + 5)} \sin x, \text{ putting } -1^2 \text{ for } D^2 \\ &= \frac{1}{2} e^{2x} \frac{1}{D+2} \sin x = \frac{1}{2} e^{2x} \frac{D-2}{(D-2)(D+2)} \sin x \\ &= \frac{1}{2} e^{2x} \frac{D-2}{(D^2 - 4)} \sin x = \frac{1}{2} e^{2x} \frac{D-2}{-1-4} \sin x \\ &= -\frac{1}{10} e^{2x} (D-2) \sin x = -\frac{1}{10} e^{2x} \{D(\sin x) - 2 \sin x\} \\ &= -\frac{1}{10} e^{2x} (\cos x - 2 \sin x).\end{aligned}$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = e^x (c_1 \cos 2x + c_2 \sin 2x) - \frac{1}{10} e^{2x} (\cos x - 2 \sin x).$$

Comprehensive Exercise 5

Solve the following differential equations :

1. $(D+1)^3 y = x^2 e^{-x}$. 2. $(D^2 - 2D + 1) y = x^2 e^x$. (Avadh 2007)
3. $(D^2 + 2D + 1) y = e^{-x}/(x+2)$. 4. $(D^2 - 3D + 2) y = xe^x$. (Bundelkhand 2008)
5. $(D^2 - 2D + 1) y = x^2 e^{2x}$. (Purvanchal 2007)
6. $(D^2 - 4D + 4) y = e^{2x} \sin 3x$. (Purvanchal 2006)
7. $(D^4 - 1) y = e^x \cos x$. (Lucknow 2010)
8. $(D^2 - 2D + 4) y = e^x \cos x$. 9. $(D^2 + 4D - 12) y = (x-1) e^{2x}$.
10. $(D^2 - 5D + 6) y = e^{2x} \sin 2x$.

Answers 5

1. $y = (c_1 + c_2x + c_3x^2) e^{-x} + \frac{1}{60} x^5 e^{-x}$ 2. $y = (c_1 + c_2x) e^x + \frac{1}{12} x^4 e^x$
3. $y = (c_1 + c_2x) e^{-x} + e^{-x} \{x \log(x+2) - x + 2 \log(x+2)\}$
4. $y = c_1 e^x + c_2 e^{2x} - e^x \left(\frac{1}{2} x^2 + x \right)$
5. $y = (c_1 + c_2x) e^x + e^{2x} (x^2 - 4x + 6)$
6. $y = (c_1 + c_2x) e^{2x} - \frac{1}{9} e^{2x} \sin 3x$
7. $y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x - \frac{1}{5} e^x \cos x$
8. $y = e^x (c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x) + \frac{1}{2} e^x \cos x$
9. $y = e^{2x} (c_1 + \frac{1}{16} x^2 - \frac{9}{64} x) + c_2 e^{-6x}$
10. $y = c_1 e^{2x} + c_2 e^{3x} + \frac{1}{10} e^{2x} (\cos 2x - 2 \sin 2x)$

4.6 P.I. when $Q = e^{ax}$ and $F(a) = 0$

In this case we use the method explained in article 4.5 to find the P.I.

We have $\text{P.I.} = \frac{1}{F(D)} e^{ax} = \frac{1}{F(D)} e^{ax} \cdot 1$, if $F(a) = 0$.

Now this is of the form $e^{ax} V$. Hence by the method of article 4.5, i.e., by putting $(D+a)$ for D in the operator and by making e^{ax} free from the operator, we have

$$\text{P.I.} = e^{ax} \frac{1}{F(D+a)} 1.$$

Now this can be evaluated by using the method for finding P.I. in the case of x^m .

Illustrative Examples

Example 22: Solve $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^x$.

Solution: The auxiliary equation is $m^2 - 3m + 2 = 0$

or $(m-1)(m-2) = 0 \quad \therefore m = 1, 2$.

Hence C.F. = $c_1 e^x + c_2 e^{2x}$.

$$\text{And P.I.} = \frac{1}{D^2 - 3D + 2} e^x = \frac{1}{(D-2)(D-1)} e^x = \frac{1}{(1-2)(D-1)} e^x$$

$$\begin{aligned} & \text{[putting 1 for } D \text{ in the factor } D-2 \text{ because it does not vanish by doing so]} \\ & = -\frac{1}{D-1} e^x \cdot 1 \end{aligned}$$

[Note that $D-1$ becomes zero by putting 1 for D ; so here we shall apply the method for $e^{ax}V$ by taking 1 for V .]

$$= -e^x \frac{1}{(D+1)-1} 1 = -e^x \frac{1}{D} 1 = -e^x x.$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

or $y = c_1 e^x + c_2 e^{2x} - x e^x$.

Note: While finding P.I. in the case of e^{ax} if $F(D)$ becomes zero by putting a for D , we factorise $F(D)$. Then we put $D = a$ in the factors which do not vanish by doing so. The remaining operator is then dealt with by using the method for $e^{ax}V$ on taking 1 for V .

Example 23: Solve $(D^3 - 7D + 6)y = e^{2x}$.

Solution: The auxiliary equation is $m^3 - 7m + 6 = 0$

or $(m-1)(m-2)(m+3) = 0 \quad \therefore m = 1, 2, -3$.

$\therefore$ C.F. = $c_1 e^x + c_2 e^{2x} + c_3 e^{-3x}$.

$$\begin{aligned} \text{And P.I.} &= \frac{1}{(D^3 - 7D + 6)} e^{2x} = \frac{1}{(D-1)(D-2)(D+3)} e^{2x} \\ &= \frac{1}{(2-1)(D-2)(2+3)} e^{2x}, \end{aligned}$$

$$\begin{aligned} & \text{putting 2 for } D \text{ in all the factors except } D-2 \\ &= \frac{1}{5} \frac{1}{D-2} e^{2x} \cdot 1 = \frac{1}{5} e^{2x} \frac{1}{(D+2)-2} 1 = \frac{1}{5} e^{2x} \frac{1}{D} 1 = \frac{1}{5} x e^{2x}. \end{aligned}$$

Hence the complete solution is

$$y = (\text{C.F.}) + (\text{P.I.}) \quad \text{or} \quad y = c_1 e^x + c_2 e^{2x} + c_3 e^{-3x} + \frac{1}{5} x e^{2x}.$$

Example 24: Solve $(D^2 + 4D + 4)y = e^{2x} - e^{-2x}$.

Solution: The auxiliary equation is $(m + 2)^2 = 0$.

$$\therefore m = -2, -2$$

$$\text{Hence C.F.} = (c_1 + c_2x)e^{-2x}.$$

$$\text{And P.I.} = \frac{1}{(D+2)^2} (e^{2x} - e^{-2x}) = \frac{1}{(D+2)^2} e^{2x} - \frac{1}{(D+2)^2} e^{-2x}.$$

$$\text{Now } \frac{1}{(D+2)^2} e^{2x} = \frac{e^{2x}}{(2+2)^2} = \frac{1}{16} e^{2x}. \quad [\because \text{here, } F(a) \neq 0].$$

$$\begin{aligned} \text{Also } \frac{1}{(D+2)^2} e^{-2x} &= \frac{1}{(D+2)^2} e^{-2x} \cdot 1, \quad [\because \text{here } F(a) = 0] \\ &= e^{-2x} \frac{1}{\{(D-2)+2\}^2} 1 = e^{-2x} \frac{1}{D^2} 1 = e^{-2x} \frac{1}{D} x = \frac{1}{2} x^2 e^{-2x}. \end{aligned}$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = (c_1 + c_2x)e^{-2x} + \frac{1}{16} e^{2x} - \frac{1}{2} x^2 e^{-2x}.$$

Example 25: Solve $D^2y - 3Dy + 2y = \cosh x$.

(Rohilkhand 2010)

Solution: The auxiliary equation is

$$m^2 - 3m + 2 = 0 \quad \text{or} \quad (m-1)(m-2) = 0 \quad \therefore m = 1, 2.$$

$$\therefore \text{C.F.} = c_1 e^x + c_2 e^{2x}.$$

$$\begin{aligned} \text{And P.I.} &= \frac{1}{D^2 - 3D + 2} \cosh x = \frac{1}{(D^2 - 3D + 2)} \left(\frac{e^x + e^{-x}}{2} \right) \\ &= \frac{1}{2} \frac{1}{D^2 - 3D + 2} e^x + \frac{1}{2} \frac{1}{D^2 - 3D + 2} e^{-x}. \end{aligned}$$

$$\begin{aligned} \text{Now } \frac{1}{2} \frac{1}{D^2 - 3D + 2} e^x &= \frac{1}{2} \frac{1}{(D-1)(D-2)} e^x \\ &= \frac{1}{2} \frac{1}{(D-1)(1-2)} e^x = -\frac{1}{2} \frac{1}{D-1} e^x \cdot 1 \\ &= -\frac{1}{2} e^x \frac{1}{D+1-1} 1 = -\frac{1}{2} e^x \frac{1}{D} 1 = -\frac{1}{2} x e^x. \end{aligned}$$

$$\begin{aligned} \text{Also } \frac{1}{2} \frac{1}{D^2 - 3D + 2} e^{-x} &= \frac{1}{2} \frac{1}{(-1)^2 - 3(-1) + 2} e^{-x}, \text{ putting } -1 \text{ for } D \\ &= \frac{1}{12} e^{-x}. \quad \therefore \text{P.I.} = -\frac{1}{2} x e^x + \frac{1}{12} e^{-x}. \end{aligned}$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = c_1 e^x + c_2 e^{2x} - \frac{1}{2} x e^x + \frac{1}{12} e^{-x}.$$

Example 26: Solve $(D^2 - 4D + 4)y = e^{2x} + \sin 2x$.

Solution: The auxiliary equation is

$$m^2 - 4m + 4 = 0 \quad \text{or} \quad (m - 2)^2 = 0. \quad \therefore m = 2, 2.$$

Hence C.F. = $(c_1 + c_2x)e^{2x}$.

And
$$\text{P.I.} = \frac{1}{(D-2)^2} e^{2x} + \frac{1}{D^2 - 4D + 4} \sin 2x.$$

Now
$$\begin{aligned} \frac{1}{(D-2)^2} e^{2x} &= \frac{1}{(D-2)^2} e^{2x} \cdot 1 = e^{2x} \frac{1}{(D+2-2)^2} 1 \\ &= e^{2x} \frac{1}{D^2} (1) = e^{2x} \frac{1}{D} (x) = e^{2x} \cdot \frac{x^2}{2} = \frac{1}{2} x^2 e^{2x}, \end{aligned}$$

and
$$\begin{aligned} \frac{1}{D^2 - 4D + 4} \sin 2x &= \frac{1}{-2^2 - 4D + 4} \sin 2x, \text{ putting } -2^2 \text{ for } D^2 \\ &= \frac{1}{-4D} \sin 2x = -\frac{1}{4} \int \sin 2x \, dx = -\frac{1}{4} \cdot \frac{1}{2} (-\cos 2x) = \frac{1}{8} \cos 2x. \end{aligned}$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

or
$$y = (c_1 + c_2x)e^{2x} + \frac{1}{2} x^2 e^{2x} + \frac{1}{8} \cos 2x.$$

Comprehensive Exercise 6

Solve the following differential equations :

- $(D^2 - 2D + 1)y = e^x.$
- $(D^2 + D - 6)y = e^{2x}.$
- $(D^2 - 4D + 3)y = 2e^{3x}.$
- $(D^3 + 3D^2 + 3D + 1)y = e^{-x}.$
- $(D^2 - a^2)y = \cosh ax.$
- $(D^3 - 5D^2 + 7D - 3)y = e^{2x} \cosh x.$
- $(D^2 - 4D + 4)y = 8(x^2 + e^{2x} + \sin 2x).$

(Rohilkhand 2005)

Answers 6

- $y = (c_1 + c_2x)e^x + \frac{1}{2} x^2 e^x$
- $y = c_1 e^{2x} + c_2 e^{-3x} + \frac{1}{5} x e^{2x}$
- $y = c_1 e^x + c_2 e^{3x} + x e^{3x}$
- $y = (c_1 + c_2x + c_3x^2)e^{-x} + \frac{1}{6} x^3 e^{-x}$
- $y = c_1 e^{ax} + c_2 e^{-ax} + (x/2a) \sinh ax$
- $y = (c_1 + c_2x)e^x + c_3 e^{3x} + \frac{1}{8} x e^{3x} - \frac{1}{8} x^2 e^x$
- $y = (c_1 + c_2x)e^{2x} + 2(x^2 + 2x + \frac{3}{2}) + 4x^2 e^{2x} + \cos 2x$

4.7 P.I. when $Q = \sin ax$ or $\cos ax$ and $F(-a^2) = 0$

To find out these types of particular integrals, it is convenient to replace $\sin ax$ or $\cos ax$ by the exponential value and then apply 4.6. Thus the particular integral of $\cos ax$

i.e., $\frac{1}{F(D^2)} \cos ax$, when $F(-a^2) = 0$, can be written as

$$= \frac{1}{F(D^2)} (\text{Real part of } e^{iax}), \quad [\because e^{iax} = \cos ax + i \sin ax]$$

$$= \text{Real Part of } \frac{1}{F(D^2)} e^{iax} \cdot 1, \text{ which is now of the form of article 4.5.}$$

Similarly, the particular integral of $\sin ax$

i.e., $\frac{1}{F(D^2)} \sin ax$, when $F(-a^2) = 0$

$$= \frac{1}{F(D^2)} (\text{Imaginary part of } e^{iax}), \quad [\because e^{iax} = \cos ax + i \sin ax]$$

$$= \text{Imaginary part of } \frac{1}{F(D^2)} e^{iax} \cdot 1,$$

which is now of the form $\frac{1}{F(D)} e^{ax}$ V of article 4.5.

Now the method of article 4.5 is to be followed.

Illustrative Examples

Example 27: Solve $\frac{d^2 y}{dx^2} + a^2 y = \cos ax$.

(Meerut 2003)

Solution: The auxiliary equation is

$$m^2 + a^2 = 0, \text{ i.e., } m = 0 \pm ai.$$

$$\therefore \text{C.F.} = e^{0x} (c_1 \cos ax + c_2 \sin ax) \\ = c_1 \cos ax + c_2 \sin ax.$$

$$\text{And P.I.} = \frac{1}{D^2 + a^2} \cos ax$$

$$= \text{the real part in } \frac{1}{D^2 + a^2} (\cos ax + i \sin ax)$$

$$= \text{the real part in } \frac{1}{D^2 + a^2} e^{iax}, \text{ by Euler's theorem of trigonometry.}$$

$$\text{Now } \frac{1}{D^2 + a^2} e^{iax} = \frac{1}{(D + ia)(D - ia)} e^{iax}$$

$$= \frac{1}{(ia + ia)(D - ia)} e^{iax}, \text{ putting } ia \text{ for } D \text{ in the factor } D + ia$$

$$\begin{aligned}
 &= \frac{1}{2ia(D - ia)} e^{iax} \cdot 1 \\
 &= \frac{1}{2ia} e^{iax} \frac{1}{(D + ia) - ia} 1 = \frac{1}{2ia} e^{iax} \frac{1}{D} 1 \\
 &= \frac{1}{2ia} e^{iax} \cdot x = \frac{x}{2ai} (\cos ax + i \sin ax), \quad [\because e^{iax} = \cos ax + i \sin ax] \\
 &= -\frac{ix}{2a} (\cos ax + i \sin ax), \quad \left[\because \frac{1}{i} = \frac{i}{i^2} = \frac{i}{-1} = -i \right] \\
 &= -i \frac{x}{2a} \cos ax + \frac{x}{2a} \sin ax, \quad [\because i^2 = -1] \\
 \therefore \quad &\frac{1}{D^2 + a^2} \cos ax = \text{the real part in } \left(\frac{x}{2a} \sin ax - i \frac{x}{2a} \cos ax \right) \\
 &= \frac{x}{2a} \sin ax. \quad (\text{Agra 2006})
 \end{aligned}$$

Hence the complete solution is

$$y = (\text{C.F.}) + (\text{P.I.})$$

or

$$y = c_1 \cos ax + c_2 \sin ax + (x/2a) \sin ax.$$

Remember : While solving the differential equation $(D^2 + a^2) y = \cos ax$, the P.I.

$$\frac{1}{D^2 + a^2} \cos ax = \frac{x}{2a} \sin ax = \frac{x}{2} \int \cos ax \, dx.$$

Example 28: Solve $(D^2 + a^2) y = \sin ax$.

(Bundelkhand 2008; Rohilkhand 10)

Solution: Here as in Ex. 1, C.F. = $c_1 \cos ax + c_2 \sin ax$.

Also
$$\text{P.I.} = \frac{1}{D^2 + a^2} \sin ax$$

$$= \text{the coefficient of } i \text{ in } \frac{1}{D^2 + a^2} (\cos ax + i \sin ax)$$

$$= \text{the coefficient of } i \text{ in } \frac{1}{D^2 + a^2} e^{iax}.$$

Now
$$\frac{1}{D^2 + a^2} e^{iax} = -i \frac{x}{2a} \cos ax + \frac{x}{2a} \sin ax. \quad [\text{Proceed as in Ex. 27}]$$

$$\therefore \quad \frac{1}{D^2 + a^2} \sin ax$$

$$= \text{the coefficient of } i \text{ in } \left(-i \frac{x}{2a} \cos ax + \frac{x}{2a} \sin ax \right) = -\frac{x}{2a} \cos ax.$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

or

$$y = c_1 \cos ax + c_2 \sin ax - (x/2a) \cos ax.$$

Remember :
$$\frac{1}{D^2 + a^2} \sin ax = -\frac{x}{2a} \cos ax = \frac{x}{2} \int \sin ax \, dx.$$

Example 29: Solve $(D^2 + 4)y = \sin^2 x$.

(Rohilkhand 2007; Gorakhpur 10; Avadh 11)

Solution: The auxiliary equation is $m^2 + 2 = 0$.

$$\therefore m = 0 \pm 2i.$$

$$\text{Hence C.F.} = c_1 \cos 2x + c_2 \sin 2x. \quad (\because e^{0x} = 1)$$

$$\begin{aligned} \text{And P.I.} &= \frac{1}{D^2 + 4} \sin^2 x = \frac{1}{D^2 + 4} \cdot \frac{1}{2} (2 \sin^2 x) = \frac{1}{2} \frac{1}{(D^2 + 4)} (1 - \cos 2x) \\ &= \frac{1}{2} \frac{1}{D^2 + 4} 1 - \frac{1}{2} \frac{1}{D^2 + 4} \cos 2x. \end{aligned}$$

$$\text{Now } \frac{1}{2} \frac{1}{D^2 + 4} 1 = \frac{1}{8} \left(1 + \frac{1}{4} D^2 \right)^{-1} 1 = \frac{1}{8} \left(1 - \frac{1}{4} D^2 + \dots \right) 1 = \frac{1}{8}.$$

$$\text{Again } \frac{1}{2} \frac{1}{D^2 + 4} \cos 2x = \frac{1}{2} \cdot \frac{1}{4} x \sin 2x, \text{ (refer Ex. 27.)}$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{8} - \frac{1}{8} x \sin 2x.$$

Example 30: Solve $(D^2 + 1)y = \sin x \sin 2x$.

Solution: The auxiliary equation is $m^2 + 1 = 0$ or $m = \pm i$.

$$\therefore \text{C.F.} = c_1 \cos x + c_2 \sin x.$$

$$\begin{aligned} \text{And P.I.} &= \frac{1}{D^2 + 1} (\sin x \sin 2x) = \frac{1}{D^2 + 1} \cdot \frac{1}{2} (2 \sin x \sin 2x) \\ &= \frac{1}{2} \frac{1}{D^2 + 1} (\cos x - \cos 3x) = \frac{1}{2} \frac{1}{D^2 + 1} \cos x - \frac{1}{2} \frac{1}{D^2 + 1} \cos 3x. \end{aligned}$$

$$\begin{aligned} \text{Now } \frac{1}{D^2 + 1} \cos x &= \text{the real part in } \frac{1}{D^2 + 1} e^{ix} \\ &= \text{the real part in } \frac{1}{(D + i)(D - i)} e^{ix} \\ &= \text{the real part in } \frac{1}{(i + i)(D - i)} e^{ix} \\ &= \text{the real part in } \frac{1}{2i(D - i)} e^{ix} \cdot 1 = \text{the real part in } \frac{1}{2i} e^{ix} \frac{1}{D + i - i} 1 \\ &= \text{the real part in } \frac{1}{2i} e^{ix} \frac{1}{D} 1 = \text{the real part in } \frac{1}{2i} e^{ix} x \\ &= \text{the real part in } \frac{x}{2i} (\cos x + i \sin x) \\ &= \text{the real part in } -\frac{1}{2} x i (\cos x + i \sin x) \\ &= \text{the real part in } -i \frac{1}{2} x \cos x + \frac{1}{2} x \sin x = \frac{1}{2} x \sin x. \end{aligned}$$

Again $\frac{1}{D^2+1} \cos 3x = \frac{1}{-3^2+1} \cos 3x = -\frac{1}{8} \cos 3x.$

$\therefore$ the P.I. $= \frac{1}{2} \cdot \frac{1}{2} x \sin x - \frac{1}{2} \cdot \left(-\frac{1}{8} \cos 3x\right) = \frac{1}{4} x \sin x + \frac{1}{16} \cos 3x.$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

or $y = c_1 \cos x + c_2 \sin x + \frac{1}{4} x \sin x + \frac{1}{16} \cos 3x.$

Comprehensive Exercise 7

Solve the following differential equations :

- $(D^2 + 4)y = \sin 2x.$ (Agra 2005; Gorakhpur 09)
- $(D^3 + a^2 D)y = \sin ax.$ (Gorakhpur 2005; Avadh 05)
- $(D^2 + 9)(D^2 + 1)y = \cos 3x.$

Answers 7

- $y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{4} x \cos 2x$
- $y = c_1 + c_2 \cos ax + c_3 \sin ax - (1/2a^2)x \sin ax$
- $y = c_1 \cos 3x + c_2 \sin 3x + c_3 \cos x + c_4 \sin x - \frac{1}{48} x \sin 3x$

4.8 P.I. when $Q = xV$, where V is any Function of x

Here P.I. $= \frac{1}{F(D)} (xV).$

By Leibnitz's theorem, we have

$$D^n (xV) = x \cdot D^n V + nD^{n-1}V = x D^n V + \left(\frac{d}{dD} D^n\right) V,$$

showing that $F(D) (xV) = xF(D) V + F'(D) V. \dots(1)$

Taking $\frac{1}{F(D)} V$ in place of V in (1) and then operating on both sides by $\frac{1}{F(D)}$, we get

after transposition

$$\frac{1}{F(D)} (xV) = x \frac{1}{F(D)} V - \frac{F'(D)}{\{F(D)\}^2} V \text{ [Remember]}$$

or $\frac{1}{F(D)} (xV) = x \frac{1}{F(D)} V + \left\{ \frac{d}{dD} \frac{1}{F(D)} \right\} V.$

Proceeding by repeated application of the above formula $\{1/F(D)\} x^m V$ can be determined.

Note: The method of article 4.8 cannot be used when V is of the form $\cos ax$ or $\sin ax$ and $F(-a^2) = 0$ i.e., $F(D^2)$ vanishes by putting $-a^2$ for D^2 . In that case P.I. is found by the method discussed in article 4.7.

Illustrative Examples

Example 31: Solve $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = x \sin x$.

(Rohilkhand 2010)

Solution: The auxiliary equation is $m^2 - 2m + 1 = 0$.

$\therefore m = 1, 1$. Hence C.F. = $(c_1 + c_2 x) e^x$.

And P.I. = $\frac{1}{(D-1)^2} x \sin x = x \frac{1}{(D-1)^2} \sin x - \frac{2}{(D-1)^3} \sin x$, by article 4.8

$$= x \frac{1}{D^2 - 2D + 1} \sin x - \frac{2}{D^3 - 3D^2 + 3D - 1} \sin x$$

$$= x \cdot \frac{1}{-2D} \sin x - \frac{2}{-D + 3 + 3D - 1} \sin x, \text{ putting } -1^2 \text{ i.e., } -1 \text{ for } D^2$$

$$= -\frac{x}{2} \cdot \frac{1}{D} \sin x - \frac{2}{2(D+1)} \sin x = \frac{x}{2} \cos x - \frac{1}{D+1} \sin x$$

$$= \frac{x}{2} \cos x - \frac{(D-1)}{(D+1)(D-1)} \sin x = \frac{x \cos x}{2} - \frac{D-1}{D^2-1} \sin x$$

$$= \frac{1}{2} x \cos x + \frac{1}{2} (D-1) \sin x = \frac{1}{2} x \cos x + \frac{1}{2} (\cos x - \sin x).$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

or $y = (c_1 + c_2 x) e^x + \frac{1}{2} x \cos x + \frac{1}{2} (\cos x - \sin x)$.

Example 32: Solve $\frac{d^2 y}{dx^2} + 4y = x \sin x$.

Solution: The auxiliary equation is $m^2 + 4 = 0$.

Therefore $m = 0 \pm 2i$.

$\therefore$ C.F. = $c_1 \cos 2x + c_2 \sin 2x$. [$\because e^{0x} = 1$]

And P.I. = $\frac{1}{D^2 + 4} x \sin x = x \frac{1}{D^2 + 4} \sin x - \frac{2D}{(D^2 + 4)^2} \sin x$, by article 4.8

$$= \frac{x \sin x}{-1^2 + 4} - \frac{2D}{(-1^2 + 4)^2} \sin x, \text{ putting } -1^2 \text{ for } D^2$$

$$= \frac{1}{3} x \sin x - \frac{2}{9} D (\sin x) = \frac{1}{3} x \sin x - \frac{2}{9} \cos x.$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{3} x \sin x - \frac{2}{9} \cos x.$$

Example 33: Solve $(D^4 - 1)y = x \sin x$.

(Lucknow 2006)

Solution: The auxiliary equation is $m^4 - 1 = 0$ or $(m^2 - 1)(m^2 + 1) = 0$.

$$\therefore m = 1, -1, 0 \pm i.$$

$$\therefore \text{C.F.} = c_1 e^x + c_2 e^{-x} + e^{0x} (c_3 \cos x + c_4 \sin x).$$

$$\text{And P.I.} = \frac{1}{D^4 - 1} x \sin x.$$

Here we cannot use the method given in 4.8 because $D^4 - 1$ vanishes by putting -1^2 i.e., -1 for D^2 . So here we shall proceed by the method given in article 4.7.

$$\therefore \text{P.I.} = \text{Imaginary part of } \frac{1}{D^4 - 1} x e^{ix}, \quad [\because e^{ix} = \cos x + i \sin x]$$

$$= \text{I. P. of } e^{ix} \frac{1}{\{(D+i)^4 - 1\}} x, \text{ by using the method for } e^{ax} V$$

$$= \text{I. P. of } e^{ix} \frac{1}{(D^4 + 4iD^3 + 6i^2D^2 + 4i^3D + i^4 - 1)} x$$

$$= \text{I. P. of } e^{ix} \frac{1}{D^4 + 4iD^3 - 6D^2 - 4iD} x, \quad [\because i^2 = -1]$$

$$= \text{I. P. of } e^{ix} \frac{1}{-4iD [1 + (3D/2i) - D^2 - (D^3/4i)]} x$$

$$= \text{I. P. of } \frac{e^{ix}}{-4i} \cdot \frac{1}{D} \left\{ 1 + \left(\frac{3D}{2i} - \dots \right) \right\}^{-1} x$$

$$= \text{I. P. of } \frac{e^{ix}}{-4i} \cdot \frac{1}{D} \left\{ 1 - \frac{3D}{2i} + \dots \right\} x$$

$$= \text{I. P. of } \frac{e^{ix}}{-4i} \frac{1}{D} \left(x - \frac{3}{2} i \right), \quad [\because D \equiv \frac{d}{dx}]$$

$$= \text{I. P. of } \frac{e^{ix}}{-4i} \left(\frac{x^2}{2} - \frac{3x}{2i} \right), \quad [\because (1/D) \text{ stands for integration w.r.t. } x]$$

$$= \text{I. P. of } \frac{1}{4} i e^{ix} \left(\frac{1}{2} x^2 + \frac{3}{2} i x \right); \quad [\because i^2 = -1 \Rightarrow 1/i = -i]$$

$$= \text{I. P. of } \frac{1}{4} i (\cos x + i \sin x) \left(\frac{1}{2} x^2 + \frac{3}{2} i x \right)$$

$$= \frac{1}{4} \left(\frac{1}{2} x^2 \cos x - \frac{3}{2} x \sin x \right)$$

$$= \frac{1}{8} (x^2 \cos x - 3x \sin x).$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x + \frac{1}{8} (x^2 \cos x - 3x \sin x).$$

Example 34: Solve $(D^4 + 2D^2 + 1)y = x^2 \cos x$. (Gorakhpur 2010; Purvanchal 07, 08)

Solution: The auxiliary equation is $m^4 + 2m^2 + 1 = 0$

$$\text{or } (m^2 + 1)^2 = 0 \text{ giving } m = \pm i, \pm i.$$

$$\therefore \text{C.F.} = (c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x. \quad [\because e^{0x} = 1]$$

$$\text{And P.I.} = \frac{1}{D^4 + 2D^2 + 1} x^2 \cos x$$

$$= \text{Real part of } \frac{1}{\{(D^2 + 1)\}^2} x^2 e^{ix}, \quad [\because e^{ix} = \cos x + i \sin x]$$

$$= \text{R. P. of } e^{ix} \frac{1}{\{(D + i)^2 + 1\}^2} x^2, \text{ by article 4.5}$$

$$= \text{R. P. of } e^{ix} \frac{1}{(D^2 + 2iD)^2} x^2, \quad [\because i^2 = -1]$$

$$= \text{R. P. of } e^{ix} \frac{1}{4i^2 D^2 [1 + (D/2i)]^2} x^2$$

$$= \text{R. P. of } -\frac{1}{4} e^{ix} \frac{1}{D^2} \left[1 + \frac{D}{2i}\right]^{-2} x^2, \quad [\because i^2 = -1]$$

$$= \text{R. P. of } -\frac{1}{4} e^{ix} \frac{1}{D^2} \left[1 - \frac{1}{2} iD\right]^{-2} x^2, \quad \left[\because \frac{1}{i} = -i\right]$$

$$= \text{R. P. of } -\frac{1}{4} e^{ix} \frac{1}{D^2} \left[1 + 2 \cdot \frac{1}{2} iD + 3 \cdot \frac{1}{4} i^2 D^2 + \dots\right] x^2,$$

expanding by binomial theorem

$$= \text{R. P. of } -\frac{1}{4} e^{ix} \frac{1}{D^2} \left[1 + iD - \frac{3}{4} D^2 + \dots\right] x^2$$

$$= \text{R. P. of } -\frac{1}{4} e^{ix} \left[\frac{1}{D^2} + \frac{i}{D} - \frac{3}{4} + \text{terms in } D, D^2, \text{ and so on}\right] x^2$$

$$= \text{R. P. of } -\frac{1}{4} e^{ix} \left[\frac{1}{3} \frac{x^4}{4} + i \frac{1}{3} x^3 - \frac{3}{4} x^2 + \text{terms in } x^1, x^0\right],$$

($\because 1/D$ stands for integration w.r.t. x)

$$= \text{R. P. of } -\frac{1}{4} (\cos x + i \sin x) \left(\frac{1}{12} x^4 + \frac{1}{3} ix^3 - \frac{3}{4} x^2 + \text{terms in } x^1, x^0\right)$$

$$= -\frac{1}{4} \left(\frac{1}{12} x^4 - \frac{3}{4} x^2\right) \cos x + \frac{1}{4} \left(\frac{1}{3} x^3\right) \sin x$$

+ terms already included in the C.F.

$$= -\frac{1}{48}(x^4 - 9x^2) \cos x + \frac{1}{12}x^3 \cdot \sin x,$$

neglecting the terms already included in the C.F.

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = (c_1 + c_2x) \cos x + (c_3 + c_4x) \sin x - \frac{1}{48}(x^4 - 9x^2) \cos x + \frac{1}{12}x^3 \sin x.$$

Example 35: Solve $(D^2 - 2D + 1)y = xe^x \sin x$.

(Purvanchal 2010; Kanpur 12; Avadh 13)

Solution: The auxiliary equation is

$$m^2 - 2m + 1 = 0 \quad \text{or} \quad (m - 1)^2 = 0. \quad \therefore m = 1, 1.$$

$$\therefore \text{C.F.} = (c_1 + c_2x)e^x.$$

$$\text{And P.I.} = \frac{1}{D^2 - 2D + 1} xe^x \sin x = \frac{1}{(D - 1)^2} e^x x \sin x$$

$$= e^x \frac{1}{\{(D + 1) - 1\}^2} x \sin x, \text{ by 4.5}$$

$$= e^x \frac{1}{D^2} (x \sin x) = e^x \cdot \frac{1}{D} \int x \sin x \, dx,$$

[$\therefore (1/D)$ stands for integration w.r.t. x]

$$= e^x \frac{1}{D} (-x \cos x + \sin x), \text{ integrating by parts}$$

$$= e^x \int (-x \cos x + \sin x) \, dx = e^x \left[\int -x \cos x \, dx + \int \sin x \, dx \right]$$

$$= e^x [-x \sin x + \int \sin x \, dx + \int \sin x \, dx] = e^x (-x \sin x - 2 \cos x).$$

Hence the complete solution is $y = (\text{C.F.}) + (\text{P.I.})$

$$\text{or } y = (c_1 + c_2x)e^x - e^x (x \sin x + 2 \cos x).$$

Comprehensive Exercise 8

Solve the following differential equations :

1. $d^2y/dx^2 + 9y = x \sin x.$

2. $(D^2 + 2D + 1)y = x \cos x.$

3. $(D^2 - 4D + 4)y = 8x^2 e^{2x} \sin 2x.$

(Gorakhpur 2011; Meerut 05B)

4. $(D^2 + 4)y = x \sin 2x.$

5. $(D^2 + D)y = x \cos x.$

6. $(D^2 - 1)y = x^2 \cos x.$

7. $(D^2 + 1)y = x^2 \sin 2x.$

(Meerut 2004, 07B; Lucknow 11)

8. $(D^4 - 1)y = x \sin 2x$.

(Meerut 2001, 06B)

9. $(D^2 + m^2)y = x \cos mx$.

10. $(d^2 y / dx^2) - y = x \sin x + (1 + x^2)e^x$.

(Bundelkhand 2009)

Answers 8

1. $y = c_1 \cos 3x + c_2 \sin 3x + \frac{1}{8}x \sin x - \frac{1}{32} \cos x$

2. $y = (c_1 + c_2 x)e^{-x} + \frac{1}{2}x \sin x - \frac{1}{2} \sin x + \frac{1}{2} \cos x$

3. $y = (c_1 + c_2 x)e^{2x} + e^{2x}(-2x^2 \sin 2x + 3 \sin 2x - 4x \cos x)$

4. $y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{8}x^2 \cos 2x + \frac{1}{16}x \sin 2x$

5. $y = c_1 + c_2 e^{-x} - \frac{1}{2}x(\cos x - \sin x) + \cos x + \frac{1}{2} \sin x$

6. $y = c_1 e^x + c_2 e^{-x} - \frac{1}{2}(x^2 - 1) \cos x + x \sin x$

7. $y = c_1 \cos x + c_2 \sin x - \frac{1}{27}[24x \cos 2x - (9x^2 - 26) \sin 2x]$

8. $y = c_1 e^x + c_2 e^{-x} + c_3 \cos x + c_4 \sin x + \frac{1}{15} \left(\frac{32}{15} \cos 2x + x \sin 2x \right)$

9. $y = c_1 \cos mx + c_2 \sin mx + (x/4m^2) \cos mx + (x^2/4m) \sin mx$

10. $y = c_1 e^x + c_2 e^{-x} - \frac{1}{2}(\cos x + x \sin x) + \frac{1}{12}e^x(2x^3 - 3x^2 + 9x)$

4.9 The Operator $\frac{1}{D - \alpha}$ and $\frac{1}{D + \alpha}$, α being a Constant

If Q is any function of x , then

$$\frac{1}{D - \alpha} Q = e^{\alpha x} \int e^{-\alpha x} Q dx \text{ and } \frac{1}{D + \alpha} Q = e^{-\alpha x} \int e^{\alpha x} Q dx$$

(Meerut 20, 03; Rohilkhand 09)

Proof. We have $\frac{1}{D - \alpha} Q = \frac{1}{D - \alpha} [e^{\alpha x} (e^{-\alpha x} Q)]$

$$= e^{\alpha x} \frac{1}{(D + \alpha) - \alpha} e^{-\alpha x} Q, \text{ by article 4.5}$$

$$= e^{\alpha x} \frac{1}{D} e^{-\alpha x} Q = e^{\alpha x} \int e^{-\alpha x} Q dx.$$

Similarly proceed for second operator proof.

Illustrative Examples

Example 36: Solve $\frac{d^2 y}{dx^2} + a^2 y = \sec ax$.

(Agra 2006; Bundelkhand 07; Purvanchal 07, 08; Lucknow 09)

Solution: The auxiliary equation is

$$m^2 + a^2 = 0, \quad \text{or} \quad m = \pm ia.$$

$\therefore$ C.F. = $c_1 \cos ax + c_2 \sin ax$.

And P.I. = $\frac{1}{D^2 + a^2} \sec ax = \frac{1}{(D + ia)(D - ia)} \sec ax$

$$= \frac{1}{2ia} \left[\frac{1}{D - ia} - \frac{1}{D + ia} \right] \sec ax \text{ [by resolving into partial fractions]}$$

$$= \frac{1}{2ia} \left[\frac{1}{D - ia} \sec ax - \frac{1}{D + ia} \sec ax \right]$$

$$= \frac{1}{2ia} \left[e^{iax} \int e^{-iax} \sec ax \, dx - e^{-iax} \int e^{iax} \sec ax \, dx \right], \text{ by article 4.9}$$

$$= \frac{1}{2ia} \left[e^{iax} \int \frac{\cos ax - i \sin ax}{\cos ax} \, dx - e^{-iax} \int \frac{\cos ax + i \sin ax}{\cos ax} \, dx \right],$$

$$(\because e^{-iax} = \cos ax - i \sin ax \text{ and } e^{iax} = \cos ax + i \sin ax)$$

$$= \frac{1}{2ia} \left[e^{iax} \left\{ x + \frac{i}{a} \log \cos ax \right\} - e^{-iax} \left\{ x - \frac{i}{a} \log \cos ax \right\} \right]$$

$$= \frac{x}{a} \left\{ \frac{e^{iax} - e^{-iax}}{2i} \right\} + \frac{1}{a^2} (\log \cos ax) \left\{ \frac{e^{iax} + e^{-iax}}{2} \right\}$$

$$= \frac{x}{a} \sin ax + \frac{1}{a^2} (\log \cos ax) \cos ax.$$

Hence the complete solution is

$$y = (\text{C.F.}) + (\text{P.I.})$$

or $y = c_1 \cos ax + c_2 \sin ax + \frac{x}{a} \sin ax + \frac{1}{a^2} \cos ax \cdot \log (\cos ax).$

Comprehensive Exercise 9

Solve the following differential equations :

1. $(d^2 y / dx^2) + 9y = \sec 3x.$

(Meerut 2007)

2. $(D^2 + a^2) y = \operatorname{cosec} ax.$

(Lucknow 2008)

3. $(D^2 + a^2) y = \tan ax.$

Answers 9

1. $y = c_1 \cos 3x + c_2 \sin 3x + \frac{1}{3} x \sin 3x + \frac{1}{9} \cos 3x \log (\cos 3x)$

2. $y = c_1 \cos ax + c_2 \sin ax + (1/a^2) \sin ax \log (\sin ax) - (x/a) \cos ax$

3. $y = c_1 \cos ax + c_2 \sin ax - (1/a^2) \cos ax \log \tan \left(\frac{1}{4} \pi + \frac{1}{2} ax \right)$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. The solution of differential equation $\frac{d^2 y}{dx^2} + y = 0$ is

(a) $c_1 e^{-x} + c_2 e^x$

(b) $c_1 \cos x + c_2 \sin x$

(c) $(c_1 + c_2 x) \cos x + (c_3 + c_4 x) \sin x$

(d) None of these.

(Bundelkhand 2001)

2. P.I. of the differential equation $(D^2 + D + 1) y = e^x$ is

(a) $\frac{1}{3} e^x$

(b) $3e^x$

(c) e^x

(d) None of these. (Avadh 2005)

3. For the differential equation $F(D)y = e^{ax}$, if $F(a) = 0$, then P.I. = $\frac{1}{F(D)} e^{ax}$ is

given by

(a) $\frac{1}{F(a)} e^{ax}$

(b) $\frac{1}{F(-a)} e^{ax}$

(c) $e^{ax} \frac{1}{F(D+a)} 1$

(d) None of these

(Rohilkhand 2005)

4. For the particular integral $\frac{1}{f(D^2)} \sin ax$ when $f(-a^2) = 0$, which one of the following is correct

$$(a) \frac{1}{f(D^2)} \sin ax = \frac{1}{f(-a^2)} \sin ax$$

$$(b) \frac{1}{D^2 + a^2} \sin ax = -\frac{x}{2a} \cos ax$$

$$(c) \frac{1}{D^2 + a^2} \sin ax = \frac{x}{2a} \cos ax$$

$$(d) \text{ None of these}$$

(Bundelkhand 2001; Agra 08)

5. The solution of the differential equation $\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^x$ is

$$(a) y = c_1 e^x + c_2 e^{3x} + x$$

$$(b) y = (c_1 + c_2) e^x - x e^x$$

$$(c) y = c_1 e^x + c_2 e^{2x} - x e^x$$

$$(d) y = c_1 x + c_2 e^x - e^{2x}$$

6. The P.I. of the differential equation $(D^2 + 1)y = \cos x$ is

$$(a) \frac{1}{2} \sin x$$

$$(b) -\frac{x}{2} \sin x$$

$$(c) \frac{x}{2} \sin x$$

$$(d) \frac{x}{2} \cos x$$

(Rohilkhand 2007)

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. The general solution of the equation $f(D)y = 0$ is called the 7 of the equation $f(D)y = Q$.

2. The solution of the differential equation $\frac{d^2 y}{dx^2} + a^2 y = 0$ is

3. The complementary function (C.F.) of the differential equation

$$\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 4y = 0 \text{ is } \dots\dots$$

4. The particular integral (P.I.) of the differential equation

$$\frac{d^3 y}{dx^3} - 4 \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} - 2y = 0 \text{ is } \dots\dots$$

5. The particular integral of the differential equation $f(D)y = Q$ will be

True or False

Write 'T' for true and 'F' for false statement.

1. A linear differential equation is an equation in which the dependent variable and its derivatives appear only in the first degree.

2. $\frac{d^4 y}{dx^4} - 5 \frac{d^3 y}{dx^3} - 2 \frac{dy}{dx} + 4xy = 0$ is a linear differential equation with constant coefficients.

(Meerut 2003)

3. $e^{-x} (c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x) + c_3 e^{2x}$ is the solution of the differential equation

$$\frac{d^3 y}{dx^3} - 8y = 0$$

Answers

Multiple Choice Questions

1. (b) 2. (a) 3. (c) 4. (b) 5. (c)
6. (c)

Fill in the Blank(s)

1. C.F. 2. $y = c_1 \cos ax + c_2 \sin ax$.
3. $c_1 e^x + c_2 e^{4x}$ 4. $\frac{2}{5}x$ 5. $\frac{1}{F(D)}Q$

True or False

1. T 2. F 3. T



Chapter

5



Homogeneous Linear Differential Equations

5.1 Homogeneous Linear Differential Equations

(Meerut 2009B)

Definition: A differential equation of the form

$$x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = Q, \quad \dots(1)$$

where $a_1, a_2, \dots, a_n$ are constants and Q is either a constant or a function of x is called a **homogeneous** linear differential equation.

5.2 Method of Solution

To solve such equations, we introduce a new independent variable z such that $x = e^z$ or $\log x = z$, so that $1/x = dz/dx$.

We have $\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \frac{1}{x} \frac{dy}{dz}$.

$\therefore x \frac{dy}{dx} = \frac{dy}{dz}$ so that $x \frac{d}{dx} \equiv \frac{d}{dz} \equiv D$, say.

Now
$$x \frac{d}{dx} \left(x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} \right) = x^n \frac{d^n y}{dx^n} + (n-1) x^{n-1} \frac{d^{n-1} y}{dx^{n-1}}$$

or
$$\begin{aligned} x^n \frac{d^n y}{dx^n} &= \left(x \frac{d}{dx} - n + 1 \right) x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} \\ &= (D - n + 1) x^{n-1} \frac{d^{n-1} y}{dx^{n-1}}. \end{aligned} \quad \dots(2)$$

Putting $n = 2, 3, 4, \dots$ etc. in (2), we have

$$\begin{aligned} x^2 \frac{d^2 y}{dx^2} &= (D-1) x \frac{dy}{dx} = (D-1) Dy \quad \left[\because x \frac{d}{dx} \equiv D \right] \\ &= D(D-1) y, \text{ because the operators can be interchanged} \\ x^3 \frac{d^3 y}{dx^3} &= (D-2) x^2 \frac{d^2 y}{dx^2} = (D-2)(D-1) Dy = D(D-1)(D-2) y. \end{aligned}$$

Whence, generalising, we have

$$x^n \frac{d^n y}{dx^n} = D(D-1)(D-2) \dots (D-n+1) y.$$

Substituting these values of $x \frac{dy}{dx}, x^2 \frac{d^2 y}{dx^2}, \dots, x^n \frac{d^n y}{dx^n}$ in (1) and thus changing the independent variable from x to z , we have

$$\begin{aligned} &[\{ D(D-1) \dots (D-n+1) \} + a_1 \{ D(D-1) \dots (D-n+2) \} \\ &\quad + \dots + a_{n-1} D + a_n] y = Q, \end{aligned}$$

or
$$f(D) y = Q, \quad \dots(3)$$

where Q has now become a function of z .

In the differential equation (3) the independent variable is z and the operator D stands for d/dz . Obviously (3) is a linear differential equation with constant coefficients and so it can be solved by the methods given in the previous chapter *i.e.*, chapter 3. Thus as in the case of the linear equations with constant coefficients, the general solution of (3) is the sum of any **particular integral** of (3) and the **complementary function** *i.e.*, the general solution of

$$f(D) y = 0. \quad \dots(4)$$

To find the Complementary function (C.F.):

- (i) If $m_1, m_2, \dots, m_n$ are the roots of the **auxiliary equation** of (4), and no two of them are equal, the general solution of (4) *i.e.*, the C.F. of the solution of (3) is easily seen to be

$$y = c_1 e^{m_1 z} + c_2 e^{m_2 z} + \dots + c_n e^{m_n z},$$

or
$$y = c_1 x^{m_1} + c_2 x^{m_2} + \dots + c_n x^{m_n}. \quad [\because e^z = x]$$

- (ii) In case there are r roots alike, each equal to m , and the rest all different, then the

$$\begin{aligned} \text{C.F.} &= (c_1 + c_2 z + \dots + c_r z^{r-1}) e^{mz} + c_{r+1} e^{m_{r+1}z} + \dots + c_n e^{m_n z} \\ &= [c_1 + c_2 \log x + \dots + c_r (\log x)^{r-1}] x^m + c_{r+1} x^{m_{r+1}} + \dots + c_n x^{m_n}. \end{aligned}$$

(iii) In case the roots are imaginary, say of the form $\alpha \pm i\beta$, then the

$$\text{C.F.} = e^{\alpha x} (c_1 \cos \beta z + c_2 \sin \beta z) = x^\alpha [c_1 \cos (\beta \log x) + c_2 \sin (\beta \log x)].$$

In this case we can also write the

$$\text{C.F.} = c_1 e^{\alpha z} \cos (\beta z + c_2) = c_1 x^\alpha \cos (\beta \log x + c_2).$$

(iv) In case the roots $\alpha \pm i\beta$ occur r times, the C.F. corresponding to these roots will be

$$\begin{aligned} &e^{\alpha z} [(c_1 + c_2 z + \dots + c_r z^{r-1}) \cos \beta z \\ &\quad + (c_1' + c_2' z + \dots + c_r' z^{r-1}) \sin \beta z] \\ &= x^\alpha [\{c_1 + c_2 \log x + \dots + c_r (\log x)^{r-1}\} \cos (\beta \log x) \\ &\quad + \{c_1' + c_2' \log x + \dots + c_r' (\log x)^{r-1}\} \sin (\beta \log x)]. \end{aligned}$$

To find the Particular integral. (P.I.):

The particular integral (P.I.) of (3) is given by

$$\frac{1}{f(D)} Q.$$

If α is a constant, we have

$$\begin{aligned} \frac{1}{D - \alpha} Q &= \frac{1}{D - \alpha} \{e^{\alpha z} (e^{-\alpha z} Q)\} \\ &= e^{\alpha z} \frac{1}{(D + \alpha) - \alpha} e^{-\alpha z} Q \\ &= e^{\alpha z} \frac{1}{D} e^{-\alpha z} Q = e^{\alpha z} \int e^{-\alpha z} Q dz. \end{aligned}$$

Methods to find the P.I:

General Methods:

(i) We resolve the operator $f(D)$ into linear factors. Thus we write

$$\begin{aligned} f(D) &= (D - m_1)(D - m_2) \dots (D - m_n). \text{ Then the} \\ \text{P.I.} &= \frac{1}{f(D)} Q = \frac{1}{(D - m_1)(D - m_2) \dots (D - m_{n-1})(D - m_n)} Q \\ &= \frac{1}{(D - m_1)(D - m_2) \dots (D - m_{n-1})} \int e^{-m_n z} Q dz, \end{aligned}$$

by operating $\frac{1}{D - m_n}$ upon Q as explained above.

Similarly we operate with other remaining factors in succession and thus we find the P.I.

(ii) First we resolve $f(D)$ into linear factors as in (i) and then we break up $\{f(D)\}^{-1}$ into partial fractions. Then the

$$\begin{aligned} \text{P.I.} &= \frac{1}{f(D)} Q = \left\{ \frac{A_1}{D - m_1} + \frac{A_2}{D - m_2} + \dots + \frac{A_n}{D - m_n} \right\} Q \\ &= A_1 e^{m_1 z} \int e^{-m_1 z} Q dz + \dots + A_n e^{m_n z} \int e^{-m_n z} Q dz. \end{aligned}$$

Special short Methods.

(i) When Q is of the form e^{az} , then

$$\text{P.I.} = \frac{1}{f(D)} e^{az} = \frac{1}{f(a)} e^{az}, \text{ provided } f(a) \neq 0.$$

(ii) When Q is of the form $\cos az$ or $\sin az$, then the P.I. is given by

$$\frac{1}{f(D^2)} \cos az = \frac{1}{f(-a^2)} \cos az,$$

$$\text{and } \frac{1}{f(D^2)} \sin az = \frac{1}{f(-a^2)} \sin az, \quad [\text{provided } f(-a^2) \neq 0]$$

(iii) If Q is of the form z^m , we have $\text{P.I.} = \frac{1}{f(D)} z^m$.

We expand $\{f(D)\}^{-1}$ in ascending powers of D retaining terms as far as D^m and operate each term on z^m .

(iv) If Q is of the form $e^{az}V$, where V is any function of z , we have the

$$\text{P.I.} = \frac{1}{f(D)} e^{az} V = e^{az} \frac{1}{f(D+a)} V.$$

(v) If Q is of the form zV , where V is any function of z , we have the

$$\text{P.I.} = \frac{1}{f(D)} (zV) = z \frac{1}{f(D)} V + \left\{ \frac{d}{dD} \frac{1}{f(D)} \right\} V.$$

Illustrative Examples

Example 1: Solve $x^2 D^2 y + 5x Dy + 4y = 0$, where $D \equiv d/dx$.

Solution: The given differential equation is

$$x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 4y = 0. \quad \dots(1)$$

Now (1) is a homogeneous linear differential equation of order 2. So putting $x = e^z$ and denoting d/dz by D' , we have

$$x \frac{dy}{dx} = D' y, \quad x^2 \frac{d^2 y}{dx^2} = D' (D' - 1) y.$$

$\therefore$ the differential equation (1) transforms to

$$\{D' (D' - 1) + 5D' + 4\} y = 0$$

$$\text{or } (D'^2 + 4D' + 4) y = 0$$

or $(D' + 2)^2 y = 0,$

which is a linear differential equation with constant coefficients, the independent variable being z .

$\therefore$ auxiliary equation is $(m + 2)^2 = 0$, giving $m = -2, -2$.

$\therefore$ the complete solution is $y = (c_1 + c_2 z) e^{-2z}$

or $y = (c_1 + c_2 z) (e^z)^{-2}$

or $y = (c_1 + c_2 \log x) x^{-2}.$

Example 2: Solve $(x^3 D^3 + 3x^2 D^2 - 2xD + 2) y = 0$, where $D \equiv d/dx$.

Solution: Putting $x = e^z$ and denoting d/dz by D' , we have

$$x Dy = x \frac{dy}{dx} = D' y, x^2 D^2 y = x^2 \frac{d^2 y}{dx^2} = D' (D' - 1) y$$

and $x^3 D^3 y = x^3 \frac{d^3 y}{dx^3} = D' (D' - 1) (D' - 2) y.$

$\therefore$ the given differential equation transforms to

$$[D' (D' - 1) (D' - 2) + 3D' (D' - 1) - 2D' + 2] y = 0$$

or $[D' (D' - 1) (D' - 2) + 3D' (D' - 1) - 2(D' - 1)] y = 0$

or $(D' - 1) [D' (D' - 2) + 3D' - 2] y = 0$

or $(D' - 1) (D'^2 + D' - 2) y = 0$

or $(D' - 1) (D' + 2) (D' - 1) y = 0$

or $(D' - 1)^2 (D' + 2) y = 0.$

$\therefore$ the auxiliary equation is $(m - 1)^2 (m + 2) = 0$, giving

$$m = 1, 1, -2.$$

$\therefore$ the general solution is

$$\begin{aligned} y &= (c_1 + c_2 z) e^z + c_3 e^{-2z} \\ &= (c_1 + c_2 z) e^z + c_3 (e^z)^{-2} \end{aligned}$$

or $y = (c_1 + c_2 \log x) x + c_3 x^{-2}, \quad [\because x = e^z \text{ and } z = \log x]$

or $y = (c_1 + c_2 \log x) x + c_3 \cdot (1/x^2).$

Example 3: Solve

$$x^4 \frac{d^3 y}{dx^3} + 2x^3 \frac{d^2 y}{dx^2} - x^2 \frac{dy}{dx} + xy = 1.$$

Solution: Dividing out by x , the given equation may be written as

$$x^3 \frac{d^3 y}{dx^3} + 2x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = \frac{1}{x}, \quad \dots(1)$$

which is a linear homogeneous equation of order 3.

Putting $x = e^z$ in (1) and denoting d/dz by D , we have

$$[D(D-1)(D-2) + 2D(D-1) - D + 1]y = e^{-z}$$

or $(D-1)^2(D+1)y = e^{-z}$.

$\therefore$ auxiliary equation is

$$(m-1)^2(m+1) = 0, \text{ giving } m = 1, 1, -1.$$

$\therefore$ C.F. = $(c_1 + c_2 z)e^z + c_3 e^{-z}$

$$= (c_1 + c_2 \log x)x + c_3 x^{-1}. \quad [\because e^z = x \text{ and } z = \log x]$$

And

$$\begin{aligned} \text{P.I.} &= \frac{1}{(D-1)^2(D+1)} e^{-z} = \frac{1}{(D+1)} \left[\frac{1}{(D-1)^2} e^{-z} \right] \\ &= \frac{1}{(D+1)} \cdot \frac{1}{4} e^{-z} = \frac{1}{4} \frac{1}{D+1} e^{-z} \cdot 1 = \frac{1}{4} e^{-z} \frac{1}{D-1+1} \cdot 1 \\ &= \frac{1}{4} e^{-z} \frac{1}{D} \cdot 1 = \frac{1}{4} z e^{-z} = \frac{1}{4x} \log x. \end{aligned}$$

Hence the complete solution is

$$y = (c_1 + c_2 \log x)x + \frac{c_3}{x} + \frac{1}{4x} \log x.$$

Example 4: Solve $x^3 \frac{d^3 y}{dx^3} + 2x^2 \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} - 3y = x^2 + x$.

(Meerut 2001, 07; Gorakhpur 05, 08)

Solution: Putting $x = e^z$ and denoting d/dz by D , the given differential equation becomes

$$[D(D-1)(D-2) + 2D(D-1) + 3D - 3]y = e^{2z} + e^z$$

or $[D(D-1)(D-2) + 2D(D-1) + 3(D-1)]y = e^{2z} + e^z$

or $(D-1)\{D(D-2) + 2D + 3\}y = e^{2z} + e^z$

or $(D-1)(D^2 + 3)y = e^{2z} + e^z$.

Hence the auxiliary equation is

$$(m-1)(m^2 + 3) = 0, \text{ giving } m = 1, 0 \pm \sqrt{3}i.$$

$\therefore$ C.F. = $c_1 e^z + e^{0z} (c_2 \cos \sqrt{3}z + c_3 \sin \sqrt{3}z)$

$$= c_1 e^z + c_2 \cos \sqrt{3}z + c_3 \sin \sqrt{3}z.$$

Also

$$\begin{aligned} \text{P.I.} &= \frac{1}{(D-1)(D^2 + 3)} e^{2z} + \frac{1}{(D-1)(D^2 + 3)} e^z \\ &= \frac{e^{2z}}{(2-1)(D^2 + 3)} + \frac{1}{(1^2 + 3)(D-1)} e^z \cdot 1 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{7} e^{2z} + \frac{1}{4} e^z \frac{1}{D+1-1} 1 = \frac{1}{7} e^{2z} + \frac{1}{4} e^z \frac{1}{D} 1 \\
 &= \frac{1}{7} e^{2z} + \frac{1}{4} e^z \cdot z.
 \end{aligned}$$

$\therefore$ the complete solution is

$$y = c_1 e^z + c_2 \cos \sqrt{3}z + c_3 \sin \sqrt{3}z + \frac{1}{7} e^{2z} + \frac{1}{4} z e^z.$$

or
$$y = c_1 x + c_2 \cos(\sqrt{3} \log x) + c_3 \sin(\sqrt{3} \log x) + \frac{1}{7} x^2 + \frac{1}{4} x \log x.$$

$$[\because e^z = x \text{ and } z = \log x]$$

Example 5: Solve $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = x^2 \log x$.

(Meerut 2004B, 10; Gorakhpur 07; Kanpur 07, 14; Lucknow 09, 11; Avadh 08; Bundelkhand 04)

Solution: Putting $x = e^z$ or $z = \log x$ and denoting d/dz by D the equation becomes

$$[D(D-1) - D - 3]y = e^{2z} \cdot z$$

or
$$(D^2 - 2D - 3)y = e^{2z} \cdot z.$$

$\therefore$ auxiliary equation is

$$(m^2 - 2m - 3) = 0$$

or
$$(m-3)(m+1) = 0 \text{ i.e., } m = -1, 3.$$

$\therefore$ C.F. is $c_1 e^{3z} + c_2 e^{-z}$.

And P.I. =
$$\begin{aligned}
 &\frac{1}{(D^2 - 2D - 3)} e^{2z} \cdot z \\
 &= e^{2z} \cdot \frac{1}{[(D+2)^2 - 2(D+2) - 3]} z \\
 &= e^{2z} \frac{1}{(D^2 + 2D - 3)} z \\
 &= e^{2z} \frac{1}{-3[1 - \frac{2}{3}D - \frac{1}{3}D^2]} z \\
 &= e^{2z} - \frac{1}{3} \left[1 - \left(\frac{2}{3}D + \frac{1}{3}D^2 \right) \right]^{-1} z \\
 &= -\frac{1}{3} e^{2z} \left(1 + \frac{2}{3}D + \frac{1}{3}D^2 + \dots \right) z = -\frac{1}{3} e^{2z} \left(z + \frac{2}{3} \right).
 \end{aligned}$$

$\therefore$ complete solution is

$$y = c_1 e^{3z} + c_2 e^{-z} - \frac{1}{3} e^{2z} \left(z + \frac{2}{3} \right)$$

or
$$y = c_1 x^3 + c_2 x^{-1} - \frac{1}{3} x^2 \left(\log x + \frac{2}{3} \right).$$

5.3 Equations Reducible to Homogeneous Form

A differential equation of the form

$$(a + bx)^n \frac{d^n y}{dx^n} + P_1 (a + bx)^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + P_{n-1} (a + bx) \frac{dy}{dx} + P_n y = Q,$$

where $P_1, P_2, \dots, P_n$ are constants and Q is a function of x , can be reduced to the homogeneous linear form by putting $a + bx = t$ and then can be solved by the method explained in article 5.2. We can also solve this differential equation directly by making the substitution.

$$e^z = a + bx$$

or $z = \log (a + bx).$

If we denote the operator d/dz by D , we can easily see that

$$(a + bx) \frac{dy}{dx} = b D y, (a + bx)^2 \frac{d^2 y}{dx^2} = b^2 D (D - 1) y, \text{ and so on.}$$

Illustrative Examples

Example 6: Solve $(x + a)^2 \frac{d^2 y}{dx^2} - 4(x + a) \frac{dy}{dx} + 6y = x$.
(Meerut 2001, 06, 11; Kanpur 11)

Solution: Putting $(x + a) = e^z$ and denoting d/dz by D , the given equation becomes

$$[D(D - 1) - 4D + 6]y = e^z - a$$

or $(D - 3)(D - 2)y = e^z - a$.

$\therefore$ auxiliary equation is $(m - 3)(m - 2) = 0$, giving $m = 3, 2$.

$\therefore$ C.F. = $c_1 e^{3z} + c_2 e^{2z} = c_1 (e^z)^3 + c_2 (e^z)^2 = c_1 (x + a)^3 + c_2 (x + a)^2$

and P.I. = $\frac{1}{(D - 3)(D - 2)} e^z - \frac{1}{(D - 3)(D - 2)} a e^{0z}$, [$\because e^{0z} = 1$]

$$= \frac{1}{(D - 3)(D - 2)} e^z - a \frac{1}{(D - 3)(D - 2)} e^{0z},$$

$$= \frac{1}{(1 - 3)(1 - 2)} e^z - a \frac{1}{(0 - 3)(0 - 2)} e^{0z}$$

$$= \frac{1}{2} e^z - \frac{1}{6} a = \frac{1}{2} (x + a) - \frac{1}{6} a.$$

Hence the complete solution is $y = \text{C.F.} + \text{P.I.}$

or $y = c_1 (x + a)^3 + c_2 (x + a)^2 + \frac{1}{2} (x + a) - \frac{1}{6} a.$

Comprehensive Exercise 1

Solve the following differential equations :

1. $x^2 (d^2 y/dx^2) - 4x (dy/dx) + 6y = x$. (Bundelkhand 2010)

2. $x^2 (d^2 y/dx^2) - 4x (dy/dx) + 6y = x^4$. (Gorakhpur 2009)

3. $x^2 (d^2 y/dx^2) - 2x (dy/dx) + 2y = 1/x$.

4. $x^2 (d^2 y/dx^2) + x (dy/dx) - 4y = x^2$. (Rohilkhand 2009)

5. $x^2 (d^2 y/dx^2) + 2x (dy/dx) - 20y = (x+1)^2$.

6. $x^2 (d^2 y/dx^2) + 4x (dy/dx) + 2y = e^x$.

(Meerut 2005; Avadh 08, 11; Rohilkhand 07, 11)

7. $x^2 (d^2 y/dx^2) + 7x (dy/dx) + 13y = \log x$. (Bundelkhand 2006)

8. (i) $x^2 (d^2 y/dx^2) - x (dy/dx) + 2y = x \log x$.

(Rohilkhand 2010; Gorakhpur 06, 10, 11; Purvanchal 09;

Avadh 06; Bundelkhand 13)

(ii) $x^3 (d^3 y/dx^3) + 3x^2 (d^2 y/dx^2) + x (dy/dx) + y = x \log x$.

(Kanpur 2009; Rohilkhand 08; Purvanchal 08)

9. $x^3 (d^3 y/dx^3) + 3x^2 (d^2 y/dx^2) + x (dy/dx) + y = x + \log x$. (Avadh 2006, 14)

10. $x^2 (d^2 y/dx^2) - x (dy/dx) + 4y = \cos(\log x) + x \sin(\log x)$.

11. $x^3 (d^3 y/dx^3) + 2x^2 (d^2 y/dx^2) + 2y = 10 [x + (1/x)]$.

(Meerut 2007B; Bundelkhand 09; Agra 2006; Lucknow 06)

12. $x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + y = \frac{\log x \sin(\log x) + 1}{x}$.

13. $(x^2 D^2 + 3xD + 1)y = \frac{1}{(1-x)^2}$.

(Rohilkhand 2008; Purvanchal 08; Bundelkhand 01)

14. Solve $(3x+2)^2 \frac{d^2 y}{dx^2} + 3(3x+2) \frac{dy}{dx} - 36y = 3x^2 + 4x + 1$.

(Bundelkhand 2009; Agra 06; Lucknow 06)

15. Solve $(1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = 4 \cos \log(1+x)$.

Answers 1

1. $y = c_1 x^2 + c_2 x^3 + \frac{1}{2} x$
2. $y = c_1 x^2 + c_2 x^3 + \frac{1}{2} x^4$
3. $y = c_1 x + c_2 x^2 + \frac{1}{6} \cdot (1/x)$
4. $y = c_1 x^2 + (c_2/x^2) + \frac{1}{4} x^2 \log x$
5. $y = c_1 x^4 + c_2 x^{-5} - \frac{1}{14} x^2 - \frac{1}{9} x - \frac{1}{20}$
6. $y = (c_1/x) + (c_2/x^2) + (e^x/x^2)$
7. $y = c_1 \{ \cos (\log x^2 + c_2) \} / x^3 + \frac{1}{13} \log x - \frac{6}{169}$
8. (i) $y = c_1 x \cos (\log x + c_2) + x \log x$
 (ii) $y = c_1 x^{-1} + c_2 \sqrt{x} \cos \{ (\sqrt{3}/2) \log x + c_3 \} + \frac{1}{2} x \log x - \frac{3}{4} x$
9. $y = c_1 x^{-1} + c_2 \sqrt{x} \cos \{ (\sqrt{3}/2) \log x + c_3 \} + \frac{1}{2} x + \log x$
10. $y = x [c_1 \cos (\sqrt{3} \log x) + c_2 \sin (\sqrt{3} \log x)]$
 $+ \frac{1}{13} [3 \cos (\log x) - 2 \sin (\log x)] + \frac{1}{2} x \sin (\log x)$
11. $y = c x^{-1} + c_2 x \cos (\log x + c_3) + 5x + 2x^{-1} \log x$
12. $y = x^2 (c_1 x^{\sqrt{3}} + c_2 x^{-\sqrt{3}}) + \frac{1}{6x}$
 $+ \frac{1}{x} \left[\log x \cdot \frac{(5 \sin \log x + 6 \cos \log x)}{61} + \frac{382 \cos \log x + 54 \sin \log x}{3721} \right]$
13. $y = (c_1 + c_2 \log x) \frac{1}{x} + \frac{1}{x} \log \frac{x}{1-x}$
14. $y = c_1 (3x+2)^2 + c_2 (3x+2)^{-2} + \frac{1}{108} [(3x+2)^2 \log (3x+2) + 1]$
15. $y = c_1 \cos \log (1+x) + c_2 \sin \log (1+x) + 2 \log (1+x) \sin \log (1+x)$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. The complementary function (C.F.) of the differential equation

$$x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = x^2 \log x \text{ is}$$

- (a) $c_1 \cos x + c_2 \sin x$ (b) $c_1 e^{3z} + c_2 e^{-z}$
(c) $c_1 e^z + c_2 e^{-3z}$ (d) None of these

2. C.F. of the differential equation $x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = e^x$ is

- (a) $c_1 + c_2 x$ (b) $c_1 x^{-1} + c_2 x^{-2}$
(c) $c_1 e^{-x} + c_2 e^x$ (d) None of these

3. C.F. of the differential equation $x^4 \frac{d^3 y}{dx^3} + 2x^3 \frac{d^2 y}{dx^2} - x^2 \frac{dy}{dx} + xy = 1$ is

- (a) $(c_1 + c_2 x) e^x + c_3 e^{-x}$ (b) $c_1 e^x + c_2 e^{-x} + c_3 e^{2x}$
(c) $(c_1 + c_2 \log x) x + \frac{c_3}{x} + \frac{1}{4x} \log x$ (d) None of these.

4. C.F. of the differential equation $(x+a)^2 \frac{d^2 y}{dx^2} - 4(x+a) \frac{dy}{dx} + 6y = x$ is

- (a) $c_1 x^2 + c_2 x^3$ (b) $c_1 (x+a)^3 + c_2 (x+a)^2$
(c) $c_1 e^{3x} + c_2 e^{2x}$ (d) None of these.

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. Homogeneous linear differential equation $x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 4y = 0$

will reduce to a linear differential equation with constant coefficients by putting $x = \dots\dots$

2. To reduce the homogeneous linear differential equation

$$(x+a)^3 \frac{d^3 y}{dx^3} - 2(x+a)^2 \frac{d^2 y}{dx^2} - 5(x+a) \frac{dy}{dx} + 3y = 2x$$

to a linear differential equation with constant coefficients, we have to put

True or False

Write 'T' for true and 'F' for false statement.

1. $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = x^2 \log x$ is a linear homogeneous differential equation of order two. (Meerut 2003)
2. $x^4 \frac{d^3 y}{dx^3} + 2x^3 \frac{d^2 y}{dx^2} - x^2 \frac{dy}{dx} + xy = 1$ is not a linear homogeneous differential equation of order 3.
3. $x^3 \frac{d^3 y}{dx^3} + 4x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} + xy = x^2$ is a linear homogeneous differential equation of order 3.

Answers

Multiple Choice Questions

1. (b)
2. (b)
3. (c)
4. (b)

Fill in the Blank(s)

1. e^z
2. $(x + a) = e^z$

True or False

1. T
2. F
3. F



Chapter

6



Ordinary Simultaneous Differential Equations

6.1 Introduction

In the present chapter, we shall discuss differential equations containing **one independent variable and two or more than two dependent variables**. To completely solve such equations we need as many simultaneous equations as there are dependent variables.

6.2 Methods of Solving Simultaneous Linear Differential Equations with Constant Coefficients

Let x and y be the two dependent variables and t be the independent variable. Thus the equations will contain differential coefficients of x, y with respect to t . Let $D \equiv d / dt$. Then such equations can be put in the form

$$f_1(D) x + f_2(D) y = T_1 \quad \dots(1)$$

and $\phi_1(D) x + \phi_2(D) y = T_2, \quad \dots(2)$

where T_1 and T_2 are some functions of the independent variable t . Here $f_1(D)$, $f_2(D)$, $\phi_1(D)$ and $\phi_2(D)$ are all rational integral functions of D with constant coefficients.

There are two methods to solve such equations.

First method: Method of Elimination (Symbolic method):

To eliminate y between (1) and (2), operating both sides of (1) by $\phi_2(D)$ and of (2) by $f_2(D)$, we get

$$\phi_2(D) f_1(D) x + \phi_2(D) f_2(D) y = \phi_2(D) T_1 \quad \dots(3)$$

$$\text{and} \quad f_2(D) \phi_1(D) x + f_2(D) \phi_2(D) y = f_2(D) T_2. \quad \dots(4)$$

Subtracting (4) from (3), we get

$$[\phi_2(D) f_1(D) - f_2(D) \phi_1(D)] x = \phi_2(D) T_1 - f_2(D) T_2, \quad \dots(5)$$

which is of the form $F(D)x = T$.

The equation (5) is a linear differential equation with constant coefficients in x and t . Solving it we can find the value of x in terms of t .

Putting this value of x in either (1) or (2), we get the value of y .

Note 1: The equations can also be solved by first finding the value of y by eliminating x between (1) and (2) and then solving the resulting linear differential equation in y and t . The value of x can now be obtained from (1) or (2) after putting the value of y .

Note 2: Since $f_2(D)$ and $\phi_2(D)$ are functions of D with constant coefficients, so

$$f_2(D) \phi_2(D) = \phi_2(D) f_2(D).$$

Second method: Method of differentiation:

Sometimes x or y can be conveniently eliminated if we differentiate the equations (1) and (2). For example suppose the two given equations (1) and (2) connect four quantities x , y , dx/dt and dy/dt . Differentiating (1) and (2) w.r.t. t , we get in all four equations containing x , y , $\frac{dx}{dt}$, $\frac{dy}{dt}$, $\frac{d^2x}{dt^2}$, $\frac{d^2y}{dt^2}$. Eliminating three quantities y , $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$ from these four equations, we get an equation of the second order with x as the dependent and t as the independent variable. Solving this equation we find the value of x in terms of t . Then the value of the other variable can be found.

6.3 Number of Arbitrary Constants

In the general solution of (1) and (2) the number of arbitrary constants is equal to the degree of D in the determinant

$$\Delta = \begin{vmatrix} f_1(D) & f_2(D) \\ \phi_1(D) & \phi_2(D) \end{vmatrix}, \text{ if } \Delta \neq 0.$$

If Δ vanishes then the system is dependent. Here we shall not consider such cases.

Illustrative Examples

Example 1: Solve the simultaneous equations

$$\frac{dx}{dt} - 7x + y = 0, \quad \frac{dy}{dt} - 2x - 5y = 0.$$

(Purvanchal 2008; Rohilkhand 09; Lucknow 10)

Solution: Writing D for d/dt , the given equations are

$$(D - 7)x + y = 0 \quad \dots(1)$$

$$\text{and} \quad -2x + (D - 5)y = 0. \quad \dots(2)$$

Let us eliminate y between (1) and (2). Operating on both sides of (1) by $D - 5$ and then subtracting (2) from it, we get

$$\{(D - 5)(D - 7) + 2\}x = 0$$

$$\text{or} \quad (D^2 - 12D + 37)x = 0, \quad \dots(3)$$

which is a linear differential equation of second order in x and t with constant coefficients.

$$\text{A.E. is } m^2 - 12m + 37 = 0 \Rightarrow m = 6 \pm i.$$

$$\therefore \quad \text{the solution of (3) is } x = e^{6t} (c_1 \cos t + c_2 \sin t) \quad \dots(4)$$

$$\therefore \quad \frac{dx}{dt} = 6e^{6t} (c_1 \cos t + c_2 \sin t) + e^{6t} (-c_1 \sin t + c_2 \cos t).$$

Putting the values of x and dx/dt in (1), we get

$$\begin{aligned} y &= 7x - Dx = 7x - (dx/dt) \\ &= 7e^{6t} (c_1 \cos t + c_2 \sin t) - 6e^{6t} (c_1 \cos t + c_2 \sin t) \\ &\quad - e^{6t} (-c_1 \sin t + c_2 \cos t) \end{aligned}$$

$$\text{or} \quad y = e^{6t} [(c_1 - c_2) \cos t + (c_1 + c_2) \sin t]. \quad \dots(5)$$

The required general solution of the given equations consists of the equations (4) and (5).

Example 2: Solve the simultaneous equations

$$\frac{d^2 y}{dt^2} - 3x - 4y = 0, \quad \frac{d^2 x}{dt^2} + x + y = 0.$$

(Kanpur 2003; Meerut 09)

Solution: Writing D for (d/dt) , the given equations are

$$(D^2 - 3)x - 4y = 0 \quad \dots(1)$$

$$\text{and} \quad x + (D^2 + 1)y = 0. \quad \dots(2)$$

Let us eliminate y from (1) and (2). Multiplying both sides of (2) by 4 and operating on both sides of (1) by $D^2 + 1$ and adding, we get

$$\{(D^2 + 1)(D^2 - 3) + 4\}x = 0$$

$$\text{or} \quad (D^4 - 2D^2 + 1)x = 0. \quad \dots(3)$$

A.E. is $m^4 - 2m^2 + 1 = 0$, or $(m^2 - 1)^2 = 0$,

$\therefore m = \pm 1, \pm 1$.

$\therefore$ the solution of (3) is $x = (c_1 + c_2 t) e^t + (c_3 + c_4 t) e^{-t}$... (4)

$\therefore \frac{dx}{dt} = (c_1 + c_2 t) e^t - (c_3 + c_4 t) e^{-t} + c_2 e^t + c_4 e^{-t}$,

and $\frac{d^2 x}{dt^2} = (c_1 + c_2 t) e^t + (c_3 + c_4 t) e^{-t} + 2c_2 e^t - 2c_4 e^{-t}$

Putting these values in (1), we get

$$\begin{aligned} 4y &= D^2 x - 3x = (d^2 x / dt^2) - 3x \\ &= (c_1 + c_2 t) e^t + (c_3 + c_4 t) e^{-t} \\ &\quad + 2c_2 e^t - 2c_4 e^{-t} - 3(c_1 + c_2 t) e^t - 3(c_3 + c_4 t) e^{-t} \\ &= -2(c_1 + c_2 t) e^t - 2(c_3 + c_4 t) e^{-t} + 2c_2 e^t - 2c_4 e^{-t} \end{aligned}$$

or $y = \frac{1}{2} e^t (c_2 - c_1 - c_2 t) - \frac{1}{2} e^{-t} (c_4 + c_3 + c_4 t)$ (5)

The required general solution of the given equations consists of the equations (4) and (5).

Example 3: Solve the simultaneous equations

$$\frac{d^2 x}{dt^2} + 4x + y = te^{3t}, \quad \frac{d^2 y}{dt^2} + y - 2x = \cos^2 t.$$

Solution: Writing D for (d/dt) , the given equations are

$$(D^2 + 4)x + y = te^{3t} \quad \dots (1)$$

and $-2x + (D^2 + 1)y = \cos^2 t \quad \dots (2)$

Eliminating y from (1) and (2), we get

$$\{(D^2 + 1)(D^2 + 4) + 2\}x = (D^2 + 1)(te^{3t}) - \cos^2 t$$

or $(D^4 + 5D^2 + 6)x = 10te^{3t} + 6e^{3t} - \cos^2 t$ (3)

A.E. is $m^4 + 5m^2 + 6 = 0$, or $(m^2 + 3)(m^2 + 2) = 0$.

$\therefore m = \pm i\sqrt{3}, \pm i\sqrt{2}$.

$\therefore$ C.F. $= (c_1 \cos \sqrt{3}t + c_2 \sin \sqrt{3}t) + (c_3 \cos \sqrt{2}t + c_4 \sin \sqrt{2}t)$.

$$\begin{aligned} \text{P.I.} &= \frac{10}{D^4 + 5D^2 + 6} te^{3t} + \frac{6}{D^4 + 5D^2 + 6} e^{3t} - \frac{1}{D^4 + 5D^2 + 6} \cos^2 t \\ &= 10e^{3t} \frac{1}{(D+3)^4 + 5(D+3)^2 + 6} t + 6 \frac{1}{3^4 + 5 \cdot 3^2 + 6} \cdot e^{3t} \\ &\quad - \frac{1}{D^4 + 5D^2 + 6} \frac{1}{2} (1 + \cos 2t) \\ &= 10e^{3t} \frac{1}{132 + 138D + 52D^2 + \dots} t + \frac{1}{22} e^{3t} \\ &\quad - \frac{1}{6 + 5D^2 + D^4} \frac{1}{2} - \frac{1}{D^4 + 5D^2 + 6} \left(\frac{1}{2} \cos 2t \right) \end{aligned}$$

$$\begin{aligned}
 &= 10e^{3t} \cdot \frac{1}{132} \left(1 + \frac{23}{22} D + \dots \right)^{-1} t + \frac{1}{22} e^{3t} \\
 &\quad - \frac{1}{2} \cdot \frac{1}{6 + 5D^2 + D^4} e^{0t} - \frac{1}{2} \frac{\cos 2t}{(-4)^2 + 5(-4) + 6} \\
 &= \frac{5e^{3t}}{66} \left(t - \frac{23}{22} \right) + \frac{e^{3t}}{22} - \frac{1}{12} - \frac{1}{4} \cos 2t.
 \end{aligned}$$

$$\begin{aligned}
 \therefore x &= (c_1 \cos \sqrt{3}t + c_2 \sin \sqrt{3}t) + (c_3 \cos \sqrt{2}t + c_4 \sin \sqrt{2}t) \\
 &\quad + \frac{5}{66} te^{3t} - \frac{49}{1452} e^{3t} - \frac{1}{4} \cos 2t - \frac{1}{12}. \quad \dots(4)
 \end{aligned}$$

Now from (4), we find $\frac{dx}{dt}$ and $\frac{d^2x}{dt^2}$.

Putting the values of x and d^2x/dt^2 in (1), we get

$$y = -\frac{d^2x}{dt^2} - 4x + te^{3t}$$

$$\begin{aligned}
 \text{or } y &= -(c_1 \cos \sqrt{3}t + c_2 \sin \sqrt{3}t) - 2(c_3 \cos \sqrt{2}t + c_4 \sin \sqrt{2}t) \\
 &\quad + \frac{1}{66} t \cdot e^{3t} - \frac{33}{2452} e^{3t} + \frac{1}{3}. \quad \dots(5)
 \end{aligned}$$

The required solution consists of the equations (4) and (5).

Example 4: Solve the simultaneous differential equations

$$\begin{aligned}
 x \frac{dy}{dx} + z &= 0 \quad \dots(1) & x \frac{dz}{dx} + y &= 0. \quad \dots(2)
 \end{aligned}$$

(Meerut 2001, 09B)

Solution: Differentiating (1) with respect to x , we get

$$x \frac{d^2y}{dx^2} + \frac{dy}{dx} + \frac{dz}{dx} = 0 \quad \text{or} \quad x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + x \frac{dz}{dx} = 0.$$

Putting the value of $x \frac{dz}{dx}$ from (2), we get

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} - y = 0, \quad \dots(3)$$

which is a linear homogeneous equation.

To solve (3), putting $x = e^t$ and denoting d/dt by D , we have

$$x \frac{dy}{dx} = Dy, \quad x^2 \frac{d^2y}{dx^2} = D(D-1)y.$$

$\therefore$ the equation (3) becomes

$$\{D(D-1) + D - 1\} y = 0 \quad \text{or} \quad (D^2 - 1)y = 0.$$

$$\therefore y = c_1 e^t + c_2 e^{-t} = c_1 x + c_2 x^{-1}. \quad \dots(4)$$

$$\therefore \frac{dy}{dx} = c_1 - \frac{c_2}{x^2} \Rightarrow x \frac{dy}{dx} = c_1 x - \frac{c_2}{x}.$$

$$\therefore \text{ from (1), we get } z = -x \frac{dy}{dx} = -c_1 x + c_2 x^{-1}. \quad \dots(5)$$

The required solution consists of the equations (4) and (5).

Example 5: Solve the simultaneous differential equations

$$\frac{d^2 x}{dt^2} + m^2 y = 0, \quad \frac{d^2 y}{dt^2} - m^2 x = 0.$$

Solution: Writing D for (d/dt) , the given equations are

$$D^2 x + m^2 y = 0 \quad \dots(1)$$

$$\text{and} \quad D^2 y - m^2 x = 0. \quad \dots(2)$$

Eliminating y from (1) and (2), we get

$$(D^4 + m^4) x = 0.$$

$$\text{A.E. is } M^4 + m^4 = 0 \Rightarrow (M^2 + m^2)^2 - 2M^2 m^2 = 0.$$

$$\text{or} \quad (M^2 - \sqrt{2} M m + m^2)(M^2 + \sqrt{2} M m + m^2) = 0.$$

$$\therefore \quad M^2 - \sqrt{2} M m + m^2 = 0, \quad \text{or} \quad M^2 + \sqrt{2} M m + m^2 = 0.$$

Solving these, we get

$$M = \frac{m}{\sqrt{2}} \pm i \frac{m}{\sqrt{2}} \quad \text{and} \quad M = -\frac{m}{\sqrt{2}} \pm i \frac{m}{\sqrt{2}}.$$

$$\therefore \quad x = c_1 e^{(m/\sqrt{2})t} \cos\left(\frac{m}{\sqrt{2}} t + c_2\right) + c_3 e^{-(m/\sqrt{2})t} \cos\left(\frac{m}{\sqrt{2}} t + c_4\right) \quad \dots(3)$$

$$\therefore \quad \frac{dx}{dt} = c_1 e^{(m/\sqrt{2})t} \cdot \frac{m}{\sqrt{2}} \left\{ \cos\left(\frac{m}{\sqrt{2}} t + c_2\right) - \sin\left(\frac{m}{\sqrt{2}} t + c_2\right) \right\} \\ - c_3 e^{-(m/\sqrt{2})t} \cdot \frac{m}{\sqrt{2}} \left\{ \cos\left(\frac{m}{\sqrt{2}} t + c_4\right) + \sin\left(\frac{m}{\sqrt{2}} t + c_4\right) \right\}$$

$$\text{and} \quad \frac{d^2 x}{dt^2} = c_1 e^{(m/\sqrt{2})t} \cdot \frac{m^2}{2} \left\{ \cos\left(\frac{m}{\sqrt{2}} t + c_2\right) - 2 \sin\left(\frac{m}{\sqrt{2}} t + c_2\right) \right. \\ \left. - \cos\left(\frac{m}{\sqrt{2}} t + c_2\right) \right\} + c_3 e^{-(m/\sqrt{2})t} \cdot \frac{m^2}{2} \left\{ \cos\left(\frac{m}{\sqrt{2}} t + c_4\right) \right. \\ \left. + 2 \sin\left(\frac{m}{\sqrt{2}} t + c_4\right) - \cos\left(\frac{m}{\sqrt{2}} t + c_4\right) \right\} \\ = m^2 \left\{ c_3 e^{-(m/\sqrt{2})t} \sin\left(\frac{m}{\sqrt{2}} t + c_4\right) - c_1 e^{(m/\sqrt{2})t} \sin\left(\frac{m}{\sqrt{2}} t + c_2\right) \right\}.$$

$$\text{From the equation (1), we get } y = -\frac{1}{m^2} \frac{d^2 x}{dt^2}.$$

Putting the value of $d^2 x/dt^2$, we have

$$y = c_1 e^{(m/\sqrt{2})t} \sin\left(\frac{m}{\sqrt{2}} t + c_2\right) - c_3 e^{-(m/\sqrt{2})t} \sin\left(\frac{m}{\sqrt{2}} t + c_4\right). \quad \dots(4)$$

The equations (3) and (4) constitute the required general solution of the given equations.

Example 6: Solve $t dx = (t - 2x) dt$, $t dy = (tx + ty + 2x - t) dt$.

(Rohilkhand 2001)

Solution: The given equations are

$$t dx = (t - 2x) dt \quad \dots(1)$$

and $t dy = (tx + ty + 2x - t) dt \quad \dots(2)$

From (1), we have $\frac{dx}{dt} + \frac{2}{t}x = 1$, which is linear.

$$\text{I.F.} = e^{\int (2/t) dt} = e^{2 \log t} = t^2.$$

$$\therefore xt^2 = \int t^2 \cdot 1 dt + c_1 = \frac{1}{3} t^3 + c_1$$

or $x = \frac{1}{3} t + c_1 t^{-2} \quad \dots(3)$

Now adding (1) and (2), we get

$$t(dx + dy) = t(x + y) dt \quad \text{or} \quad \frac{dx + dy}{x + y} = dt.$$

Integrating, $\log(x + y) = t + \log c_2$.

$$\therefore x + y = c_2 e^t \quad \text{or} \quad y = c_2 e^t - x \quad \dots(4)$$

Putting in (4) the value of x found in (3), we get

$$y = c_2 e^t - \frac{1}{3} t - c_1 t^{-2} \quad \dots(5)$$

The equations (3) and (5) give the required general solution of the given equations.

Example 7: Solve the simultaneous equations

$$t^2 \frac{d^2 x}{dt^2} + t \frac{dx}{dt} + 2y = 0, \quad t^2 \frac{d^2 y}{dt^2} + t \frac{dy}{dt} - 2x = 0.$$

Solution: Here both the given equations are linear homogeneous equations. First we shall change them into linear differential equations with constant coefficients.

For this, we put $t = e^z$.

Then denoting d/dz by D , we have

$$t \frac{dx}{dt} = Dx, \quad t^2 \frac{d^2 x}{dt^2} = D(D-1)x, \quad t \frac{dy}{dt} = Dy, \quad t^2 \frac{d^2 y}{dt^2} = D(D-1)y.$$

$\therefore$ the given equations transform to

$$\{D(D-1) + D\}x + 2y = 0$$

i.e., $D^2x + 2y = 0 \quad \dots(1)$

and $\{D(D-1) + D\}y - 2x = 0$

i.e., $D^2y - 2x = 0 \quad \dots(2)$

Eliminating y from (1) and (2), we get

$$(D^4 + 4)x = 0.$$

A.E. is $m^4 + 4 = 0$, or $(m^2 + 2)^2 - 4m^2 = 0$

or $(m^2 - 2m + 2)(m^2 + 2m + 2) = 0.$

$$\therefore m^2 - 2m + 2 = 0 \quad \text{or} \quad m^2 + 2m + 2 = 0.$$

$$\therefore m = \frac{2 \pm \sqrt{4-8}}{2} \quad \text{and} \quad m = \frac{-2 \pm \sqrt{4-8}}{2}$$

$$\text{i.e.,} \quad m = 1 \pm i, -1 \pm i.$$

$$\therefore x = e^z (c_1 \cos z + c_2 \sin z) + e^{-z} (c_3 \cos z + c_4 \sin z) \quad \dots(3)$$

$$\therefore \frac{dx}{dz} = e^z (c_1 \cos z + c_2 \sin z) + e^z (-c_1 \sin z + c_2 \cos z) - e^{-z} (c_3 \cos z + c_4 \sin z) + e^{-z} (-c_3 \sin z + c_4 \cos z)$$

$$\begin{aligned} \text{and} \quad \frac{d^2x}{dz^2} &= e^z (c_1 \cos z + c_2 \sin z) + 2e^z (-c_1 \sin z + c_2 \cos z) \\ &\quad - e^z (c_1 \cos z + c_2 \sin z) + e^{-z} (c_3 \cos z + c_4 \sin z) \\ &\quad - 2e^{-z} (-c_3 \sin z + c_4 \cos z) - e^{-z} (c_3 \cos z + c_4 \sin z) \\ &= 2e^z (-c_1 \sin z + c_2 \cos z) - 2e^{-z} (-c_3 \sin z + c_4 \cos z). \end{aligned}$$

From (1), we have

$$y = -\frac{1}{2} D^2 x = -\frac{1}{2} \frac{d^2x}{dz^2}.$$

$$\therefore y = e^z (c_1 \sin z - c_2 \cos z) + e^{-z} (-c_3 \sin z + c_4 \cos z). \quad \dots(4)$$

Now changing the variable z into t in the equations (3) and (4) by the relation $t = e^z$ or $z = \log t$, we get

$$x = t (c_1 \cos \log t + c_2 \sin \log t) + t^{-1} (c_3 \cos \log t + c_4 \sin \log t)$$

$$\text{and} \quad y = t (c_1 \sin \log t - c_2 \cos \log t) + t^{-1} (-c_3 \sin \log t + c_4 \cos \log t).$$

These equations give the required general solution of the given equations.

Example 8: Solve $\frac{dx}{dt} = ny - mz$, $\frac{dy}{dt} = lz - nx$, $\frac{dz}{dt} = mx - ly$. (Kanpur 2012)

Solution: Multiplying the given equations by x, y, z respectively and then adding, we get

$$x \frac{dx}{dt} + y \frac{dy}{dt} + z \frac{dz}{dt} = 0.$$

Integrating, we get

$$\frac{1}{2} x^2 + \frac{1}{2} y^2 + \frac{1}{2} z^2 = k$$

$$\text{or} \quad x^2 + y^2 + z^2 = c_1. \quad \dots(1)$$

Now multiplying the given equations by l, m, n respectively and then adding, we get

$$l \frac{dx}{dt} + m \frac{dy}{dt} + n \frac{dz}{dt} = 0.$$

$$\text{Integrating, we get } lx + my + nz = c_2. \quad \dots(2)$$

The equations (1) and (2) constitute the complete solution of the given differential equations.

Example 9: Solve the simultaneous equations

$$3 \frac{dx}{dt} + 2 \frac{dy}{dt} - 4x + 3y = 8e^{-3t}, 4 \frac{dx}{dt} + \frac{dy}{dt} + 3x + 4y = 8e^{-3t}$$

given that $x = \frac{1}{5}, y = 0$, when $t = 0$.

Solution: Writing D for (d/dt) , the given equations can be written as

$$(3D - 4)x + (2D + 3)y = 8e^{-3t} \quad \dots(1)$$

$$\text{and} \quad (4D + 3)x + (D + 4)y = 8e^{-3t} \quad \dots(2)$$

Eliminating y from (1) and (2), we get

$$[(D + 4)(3D - 4) - (2D + 3)(4D + 3)]x = 8[(D + 4)e^{-3t} - (2D + 3)e^{-3t}]$$

$$\text{or} \quad -5(D^2 + 2D + 5)x = 8(-3e^{-3t} + 4e^{-3t} + 6e^{-3t} - 3e^{-3t})$$

$$\text{or} \quad (D^2 + 2D + 5)x = -\frac{32}{5}e^{-3t}.$$

A.E. is $m^2 + 2m + 5 = 0$, giving $m = -1 \pm 2i$.

$$\therefore \text{C.F.} = e^{-1} (c_1 \cos 2t + c_2 \sin 2t).$$

$$\text{Also P.I.} = \frac{1}{D^2 + 2D + 5} \left(-\frac{32}{5} e^{-3t} \right) = -\frac{4}{5} e^{-3t}.$$

$$\therefore x = e^{-t} (c_1 \cos 2t + c_2 \sin 2t) - \frac{4}{5} e^{-3t}.$$

Given that $x = 1/5$, when $t = 0$, we have

$$\frac{1}{5} = c_1 - \frac{4}{5}, \quad \text{or} \quad c_1 = 1.$$

$$\therefore x = e^{-t} (\cos 2t + c_2 \sin 2t) - \frac{4}{5} e^{-3t}$$

$$\Rightarrow \frac{dx}{dt} = -e^{-t} (\cos 2t + c_2 \sin 2t) + e^{-t} (-2 \sin 2t + 2c_2 \cos 2t) + \frac{12}{5} e^{-3t}.$$

Now multiplying (2) by 2 and then subtracting (1) from it, we get

$$5y = -5 \frac{dx}{dt} - 10x + 8e^{-3t}.$$

Putting the values of x and dx/dt , we get

$$y = e^{-t} (-\cos 2t - 2c_2 \cos 2t + 2 \sin 2t - c_2 \sin 2t) + \frac{4}{5} e^{-3t}.$$

Given that $y = 0$, when $t = 0$, we have

$$0 = -1 - 2c_2 + \frac{4}{5}, \quad \text{or} \quad c_2 = -\frac{1}{10}.$$

$$\therefore x = \frac{1}{10} e^{-t} (10 \cos 2t - \sin 2t) - \frac{4}{5} e^{-3t} \quad \dots(3)$$

$$\text{and} \quad y = \frac{1}{10} e^{-t} (21 \sin 2t - 8 \sin 2t) + \frac{4}{5} e^{-3t}. \quad \dots(4)$$

The required solution consists of equations (3) and (4).

Comprehensive Exercise 1

Solve the following simultaneous differential equations :

1. $\frac{dx}{dt} + 5x + y = e^t, \frac{dy}{dt} - x + 3y = e^{2t}.$ (Meerut 2004B)
2. $\frac{dx}{dt} + 2\frac{dy}{dt} - 2x + 2y = 3e^t, 3\frac{dx}{dt} + \frac{dy}{dt} + 2x + y = 4e^{2t}.$ (Meerut 2007)
3. $\frac{dx}{dt} + 2x - 3y = t, \frac{dy}{dt} - 3x + 2y = e^{2t}.$ (Lucknow 2011)
4. $\left(\frac{d}{dt} + 2\right)x + 3y = 0, 3x + \left(\frac{d}{dt} + 2\right)y = 2e^{3t}.$
5. $\frac{dx}{dt} = 3x + 2y, \frac{dy}{dt} + 5x + 3y = 0.$ (Gorakhpur 2009)
6. $\frac{dx}{dt} = ax + by, \frac{dy}{dt} = a'x + b'y.$
7. $\frac{dx}{dt} + \omega y = 0, \frac{dy}{dt} - \omega x = 0.$ (Kanpur 2003, 05)
8. $\frac{dx}{dt} + 4x + 3y = t, \frac{dy}{dt} + 2x + 5y = e^t.$
9. $\frac{dx}{dt} + \frac{dy}{dt} - 2y = 2\cos t - 7\sin t, \frac{dx}{dt} - \frac{dy}{dt} + 2x = 4\cos t - 3\sin t.$ (Lucknow 2005, 09; Avadh 10)
10. $(D - 17)y + (2D - 8)z = 0, (13D - 53)y - 2z = 0.$
11. $4\frac{dx}{dt} + 9\frac{dy}{dt} + 11x + 31y = e^t, 3\frac{dx}{dt} + 7\frac{dy}{dt} + 8x + 24y = e^{2t}.$
12. $\frac{dx}{dt} + 2\frac{dy}{dt} + x + 7y = e^t - 3, \frac{dy}{dt} - 2x + 3y = 12 - 3e^t.$
13. $\frac{d^2x}{dt^2} + 4x + y = te^t, \frac{d^2y}{dt^2} + y - 2x = \sin^2 t.$
14. $\frac{d^2x}{dt^2} - \frac{dy}{dt} = 2x + 2t, \frac{dx}{dt} + 4\frac{dy}{dt} = 3y.$
15. $\frac{dx}{dt} + \frac{2}{t}(x - y) = 1, \frac{dy}{dt} + \frac{1}{t}(x + 5y) = t.$

Answers 1

1. $x = (c_1 + c_2t)e^{-4t} + \frac{4}{25}e^t - \frac{1}{36}e^{2t}, y = -(c_1 + c_2 + c_2t)e^{-4t} + \frac{7}{36}e^{2t} + \frac{1}{25}e^t$
2. $x = \frac{1}{2}e^{2t} - \frac{3}{11}e^t + c_1e^{-6t/5}, y = \frac{15}{22}e^t - 8c_1e^{-6t/5} + c_2e^{-t}$
3. $x = c_1e^{-5t} + c_2e^t + \frac{3}{7}e^{2t} - \frac{2}{5}t - \frac{13}{25}, y = -c_1e^{-5t} + c_2e^t + \frac{4}{7}e^{2t} - \frac{3}{5}t - \frac{12}{25}$

$$4. \quad x = c_1 e^t + c_2 e^{-5t} - \frac{3}{8} e^{3t}, y = -c_1 e^t + c_2 e^{-5t} + \frac{5}{8} e^{3t}$$

$$5. \quad x = c_1 \cos t + c_2 \sin t, y = \frac{1}{2} (c_2 - 3c_1) \cos t - \frac{1}{2} (c_1 + 3c_2) \sin t$$

$$6. \quad x = c_1 e^{m_1 t} + c_2 e^{m_2 t}, y = \frac{1}{b} \left\{ (m_1 - a) c_1 e^{m_1 t} + (m_2 - a) c_2 e^{m_2 t} \right\}, \text{ where}$$

$$m_1 = \frac{(a + b') + \sqrt{[(a - b')^2 + 4a'b]}}{2} \text{ and } m_2 = \frac{(a + b') - \sqrt{[(a - b')^2 + 4a'b]}}{2}$$

$$7. \quad x = c_1 \cos \omega t + c_2 \sin \omega t, y = c_1 \sin \omega t - c_2 \cos \omega t$$

$$8. \quad x = c_1 e^{-2t} + c_2 e^{-7t} + \frac{5}{14} t - \frac{1}{8} e^t - \frac{31}{196},$$

$$y = \frac{1}{3} \left\{ -\frac{3}{7} t - 2c_1 e^{-2t} + 3c_2 e^{-7t} + \frac{5}{8} e^t + \frac{27}{98} \right\}$$

$$9. \quad x = c_1 e^{\sqrt{2}t} + c_2 e^{-\sqrt{2}t} + 3 \cos t,$$

$$y = (\sqrt{2} + 1) c_1 e^{\sqrt{2}t} + (1 - \sqrt{2}) c_2 e^{-\sqrt{2}t} + 2 \sin t$$

$$10. \quad y = c_1 e^{3x} + c_2 e^{5x}, z = -7c_1 e^{3x} + 6c_2 e^{5x}$$

$$11. \quad y = (c_1 + c_2 t) e^{-4t} + \frac{31}{25} e^t - \frac{49}{36} e^{2t}, y = -(c_2 + c_1 + c_2 t) e^{-4t} + \frac{19}{36} e^{2t} - \frac{11}{25} e^t.$$

$$12. \quad x = e^{-4t} (c_1 \cos t + c_2 \sin t) + \frac{31}{26} e^t - \frac{93}{17}.$$

$$y = -(c_1 + c_2) e^{-4t} \cos t + (c_1 - c_2) e^{-4t} \sin t - \frac{2}{13} e^t + \frac{6}{17}.$$

$$13. \quad x = c_1 \cos(\sqrt{3}t + c_2) + c_3 \cos(\sqrt{2}t + c_4) + \frac{1}{6} e^t \left(t - \frac{1}{6} \right) - \frac{1}{12} + \frac{1}{4} \cos 2t,$$

$$y = -c_1 \cos(\sqrt{3}t + c_2) - 2c_3 \cos(\sqrt{2}t + c_4) + \frac{1}{6} e^t \left(t - \frac{7}{6} \right) + \frac{1}{3}.$$

$$14. \quad x = (c_1 + c_2 t) e^t + c_3 e^{-3/2t} - t, y = (3c_2 - c_1 - c_2 t) e^t - \frac{1}{6} c_3 e^{-(3/2)t} - \frac{1}{3}.$$

$$15. \quad x = \frac{c_1}{t^3} + \frac{c_2}{t^4} + \frac{t^2}{15} + \frac{3t}{10}, y = -\frac{c_1}{2t^3} - \frac{c_2}{t^4} + \frac{2t^2}{15} - \frac{1}{20} t.$$

6.4 Simultaneous Equations of the Form

$$P_1 dx + Q_1 dy + R_1 dz = 0 \quad \dots(1)$$

$$P_2 dx + Q_2 dy + R_2 dz = 0, \quad \dots(2)$$

where the coefficients P_1, P_2, Q_1, Q_2, R_1 and R_2 are functions of x, y, z .

Above is the general form of a system of simultaneous equations of the first order in three variables.

Taking z as the independent variable, the equations (1) and (2) can be written in the form

$$P_1 \frac{dx}{dz} + Q_1 \frac{dy}{dz} + R_1 = 0$$

$$P_2 \frac{dx}{dz} + Q_2 \frac{dy}{dz} + R_2 = 0.$$

Solving these equations for dx/dz and dy/dz by the method of cross-multiplication, we get

$$\frac{dx/dz}{Q_1 R_2 - Q_2 R_1} = \frac{dy/dz}{R_1 P_2 - R_2 P_1} = \frac{1}{P_1 Q_2 - P_2 Q_1}$$

or

$$\frac{dx}{Q_1 R_2 - Q_2 P_1} = \frac{dy}{R_1 P_2 - R_2 P_1} = \frac{dz}{P_1 Q_2 - P_2 Q_1},$$

which is of the form

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}, \quad \dots(3)$$

where P, Q, R are functions of x, y and z .

Thus the equations (1) and (2) can always be expressed in the form (3). To solve the simultaneous equations in the form (3) we can use the following methods.

First Method: Suppose one variable is absent from two members of (3), say, z is absent from P and Q . Then the solution of

$$\frac{dx}{P} = \frac{dy}{Q}$$

gives a relation between x and y and we thus get one equation constituting the complete solution of the given equations. If necessary, we can use this relation to eliminate x or y from one of the remaining equations in (3) and we can find the other integral relation which gives us the second equation of the complete solution.

Second Method: By a property of ratio and proportion in algebra, we know that each of the equal fractions $\frac{dx}{P}, \frac{dy}{Q}, \frac{dz}{R}$ in (3), is also equal to

$$\frac{l dx + m dy + n dz}{lP + mQ + nR}, \frac{L dx + M dy + N dz}{LP + MQ + NR}, \text{ etc.}$$

$$\therefore \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l dx + m dy + n dz}{lP + mQ + nR} = \frac{L dx + M dy + N dz}{LP + MQ + NR}. \quad \dots(4)$$

The relations between x, y, z that satisfy (3) also satisfy (4).

By a proper choice of multipliers $l, m, n; L, M, N$ etc., it is possible to find equations which can be easily solved. Particularly l, m, n can be so chosen that

$$lP + mQ + nR = 0.$$

which makes $l dx + m dy + n dz = 0$(5)

Let $l dx + m dy + n dz$ be an exact differential, say du .

Then $(5) \Rightarrow du = 0 \Rightarrow u = a$, which gives one equation constituting the complete solution.

Again L, M, N may be found such that

$$LP + MQ + NR = 0 \Rightarrow L dx + M dy + N dz = 0.$$

If $L dx + M dy + N dz$ be an exact differential, say dv , then $dv = 0 \Rightarrow v = b$, which gives the second equation constituting the complete solution.

Hence the complete solution consists of the equations $u = a$ and $v = b$ which must of course be independent of each other.

Note 1: Sometimes only one set of multipliers may serve the purpose.

Note 2: We can find one relation, say $u = a$, by the first method and the second relation, say $v = b$, by the second method.

6.5 Geometrical Interpretation of the Differential

Equations $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}.$

(Gorakhpur 2005)

From solid geometry, we know that the direction cosines of the tangent to a curve in three dimensional Euclidean space at any point (x, y, z) are

$$\frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds}$$

i.e., the direction cosines are in the ratio of $dx : dy : dz$. Hence the given differential equations represent a system of curves in space, the direction cosines of the tangent to any member of this system at any point (x, y, z) are proportional to P, Q and R . Let $u = a$ and $v = b$ be two component solutions of the given equations. Then the curves are obtained by the intersection of the surfaces $u = a$ and $v = b$. Since the arbitrary constants a and b can take infinite values, the curves are doubly infinite in number.

Illustrative Examples

Example 10: Solve $\frac{x dx}{z^2 - 2yz - y^2} = \frac{dy}{y + z} = \frac{dz}{y - z}.$

(Meerut 2001, 04; Avadh 08)

Solution: Taking the last two members of the given equations, we get

$$(y - z) dy = (y + z) dz \quad \text{or} \quad y dy - z dz - (y dz + z dy) = 0$$

$$\text{or} \quad 2y dy - 2z dz - 2(y dz + z dy) = 0.$$

Integrating, we get

$$y^2 - z^2 - 2yz = c_1. \quad \dots(1)$$

Again choosing 1, y, z as multipliers, we get

$$\begin{aligned}\frac{x \, dx}{z^2 - 2yz - y^2} &= \frac{dy}{y+z} = \frac{dz}{y-z} \\ &= \frac{x \, dx + y \, dy + z \, dz}{1 \cdot (z^2 - 2yz - y^2) + y(y+z) + z(y-z)} = \frac{x \, dx + y \, dy + z \, dz}{0}.\end{aligned}$$

$$\therefore x \, dx + y \, dy + z \, dz = 0.$$

$$\text{Integrating, we get } x^2 + y^2 + z^2 = c_2. \quad \dots(2)$$

The required solution consists of the equations (1) and (2).

Example 11: Solve $\frac{dx}{y^2 + z^2 - x^2} = \frac{dy}{-2xy} = \frac{dz}{-2xz}$.

Solution: Taking the last two members of the given equations, we get

$$\frac{dy}{y} = \frac{dz}{z}.$$

Integrating, we get

$$\log y = \log z + \log c_1$$

$$\text{or } \log y = \log (c_1 z) \text{ or } y = c_1 z. \quad \dots(1)$$

Again choosing x, y, z as multipliers, we get

$$\frac{dx}{y^2 + z^2 - x^2} = \frac{dy}{-2xy} = \frac{dz}{-2xz} = \frac{x \, dx + y \, dy + z \, dz}{-x(x^2 + y^2 + z^2)}.$$

Taking the last two fractions, we get

$$\frac{dz}{z} = \frac{2x \, dx + 2y \, dy + 2z \, dz}{x^2 + y^2 + z^2}.$$

$$\text{Integrating, } \log z + \log c_2 = \log (x^2 + y^2 + z^2)$$

$$\text{or } x^2 + y^2 + z^2 = zc_2. \quad \dots(2)$$

The required solution consists of the equations (1) and (2).

Example 12: Solve $\frac{dx}{1} = \frac{dy}{-2} = \frac{dz}{3x^2 \sin(y+2x)}$.

(Meerut 2001, 10B; Gorakhpur 07)

Solution: From the first two members, we get

$$2x + y = c_1. \quad \dots(1)$$

Taking the first and the last members, we get

$$\frac{dx}{1} = \frac{dz}{3x^2 \sin(y+2x)} = \frac{dz}{dx^2 \sin c_1}, \text{ using (1).}$$

$$\therefore dz = 3x^2 \sin c_1 \, dx.$$

$$\text{Integrating, } z = x^3 \sin c_1 + c_2$$

$$\text{or } z - x^3 \sin(y+2x) = c_2. \quad \dots(2)$$

The required solution consists of the equations (1) and (2).

Example 13: Solve $\frac{dx}{x(y^2 - z^2)} = \frac{dy}{-y(z^2 + x^2)} = \frac{dz}{z(x^2 + y^2)}$.

(Meerut 2005, 11)

Solution: Choosing $\frac{1}{x}, -\frac{1}{y}, -\frac{1}{z}$ as multipliers, we have each fraction

$$= \frac{\frac{dx}{x} - \frac{dy}{y} - \frac{dz}{z}}{0}.$$

$$\therefore \frac{dx}{x} - \frac{dy}{y} - \frac{dz}{z} = 0 \quad \Rightarrow \quad \frac{dx}{x} = \frac{dy}{y} + \frac{dz}{z}.$$

Integrating, $\log x + \log c_1 = \log y + \log z$

$$\text{or} \quad yz = c_1 x. \quad \dots(1)$$

Again choosing x, y, z as multipliers, we get

$$\text{each fraction} = \frac{x dx + y dy + z dz}{0}.$$

$$\therefore x dx + y dy + z dz = 0.$$

$$\text{Integrating, } x^2 + y^2 + z^2 = c_2. \quad \dots(2)$$

The equations (1) and (2) give the required solution of the given equations.

Example 14: Solve $\frac{dx}{\cos(x+y)} = \frac{dy}{\sin(x+y)} = \frac{dz}{z}$.

(Meerut 2006, 07B; Gorakhpur 08, 11)

Solution: From the given equations, we have

$$\frac{dx + dy}{\cos(x+y) + \sin(x+y)} = \frac{dx - dy}{\cos(x+y) - \sin(x+y)} = \frac{dz}{z}.$$

Taking the first two members, we get

$$\frac{\cos(x+y) - \sin(x+y)}{\cos(x+y) + \sin(x+y)} (dx + dy) = dx - dy.$$

Integrating, $\log [\cos(x+y) + \sin(x+y)] = x - y + \log c_1$

$$\text{or} \quad [\cos(x+y) + \sin(x+y)] e^{y-x} = c_1. \quad \dots(1)$$

Again taking the first and the third members, we get

$$\frac{dx + dy}{\cos(x+y) + \sin(x+y)} = \frac{dz}{z}$$

$$\text{or} \quad \frac{dz}{z} = \frac{(1/\sqrt{2})(dx + dy)}{\sin(x+y + \frac{1}{4}\pi)}$$

$$\text{or} \quad \sqrt{2} \frac{dz}{z} = \operatorname{cosec}(x+y + \frac{1}{4}\pi) (dx + dy).$$

$$\text{Integrating, } \sqrt{2} \log z = \log \tan \frac{1}{2}(x+y + \frac{1}{4}\pi) + \log c_2$$

$$\text{or} \quad z^{\sqrt{2}} \cot\left(\frac{x}{2} + \frac{y}{2} + \frac{\pi}{8}\right) = c_2. \quad \dots(2)$$

The required solution consists of the equations (1) and (2).

Example 15: Solve $\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z - a\sqrt{(x^2 + y^2 + z^2)}}$. (Meerut 2005B, 10B)

Solution: From the first two members of the given differential equations, we have

$$x = c_1 y. \quad \dots(1)$$

Again choosing x, y, z as multipliers, we have

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z - a\sqrt{(x^2 + y^2 + z^2)}} = \frac{x dx + y dy + z dz}{x^2 + y^2 + z^2 - az\sqrt{(x^2 + y^2 + z^2)}}$$

$$\text{or} \quad \frac{dy}{y} = \frac{dz}{z - au} = \frac{u du}{u^2 - azu}, \text{ putting } x^2 + y^2 + z^2 = u^2$$

$$\text{or} \quad \frac{dy}{y} = \frac{dz}{z - au} = \frac{du}{u - az} = \frac{du + dz}{(u + z)(1 - a)}.$$

Taking the first and the last members and integrating, we get

$$(1 - a) \log y = \log(u + z) + \log c_2$$

$$\text{or} \quad y^{(1-a)} = (u + z) c_2$$

$$\text{or} \quad y^{(1-a)} = c_2 \{\sqrt{(x^2 + y^2 + z^2)} + z\}. \quad \dots(2)$$

The required solution consists of the equations (1) and (2).

Example 16. Solve $\frac{dx}{y^3 x - 2x^4} = \frac{dy}{2y^4 - x^3 y} = \frac{dz}{9z(x^3 - y^3)}$.

Solution: From the first two members of the given differential equations, we get

$$(2y^4 - x^3 y) dx = (y^3 x - 2x^4) dy$$

$$\text{or} \quad \left(\frac{2y}{x^3} - \frac{1}{y^2}\right) dx = \left(\frac{1}{x^2} - \frac{2x}{y^3}\right) dy, \text{ dividing by } x^3 y^3$$

$$\text{or} \quad \left(\frac{1}{x^2} dy - \frac{2y}{x^3} dx\right) + \left(\frac{1}{y^2} dx - \frac{2x}{y^3} dy\right) = 0.$$

$$\text{Integrating,} \quad \frac{y}{x^2} + \frac{x}{y^2} = c_1. \quad \dots(1)$$

Again taking $\frac{1}{x}, \frac{1}{y}, \frac{1}{3z}$ as multipliers, we get

$$\text{each fraction} = \frac{\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{3z} dz}{(y^3 - 2x^3) + (2y^3 - x^3) + 3(x^3 - y^3)} = \frac{\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{3z} dz}{0}.$$

$$\therefore \quad \frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{3z} dz = 0.$$

Integrating, $\log x + \log y + \frac{1}{3} \log z = \log c_2$

or $xyz^{1/3} = c_2. \quad \dots(2)$

The required solution consists of the equations (1) and (2).

Example 17: Solve $\frac{dx}{xy} = \frac{dy}{y^2} = \frac{dz}{zy - 2x^2}.$

(Purvanchal 2006)

Solution: From the first two members of the given differential equations, we get

$$\frac{dx}{x} = \frac{dy}{y}.$$

Integrating, $\log x = \log y + \log c_1$

or $x = c_1 y. \quad \dots(1)$

Again taking the last two members, we get

$$\frac{dy}{y^2} = \frac{dz}{xyz - 2x^2} \quad \text{or} \quad \frac{dy}{y^2} = \frac{dz}{c_1 y^2 z - 2c_1^2 y^2} \quad [\because x = c_1 y]$$

or $dy = \frac{dz}{c_1 z - 2c_1^2} \quad \text{or} \quad c_1 dy = \frac{dz}{z - 2c_1}.$

Integrating, $c_1 y = \log(z - 2c_1) + c_2$

or $x = \log\left(z - 2\frac{x}{y}\right) + c_2 \quad [\because x = c_1 y]$

or $x = \log(yz - 2x) - \log y + c_2. \quad \dots(2)$

The required solution consists of the equations (1) and (2).

Example 18: Solve $\frac{dx}{x^2 + y^2} = \frac{dy}{2xy} = \frac{dz}{(x + y)z}.$

Solution: From the given equations, we have

$$\frac{dx + dy}{x^2 + y^2 + 2xy} = \frac{dx - dy}{x^2 + y^2 - 2xy} = \frac{dz}{(x + y)z}.$$

Taking the first two members, we have

$$\frac{dx + dy}{(x + y)^2} = \frac{dx - dy}{(x - y)^2}.$$

Integrating, $-(x + y)^{-1} = -(x - y)^{-1} + c_1$

or $\frac{1}{x - y} - \frac{1}{x + y} = c_1, \quad \text{or} \quad \frac{2y}{x^2 - y^2} = c_1$

or $2y = c_1 (x^2 - y^2) \quad \dots(1)$

Taking the first and the last members, we have

$$\frac{dx + dy}{(x + y)^2} = \frac{dz}{(x + y)z} \quad \text{or} \quad \frac{dx + dy}{x + y} = \frac{dz}{z}.$$

Integrating, $\log (x+y) = \log z + \log c_2$

or $(x+y) = c_2 z. \quad \dots(2)$

The required solution consists of the equations (1) and (2).

Example 19: Solve $\frac{dx}{x^2} = \frac{dy}{y^2} = \frac{dz}{nxy}$.

Solution: We have $\frac{dx}{x^2} = \frac{dy}{y^2}$.

Integrating, $\frac{1}{x} = \frac{1}{y} + c_1$

or $y - x = c_1 xy. \quad \dots(1)$

Also using $1/x, -1/y$ and c_1/n as multipliers, each fraction

$$\begin{aligned} &= \frac{(1/x) dx - (1/y) dy + (c_1/n) dz}{x - y + c_1 xy} \\ &= \frac{(1/x) dx - (1/y) dy + (c_1/n) dz}{0}, \text{ using (1).} \end{aligned}$$

$\therefore (1/x) dx - (1/y) dy + (c_1/n) dz = 0.$

Integrating, $\log x - \log y + (c_1/n) z = c_2$

or $\frac{c_1}{n} z = \log \left(\frac{y}{x} \right) + c_2 \quad \text{or} \quad z = \frac{nxy}{y-x} \log \left(\frac{y}{x} \right) + c_2. \quad \dots(2)$

The required solution consists of the equations (1) and (2).

Comprehensive Exercise 2

Solve the following simultaneous differential equations :

1. $\frac{dx}{yz} = \frac{dy}{zx} = \frac{dz}{xy}.$

2. $\frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx}.$

Also give geometrical interpretation of the solution of these equations.

3. $\frac{dx}{z} = \frac{dy}{-z} = \frac{dz}{z^2 + (x+y)^2}.$

4. $\frac{a dx}{(b-c) yz} = \frac{b dy}{(c-a) zx} = \frac{c dz}{(a-b) xy}.$

(Lucknow 2006)

5. $\frac{dx}{xz(z^2 + xy)} = \frac{dy}{-yz(z^2 + xy)} = \frac{dz}{x^4}.$

6. $\frac{dx}{z(x+y)} = \frac{dy}{z(x-y)} = \frac{dz}{x^2 + y^2}.$

$$7. \frac{-dx}{x(x+y)} = \frac{dy}{y(x+y)} = \frac{dz}{(x-y)(2x+2y+z)}.$$

$$8. \frac{dx}{1} = \frac{dy}{3} = \frac{dz}{5z + \tan(y-3x)}.$$

(Meerut 2006B; Gorakhpur 08, 11)

$$9. \frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)}.$$

$$10. \frac{dx}{x(y^2-z^2)} = \frac{dy}{y(z^2-x^2)} = \frac{dz}{z(x^2-y^2)}.$$

(Gorakhpur 2005, 09)

$$11. \frac{dx}{1+y} = \frac{dy}{1+x} = \frac{dz}{z}.$$

$$12. \frac{dx}{y^2 + yz + z^2} = \frac{dy}{z^2 + zx + x^2} = \frac{dz}{x^2 + xy + y^2}.$$

(Avadh 2007)

$$\left[\text{Hint. } \frac{dy - dx}{(x-y)(x+y+z)} = \frac{dz - dy}{(y-z)(x+y+z)} = \frac{dz - dx}{(x-z)(x+y+z)} \right]$$

$$13. \frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy}.$$

(Gorakhpur 2006)

$$\left[\text{Hint. } \frac{dx - dy}{(x-y)(x+y+z)} = \frac{dy - dz}{(y-z)(x+y+z)} = \frac{dz - dx}{(z-x)(x+y+z)} \right].$$

$$14. \frac{dx}{x^2 + y^2 + yz} = \frac{dy}{x^2 + y^2 - xz} = \frac{dz}{z(x+y)}.$$

(Kanpur 2009)

$$\left[\text{Hint. } \frac{dx - dy}{z(x+y)} = \frac{dz}{z(x+y)} = \frac{x dx + y dy}{(x+y)(x^2 + y^2)} \right]$$

$$15. \frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y}.$$

$$\left[\text{Hint. } \frac{dx - dy}{y-x} = \frac{dy - dz}{z-x} = \frac{dx + dy + dz}{2(x+y+z)} \right].$$

Answers 2

$$1. x^2 - y^2 = c_1, x^2 - z^2 = c_2$$

$$2. lx + my + nz = c_1, x^2 + y^2 + z^2 = c_2$$

These equations constituting the general solution of the given differential equations represent a family of circles.

$$3. x + y = c_1, \log \{z^2 + (x+y)^2\} - 2x = c_2$$

4. $a^2x^2 + b^2y^2 + c^2z^2 = c_1, ax^2 + by^2 + cz^2 = c_2$
5. $xy = c_1, (z^2 + xy)^2 - x^4 = c_2$
6. $x^2 - y^2 - z^2 = c_1, 2xy - z^2 = c_2$
7. $xy = c_1, x^2 + y^2 + (x + y)z = c_2$
8. $y - 3x = c_1, 5z + \tan(y - 3x) = c_2 e^{5x}$
9. $x + y + z = c_1, xyz = c_2$
10. $x^2 + y^2 + z^2 = c_1, xyz = c_2$
11. $z(x - y) = c_1, z = c_2(x + y + 2)$
12. $(y - x) = c_1(z - x), (y - x) = c_2(z - y)$
13. $(x - y) = c_1(y - z), (y - z) = c_2(z - x)$
14. $x - y - z = c_1, x^2 + y^2 = z^2 c_2$
15. $(y - x) = c_1(z - y), (x - y)^2(x + y + z) = c_2$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. Solving the equations $\frac{dx}{dt} + 5x + y = e^t, \frac{dy}{dt} - x + 3y = e^{2t}$, for x , we get $x =$
 - (a) $(c_1 + c_2 t)e^{-4t} + \frac{4}{25}e^t - \frac{1}{36}e^{2t}$
 - (b) $c_1 e^{-4t} + c_2 e^t - \frac{2}{5}$
 - (c) $c_1 e^t + c_2 e^{-4t} + c_3 e^{-2t}$
 - (d) None of these.
2. Solving the equations $\frac{dx}{dt} + \frac{dy}{dt} + 2x + y = 0, \frac{dy}{dt} + 5x + 3y = 0$, we get $x =$
 - (a) $c_1 e^t + c_2 e^{-t}$
 - (b) $c_1 \cos t + c_2 \sin t$
 - (c) $(c_1 + c_2 t) \cos t$
 - (d) $c_1 e^t + c_2 e^{-t}$.
3. Solving the equations $\frac{dx}{dt} + \omega y = 0, \frac{dy}{dt} - \omega x = 0$ for x , we get $x =$
 - (a) $c_1 e^{\omega t} + c_2 e^{-\omega t}$
 - (b) $(c_1 + c_2 t) e^{\omega t}$
 - (c) $c_1 \cos \omega t + c_2 \sin \omega t$
 - (d) None of these.

4. Solution set of the equations $\frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y}$ is
- (a) $(y-x) = c_1 (z-y)$ (b) $x+y = c_1 (y+z)$
 $(x-y)^2 (x+y+z) = c_2$ $(x-y)^2 (x+y+z) = c_2$.
- (c) $(y-z) = c_1 (x-y)$ (d) None of these.
 $(x+y)^2 (x-y+z) = c_2$
5. Solution set of the equations $\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$ is
- (a) $xy = c_1, yz = c_2$ (b) $x = c_1 z, y = c_2 z$
(c) $x = c_1 y, y = c_2 z$ (d) None of these.
6. Solution set of the equations $\frac{dx}{yz} = \frac{dy}{zx} = \frac{dz}{xy}$ is
- (a) $x^2 + y^2 = c_1, y^2 - z^2 = c_2$ (b) $x^2 - z^2 = c_1, y^2 + z^2 = c_2$
(c) $x^2 - y^2 = c_1, x^2 - z^2 = c_2$ (d) None of these.
7. Solution set of the simultaneous equations $\frac{dx}{y} = \frac{dy}{x} = \frac{dz}{z}$ is
- (a) $x^2 - y^2 = c_1, x + y = c_2 z$ (b) $x^2 + y^2 = c_1, x + y = c_2 z$
(c) $x^2 - y^2 = c_1, x - y = c_2 z$ (d) $x^2 - y^2 = c_1, y^2 - z^2 = c_2$.

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. Eliminating y between the equations

$$\frac{dx}{dt} - 7x + y = 0, \quad \frac{dy}{dt} - 2x - 5y = 0$$

we obtain the differential equation.....

2. Eliminating y between the simultaneous equations

$$\frac{d^2 x}{dt^2} + m^2 y = 0, \quad \frac{d^2 y}{dt^2} - m^2 x = 0$$

we obtain the differential equation

(Meerut 2003)

3. The complete solution of the simultaneous equations

$$\frac{dx}{dt} = ny - mz, \quad \frac{dy}{dt} = lz - nx, \quad \frac{dz}{dt} = mx - ly \text{ is}$$

Answers

Multiple Choice Questions

- | | |
|--------|--------|
| 1. (a) | 2. (b) |
| 3. (c) | 4. (a) |
| 5. (b) | 6. (c) |
| 7. (a) | |

Fill in the Blank(s)

- $(D^2 - 12D + 37)x = 0$
- $(D^4 + m^4)x = 0$
- $x^2 + y^2 + z^2 = c_1, lx + my + nz = c_2$



Chapter

7



Linear Equations of Second Order With Variable Coefficients

7.1 Linear Equation of Second Order

An equation of the form

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R,$$

where P, Q, R are functions of x only, is called the ‘linear equation of second order’.

There is no general method for solving such equations. Here we shall discuss certain methods by which the solutions of such equations can be found.

7.2 The Complete Solution in Terms of a Known Integral

The following theorem relating to the linear differential equation of the second order is of great importance :

If an integral included in the complementary function of such an equation be known, then the complete primitive (or the general solution) can be found in terms of the known integral.

Let $y = u$ be a known integral in the complementary function of

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R \quad \dots(1)$$

i.e., it is a solution of

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = 0.$$

$$\therefore \frac{d^2 u}{dx^2} + P \frac{du}{dx} + Qu = 0. \quad \dots(2)$$

Let $y = uv$ be the solution of (1).

Putting $y = uv$, we get $\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$

$$\text{and} \quad \frac{d^2 y}{dx^2} = v \frac{d^2 u}{dx^2} + 2 \frac{du}{dx} \frac{dv}{dx} + u \frac{d^2 v}{dx^2}.$$

Substituting these values the equation (1) becomes

$$\left(v \frac{d^2 u}{dx^2} + 2 \frac{du}{dx} \frac{dv}{dx} + u \frac{d^2 v}{dx^2} \right) + P \left(v \frac{du}{dx} + u \frac{dv}{dx} \right) + Qvu = R$$

$$\text{or} \quad u \frac{d^2 v}{dx^2} + \frac{dv}{dx} \left(2 \frac{du}{dx} + Pu \right) + v \left(\frac{d^2 u}{dx^2} + P \frac{du}{dx} + Qu \right) = R$$

$$\text{or} \quad u \frac{d^2 v}{dx^2} + \frac{dv}{dx} \left(2 \frac{du}{dx} + Pu \right) + v \cdot 0 = R, \text{ using (2)}$$

$$\text{or} \quad \frac{d^2 v}{dx^2} + \left(P + \frac{2}{u} \frac{du}{dx} \right) \frac{dv}{dx} = \frac{R}{u}. \quad \dots(3)$$

Putting $\frac{dv}{dx} = p$, $\frac{d^2 v}{dx^2} = \frac{dp}{dx}$, (3) becomes

$$\frac{dp}{dx} + \left(P + \frac{2}{u} \frac{du}{dx} \right) p = \frac{R}{u}, \quad \dots(4)$$

which is linear with p as dependent variable.

$$\begin{aligned} \text{I.F.} &= e^{\int \left(P + \frac{2}{u} \frac{du}{dx} \right) dx} = e^{\int \left(P dx + \frac{2}{u} du \right)} \\ &= e^2 \log u + \int P dx = u^2 e^{\int P dx} \end{aligned}$$

Hence solution of (4) is

$$pu^2 e^{\int P dx} = \int \left[\frac{R}{u} u^2 e^{\int P dx} \right] dx + c_1. \quad \dots(5)$$

$$\therefore p = \frac{dv}{dx} = \frac{c_1 e^{-\int P dx}}{u^2} + \frac{e^{-\int P dx}}{u^2} \int u R e^{\int P dx} dx.$$

Integrating this, we get

$$y = c_2 + c_1 \int \frac{e^{-\int P dx}}{u^2} dx + \int \left[\frac{e^{-\int P dx}}{u^2} \int u R e^{\int P dx} dx \right] dx.$$

Hence the solution of (1) is

$$y = uv = c_2 u + c_1 u \int \frac{e^{-\int P dx}}{u^2} dx + u \int \left[\frac{e^{-\int P dx}}{u^2} \cdot \int u R e^{\int P dx} dx \right] dx. \dots(6)$$

It contains the given solution $y = u$ and since it contains two arbitrary constants so it is the complete primitive or the general solution of (1). It is evident from (6) that the second part of the complementary function is

$$u \int \frac{e^{-\int P dx}}{u^2} dx \text{ and the particular integral is } u \int \left[\frac{e^{-\int P dx}}{u^2} \cdot \int u \operatorname{Re}^{\int P dx} dx \right] dx.$$

7.3 To Find one Integral In C.F. by Inspection

Consider
$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = 0. \quad \dots(1)$$

1. $y = e^{mx}$ is a solution of (1) if

$$m^2 + Pm + Q = 0.$$

We have $y = e^{mx} \Rightarrow \frac{dy}{dx} = me^{mx}, \frac{d^2 y}{dx^2} = m^2 e^{mx}.$

If $y = e^{mx}$ is a solution of (1), then

$$(m^2 + Pm + Q) e^{mx} = 0 \Rightarrow m^2 + Pm + Q = 0.$$

Deductions. (i) $y = e^x$ is a solution of (1) if $1 + P + Q = 0.$

(ii) $y = e^{-x}$ is a solution of (1) if $1 - P + Q = 0.$

2. $y = x^m$ is a solution of (1) if $m(m-1) + Pmx + Qx^2 = 0.$

We have $y = x^m \Rightarrow \frac{dy}{dx} = mx^{m-1}, \frac{d^2 y}{dx^2} = m(m-1)x^{m-2}.$

If $y = x^m$ is a solution of (1), then

$$m(m-1)x^{m-2} + Pmx^{m-1} + Qx^m = 0$$

or $m(m-1) + Pmx + Qx^2 = 0.$

Deductions. (i) $y = x$ is a solution of (1) if $P + Qx = 0.$

(ii) $y = x^2$ is a solution of (1) if $2 + 2Px + Qx^2 = 0.$

All the above results are summarised below :

(i) $y = e^x$ is a part of C.F. if $1 + P + Q = 0.$

(ii) $y = e^{-x}$ is a part of C.F. if $1 - P + Q = 0.$

(iii) $y = e^{mx}$ is a part of C.F. if $m^2 + mP + Q = 0.$

(iv) $y = x$ is a part of C.F. if $P + Qx = 0.$

(v) $y = x^2$ is a part of C.F. if $2 + 2Px + Qx^2 = 0.$

(vi) $y = x^m$ is a part of C.F. if $m(m-1) + Pmx + Qx^2 = 0.$

Illustrative Examples

Example 1: Solve $(x+2) \frac{d^2 y}{dx^2} - (2x+5) \frac{dy}{dx} + 2y = (x+1) e^x$.

(Gorakhpur 2009; Lucknow 07; Rohilkhand 09; Kanpur 07, 09)

Solution: The given equation can be written as

$$\frac{d^2 y}{dx^2} - \frac{(2x+5)}{x+2} \frac{dy}{dx} + \frac{2}{x+2} y = \frac{(x+1)}{x+2} e^x. \quad \dots(1)$$

Comparing (1) with the standard form

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R, \text{ we have}$$

$$P = -\frac{2x+5}{x+2}, Q = \frac{2}{x+2}, R = \frac{x+1}{x+2} e^x.$$

Here $2^2 + 2P + Q = 0$,

$\therefore y = e^{2x}$ is a part of the C.F. of the solution of the equation (1).

Putting $y = v e^{2x}$, $\frac{dy}{dx} = e^{2x} \frac{dv}{dx} + 2v e^{2x}$,

and $\frac{d^2 y}{dx^2} = e^{2x} \frac{d^2 v}{dx^2} + 4e^{2x} \frac{dv}{dx} + 4v e^{2x}$ in (1), we get

$$\frac{d^2 v}{dx^2} + \frac{2x+3}{x+2} \frac{dv}{dx} = \frac{x+1}{x+2} e^{-x}$$

or $\frac{dp}{dx} + \frac{2x+3}{x+2} p = \frac{x+1}{x+2} e^{-x}, \quad \left(\text{where } p = \frac{dv}{dx} \right)$

which is a linear equation with p as dependent variable.

$$\text{I.F.} = e^{\int \frac{2x+3}{x+2} dx} = e^{\int \left(2 - \frac{1}{x+2} \right) dx} = e^{2x - \log(x+2)} = \frac{e^{2x}}{x+2}$$

$$\begin{aligned} \therefore p \cdot \frac{e^{2x}}{x+2} &= \int \frac{x+1}{x+2} e^{-x} \cdot \frac{e^{2x}}{x+2} dx + c_1 \\ &= \int \frac{x+1}{(x+2)^2} e^x dx + c_1 = \int \left\{ \frac{1}{x+2} - \frac{1}{(x+2)^2} \right\} e^x dx + c_1 \\ &= \frac{1}{x+2} e^x + \int e^x \cdot \frac{1}{(x+2)^2} dx - \int e^x \frac{1}{(x+2)^2} dx + c_1, \end{aligned}$$

integrating the first integral by parts.

$$\therefore p = \frac{dv}{dx} = e^{-x} + c_1 e^{-2x} (x+2).$$

Integrating this, we get

$$v = -e^{-x} - \frac{1}{2} c_1 e^{-2x} (x+2) - \frac{1}{4} c_1 e^{-2x} + c_2$$

$$= -e^{-x} - \frac{1}{4} c_1 (2x + 5) e^{-2x} + c_2.$$

∴ The general solution of (1) is

$$y = v e^{2x} = -e^x - \frac{1}{4} c_1 (2x + 5) + c_2 e^{2x}.$$

Example 2: Solve $x \frac{d^2 y}{dx^2} - (2x - 1) \frac{dy}{dx} + (x - 1) y = 0$.

(Meerut 2010; Gorakhpur 08)

Solution: The given equation can be written in the standard form as

$$\frac{d^2 y}{dx^2} - \left(2 - \frac{1}{x}\right) \frac{dy}{dx} + \left(1 - \frac{1}{x}\right) y = 0. \quad \dots(1)$$

Here $1 + P + Q = 0$,

∴ $y = e^x$ is a part of the C.F. of the solution of (1).

Putting $y = v e^x$, $\frac{dy}{dx} = \frac{dv}{dx} e^x + v e^x$,

and $\frac{d^2 y}{dx^2} = \frac{d^2 v}{dx^2} e^x + 2 \frac{dv}{dx} e^x + v e^x$

in (1), we get

$$\frac{d^2 v}{dx^2} + \frac{1}{x} \frac{dv}{dx} = 0 \Rightarrow \frac{dp}{dx} + \frac{p}{x} = 0, \text{ where } p = \frac{dv}{dx}$$

$$\Rightarrow \frac{dp}{p} = -\frac{dx}{x} \Rightarrow \log p = -\log x + \log c_1$$

$$\Rightarrow p = \frac{dv}{dx} = \frac{c_1}{x} \Rightarrow v = c_1 \log x + c_2.$$

∴ The complete solution of (1) is

$$y = v e^x = (c_1 \log x + c_2) e^x.$$

Example 3: Solve $x^2 \frac{d^2 y}{dx^2} - (x^2 + 2x) \frac{dy}{dx} + (x + 2) y = x^3 e^x$.

(Meerut 2010B; Bundelkhand 01; Gorakhpur 06; Purvanchal 07)

Solution: The standard form of the given equation is

$$\frac{d^2 y}{dx^2} - \left(1 + \frac{2}{x}\right) \frac{dy}{dx} + \left(\frac{1}{x} + \frac{2}{x^2}\right) y = x e^x. \quad \dots(1)$$

Here $P + Qx = -\left(1 + \frac{2}{x}\right) + \left(\frac{1}{x} + \frac{2}{x^2}\right) x = 0$,

∴ $y = x$ is a part of the C.F. of the solution of (1).

Putting $y = vx$, $\frac{dy}{dx} = x \frac{dv}{dx} + v$, $\frac{d^2 y}{dx^2} = x \frac{d^2 v}{dx^2} + 2 \frac{dv}{dx}$

in (1), we get

$$\frac{d^2 v}{dx^2} - \frac{dv}{dx} = e^x \quad \text{or} \quad \frac{dp}{dx} - p = e^x, \quad \left(\text{where } p = \frac{dv}{dx}\right)$$

which is linear in p .

$$\text{I.F.} = e^{\int -dx} = e^{-x}.$$

$$\therefore p e^{-x} = \int e^x \cdot e^{-x} dx + c_1 = x + c_1.$$

$$\therefore p = \frac{dv}{dx} = x e^x + c_1 e^x.$$

Integrating this, we get

$$v = x e^x - e^x + c_1 e^x + c_2.$$

$\therefore$ The complete solution of (1) is

$$y = v x = x^2 e^x - x e^x + c_1 x e^x + c_2 x.$$

Example 4: Solve $\frac{d^2 y}{dx^2} - \cot x \frac{dy}{dx} - (1 - \cot x) y = e^x \sin x$.

(Meerut 2007B)

Solution: The given differential equation is

$$\frac{d^2 y}{dx^2} - \cot x \frac{dy}{dx} - (1 - \cot x) y = e^x \sin x.$$

Here $1 + P + Q = 0$,

$\therefore y = e^x$ is a part of the C.F. of (1).

Putting $y = v e^x$, $\frac{dy}{dx} = \frac{dv}{dx} e^x + v e^x$,

$$\text{and } \frac{d^2 y}{dx^2} = \frac{d^2 v}{dx^2} e^x + 2 \frac{dv}{dx} e^x + v e^x$$

in (1), we get

$$\frac{d^2 v}{dx^2} + (2 - \cot x) \frac{dv}{dx} = \sin x$$

$$\text{or } \frac{dp}{dx} + (2 - \cot x) p = \sin x, \quad \left(\text{where } p = \frac{dv}{dx} \right)$$

which is linear in p .

$$\text{I.F.} = e^{\int (2 - \cot x) dx} = e^{2x - \log \sin x} = \frac{e^{2x}}{\sin x}.$$

$$\therefore p \frac{e^{2x}}{\sin x} = \int \frac{e^{2x}}{\sin x} \sin x dx + c_1 = \frac{1}{2} e^{2x} + c_1.$$

$$\therefore p = \frac{dv}{dx} = \frac{1}{2} \sin x + c_1 e^{-2x} \sin x.$$

Integrating this, we get

$$v = -\frac{1}{2} \cos x + \frac{c_1}{5} e^{-2x} (-2 \sin x - \cos x) + c_2.$$

$\therefore$ The complete solution of (1) is

$$y = v e^x = -\frac{1}{2} e^x \cos x - \frac{c_1}{5} e^{-x} (2 \sin x + \cos x) + c_2 e^x.$$

Example 5: Solve $x^2 \frac{d^2 y}{dx^2} - 2x(1+x) \frac{dy}{dx} + 2(1+x)y = x^3$.

(Rohilkhand 01; Gorakhpur 10; Lucknow 05, 06)

Solution: The given equation can be written in the standard form as

$$\frac{d^2 y}{dx^2} - 2\left(\frac{1}{x} + 1\right) \frac{dy}{dx} + 2\left(\frac{1}{x^2} + \frac{1}{x}\right)y = x. \quad \dots(1)$$

Here $P + Qx = 0$,

$\therefore y = x$ is a part of the C.F. of (1).

Putting $y = vx$, $\frac{dy}{dx} = x \frac{dv}{dx} + v$

and $\frac{d^2 y}{dx^2} = x \frac{d^2 v}{dx^2} + 2 \frac{dv}{dx}$

in (1), we have

$$\frac{d^2 v}{dx^2} - 2 \frac{dv}{dx} = 1 \quad \text{or} \quad \frac{dp}{dx} - 2p = 1, \quad \left(\text{where } p = \frac{dv}{dx} \right)$$

which is linear in p .

$$\text{I.F.} = e^{\int -2dx} = e^{-2x}.$$

$$\therefore pe^{-2x} = \int 1 \cdot e^{-2x} dx + c_1 = -\frac{1}{2}e^{-2x} + c_1.$$

$$\therefore p = \frac{dv}{dx} = -\frac{1}{2} + c_1 e^{2x}.$$

$$\text{Integrating, we get } v = -\frac{1}{2}x + \frac{c_1}{2}e^{2x} + c_2.$$

$\therefore$ The complete solution of (1) is

$$y = vx = -\frac{1}{2}x^2 + \frac{c_1}{2}xe^{2x} + c_2x.$$

Example 6: Solve $\sin^2 x \frac{d^2 y}{dx^2} = 2y$, given $y = \cot x$ is a solution.

(Purvanchal 2010; Lucknow 08; Kanpur 07, 08)

Solution: Putting $y = v \cot x$, $\frac{dy}{dx} = \frac{dv}{dx} \cot x - v \operatorname{cosec}^2 x$,

$$\frac{d^2 y}{dx^2} = \frac{d^2 v}{dx^2} \cot x - 2 \frac{dv}{dx} \operatorname{cosec}^2 x + 2v \operatorname{cosec}^2 x \cot x,$$

the given differential equation becomes

$$\sin^2 x \cot x \frac{d^2 v}{dx^2} - 2 \frac{dv}{dx} = 0$$

$$\text{or} \quad \frac{d^2 v}{dx^2} - \frac{2}{\sin x \cos x} \frac{dv}{dx} = 0$$

$$\text{or} \quad \frac{dp}{dx} - \frac{2}{\sin x \cos x} p = 0, \text{ where } p = \frac{dv}{dx}$$

or
$$\frac{dp}{p} = \frac{2}{\sin x \cos x} dx = 4 \operatorname{cosec} 2x dx.$$

Integrating, we get

$$\log p = 4 \cdot \frac{1}{2} \log \tan x + \log c_1$$

or
$$p = \frac{dv}{dx} = c_1 \tan^2 x = c_1 (\sec^2 x - 1).$$

$\therefore v = c_1 \int (\sec^2 x - 1) dx + c_2 = c_1 (\tan x - x) + c_2.$

$\therefore$ the complete solution of the given equation is

$$y = v \cot x = c_1 (1 - x \cot x) + c_2 \cot x.$$

Example 7: Solve $(x \sin x + \cos x) \frac{d^2 y}{dx^2} - x \cos x \frac{dy}{dx} + y \cos x = \sin x (x \sin x + \cos x)^2.$

Solution: The given equation in the standard form is

$$\frac{d^2 y}{dx^2} - \frac{x \cos x}{x \sin x + \cos x} \frac{dy}{dx} + \frac{\cos x}{x \sin x + \cos x} y = \sin x (x \sin x + \cos x).$$

...(1)

Here $P + Qx = 0$,

$\therefore y = x$ is a part of the C.F. of the solution of (1).

Putting $y = vx$ and the corresponding values of $\frac{dy}{dx}, \frac{d^2 y}{dx^2}$ in (1), we get

$$\frac{d^2 v}{dx^2} + \left(\frac{2}{x} - \frac{x \cos x}{x \sin x + \cos x} \right) \frac{dv}{dx} = \frac{\sin x (x \sin x + \cos x)}{x}$$

or
$$\frac{dp}{dx} + \left(\frac{2}{x} - \frac{x \cos x}{x \sin x + \cos x} \right) p = \frac{\sin x (x \sin x + \cos x)}{x}.$$

which is linear in p where $p = \frac{dv}{dx}.$

$$\begin{aligned} \text{I. F.} &= e^{\int \left(\frac{2}{x} - \frac{x \cos x}{x \sin x + \cos x} \right) dx} \\ &= e^{2 \log x - \log (x \sin x + \cos x)} = \frac{x^2}{x \sin x + \cos x} \end{aligned}$$

$\therefore p \cdot \frac{x^2}{x \sin x + \cos x} = \int x \sin x dx + c_1 = -x \cos x + \sin x + c_1.$

$\therefore p = \frac{dv}{dx} = \frac{1}{x^2} (-x \cos x + \sin x) (x \sin x + \cos x) + \frac{c_1}{x^2} (x \sin x + \cos x)$

or
$$\frac{dv}{dx} = -\sin x \cos x - \frac{1}{x} \cos 2x + \frac{1}{x^2} \sin x \cos x + c_1 \left(\frac{1}{x} \sin x + \frac{1}{x^2} \cos x \right)$$

$$\begin{aligned}
 \text{or} \quad v &= -\frac{1}{2} \int \sin 2x \, dx - \frac{1}{2} \int \left(\frac{2}{x} \cos 2x - \frac{1}{x^2} \sin 2x \right) dx \\
 &\quad + c_1 \int \left(\frac{\sin x}{x} + \frac{\cos x}{x^2} \right) dx + c_2 \\
 &= \frac{1}{4} \cos 2x - \frac{1}{2} \int \frac{d}{dx} \left(\frac{\sin 2x}{x} \right) dx - c_1 \int \frac{d}{dx} \left(\frac{\cos x}{x} \right) dx + c_2 \\
 &= \frac{1}{4} \cos 2x - \frac{1}{2} \frac{\sin 2x}{x} - c_1 \frac{\cos x}{x} + c_2.
 \end{aligned}$$

∴ The complete solution of (1) is

$$y = vx = \frac{1}{4} x \cos 2x - \frac{1}{2} \sin 2x - c_1 \cos x + c_2 x.$$

Example 8: Solve $x \frac{dy}{dx} - y = (x-1) \left(\frac{d^2 y}{dx^2} - x + 1 \right)$.

Solution: The given equation in the standard form is

$$\frac{d^2 y}{dx^2} - \frac{x}{x-1} \frac{dy}{dx} + \frac{y}{x-1} = x-1. \quad \dots(1)$$

Here $P + Qx = 0$,

∴ $y = x$ is a part of the C.F. of the solution of (1).

Putting $y = vx$ and the corresponding values of $\frac{dy}{dx}$ and $\frac{d^2 y}{dx^2}$ in (1), we get

$$\frac{d^2 v}{dx^2} + \left(\frac{2}{x} - \frac{x}{x-1} \right) \frac{dv}{dx} = \frac{x-1}{x}$$

$$\text{or} \quad \frac{dp}{dx} + \left(\frac{2}{x} - \frac{x}{x-1} \right) p = \frac{x-1}{x}, \quad \left(\text{where } p = \frac{dv}{dx} \right)$$

which is linear in p .

$$\begin{aligned}
 \text{I.F.} &= e^{\int \left(\frac{2}{x} - \frac{x}{x-1} \right) dx} = e^{\int \left(\frac{2}{x} - 1 - \frac{1}{x-1} \right) dx} \\
 &= e^{2 \log x - x - \log(x-1)} = \frac{x^2}{x-1} e^{-x}.
 \end{aligned}$$

$$\begin{aligned}
 \therefore p \cdot \frac{x^2}{(x-1)} e^{-x} &= \int \frac{x-1}{x} \cdot \frac{x^2}{x-1} e^{-x} dx + c_1 \\
 &= \int x e^{-x} dx + c_1 = -x e^{-x} - e^{-x} + c_1.
 \end{aligned}$$

$$\therefore p = \frac{dv}{dx} = -\frac{x-1}{x} - \frac{(x-1)}{x^2} + \frac{c_1 (x-1)}{x^2} e^x$$

$$\text{or} \quad \frac{dv}{dx} = -1 + \frac{1}{x^2} + c_1 \left(\frac{1}{x} - \frac{1}{x^2} \right) e^x.$$

Integrating, we get

$$v = -x - \frac{1}{x} + c_1 \frac{1}{x} e^x + c_2.$$

∴ The complete solution of (1) is

$$y = vx = -x^2 - 1 + c_1 e^x + c_2 x$$

or $y = c_1 e^x + c_2 x - (1 + x^2).$

Comprehensive Exercise 1

Solve the following differential equations :

1. $(x+1) \frac{d^2 y}{dx^2} - 2(x+3) \frac{dy}{dx} + (x+5) y = e^x.$
2. $x \frac{d^2 y}{dx^2} - 2(x+1) \frac{dy}{dx} + (x+2) y = (x-2) e^x.$
3. $x \frac{d^2 y}{dx^2} - (2x+1) \frac{dy}{dx} + (x+1) y = (x^2 + x - 1) e^{2x}.$
4. $\frac{d^2 y}{dx^2} + (1 - \cot x) \frac{dy}{dx} - y \cot x = \sin^2 x.$
5. $(x-x^2) \frac{d^2 y}{dx^2} - (1-2x) \frac{dy}{dx} + (1-3x+x^2) y = (1-x)^3.$
6. $(3-x) \frac{d^2 y}{dx^2} - (9-4x) \frac{dy}{dx} + (6-3x) y = 0.$
7. $x \frac{d^2 y}{dx^2} + (x-2) \frac{dy}{dx} - 2y = x^3.$ (Kanpur 2011)
8. $(2x-1) \frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + (3-2x) y = 2e^x.$
9. $x \frac{d^2 y}{dx^2} + (1-x) \frac{dy}{dx} - y = e^x.$
10. Solve $x^2 y_2 + xy_1 - 9y = 0$, given that $y = x^3$ is a solution.
11. Solve $\frac{d^2 y}{dx^2} + \left(1 + \frac{2}{x} \cot x - \frac{2}{x^2}\right) y = x \cos x,$ (Gorakhpur 2008, 11)
given that $\frac{\sin x}{x}$ is an integral included in the C.F.
12. Solve $(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - a^2 y = 0$, given that $y = ce^{a \sin^{-1} x}$ is an integral.

Answers 1

1. $y = -\frac{1}{4} x e^x + \frac{1}{5} c_1 e^x (x+1)^5 + c_2 e^x$
2. $y = -\frac{1}{2} x^2 e^x + x e^x + \frac{1}{3} c_1 x^3 e^x + c_2 e^x$
3. $y = x e^{2x} + c_1 x^2 e^x + c_2 e^x$
4. $y = -\frac{1}{10} (\sin 2x - 2 \cos 2x) + c_1 (\sin x - \cos x) + c_2 e^{-x}$
5. $y = -x + \frac{1}{2} c_1 x^2 e^{-x} + c_2 e^x$
6. $y = c_1 e^{3x} (4x^3 - 42x^2 + 150x - 183) + c_2 e^x$
7. $y = x^3 + (c_1 - 3) x^2 - 2 (c_1 - 3) (x - 1) + c_2 e^{-x}$
8. $y = -c_1 x e^{-x} - x e^x + e^x \int (2x - 1) \log (2x - 1) dx$
 $- 2e^x \int [e^{-2x} (2x - 1) \int e^{2x} \log (2x - 1) dx] dx + c_2 e^x$
9. $y = e^x \log x + c_1 e^x \int \frac{e^{-x}}{x} dx + c_2 e^x$
10. $y = c_1 x^{-3} + c_2 x^3$
11. $y = \frac{x^2 \sin x}{6} + c_1 \left[-x \cos x + 2 \sin x \log \sin x - \frac{2 \sin x}{x} \int \log \sin x dx \right]$
 $+ c_2 \frac{\sin x}{x}$
12. $y = \left[-\frac{c_1}{2a} e^{-2a \sin^{-1} x} + c_2 \right] \cdot e^{a \sin^{-1} x}$

7.4 Removal of the First Derivative (Reduction to Normal Form)

(Agra 2007; Gorakhpur 05)

If we are unable to obtain a part of the C.F. of the solution of the differential equation

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R, \quad \dots(1)$$

then we cannot solve (1) by the method given in article 7.2. In such cases the equation (1) can sometimes be solved by reducing it into the form in which the term containing the first derivative is absent. For this we first change the dependent variable from y to v in the equation (1) by putting $y = uv$, where u is some function of x .

Then
$$\begin{aligned}\frac{dy}{dx} &= v \frac{du}{dx} + u \frac{dv}{dx}, \quad \frac{d^2 y}{dx^2} \\ &= v \frac{d^2 u}{dx^2} + 2 \frac{dv}{dx} \cdot \frac{du}{dx} + u \frac{d^2 v}{dx^2}.\end{aligned}$$

Putting the values of $y, \frac{dy}{dx}, \frac{d^2 y}{dx^2}$ in (1), we get

$$\begin{aligned}&\left[v \frac{d^2 u}{dx^2} + 2 \frac{dv}{dx} \cdot \frac{du}{dx} + u \frac{d^2 v}{dx^2} \right] + P \left[v \frac{du}{dx} + u \frac{dv}{dx} \right] + Q vu = R \\ \text{or} \quad &u \frac{d^2 v}{dx^2} + u \left[P + \frac{2}{u} \frac{du}{dx} \right] \frac{dv}{dx} + v \left[\frac{d^2 u}{dx^2} + P \frac{du}{dx} + Qu \right] = R. \quad \dots(2)\end{aligned}$$

To remove the term of the first derivative $\frac{dv}{dx}$ we choose u such that

$$\begin{aligned}P + \frac{2}{u} \frac{du}{dx} &= 0 \quad \text{or} \quad \frac{du}{u} = -\frac{1}{2} P dx \\ \text{or} \quad \log u &= -\frac{1}{2} \int P dx \quad \text{or} \quad u = e^{-\frac{1}{2} \int P dx} \quad \dots(3)\end{aligned}$$

Now the equation (2) becomes

$$\frac{d^2 v}{dx^2} + \frac{v}{u} \left[\frac{d^2 u}{dx^2} + P \frac{du}{dx} + Qu \right] = \frac{R}{u}. \quad \dots(4)$$

But from (3), $\frac{du}{dx} = -\frac{1}{2} Pu, \frac{d^2 u}{dx^2} = -\frac{1}{2} \left[P \frac{du}{dx} + u \frac{dP}{dx} \right]$

$$\begin{aligned}&= -\frac{1}{2} \left[P \left(-\frac{1}{2} Pu \right) + u \frac{dP}{dx} \right] \\ &= \frac{1}{4} P^2 u - \frac{1}{2} u \frac{dP}{dx}.\end{aligned}$$

Putting these values in (4), we get

$$\begin{aligned}&\frac{d^2 v}{dx^2} + v \left[\frac{1}{4} P^2 - \frac{1}{2} \frac{dP}{dx} - P \cdot \frac{1}{2} P + Q \right] = \frac{R}{e^{-\frac{1}{2} \int P dx}} \\ \text{or} \quad &\frac{d^2 v}{dx^2} + v \left[Q - \frac{P^2}{4} - \frac{1}{2} \frac{dP}{dx} \right] = R e^{\frac{1}{2} \int P dx} \\ \text{or} \quad &\frac{d^2 v}{dx^2} + Xv = Y,\end{aligned}$$

where $X = Q - \frac{1}{2} \frac{dP}{dx} - \frac{1}{4} P^2$ and $Y = R e^{\frac{1}{2} \int P dx}$.

The reduced equation (5) may easily be integrated. The equation (5) is called the **normal form** of the equation (1).

Note: The students should remember the values of u, X and Y . They can write the reduced equation (5) directly.

Illustrative Examples

Example 9: Solve $\frac{d^2 y}{dx^2} - 4x \frac{dy}{dx} + (4x^2 - 1)y = -3e^{x^2} \sin 2x$.

(Rohilkhand 2001; Meerut 05; Agra 07; Gorakhpur 09)

Solution: Here $P = -4x$, $Q = 4x^2 - 1$, $R = -3e^{x^2} \sin 2x$.

We choose $u = e^{-\frac{1}{2} \int P dx} = e^{-\frac{1}{2} \int -4x dx} = e^{x^2}$.

Putting $y = uv$ in the given equation, it reduces to its normal form

$$\frac{d^2 v}{dx^2} + Xv = Y, \quad \dots(1)$$

where $X = Q - \frac{1}{2} \frac{dP}{dx} - \frac{1}{4} P^2 = 4x^2 - 1 - \frac{1}{2}(-4) - \frac{1}{4} \cdot 16x^2 = 1$,

and $Y = Re^{\frac{1}{2} \int P dx} = -3e^{x^2} \sin 2x \cdot e^{-x^2} = -3 \sin 2x$.

Hence the normal form (1) of the given equation is

$$\frac{d^2 v}{dx^2} + v = -3 \sin 2x \quad \text{or} \quad (D^2 + 1)v = -3 \sin 2x. \quad \dots(2)$$

Now (2) is a linear differential equation with constant coefficients.

A.E. is $m^2 + 1 = 0 \Rightarrow m^2 = -1 \Rightarrow m = \pm i$.

$\therefore$ C.F. of the solution of (2) = $c_1 \cos x + c_2 \sin x$.

$$\text{P.I.} = \frac{1}{D^2 + 1} (-3 \sin 2x) = \frac{-3}{-2^2 + 1} \sin 2x = \sin 2x.$$

$\therefore$ the solution of (2) is $v = c_1 \cos x + c_2 \sin x + \sin 2x$.

Hence the general solution of the given equation is

$$y = uv = e^{x^2} (c_1 \cos x + c_2 \sin x + \sin 2x).$$

Example 10: Solve $\frac{d^2 y}{dx^2} - 2 \tan x \frac{dy}{dx} + 5y = \sec x \cdot e^x$.

(Agra 2006; Gorakhpur 06; Kanpur 09)

Solution: Here $P = -2 \tan x$, $Q = 5$, $R = e^x \sec x$.

We choose $u = e^{-\frac{1}{2} \int P dx} = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$.

Substituting $y = uv$ in the given equation, it reduces to its normal form

$$\frac{d^2 v}{dx^2} + Xv = Y, \quad \dots(1)$$

where $X = Q - \frac{1}{2} \frac{dP}{dx} - \frac{1}{4} P^2 = 5 - \frac{1}{2}(-2 \sec^2 x) - \tan^2 x = 6$

and $Y = Re^{\frac{1}{2} \int P dx} = e^x \sec x (\sec x)^{-1} = e^x$.

Hence the equation (1) is

$$\frac{d^2 v}{dx^2} + 6v = e^x \quad \text{or} \quad (D^2 + 6)v = e^x. \quad \dots(2)$$

A.E. is $m^2 + 6 = 0 \Rightarrow m = \pm \sqrt{6}i$.

$\therefore$ C.F. of the solution of (2) = $c_1 \cos \sqrt{6}x + c_2 \sin \sqrt{6}x$.

$$\text{P.I.} = \frac{1}{D^2 + 6} e^x = \frac{1}{7} e^x.$$

$\therefore$ the solution of (2) is $v = c_1 \cos \sqrt{6}x + c_2 \sin \sqrt{6}x + \frac{1}{7} e^x$.

Hence the complete solution of the given equation is

$$y = uv = \sec x (c_1 \cos \sqrt{6}x + c_2 \sin \sqrt{6}x + \frac{1}{7} e^x).$$

Example 11: Solve $\frac{d^2 y}{dx^2} + \frac{1}{x^{1/3}} \frac{dy}{dx} + \left(\frac{1}{4x^{2/3}} - \frac{1}{6x^{4/3}} - \frac{6}{x^2} \right) y = 0$.

Solution: Here $P = x^{-1/3}$, $Q = \frac{1}{4x^{2/3}} - \frac{1}{6x^{4/3}} - \frac{6}{x^2}$, $R = 0$.

We choose $u = e^{-\frac{1}{2} \int P dx} = e^{-\frac{1}{2} \int x^{-1/3} dx} = e^{-\frac{3}{4} x^{2/3}}$.

Substituting $y = uv$ in the given equation, it reduces to its normal form

$$\frac{d^2 v}{dx^2} + Xv = Y, \quad \dots(1)$$

where

$$\begin{aligned} X &= Q - \frac{1}{2} \frac{dP}{dx} - \frac{1}{4} P^2 \\ &= \frac{1}{4x^{2/3}} - \frac{1}{6x^{4/3}} - \frac{6}{x^2} - \frac{1}{2} \left(-\frac{1}{3} x^{-4/3} \right) - \frac{1}{4} x^{-2/3} = -\frac{6}{x^2}, \end{aligned}$$

and

$$Y = R e^{\frac{1}{2} \int P dx} = 0.$$

Hence the normal form (1) of the given equation is

$$\frac{d^2 v}{dx^2} - \frac{6}{x^2} v = 0 \quad \text{or} \quad x^2 \frac{d^2 v}{dx^2} - 6v = 0, \quad \dots(2)$$

which is a homogeneous linear equation.

In order to solve it, putting $x = e^z$, the differential equation (2) becomes

$$[D(D-1) - 6]v = 0, \text{ where } D \equiv d/dz$$

or

$$(D^2 - D - 6)v = 0.$$

A.E. is $m^2 - m - 6 = 0$, giving $m = -2, 3$.

$\therefore$ the solution of (2) is $v = c_1 e^{-2z} + c_2 e^{3z} = c_1 x^{-2} + c_2 x^3$.

Hence the complete solution of the given equation is

$$y = uv = e^{(-3/4)x^{2/3}} (c_1 x^{-2} + c_2 x^3).$$

Example 12: Solve $\frac{d^2 y}{dx^2} - 4x \frac{dy}{dx} + (4x^2 - 3)y = e^{x^2}$. (Gorakhpur 2005, 07; Lucknow 09)

Solution: Here $P = -4x$, $Q = 4x^2 - 3$, $R = e^{x^2}$.

We choose $u_1 = e^{-\frac{1}{2} \int P dx} = e^{\int 2x dx} = e^{x^2}$.

Substituting $y = uv$ in the given equation, it reduces to its normal form

$$\frac{d^2 v}{dx^2} + Xv = Y, \quad \dots(1)$$

where $X = Q - \frac{1}{2} \frac{dP}{dx} - \frac{1}{4} P^2 = -1$

and $Y = R e^{\frac{1}{2} \int P dx} = 1$.

Hence the normal form (1) of the given equation is

$$\frac{d^2 v}{dx^2} - v = 1 \text{ or } (D^2 - 1)v = 1. \quad \dots(2)$$

A.E. is $m^2 - 1 = 0$, giving $m = \pm 1$.

$\therefore$ C.F. of the solution of (2) = $c_1 e^x + c_2 e^{-x}$.

$$\text{P.I.} = \frac{1}{D^2 - 1} (1) = \frac{1}{D^2 - 1} e^{0x} = \frac{1}{0^2 - 1} e^{0x} = -1.$$

$\therefore$ the solution of (2) is $v = c_1 e^x + c_2 e^{-x} - 1$.

Hence the complete solution of the given equation is

$$y = uv = e^{x^2} (c_1 e^x + c_2 e^{-x} - 1).$$

Example 13: Solve $\frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} + (x^2 + 1)y = x^3 + 3x$.

Solution: Here $P = 2x$, $Q = x^2 + 1$, $R = x(x^2 + 3)$.

We choose $u = e^{-\frac{1}{2} \int P dx} = e^{-x^2/2}$.

Substituting $y = uv$ in the given equation, it reduces to its normal form

$$\frac{d^2 v}{dx^2} + Xv = Y. \quad \dots(1)$$

where $X = Q - \frac{1}{2} \frac{dP}{dx} - \frac{1}{4} P^2 = 0$

and $Y = R e^{\frac{1}{2} \int P dx} = x(x^2 + 3)e^{x^2/2}$.

Hence the normal form (1) of the given equation is

$$\frac{d^2 v}{dx^2} = x(x^2 + 3)e^{x^2/2}. \quad \dots(2)$$

Integrating (2), we get

$$\begin{aligned}\frac{dv}{dx} &= \int x^3 e^{x^2/2} dx + 3 \int x e^{x^2/2} dx + c_1 \\ &= \int x^2 \left(x e^{x^2/2} \right) dx + 3 \int x e^{x^2/2} dx + c_1 \\ &= x^2 e^{x^2/2} - 2 \int x e^{x^2/2} dx + 3 \int x e^{x^2/2} dx + c_1,\end{aligned}$$

integrating the first integral by parts

$$= x^2 e^{x^2/2} + e^{x^2/2} + c_1.$$

Integrating again, we get

$$v = \int x^2 e^{x^2/2} dx + \int e^{x^2/2} dx + c_1 x + c_2 = x e^{x^2/2} + c_1 x + c_2.$$

Hence the complete solution of the given equation is

$$y = uv = x + (c_1 x + c_2) e^{-x^2/2}.$$

Example 14: Solve $\frac{d^2 y}{dx^2} - \frac{1}{x^{1/2}} \frac{dy}{dx} + \frac{1}{4x^2} (x + x^{1/2} - 8) y = 0$.

Solution: Here $P = -\frac{1}{x^{1/2}}$, $Q = \frac{1}{4x^2} (x + x^{1/2} - 8)$, $R = 0$.

We choose $u = e^{-\frac{1}{2} \int P dx} = e^{\frac{1}{2} \int x^{-1/2} dx} = e^{x^{1/2}} = e^{\sqrt{x}}$.

Substituting $y = uv$ in the given equation, it reduces to its normal form

$$\frac{d^2 v}{dx^2} + Xv = Y, \quad \dots(1)$$

where $X = Q - \frac{1}{2} \frac{dP}{dx} - \frac{1}{4} P^2 = -\frac{2}{x^2}$

and $Y = R e^{\frac{1}{2} \int P dx} = 0$.

Hence the normal form (1) of the given equation is

$$\frac{d^2 v}{dx^2} + v \left(-\frac{2}{x^2} \right) = 0$$

or $x^2 \frac{d^2 v}{dx^2} - 2v = 0. \quad \dots(2)$

This is a homogeneous linear equation.

Putting $x = e^z$, the differential equation (2) becomes

$$\{D(D-1) - 2\} v = 0 \quad \text{or} \quad (D^2 - D - 2) v = 0,$$

where $D \equiv d/dz$.

A.E. is $m^2 - m - 2 = 0 \quad \text{or} \quad (m-2)(m+1) = 0.$

$\therefore m = 2, -1.$

Thus the solution of (2) is $v = c_1 e^{2z} + c_2 e^{-z} = c_1 x^2 + c_2 x^{-1}$.

Hence the complete solution of the given equation is

$$y = uv = e^{\sqrt{x}}(c_1 x^2 + c_2 x^{-1}).$$

Comprehensive Exercise 2

Solve the following equations by removing the first derivative :

- $x^2 \frac{d^2 y}{dx^2} - 2(x^2 + x) \frac{dy}{dx} + (x^2 + 2x + 2)y = 0$. (Kanpur 2008; Bundelkhand 01)
- $\frac{d^2 y}{dx^2} - 4x \frac{dy}{dx} + (4x^2 - 1)y = -3e^{x^2} (\sin 2x + 5e^{-2x} + 6)$.
- $\frac{d^2 y}{dx^2} - \frac{2}{x} \frac{dy}{dx} + \left(1 + \frac{2}{x^2}\right)y = xe^x$. (Kanpur 2009)
- $\frac{d}{dx} \left(\cos^2 x \frac{dy}{dx} \right) + y \cos^2 x = 0$. (Rohilkhand 2007)
- $\frac{d^2 y}{dx^2} + \frac{2}{x} \frac{dy}{dx} \pm n^2 y = 0$. (Agra 2008)
- $x \frac{d}{dx} \left(x \frac{dy}{dx} - y \right) - 2x \frac{dy}{dx} + 2y + x^2 y = 0$. (Rohilkhand 2008)
- $\frac{d^2 y}{dx^2} - \frac{2}{x} \frac{dy}{dx} + \left(a^2 + \frac{2}{x^2}\right)y = 0$.
- $\left(\frac{d^2 y}{dx^2} + y \right) \cot x + 2 \left(\frac{dy}{dx} + y \tan x \right) = \sec x$.
- $x^2 (\log x)^2 \frac{d^2 y}{dx^2} - 2x \log x \frac{dy}{dx} + [2 + \log x - 2(\log x)^2]y = x^2 (\log x)^3$.
- $\frac{d^2 y}{dx^2} - 2 \tan x \frac{dy}{dx} + 5y = 0$. (Agra 2008)

Answers 2

- $y = xe^x (c_1 x + c_2)$
- $y = e^{x^2} (c_1 \cos x + c_2 \sin x + \sin 2x - 3e^{-2x} - 18)$
- $y = x (c_1 \cos x + c_2 \sin x + \frac{1}{2} e^x)$
- $y = (c_1 \cos \sqrt{2x} + c_2 \sin \sqrt{2x}) \sec x$

$$5. \quad y = \frac{1}{x} c_1 \cos (nx + c_2) \quad \text{or} \quad y = \frac{1}{x} (c_1 e^{nx} + c_2 e^{-nx})$$

$$6. \quad y = x (c_1 \cos x + c_2 \sin x)$$

$$7. \quad y = x c_1 \cos (ax + c_2)$$

$$8. \quad y = \frac{1}{2} \sin x + (c_1 x + c_2) \cos x$$

$$9. \quad y = \{c_1 x^2 + c_2 x^{-1} + \frac{1}{3} x^2 \log x\} \cdot \log x$$

$$10. \quad y = (\sec x) \cdot \{c_1 \cos \sqrt{6}x + c_2 \sin \sqrt{6}x\}$$

7.5 Transformation of the Equation by Changing the Independent Variable

Sometimes the equation may become easily integrable by changing the independent variable.

Let the linear equation of second order be

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Q y = R. \quad \dots(1)$$

Let the independent variable be changed from x to z with the help of a relation of the form $z = f(x)$.

Then
$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx}, \quad \frac{d^2 y}{dx^2} = \frac{d^2 y}{dz^2} \left(\frac{dz}{dx} \right)^2 + \frac{dy}{dz} \cdot \frac{d^2 z}{dx^2}.$$

Substituting these values in (1), we get

$$\frac{d^2 y}{dz^2} \left(\frac{dz}{dx} \right)^2 + \frac{dy}{dz} \cdot \frac{d^2 z}{dx^2} + P \left[\frac{dy}{dz} \cdot \frac{dz}{dx} \right] + Q y = R$$

or
$$\frac{d^2 y}{dz^2} \left(\frac{dz}{dx} \right)^2 + \frac{dy}{dz} \left(\frac{d^2 z}{dx^2} + P \frac{dz}{dx} \right) + Q y = R$$

or
$$\frac{d^2 y}{dz^2} + \frac{\frac{d^2 z}{dx^2} + P \frac{dz}{dx}}{\left(\frac{dz}{dx} \right)^2} \cdot \frac{dy}{dz} + \frac{Q}{\left(\frac{dz}{dx} \right)^2} y = \frac{R}{\left(\frac{dz}{dx} \right)^2}$$

or
$$\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R_1, \quad \dots(2)$$

where
$$P_1 = \left(\frac{d^2 z}{dx^2} + P \frac{dz}{dx} \right) / \left(\frac{dz}{dx} \right)^2, \quad Q_1 = Q / \left(\frac{dz}{dx} \right)^2 \quad \text{and} \quad R_1 = R / \left(\frac{dz}{dx} \right)^2.$$

Here P_1 , Q_1 and R_1 are functions of x and may be expressed as functions of z with the help of the relation $z = f(x)$.

How to choose z ? We would like to choose z in such a way that the equation (2) can be easily integrated.

Case I: Let us choose z in such a way that P_1 vanishes *i.e.*,

$$\begin{aligned} \frac{d^2 z}{dx^2} + P \frac{dz}{dx} &= 0 \\ \Rightarrow \frac{d}{dx} \left(\frac{dz}{dx} \right) + P \frac{dz}{dx} &= 0 \\ \Rightarrow \frac{dz}{dx} = e^{-\int P dx} \Rightarrow z &= \int [e^{-\int P dx}] dx. \end{aligned}$$

Then the equation (2) is reduced to $\frac{d^2 y}{dz^2} + Q_1 y = R_1$.

Now this equation can be easily solved, if

- (i) Q_1 is a constant because then it is a linear equation with constant coefficients,
- (ii) Q_1 is of the form (constant)/ z^2 because then it is a linear homogeneous equation with variable coefficients.

Case II: Suppose we choose z such that

$$Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} = \pm a^2 \text{ (i.e., a constant)}$$

or $a \frac{dz}{dx} = \sqrt{(\pm Q)} \quad \text{or} \quad az = \int \sqrt{(\pm Q)} dx,$

(+ ive or - ive sign is taken to make the expression under the radical sign + ive).

With this choice of z , the equation (2) is reduced to

$$\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + a^2 y = R_1.$$

This equation can be easily solved if P_1 comes out to be a constant because then it is a linear equation with constant coefficients.

Note 1: Students should remember the values of P_1 , Q_1 and R_1 in equation (2).

Note 2: There are only two choices for z , either $P_1 = 0$ or $Q_1 = a^2$. Sometimes it is possible to make both the choices to get the solution of the given equation.

Illustrative Examples

Example 15: Solve $x \frac{d^2 y}{dx^2} - \frac{dy}{dx} - 4x^3 y = 8x^3 \sin x^2$.

(Meerut 2004, 13B; Garhwal 09; Agra 05; Gorakhpur 09)

Solution: The given equation written in the standard form is

$$\frac{d^2 y}{dx^2} - \frac{1}{x} \frac{dy}{dx} - 4x^2 y = 8x^2 \sin x^2. \quad \dots(1)$$

Here $P = -1/x$, $Q = -4x^2$, $R = 8x^2 \sin x^2$.

Changing the independent variable from x to z by a relation of the form $z = f(x)$, the given equation is transformed into

$$\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R_1, \quad \dots(2)$$

where
$$P_1 = \frac{\frac{d^2 z}{dx^2} + P \frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2}, \quad Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2}, \quad R_1 = \frac{R}{\left(\frac{dz}{dx}\right)^2}.$$

Choosing z such that $Q_1 = \frac{-4x^2}{(dz/dx)^2} = \text{constant} = -1$ (say), we have

$$(dz/dx)^2 = 4x^2. \quad \therefore \quad dz/dx = 2x \Rightarrow z = x^2.$$

Now
$$P_1 = \frac{2 - \frac{1}{x} \cdot 2x}{4x^2} = 0, \quad R_1 = 2 \sin x^2 = 2 \sin z.$$

$\therefore$ The transformed equation (2) is

$$\frac{d^2 y}{dz^2} - y = 2 \sin z \quad \text{or} \quad (D^2 - 1)y = 2 \sin z. \quad \dots(3)$$

A.E. is $m^2 - 1 = 0 \Rightarrow m^2 = 1 \Rightarrow m = \pm 1$.

$\therefore$ C.F. of the solution of (3) = $c_1 e^z + c_2 e^{-z}$.

$$\text{P.I.} = \frac{1}{D^2 - 1} 2 \sin z = \frac{2}{-1 - 1} \sin z = -\sin z.$$

$\therefore$ Solution of the equation (3) is

$$y = c_1 e^z + c_2 e^{-z} - \sin z.$$

Hence the complete solution of the equation (1) is

$$y = c_1 e^{x^2} + c_2 e^{-x^2} - \sin x^2.$$

Example 16: Solve $\frac{d^2 y}{dx^2} + \cot x \frac{dy}{dx} + 4y \operatorname{cosec}^2 x = 0$.
(Meerut 2001; Agra 06; Gorakhpur 05)

Solution: Here $P = \cot x$, $Q = 4 \operatorname{cosec}^2 x$, $R = 0$.

Changing the independent variable from x to z by a relation of the form $z = f(x)$, the given equation is transformed into

$$\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R_1, \quad \dots(1)$$

where
$$P_1 = \frac{\frac{d^2 z}{dx^2} + P \frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2}, Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2}, R_1 = \frac{R}{\left(\frac{dz}{dx}\right)^2}.$$

Choosing z such that $Q_1 = \frac{4 \operatorname{cosec}^2 x}{(dz/dx)^2} = \text{constant} = 1$ (say), we have

$$(dz/dx)^2 = 4 \operatorname{cosec}^2 x$$

or
$$\frac{dz}{dx} = 2 \operatorname{cosec} x, \quad \text{or} \quad z = 2 \log \tan \frac{x}{2}.$$

Now $P_1 = \frac{-2 \operatorname{cosec} x \cot x + 2 \operatorname{cosec} x \cot x}{4 \operatorname{cosec}^2 x} = 0, R_1 = 0$

$\therefore$ the transformed equation (1) is

$$\frac{d^2 y}{dz^2} + y = 0 \quad \text{or} \quad (D^2 + 1)y = 0. \quad \dots(2)$$

A.E. is $m^2 + 1 = 0 \Rightarrow m^2 = -1 \Rightarrow m = \pm i$.

$\therefore$ Solution of the equation (2) is

$$y = c_1 \cos z + c_2 \sin z.$$

Hence the complete solution of the given equation is

$$y = c_1 \cos \left(2 \log \tan \frac{x}{2} \right) + c_2 \sin \left(2 \log \tan \frac{x}{2} \right).$$

Example 17: Solve $\cos x \frac{d^2 y}{dx^2} + \sin x \frac{dy}{dx} - 2y \cos^3 x = 2 \cos^5 x$.

(Meerut 2006; Rohilkhand 07; Agra 07; Avadh 11; Purvanchal 06, 10)

Solution: The given equation can be written in the standard form as

$$\frac{d^2 y}{dx^2} + \tan x \frac{dy}{dx} - 2y \cos^2 x = 2 \cos^4 x. \quad \dots(1)$$

Here $P = \tan x, Q = -2 \cos^2 x, R = 2 \cos^4 x$.

Changing the independent variable from x to z by a relation of the form $z = f(x)$, the given equation is transformed into

$$\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R_1, \quad \dots(2)$$

where
$$P_1 = \frac{\frac{d^2 z}{dx^2} + P \frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2}, Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2}, R_1 = \frac{R}{\left(\frac{dz}{dx}\right)^2}.$$

Let us choose z such that $Q_1 = \frac{-2 \cos^2 x}{(dz/dx)^2} = \text{constant} = -2$ (say).

Then $\left(\frac{dz}{dx}\right)^2 = \cos^2 x$, or $\frac{dz}{dx} = \cos x$ or $z = \sin x$.

Now $P_1 = 0$, $R_1 = 2 \cos^2 x = 2(1 - \sin^2 x) = 2(1 - z^2)$.

$\therefore$ The transformed equation (2) is

$$\frac{d^2 y}{dz^2} - 2y = 2(1 - z^2). \quad \dots(3)$$

A.E. is $m^2 - 2 = 0$, giving $m = \pm \sqrt{2}$.

$\therefore$ C.F. of the solution of (3) $= c_1 e^{\sqrt{2}z} + c_2 e^{-\sqrt{2}z}$.

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 - 2} 2(1 - z^2) = -\left(1 - \frac{D^2}{2}\right)^{-1} (1 - z^2) \\ &= -(1 + \frac{1}{2} D^2 + \dots)(1 - z^2) = -(1 - z^2) + \frac{1}{2} \cdot 2 = z^2. \end{aligned}$$

$\therefore$ The solution of the equation (3) is

$$y = c_1 e^{\sqrt{2}z} + c_2 e^{-\sqrt{2}z} + z^2.$$

Hence the complete solution of the given equation is

$$y = c_1 e^{\sqrt{2} \sin x} + c_2 e^{-\sqrt{2} \sin x} + \sin^2 x.$$

Example 18: Solve $x^6 \frac{d^2 y}{dx^2} + 3x^5 \frac{dy}{dx} + a^2 y = \frac{1}{x^2}$.

Solution: The given equation can be written in the standard form as

$$\frac{d^2 y}{dx^2} + \frac{3}{x} \frac{dy}{dx} + \frac{a^2}{x^6} y = \frac{1}{x^8}.$$

Here $P = \frac{3}{x}$, $Q = \frac{a^2}{x^6}$, $R = \frac{1}{x^8}$.

To solve this differential equation we change the independent variable from x to z by choosing z such that

$$\frac{Q}{(dz/dx)^2} = \frac{a^2/x^6}{(dz/dx)^2} = \text{constant} = a^2 \text{ (say)}.$$

Then $\left(\frac{dz}{dx}\right)^2 = \frac{1}{x^6}$, or $\frac{dz}{dx} = \frac{1}{x^3}$ or $z = -1/(2x^2)$.

Now by the substitution $z = -1/(2x^2)$, the given differential equation is transformed into

$$\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R_1, \text{ where}$$

$$P_1 = \frac{\frac{d^2 z}{dx^2} + P \frac{dz}{dx}}{(dz/dx)^2} = \frac{(-3/x^4) + (3/x)(1/x^3)}{(dz/dx)^2} = 0,$$

$$Q_1 = \frac{Q}{(dz/dx)^2} = a^2, R_1 = \frac{R}{(dz/dx)^2} = \frac{1/x^8}{1/x^6} = \frac{1}{x^2} = -2z.$$

∴ the transformed equation is

$$\frac{d^2 y}{dz^2} + a^2 y = -2z \quad \text{or} \quad (D^2 + a^2) y = -2z. \quad \dots(1)$$

The C.F. of the solution of (1) = $c_1 \cos az + c_2 \sin az$.

$$\begin{aligned} \text{P.I.} &= \frac{1}{D^2 + a^2} (-2z) = \frac{1}{a^2} \left(1 + \frac{D^2}{a^2} \right)^{-1} (-2z) \\ &= \frac{1}{a^2} \left(1 - \frac{1}{a^2} D^2 + \dots \right) (-2z) = \frac{1}{a^2} (-2z). \end{aligned}$$

∴ the solution of the equation (1) is

$$y = c_1 \cos az + c_2 \sin az - \frac{1}{a^2} \cdot 2z.$$

Hence the complete solution of the given equation is

$$y = c_1 \cos \frac{a}{2x^2} - c_2 \sin \frac{a}{2x^2} + \frac{1}{a^2 x^2}$$

$$\text{or} \quad y = c_1 \cos \frac{a}{2x^2} + c_2 \sin \frac{a}{2x^2} + \frac{1}{a^2 x^2}.$$

Example 19: Solve $\frac{d^2 y}{dx^2} + (\tan x - 1)^2 \frac{dy}{dx} - n(n-1)y \sec^4 x = 0$.

Solution: Comparing the given differential equation with the standard form

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R, \text{ we have}$$

$$P = (\tan x - 1)^2, Q = -n(n-1)\sec^4 x, R = 0.$$

To solve the given differential equation we change the independent variable from x to z by choosing z such that

$$\frac{Q}{(dz/dx)^2} = \frac{-n(n-1)\sec^4 x}{(dz/dx)^2} = \text{constant} = -n(n-1), \text{ say.}$$

$$\text{Then} \quad \left(\frac{dz}{dx} \right)^2 = \sec^4 x, \quad \text{or} \quad \frac{dz}{dx} = \sec^2 x, \quad \text{or} \quad z = \tan x.$$

Now by the substitution $z = \tan x$, the given differential equation is transformed into

$$\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R_1, \text{ where}$$

$$P_1 = \frac{\frac{d^2 z}{dx^2} + P \frac{dz}{dx}}{(dz/dx)^2} = \frac{2 \sec^2 x \tan x + (\tan x - 1)^2 \sec^2 x}{\sec^4 x} = 1,$$

$$Q_1 = \frac{Q}{(dz/dx)^2} = -n(n-1), \quad R_1 = \frac{R}{(dz/dx)^2} = 0.$$

$\therefore$ the transformed equation is

$$\frac{d^2 y}{dz^2} + \frac{dy}{dz} - n(n-1)y = 0$$

$$\text{or} \quad \{D^2 + D - n(n-1)\} y = 0. \quad \dots(1)$$

$$\text{A.E. is } m^2 + m - n(n-1) = 0$$

$$\text{or} \quad (m+n)\{m-(n-1)\} = 0.$$

$$\therefore \quad m = -n, n-1.$$

$\therefore$ the solution of the equation (1) is

$$y = c_1 e^{-nz} + c_2 e^{(n-1)z}.$$

Hence the complete solution of the given equation is

$$y = c_1 e^{-n \tan x} + c_2 e^{(n-1) \tan x}.$$

Comprehensive Exercise 3

Solve the following equations :

$$1. \quad \frac{d^2 y}{dx^2} + (3 \sin x - \cot x) \frac{dy}{dx} + 2y \sin^2 x = e^{-\cos x} \sin^2 x.$$

$$2. \quad \frac{d^2 y}{dx^2} - (1 + 4e^x) \frac{dy}{dx} + 3e^{2x} y = e^{2(x+e^x)}.$$

(Gorakhpur 2008, 11)

$$3. \quad \frac{d^2 y}{dx^2} + \left(1 - \frac{1}{x}\right) \frac{dy}{dx} + 4x^2 y e^{-2x} = 4(x^2 + x^3) e^{-3x}.$$

(Rohilkhand 2006)

$$4. \quad (a^2 - x^2) \frac{d^2 y}{dx^2} - \frac{a^2}{x} \frac{dy}{dx} + \frac{x^2}{a} y = 0.$$

(Bundelkhand 2001)

$$5. \quad \frac{d^2 y}{dx^2} - \frac{1}{x} \frac{dy}{dx} + 4x^2 y = x^4.$$

$$6. \quad x \frac{d^2 y}{dx^2} + (4x^2 - 1) \frac{dy}{dx} + 4x^3 y = 2x^3.$$

$$7. \quad (1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = 4 \cos \log(1+x).$$

(Gorakhpur 2010)

$$8. \quad \frac{d^2 y}{dx^2} - \cot x \frac{dy}{dx} - y \sin^2 x = 0.$$

9. $x^4 \frac{d^2 y}{dx^2} + 2x^3 \frac{dy}{dx} + n^2 y = 0$.
10. $(1+x^2)^2 \frac{d^2 y}{dx^2} + 2x(1+x^2) \frac{dy}{dx} + 4y = 0$.

Answers 3

1. $y = c_1 e^{\cos x} + c_2 e^{2 \cos x} + \frac{1}{6} e^{-\cos x}$
2. $y = c_1 e^{3e^x} + c_2 e^{e^x} - e^{2e^x}$
3. $y = c_1 \cos \{2e^{-x} (1+x)\} - c_2 \sin \{2e^{-x} (1+x)\} + e^{-x} (1+x)$
4. $y = c_1 \cos \left\{ \frac{\sqrt{(a^2 - x^2)}}{\sqrt{a}} \right\} + c_2 \sin \left\{ \frac{\sqrt{(a^2 - x^2)}}{\sqrt{a}} \right\}$
5. $y = c_1 \cos (x^2 + c_2) + \frac{1}{4} x^2$
6. $y = (c_1 + c_2 x^2) e^{-x^2} + \frac{1}{2}$
7. $y = c_1 \cos \{\log (1+x) + c_2\} + 2 \log (1+x) \cdot \sin \log (1+x)$
8. $y = c_1 e^{-\cos x} + c_2 e^{\cos x}$
9. $y = c_1 \cos \left(\frac{n}{x} + c_2 \right)$
10. $y = c_1 \cos (2 \tan^{-1} x) + c_2 \sin (2 \tan^{-1} x)$, or $y(1+x^2) = c_1(1-x^2) + 2c_2 x$.

7.6 Method of Variation of Parameters

Now we shall discuss an elegant but somewhat artificial method for finding the complete primitive of a linear equation whose complementary function is known.

Let the linear equation of second order be

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R. \quad \dots(1)$$

Suppose $y = Au + Bv$ is the complementary function of (1), where A and B are arbitrary constants and u and v are functions of x . Then $y = Au + Bv$ is the solution of

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = 0.$$

$$\therefore (Au_2 + Bv_2) + P(Au_1 + Bv_1) + Q(Au + Bv) = 0$$

$$\text{or } A(u_2 + Pu_1 + Qu) + B(v_2 + Pv_1 + Qv) = 0$$

$$\Rightarrow u_2 + Pu_1 + Qu = 0 \text{ and } v_2 + Pv_1 + Qv = 0. \quad \dots(2)$$

Now, suppose that

$$y = Au + Bv \quad \dots(3)$$

is the complete primitive of (1) where A and B are not constants but functions of x , so chosen that (1) will be satisfied.

Differentiating (3),

$$\frac{dy}{dx} = A_1 u + B_1 v + Au_1 + Bv_1.$$

Let us choose A and B such that

$$A_1 u + B_1 v = 0. \quad \dots(4)$$

Then $\frac{dy}{dx} = Au_1 + Bv_1$

and $\frac{d^2 y}{dx^2} = Au_2 + Bv_2 + A_1 u_1 + B_1 v_1.$

Putting these values in (1), we get

$$(Au_2 + Bv_2 + A_1 u_1 + B_1 v_1) + P(Au_1 + Bv_1) + Q(Au + Bv) = R$$

or $A(u_2 + Pu_1 + Qu) + B(v_2 + Pv_1 + Qv) + A_1 u_1 + B_1 v_1 = R$

or $A_1 u_1 + B_1 v_1 = R_1, \quad [\text{using 2}] \quad \dots(5)$

Solving (4) and (5), we get

$$A_1 = \frac{vR}{u_1 v - uv_1} \text{ and } B_1 = \frac{uR}{v_1 u - vu_1}, \text{ which on integration give } A \text{ and } B.$$

$$A = \int \frac{v R dx}{u_1 v - uv_1} + c_1 \text{ and } B = \int \frac{u R dx}{v_1 u - vu_1} + c_2,$$

where c_1 and c_2 are parameters.

Substituting these values of A and B in (3), we get the complete primitive of (1).

Note : Since the general solution is obtained by varying the arbitrary constants of the complementary function, therefore the method is known as *method of variation of parameters*.

Alternate method to find C.S:

Consider the linear equation of second order

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R. \quad \dots(1)$$

The P.I. of (1) is given by

$$\text{P.I.} = -u \int \frac{vR}{W} dx + v \int \frac{uR}{W} dx = -u \int \frac{vR}{uv_1 - vu_1} dx + v \int \frac{uR}{uv_1 - vu_1} dx, \quad \dots(2)$$

where u and v are defined by the C.F. of

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = 0 \quad \dots(3)$$

i.e., by $y = c_1 u + c_2 v$, where c_1 and c_2 are arbitrary constants

and $W = \begin{vmatrix} u & v \\ u_1 & v_1 \end{vmatrix}$ is called the *Wronskian* of u and v

$\therefore$ C. S. = C. F. + P. I.

Note: The method of variation of parameters must be used, if instructed to do so in the question.

Illustrative Examples

Example 20: Solve by the method of variation of parameters

$$\frac{d^2 y}{dx^2} + 4y = 4 \tan 2x.$$

(Rohilkhand 2001, 11; Lucknow 09; Avadh 05; Meerut 04B)

Solution: The C.F. of the given equation i.e., the solution of the equation

$$\frac{d^2 y}{dx^2} + 4y = 0 \text{ is}$$

$$y = c_1 \cos 2x + c_2 \sin 2x, \text{ where } c_1 \text{ and } c_2 \text{ are constants.}$$

$$\text{Let } y = A \cos 2x + B \sin 2x \quad \dots(1)$$

be the complete primitive of the given equation where A and B are functions of x , so chosen that the given equation will be satisfied.

$$\text{Then } \frac{dy}{dx} = -2A \sin 2x + 2B \cos 2x + \frac{dA}{dx} \cos 2x + \frac{dB}{dx} \sin 2x.$$

Let us choose A and B such that

$$\frac{dA}{dx} \cos 2x + \frac{dB}{dx} \sin 2x = 0. \quad \dots(2)$$

$$\text{Then } \frac{dy}{dx} = -2A \sin 2x + 2B \cos 2x$$

$$\text{and } \frac{d^2 y}{dx^2} = -2 \frac{dA}{dx} \sin 2x + 2 \frac{dB}{dx} \cos 2x - 4A \cos 2x - 4B \sin 2x.$$

Putting these values in the given equation, we get

$$-2 \sin 2x \frac{dA}{dx} + 2 \cos 2x \frac{dB}{dx} = 4 \tan 2x$$

$$\text{or } -\sin 2x \frac{dA}{dx} + \cos 2x \frac{dB}{dx} = 2 \tan 2x. \quad \dots(3)$$

Solving (2) and (3), we get

$$\frac{dA}{dx} = -\frac{2 \sin^2 2x}{\cos 2x}, \quad \frac{dB}{dx} = 2 \sin 2x.$$

Integrating these, we get

$$A = -2 \int \frac{(1 - \cos^2 2x)}{\cos 2x} dx + c_1$$

$$= -2 \int (\sec 2x - \cos 2x) dx + c_1$$

$$= -\log (\sec 2x + \tan 2x) + \sin 2x + c_1$$

and

$$B = -\cos 2x + c_2.$$

Putting the values of A and B in (1), the complete primitive of the given equation is

$$y = c_1 \cos 2x + c_2 \sin 2x - [\log (\sec 2x + \tan 2x)] \cdot \cos 2x.$$

Alternate method to find P.I.:

Here $u = \cos 2x$, $v = \sin 2x$, $R = 4 \tan 2x$

$$\text{and } W = \begin{vmatrix} u & v \\ u_1 & v_1 \end{vmatrix} = \begin{vmatrix} \cos 2x & \sin 2x \\ -2 \sin 2x & 2 \cos 2x \end{vmatrix} = 2.$$

$$\text{Thus, P.I.} = -u \int \frac{vR}{W} dx + v \int \frac{uR}{W} dx$$

$$= -\cos 2x \int 2 \sin 2x \tan 2x dx + \sin 2x \int 2 \cos 2x \tan 2x dx$$

$$= -2 \cos 2x \int (\sec 2x - \cos 2x) dx + 2 \sin 2x \int \sin 2x dx$$

$$= -\cos 2x [\log (\sec 2x + \tan 2x) - \sin 2x] - \sin 2x \cos 2x.$$

$$= -\cos 2x \log (\sec 2x + \tan 2x).$$

Hence, the C.S. is $y = c_1 \cos 2x + c_2 \sin 2x - (\cos 2x) \log (\sec 2x + \tan 2x)$.

Example 21: (i) Solve by the method of variation of parameters

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = e^x \log x.$$

Solution: First we shall find the C.F. of the given equation i.e., the solution of the equation

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0 \quad \text{or} \quad (D^2 - 2D + 1) y = 0.$$

$$\text{A.E. is } m^2 - 2m + 1 = 0 \quad \text{or} \quad (m - 1)^2 = 0, \text{ giving } m = 1, 1.$$

$\therefore$

$$\text{C.F. is } y = (c_1 + c_2 x) e^x.$$

Now assume A and B as the functions of x in such a way that the given equation is satisfied by

$$y = (A + Bx) e^x = Ae^x + Bxe^x. \quad \dots(1)$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= Ae^x + B(e^x + xe^x) + e^x \frac{dA}{dx} + xe^x \frac{dB}{dx} \\ &= Ae^x + B(e^x + xe^x), \text{ assuming } e^x \frac{dA}{dx} + xe^x \frac{dB}{dx} = 0. \quad \dots(2) \end{aligned}$$

$$\therefore \frac{d^2 y}{dx^2} = Ae^x + B(2e^x + xe^x) + e^x \frac{dA}{dx} + (e^x + xe^x) \frac{dB}{dx}.$$

Now putting these values in the given equation, we get

$$e^x \frac{dA}{dx} + e^x (1+x) \frac{dB}{dx} = e^x \log x. \quad \dots(3)$$

Solving (2) and (3), we get

$$\frac{dA}{dx} = -x \log x \text{ and } \frac{dB}{dx} = \log x.$$

Integrating these, we get

$$\begin{aligned} A &= - \int x \log x \, dx + c_1 = - \left(\frac{x^2}{2} \log x - \int \frac{1}{x} \cdot \frac{x^2}{2} \, dx \right) + c_1 \\ &= - \left(\frac{x^2}{2} \log x - \frac{x^2}{4} \right) + c_1, \end{aligned}$$

$$B = \int \log x \, dx + c_2 = x \log x - \int \frac{1}{x} \cdot x \, dx + c_2 = x \log x - x + c_2.$$

Putting these values of A and B in (1), the general solution of the given equation is

$$\begin{aligned} y &= -e^x \left(\frac{x^2}{2} \log x - \frac{x^2}{4} + c_1 \right) + x e^x (x \log x - x + c_2) \\ &= (c_1 + c_2 x) e^x + \frac{1}{4} x^2 e^x (2 \log x - 3). \end{aligned}$$

Alternate method to find P.I.

Here $u = e^x$, $v = x e^x$, $R = e^x \log x$

and
$$W = \begin{vmatrix} u & v \\ u_1 & v_1 \end{vmatrix} = \begin{vmatrix} e^x & x e^x \\ e^x & (1+x) e^x \end{vmatrix} = e^{2x}.$$

Thus
$$\begin{aligned} \text{P.I.} &= -u \int \frac{uR}{W} \, dx + v \int \frac{uR}{W} \, dx \\ &= -e^x \int \frac{x e^x \cdot e^x \log x}{e^{2x}} \, dx + x e^x \int \frac{e^x \cdot e^x \log x}{e^{2x}} \, dx \\ &= -e^x \int x \log x \, dx + x e^x \int \log x \, dx \\ &= -e^x \left(\frac{x^2}{2} \log x - \int \frac{1}{x} \cdot \frac{x^2}{2} \, dx \right) + x \cdot e^x \left(x \log x - \int \frac{1}{x} \cdot x \, dx \right) \\ &= -e^x \left(\frac{x^2}{2} \log x - \frac{x^2}{4} \right) + x e^x (x \log x - x) = \frac{1}{4} x^2 e^x (2 \log x - 3). \end{aligned}$$

Hence, the C.S. is $y = (c_1 + c_2 x) e^x + \frac{1}{4} x^2 e^x (2 \log x - 3).$

Example 22: (ii) Solve by the method of variation of parameters

$$\frac{d^2 y}{dx^2} + n^2 y = \sec nx.$$

(Meerut 2003, 07; Gorakhpur 11; Lucknow 10; Avadh 09, 10, 13)

Solution: The C.F. of the given equation i.e. the solution of the equation

$$\frac{d^2 y}{dx^2} + n^2 y = 0 \text{ is}$$

$y = c_1 \cos nx + c_2 \sin nx$, where c_1 and c_2 are constants.

$$\text{Let } y = A \cos nx + B \sin nx \quad \dots(1)$$

be the general solution of the given equation where A and B are functions of x , so chosen that the given equation will be satisfied.

$$\text{Then } \frac{dy}{dx} = -An \sin nx + Bn \cos nx + \frac{dA}{dx} \cos nx + \frac{dB}{dx} \sin nx.$$

Let us choose A and B such that

$$\frac{dA}{dx} \cos nx + \frac{dB}{dx} \sin nx = 0 \quad \dots(2)$$

$$\text{Then } \frac{dy}{dx} = -An \sin nx + Bn \cos nx$$

$$\text{and } \frac{d^2 y}{dx^2} = -An^2 \cos nx - Bn^2 \sin nx - n \frac{dA}{dx} \sin nx + n \frac{dB}{dx} \cos nx.$$

Putting these values in the given equation, we get

$$-n \frac{dA}{dx} \sin nx + n \frac{dB}{dx} \cos nx = \sec nx. \quad \dots(3)$$

Solving (2) and (3), we get

$$\frac{dA}{dx} = -\frac{1}{n} \tan nx \quad \text{and} \quad \frac{dB}{dx} = \frac{1}{n}.$$

Integrating these, we get

$$A = \frac{1}{n^2} \log \cos nx + c_1, \quad B = \frac{x}{n} + c_2.$$

Putting these values of A and B in (1), the general solution of the given equation is

$$y = c_1 \cos nx + c_2 \sin nx + \frac{1}{n^2} \cos nx \log \cos nx + \frac{x}{n} \sin nx.$$

Example 23: Apply the method of variation of parameters to solve

$$\frac{d^2 y}{dx^2} + y = \tan x.$$

Solution: The given differential equation is

$$\frac{d^2 y}{dx^2} + y = \tan x \quad \text{or} \quad (D^2 + 1)y = \tan x \quad \dots(1)$$

$$\text{A.E. is } m^2 = -1 \quad \text{or} \quad m = \pm i$$

$$\therefore \text{C.F. is } y = A \cos x + B \sin x$$

Here $u = \cos x$, $v = \sin x$, $R = \tan x$

$$\text{and } W = u_1 v_2 - u_2 v_1 = \cos^2 x + \sin^2 x = 1$$

$$\therefore \text{P.I.} = - \int \frac{v R}{W} dx + \int \frac{u R}{W} dx.$$

$$\begin{aligned} \text{We have } \int \frac{-vR}{W} dx &= - \int \frac{\sin x \tan x}{1} dx = - \int \frac{\sin^2 x}{\cos x} dx = - \int \frac{1 - \cos^2 x}{\cos x} dx \\ &= \int (\cos x - \sec x) dx = \sin x - \log (\sec x + \tan x) \end{aligned}$$

$$\text{and } \int \frac{uR}{W} dx = \int \frac{\cos x \cdot \tan x}{1} dx = \int \sin x dx = -\cos x$$

$$\begin{aligned} \therefore \text{P.I.} &= [\sin x - \log (\sec x + \tan x)] \cos x - \cos x \sin x \\ &= -\cos x \log (\sec x + \tan x) \end{aligned}$$

Hence the complete solution is $y = A \cos x + B \sin x - \cos x \log (\sec x + \tan x)$

Example 24: Solve by the method of variation of parameters

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^2 e^x.$$

(Rohilkhand 2006, 11; Meerut 07;

Gorakhpur 06, 10; Purvanchal 06; Lucknow 08, 11; Avadh 14)

Solution: First we shall find the C.F. of the given equation *i.e.*, the solution of the equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 0. \quad \dots(1)$$

This is a homogeneous linear equation. Putting $x = e^z$, the differential equation (1) becomes

$$\{D(D-1) + D - 1\} y = 0, \text{ where } D \equiv \frac{d}{dz}$$

$$\text{or } (D^2 - 1) y = 0$$

whose solution is $y = c_1 e^z + c_2 e^{-z}$.

$$\therefore \text{ solution of (1) is } y = c_1 x + c_2 \frac{1}{x}.$$

$$\text{Let } y = Ax + \frac{B}{x} \quad \dots(2)$$

be the complete primitive of the given equation where A and B are functions of x , so chosen that the given equation will be satisfied.

$$\text{Then } \frac{dy}{dx} = A - \frac{B}{x^2} + \frac{dA}{dx} x + \frac{1}{x} \cdot \frac{dB}{dx}.$$

Let us choose A and B such that

$$x \frac{dA}{dx} + \frac{1}{x} \cdot \frac{dB}{dx} = 0. \quad \dots(3)$$

$$\text{Then } \frac{dy}{dx} = A - \frac{B}{x^2} \quad \text{and} \quad \frac{d^2 y}{dx^2} = \frac{dA}{dx} - \frac{1}{x^2} \frac{dB}{dx} + \frac{2}{x^3} B.$$

Putting these values in the given equation, we get

$$\frac{dA}{dx} - \frac{1}{x^2} \frac{dB}{dx} = e^x. \quad \dots(4)$$

Solving (3) and (4), we get

$$\frac{dA}{dx} = \frac{1}{2} e^x \text{ and } \frac{dB}{dx} = -\frac{x^2}{2} e^x.$$

Integrating these, we get

$$A = \frac{1}{2} e^x + c_1 \quad \text{and} \quad B = -\frac{1}{2} x^2 e^x + x e^x - e^x + c_2.$$

Putting the values of A and B in (2), the complete primitive of the given equation is

$$y = c_1 x + \frac{c_2}{x} + e^x - \frac{1}{x} e^x.$$

Example 25: Solve by the method of variation of parameters

$$(1-x) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = (1-x)^2.$$

Solution: First we shall find the C.F. of the given equation *i.e.*, the solution of the equation

$$(1-x) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 0$$

$$\text{or} \quad \frac{d^2 y}{dx^2} + \frac{x}{1-x} \frac{dy}{dx} - \frac{y}{1-x} = 0. \quad \dots(1)$$

Here $P + Qx = 0$.

$\therefore$ $y = x$ is a part of the C.F.

Putting $y = vx$ in (1), the reduced equation is

$$\frac{d^2 v}{dx^2} + \left(\frac{x}{1-x} + \frac{2}{x} \right) \frac{dv}{dx} = 0$$

$$\text{or} \quad \frac{dp}{dx} + \left(\frac{x}{1-x} + \frac{2}{x} \right) p = 0, \text{ where } p = \frac{dv}{dx}$$

$$\text{or} \quad \frac{dp}{dx} + \left(-1 - \frac{1}{x-1} + \frac{2}{x} \right) p = 0 \quad \text{or} \quad \frac{dp}{p} = \left(1 + \frac{1}{x-1} - \frac{2}{x} \right) dx.$$

Integrating, $\log p = x + \log(x-1) - 2 \log x + \log c_1$

$$\text{or} \quad p = \frac{dv}{dx} = \frac{c_1 (x-1) e^x}{x^2} = c_1 \left(\frac{1}{x} - \frac{1}{x^2} \right) e^x.$$

Integrating again, $v = c_1 \int \left(\frac{1}{x} - \frac{1}{x^2} \right) e^x dx + c_2$

$$\text{or} \quad v = \frac{c_1}{x} e^x + c_2.$$

$\therefore$ solution of (1) is $y = vx = c_1 e^x + c_2 x$.

$$\text{Let} \quad y = Ae^x + Bx \quad \dots(2)$$

be the complete primitive of the given equation where A and B are functions of x , so chosen that the given equation will be satisfied.

$$\text{Then } \frac{dy}{dx} = Ae^x + B + e^x \frac{dA}{dx} + x \frac{dB}{dx}.$$

Let us choose A and B such that

$$e^x \frac{dA}{dx} + x \frac{dB}{dx} = 0. \quad \dots(3)$$

$$\text{Then } \frac{dy}{dx} = Ae^x + B$$

$$\text{and } \frac{d^2y}{dx^2} = e^x \frac{dA}{dx} + \frac{dB}{dx} + e^x A.$$

Putting these values in the given equation, we get

$$e^x \frac{dA}{dx} + \frac{dB}{dx} = 1 - x. \quad \dots(4)$$

Solving (3) and (4), we get

$$\frac{dA}{dx} = -xe^{-x} \text{ and } \frac{dB}{dx} = 1.$$

Integrating these, we get

$$A = (1+x)e^{-x} + c_1 \text{ and } B = x + c_2.$$

Putting the values of A and B in (2) the general solution of the given equation is

$$y = c_1 e^x + c_2 x + 1 + x + x^2.$$

Example 26: Solve by the method of variation of parameters

$$\frac{d^2y}{dx^2} + (1 - \cot x) \frac{dy}{dx} - y \cot x = \sin^2 x.$$

Solution: First we shall find the C.F. of the given equation i.e., the solution of the equation

$$\frac{d^2y}{dx^2} + (1 - \cot x) \frac{dy}{dx} - y \cot x = 0. \quad \dots(1)$$

Here $1 - P + Q = 0$, $\therefore y = e^{-x}$ is a part of the C.F.

Putting $y = ve^{-x}$, the equation (1) reduces to

$$\frac{d^2v}{dx^2} - (1 + \cot x) \frac{dv}{dx} = 0$$

$$\text{or } \frac{dp}{dx} - (1 + \cot x) p = 0, \text{ where } p = \frac{dv}{dx}$$

$$\text{or } \frac{dp}{p} = (1 + \cot x) dx.$$

Integrating, $\log p = x + \log \sin x + \log c_1$.

$$\therefore p = \frac{dv}{dx} = c_1 e^x \sin x$$

$$\text{or } v = c_1 \int e^x \sin x dx + c_2 = c_1 \cdot \frac{1}{2} e^x (\sin x - \cos x) + c_2.$$

$\therefore$ the solution of (1) i.e., the C.F. of the given equation is

$$y = ve^{-x} = c_1 \cdot \frac{1}{2} (\sin x - \cos x) + c_2 e^{-x}.$$

Let $y = A (\sin x - \cos x) + Be^{-x}$... (2)

be the complete solution of the given equation where A and B are functions of x , so chosen that the given equation will be satisfied.

$$\therefore \frac{dy}{dx} = A (\cos x + \sin x) - Be^{-x} + \frac{dA}{dx} (\sin x - \cos x) + \frac{dB}{dx} e^{-x}.$$

Let us choose A and B such that

$$\frac{dA}{dx} (\sin x - \cos x) + \frac{dB}{dx} e^{-x} = 0. \quad \dots (3)$$

$$\therefore \frac{dy}{dx} = A (\cos x + \sin x) - Be^{-x}$$

and $\frac{d^2y}{dx^2} = \frac{dA}{dx} (\cos x + \sin x) - \frac{dB}{dx} e^{-x} + A (-\sin x + \cos x) + Be^{-x}.$

Putting these values in the given equation, we get

$$\frac{dA}{dx} (\cos x + \sin x) - \frac{dB}{dx} e^{-x} = \sin^2 x. \quad \dots (4)$$

Solving (3) and (4), we get

$$\frac{dA}{dx} = \frac{1}{2} \sin x$$

and $\frac{dB}{dx} = \frac{1}{2} e^x (\sin x \cos x - \sin^2 x).$

Integrating these, we get

$$A = -\frac{1}{2} \cos x + c_1$$

and $B = \frac{1}{4} \int e^x (\sin 2x - 1 + \cos 2x) dx + c_2$

$$= \frac{1}{4} \cdot \frac{e^x}{5} (\sin 2x - 2 \cos 2x) - \frac{e^x}{4} + \frac{1}{4} \cdot \frac{e^x}{5} (\cos 2x + 2 \sin 2x) + c_2$$

$$= \frac{e^x}{20} (3 \sin 2x - \cos 2x) - \frac{e^x}{4} + c_2.$$

Putting the values of A and B in (2), the general solution of the given equation is

$$y = c_1 (\sin x - \cos x) + c_2 e^{-x} - \frac{1}{10} (\sin 2x - 2 \cos 2x).$$

Comprehensive Exercise 4

Solve the following differential equations by the method of variation of parameters:

1. $\frac{d^2y}{dx^2} + y = x.$ (Agra 2007)

2. $\frac{d^2y}{dx^2} + a^2y = \operatorname{cosec} ax.$

$$3. \quad x^2 \frac{d^2 y}{dx^2} - 2x(1+x) \frac{dy}{dx} + 2(x+1)y = x^3.$$

(Gorakhpur 2007)

$$4. \quad (1-x^2) \frac{d^2 y}{dx^2} - 4x \frac{dy}{dx} - (1+x^2)y = x.$$

$$5. \quad (1-x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x(1-x^2)^{3/2}.$$

$$6. \quad (x+2) \frac{d^2 y}{dx^2} - (2x+5) \frac{dy}{dx} + 2y = (x+1)e^x.$$

$$7. \quad \frac{d^2 y}{dx^2} + y = \operatorname{cosec} x.$$

(Purvanchal 2007, 10)

$$8. \quad \frac{d^2 y}{dx^2} - y = \frac{2}{1+e^x}.$$

(Agra 2008; Gorakhpur 05; Purvanchal 09; Kanpur 07, 14)

$$9. \quad xy_1 - y = (x-1)(y_2 - x + 1).$$

Answers 4

$$1. \quad y = c_1 \cos x + c_2 \sin x + x$$

$$2. \quad y = c_1 \cos ax + c_2 \sin ax - \frac{x}{a} \cos ax + \frac{1}{a^2} \sin ax \log \sin ax$$

$$3. \quad y = c_1 x + c_2 x e^{2x} - \frac{1}{2} x^2 - \frac{1}{4} x$$

$$4. \quad y = \frac{1}{(1-x^2)} \cdot (c_1 \cos x + c_2 \sin x + x)$$

$$5. \quad y = c_1 [\sqrt{1-x^2} + x \sin^{-1} x] + c_2 x - (x/9)(1-x^2)^{3/2}$$

$$6. \quad y = c_1 (2x+5) + c_2 e^{2x} - e^x$$

$$7. \quad y = c_1 \cos x + c_2 \sin x - x \cos x + \sin x \log \sin x$$

$$8. \quad y = c_1 e^x + c_2 e^{-x} + e^x \log \left(\frac{1+e^x}{e^x} \right) - 1 - e^{-x} \log (1+e^x)$$

$$9. \quad y = c_1 e^x + c_2 x - (1+x^2)$$

7.7 Method of Operational Factors

In some cases the method of operational factors easily solves the equation.

Let the linear equation of second order

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R,$$

be expressed in the form $f(D)y = R$

or $f_1(D) f_2(D) y = R$

i.e., $f(D)$ can be resolved into a product of two factors $f_1(D)$ and $f_2(D)$, such that, if $f_2(D)$ operates upon y , and then $f_1(D)$ operates upon the result of the operation, the same result is obtained as if $f(D)$ operates upon y .

Note: With the exception of the classes of equations, the factors of $f(D)$ are generally not commutative. Hence a great care is to be taken in writing them in the right order.

Illustrative Examples

Example 27: Solve $x \frac{d^2 y}{dx^2} + (1-x) \frac{dy}{dx} - y = e^x$.

Solution: In the symbolic form, the given equation is

$$[xD^2 + (1-x)D - 1]y = e^x, \text{ where } D \equiv d/dx$$

$$\text{or } (xD + 1)(D - 1)y = e^x. \quad \dots(1)$$

[Note that here the factors are not commutative because on expansion $\{(D-1)(xD+1)\}y$ gives $\{xD^2 + (2-x)D - 1\}y$].

$$\text{Let } (D-1)y = v. \quad \dots(2)$$

Then (1) gives $(xD+1)v = e^x$.

$$\text{or } x \frac{dv}{dx} + v = e^x \quad \text{or} \quad \frac{dv}{dx} + \frac{1}{x}v = \frac{e^x}{x},$$

which is a linear differential equation of the first order.

$$\text{Now I.F.} = e^{\int (1/x) dx} = e^{\log x} = x.$$

$$\therefore vx = \int x \frac{e^x}{x} dx + c_1 = e^x + c_1 \quad \text{or} \quad v = e^x x^{-1} + c_1 x^{-1}.$$

Putting the value of v in (2), we get

$$(D-1)y = e^x x^{-1} + c_1 x^{-1}$$

$$\text{or } \frac{dy}{dx} - y = e^x x^{-1} + c_1 x^{-1}, \text{ which is again linear.}$$

$$\text{Here I.F.} = e^{\int -dx} = e^{-x}.$$

$$\begin{aligned} \therefore ye^{-x} &= \int (e^x x^{-1} + c_1 x^{-1}) e^{-x} dx + c_2 \\ &= \int \frac{1}{x} dx + c_1 \int \frac{e^{-x}}{x} dx + c_2 = \log x + c_1 \int \frac{e^{-x}}{x} dx + c_2. \end{aligned}$$

$$\therefore y = e^x \log x + c_1 e^x \int \frac{e^{-x}}{x} dx + c_2 e^x,$$

which is the required general solution of the given differential equation.

Example 28: Solve $3x^2 \frac{d^2 y}{dx^2} + (2 + 6x - 6x^2) \frac{dy}{dx} - 4y = 0$.

Solution: In the symbolic form, the given equation is

$$[3x^2 D^2 + (2 + 6x - 6x^2) D - 4] y = 0$$

$$\text{or} \quad [(3x^2 D^2 + 6x D + 2D) - (6x^2 D + 4)] y = 0$$

$$\text{or} \quad [D(3x^2 D + 2) - 2(3x^2 D + 2)] y = 0$$

$$\text{or} \quad (D - 2)(3x^2 D + 2) y = 0. \quad \dots(1)$$

[Here factors are not commutative because on expansion

$$(3x^2 D + 2)(D - 2) y \text{ gives } \{3x^2 D^2 + (2 - 6x^2) D - 4\} y].$$

$$\text{Let} \quad (3x^2 D + 2) y = v. \quad \dots(2)$$

$$\text{Then (1) gives } (D - 2)v = 0 \Rightarrow \frac{dv}{dx} - 2v = 0$$

$$\frac{dv}{dx} = 2v \Rightarrow v = c_1 e^{2x}.$$

Putting the value of v in (2), we get

$$(3x^2 D + 2) y = c_1 e^{2x}$$

$$\text{or} \quad 3x^2 \frac{dy}{dx} + 2y = c_1 e^{2x} \Rightarrow \frac{dy}{dx} + \frac{2}{3x^2} y = \frac{c_1}{3x^2} e^{2x}.$$

$$\text{I.F.} = e^{\int \frac{2}{3x^2} dx} = e^{\frac{2}{3}(-\frac{1}{x})} = e^{-\frac{2}{3x}}.$$

$$\therefore y e^{-\frac{2}{3x}} = \frac{c_1}{3} \int x^{-2} e^{2x} \cdot e^{-\frac{2}{3x}} dx + c_2$$

$$\text{or} \quad y = c_2 e^{\frac{2}{3x}} + \frac{c_1}{3} e^{\frac{2}{3x}} \int x^{-2} e^{2x - \frac{2}{3x}} dx,$$

which is the required solution.

Example 29: Solve $x \frac{d^2 y}{dx^2} - (x + 2) \frac{dy}{dx} + 2y = x^3$.

Solution: In the symbolic form, the given equation is

$$[xD^2 - (x + 2) D + 2] y = x^3$$

$$\text{or} \quad (xD - 2)(D - 1) y = x^3 \quad \dots(1)$$

[Here factors are not commutative because on expansion

$$\{(D - 1)(xD - 2)\} y \text{ gives } \{xD^2 - (x + 1) D + 2\} y].$$

$$\text{Let} \quad (D - 1) y = v. \quad \dots(2)$$

$$\text{Then (1) gives } (xD - 2)v = x^3$$

$$\text{or} \quad x \frac{dv}{dx} - 2v = x^3$$

or $\frac{dv}{dx} - \frac{2}{x} v = x^2$, which is linear.

$$\text{I.F.} = e^{\int (-2/x) dx} = e^{-2 \log x} = \frac{1}{x^2}.$$

$$\therefore v \cdot \frac{1}{x^2} = \int \left(x^2 \cdot \frac{1}{x^2} \right) dx + c_1 = x + c_1.$$

$$\therefore v = x^3 + c_1 x^2.$$

Putting the value of v in (2), we get

$$(D-1)y = x^3 + c_1 x^2$$

or $\frac{dy}{dx} - y = x^3 + c_1 x^2$, which is again linear.

$$\text{I.F.} = e^{\int -dx} = e^{-x}.$$

$$\begin{aligned} \therefore y e^{-x} &= \int e^{-x} (x^3 + c_1 x^2) dx + c_2 \\ &= (x^3 + c_1 x^2) e^{-x} - (3x^2 + 2c_1 x) e^{-x} - (6x + 2c_1) e^{-x} - 6e^{-x} + c_2 \end{aligned}$$

$$\begin{aligned} \text{or } y &= (x^3 + c_1 x^2) - (3x^2 + 2c_1 x) - (6x + 2c_1) - 6 + c_2 e^x \\ &= -x^3 - (c_1 + 3)x^2 - 2(c_1 + 3)x - 2(c_1 + 3) + c_2 e^x \end{aligned}$$

$$\text{or } y = -x^3 - (c_1 + 3)\{x^2 + 2x + 2\} + c_2 e^x.$$

which is the required general solution of the given differential equation.

Example 30: Solve $[(x+3)D^2 - (2x+7)D + 2]y = (x+3)^2 e^x$.

Solution: The given equation can be written as

$$\{(x+3)D - 1\}(D-2)y = (x+3)^2 e^x. \quad \dots(1)$$

[Here factors are not commutative because on expansion

$$(D-2)\{(x+3)D - 1\}y \text{ gives } \{(x+3)D^2 - (2x+5)D + 2\}y]$$

$$\text{Let } (D-2)y = v. \quad \dots(2)$$

Then (1) gives $\{(x+3)D - 1\}v = (x+3)^2 e^x$

$$\text{or } (x+3)\frac{dv}{dx} - v = (x+3)^2 e^x,$$

$$\text{or } \frac{dv}{dx} - \frac{1}{x+3}v = (x+3)e^x, \text{ which is linear.}$$

$$\text{I.F.} = e^{-\int \frac{dx}{x+3}} = e^{-\log(x+3)} = \frac{1}{x+3}.$$

$$\therefore v \cdot \frac{1}{x+3} = \int \frac{1}{x+3} \cdot (x+3)e^x dx + c_1 = e^x + c_1.$$

$$\therefore v = e^x(x+3) + c_1(x+3).$$

Putting the value of v in (2), we get

$$(D - 2) y = e^x (x + 3) + c_1 (x + 3)$$

or $\frac{dy}{dx} - 2y = e^x (x + 3) + c_1 (x + 3)$, which is again linear.

$$\text{I. F.} = e^{\int -2 dx} = e^{-2x}.$$

$$\begin{aligned} \therefore y e^{-2x} &= \int e^{-2x} (x + 3) dx + c_1 \int e^{-2x} (x + 3) dx + c_2 \\ &= - (x + 3) e^{-x} - e^{-x} - \frac{1}{2} c_1 (x + 3) e^{-2x} - \frac{1}{4} c_1 e^{-2x} + c_2 \end{aligned}$$

$$\text{or } y = - (x + 3) e^x - e^x - \frac{1}{2} c_1 (x + 3) - \frac{1}{4} c_1 + c_2 e^{2x}$$

$$\text{or } y = - x e^x - 4 e^x - \frac{1}{4} c_1 (2x + 7) + c_2 e^{2x}$$

$$\text{or } y = - x e^x - 4 e^x + A (2x + 7) + B e^{2x},$$

which is the required general solution of the given differential equation.

Comprehensive Exercise 5

Solve the following equations by the method of factorization of the operator:

1. $x \frac{d^2 y}{dx^2} + (x - 2) \frac{dy}{dx} - 2y = x^3.$
2. $x \frac{d^2 y}{dx^2} + (x - 1) \frac{dy}{dx} - y = x^2.$
3. $3x^2 D^2 y + (2 - 6x^2) Dy - 4y = 0.$
4. $x^2 \frac{d^2 y}{dx^2} + \frac{dy}{dx} - (1 + x^2) y = e^{-x}.$
5. $[xD^2 + (1 - x)D - 2(1 + x)] y = e^{-x} (1 - 6x).$

Answers 5

1. $y = x^3 + a (x^2 - 2x + 2) + b e^{-x}$
2. $y = c_1 (x - 1) + c_2 e^{-x} + x^2$
3. $y = c_1 e^{2x} \int e^{(-2x)+2/(3x)} dx + c_2 e^{2x}$

$$4. \quad y = c_1 e^x \int e^{-2x + (1/x)} dx + c_2 e^x - \frac{1}{2} e^{-x}$$

$$5. \quad y = x e^{-x} + c_1 e^{2x} \int \frac{1}{x} e^{-3x} dx + c_2 e^{2x}$$

Objective Type Questions

Fill in The Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. $y = x^m$ is a part of C.F. of equation

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R \text{ if }$$

(Meerut 2003)

2. $y = e^x$ is a solution of

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R \text{ if }$$

3. $y = e^{-x}$ is a part of C.F. of

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R \text{ if }$$

4. $y = e^{mx}$ is a part of C.F. of

$$\left(\frac{d^2 y}{dx^2} \right) + P \left(\frac{dy}{dx} \right) + Qy = R \text{ if }$$

5. $y = x$ is a solution of

$$\left(\frac{d^2 y}{dx^2} \right) + P \left(\frac{dy}{dx} \right) + Qy = R \text{ if }$$

6. $y = x^2$ is a part of C.F. of equation

$$\left(\frac{d^2 y}{dx^2} \right) + P \left(\frac{dy}{dx} \right) + Qy = R \text{ if }$$

True or False

Write 'T' for true and 'F' for false statement.

1. $y = x^2$ is a part of C.F. of the solution of equation

$$(1-x) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = (1-x)^2.$$

2. $y = e^x$ is a part of C.F. of the solution of equation

$$x \frac{d^2 y}{dx^2} - (2x - 1) \frac{dy}{dx} + (x - 1) y = 0.$$

3. $y = x$ is a part of C.F. of the solution of equation

$$x \frac{dy}{dx} - y = (x - 1) \left(\frac{d^2 y}{dx^2} - x + 1 \right).$$

4. $\frac{d^2 y}{dx^2} + \left(1 + \frac{2}{x} \cot x - \frac{2}{x^2} \right) y = x \cos x$ is a linear differential equation of second order.

Answers

Fill in the Blank(s)

- | | |
|--------------------------------|-------------------------|
| 1. $m(m - 1) + Pmx + Qx^2 = 0$ | 2. $1 + P + Q = 0$ |
| 3. $1 - P + Q = 0$ | 4. $m^2 + Pm + Q = 0$ |
| 5. $P + Qx = 0$ | 6. $2 + 2Px + Qx^2 = 0$ |

True or False

- | | |
|--------|--------|
| 1. F | 2. T |
| 3. T | 4. T |

7.8 Guidelines of the Procedure for the Solution of Linear Differential Equations of Second Order

Before closing this chapter we give the guidelines of the procedure to be applied for the solution of linear differential equation of the second order.

1. First of all put the differential equation in the standard form

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R.$$

2. Then try to find by inspection an integral belonging to the complementary function of the given differential equation. If it is found, then proceed as in article 7.2.
3. If a part of the C.F. is not found by inspection, then find the value of $Q - \frac{1}{2} \frac{dP}{dx} - \frac{1}{4} P^2$. If it is a constant or a constant divided by x^2 , then the normal form of the equation is easily integrable and so proceed as in article 7.4.

4. In case the methods given in (2) and (3) do not succeed, the method of change of independent variable given in article 7.5 may be tried.

In this method we often choose z such that $\frac{Q}{(dz/dx)^2} = \text{some suitable constant}$.

5. In some cases the method of operational factors solves the equation easily.
6. If the students are instructed in the question to solve the equation by the method of variation of parameters, then only the method of variation of parameters must be applied.



Chapter

8



Partial Differential Equations of The First Order

8.1 Partial Differential Equations

E*quations which contain one or more partial derivatives are called* **Partial Differential Equations**. Such equations arise in geometry and physics when the number of independent variables in the problem under consideration is two or more. Whenever we consider the case of two independent variables, x and y will usually be taken as the independent variables and z as the dependent variable. The partial differential coefficients $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$ will be denoted by p and q respectively. The second order partial

derivatives are denoted by r, s, t , so that

$$\frac{\partial^2 z}{\partial x^2} = r, \quad \frac{\partial^2 z}{\partial x \partial y} = s, \quad \frac{\partial^2 z}{\partial y^2} = t.$$

Some examples of partial differential equations are :

$$pz - qz = z^2 + (x + y)^2, \quad \dots(1)$$

$$p \tan x + q \tan y = \tan z, \quad \dots(2)$$

$$p^2 + q^2 - 2px - 2qy + 2xy = 0, \quad \dots(3)$$

$$p^2 - qy^2 = y^2 - x^2, \quad \dots(4)$$

$$r - 2s + t = \sin(2x + 3y), \quad \dots(5)$$

$$y^2 \frac{\partial^2 z}{\partial x^2} - 2xy \frac{\partial^2 z}{\partial x \partial y} + x^2 \frac{\partial^2 z}{\partial y^2} = \frac{y^2}{x} \frac{\partial z}{\partial x} + \frac{x^2}{y} \frac{\partial z}{\partial y}. \quad \dots(6)$$

Order and Degree of a Partial Differential equation.

As in the case of ordinary differential equations, we define the **order** of a partial differential equation to be the order of the highest order derivative occurring in the equation.

The **degree** of a partial differential equation is the degree of the highest order derivative which occurs in it after the differential equation has been rationalised (i.e., made free from radicals and fractions so far as derivatives are concerned).

In the above mentioned examples equations (1) and (2) are of first order and first degree.

Equations (3) and (4) are of first order and second degree. Equations (5) and (6) are of second order and first degree.

In the present chapter we shall discuss the partial differential equations of the first order.

8.2 Derivation of a Partial Differential Equation

Partial differential equations can be derived in two ways :

- (a) By the elimination of arbitrary constants from a relation between x , y and z
- (b) By the elimination of arbitrary functions of these variables.

Now we illustrate these methods.

(a) By the elimination of arbitrary constants:

Let z be a function of x and y such that

$$f(x, y, z, a, b) = 0, \quad \dots(1)$$

where a and b denote arbitrary constants.

Differentiating (1) partially with respect to x and y , we get the relations

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial x} = 0 \quad \text{i.e.,} \quad \frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} p = 0 \quad \dots(2)$$

and
$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial y} = 0 \quad \text{i.e.,} \quad \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} q = 0. \quad \dots(3)$$

By means of the three equations (1), (2) and (3) two constants a and b can be eliminated and we obtain a relation of the form

$$F(x, y, z, p, q) = 0. \quad \dots(4)$$

This shows that the system of surfaces (1) gives rise to a partial differential equation of the first order given by (4).

Note: We observe that if the number of constants to be eliminated is equal to the number of independent variables then the derived partial differential equation is of the first order. But if the number of constants to be eliminated is greater than the number of independent variables then, in general, the derived partial differential equations will be of the second or higher order.

(b) By the elimination of arbitrary functions:

Suppose we have a relation between x, y and z of the type

$$f(u, v) = 0, \quad \dots(1)$$

where u and v are known functions of x, y and z and f is an arbitrary function of u and v .

This relation can also be expressed in the form $u = \phi(v)$, where ϕ is arbitrary.

Differentiating (1) partially with respect to each of the independent variables x and y regarding z as dependent variable, we get

$$\frac{\partial f}{\partial u} \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial z} p \right) + \frac{\partial f}{\partial v} \left(\frac{\partial v}{\partial x} + \frac{\partial v}{\partial z} p \right) = 0 \quad \dots(2)$$

$$\text{and} \quad \frac{\partial f}{\partial u} \left(\frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} q \right) + \frac{\partial f}{\partial v} \left(\frac{\partial v}{\partial y} + \frac{\partial v}{\partial z} q \right) = 0. \quad \dots(3)$$

Eliminating $\frac{\partial f}{\partial u}$ and $\frac{\partial f}{\partial v}$ between (2) and (3), we get

$$\left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right) \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right) = \left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right) \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right)$$

$$\text{or} \quad \left(\frac{\partial u}{\partial y} \frac{\partial v}{\partial z} - \frac{\partial v}{\partial y} \frac{\partial u}{\partial z} \right) p + \left(\frac{\partial v}{\partial x} \frac{\partial u}{\partial z} - \frac{\partial u}{\partial x} \frac{\partial v}{\partial z} \right) q = \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x}$$

$$\text{or} \quad Pp + Qq = R \quad \dots(4)$$

$$\text{where} \quad P = \frac{\partial u}{\partial y} \frac{\partial v}{\partial z} - \frac{\partial v}{\partial y} \frac{\partial u}{\partial z} = \frac{\partial(u, v)}{\partial(y, z)},$$

$$Q = \frac{\partial v}{\partial x} \frac{\partial u}{\partial z} - \frac{\partial u}{\partial x} \frac{\partial v}{\partial z} = \frac{\partial(u, v)}{\partial(z, x)},$$

$$\text{and} \quad R = \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} = \frac{\partial(u, v)}{\partial(x, y)}.$$

The equation (4) is a partial differential equation of the first order.

Note 1: If the given relation between x, y, z contains two arbitrary functions then the derived partial differential equation will contain partial derivatives of an order higher than two, except in particular cases.

Note 2: It should be observed that the partial differential equation (4) derived in (b) is a linear equation *i.e.*, powers of p and q are both unity while the partial differential equation (4) derived in (a) need not be linear.

Illustrative Examples

Example 1: Form a partial differential equation by the elimination of the constants h and k from

$$(x-h)^2 + (y-k)^2 + z^2 = c^2. \quad \dots(1)$$

Solution: Differentiating (1) partially w.r.t. x and y , we get

$$x-h+zp=0 \quad \text{and} \quad y-k+zq=0.$$

Putting the values of $x-h$ and $y-k$ from the last two equations in the given equation (1), we get

$$z^2 (p^2 + q^2 + 1) = c^2,$$

which is the required partial differential equation.

Example 2: Form a partial differential equation by eliminating a, b, c from

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1. \quad \dots(1)$$

Solution: Differentiating (1) partially w.r.t. x and y , we get

$$\frac{x}{a^2} + \frac{z}{c^2} p = 0 \quad \dots(2)$$

and $\frac{y}{b^2} + \frac{z}{c^2} q = 0. \quad \dots(3)$

Since the relations (1), (2) and (3) are not sufficient to eliminate the constants a, b and c so we require one more relation.

Differentiating (2) partially w.r.t. x , we get

$$\frac{1}{a^2} + \frac{p^2}{c^2} + \frac{z}{c^2} r = 0. \quad \dots(4)$$

Multiplying (4) by x and then subtracting (2) from it, we get

$$\frac{1}{c^2} \{xZR + xP^2 - pz\} = 0$$

or $pz = xP^2 + xZR.$

Thus after the elimination of a, b and c we obtain a partial differential equation of order 2.

Note: In this case one more partial differential equation can also be obtained.

Differentiating (3) partially w.r.t. y , we get

$$\frac{1}{b^2} + \frac{q^2}{c^2} + \frac{z}{c^2} t = 0.$$

Multiplying it by y and then subtracting (3) from it, we get

$$qz = yq^2 + yzt,$$

which is also a partial differential equation of order 2.

Example 3: Form a partial differential equation by eliminating the arbitrary function ϕ from

$$z = e^{ny} \phi(x - y). \quad \dots(1)$$

Solution: Differentiating (1) partially w.r.t. x and y , we get

$$p = e^{ny} \phi'(x - y), \quad \dots(2)$$

$$\text{and} \quad q = n e^{ny} \phi(x - y) - e^{ny} \phi'(x - y). \quad \dots(3)$$

From (1), (2) and (3), we get

$$q = nz - p \quad \text{or} \quad p + q = nz,$$

which is the required partial differential equation of order one.

Example 4: Form a partial differential equation by eliminating the functions f and F from

$$z = f(x + iy) + F(x - iy). \quad \dots(1)$$

Solution: Differentiating (1) partially w.r.t. x and y , we get

$$\frac{\partial z}{\partial x} = f'(x + iy) + F'(x - iy) \quad \dots(2)$$

$$\text{and} \quad \frac{\partial z}{\partial y} = if'(x + iy) - iF'(x - iy) \quad \dots(3)$$

Differentiating (2) and (3) partially w.r.t. x and y respectively, we get

$$\frac{\partial^2 z}{\partial x^2} = f''(x + iy) + F''(x - iy),$$

$$\text{and} \quad \frac{\partial^2 z}{\partial y^2} = -f''(x + iy) - F''(x - iy).$$

$$\text{Hence} \quad \frac{\partial^2 z}{\partial y^2} = -\frac{\partial^2 z}{\partial x^2} \quad \text{or} \quad \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0,$$

which is the required partial differential equation of the second order.

Comprehensive Exercise 1

Form partial differential equations by eliminating arbitrary constants from the following relations :

1. $z = (x + a)(y + b).$ (Lucknow 2011; Bundelkhand 13)

2. $z = ax + a^2 y^2 + b.$ (Lucknow 2008)

3. $z = ax e^y + \frac{1}{2} a^2 e^{2y} + b.$

4. $z = A e^{pt} \sin px.$

Form partial differential equations by eliminating the arbitrary functions from the following equations :

5. $z = f(y/x).$ (Kanpur 2010)

6. $z = f(x + ay) + F(x - ay)$. (Lucknow 2009)
7. $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$.
8. $f(x + y + z, x^2 + y^2 - z^2) = 0$. (Agra 2001)

Answers 1

1. $z = pq$ 2. $q = 2yp^2$ 3. $q = px + p^2$
4. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial t^2} = 0$ 5. $px + qy = 0$ 6. $\frac{\partial^2 z}{\partial y^2} = a^2 \frac{\partial^2 z}{\partial x^2}$
7. $x^2 \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 2y^2$ 8. $(y + z)p - (z + x)q = x - y$

8.3 Linear Partial Differential Equation

Definition: A partial differential equation is said to be **linear** if the dependent variable z and all its partial differential coefficients occur in it in first degree.

A partial differential equation is said to be **non-linear** if some or all the partial differential coefficients appearing in it do not occur in first degree.

Every linear partial differential equation is necessarily of first degree but a partial differential equation of first degree may or may not be linear. For example, the partial differential equation

$$\frac{\partial^2 z}{\partial x^2} + 4\left(\frac{\partial z}{\partial x}\right)^2 + 5\frac{\partial z}{\partial y} + 6z = 9$$

is of first degree but it is not linear.

Below we give some examples of linear partial differential equations.

(i) The partial differential equation

$$(x^2 + y^2) \frac{\partial z}{\partial x} + 3xy \frac{\partial z}{\partial y} + (x + y)z = e^{x+y}$$

is a linear partial differential equation of first order.

(ii) The partial differential equation

$$x \frac{\partial^2 z}{\partial x^2} + (3x + 4y) \frac{\partial^2 z}{\partial x \partial y} + e^x \frac{\partial^2 z}{\partial y^2} + 5x \frac{\partial z}{\partial x} + 6y \frac{\partial z}{\partial y} + 9z = x^2 + y^2$$

is a linear partial differential equation of second order.

(iii) The partial differential equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

is a linear partial differential equation of second order.

- (iv) The partial differential equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

is a linear partial differential equation of second order.

- (v) The partial differential equation

$$x^4 \frac{\partial^3 z}{\partial x^3} + y^2 \frac{\partial^2 z}{\partial x \partial y} + x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} + 6z = 9(x^2 + y^2)$$

is a linear partial differential equation of third order.

- (vi) The partial differential equation

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 3x + 4y$$

is a linear partial differential equation of first order.

- (vii) The partial differential equation

$$\frac{\partial^4 z}{\partial x^4} - \frac{\partial^4 z}{\partial y^4} = 0$$

is a linear partial differential equation of order 4.

- (viii) The partial differential equation

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$$

is a first-order linear partial differential equation in three variables.

Here are some examples of partial differential equations which are not linear.

- (i) The partial differential equation $p^2 x + q^2 y = z$ is not linear.
 (ii) The partial differential equation $(x^2 - y^2) pq - x y (p + q) - 1 = 0$ is not linear.
 (iii) The partial differential equation $\frac{\partial^2 z}{\partial x^2} + 4 \frac{\partial z}{\partial x} \cdot \frac{\partial z}{\partial y} + 5 \frac{\partial^2 z}{\partial y^2} = 0$ is not linear.

However the partial differential equation

$$\frac{\partial^2 z}{\partial x^2} + 4 \frac{\partial^2 z}{\partial x \partial y} + 5 \frac{\partial^2 z}{\partial y^2} = 0 \text{ is linear.}$$

8.4 Classification of Partial Differential Equations of First Order into Linear, Semi-Linear, Quasi-Linear and Non-Linear

A first order partial differential equation in two variables in its most general form is given by

$$F(x, y, z, p, q) = 0, \quad \dots(1)$$

where $p = \frac{\partial z}{\partial x}$

and $q = \frac{\partial z}{\partial y}$.

In this differential equation z is dependent variable and x, y are independent variables.

When the function F is not a linear expression in p and q , the equation (1) is said to be *non-linear*.

(a) Non-linear partial differential equation of first order:

A first order partial differential equation in which dependent variable is z and is a function of two independent variables x and y is called a *non-linear equation* if the partial derivatives $\frac{\partial z}{\partial x}$ i.e., p and $\frac{\partial z}{\partial y}$ i.e., q do not occur in it in first degree.

For example, the partial differential equations of the first order such as

$$\begin{aligned}(x+y)(p+q)^2 + (x-y)(p-q)^2 &= 1, \quad x^2 y^3 p^2 q = z^3, \\ (x^2 + y^2)(p^2 + q^2) &= 1, \quad z(p^2 - q^2) = x - y, \\ z^2(p^2 + q^2) &= x^2 + y^2 \text{ etc., are all non-linear equations.}\end{aligned}$$

(b) Quasi-linear partial differential equation of first order:

A first order partial differential equation $F(x, y, z, p, q) = 0$ in which dependent variable is z and is a function of two independent variables x and y is called a *quasi-linear equation* if the function F is a linear expression in p and q but not necessarily linear in z .

A quasi-linear partial differential equation of first order is of the form

$$f(x, y, z) \frac{\partial z}{\partial x} + g(x, y, z) \frac{\partial z}{\partial y} = h(x, y, z)$$

where the functions f and g depend on z also.

Thus the first order partial differential equations such as

$$\begin{aligned}(x+y+z) \frac{\partial z}{\partial x} + xy \frac{\partial z}{\partial y} + xz &= 3x^2 + 5y^2 + 6z^2, \\ (x^2 + y^2) \frac{\partial z}{\partial x} + 4xyz \frac{\partial z}{\partial y} &= 3z + e^{x+y}, \\ 2z \frac{\partial z}{\partial x} + 5y \frac{\partial z}{\partial y} &= 6z^2 + \log x + e^y \text{ etc.,}\end{aligned}$$

are all quasi-linear equations.

(c) Semi-linear partial differential equation of first order:

A first order partial differential equation $F(x, y, z, p, q) = 0$ in which dependent variable is z and is a function of two independent variables x and y is called a *semi-linear equation* if it is of the form

$$f(x, y) \frac{\partial z}{\partial x} + g(x, y) \frac{\partial z}{\partial y} = h(x, y, z).$$

In a semi-linear partial differential equation of first order the coefficients of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ are functions of x and y only and they do not depend on z . The terms that do not involve $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ contain some terms that are not of first degree in z .

Thus the first order partial differential equations such as

$$(x + y^2) \frac{\partial z}{\partial x} + x \log y \frac{\partial z}{\partial y} = 2z^2 x + xy + e^x,$$

$$p \cos (x + y) + q \sin (x + y) = z^3 + \sin x + e^y,$$

$$(y^3 x - 2x^4) \frac{\partial z}{\partial x} + (2y^4 - x^3 y) \frac{\partial z}{\partial y} = 9(x^2 - y^2) \log z, \text{ etc.,}$$

are all semi-linear equations.

(d) Linear partial differential equation of first order:

A first order partial differential equation $F(x, y, z, p, q) = 0$ in which dependent variable is z and is a function of two independent variables x and y is called a *linear equation* if in this differential equation the dependent variable z and its partial differential coefficients $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ all occur in first degree.

Thus a linear partial differential equation of first order is of the form

$$f(x, y) \frac{\partial z}{\partial x} + g(x, y) \frac{\partial z}{\partial y} + h(x, y) z = c(x, y)$$

where $f(x, y)$, $g(x, y)$, $h(x, y)$ and $c(x, y)$ are functions of x and y only and they do not contain any term of z .

For example, the first order partial differential equations such as

$$(x^2 + y^3) \frac{\partial z}{\partial x} + (3x + 5y^2) \frac{\partial z}{\partial y} + (x - y) z = \sin(x + y),$$

$$p \cos(x + y) + q \sin(x + y) = z + e^y \sin x,$$

$$\log x \frac{\partial z}{\partial x} + e^y \frac{\partial z}{\partial y} = x(y - z),$$

$$p + 3q = 5z + \tan(y - 3x),$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 5u + \frac{xy}{z}, x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^3 \frac{\partial u}{\partial z} = u + xyz,$$

$$(y^3 x - 2x^4) p + (2y^4 - x^3 y) q = 9z(x^3 - y^3), \text{ etc.,}$$

are all linear equations.

Remark: In many standard text books on partial differential equations the concepts of quasi-linear and semi-linear differential equations of first order are not introduced. There we find only two categories of first order partial differential equations— one

linear and the other non-linear. According to this classification a partial differential equation is said to be linear if all the partial derivatives occurring in it appear only in first degree and there is no restriction on the dependent variable, it may or may not occur in first degree. Also, a differential equation is said to be non-linear if it is not linear. Thus a first order partial differential equation $F(x, y, z, p, q) = 0$ is linear if in this differential equation the partial derivatives p and q occur only in first degree and there is no restriction on the dependent variable z , it may occur in any form. Accordingly every first order partial differential equation of the form

$$f(x, y, z) \frac{\partial z}{\partial x} + g(x, y, z) \frac{\partial z}{\partial y} = h(x, y, z)$$

is linear where f, g, h are any functions of x, y, z . Thus the partial differential equations such as

$$(x^2 - yz)p + (y^2 - zx)q = z^2 - xy,$$

$$(mz - ny)p + (nx - lz)q = ly - mx,$$

$$(y^2 + z^2 - x^2)p - 2xyq + 2zx = 0,$$

$$x(y - z)p + y(z - x)q = z(x - y), \text{ etc., are all linear.}$$

On the other hand the partial differential equations such as

$$x^2 p^2 + y^2 q^2 = z^2, p^2 + q^2 = 3pq,$$

$$pq = 5, (y - x)(qy - px) = (p - q)^2,$$

$$p^2 = z^2(1 - pq), p^3 + q^3 = 27z \text{ etc., are all non-linear.}$$

8.5 Lagrange's Linear Partial Differential Equation

The partial differential equation $Pp + Qq = R$, where P, Q and R are any functions of x, y , and z is called Lagrange's linear partial differential equation of first order.

Thus Lagrange's linear partial differential equation is of the form

$$f(x, y, z) \frac{\partial z}{\partial x} + g(x, y, z) \frac{\partial z}{\partial y} = h(x, y, z),$$

where f, g, h are any functions of x, y, z .

The partial differential equations such as

$$(y^2 + z^2 - x^2)p - 2xyq + 2zx = 0,$$

$$p \cos(x + y) + q \sin(x + y) = z,$$

$$x(y - z) \frac{\partial z}{\partial x} + y(z - x) \frac{\partial z}{\partial y} = z(x - y), \text{ etc.,}$$

are all Lagrange's linear partial differential equations of first order.

Comprehensive Exercise 2

Tell whether the following partial differential equations are linear or non-linear:

1. $x^2 \frac{\partial^2 u}{\partial x^2} + y^2 \frac{\partial^2 u}{\partial y^2} + z^2 \frac{\partial^2 u}{\partial z^2} = 3u.$
2. $\frac{\partial^3 z}{\partial x^3} + 3 \frac{\partial^2 z}{\partial x^2} + 4 \frac{\partial z}{\partial x} \cdot \frac{\partial z}{\partial y} + 5z = 9e^x.$
3. $4 \frac{\partial^2 z}{\partial x^2} + x \frac{\partial^2 z}{\partial y^2} + 9 \frac{\partial^2 z}{\partial x \partial y} + 5 \frac{\partial z}{\partial x} + 6 \frac{\partial z}{\partial y} = \log x.$
4. $\frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x \partial y} = \frac{4x}{y^2} - \frac{y}{x^2}.$
5. $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial x \partial y} - 2 \left(\frac{\partial z}{\partial y} \right)^2 = (y - 1) e^x.$

Which of the following partial differential equations of first order are non-linear, quasi-linear, semi-linear or linear ?

6. $(x^2 + y^2) \frac{\partial z}{\partial x} + (x - y) \frac{\partial z}{\partial y} + xz = 3x + 4y.$
7. $zy^2 p - xyq = x(z - 2y).$
8. $(y^2 - x^2) \frac{\partial z}{\partial x} - xy \frac{\partial z}{\partial y} = x^2 + y^2 - xz^2.$
9. $(y^3 + x^3) \frac{\partial z}{\partial x} + 4x^2 y \frac{\partial z}{\partial y} = x^3 z - 4y^2 + yz.$
10. $4 \frac{\partial z}{\partial x} \cdot \frac{\partial z}{\partial y} + 5 \frac{\partial z}{\partial x} + 6 \frac{\partial z}{\partial y} + 3z = 9(x^2 + y^2).$
11. $(x^2 - yz) p + (y^2 - zx) q = z^2 - xy.$
12. $(mx - ny) p + (nx + my) q = z + ly - mx^2.$
13. $x^2 \frac{\partial z}{\partial x} + (y^3 + x^3) \frac{\partial z}{\partial y} = x^2 + y^2 + z^2.$
14. $x \frac{\partial u}{\partial x} + y \left(\frac{\partial u}{\partial y} \right)^2 + 3 \frac{\partial u}{\partial z} = x^2 + y^2 + z^2.$
15. $(z^2 - 2yz - y^2) p + x(y + z) q = x(y - z).$

Answers 2

- | | | |
|-----------------|------------------|------------------|
| 1. Linear | 2. Non-linear | 3. Linear |
| 4. Linear | 5. Non-linear | 6. Linear |
| 7. Quasi-linear | 8. Semi-linear | 9. Linear |
| 10. Non-linear | 11. Quasi-linear | 12. Linear |
| 13. Semi-linear | 14. Non-linear | 15. Quasi-linear |

8.6 Equation Equivalent to the Linear Equation

A partial differential equation which is linear in p and q is of the type

$$Pp + Qq = R, \quad \dots(1)$$

where P, Q, R are functions of x, y, z .

Let any relation $u = a$ be an integral of (1). Differentiating it partially with respect to x and y , we get

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial z} p = 0 \quad \text{and} \quad \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} q = 0.$$

These give
$$p = -\frac{\partial u}{\partial x} / \frac{\partial u}{\partial z}, \quad q = -\frac{\partial u}{\partial y} / \frac{\partial u}{\partial z}.$$

Putting these values of p and q in (1) it changes to

$$P \frac{\partial u}{\partial x} + Q \frac{\partial u}{\partial y} + R \frac{\partial u}{\partial z} = 0. \quad \dots(2)$$

Hence, if $u = a$ satisfies (1), $u = a$ also satisfies (2).

Conversely, dividing (2) by $\frac{\partial u}{\partial z}$ and substituting p and q for their values given above we see that if $u = a$ is an integral of (2), it is also an integral of (1).

Thus equation (2) can be taken as equivalent to equation (1).

8.7 Lagrange's Solution of the Linear Partial Differential Equation of First Order

The first systematic theory of the linear partial differential equations was given by **Lagrange**. For that reason the equation $Pp + Qq = R$ is referred to as **Lagrange's equation**. The method of solving a linear equation of this form is contained in the theorem given below :

Theorem: *The general solution of the linear partial differential equation*

$$Pp + Qq = R \quad \dots(1)$$

is
$$f(u, v) = 0, \quad \dots(2)$$

where f is an arbitrary function and

$$u(x, y, z) = c_1 \text{ and } v(x, y, z) = c_2 \quad \dots(3)$$

form a solution of the equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}. \quad \dots(4)$$

(Lucknow 2010)

Proof: We shall give a purely analytic proof of this theorem. If the relation $u(x, y, z) = c_1$ satisfies the equations (4) then the equations

$$\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz = 0 \quad \text{and} \quad \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

must be compatible i.e., we should have

$$P \frac{\partial u}{\partial x} + Q \frac{\partial u}{\partial y} + R \frac{\partial u}{\partial z} = 0. \quad \dots(5)$$

Similarly we should have

$$P \frac{\partial v}{\partial x} + Q \frac{\partial v}{\partial y} + R \frac{\partial v}{\partial z} = 0. \quad \dots(6)$$

Solving the equations (5) and (6), we get

$$\frac{P}{\frac{\partial u}{\partial y} \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \frac{\partial v}{\partial y}} = \frac{Q}{\frac{\partial u}{\partial z} \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \frac{\partial v}{\partial z}} = \frac{R}{\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x}}. \quad \dots(7)$$

Earlier we have seen that the relation $f(u, v) = 0$ gives the partial differential equation

$$\left(\frac{\partial u}{\partial y} \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \frac{\partial v}{\partial y} \right) p + \left(\frac{\partial u}{\partial z} \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \frac{\partial v}{\partial z} \right) q = \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x}. \quad \dots(8)$$

Substituting from the equations (7) into the equation (8), we find that (2) is a solution of the equation (1) if u and v are given by the equations (3).

Note: In place of $f(u, v) = 0$ the functional relation can be written as $u = \phi(v)$, where ϕ denotes an arbitrary function.

It is called **Lagrange's solution** of the linear equation (1) and the equations (4) are called **Lagrange's auxiliary equations** or **Lagrange's subsidiary equations**.

8.8 The Linear Equation Containing more than two Independent Variables

The generalisation of Lagrange's method is as follows :

Let the linear equation with n independent variables $x_1, x_2, \dots, x_n$ be

$$P_1 p_1 + P_2 p_2 + P_3 p_3 + \dots + P_n p_n = R. \quad \dots(1)$$

where $P_1, P_2, \dots, P_n$ and R are functions of $x_1, x_2, \dots, x_n$ and z . Here p_i denotes $\frac{\partial z}{\partial x_i}$ for

$i = 1, 2, \dots, n$.

Then the general solution of (1) is given by

$$f(u_1, u_2, \dots, u_n) = 0$$

where

$$u_i(x_1, x_2, \dots, x_n, z) = c_i, \quad i = 1, 2, \dots, n$$

are any independent integrals of the auxiliary equations

$$\frac{dx_1}{P_1} = \frac{dx_2}{P_2} = \dots = \frac{dx_n}{P_n} = \frac{dz}{R}.$$

8.9 Geometrical Interpretation of Lagrange's Linear Equation

(Lucknow 2011)

Lagrange's linear equation is

$$Pp + Qq = R \quad \text{or} \quad Pp + Qq + (-1)R = 0. \quad \dots(1)$$

Since d.r.'s of the normal at the point (x, y, z) on the surface $f(x, y, z) = c$ are given by

$$\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \quad \text{or} \quad -\frac{\partial f}{\partial x} / \frac{\partial f}{\partial z}, -\frac{\partial f}{\partial y} / \frac{\partial f}{\partial z}, -1$$

$$\text{i.e.,} \quad \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, -1 \quad \text{or} \quad p, q, -1,$$

hence the equation (1) shows that the normal to a certain surface is perpendicular to the line whose d.r.'s are P, Q, R .

But we know that the simultaneous equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \dots(2)$$

represent a family of curves in space such that the direction ratios of the tangent to any member of this family at any point (x, y, z) are P, Q, R . If $u = a$ and $v = b$ are two independent integrals of (2) then $f(u, v) = 0$ represents a surface through such curves. Through every point of such a surface passes a curve of the family, lying wholly on the surface.

Thus the general solution of (1) is the family of surfaces such that the normal to a surface at any point must be perpendicular to the tangent to a curve of the family represented by (2).

8.10 The Equation $Pp + Qq = R$ Represents a Family of Surfaces Orthogonal to the Family of Surfaces Represented by $P dx + Q dy + R dz = 0$ (If Integrable)

We know that d.r.'s of the normal at the point (x, y, z) to a surface of the family represented by $Pp + Qq = R$ are $p, q, -1$.

Also d.r.'s of the normal at the point (x, y, z) to a surface of the family represented by $P dx + Q dy + R dz = 0$ are P, Q, R .

Since the equation $Pp + Qq + (-1)R = 0$ shows that the two lines whose d.r.'s are $p, q, -1$ and P, Q, R are perpendicular, hence the surfaces represented by $Pp + Qq = R$ are orthogonal to the surfaces represented by $P dx + Q dy + R dz = 0$.

8.11 Integral Surfaces Passing Through a Given Curve

Now we shall indicate how the general solution of a linear partial differential equation may be used to find the integral surface passing through a given curve. Suppose that we have obtained two solutions

$$u = a \text{ and } v = b \quad \dots(1)$$

of the auxiliary equations (4) of 8.7. Then we know that the general solution of the corresponding linear equation is of the form

$$f(u, v) = 0 \quad \dots(2)$$

arising from a relation

$$f(a, b) = 0 \quad \dots(3)$$

between the constants a and b . We have to consider the problem of determining the function f in special cases.

If we want to find the integral surface passing through the curve whose parametric equations are $x = x(t), y = y(t), z = z(t)$,

t being a parameter, then the solutions in (1) must be such that

$$u(x(t), y(t), z(t)) = a, \quad v(x(t), y(t), z(t)) = b.$$

Eliminating the single variable t from these two equations we find a relation of the type (3). The required solution is then given by the equation (2).

8.12 Working Rule to Find the Solution of Lagrange's Equation $Pp + Qq = R$

Form the auxiliary equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}.$$

Find two independent integrals of these auxiliary equations, say $u = a$ and $v = b$. Then the general solution of the partial differential equation $Pp + Qq = R$ is given by $f(u, v) = 0$, where f is an arbitrary function or it can also be written as $u = \phi(v)$, where ϕ is an arbitrary function.

Illustrative Examples

Example 5: Solve $(z^2 - 2yz - y^2)p + (xy + zx)q = xy - zx$(1)
(Lucknow 2007)

Solution: The Lagrange's auxiliary equations of the given differential equation (1) are

$$\frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{xy + zx} = \frac{dz}{xy - zx}.$$

Taking the last two members, we get

$$(y - z)dy = (y + z)dz \quad \text{or} \quad ydy - (zdy + ydz) - zdz = 0.$$

Integrating, $y^2 - 2yz - z^2 = c_1$.

Again choosing x, y, z as multipliers, we get

$$\text{each fraction} = \frac{x dx + y dy + z dz}{0} \Rightarrow x dx + y dy + z dz = 0.$$

Integrating, $x^2 + y^2 + z^2 = c_2$.

Hence the general solution of (1) is given by

$$f(y^2 - 2yz - z^2, x^2 + y^2 + z^2) = 0$$

or $y^2 - 2yz - z^2 = \phi(x^2 + y^2 + z^2)$.

Example 6: Solve $(mz - ny)p + (nx - lz)q = ly - mx$(1)
(Rohilkhand 2008; Lucknow 09; Purvanchal 10; Avadh 10)

Solution: The Lagrange's auxiliary equations of the given differential equation (1) are

$$\frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{ly - mx}.$$

Choosing x, y, z as multipliers, we get

$$\text{each fraction} = \frac{x dx + y dy + z dz}{0} \Rightarrow x dx + y dy + z dz = 0.$$

Integrating, $x^2 + y^2 + z^2 = c_1$.

Again choosing l, m, n as multipliers, we get

$$\text{each fraction} = \frac{l dx + m dy + n dz}{0} \Rightarrow l dx + m dy + n dz = 0.$$

Integrating, $lx + my + nz = c_2$.

Hence the general solution of (1) is given by

$$f(x^2 + y^2 + z^2, lx + my + nz) = 0,$$

where f is an arbitrary function.

Example 7: Solve $(y^2 + z^2 - x^2)p - 2xyq + 2zx = 0$(1)

(Lucknow 2009; Avadh 10, 11)

Solution: The Lagrange's auxiliary equations of (1) are

$$\frac{dx}{y^2 + z^2 - x^2} = \frac{dy}{-2xy} = \frac{dz}{-2zx}.$$

Taking the last two members, we get

$$\frac{dy}{y} = \frac{dz}{z}.$$

Integrating, $\log \frac{y}{z} = \log c_1$ or $\frac{y}{z} = c_1$.

Again choosing x, y, z as multipliers, we get

$$\text{each fraction} = \frac{x dx + y dy + z dz}{-x(x^2 + y^2 + z^2)}.$$

$$\therefore \frac{x dx + y dy + z dz}{-x(x^2 + y^2 + z^2)} = \frac{dz}{-2zx} \Rightarrow 2 \frac{x dx + y dy + z dz}{x^2 + y^2 + z^2} = \frac{dz}{z}.$$

Integrating, $\log (x^2 + y^2 + z^2) = \log z + \log c_2$ or $x^2 + y^2 + z^2 = zc_2$.

Hence the general solution of (1) is

$$f \left(\frac{y}{z}, \frac{x^2 + y^2 + z^2}{z} \right) = 0, \text{ or } x^2 + y^2 + z^2 = z\phi \left(\frac{y}{z} \right),$$

where f and ϕ are arbitrary functions.

Example 8: Solve $(y + z)p + (z + x)q = x + y$ (1)

(Rohilkhand 2008; Kanpur 08; Purvanchal 07; Bundelkhand 13)

Solution: The Lagrange's auxiliary equations of (1) are

$$\frac{dx}{y + z} = \frac{dy}{z + x} = \frac{dz}{x + y}.$$

$$\therefore \frac{dx - dy}{y - x} = \frac{dy - dz}{z - y} = \frac{dx + dy + dz}{2(x + y + z)}.$$

Taking the first two members, we get

$$\log (y - x) = \log (z - y) + \log c_1$$

$$\text{or } y - x = c_1 (z - y).$$

Again taking the first and the last members, we get

$$-2 \log (x - y) = \log (x + y + z) - \log c_2$$

$$\text{or } (x - y)^2 (x + y + z) = c_2.$$

Hence the general solution of (1) is given by

$$f \left[\frac{y - x}{z - y}, (x - y)^2 (x + y + z) \right] = 0.$$

Example 9: Solve $\frac{(y - z)}{yz} p + \frac{(z - x)}{zx} q = \frac{x - y}{xy}$ (1)

Solution: The auxiliary equations of (1) are (Rohilkhand 2009)

$$\frac{dx}{x(y - z)} = \frac{dy}{y(z - x)} = \frac{dz}{z(x - y)}.$$

Each fraction = $\frac{dx + dy + dz}{0}$ and also = $\frac{\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz}{0}$.

$\therefore dx + dy + dz = 0$ and $\frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz = 0$.

Integrating, $x + y + z = c_1$ and $xyz = c_2$.

Hence the general solution of (1) is given by

$$f(x + y + z, xyz) = 0.$$

Example 10: Solve $p + 3q = 5z + \tan(y - 3x)$ (1)

(Rohilkhand 2007, 10; Avadh 14)

Solution: The auxiliary equations of (1) are

$$\frac{dx}{1} = \frac{dy}{3} = \frac{dz}{5z + \tan(y - 3x)}.$$

Taking the first two members, we get

$$dy - 3dx = 0.$$

$\therefore y - 3x = c_1.$

Again taking the first and the last members, we get

$$dx = \frac{dz}{5z + \tan c_1}.$$

$\therefore \log(5z + \tan c_1) = 5x + \log c_2$

or $e^{-5x} \{5z + \tan(y - 3x)\} = c_2.$

Hence the general solution of (1) is given by

$$e^{-5x} \{5z + \tan(y - 3x)\} = f(y - 3x).$$

Example 11: Solve $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$ (1) (Agra 2001)

Solution: The auxiliary equations of (1) are

$$\frac{dx}{x^2(y - x)} = \frac{dy}{y^2(z - x)} = \frac{dz}{z^2(x - y)}.$$

Each fraction = $\frac{\frac{dx}{x^2} + \frac{dy}{y^2} + \frac{dz}{z^2}}{0}$ and also = $\frac{\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z}}{0}$.

$\therefore \frac{dx}{x^2} + \frac{dy}{y^2} + \frac{dz}{z^2} = 0$ and $\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0.$

Integrating, $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = c_1$ and $xyz = c_2.$

Hence the general solution of (1) is given by

$$f\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}, xyz\right) = 0.$$

Example 12: Solve $x(y^2 + z)p - y(x^2 + z)q = z(x^2 - y^2)$ (1)
(Lucknow 2011; Purvanchal 07, 11, 14)

Solution: The auxiliary equations of (1) are

$$\frac{dx}{x(y^2 + z)} = \frac{dy}{-y(x^2 + z)} = \frac{dz}{z(x^2 - y^2)}.$$

Choosing $1/x, 1/y, 1/z$ as multipliers, we get

$$\frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} = 0.$$

Integrating, $\log x + \log y + \log z = \log c_1$

or $xyz = c_1$.

Again choosing $x, y, -1$ as multipliers, we get

$$x dx + y dy - dz = 0.$$

Integrating, $x^2 + y^2 - 2z = c_2$.

Hence the general solution of (1) is given by $f(xyz, x^2 + y^2 - 2z) = 0$.

Example 13: Find the equation of the integral surface of the differential equation

$$2y(z - 3)p + (2x - z)q = y(2x - 3), \quad \dots (1)$$

which passes through the circle $z = 0, x^2 + y^2 = 2x$.

Solution: The auxiliary equations of (1) are

$$\frac{dx}{2y(z - 3)} = \frac{dy}{2x - z} = \frac{dz}{y(2x - 3)}.$$

Taking the first and the third members, we get

$$(2x - 3)dx = 2(z - 3)dz.$$

Integrating, $x^2 - 3x - z^2 + 6z = a$ (2)

Again, using $\frac{1}{2}, y, -1$ as multipliers, we get

$$\frac{1}{2}dx + ydy - dz = 0.$$

Integrating, $\frac{1}{2}x + \frac{1}{2}y^2 - z = \frac{1}{2}b$ or $x + y^2 - 2z = b$ (3)

The parametric equations of the circle are

$$x = t, y = \sqrt{2t - t^2}, z = 0.$$

Putting these values in the equations (2) and (3), we get

$$t^2 - 3t = a \text{ and } t + (2t - t^2) = b.$$

Eliminating t from these, we find the relation

$$a + b = 0,$$

showing that the required integral surface is

$$(x^2 - 3x - z^2 + 6z) + (x + y^2 - 2z) = 0$$

or $x^2 + y^2 - z^2 - 2x + 4z = 0$.

Comprehensive Exercise 3

Solve the following equations :

1. $xzp + yzq = xy$. (Avadh 2006; Lucknow 10)
2. $p \tan x + q \tan y = \tan z$. (Kanpur 2007; Avadh 08, 10)
3. $x^2 p + y^2 q = z^2$. (Avadh 2007)
4. $yzp + zxq = xy$. (Lucknow 2011)
5. $\frac{y^2 z}{x} p + zxq = y^2$. (Lucknow 2007; Avadh 10, 11)
6. $(x^2 - yz) p + (y^2 - zx) q = z^2 - xy$. (Lucknow 2010)
7. $z - px - qy = a \sqrt{(x^2 + y^2 + z^2)}$. (Rohilkhand 2011)
8. $z(xp - yq) = y^2 - x^2$ (Avadh 2014)
9. $(3x + y - z) p + (x + y - z) q = 2(z - y)$.

Hint. $\frac{dx}{3x + y - z} = \frac{dy}{x + y - z} = \frac{dz}{2(z - y)}$

(i) Choose -1, 3, 1 as multipliers.

(ii) $\left[\frac{dx - dy + dz}{\dots} = \frac{dx + dy - dz}{\dots} \right]$
10. $p_2 + p_3 = 1 + p_1$.
11. $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = xyz$. (Lucknow 2006; Kanpur 12)
12. Find the integral surface of the linear partial differential equation

$$x(y^2 + z)p - y(x^2 + z)q = (x^2 - y^2)z$$

which contains the line $x + y = 0, z = 1$.
13. Find the general integral of the partial differential equation

$$(2xy - 1)p + (z - 2x^2)q = 2(x - yz)$$

and also the particular integral which passes through the line
 $x = 1, y = 0$.
14. Find the family of surfaces orthogonal to the family of surfaces given by the differential equation

$$(y + z)p + (z + x)q = x + y$$
15. Find the integral surface of the partial differential equation

$$x^2 p + y^2 q + z^2 = 0$$

which passes through the hyperbola
 $xy = x + y, z = 1$. (Avadh 2012)

Answers 3

1. $f\left(z^2 - xy, \frac{y}{x}\right) = 0$
2. $\frac{\sin z}{\sin y} = f\left(\frac{\sin x}{\sin y}\right)$
3. $\frac{1}{x} - \frac{1}{y} = f\left(\frac{1}{x} - \frac{1}{z}\right)$
4. $f(x^2 - y^2, x^2 - z^2) = 0$
5. $f(x^3 - y^3, x^2 - y^2) = 0$
6. $f\left(\frac{x-y}{y-z}, \frac{z-x}{y-z}\right) = 0$
7. $z^{1-a} = \{z + \sqrt{(x^2 + y^2 + z^2)}\} f\left(\frac{x}{y}\right)$
8. $f(xy, x^2 + y^2 + z^2) = 0$
9. $(x - y + z)^2 = (x + y - z) f(x - 3y - z)$
10. $f(x_1 + z, x_1 + x_2, x_1 + x_3) = 0$
11. $f\left(\frac{x}{y}, \frac{y}{z}, xyz - 3u\right) = 0$
12. $x^2 + y^2 + 2xyz - 2z + 2 = 0$
13. $x^2 + y^2 - xz - y - z - 1 = 0$
14. $xy + yz + zx = c$
15. $\frac{1}{x} + \frac{1}{y} + \frac{2}{z} = 3$

8.13 The Integrals of the Non-Linear Equation

The complete and particular integrals: We have seen that the relation of the type

$$f(x, y, z, a, b) = 0 \quad \dots(1)$$

gives rise to a partial differential equation of the first order of the form

$$F(x, y, z, p, q) = 0 \quad \dots(2)$$

on the elimination of arbitrary constants a and b . Here x, y are independent variables and z is dependent variable. The relation (1) is a solution of (2). *Any such relation which contains as many arbitrary constants as there are independent variables and is a solution of a partial differential equation of the first order is called a complete solution or a complete integral of that equation.*

A particular integral of (2) can be obtained by giving particular values to a and b in (1).

The singular integral: If the envelope of the doubly infinite system of surfaces represented by (1) exists, it is also a solution of the equation (2). The reason is that the envelope of all the surfaces represented by (1) is touched at each of its points by some one of these surfaces. Hence the coordinates of any point on the envelope of surfaces (1) with the corresponding values of p and q , being identical with the x, y, z, p, q of some point on one of these surfaces, must satisfy (2). The equation of the envelope of the

surfaces represented by (1) can be obtained by eliminating a and b between the three equations

$$f = 0, \frac{\partial f}{\partial a} = 0 \text{ and } \frac{\partial f}{\partial b} = 0.$$

This equation of the envelope is called the **singular integral** of the differential equation (2).

It differs from a particular integral in the sense that it is not contained in the complete integral, i.e., it cannot be obtained from the complete integral by giving particular values to the constants.

The general integral: If in the equation (1), one of the constants is a function of the other, say $b = \phi(a)$, then this equation becomes

$$f(x, y, z, a, \phi(a)) = 0. \quad \dots(3)$$

It is a one-parameter subfamily of the family (1). The equation of the envelope of the family of surfaces represented by (3) is also a solution of the equation (2).

It is called the *general integral of (2) corresponding to the complete integral (1)*.

The equation of the envelope of the surfaces represented by (3) is obtained by eliminating a between

$$f(x, y, z, a, \phi(a)) = 0 \text{ and } \frac{\partial f}{\partial a} = 0.$$

8.14 General Method of Solution of a Non-Linear Partial Differential Equation of Order One with two Independent Variables. (Charpit's Method)

(Lucknow 2006, 11; Avadh 06)

Let the given partial differential equation be

$$f(x, y, z, p, q) = 0. \quad \dots(1)$$

Since z depends upon x and y , it gives that

$$dz = p dx + q dy. \quad \dots(2)$$

The fundamental idea in Charpit's method is to introduce a second partial differential equation of the first order $F(x, y, z, p, q, a) = 0$,

$\dots(3)$

containing an arbitrary constant a and which is such that :

(i) The equations (1) and (3) can be solved to find

$$p = p(x, y, z, a), q = q(x, y, z, a).$$

(ii) Substituting these values of p and q in (2), the equation

$$dz = p(x, y, z, a) dx + q(x, y, z, a) dy \quad \dots(4)$$

is integrable.

If such a relation (3) has been found, the solution of equation (4),

$$\phi(x, y, z, a, b) = 0 \quad \dots(5)$$

containing two arbitrary constants a and b will be a solution of the equation (1). Also it is a complete integral of the equation (1).

Now the main problem is to devise a method of finding the relation (3). Let us assume that (3) is the relation which when taken along with (1) gives those values of p and q which make (2) integrable. Differentiating (1) and (3) with respect to x , we get

$$\left. \begin{aligned} \frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} p + \frac{\partial f}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} &= 0 \\ \frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} p + \frac{\partial F}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial F}{\partial q} \frac{\partial q}{\partial x} &= 0 \end{aligned} \right\} \quad \text{and} \quad \dots(6)$$

Again, differentiating (1) and (3) w.r.t. y , we get

$$\left. \begin{aligned} \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} q + \frac{\partial f}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial y} &= 0 \\ \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z} q + \frac{\partial F}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial F}{\partial q} \frac{\partial q}{\partial y} &= 0 \end{aligned} \right\} \quad \text{and} \quad \dots(7)$$

Eliminating $\frac{\partial p}{\partial x}$ from the equations in (6) and $\frac{\partial q}{\partial y}$ from the equations in (7),

we get

$$\left(\frac{\partial f}{\partial x} \frac{\partial F}{\partial p} - \frac{\partial f}{\partial p} \frac{\partial F}{\partial x} \right) + p \left(\frac{\partial f}{\partial z} \frac{\partial F}{\partial p} - \frac{\partial f}{\partial p} \frac{\partial F}{\partial z} \right) + \frac{\partial q}{\partial x} \left(\frac{\partial f}{\partial q} \frac{\partial F}{\partial p} - \frac{\partial f}{\partial p} \frac{\partial F}{\partial q} \right) = 0, \quad \dots(8)$$

$$\text{and} \quad \left(\frac{\partial f}{\partial y} \frac{\partial F}{\partial q} - \frac{\partial f}{\partial q} \frac{\partial F}{\partial y} \right) + q \left(\frac{\partial f}{\partial z} \frac{\partial F}{\partial q} - \frac{\partial f}{\partial q} \frac{\partial F}{\partial z} \right) + \frac{\partial p}{\partial y} \left(\frac{\partial f}{\partial p} \frac{\partial F}{\partial q} - \frac{\partial f}{\partial q} \frac{\partial F}{\partial p} \right) = 0. \quad \dots(9)$$

Since $\frac{\partial q}{\partial x} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial p}{\partial y}$, hence adding (8) and (9) and re-arranging, we get

$$\begin{aligned} \left(\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z} \right) \frac{\partial F}{\partial p} + \left(\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z} \right) \frac{\partial F}{\partial q} + \left(-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q} \right) \frac{\partial F}{\partial z} \\ + \left(-\frac{\partial f}{\partial p} \right) \frac{\partial F}{\partial x} + \left(-\frac{\partial f}{\partial q} \right) \frac{\partial F}{\partial y} = 0. \quad \dots(10) \end{aligned}$$

This is a linear partial differential equation of the first order with x, y, z, p, q as independent variables and F as dependent variable. Lagrange's auxiliary equations of (10) are

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dF}{0}. \quad \dots(11)$$

These equations are known as **Charpit's auxiliary equations**.

Any of the integrals of (11) will satisfy (10). If such an integral contains p or q , it can be taken as the required second relation. It should be noted that not all of Charpit's equations (11) need be used, but that p or q must occur in the solution obtained. Of course, the simpler the integral containing p or q , or both p and q that is derived from (11), the easier will be the subsequent labour in finding the solution of (1).

Illustrative Examples

Example 14: Find a complete integral of the equation

$$2zx - px^2 - 2qxy + pq = 0.$$

Solution: The given differential equation is

$$f(x, y, z, p, q) \equiv 2zx - px^2 - 2qxy + pq = 0. \quad \dots(1)$$

Charpit's auxiliary equations are

$$\frac{dp}{2z - 2qy} = \frac{dq}{0} = \frac{dz}{px^2 - pq + 2qxy - pq} = \frac{dx}{x^2 - q} = \frac{dy}{2xy - p}$$

from which it follows that $dq = 0$ or $q = a$ (a constant). ... (2)

Solving the equations (1) and (2) for p and q , we get

$$p = \frac{2x(z - ay)}{x^2 - a}, \quad q = a.$$

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = \frac{2x(z - ay)}{x^2 - a} dx + a dy$$

or
$$\frac{dz - a dy}{z - ay} = \frac{2x}{x^2 - a} dx.$$

Integrating, $\log(z - ay) = \log(x^2 - a) + \log b$

or $z - ay = b(x^2 - a) \quad \text{or} \quad z = ay + b(x^2 - a),$

which is a complete integral of (1).

Example 15: Find a complete integral of $px + qy = pq$.

(Rohilkhand 08, 10, 14; Avadh 10, 11; Kanpur 12; Purvanchal 09)

Solution: Here $f(x, y, z, p, q) \equiv px + qy - pq = 0$ (1)

Charpit's auxiliary equations are

$$\frac{dp}{p} = \frac{dq}{q} = \frac{dz}{-p(x - q) - q(y - p)} = \frac{dx}{-(x - q)} = \frac{dy}{-(y - p)}$$

from which it follows that

$$\frac{dp}{p} = \frac{dq}{q}, \quad \text{or} \quad p = aq. \quad \dots(2)$$

Solving the equations (1) and (2) for p and q , we get

$$q = \frac{y + ax}{a}, \quad p = y + ax.$$

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = (y + ax) dx + \frac{(y + ax)}{a} dy$$

or $a dz = (y + ax)(dy + a dx).$

Integrating, $az = \frac{1}{2}(y + ax)^2 + b$, which is a complete integral of (1).

Example 16: Find a complete integral of $(p^2 + q^2) y = qz$.

(Lucknow 2007, 08; Purvanchal 10, 14; Avadh 12; Kanpur 14)

Solution: The given differential equation is

$$f \equiv (p^2 + q^2) y - qz = 0. \quad \dots(1)$$

Charpit's auxiliary equations are

$$\frac{dp}{-pq} = \frac{dq}{(p^2 + q^2) - q^2} = \frac{dz}{-2p^2 y - 2q^2 y + qz} = \frac{dx}{-2py} = \frac{dy}{-2qy + z}$$

from which it follows that

$$\frac{dp}{-pq} = \frac{dq}{p^2} \quad \text{or} \quad p dp + q dq = 0$$

$$\text{or} \quad p^2 + q^2 = a^2 \text{ (say)} \quad \dots(2)$$

Solving the equations (1) and (2) for p and q , we get

$$q = \frac{a^2 y}{z}, \quad p = \frac{q}{z} \sqrt{(z^2 - a^2 y^2)}.$$

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = \frac{a}{z} \sqrt{(z^2 - a^2 y^2)} dx + \frac{a^2 y}{z} dy$$

$$\text{or} \quad \frac{z dz - a^2 y dy}{\sqrt{(z^2 - a^2 y^2)}} = a dx.$$

$$\text{Integrating, } \sqrt{(z^2 - a^2 y^2)} = ax + b$$

$$\text{or} \quad z^2 = a^2 y^2 + (ax + b)^2, \quad \text{which is a complete integral of (1).}$$

Example 17: Find a complete integral of $p = (qy + z)^2$.

(Meerut 2008; Lucknow 11; Purvanchal 08, 11; Bundelkhand 13)

Solution: The given differential equation is $f \equiv p - (qy + z)^2 = 0. \quad \dots(1)$

Charpit's auxiliary equations are

$$\frac{dp}{2p(qy + z)} = \frac{dq}{4q(qy + z)} = \frac{dy}{-2y(qy + z)} = \dots\dots$$

$$\text{from which it follows that } \frac{dp}{p} = -\frac{dy}{y}.$$

$$\text{Integrating, } py = a. \quad \dots(2)$$

Putting $p = a/y$ in (1), we get

$$(qy + z)^2 = \frac{a}{y}.$$

$$\therefore \quad q = \frac{1}{y} \left\{ \sqrt{\left(\frac{a}{y} \right)} - z \right\}.$$

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = \frac{a}{y} dx + \frac{1}{y} \left\{ \sqrt{\left(\frac{a}{y}\right)} - z \right\} dy$$

or $y dz + z dy = a dx + \sqrt{\left(\frac{a}{y}\right)} dy.$

Integrating, $yz = ax + 2\sqrt{ay} + b$,

which is a complete integral of (1).

Example 18: Find a complete integral of $px + qy = z(1 + pq)^{1/2}$.

(Avadh 2013; Kanpur 14)

Solution: Here $f \equiv px + qy - z(1 + pq)^{1/2} = 0$.

...(1)

Charpit's auxiliary equations are

$$\frac{dp}{p - p(1 + pq)^{1/2}} = \frac{dq}{q - q(1 + pq)^{1/2}} = \dots\dots$$

from which it follows that

$$\frac{dp}{p} = \frac{dq}{q} \Rightarrow p = aq. \quad \dots(2)$$

Putting $p = aq$ in (1), we get

$$q(ax + y) = z(1 + aq^2)^{1/2}$$

or $q^2[(ax + y)^2 - az^2] = z^2.$

$$\therefore q = \frac{z}{[(ax + y)^2 - az^2]^{1/2}} \quad \text{and} \quad p = aq = \frac{az}{[(ax + y)^2 - az^2]^{1/2}}.$$

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = \frac{z(a dx + dy)}{\sqrt{\{(ax + y)^2 - az^2\}}}$$

or $\frac{dz}{z} = \frac{a dx + dy}{\sqrt{\{(ax + y)^2 - az^2\}}}$

Let $ax + y = \sqrt{a}u$ so that $a dx + dy = \sqrt{a} du$.

$$\therefore \frac{dz}{z} = \frac{\sqrt{a} du}{\sqrt{(au^2 - az^2)}}$$

or $\frac{du}{dz} = \frac{\sqrt{(u^2 - z^2)}}{z}$

To solve this, putting $u = vz$, we get $v + z \frac{dv}{dz} = \frac{1}{z} \sqrt{(v^2 z^2 - z^2)}$

or $z \frac{dv}{dz} = \sqrt{(v^2 - 1)} - v$

or $\frac{dz}{z} = \frac{dv}{\sqrt{(v^2 - 1)} - v}$

or $\frac{dz}{z} = -\{\sqrt{(v^2 - 1)} + v\} dv.$

Integrating,

$$\log z = - \left[\frac{v}{2} \sqrt{(v^2 - 1)} - \frac{1}{2} \log \{v + \sqrt{(v^2 - 1)}\} \right] - \frac{v^2}{2} + b$$

or
$$\log z + \frac{v^2}{2} + \frac{v}{2} \sqrt{(v^2 - 1)} - \frac{1}{2} \log \{v + \sqrt{(v^2 - 1)}\} = b,$$

which is a complete integral of (1), where $v = \frac{u}{z} = \frac{ax + y}{z \sqrt{a}}$.

Example 19: Find a complete integral of $p^2 x + q^2 y = z$.

(Lucknow 2009)

Solution: The given differential equation is

$$f \equiv p^2 x + q^2 y - z = 0. \quad \dots(1)$$

Charpit's auxiliary equations are

$$\frac{dp}{-p + p^2} = \frac{dq}{-q + q^2} = \frac{dx}{-2px} = \frac{dy}{-2qy} = \dots$$

from which it follows that

$$\frac{p^2 dx + 2px dp}{p^2 x} = \frac{q^2 dy + 2qy dq}{q^2 y}.$$

Integrating, $\log p^2 x = \log q^2 y + \log a$.

$\therefore p^2 x = a q^2 y. \quad \dots(2)$

Solving (1) and (2) simultaneously for p and q , we get

$$q = \left\{ \frac{z}{(1+a)y} \right\}^{1/2} \quad \text{and} \quad p = \left\{ \frac{az}{(1+a)x} \right\}^{1/2}.$$

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = \left\{ \frac{az}{(1+a)x} \right\}^{1/2} dx + \left\{ \frac{z}{(1+a)y} \right\}^{1/2} dy$$

or
$$\sqrt{1+a} \frac{dz}{\sqrt{z}} = \sqrt{a} \frac{dx}{\sqrt{x}} + \frac{dy}{\sqrt{y}}.$$

Integrating, $\sqrt{\{(1+a)z\}} = \sqrt{ax} + \sqrt{y} + b$,

which is a complete integral of (1).

Comprehensive Exercise 4

Apply Charpit's method to find the complete integrals of the following equations:

- $z^2 (p^2 z^2 + q^2) = 1.$ (Lucknow 2006, 07, 08; Meerut 07; Purvanchal 09, 11)
- $(p+q)(px+qy) = 1.$ (Kanpur 2010; Purvanchal 07)
- $pxy + pq + qy = yz.$ (Avadh 2009)

4. $z = px + qy + p^2 + q^2$. (Kanpur 2009; Purvanchal 07)
5. $z = pq$. (Rohilkhand 2009)
6. $\sqrt{p} + \sqrt{q} = 2x$. (Rohilkhand 2004)
7. $q = -px + p^2$.
8. $q = 3p^2$. (Rohilkhand 2014)
9. $yz p^2 - q = 0$.
10. $zpq = p + q$.
11. $2(z + px + qy) = yp^2$. (Lucknow 2011)
12. $z^2 = pqxy$. (Meerut 2006; Lucknow 10)

Answers 4

1. $(a^2 z^2 + 1)^3 = 9a^4 (ax + y + b)^2$
2. $\sqrt{(1+a)z} = 2\sqrt{(x+ay)+b}$
3. $(z - ax)(y + a)^a = be^y$
4. $z = ax + by + a^2 + b^2$
5. $2\sqrt{z} = \sqrt{ax} + (1/\sqrt{a})y + b$
6. $z = \frac{1}{6}(2x - a)^3 + a^2 y + b$
7. $z = \frac{x^2}{4} \pm \frac{1}{2} \left[\frac{x}{2} \sqrt{(x^2 + 4a)} + 2a \log \{x + \sqrt{(x^2 + 4a)}\} \right] + ay + b$
8. $z = ax + 3x^2 y + b$
9. $z^2 = 2ax + a^2 y^2 + b$
10. $z^2 = 2(a+1)\{x + (y/a)\} + b$
11. $z = \frac{ax}{y^2} - \frac{a^2}{4y^3} + \frac{b}{y}$
12. $z = bx^a y^{1/a}$

8.15 Special Methods of Solution Applicable to Certain Standard Forms.

There are a few standard forms to which many first order differential equations are reducible and which can be integrated by methods which are sometimes shorter than the general method.

Standard I: Equations involving only p and q and no x, y, z

Let the equation be written as

$$f(p, q) = 0. \quad \dots(1)$$

The complete integral is given by

$$z = ax + by + c, \quad \dots(2)$$

where a and b are connected by

$$f(a, b) = 0. \quad \dots(3)$$

Obviously, from (2) we have

$$p = \frac{\partial z}{\partial x} = a$$

and

$$q = \frac{\partial z}{\partial y} = b.$$

Substituting $p = a$ and $q = b$ in (3), we get (1).

From (3) we may find b in terms of a

i.e., $b = \phi(a)$ say.

The complete integral of (1) is

$$z = ax + \phi(a) \cdot y + c. \quad \dots(4)$$

General Integral: Putting $c = \psi(a)$ in (4), where ψ denotes an arbitrary function, we get

$$z = ax + \phi(a)y + \psi(a). \quad \dots(5)$$

Differentiating (5) partially w.r.t. a , we get

$$0 = x + \phi'(a)y + \psi'(a). \quad \dots(6)$$

The *general integral* is obtained by eliminating a between (5) and (6).

Singular Integral: The singular integral, if it exists, is obtained by eliminating a and c between the complete integral (4) and the equations formed by differentiating (4) partially w.r.t. a and c i.e., between the equations

$$z = ax + \phi(a)y + c,$$

$$0 = x + \phi'(a) \cdot y$$

and

$$0 = 1.$$

Since $1 = 0$ is inconsistent, therefore, in this case there is no singular integral.

Note. In many cases, using some transformations, equations can be reduced to the form of the standard I.

Illustrative Examples

Example 20: Solve $p^2 + q^2 = m^2$ (1)

Solution: The given equation is of the form $f(p, q) = 0$.

Therefore a complete integral is given by

$$z = ax + by + c \quad \text{where} \quad a^2 + b^2 = m^2$$

i.e., $z = ax + \sqrt{(m^2 - a^2)} \cdot y + c$ (2)

To find the general integral put $c = \phi(a)$ in (2).

Then $z = ax + \sqrt{(m^2 - a^2)} \cdot y + \phi(a)$ (3)

Differentiating (3) partially w.r.t. a , we get

$$0 = x - \frac{a}{\sqrt{(m^2 - a^2)}} y + \phi'(a). \quad \dots(4)$$

Eliminating a from (3) and (4) the general integral is obtained.

Example 21: Find a complete integral of

$$(x + y)(p + q)^2 + (x - y)(p - q)^2 = 1. \quad \dots(1)$$

Solution: Put $x + y = X^2$, $x - y = Y^2$

so that
$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial X} \cdot \frac{\partial X}{\partial x} + \frac{\partial z}{\partial Y} \cdot \frac{\partial Y}{\partial x} = \frac{1}{2X} \cdot \frac{\partial z}{\partial X} + \frac{1}{2Y} \cdot \frac{\partial z}{\partial Y}$$

and
$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial X} \cdot \frac{\partial X}{\partial y} + \frac{\partial z}{\partial Y} \cdot \frac{\partial Y}{\partial y} = \frac{1}{2X} \frac{\partial z}{\partial X} - \frac{1}{2Y} \frac{\partial z}{\partial Y}.$$

These give $p + q = \frac{1}{X} \frac{\partial z}{\partial X}$, $p - q = \frac{1}{Y} \frac{\partial z}{\partial Y}$.

Substituting these values in the equation (1), we get

$$\left(\frac{\partial z}{\partial X}\right)^2 + \left(\frac{\partial z}{\partial Y}\right)^2 = 1, \quad \dots(2)$$

which is of the form of standard I.

Hence, a complete integral of (2) is given by

$$z = aX + bY + c, \text{ where } a^2 + b^2 = 1.$$

$\therefore$ a complete integral of (1) is given by

$$z = a\sqrt{x + y} + \sqrt{1 - a^2}\sqrt{x - y} + c,$$

where a and c are arbitrary constants.

Example 22: Solve $x^2 p^2 + y^2 q^2 = z^2$.

(Purvanchal 2009)

Solution: The given equation can be written as

$$\left(\frac{x}{z} \frac{\partial z}{\partial x}\right)^2 + \left(\frac{y}{z} \frac{\partial z}{\partial y}\right)^2 = 1. \quad \dots(1)$$

Put $\frac{dx}{x} = dX$, $\frac{dy}{y} = dY$, $\frac{dz}{z} = dZ$

i.e., $X = \log x$, $Y = \log y$, $Z = \log z$

so that
$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial X} \cdot \frac{\partial X}{\partial x} + \frac{\partial z}{\partial Y} \cdot \frac{\partial Y}{\partial x} = \frac{\partial z}{\partial X} \cdot \frac{1}{x}$$

$\therefore \frac{\partial Z}{\partial X} = \frac{\partial Z}{\partial z} \cdot \frac{\partial z}{\partial X} = \frac{1}{z} \cdot p \cdot x = \frac{x}{z} \cdot \frac{\partial z}{\partial x}$

Similarly
$$\frac{\partial Z}{\partial Y} = \frac{y}{z} \frac{\partial z}{\partial y}.$$

Substituting these values in (1), we get

$$\left(\frac{\partial Z}{\partial X}\right)^2 + \left(\frac{\partial Z}{\partial Y}\right)^2 = 1, \quad \dots(2)$$

which is of the form of standard I i.e., of the form

$$f\left(\frac{\partial Z}{\partial X}, \frac{\partial Z}{\partial Y}\right) = 0.$$

Hence, a complete integral of (2) is given by

$$Z = aX + bY + c_1, \text{ where } a^2 + b^2 = 1.$$

$\therefore$ a complete integral of (1) is given by

$$\log z = a \log x + \sqrt{1 - a^2} \log y + c_1.$$

If we take $a = \cos \alpha, c_1 = \log c$ then complete integral can be written as

$$\log z = \cos \alpha \log x + \sin \alpha \log y + \log c$$

or

$$z = c x^{\cos \alpha} \cdot y^{\sin \alpha}, \text{ where } \alpha \text{ and } c \text{ are arbitrary constants.}$$

General integral is obtained by eliminating α from

$$z = \phi(\alpha) x^{\cos \alpha} \cdot y^{\sin \alpha}, \text{ where } c = \phi(\alpha)$$

and

$$0 = \phi'(\alpha) x^{\cos \alpha} \cdot y^{\sin \alpha} + \phi(\alpha) \{x^{\cos \alpha} \cdot \log x \cdot (-\sin \alpha) \cdot y^{\sin \alpha} + x^{\cos \alpha} \cdot y^{\sin \alpha} \cdot \log y \cdot \cos \alpha\}$$

Singular integral is obtained by eliminating α and c between the equations

$$z = c x^{\cos \alpha} \cdot y^{\sin \alpha},$$

$$0 = c \{x^{\cos \alpha} \cdot \log x \cdot (-\sin \alpha) \cdot y^{\sin \alpha} + x^{\cos \alpha} \cdot y^{\sin \alpha} \cdot \log y \cdot \cos \alpha\}$$

and

$$0 = x^{\cos \alpha} \cdot y^{\sin \alpha}.$$

Hence the singular integral is $z = 0$.

Example 23: Find a complete integral of $(y - x)(qy - px) = (p - q)^2$.

(Agra 2002; Kanpur 07, 08; Purvanchal 11)

Solution: Put $x + y = X, xy = Y$, so that

$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial X} \cdot \frac{\partial X}{\partial x} + \frac{\partial z}{\partial Y} \cdot \frac{\partial Y}{\partial x} = \frac{\partial z}{\partial X} \cdot 1 + \frac{\partial z}{\partial Y} \cdot y$$

and

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial X} \cdot \frac{\partial X}{\partial y} + \frac{\partial z}{\partial Y} \cdot \frac{\partial Y}{\partial y} = \frac{\partial z}{\partial X} \cdot 1 + \frac{\partial z}{\partial Y} \cdot x.$$

Substituting these values of p and q in the given equation, we get

$$(y - x) \left[\left(\frac{\partial z}{\partial X} + x \frac{\partial z}{\partial Y} \right) y - \left(\frac{\partial z}{\partial X} + y \frac{\partial z}{\partial Y} \right) x \right] = (y - x)^2 \left(\frac{\partial z}{\partial Y} \right)^2$$

or

$$(y - x)^2 \frac{\partial z}{\partial X} = (y - x)^2 \left(\frac{\partial z}{\partial Y} \right)^2$$

or

$$\frac{\partial z}{\partial X} = \left(\frac{\partial z}{\partial Y} \right)^2, \text{ which is of the form of standard I.}$$

Hence, a complete integral is given by

$$z = aX + bY + c, \text{ where } a = b^2.$$

$\therefore$ a complete integral of the given equation is

$$z = a(x + y) + \sqrt{a}(xy) + c,$$

where a and c are arbitrary constants.

Example 24: Find a complete integral of $pq = x^m y^n z^{2l}$.

(Avadh 2014)

Solution: The given equation can be written as

$$\frac{pz^{-l}}{x^m} \cdot \frac{qz^{-l}}{y^n} = 1.$$

Put $X = \frac{x^{m+1}}{m+1}$, $Y = \frac{y^{n+1}}{n+1}$, $Z = \frac{z^{1-l}}{1-l}$,

so that $\frac{\partial Z}{\partial X} = \frac{\partial Z}{\partial x} \cdot \frac{dx}{dX} = z^{-l} \cdot p \frac{1}{x^m}$, $\frac{\partial Z}{\partial Y} = \frac{\partial Z}{\partial y} \cdot \frac{dy}{dY} = z^{-l} q \frac{1}{y^n}$.

Putting these values in (1), we get

$$\frac{\partial Z}{\partial X} \cdot \frac{\partial Z}{\partial Y} = 1, \text{ which is of the form of standard I.}$$

Hence a complete integral is given by

$$Z = aX + bY + c, \text{ where } ab = 1.$$

$\therefore$ a complete integral of the given equation is

$$\frac{z^{1-l}}{1-l} = a \frac{x^{m+1}}{m+1} + \frac{y^{n+1}}{a(n+1)} + c,$$

where a and c are arbitrary constants.

Standard II: Equations involving only p, q and z i.e., equations of the form

$$f(z, p, q) = 0. \quad \dots(1)$$

Let us assume $z = f(x + ay)$ as a trial solution of given equation (1), where a is an arbitrary constant.

$\therefore z = f(X)$ where $X = x + ay$.

$\therefore p = \frac{\partial z}{\partial x} = \frac{dz}{dX} \frac{\partial X}{\partial x} = \frac{dz}{dX}$

and $q = \frac{\partial z}{\partial y} = \frac{dz}{dX} \frac{\partial X}{\partial y} = a \frac{dz}{dX}.$

$\therefore$ Equation (1) reduces to the form $f\left(z, \frac{dz}{dX}, a \frac{dz}{dX}\right) = 0$

which is an ordinary differential equation of order one. Integrating it we may get the complete integral.

The general and the singular integrals are to be found in the usual way.

Rule: The method of solving the equations of the Standard II can be formulated in the following rule :

Put $\frac{dz}{dX}$ for p , $a \frac{dz}{dX}$ for q , where $X = x + ay$.

Now solve the resulting ordinary differential equation in the variables z and X . Then substitute $x + ay$ for X .

This gives a complete solution.

Note: Sometimes using transformations equations reduce to the form of standard II.

Illustrative Examples

Example 25: Find a complete integral of

$$9(p^2z + q^2) = 4.$$

Solution: The given equation is of the form $f(z, p, q) = 0$.

Putting $\frac{dz}{dX}$ for p , $a \frac{dz}{dX}$ for q , the given equation becomes

$$9 \left[z \left(\frac{dz}{dX} \right)^2 + a^2 \left(\frac{dz}{dX} \right)^2 \right] = 4, \text{ where } X = x + ay$$

$$\text{or } 9(z + a^2) \left(\frac{dz}{dX} \right)^2 = 4, \quad \text{or } \frac{dz}{dX} = \frac{2}{3\sqrt{(z + a^2)}}$$

$$\text{or } \frac{3}{2} \sqrt{(z + a^2)} dz = dX.$$

$$\text{Integrating, } (z + a^2)^{3/2} = X + b$$

$$\text{or } (z + a^2)^{3/2} = x + ay + b$$

$$\text{or } (z + a^2)^3 = (x + ay + b)^2,$$

which is a complete integral of the given equation.

Example 26: Find a complete integral of

$$z^2(p^2z^2 + q^2) = 1.$$

(Lucknow 2006; Meerut 08)

Solution: Putting $\frac{dz}{dX}$ for p , $a \frac{dz}{dX}$ for q , the given equation becomes

$$z^2 \left[\left(\frac{dz}{dX} \right)^2 \cdot z^2 + a^2 \left(\frac{dz}{dX} \right)^2 \right] = 1, \text{ where } X = x + ay$$

$$\text{or } z^2(z^2 + a^2) \left(\frac{dz}{dX} \right)^2 = 1$$

$$\text{or } z \sqrt{(z^2 + a^2)} dz = dX.$$

$$\text{Integrating, } \frac{1}{3} (z^2 + a^2)^{3/2} = X + b$$

$$\text{or } 9(x + ay + b)^2 = (z^2 + a^2)^3,$$

which is a complete integral of the given equation.

Example 27: Find a complete integral of $pz = 1 + q^2$.

Solution: The given equation is of the form $f(p, q, z) = 0$.

Putting $\frac{dz}{dX}$ for p , $a \frac{dz}{dX}$ for q , the given equation becomes

$$z \frac{dz}{dX} = 1 + a^2 \left(\frac{dz}{dX} \right)^2, \text{ where } X = x + ay$$

$$\text{or } a^2 \left(\frac{dz}{dX} \right)^2 - z \frac{dz}{dX} + 1 = 0.$$

$$\therefore \frac{dz}{dX} = \frac{z \pm \sqrt{(z^2 - 4a^2)}}{2a^2}$$

$$\text{or } \frac{dz}{z \pm \sqrt{(z^2 - 4a^2)}} = \frac{dX}{2a^2} \quad \text{or } \frac{z \mp \sqrt{(z^2 - 4a^2)}}{4a^2} dz = \frac{dX}{2a^2}$$

$$\text{or } \{z \mp \sqrt{(z^2 - 4a^2)}\} dz = 2 dX.$$

Integrating, we get

$$\frac{z^2}{2} \mp \left[\frac{z}{2} \sqrt{(z^2 - 4a^2)} - \frac{1}{2} \cdot 4a^2 \log \{z \pm \sqrt{(z^2 - 4a^2)}\} \right] = 2X + b$$

$$\text{or } z^2 \mp [z \sqrt{(z^2 - 4a^2)} - 4a^2 \log \{z \pm \sqrt{(z^2 - 4a^2)}\}] = 4(x + ay) + b,$$

which is a complete integral of the given equation.

Example 28: Find a complete integral of $p^2 = z^2 (1 - pq)$.

Solution: The given equation is of the form $f(p, q, z) = 0$.

Putting $\frac{dz}{dX}$ for p , $a \frac{dz}{dX}$ for q , the given equation becomes

$$\left(\frac{dz}{dX} \right)^2 = z^2 \left\{ 1 - \frac{dz}{dX} a \frac{dz}{dX} \right\}, \text{ where } X = x + ay$$

$$\text{or } \left(\frac{dz}{dX} \right)^2 (1 + az^2) = z^2$$

$$\text{or } \frac{\sqrt{(1 + az^2)}}{z} dz = dX$$

$$\text{or } \frac{1 + az^2}{z \sqrt{(1 + az^2)}} dz = dX$$

$$\text{or } \left\{ \frac{1}{z \sqrt{(1 + az^2)}} + \frac{az}{\sqrt{(1 + az^2)}} \right\} dz = dX.$$

$$\text{Integrating, } \frac{1}{\sqrt{a}} \log [z \sqrt{a} + \sqrt{(1 + az^2)}] + \sqrt{(1 + az^2)} = X + b$$

$$\text{or } \frac{1}{\sqrt{a}} \log [z \sqrt{a} + \sqrt{(1 + az^2)}] + \sqrt{(1 + az^2)} = x + ay + b,$$

which is a complete integral of the given equation.

Standard III: Equation of the form $f(x, p) = F(y, q)$.

As a trial solution let us put each side equal to an arbitrary constant

$$\text{i.e., } f(x, p) = F(y, q) = a \quad \dots(1)$$

from which we obtain

$$p = f_1(x, a) \quad \text{and} \quad q = f_2(y, a).$$

Now from $dz = p dx + q dy$, we have $dz = f_1(x, a) dx + f_2(y, a) dy$.

$$\text{Integrating, we get } z = \int f_1(x, a) dx + \int f_2(y, a) dy + b$$

which is the complete integral.

The general integral can be obtained in the usual way. As in the case of standard I, there is no singular integral.

Rule: Write the differential equation in the form (1). Put both the sides of the equation equal to an arbitrary constant. Solving them find the values of p and q . Substitute the values of p and q in $dz = p dx + q dy$ and integrate to find a complete integral.

Note: Sometimes using transformations equations reduce to the form of standard III.

Illustrative Examples

Example 29: Find a complete integral of $p^2 + q^2 = x + y$. (Meerut 2007; Rohilkhand 03)

Solution: Separating q and y from p and x , the given equation can be written as

$$p^2 - x = y - q^2 = a, \text{ (say).}$$

$$\therefore p = \sqrt{x + a} \text{ and } q = \sqrt{y - a}.$$

Putting the values of p and q in $dz = p dx + q dy$, we get

$$dz = \sqrt{x + a} dx + \sqrt{y - a} dy.$$

$$\text{Integrating, } z = \frac{2}{3} (x + a)^{3/2} + \frac{2}{3} (y - a)^{3/2} + b,$$

which is a complete integral of the given equation.

Example 30: Find a complete integral of $yp = 2yx + \log q$.

(Lucknow 2010)

Solution: The given equation can be written as

$$p = 2x + \frac{1}{y} \log q$$

$$\text{or } p - 2x = \frac{1}{y} \log q = a, \text{ (say).}$$

$$\therefore p = 2x + a, \log q = ay \text{ i.e., } q = e^{ay}.$$

Putting the values of p and q in $dz = p dx + q dy$, we get

$$dz = (2x + a) dx + e^{ay} dy.$$

$$\text{Integrating, } z = x^2 + ax + \frac{1}{a} e^{ay} + b,$$

which is a complete integral of the given equation.

Example 31: Find a complete integral of

$$z^2 (p^2 + q^2) = x^2 + y^2.$$

(Avadh 2006)

Solution: Put $z dz = dZ$; i.e., $Z = \frac{1}{2} z^2$, so that

$$z \frac{\partial z}{\partial x} = \frac{\partial Z}{\partial x} = P \text{ (say), } z \frac{\partial z}{\partial y} = \frac{\partial Z}{\partial y} = Q \text{ (say).}$$

Putting these values the given equation becomes

$$P^2 + Q^2 = x^2 + y^2 \text{ or } P^2 - x^2 = y^2 - Q^2 = a^2 \text{ (say).}$$

$$\therefore P = \sqrt{(a^2 + x^2)}, Q = \sqrt{(y^2 - a^2)}.$$

$$\text{Now } dZ = P dx + Q dy = \sqrt{(a^2 + x^2)} dx + \sqrt{(y^2 - a^2)} dy.$$

Integrating ,

$$Z = \frac{1}{2} x \sqrt{(a^2 + x^2)} + \frac{a^2}{2} \log \{x + \sqrt{(a^2 + x^2)}\} \\ + \frac{1}{2} y \sqrt{(y^2 - a^2)} - \frac{a^2}{2} \log \{y + \sqrt{(y^2 - a^2)}\} + b$$

$$\text{or } z^2 = x \sqrt{(a^2 + x^2)} + a^2 \log \{x + \sqrt{(a^2 + x^2)}\} + y \sqrt{(y^2 - a^2)} \\ - a^2 \log \{y + \sqrt{(y^2 - a^2)}\} + b,$$

which is a complete integral of the given equation.

Example 32: Find a complete integral of $x^2 y^3 p^2 q = z^3$.

Solution: The given equation can be written as

$$x^2 y^3 \left(\frac{1}{z} \frac{\partial z}{\partial x} \right)^2 \left(\frac{1}{z} \frac{\partial z}{\partial y} \right) = 1. \quad \dots(1)$$

Put $\frac{1}{z} dz = dZ$ i.e., $Z = \log z$. Then

$$\frac{1}{z} \frac{\partial z}{\partial x} = \frac{\partial Z}{\partial x} = P \text{ (say), } \frac{1}{z} \frac{\partial z}{\partial y} = \frac{\partial Z}{\partial y} = Q \text{ (say).}$$

The equation (1) reduces to $x^2 y^3 P^2 Q = 1$

$$\text{or } x^2 P^2 = \frac{1}{Q y^3} = a^2, \text{ (say).}$$

$$\therefore P = \frac{a}{x}, Q = \frac{1}{a^2 y^3}.$$

$$\text{Now } dZ = P dx + Q dy = \frac{a}{x} dx + \frac{1}{a^2 y^3} dy.$$

$$\text{Integrating, } Z = a \log x - \frac{1}{2a^2 y^2} + b$$

$$\text{or } \log z = a \log x - \frac{1}{2a^2 y^2} + b,$$

which is a complete integral of the given equation.

Example 33: Find a complete integral of $z(p^2 - q^2) = x - y$.

(Lucknow 2006)

Solution: The given equation can be written as

$$\left[\left(\sqrt{z} \frac{\partial z}{\partial x} \right)^2 - \left(\sqrt{z} \frac{\partial z}{\partial y} \right)^2 \right] = x - y. \quad \dots(1)$$

Putting $\sqrt{z} dz = dZ$ i.e., $Z = \frac{2}{3} z^{3/2}$,

so that $\frac{\partial Z}{\partial x} = \sqrt{z} \frac{\partial z}{\partial x} = P$ (say), $\frac{\partial Z}{\partial y} = \sqrt{z} \frac{\partial z}{\partial y} = Q$ (say),

the equation (1) reduces to

$$P^2 - Q^2 = x - y \text{ or } P^2 - x = Q^2 - y = a \text{ (say).}$$

$\therefore P = \sqrt{a+x}, Q = \sqrt{a+y}$.

Now $dZ = P dx + Q dy = \sqrt{a+x} dx + \sqrt{a+y} dy$.

Integrating, $Z = \frac{2}{3} (a+x)^{3/2} + \frac{2}{3} (a+y)^{3/2} + b$

or $z^{3/2} = (a+x)^{3/2} + (a+y)^{3/2} + b$,

which is a complete integral of the given equation.

Standard IV: Equation of the form

$$z = px + qy + f(p, q) \quad \dots(1)$$

(analogous to Clairaut's form)

We know that the solution of Clairaut's equation

$$y = px + f(p) \text{ where } p = \frac{dy}{dx} \text{ is } y = cx + f(c).$$

Similarly the complete integral of Clairaut's equation (1) is

$$z = ax + by + f(a, b).$$

Rule: To get the complete integral of the equation of this type replace p and q by a, b (two arbitrary constants) respectively.

General Integral is obtained as in other cases.

Singular Integral. The complete integral is

$$F = z - ax - by - f(a, b) = 0 \quad \dots(1)$$

$$\frac{\partial F}{\partial a} = 0, \text{ gives } x + \frac{\partial f}{\partial a} = 0, \quad \dots(2)$$

$$\text{and } \frac{\partial F}{\partial b} = 0, \text{ gives } y + \frac{\partial f}{\partial b} = 0. \quad \dots(3)$$

Singular integral is obtained by eliminating a, b from (1), (2) and (3).

Illustrative Examples

Example 34: Find a complete integral of

$$z = px + qy + p^2 + q^2.$$

(Kanpur 2009)

Solution: The given equation is of the form of standard IV i.e., of the form

$$z = px + qy + f(p, q).$$

Hence a complete integral is given by

$$z = ax + by + a^2 + b^2.$$

Example 35: Find the singular integral of

$$z = px + qy + \log pq.$$

(Lucknow 2009)

Solution: The complete integral of the given equation is

$$z = ax + by + \log ab. \quad \dots(1)$$

Differentiating (1) partially w.r.t. a and b , we get

$$0 = x + \frac{1}{a}$$

and $0 = y + \frac{1}{b}.$

$\therefore a = -1/x$

and $b = -1/y. \quad \dots(2)$

Eliminating a and b between (1) and the equations (2), we get

$$z = x \left(-\frac{1}{x} \right) + y \left(-\frac{1}{y} \right) + \log \left(\frac{1}{xy} \right)$$

or $z = -2 - \log xy$, which is the required singular integral.

Example 36: Find a complete integral and the singular integral of

$$4xyz = pq + 2px^2y + 2qxy^2.$$

Solution: Put $x^2 = X$, $y^2 = Y$, so that

$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial X} \frac{dX}{dx} = 2X^{1/2} \frac{\partial z}{\partial X}, \quad q = 2Y^{1/2} \frac{\partial z}{\partial Y}.$$

The given equation then reduces to

$$z = X \frac{\partial z}{\partial X} + Y \frac{\partial z}{\partial Y} + \frac{\partial z}{\partial X} \cdot \frac{\partial z}{\partial Y}.$$

$\therefore$ a complete integral is $z = aX + bY + ab$

or $z = ax^2 + by^2 + ab. \quad \dots(1)$

Differentiating (1) partially w.r.t. a and b , we get

$$0 = x^2 + b \quad \text{and} \quad 0 = y^2 + a. \quad \dots(2)$$

Eliminating a and b between (1) and the equations (2), the singular integral is

$$z = -x^2y^2 - x^2y^2 + x^2y^2 \text{ i.e., } z + x^2y^2 = 0.$$

Example 37: Find the singular integral of

$$z = px + qy + c \sqrt{(1+p^2+q^2)}. \quad (\text{Rohilkhand 2011; Avadh 14})$$

Solution: The complete integral of the given equation is

$$z = ax + by + c \sqrt{(1+a^2+b^2)}. \quad \dots(1)$$

Singular integral. Differentiating (1) partially w.r.t. a and b , we get

$$0 = x + \frac{ac}{\sqrt{(1+a^2+b^2)}}, \quad 0 = y + \frac{bc}{\sqrt{(1+a^2+b^2)}}, \quad \dots(2)$$

so that
$$x^2 + y^2 = \frac{a^2 c^2 + b^2 c^2}{1 + a^2 + b^2}$$

i.e.,
$$c^2 - x^2 - y^2 = \frac{c^2}{1 + a^2 + b^2}$$

i.e.,
$$1 + a^2 + b^2 = \frac{c^2}{c^2 - x^2 - y^2} \quad \dots(3)$$

Also, from (2)

$$a = -\frac{x \sqrt{1 + a^2 + b^2}}{c} = \frac{-x}{\sqrt{c^2 - x^2 - y^2}} \quad \dots(4)$$

and
$$b = -\frac{y \sqrt{1 + a^2 + b^2}}{c} = \frac{-y}{\sqrt{c^2 - x^2 - y^2}} \quad \dots(5)$$

Putting the values from (3), (4) and (5) in (1), we get the singular solution as

$$z = -\frac{x^2}{\sqrt{c^2 - x^2 - y^2}} - \frac{y^2}{\sqrt{c^2 - x^2 - y^2}} + \frac{c^2}{\sqrt{c^2 - x^2 - y^2}}$$

or
$$z = \frac{c^2 - x^2 - y^2}{\sqrt{c^2 - x^2 - y^2}} \quad \text{i.e., } z^2 = c^2 - x^2 - y^2 \quad \text{i.e., } x^2 + y^2 + z^2 = c^2.$$

Comprehensive Exercise 5

Find complete integrals of the following equations :

1. $q = e^{-p/\alpha}.$
2. $p^2 - q^2 = \lambda.$
3. $\sqrt{p} + \sqrt{q} = 1.$
4. $p^2 = zq.$
5. $p^3 + q^3 - 3pqz = 0.$
6. $p(1 + q) = qz.$
7. $\sqrt{p} + \sqrt{q} = 2x.$
8. $pq = xy.$
9. $p - 3x^3 = q^2 - y.$
10. $z = px + qy + pq.$
11. $z = px + qy - 2\sqrt{pq}.$ (Lucknow 2010)
12. $z = px + qy - p^2q.$ (Lucknow 2008)
13. $p^2 + q^2 = z^2(x + y).$
14. $z = px + qy + \frac{q}{p} - p.$
15. $z = px + qy + 3p^{1/3}q^{1/3}.$

Answers 5

1. $z = ax + ye^{-a/\alpha} + c$
2. $z = ax + (a^2 - \lambda)^{1/2} + c$
3. $z = ax + (1 - \sqrt{a})^2 y + c$
4. $z = ax + (a^2 - \lambda)^{1/2} + c$

5. $3a(x + ay) + b = (1 + a^3) \log z$ 6. $az - 1 = be^{x+ay}$
 7. $z = \frac{1}{6}(a + 2x)^3 + a^2y + b$ 8. $2az = a^2x^2 + y^2 + 2ab$
 9. $z = x^3 + ax \pm \frac{2}{3}(y + a)^{3/2} + b$ 10. $z = ax + by + ab$
 11. $z = ax + by - 2\sqrt{ab}$ 12. $z = ax + by - a^2b$
 13. $\frac{3}{2} \log z = (a + x)^{3/2} + (y - a)^{3/2} + b$
 14. $z = ax + by + \frac{b}{a} - a$ 15. $z = ax + by + 3(ab)^{1/3}$

8.16 Compatible Systems of First Order Partial Differential Equations

Definition: If every solution of the first order partial differential equation

$$f(x, y, z, p, q) = 0 \quad \dots(1)$$

is also a solution of the first order partial differential equation

$$g(x, y, z, p, q) = 0, \quad \dots(2)$$

then the differential equations (1) and (2) are said to be **compatible**.

To find the condition for the pair of equations (1) and (2) to be compatible.

If $J = \frac{\partial(f, g)}{\partial(p, q)} \neq 0$, then the equations (1) and (2) can be solved to obtain the explicit

$$\text{expressions } p = \phi(x, y, z), q = \psi(x, y, z) \quad \dots(3)$$

for p and q .

The condition that the pair of equations (1) and (2) should be compatible reduces then to the condition that the system of equations (3) should be completely integrable.

If z is a function of x and y , then we know that

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = p dx + q dy.$$

Therefore, the system of equations (3) will be completely integrable if and only if the single differential equation

$$dz = \phi dx + \psi dy \quad \text{i.e., } \phi dx + \psi dy - dz = 0 \quad \dots(4)$$

is integrable.

We know that the necessary and sufficient condition for the single differential equation $P dx + Q dy + R dz = 0$ to be integrable is that

$$P \left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y} \right) + Q \left(\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z} \right) + R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) = 0.$$

Therefore, the single differential equation (4) is integrable if and only if

$$\phi(\psi_z - 0) + \psi(0 - \phi_z) - 1(\phi_y - \psi_x) = 0 \quad [\because P = \phi, Q = \psi, R = -1]$$

i.e., if and only if $\psi_x + \phi \psi_z = \phi_y + \psi \phi_z$ (5)

Substituting from equations (3) into equation (1) and differentiating with respect to x and z , respectively, we obtain the equations

$$f_x + f_p \phi_x + f_q \psi_x = 0 \quad \dots (6)$$

$$\text{and} \quad f_z + f_p \phi_z + f_q \psi_z = 0. \quad \dots (7)$$

Multiplying (7) by ϕ and adding to (6), we get

$$f_x + \phi f_z + f_p (\phi_x + \phi \phi_z) + f_q (\psi_x + \phi \psi_z) = 0. \quad \dots (8)$$

In a similar manner, equation (2) will give

$$g_x + \phi g_z + g_p (\phi_x + \phi \phi_z) + g_q (\psi_x + \phi \psi_z) = 0. \quad \dots (9)$$

Now solving the equations (8) and (9), we obtain

$$\psi_x + \phi \psi_z = \frac{1}{J} \left[\frac{\partial (f, g)}{\partial (x, p)} + \phi \frac{\partial (f, g)}{\partial (z, p)} \right] \quad \dots (10)$$

$$\text{where} \quad J = \frac{\partial (f, g)}{\partial (p, q)}.$$

If we had differentiated the given pair of equations (1) and (2) with respect to y and z after substituting in them from equations (3), then we would have obtained

$$\phi_y + \psi \phi_z = -\frac{1}{J} \left[\frac{\partial (f, g)}{\partial (y, q)} + \psi \frac{\partial (f, g)}{\partial (z, q)} \right]. \quad \dots (11)$$

Substituting from equations (10) and (11) in (5), we get the desired condition that the two equations (1) and (2) should be compatible as

$$[f, g] = 0, \quad \dots (12)$$

$$\text{where} \quad [f, g] = \frac{\partial (f, g)}{\partial (x, p)} + p \frac{\partial (f, g)}{\partial (z, p)} + \frac{\partial (f, g)}{\partial (y, q)} + q \frac{\partial (f, g)}{\partial (z, q)}. \quad \dots (13)$$

$$[\because p = \phi \text{ and } q = \psi]$$

Particular Case:

Theorem: The first order partial differential equations

$$p = P(x, y), \quad q = Q(x, y)$$

are compatible if and only if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

Proof: We know that if z is a function of two independent variables x and y , then

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = p dx + q dy.$$

So, the partial differential equations $p = P(x, y), q = Q(x, y)$ are compatible if and only if the single differential equation

$$dz = P dx + Q dy$$

is integrable.

Now we know that if P and Q are functions of two variables x and y , then $P dx + Q dy$ is an exact differential $d\phi(x, y)$ if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

$\therefore$ The single differential equation $dz = P dx + Q dy$ is integrable if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Hence, the differential equations $p = P(x, y), q = Q(x, y)$ are compatible if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Illustrative Examples

Example 38: Show that the equations

$$xp - yq = x, \quad x^2 p + q = xz$$

are compatible and find their solution.

Solution: We know that the first order partial differential equations

$$f(x, y, z, p, q) = 0$$

and

$$g(x, y, z, p, q) = 0$$

are compatible if and only if $[f, g] = 0$,

$$\text{where } [f, g] = \frac{\partial(f, g)}{\partial(x, p)} + p \frac{\partial(f, g)}{\partial(z, p)} + \frac{\partial(f, g)}{\partial(y, q)} + q \frac{\partial(f, g)}{\partial(z, q)}.$$

Here, the given equations are $f(x, y, z, p, q) \equiv xp - yq - x = 0$

and

$$g(x, y, z, p, q) \equiv x^2 p + q - xz = 0.$$

$$\therefore \frac{\partial(f, g)}{\partial(x, p)} = \begin{vmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial p} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial p} \end{vmatrix} = \begin{vmatrix} p-1 & x \\ 2xp-z & x^2 \end{vmatrix} = x^2(p-1) - x(2xp-z),$$

$$\frac{\partial(f, g)}{\partial(z, p)} = \begin{vmatrix} \frac{\partial f}{\partial z} & \frac{\partial f}{\partial p} \\ \frac{\partial g}{\partial z} & \frac{\partial g}{\partial p} \end{vmatrix} = \begin{vmatrix} 0 & x \\ -x & x^2 \end{vmatrix} = 0 - (-x^2) = x^2,$$

$$\frac{\partial(f, g)}{\partial(y, q)} = \begin{vmatrix} \frac{\partial f}{\partial y} & \frac{\partial f}{\partial q} \\ \frac{\partial g}{\partial y} & \frac{\partial g}{\partial q} \end{vmatrix} = \begin{vmatrix} -q & -y \\ 0 & 1 \end{vmatrix} = -q,$$

and
$$\frac{\partial (f, g)}{\partial (z, q)} = \begin{vmatrix} \frac{\partial f}{\partial z} & \frac{\partial f}{\partial q} \\ \frac{\partial g}{\partial z} & \frac{\partial g}{\partial q} \end{vmatrix} = \begin{vmatrix} 0 & -y \\ -x & 1 \end{vmatrix} = -xy.$$

Therefore,
$$\begin{aligned} [f, g] &= x^2 (p - 1) - 2x^2 p + xz + px^2 - q - qxy \\ &= x^2 p - x^2 - 2x^2 p + xz + px^2 - q - qxy \\ &= -x^2 + xz - q - qxy \\ &= -x^2 + x^2 p + q - q - qxy \\ &\quad [\because \text{from the given differential equations, } xz = x^2 p + q] \\ &= -x^2 + x(xp - yq) \\ &= -x^2 + x \cdot x \\ &\quad [\because \text{from the given differential equations, } xp - yq = x] \\ &= -x^2 + x^2 = 0. \end{aligned}$$

Hence, the given equations are compatible.

Now, let us find p and q by solving the equations

$$xp - yq = x \quad \dots(1)$$

and
$$x^2 p + q = xz \quad \dots(2)$$

Multiplying (2) by y and adding to (1), we get

$$yx^2 p + xp = xyz + x$$

or
$$xp(1 + xy) = x(1 + yz)$$

or
$$p = \frac{1 + yz}{1 + xy}.$$

$\therefore$ from (2),
$$\begin{aligned} q &= xz - x^2 p = xz - \frac{x^2 + x^2 yz}{1 + xy} \\ &= \frac{xz(1 + xy) - x^2 - x^2 yz}{1 + xy} \\ &= \frac{x(z - x)}{1 + xy}. \end{aligned}$$

Thus,
$$p = \frac{1 + yz}{1 + xy}, q = \frac{x(z - x)}{1 + xy}.$$

Now
$$\begin{aligned} dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = p dx + q dy \\ &= \left(\frac{1 + yz}{1 + xy} \right) dx + \left[\frac{x(z - x)}{1 + xy} \right] dy. \end{aligned}$$

So we have to solve the single differential equation

$$\begin{aligned}
 & (1 + xy) dz - (1 + yz) dx - x(z - x) dy = 0 \\
 \Rightarrow & (1 + xy) dz - (1 + xy) dx + (1 + xy) dx - (1 + yz) dx - x(z - x) dy = 0 \\
 \Rightarrow & (1 + xy) (dz - dx) - y(z - x) dx - x(z - x) dy = 0 \\
 \Rightarrow & \frac{(1 + xy) (dz - dx) - (z - x) (y dx + x dy)}{(1 + xy)^2} = 0 \\
 \Rightarrow & \frac{(1 + xy) (dz - dx) - (z - x) d(1 + xy)}{(1 + xy)^2} = 0 \\
 \Rightarrow & d\left(\frac{z - x}{1 + xy}\right) = 0 \\
 \Rightarrow & \frac{z - x}{1 + xy} = c, \text{ where } c \text{ is an arbitrary constant.}
 \end{aligned}$$

Hence, the required solution of the given equations is

$$z - x = c(1 + xy)$$

or $z = x + c(1 + xy)$, where c is a constant.

Example 39: Show that the differential equations $p = 5x - 4y + 3, q = 4x + 5y + 2$ do not possess any common solution.

Solution: We know that the first order partial differential equations

$$p = P(x, y), \quad q = Q(x, y)$$

are compatible if and only if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

Here, $P = 5x - 4y + 3, Q = 4x + 5y + 2$.

We have $\frac{\partial P}{\partial y} = -4$ and $\frac{\partial Q}{\partial x} = 4$.

Since $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$, therefore the given differential equations are not compatible.

Hence, the given differential equations do not possess any common solution.

Example 40: Show that the differential equations

$$\frac{\partial z}{\partial x} = 6x + 3y, \quad \frac{\partial z}{\partial y} = 3x - 4y$$

are compatible and find their solution.

Solution: We know that the first order partial differential equations

$$p = P(x, y), \quad q = Q(x, y)$$

are compatible if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Here, $P(x, y) = 6x + 3y, Q(x, y) = 3x - 4y$.

We have $\frac{\partial P}{\partial y} = 3$

and $\frac{\partial Q}{\partial x} = 3.$

Since $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, therefore the given differential equations are compatible.

$$\begin{aligned}\text{Now } dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = p dx + q dy \\ &= (6x + 3y) dx + (3x - 4y) dy \\ &= 6x dx + 3(y dx + x dy) - 4y dy \\ &= 6x dx + 3d(xy) - 4y dy.\end{aligned}$$

Integrating both sides, we get

$$z = 6 \cdot \frac{x^2}{2} + 3xy - 4 \cdot \frac{y^2}{2} + c$$

or $z = 3x^2 + 3xy - 2y^2 + c$

as the required solution of the given differential equations.

Example 41: Show that the differential equations

$$p = x^2 - ay, q = y^2 - ax$$

are compatible and find their common solution.

Solution: We know that the first order partial differential equations

$$p = P(x, y), q = Q(x, y)$$

are compatible if and only if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

Here, $P(x, y) = x^2 - ay, Q(x, y) = y^2 - ax$.

We have $\frac{\partial P}{\partial y} = -a$ and $\frac{\partial Q}{\partial x} = -a$.

Since $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, therefore the given differential equations are compatible.

$$\begin{aligned}\text{Now } dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = p dx + q dy \\ &= (x^2 - ay) dx + (y^2 - ax) dy \\ &= x^2 dx + y^2 dy - a(x dy + y dx) \\ &= x^2 dx + y^2 dy - ad(xy).\end{aligned}$$

Integrating both sides, we get

$$z = \frac{x^3}{3} + \frac{y^3}{3} - axy + c$$

as the required common solution of the given differential equations.

Example 42: Solve the simultaneous differential equations

$$\frac{\partial z}{\partial x} = x^4 - 2xy^2 + y^4, \quad \frac{\partial z}{\partial y} = -(2x^2y - 4xy^3 + \sin y).$$

Solution: We know that the first order partial differential equations

$$p = P(x, y), \quad q = Q(x, y)$$

are compatible if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Here, $P = x^4 - 2xy^2 + y^4, Q = -2x^2y + 4xy^3 - \sin y.$

We have $\frac{\partial P}{\partial y} = -4xy + 4y^3, \quad \frac{\partial Q}{\partial x} = -4xy + 4y^3.$

Since $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, therefore the given differential equations are compatible.

Now
$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = p dx + q dy$$

$$= P dx + Q dy = (x^4 - 2xy^2 + y^4) dx + (-2x^2y + 4xy^3 - \sin y) dy.$$

Integrating both sides, we get

$$z = \int P dx + \int Q dy + c$$

(treating y as a constant) (taking in Q only those terms which do not contain x)

$$= \int (x^4 - 2xy^2 + y^4) dx + \int (-\sin y) dy + c$$

(treating y as a constant)

$$= \frac{1}{5} x^5 - x^2 y^2 + x y^4 + \cos y + c.$$

Hence, the required common solution of the given simultaneous equations is

$$z = \frac{1}{5} x^5 - x^2 y^2 + xy^4 + \cos y + c,$$

where c is an arbitrary constant.

Comprehensive Exercise 6

1. Show that the differential equations

$$\frac{\partial z}{\partial x} = 7x + 8y - 5, \quad \frac{\partial z}{\partial y} = 9x + 11y + 3$$

are not compatible.

2. Show that the differential equations

$$p = 4x + 3y + 1, \quad q = 3x + 2y + 1$$

are compatible and find their solution.

3. Show that the differential equations

$$p = 1 + 4xy + 2y^2, \quad q = 1 + 4xy + 2x^2$$

are compatible and find their common solution.

4. Show that the differential equations in each of the following systems are compatible and find their solution :

(i) $p = ax + hy + g, \quad q = hx + by + f$

(ii) $p = 2x - y, \quad q = 5 - x - 2y$

(iii) $p = (2ax + by) y, \quad q = (ax + 2by) x$

(iv) $p = x - \frac{y}{x^2 + y^2}, \quad q = y + \frac{x}{x^2 + y^2}$

(v) $p = 1 + e^{x/y}, \quad q = e^{x/y} \left(1 - \frac{x}{y} \right)$

(vi) $p = y^2 e^{xy^2} + 4x^3, \quad q = 2xy e^{xy^2} - 3y^2$

(vii) $p = (e^y + 1) \cos x, \quad q = e^y \sin x$

(viii) $p = \sin x \cos y + e^{2x}, \quad q = \cos x \sin y + \tan y.$

5. Show that the equations $f(x, y, p, q) = 0, \quad g(x, y, p, q) = 0$

are compatible if $\frac{\partial(f, g)}{\partial(x, p)} + \frac{\partial(f, g)}{\partial(y, q)} = 0.$

Answers 6

1. $z = 2x^2 + 3xy + x + y^2 + y + c$

2. $z = x + 2x^2 y + 2xy^2 + y + c$

4. (i) $z = \frac{1}{2} ax^2 + hxy + gx + \frac{1}{2} by^2 + fy + c$

(ii) $z = x^2 - xy - y^2 + 5y + c$

(iii) $z = ayx^2 + bxy^2 + c$

(iv) $z = \frac{1}{2} x^2 - \tan^{-1} \left(\frac{x}{y} \right) + \frac{1}{2} y^2 + c$

(v) $z = x + ye^{x/y} + c$

(vi) $z = e^{xy^2} + x^4 - y^3 + c$

(vii) $z = (e^y + 1) \sin x + c.$

(viii) $z = -\cos x \cos y + \frac{1}{2} e^{2x} + \log \sec y + c.$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- Equation $p \tan y + q \tan x = \sec^2 z$ is of order
 - one
 - two
 - zero
 - None of these.
- Equation $r^2 + 2s - t^2 = 0$ is of order
 - one
 - two
 - three
 - None of these.

(Rohilkhand 2003)
- Equation $\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} + \left(\frac{\partial z}{\partial y}\right)^3 = 0$ is of degree
 - one
 - two
 - three
 - None of these.
- Equation $p^3 + qx^2 + z^4 = 0$ is of degree
 - two
 - three
 - four
 - one.
- The equation $Pp + Qq = R$ is known as
 - Charpit's equation
 - Lagrange's equation
 - Bernoulli's equation
 - Clairaut's equation.

(Rohilkhand 2003)
- Out of the following four partial differential equations, the differential equation which is linear is
 - $\frac{\partial^3 z}{\partial x^3} - 3 \frac{\partial^2 z}{\partial x^2} \cdot \frac{\partial z}{\partial y} + 8 \frac{\partial^2 z}{\partial y^2} = \sin x$
 - $\frac{\partial^2 z}{\partial x^2} + \left(\frac{\partial z}{\partial x}\right)^2 + \frac{\partial z}{\partial y} + 9z = 0$
 - $4 \frac{\partial^2 z}{\partial x^2} + 5 \frac{\partial^2 z}{\partial x \partial y} + 6 \frac{\partial^2 z}{\partial y^2} + 7 \frac{\partial z}{\partial x} + 8 \frac{\partial z}{\partial y} + 3z = (x^3 + y^3) \sin x$
 - $\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = x^2 + y^2.$
- The differential equation $(2x + 3y)p + 4xq - 8pq = x + y$ is
 - linear
 - non-linear
 - quasi-linear
 - semi-linear.

8. The differential equation $(x^2 + y^2) \frac{\partial z}{\partial x} + (x + y) \frac{\partial z}{\partial y} + (3x - 4y)z = x^2 + y^2$ is
- (a) linear (b) quasi-linear
(c) semi-linear (d) non-linear.
9. The differential equation $(x + y - 3z) \frac{\partial z}{\partial x} + (3x + 4y) \frac{\partial z}{\partial y} + 2z = x + y$ is
- (a) linear (b) quasi-linear
(c) semi-linear (d) non-linear.
10. The differential equation $x^2 \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = (x + y)z^2 + 4x$ is
- (a) linear (b) quasi-linear
(c) semi-linear (d) non-linear.
11. Out of the following four pairs of first order partial differential equations, mention the pair in which the differential equations are compatible :
- (a) $p = 4y - 7x + 3, q = 7x + 4y + 2$ (b) $p = x + y, q = y - x$
(c) $p = 3x + 8y, q = 8x - 3y$ (d) $p = x + y + 1, q = 2x - y + 1$.
12. Lagrange's auxiliary equations of $Pp + Qq = R$ are given by
- (a) $\frac{dx}{R} = \frac{dy}{Q} = \frac{dz}{P}$ (b) $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$
(c) $\frac{dx}{Q} = \frac{dy}{R} = \frac{dz}{P}$ (d) None of these.
- (Rohilkhand 2004)

Fill In The Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. The order of the differential equation

$$4 \frac{\partial^3 z}{\partial x^2 \partial y} + 9 \frac{\partial^2 z}{\partial x^2} + 8 \frac{\partial^2 z}{\partial y^2} + 6 \frac{\partial z}{\partial x} + 3 \frac{\partial z}{\partial y} + 8 = 0$$

is

2. In a linear partial differential equation all the partial derivatives occurring in it are in ... degree.
3. Lagrange's linear equation is of the form
4. Lagrange's auxiliary equations or Lagrange's subsidiary equations are
5. Lagrange's auxiliary equations of $(y^2 + z^2 - x^2)p - 2xyq + 2zx = 0$ are
6. Lagrange's subsidiary equations of $z(xp - yq) = y^2 - x^2$ are
7. Lagrange's auxiliary equations of $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} + t \frac{\partial z}{\partial t} = az + \frac{xy}{t}$ are
8. Equations $\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}} = \frac{dF}{0}$ are known as

9. For the differential equation $(p^2 + q^2)y = qz$, Charpit's auxiliary equations are...
10. For the differential equation $p^2x + q^2y = z$, Charpit's auxiliary equations are ...
11. The first order partial differential equations $\frac{\partial z}{\partial x} = P(x, y), \frac{\partial z}{\partial y} = Q(x, y)$ are compatible if $\frac{\partial P}{\partial y} = \dots$
12. If every solution of the differential equation $f(x, y, z, p, q) = 0$ is also a solution of the differential equation $g(x, y, z, p, q) = 0$, then the two differential equations are said to be ...

True or False

Write 'T' for true and 'F' for false statement.

1. The differential equation $\frac{\partial z}{\partial x} \cdot \frac{\partial z}{\partial y} = 3xy$ is non-linear.
2. The differential equation $\frac{\partial^2 u}{\partial x^2} + 2 \frac{\partial^2 u}{\partial x \partial y} + 5 \frac{\partial u}{\partial x} + 6 \frac{\partial u}{\partial y} + 7u = x^2$ is linear.
3. The differential equation $\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$ is non-linear.
4. The differential equation $z = px + qy$ is linear.
5. The differential equation $p^2 + q^2 = (x^2 + y^2)^2$ is quasi-linear.
6. A complete integral of the equation $z = px + qy + p^2 + q^2$ is given by $z = ax + by + a^2 + b^2$.
7. A complete integral of $q = 3p^2$ is $z = ax + 3a^2y + c$, a and c are arbitrary constants.
8. $p(1 + q) = qz$ is a linear equation.
9. $pz - qy = z^2 + (x + y)^2$ is a linear differential equation.
10. The differential equations $\frac{\partial z}{\partial x} = 5x - 7y, \frac{\partial z}{\partial y} = 6x + 8y$ possess a common solution.
11. The differential equations $p = 12x + 7y + 1, q = 7x + 4y + 1$ are compatible.
12. The differential equations $p = \frac{y}{x^2 + y^2} - 1, q = -\frac{x}{x^2 + y^2}$ are compatible.
13. The differential equations $\frac{\partial z}{\partial x} = y \sin 2x, \frac{\partial z}{\partial y} = -(1 + y^2 + \cos^2 x)$ are not compatible.
14. The differential equations $p = \cos x (\cos x - \sin \alpha \sin y), q = \cos y (\cos y - \sin \alpha \sin x)$ are compatible.

15. The differential equations $p = P(x, y), q = Q(x, y)$ are compatible if $\frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y}$.
16. Singular integral of a differential equation is obtained by giving particular values to arbitrary constants in its general solution.
17. The equation of envelope to a surface is the singular integral of its differential equation.
18. Any relation which contains as many arbitrary constants as there are independent variables and is a solution of a partial differential equation of the first order is called a complete integral of that equation.

Answers

Multiple Choice Questions

- | | | |
|---------|---------|---------|
| 1. (a) | 2. (b) | 3. (a) |
| 4. (b) | 5. (b) | 6. (c) |
| 7. (b) | 8. (a) | 9. (b) |
| 10. (c) | 11. (c) | 12. (b) |

Fill in the Blank(s)

1. 3
2. first
3. $Pp + Qq = R$
4. $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$
5. $\frac{dx}{y^2 + z^2 - x^2} = \frac{dy}{-2xy} = \frac{dz}{-2zx}$
6. $\frac{dx}{zx} = \frac{dy}{-yz} = \frac{dz}{y^2 - x^2}$
7. $\frac{dx}{x} = \frac{dy}{y} = \frac{dt}{t} = \frac{dz}{az + (xy/t)}$
8. Charpit's auxiliary equations.
9. $\frac{dp}{-pq} = \frac{dq}{p^2} = \frac{dz}{-2p^2y - 2q^2y + qz} = \frac{dx}{-2py} = \frac{dy}{-2qy + z}$
10. $\frac{dp}{-p + p^2} = \frac{dq}{-q + q^2} = \frac{dz}{-2p^2x - 2q^2y} = \frac{dx}{-2px} = \frac{dy}{-2qy}$
11. $\frac{\partial P}{\partial x}$
12. compatible.

True or False

- | | | |
|---------|---------|---------|
| 1. T | 2. T | 3. F |
| 4. T | 5. F | 6. T |
| 7. T | 8. F | 9. T |
| 10. F | 11. T | 12. T |
| 13. F | 14. T | 15. F |
| 16. F | 17. T | 18. T |



Chapter

9



Linear Partial Differential Equations of Second and Higher Order with Constant Coefficients

9.1 The General Linear Partial Differential Equation of an Order Higher than the First

A partial differential equation in which the dependent variable and its derivatives appear only in the first degree and are not multiplied together, their coefficients all being constants or functions of x and y , is called a **linear partial differential equation**. The general form of such an equation is

$$\begin{aligned} \frac{\partial^n z}{\partial x^n} + A_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + \dots + A_n \frac{\partial^n z}{\partial y^n} + B_0 \frac{\partial^{n-1} z}{\partial x^{n-1}} + \dots \\ + M \frac{\partial z}{\partial x} + N \frac{\partial z}{\partial y} + P z = f(x, y), \quad \dots (1) \end{aligned}$$

where the coefficients $A_1, \dots, A_n, B_0, \dots, M, N, P$ are constants or functions of x and y . If the coefficients of various terms are constants then it is called a **linear partial differential equation with constant coefficients**.

9.2 The Homogeneous Linear Partial Differential Equation with Constant Coefficients

In this equation all the partial derivatives appearing in the equation are of the same order. A linear homogeneous partial differential equation of order n with constant coefficients is of the form

$$\frac{\partial^n z}{\partial x^n} + A_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + \dots + A_n \frac{\partial^n z}{\partial y^n} = f(x, y), \quad \dots(1)$$

where $A_1, \dots, A_n$ are constants.

Denoting the operators $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$ by D, D' respectively, the equation (1) can also be

written as $(D^n + A_1 D^{n-1} D' + \dots + A_n D'^n) z = f(x, y)$

$$\text{or} \quad F(D, D') z = f(x, y), \quad \dots(2)$$

where $F(D, D') \equiv D^n + A_1 D^{n-1} D' + \dots + A_n D'^n$.

Note that $F(D, D')$ is a homogeneous function in D, D' of degree n .

9.3 Solution of a Linear Homogeneous Partial Differential Equation with Constant Coefficients.

As in the case of ordinary linear differential equations the basic theorem is :

Theorem 1: If u is the complementary function and z_1 a particular integral of a linear partial differential equation $F(D, D') z = f(x, y)$ then $u + z_1$ is a general solution of the equation.

Proof: The complementary function of the equation

$$F(D, D') z = f(x, y) \quad \dots(1)$$

is the most general solution of the equation $F(D, D') z = 0$. \dots(2)

It must contain as many arbitrary constants as is the order of the differential equation (2).

Any solution of (1) is called a particular integral of (1). It does not contain any arbitrary constant.

Since the equations (1) and (2) are of the same order, the solution $u + z_1$ will contain as many arbitrary constants as the general solution of (1) requires. Also

$$F(D, D') u = 0,$$

$$F(D, D') z_1 = f(x, y)$$

so that $F(D, D') (u + z_1) = f(x, y)$.

This shows that, in fact, $u + z_1$ is a solution of (1). This completes the proof.

The next result is of extensive use in the solution of the partial differential equations.

Theorem 2: If $u_1, u_2, \dots, u_n$, are solutions of the homogeneous linear partial differential equation $F(D, D')z = 0$ then $\sum_{r=1}^n c_r u_r$ is also a solution, where c_r 's are arbitrary constants.

Proof: The proof of this theorem is obvious.

We have $F(D, D')(c_r u_r) = c_r F(D, D')u_r$.

Also $F(D, D') \sum_{r=1}^n v_r = \sum_{r=1}^n F(D, D')v_r$, for any set of functions v_r .

$$\begin{aligned} \text{Hence } F(D, D') \sum_{r=1}^n c_r u_r &= \sum_{r=1}^n F(D, D')(c_r u_r) \\ &= \sum_{r=1}^n c_r F(D, D')u_r = \sum_{r=1}^n c_r \cdot 0 = 0. \end{aligned}$$

9.4 Determination of the Complementary Function (C.F.) of the Linear Homogeneous Partial Differential Equation with Constant Coefficients

Consider a linear homogeneous n th order partial differential equation with constant coefficients of the form $F(D, D')z = f(x, y)$ (1)

The complementary function of (1) is the general solution of

$$\begin{aligned} &F(D, D')z = 0 \\ \text{i.e., } &(D^n + A_1 D^{n-1} D' + A_2 D^{n-2} D'^2 + \dots + A_n D'^n)z = 0 \end{aligned} \quad \dots (2)$$

This is equivalent to

$$[(D - m_1 D')(D - m_2 D') \dots (D - m_n D')]z = 0, \quad \dots (3)$$

where $m_1, m_2, \dots, m_n$ are some constants.

The solution of any one of the equations

$$(D - m_1 D')z = 0, (D - m_2 D')z = 0, \dots, (D - m_n D')z = 0 \quad \dots (4)$$

is also a solution of (3) and we know that the general solution of $(D - mD')z = 0$ is $z = \phi(y + mx)$, where ϕ is an arbitrary function. Hence we can assume that a solution of the equation (3) is of the form $z = \phi(y + mx)$.

Differentiation will give

$$Dz = m\phi'(y + mx), D^n z = m^n \phi^{(n)}(y + mx), D'^n z = \phi^{(n)}(y + mx)$$

and, in general,

$$D^r D'^s z = m^r \phi^{(r+s)}(y + mx).$$

Therefore, the substitution of $\phi(y + mx)$ for z in (2) gives

$$(m^n + A_1 m^{n-1} + \dots + A_n) \phi^{(n)}(y + mx) = 0.$$

This is true if m is a root of the equation

$$m^n + A_1 m^{n-1} + \dots + A_n = 0. \quad \dots(5)$$

The equation (5) is called the **auxiliary equation (A.E.)** and is obtained by putting $D = m, D' = 1$ in $F(D, D') = 0$.

It will give in general n roots, say, $m_1, m_2, \dots, m_n$. Each value of m will give a solution of (2). Hence if all the roots of the Auxiliary equation are distinct, the general solution of (2) i.e., the complementary function of (1) is

$$z = \phi_1(y + m_1 x) + \phi_2(y + m_2 x) + \dots + \phi_n(y + m_n x) \quad \dots(6)$$

where $\phi_1, \phi_2, \dots, \phi_n$ are arbitrary functions.

Solution when the auxiliary equation has equal roots

i.e., the roots of the A.E. are repeated.

The equation corresponding to two repeated roots each equal to m is

$$(D - mD')(D - mD')z = 0.$$

Putting $(D - mD')z = u$, this becomes $(D - mD')u = 0$, the solution of which is $u = \phi(y + mx)$.

Hence $(D - mD')z = \phi(y + mx)$

or $p - mq = \phi(y + mx)$.

Lagrange's auxiliary equations of this linear equation are

$$\frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{\phi(y + mx)}.$$

Taking the first two members, we get

$$dy + m dx = 0.$$

$\therefore y + mx = a.$

Taking the first and the third members, we get

$$dz = \phi(y + mx) dx = \phi(a) dx.$$

$\therefore z = \phi(a).x + b.$

Hence $z = x\phi(y + mx) + \psi(y + mx).$

Proceeding in the same way it can be shown that when a root m is repeated r times, the corresponding part of the complementary function is

$$\phi_1(y + mx) + x \phi_2(y + mx) + x^2 \phi_3(y + mx) + \dots + x^{r-1} \phi_r(y + mx).$$

Illustrative Examples

Example 1: Solve $2r + 5s + 2t = 0$.

$\dots(1)$

Solution: We know that

$$r = \frac{\partial^2 z}{\partial x^2} = D^2 z, s = \frac{\partial^2 z}{\partial x \partial y} = DD' z, t = \frac{\partial^2 z}{\partial y^2} = D'^2 z.$$

Hence the given equation can be written as

$$(2D^2 + 5DD' + 2D'^2)z = 0.$$

$$\text{A.E. is } 2m^2 + 5m + 2 = 0 \quad \text{or} \quad (2m + 1)(m + 2) = 0.$$

$$\therefore m = -\frac{1}{2}, -2.$$

Therefore the general solution of (1) is

$$z = f\left(y - \frac{1}{2}x\right) + \psi(y - 2x) \quad \text{or} \quad z = \phi(2y - x) + \psi(y - 2x).$$

Example 2: Solve $(D^3 - 6D^2D' + 11DD'^2 - 6D'^3)z = 0$(1)

Solution: The auxiliary equation is

$$m^3 - 6m^2 + 11m - 6 = 0 \quad \text{or} \quad (m - 1)(m - 2)(m - 3) = 0.$$

$$\therefore m = 1, 2, 3.$$

Therefore the general solution of (1) is

$$z = \phi_1(y + x) + \phi_2(y + 2x) + \phi_3(y + 3x).$$

Example 3: Solve $(D^3 - 3D^2D' + 2DD'^2)z = 0$. (Lucknow 2010)

Solution: The auxiliary equation is

$$m^3 - 3m^2 + 2m = 0 \quad \text{or} \quad m(m - 1)(m - 2) = 0.$$

$$\therefore m = 0, 1, 2.$$

Therefore the general solution of (1) is

$$z = \phi_1(y) + \phi_2(y + x) + \phi_3(y + 2x).$$

Example 4: Solve $\frac{\partial^4 z}{\partial x^4} - \frac{\partial^4 z}{\partial y^4} = 0$. (Avadh 2012) ... (1)

Solution: The equation (1) can be written as $(D^4 - D'^4)z = 0$.

$$\text{A.E. is } m^4 - 1 = 0 \quad \text{or} \quad (m - 1)(m + 1)(m^2 + 1) = 0.$$

$$\therefore m = 1, -1, \pm i.$$

Therefore the general solution of (1) is

$$z = \phi_1(y + x) + \phi_2(y - x) + \phi_3(y + ix) + \phi_4(y - ix).$$

Example 5: Solve $(D^4 - 2D^3D' + 2DD'^3 - D'^4)z = 0$. (Lucknow 2006, 09) ... (1)

Solution: The auxiliary equation is

$$m^4 - 2m^3 + 2m - 1 = 0 \quad \text{or} \quad (m + 1)(m - 1)^3 = 0.$$

$$\therefore m = -1, 1, 1, 1.$$

Therefore the general solution of (1) is

$$z = \phi_1(y - x) + \phi_2(y + x) + x\phi_3(y + x) + x^2\phi_4(y + x).$$

Example 6: Solve $25r - 40s + 16t = 0$(1)

Solution: The equation (1) can be written as

$$(25D^2 - 40DD' + 16D'^2)z = 0.$$

A.E. is $25m^2 - 40m + 16 = 0$ or $(5m - 4)^2 = 0.$

$\therefore m = 4/5, 4/5.$

Therefore the general solution of (1) is

$$z = \phi_1 \left(y + \frac{4}{5}x \right) + x \phi_2 \left(y + \frac{4}{5}x \right)$$

or $z = f_1(5y + 4x) + x f_2(5y + 4x).$

Comprehensive Exercise 1

- Solve $r = a^2t$. (Rohilkhand 2010, Avadh 10)
- Solve $(D^3 - 4D^2D' + 4DD'^2)z = 0.$
- Solve $\frac{\partial^3 z}{\partial x^3} - 7 \frac{\partial^3 z}{\partial x \partial y^2} + 6 \frac{\partial^3 z}{\partial y^3} = 0.$
- Solve $(D^2 - 3aDD' + 2a^2D'^2)z = 0.$
- Solve $2 \frac{\partial^2 z}{\partial x^2} - 3 \frac{\partial^2 z}{\partial x \partial y} - 2 \frac{\partial^2 z}{\partial y^2} = 0.$
- Solve $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = 0.$
- Solve $r + t + 2s = 0.$ (Kanpur 2009)
- Solve $(4D^2 + 12DD' + 9D'^2)z = 0.$
- Solve $(D^4 + D'^4 - 2D^2D'^2)z = 0.$
- Solve $(D^4 + D'^4)z = 0.$

Answers 1

- $z = \phi_1(y + ax) + \phi_2(y - ax)$
- $z = \phi_1(y) + \phi_2(y + 2x) + x \phi_3(y + 2x)$
- $z = \phi_1(y + x) + \phi_2(y + 2x) + \phi_3(y - 3x)$
- $z = \phi_1(y + ax) + \phi_2(y + 2ax)$
- $z = \phi_1(2y - x) + \phi_2(y + 2x)$
- $z = \phi_1(y + x) + \phi_2(y - x)$
- $z = \phi_1(y - x) + x \phi_2(y - x)$
- $z = \phi_1(2y - 3x) + x \phi_2(2y - 3x)$

9. $z = \phi_1 (y + x) + x \phi_2 (y + x) + \phi_3 (y - x) + x \phi_4 (y - x)$
10. $z = \phi_1 (y + \alpha x) + \phi_2 (y + \bar{\alpha} x) + \phi_3 (y + \beta x) + \phi_4 (y + \bar{\beta} x)$
- where $\alpha = \frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$ and $\beta = -\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}$.

9.5 Determination of the Particular Integral (P.I.)

The particular integral of the equation (1) of 9.4 will be denoted by $\frac{1}{F(D, D')} f(x, y)$.

$\frac{1}{F(D, D')} V$ is defined as the function which gives V when it is operated upon by $F(D, D')$.

The symbolic function $F(D, D')$ can be treated as an algebraic function of D and D' . It can be factorised, resolved into partial fractions or can be expanded in ascending powers of D or D' .

Note: $\frac{1}{D}$ means integration w.r.t. x , $\frac{1}{D'}$ means integration w.r.t. y , and so on and

particular integral would be different according as $F(D, D')$ is expanded in ascending powers of D or D' .

Illustrative Examples

Example 7: Solve $(D^2 + 3DD' + 2D'^2)z = x + y$. (Avadh 2012; Kanpur 14)

Solution: The auxiliary equation is

$$m^2 + 3m + 2 = 0 \quad \text{or} \quad (m + 2)(m + 1) = 0.$$

$$\therefore m = -1, -2.$$

$$\therefore \text{C.F.} = \phi_1 (y - x) + \phi_2 (y - 2x).$$

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{D^2 + 3DD' + 2D'^2} (x + y) = \frac{1}{D^2} \left(1 + \frac{3D'}{D} + \frac{2D'^2}{D^2} \right)^{-1} (x + y) \\ &= \frac{1}{D^2} \left(1 - \frac{3D'}{D} + \dots \right) (x + y) = \frac{1}{D^2} (x + y) - \frac{3}{D^3} D' (x + y) \\ &= \frac{x^3}{6} + y \cdot \frac{x^2}{2} - \frac{3}{D^3} 1 = \frac{x^3}{6} + \frac{1}{2} x^2 y - 3 \cdot \frac{x^3}{6} = -\frac{1}{3} x^3 + \frac{1}{2} x^2 y. \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1 (y - x) + \phi_2 (y - 2x) - \frac{1}{3} x^3 + \frac{1}{2} x^2 y.$$

Example 8: Solve $(D^2 - 6DD' + 9D'^2)z = 12x^2 + 36xy$.

Solution: A.E. is $m^2 - 6m + 9 = 0$ or $(m - 3)^2 = 0$.

$$\therefore m = 3, 3.$$

$$\therefore \text{C.F.} = \phi_1 (y + 3x) + x \phi_2 (y + 3x).$$

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{D^2 - 6DD' + 9D'^2} (12x^2 + 36xy) \\ &= \frac{1}{(D - 3D')^2} (12x^2 + 36xy) \\ &= \frac{1}{D^2} \left(1 - \frac{3D'}{D} \right)^{-2} (12x^2 + 36xy) \\ &= \frac{1}{D^2} \left\{ 1 + \frac{6D'}{D} + 27 \frac{D'^2}{D^2} + \dots \right\} (12x^2 + 36xy) \\ &= \frac{1}{D^2} (12x^2 + 36xy) + \frac{6}{D^3} D' (12x^2 + 36xy) \\ &= x^4 + 6x^3y + \frac{6}{D^3} (36x) \\ &= x^4 + 6x^3y + 9x^4 = 10x^4 + 6x^3y. \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1 (y + 3x) + x \phi_2 (y + 3x) + 10x^4 + 6x^3y.$$

Example 9: Solve $r + (a + b)s + abt = xy$.

(Purvanchal 2014)

Solution: The given equation can be written as

$$\{D^2 + (a + b)DD' + abD'^2\}z = xy.$$

$$\text{A.E. is } m^2 + (a + b)m + ab = 0 \quad \text{or} \quad (m + a)(m + b) = 0.$$

$$\therefore m = -a, -b.$$

$$\therefore \text{C.F.} = \phi_1 (y - ax) + \phi_2 (y - bx).$$

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{D^2 + (a + b)DD' + abD'^2} xy \\ &= \frac{1}{D^2} \left\{ 1 + (a + b) \frac{D'}{D} + ab \frac{D'^2}{D^2} \right\}^{-1} xy \\ &= \frac{1}{D^2} \left\{ 1 - (a + b) \frac{D'}{D} \dots \right\} (xy) = \frac{1}{D^2} (xy) - (a + b) \frac{1}{D^3} \{D' (xy)\} \\ &= \frac{x^3y}{6} - (a + b) \frac{x^4}{24}. \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1 (y - ax) + \phi_2 (y - bx) + \frac{x^3y}{6} - (a + b) \frac{x^4}{24}.$$

Example 10: Solve

$$(2D^2 - 5DD' + 2D'^2)z = 24(y - x).$$

Solution: A.E. is $2m^2 - 5m + 2 = 0$

$$\text{or} \quad (2m-1)(m-2)=0.$$

$$\therefore m = \frac{1}{2}, 2.$$

$$\therefore \text{C.F.} = \phi_1(2y+x) + \phi_2(y+2x).$$

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{2D^2 - 5DD' + 2D'^2} 24(y-x) \\ &= \frac{1}{2D^2} \left(1 - \frac{5D'}{2D} + \frac{D'^2}{D^2} \right)^{-1} 24(y-x) \\ &= \frac{1}{2D^2} \left(1 + \frac{5D'}{2D} \dots \right) 24(y-x) = \frac{1}{2D^2} 24(y-x) + \frac{5}{4D^3} 24 \\ &= 12 \left(y \frac{x^2}{2} - \frac{x^3}{6} \right) + \frac{5}{4} \cdot 24 \cdot \frac{x^3}{6} = 6x^2 y + 3x^3. \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1(2y+x) + \phi_2(y+2x) + 6x^2 y + 3x^3.$$

Example 11: Solve $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = x - y$.

(Lucknow 2008)

Solution: The given equation can be written as

$$(D^2 - D'^2)z = x - y.$$

A.E. is $m^2 - 1 = 0$.

$$\therefore m = 1, -1.$$

$$\therefore \text{C.F.} = \phi_1(y+x) + \phi_2(y-x).$$

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{D^2 - D'^2} (x-y) = \frac{1}{D^2} \left(1 - \frac{D'^2}{D^2} \right)^{-1} (x-y) \\ &= \frac{1}{D^2} \left(1 + \frac{D'^2}{D^2} + \dots \right) (x-y) \\ &= \frac{1}{D^2} (x-y) = \frac{1}{6} x^3 - y \cdot \frac{1}{2} x^2. \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1(y+x) + \phi_2(y-x) + \frac{1}{6} x^3 - \frac{1}{2} yx^2.$$

Example 12: Solve

$$\frac{\partial^3 u}{\partial x^3} + \frac{\partial^3 u}{\partial y^3} + \frac{\partial^3 u}{\partial z^3} - 3 \frac{\partial^3 u}{\partial x \partial y \partial z} = x^3 + y^3 + z^3 - 3xyz.$$

(Avadh 2012)

...(1)

Solution: The given equation can be written as

$$(D_1^3 + D_2^3 + D_3^3 - 3 D_1 D_2 D_3) u = x^3 + y^3 + z^3 - 3xyz$$

$$\text{or} \quad (D_1 + D_2 + D_3)(D_1^2 + D_2^2 + D_3^2 - D_1 D_2 - D_2 D_3 - D_3 D_1) u = x^3 + y^3 + z^3 - 3xyz$$

$$\text{or} \quad (D_1 + D_2 + D_3)(D_1 + \omega D_2 + \omega^2 D_3)(D_1 + \omega^2 D_2 + \omega D_3) u = x^3 + y^3 + z^3 - 3xyz$$

where ω is an imaginary cube root of unity.

$$\text{Now consider } (D_1 + \omega D_2 + \omega^2 D_3) u = 0. \quad \dots(2)$$

Auxiliary equations are

$$\frac{dx}{1} = \frac{dy}{\omega} = \frac{dz}{\omega^2}.$$

These give $y - \omega x = a$, $z - \omega^2 x = b$.

Hence solution of (2) is $\phi_1(y - \omega x, z - \omega^2 x) = 0$.

Similarly $(D_1 + D_2 + D_3) u = 0$ and $(D_1 + \omega^2 D_2 + \omega D_3) u = 0$ give

$$\phi_2(y - x, z - x) = 0 \text{ and } \phi_3(y - \omega^2 x, z - \omega x) = 0 \text{ respectively.}$$

$$\therefore \text{ C.F. of (1)} = \phi_1(y - \omega x, z - \omega^2 x) + \phi_2(y - x, z - x) + \phi_3(y - \omega^2 x, z - \omega x).$$

Now P.I. corresponding to x^3 is

$$\begin{aligned} &= \frac{1}{D_1^3 + D_2^3 + D_3^3 - 3D_1 D_2 D_3} x^3 \\ &= \frac{1}{D_1^3} \left(1 + \frac{D_2^3}{D_1^3} + \dots \right)^{-1} x^3 = \frac{1}{D_1^3} x^3 = \frac{x^6}{4 \cdot 5 \cdot 6} = \frac{x^6}{120}. \end{aligned}$$

Similarly particular integrals corresponding to y^3 and z^3 are $\frac{y^6}{120}$ and $\frac{z^6}{120}$ respectively

and P.I. corresponding to $(-3xyz)$

$$\begin{aligned} &= \frac{1}{D_1^3 + D_2^3 + D_3^3 - 3D_1 D_2 D_3} (-3xyz) \\ &= \frac{1}{-3D_1 D_2 D_3} \left(1 - \frac{D_1^2}{D_2 D_3} \dots \right)^{-1} (-3xyz) \\ &= -\frac{1}{3D_1 D_2 D_3} (-3xyz) = \frac{x^2 y^2 z^2}{8}. \end{aligned}$$

Hence the general solution of the equation (1) is

$$z = \text{C.F.} + \text{P.I.}$$

Comprehensive Exercise 2

1. Solve $(D^2 - 2DD' + D'^2)z = 12xy$. (Lucknow 2008; Avadh 10, 11; Kanpur 12)

2. Solve $\frac{\partial^3 z}{\partial x^3} - \frac{\partial^3 z}{\partial y^3} = x^3 y^3$.

3. Solve $(D^2 - a^2 D'^2)z = x^2$.
4. Solve $\frac{\partial^2 z}{\partial x^2} + 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = 2x + 3y$.
5. Solve $\frac{\partial^2 z}{\partial x^2} + 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = 6(x + y)$.
6. Solve $(D^2 - DD' - 6D'^2)z = xy$.

Answers 2

1. $z = \phi_1(y + x) + x \phi_2(y + x) + 2x^3 y + x^4$
2. $z = \phi_1(y + x) + \phi_2(y + \omega x) + \phi_3(y + \omega^2 x) + \frac{1}{120} x^6 y^3 + \frac{x^9}{10080}$
3. $z = \phi_1(y + ax) + \phi_2(y - ax) + \frac{1}{12} x^4$
4. $z = \phi_1(y - x) + \phi_2(y - 2x) - \frac{7}{6} x^3 + \frac{3}{2} x^2 y$
5. $z = \phi_1(y - x) + \phi_2(y - 2x) + 3x^2 y - 2x^3$
6. $z = \phi_1(y - 2x) + \phi_2(y + 3x) + \frac{1}{6} x^3 y + \frac{1}{24} x^4$

9.6 Short Method for Finding the Particular Integral

When $f(x, y)$ is a function of $ax + by$, we have a shorter method for determining the particular integral.

Let $f(x, y) = \phi(ax + by)$.

Then $D^r \phi(ax + by) = a^r \phi^{(r)}(ax + by)$, $D'^r \phi(ax + by) = b^r \phi^{(r)}(ax + by)$,

and $D^r D'^s \phi(ax + by) = a^r b^s \phi^{(r+s)}(ax + by)$,

where $\phi^{(r)}$ is the r th derivative of ϕ w.r.t. $ax + by$ as a whole.

Since $F(D, D')$ is homogeneous in D and D' of degree n ,

hence $F(D, D') \phi(ax + by) = F(a, b) \phi^{(n)}(ax + by)$

or $\frac{1}{F(D, D')} \phi^{(n)}(ax + by) = \frac{1}{F(a, b)} \phi^{(n)}(ax + by)$,

provided that $F(a, b) \neq 0$.

Putting $ax + by = t$, we get $\frac{1}{F(D, D')} \phi^{(n)}(t) = \frac{1}{F(a, b)} \phi^{(n)}(t)$.

Now integrating both the sides n times w.r.t. ' t ', we get

$$\frac{1}{F(D, D')} \phi(t) = \frac{1}{F(a, b)} \int \int \dots \int \phi(t) dt \dots dt, \text{ where } t = ax + by.$$

Working Rule: To find the particular integral of an equation $F(D, D')z = \phi(ax + by)$, where $F(D, D')$ is a homogeneous function of D, D' of degree n , proceed as follows :

- (i) Put $ax + by = t$ and integrate $\phi(t)$, n times with respect to t .
- (ii) Find $F(a, b)$, replacing D, D' by a, b respectively in $F(D, D')$.
- (iii) Now P.I. = $\frac{1}{F(a, b)} \times n$ th integral of $\phi(t)$ w.r.t. ' t ', where $t = ax + by$.

Exceptional case when $F(a, b) = 0$.

If $F(a, b) = 0$ then the above method fails.

Now $F(a, b) = 0$ if and only if $(bD - aD')$ is a factor of $F(D, D')$.

Let us consider the equation $(bD - aD')z = x^r \phi(ax + by)$... (1)

or $bp - aq = x^r \phi(ax + by)$.

Lagrange's auxiliary equations are

$$\frac{dx}{b} = \frac{dy}{-a} = \frac{dz}{x^r \phi(ax + by)}.$$

Taking the first two members, we get

$$a dx + b dy = 0.$$

$\therefore ax + by = c$ (constant).

Again taking the first and the last members, we get

$$dz = \frac{x^r}{b} \phi(ax + by) dx = \frac{x^r}{b} \phi(c) dx.$$

$\therefore z = \frac{x^{r+1}}{b(r+1)} \phi(c) = \frac{x^{r+1}}{b(r+1)} \phi(ax + by).$... (2)

It gives the solution of (1).

From (1), $z = \frac{1}{bD - aD'} x^r \phi(ax + by).$... (3)

Thus $\frac{1}{(bD - aD')} x^r \phi(ax + by) = \frac{x^{r+1}}{b(r+1)} \phi(ax + by).$... (4)

Hence if $z = \frac{1}{(bD - aD')^n} \phi(ax + by),$

then
$$z = \frac{1}{(bD - aD')^{n-1}} \left[\frac{1}{(bD - aD')} \phi(ax + by) \right]$$

$$= \frac{1}{(bD - aD')^{n-1}} \frac{x}{b} \phi(ax + by), \text{ using (4)}$$

$$= \frac{x^n}{h^n n!} \phi(ax + by), \text{ by repeated application of (4).}$$

$$\text{Thus } \frac{1}{(bD - aD')^n} \phi(ax + by) = \frac{x^n}{b^n n!} \phi(ax + by).$$

Illustrative Examples

Example 13: Solve $4r - 4s + t = 16 \log(x + 2y)$. (Lucknow 2007; Purvanchal 09; Rohilkhand 14)

Solution: The given equation can be written as

$$(4D^2 - 4DD' + D'^2)z = 16 \log(x + 2y).$$

A.E. is $4m^2 - 4m + 1 = 0$ or $(2m - 1)^2 = 0$.

$$\therefore \text{C.F.} = \phi_1 (2y + x) + x \phi_2 (2y + x).$$

Now P.I. = $\frac{1}{(2D - D')^2} 16 \log(x + 2y)$

$$= \frac{x^2}{2^2 \cdot 2!} 16 \log(x+2y) = 2x^2 \log(x+2y). \quad [\because F(a,b)=0]$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1 (2y + x) + x \phi_2 (2y + x) + 2x^2 \log (x + 2y).$$

Example 14: Solve $(D^3 - 4D^2D' + 4DD'^2)z = 4 \sin(2x + y)$. (Rohilkhand 2011)

Solution: A.E. is $m^3 - 4m^2 + 4m = 0$ or $m(m-2)^2 = 0$.

$$\therefore m = 0, 2, 2.$$

$$\therefore \text{C.F.} = \phi_1(y) + \phi_2(y + 2x) + x\phi_3(y + 2x).$$

Now P.I. = $\frac{1}{D^3 - 4D^2D' + 4DD'^2} 4 \sin(2x + y)$

$$= \frac{1}{(D - 2D')^2} \cdot \frac{1}{D} 4 \sin(2x + y)$$

$$= \frac{1}{(D - 2D')^2} \{-2 \cos(2x + y)\}$$

$$= -2 \frac{1}{(D - 2D')^2} \cos(2x + y)$$

$$= -2 \cdot \frac{x^2}{2!} \cos(2x + y) = -x^2 \cos(2x + y). \quad [\because F(a, b) = 0]$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.}$$

$$= \phi_1(y) + \phi_2(y + 2x) + x \phi_3(y + 2x) - x^2 \cos(2x + y).$$

Example 15: Solve $(D^2 - 2DD' + D'^2)z = e^{x+2y} + x^3$.

(Lucknow 2010; Rohilkhand 14)

Solution: A.E. is $m^2 - 2m + 1 = 0$ or $(m - 1)^2 = 0$.

$$\therefore m = 1, 1.$$

$$\therefore \text{C.F.} = \phi_1(y + x) + x \phi_2(y + x).$$

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{(D - D')^2} e^{x+2y} + \frac{1}{(D - D')^2} x^3 \\ &= \frac{1}{(1 - 2)^2} \cdot e^{x+2y} + \frac{1}{D^2} \left(1 - \frac{D'}{D}\right)^{-2} x^3 \\ &= e^{x+2y} + \frac{1}{D^2} \left(1 + \frac{2D'}{D} + \dots\right) x^3 = e^{x+2y} + \frac{1}{D^2} x^3 = e^{x+2y} + \frac{1}{20} x^5. \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1(y + x) + x \phi_2(y + x) + e^{x+2y} + \frac{1}{20} x^5.$$

Example 16: Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \cos mx \cos ny$. (Lucknow 2006; Kanpur 07; Avadh 14)

Solution: The given equation can be written as

$$(D^2 + D'^2)z = \frac{1}{2} [\cos(mx + ny) + \cos(mx - ny)].$$

$$\text{A.E. is } m^2 + 1 = 0.$$

$$\therefore m = \pm i.$$

$$\therefore \text{C.F.} = \phi_1(y + ix) + \phi_2(y - ix).$$

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{2} \cdot \frac{1}{D^2 + D'^2} \cos(mx + ny) + \frac{1}{2} \cdot \frac{1}{D^2 + D'^2} \cos(mx - ny) \\ &= \frac{1}{2} \cdot \frac{1}{m^2 + n^2} \{-\cos(mx + ny)\} + \frac{1}{2} \cdot \frac{1}{m^2 + n^2} \{-\cos(mx - ny)\} \\ &= -\frac{1}{2(m^2 + n^2)} \{\cos(mx + ny) + \cos(mx - ny)\}. \end{aligned}$$

Hence the general solution of the given equation is

$$\begin{aligned} z = \text{C.F.} + \text{P.I.} &= \phi_1(y + ix) + \phi_2(y - ix) \\ &\quad - \frac{1}{2(m^2 + n^2)} [\cos(mx + ny) + \cos(mx - ny)]. \end{aligned}$$

Example 17: Solve $2r - s - 3t = 5e^x/e^y$.

(Rohilkhand 2009)

Solution: The given equation can be written as

$$(2D^2 - DD' - 3D'^2)z = 5e^{x-y}.$$

A.E. is $2m^2 - m - 3 = 0$ or $(2m - 3)(m + 1) = 0$.

$\therefore$ C.F. = $\phi_1(2y + 3x) + \phi_2(y - x)$.

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{2D^2 - DD' - 3D'^2} 5e^{x-y} = \frac{1}{(D + D')(2D - 3D')} 5e^{x-y} \\ &= \frac{1}{D + D'} \cdot \frac{1}{2 + 3} \cdot 5e^{x-y} = \frac{1}{D + D'} e^{x-y} \\ &= \frac{x}{1!} e^{x-y} = x e^{x-y}. \quad [\because F(a, b) = 0] \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1(2y + 3x) + \phi_2(y - x) + x e^{x-y}.$$

Example 18: Solve $(D^2 + 3DD' + 2D'^2)z = x + y$.

(Rohilkhand 2010; Agra 02; Avadh 14; Purvanchal 14)

Solution: A.E. is $m^2 + 3m + 2 = 0$ or $(m + 2)(m + 1) = 0$.

$\therefore m = -1, -2$.

$\therefore$ C.F. = $\phi_1(y - x) + \phi_2(y - 2x)$.

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{D^2 + 3DD' + 2D'^2} (x + y) \\ &= \frac{1}{(1 + 3 \cdot 1 \cdot 1 + 2 \cdot 1)} \cdot \frac{(x + y)^3}{6} = \frac{(x + y)^3}{36}. \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1(y - x) + \phi_2(y - 2x) + \frac{(x + y)^3}{36}.$$

Example 19: Solve $(D^2 - 5DD' + 4D'^2)z = \sin(4x + y)$.

(Lucknow 2006)

Solution: A.E. is $m^2 - 5m + 4 = 0$

or $(m - 1)(m - 4) = 0 \therefore m = 1, 4$.

$\therefore$ C.F. = $\phi_1(y + x) + \phi_2(y + 4x)$.

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{D^2 - 5DD' + 4D'^2} \sin(4x + y) \\ &= \frac{1}{(D - 4D')} \cdot \frac{1}{(D - D')} \sin(4x + y) \\ &= \frac{1}{(D - 4D')} \cdot \frac{1}{(4 - 1)} \{-\cos(4x + y)\} \\ &= \frac{1}{D - 4D'} \left\{-\frac{1}{3} \cos(4x + y)\right\} \end{aligned}$$

$$= -\frac{1}{3} \cdot x \cos (4x + y). \quad [\because F(a, b) = 0]$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1(y + x) + \phi_2(y + 4x) - \frac{1}{3} x \cos (4x + y).$$

Comprehensive Exercise 3

1. Solve $(D^2 - 6DD' + 9D'^2)z = 6x + 2y$.
2. Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 12(x + y)$.
3. Solve $(D^3 - 4D^2D' + 4DD'^2)z = \cos(y + 2x)$. (Purvanchal 2011)
4. Solve $r - 2s + t = \sin(2x + 3y)$. (Kanpur 2010; Meerut 13)
5. Solve $(2D^2 - 5DD' + 2D'^2)z = 5 \sin(2x + y)$.
6. Solve $(D^2 + 2DD' + D'^2)z = e^{2x+3y}$. (Purvanchal 2007)
7. Solve $(D^2 + D'^2)z = 30(2x + y)$.
8. Solve $(D^3 - 2D^2D' - DD'^2 + 2D'^3)z = e^{x+y}$.
9. Solve $(D^3 - 7DD'^2 - 6D'^3)z = x^2 + xy^2 + y^3 + \cos(x - y)$.
10. Solve $r + s - 2t = \sqrt{2x + y}$. (Lucknow 2010)
11. Solve $(D^3 - 4D^2D' + 5DD'^2 - 2D'^3)z = e^{y+2x} + (y + x)^{1/2}$.
12. Solve $(D^2 - 3DD' + 2D'^2)z = e^{2x-y} + e^{x+y} + \cos(x + 2y)$.
13. Solve $\log s = x + y$.
14. Solve $(D_x^3 - 7D_x D_y^2 - 6D_y^3)z = \sin(x + 2y) + e^{3x+y}$. (Meerut 2013)

Answers 3

1. $z = \phi_1(y + 3x) + x\phi_2(y + 3x) + \frac{1}{4}x^2(3x + y)$
2. $z = \phi_1(y + ix) + \phi_2(y - ix) + (x + y)^3$
3. $z = \phi_1(y) + \phi_2(y + 2x) + x\phi_3(y + 2x) + \frac{1}{4}x^2 \sin(y + 2x)$
4. $z = \phi_1(y + x) + x\phi_2(y + x) - \sin(2x + 3y)$
5. $z = \phi_1(2y + x) + \phi_2(y + 2x) - \frac{1}{3} \cdot 5x \cos(2x + y)$
6. $z = \phi_1(y - x) + x\phi_2(y - x) + \frac{1}{25}e^{2x+3y}$

7. $z = \phi_1 (y + ix) + \phi_2 (y - ix) + (2x + y)^3$
8. $z = \phi_1 (y - x) + \phi_2 (y + x) + \phi_3 (y + 2x) - \frac{1}{2} x e^{x+y}$
9. $z = \phi_1 (y - x) + \phi_2 (y - 2x) + \phi_3 (y + 3x) + \frac{5}{72} x^6 + \frac{1}{60} x^5 + \frac{7}{20} x^5 y$
 $+ \frac{1}{24} x^4 y^2 + \frac{1}{6} x^3 y^3 - \frac{1}{4} x \cos (x - y).$
10. $z = \phi_1 (y + x) + \phi_2 (y - 2x) + \frac{1}{15} (2x + y)^{5/2}$
11. $z = \phi_1 (y + x) + x \phi_2 (y + x) + \phi_3 (y + 2x) + x e^{y+2x} - \frac{1}{3} x^2 (y + x)^{3/2}$
12. $z = \phi_1 (y + x) + \phi_2 (y + 2x) + \frac{1}{12} e^{2x-y} - x e^{x+y} - \frac{1}{3} \cos (x + 2y)$
13. $z = \phi_1 (x) + \phi_2 (y) + e^{x+y}$
14. $z = \phi_1 (y - x) + \phi_2 (y - 2x) + \phi_3 (y + 3x) + \frac{1}{75} \cos (x + 2y) + \frac{x}{20} e^{3x+y}$

9.7 A General Method of Finding the Particular Integral

Consider the equation $(D - mD')z = \phi(x, y)$ or $p - mq = \phi(x, y)$.

Lagrange's auxiliary equations are

$$\frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{\phi(x, y)}.$$

Taking the first two members, we get $dy + m dx = 0$.

$\therefore y + mx = a$ (a constant).

Again taking the first and the last members, we get

$$dz = \phi(x, y) dx = \phi(x, a - mx) dx.$$

$$z = \int \phi(x, a - mx) dx.$$

$$\text{Thus } z = \frac{1}{D - mD'} \phi(x, y) = \int \phi(x, a - mx) dx,$$

where after integration the constant a is to be replaced by $y + mx$, as the particular integral does not contain any arbitrary constant.

Now if the equation is $F(D, D')z = \phi(x, y)$

where $F(D, D') = (D - m_1 D')(D - m_2 D') \dots (D - m_n D')$,

then $\text{P.I.} = \frac{1}{F(D, D')} \phi(x, y)$

$$= \frac{1}{D - m_1 D'} \cdot \frac{1}{D - m_2 D'} \dots \frac{1}{D - m_n D'} \phi(x, y),$$

which can be evaluated by the repeated application of the above method.

Illustrative Examples

Example 20: Solve $r + s - 6t = y \cos x$. (Avadh 2009, 10, 11; Purvanchal 07)

Solution: The given equation can be written as

$$(D^2 + DD' - 6D'^2)z = y \cos x.$$

A.E. is $m^2 + m - 6 = 0$ or $(m + 3)(m - 2) = 0$.

$$\therefore m = 2, -3.$$

$$\therefore \text{C.F.} = \phi_1(y + 2x) + \phi_2(y - 3x).$$

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{D^2 + DD' - 6D'^2} y \cos x = \frac{1}{(D - 2D')(D + 3D')} y \cos x \\ &= \frac{1}{D - 2D'} \int (a + 3x) \cos x \, dx, \text{ where } y - 3x = a \\ &= \frac{1}{D - 2D'} [a \sin x + 3x \sin x + 3 \cos x] \\ &= \frac{1}{D - 2D'} [(y - 3x) \sin x + 3x \sin x + 3 \cos x] \\ &= \frac{1}{(D - 2D')} [y \sin x + 3 \cos x] \\ &= \int [(b - 2x) \sin x + 3 \cos x] \, dx, \text{ where } y + 2x = b \\ &= -b \cos x - 2(-x \cos x + \sin x) + 3 \sin x \\ &= -(y + 2x) \cos x + 2x \cos x + \sin x = -y \cos x + \sin x. \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1(y + 2x) + \phi_2(y - 3x) - y \cos x + \sin x.$$

Example 21: Solve $(D^2 + 2DD' + D'^2)z = 2 \cos y - x \sin y$. (Lucknow 2011)

Solution: A.E. is $m^2 + 2m + 1 = 0$ or $(m + 1)^2 = 0$.

$$\therefore m = -1, -1.$$

$$\therefore \text{C.F.} = \phi_1(y - x) + x \phi_2(y - x).$$

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{D^2 + 2DD' + D'^2} (2 \cos y - x \sin y) \\ &= \frac{1}{(D + D')(D + D')} (2 \cos y - x \sin y) \\ &= \frac{1}{D + D'} \int [2 \cos(x + a) - x \sin(x + a)] \, dx, \text{ where } y - x = a \\ &= \frac{1}{D + D'} [2 \sin(x + a) - \{-x \cos(x + a) + \sin(x + a)\}] \\ &= \frac{1}{D + D'} [\sin(x + a) + x \cos(x + a)] \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{D+D'} [\sin y + x \cos y] \\
 &= \int [\sin (x+b) + x \cos (x+b)] dx, \text{ where } y-x=b \\
 &= -\cos (x+b) + x \sin (x+b) + \cos (x+b) = x \sin (x+b) = x \sin y.
 \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1 (y-x) + x \phi_2 (y-x) + x \sin y.$$

Example 22: Solve $(D^2 - 2DD' - 15D'^2)z = 12xy$. (Purvanchal 2010)

Solution: A.E. is $m^2 - 2m - 15 = 0$ or $(m-5)(m+3) = 0$.

$$\therefore m = 5, -3.$$

$$\therefore \text{C.F.} = \phi_1 (y+5x) + \phi_2 (y-3x).$$

$$\begin{aligned}
 \text{Now P.I.} &= \frac{1}{D^2 - 2DD' - 15D'^2} 12xy = \frac{1}{(D+3D')(D-5D')} 12xy \\
 &= \frac{12}{D+3D'} \int x(a-5x) dx, \text{ where } y+5x=a \\
 &= \frac{12}{D+3D'} \left\{ \frac{1}{2} ax^2 - \frac{5}{3} x^3 \right\} \\
 &= \frac{12}{D+3D'} \left\{ \frac{1}{2} (y+5x)x^2 - \frac{5}{3} x^3 \right\} \\
 &= \frac{2}{D+3D'} \{3yx^2 + 5x^3\} \\
 &= 2 \int [3x^2 (3x+b) + 5x^3] dx, \text{ where } y-3x=b \\
 &= 2 \int (14x^3 + 3x^2b) dx \\
 &= 2 \left(14 \cdot \frac{x^4}{4} + 3 \cdot \frac{x^3}{3} b \right) = 7x^4 + 2x^3b \\
 &= 7x^4 + 2x^3 (y-3x) = x^4 + 2x^3y.
 \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1 (y+5x) + \phi_2 (y-3x) + x^4 + 2x^3y.$$

Example 23: Solve $r - t = \tan^3 x \tan y - \tan x \tan^3 y$. (Lucknow 2006; Avadh 10)

Solution: The given equation can be written as

$$(D^2 - D'^2)z = \tan^3 x \tan y - \tan x \tan^3 y.$$

$$\text{A.E. is } m^2 - 1 = 0.$$

$$\therefore m = 1, -1.$$

$$\therefore \text{C.F.} = \phi_1 (y+x) + \phi_2 (y-x).$$

$$\begin{aligned}
 \text{Now P.I.} &= \frac{1}{D^2 - D'^2} (\tan^3 x \tan y - \tan x \tan^3 y) \\
 &= \frac{1}{(D+D')(D-D')} (\tan^3 x \tan y - \tan x \tan^3 y)
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{D+D'} \int [\tan^3 x \tan(a-x) - \tan x \tan^3(a-x)] dx, \\
&\quad \text{where } y+x=a \\
&= \frac{1}{D+D'} \int [\{-1+\sec^2 x\} \tan x \tan(a-x) \\
&\quad - \tan x \tan(a-x) \{-1+\sec^2(a-x)\}] dx \\
&= \frac{1}{D+D'} \left[\int \tan(a-x) \tan x \sec^2 x dx \right. \\
&\quad \left. - \int \tan x \tan(a-x) \sec^2(a-x) dx \right] \\
&= \frac{1}{D+D'} \left[\tan(a-x) \cdot \frac{\tan^2 x}{2} + \frac{1}{2} \int \sec^2(a-x) \cdot \tan^2 x dx \right. \\
&\quad \left. + \tan x \cdot \frac{\tan^2(a-x)}{2} - \frac{1}{2} \int \sec^2 x \tan^2(a-x) dx \right] \\
&= \frac{1}{2(D+D')} [\tan^2 x \tan(a-x) + \tan x \tan^2(a-x) \\
&\quad - \int \sec^2 x \{\sec^2(a-x) - 1 - \sec^2(a-x) \cdot (\sec^2 x - 1)\} dx] \\
&= \frac{1}{2(D+D')} [\tan^2 x \tan(a-x) + \tan x \tan^2(a-x) \\
&\quad + \int \{\sec^2 x - \sec^2(a-x)\} dx] \\
&= \frac{1}{2(D+D')} [\tan^2 x \tan y + \tan x \tan^2 y + (\tan x + \tan y)] \\
&= \frac{1}{2(D+D')} [\tan x \cdot \sec^2 y + \tan y \cdot \sec^2 x] \\
&= \frac{1}{2} \int [\tan x \sec^2(b+x) + \tan(b+x) \sec^2 x] dx, \text{ where } y-x=b \\
&= \frac{1}{2} \tan x \tan(b+x) - \frac{1}{2} \int \sec^2 x \tan(b+x) dx \\
&\quad + \frac{1}{2} \int \tan(b+x) \sec^2 x dx \\
&= \frac{1}{2} \tan x \tan(b+x) = \frac{1}{2} \tan x \tan y.
\end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \phi_1(y+x) + \phi_2(y-x) + \frac{1}{2} \tan x \tan y.$$

Comprehensive Exercise 4

1. Solve $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = \sin x$.

2. Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6 \frac{\partial^2 z}{\partial y^2} = y \sin x$.

(Kanpur 2011)

$$3. \quad \text{Solve } \frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial y^2} = \frac{4x}{y^2} - \frac{y}{x^2}.$$

(Rohilkhand 2000; Agra 03)

$$4. \quad \text{Solve } r - s - 2t = (2x^2 + xy - y^2) \sin xy - \cos xy.$$

$$5. \quad \text{Solve } (D^2 - DD' - 2D'^2)z = (y - 1)e^x.$$

Answers 4

$$1. \quad z = \phi_1(y - x) - \cos x$$

$$2. \quad z = \phi_1(y + 2x) + \phi_2(y - 3x) - (y \sin x + \cos x)$$

$$3. \quad z = \phi_1(y + 2x) + \phi_2(y - 2x) + x \log y + y \log x + 3x.$$

$$4. \quad z = \phi_1(y + 2x) + \phi_2(y - x) + \sin xy.$$

$$5. \quad z = \phi_1(y + 2x) + \phi_2(y - x) + ye^x.$$

9.8 Non-Homogeneous Linear Equations with Constant Coefficients

A linear partial differential equation which is not homogeneous is called a non-homogeneous linear equation.

Consider the differential equation $F(D, D')z = f(x, y)$ (1)

When $F(D, D')$ is a homogeneous function in D, D' it can always be resolved into linear factors. But the result is not always true when $F(D, D')$ is non-homogeneous. We classify linear differential operators $F(D, D')$ into two main types. These are :

(i) $F(D, D')$ is **reducible** if it can be written as the product of linear factors of the form $D + aD' + b$, with a, b constants.

(ii) $F(D, D')$ is **irreducible** if it cannot be so written.

Complementary function of non-homogeneous linear equation:

When $F(D, D')$ can be resolved into linear factors.

The complementary function of non-homogeneous linear equation (1) is the general solution of the equation

$$F(D, D')z = 0. \quad \dots (2)$$

Let us consider a simple non-homogeneous equation

$$(D - mD' - k)z = 0 \quad \dots (3)$$

or $p - mq = kz.$

Lagrange's auxiliary equations for it are

$$\frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{kz}.$$

The first two members give

$$dy + m dx = 0.$$

$$\therefore y + mx = a \text{ (a constant).}$$

Again taking the first and the third members, we get

$$\frac{dz}{z} = k dx.$$

$$\therefore \log z = k x + \log b$$

$$\text{or } z = be^{kx}.$$

$$\therefore z = e^{kx} \phi(y + mx), \text{ which is the solution of (3).}$$

If $F(D, D')$ can be factorised into **non-repeated** linear factors

$$(D - m_1 D' - k_1), (D - m_2 D' - k_2), \dots, (D - m_n D' - k_n),$$

then the equation (2) is equivalent to

$$[(D - m_1 D' - k_1)(D - m_2 D' - k_2) \dots (D - m_n D' - k_n)] z = 0 \quad \dots(4)$$

The complete solution of (2) or (4) is made up of the solutions of

$$(D - m_1 D' - k_1) z = 0, (D - m_2 D' - k_2) z = 0, \dots, (D - m_n D' - k_n) z = 0.$$

Hence the general solution (complete solution) of (2) is

$$z = e^{k_1 x} \phi_1(y + m_1 x) + e^{k_2 x} \phi_2(y + m_2 x) + \dots + e^{k_n x} \phi_n(y + m_n x). \quad \dots(5)$$

Note 1: If the operator $F(D, D')$ is reducible, the order in which the linear factors occur is immaterial.

Note 2: It can be shown that if the equation is

$$(\alpha D + \beta D' + \gamma) z = 0, \text{ then } z = e^{(-\gamma x/\alpha)} \phi(\beta x - \alpha y).$$

$F(D, D')$ has repeated factors.

Let a factor $(D - mD' - k)$ occur twice in $F(D, D')$.

$$\text{Consider the equation } (D - mD' - k)^2 z = 0. \quad \dots(6)$$

Let $(D - mD' - k) z = u$. Then (6) reduces to

$$(D - mD' - k) u = 0.$$

$$\text{This gives } u = e^{kx} \phi_1(y + mx).$$

$$\text{Hence } (D - mD' - k) z = e^{kx} \phi_1(y + mx)$$

$$\text{or } p - mq = kz + e^{kx} \phi_1(y + mx).$$

Lagrange's auxiliary equations for this are

$$\frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{kz + e^{kx} \phi_1(y + mx)}.$$

The first two members give, $y + mx = a$ (a constant).

From the first and the last members, we get

$$\frac{dz}{dx} - kz = e^{kx} \phi_1(y + mx) = e^{kx} \phi_1(a),$$

which is an ordinary linear equation.

$$\text{I.F.} = e^{\int -k dx} = e^{-kx}.$$

$$\therefore ze^{-kx} = \int e^{-kx} e^{kx} \phi_1(a) dx + b = x \phi_1(a) + b.$$

$$\therefore z = e^{kx} [x \phi_1(y + mx) + \phi_2(y + mx)], \quad \dots(7)$$

which is the general solution of (6).

Hence if $(D - mD' - k)$ occurs twice in $F(D, D')$ then the C.F. corresponding to this factor is

$$e^{kx} [x \phi_1(y + mx) + \phi_2(y + mx)] \quad \dots(8)$$

It can be shown that the general solution of

$$(D - mD' - k)^r z = 0$$

$$\text{is } z = e^{kx} [\phi_1(y + mx) + x \phi_2(y + mx) + \dots + x^{r-1} \phi_r(y + mx)].$$

Hence if $(D - mD' - k)$ occurs r times in $F(D, D')$ then the C.F. corresponding to this factor is

$$e^{kx} [\phi_1(y + mx) + x \phi_2(y + mx) + \dots + x^{r-1} \phi_r(y + mx)]. \quad \dots(9)$$

Case when linear factors of $F(D, D')$ are not possible:

In case $F(D, D')$ is irreducible, i.e., it cannot be resolved into linear factors in D and D' , the above methods of finding the complementary function fail. In such cases a trial method is used to find solutions.

9.9 Particular Integral

The complete solution of $F(D, D')z = f(x, y)$

$$\text{is } z = \text{C.F.} + \text{P.I.}$$

$$\text{where } \text{P.I.} = \frac{1}{F(D, D')} f(x, y).$$

The methods of obtaining particular integrals of non-homogeneous partial differential equations are very similar to those of ordinary linear equations with constant coefficients. We give some cases of finding the particular integrals.

$$\text{Case I: } \frac{1}{F(D, D')} e^{ax+by} = \frac{1}{F(a, b)} e^{ax+by}, \text{ if } F(a, b) \neq 0.$$

$$\text{We have } D^r (e^{ax+by}) = a^r e^{ax+by},$$

$$D'^s (e^{ax+by}) = b^s e^{ax+by}$$

$$\text{and } (D^r D'^s) (e^{ax+by}) = a^r b^s e^{ax+by}.$$

$$\therefore F(D, D') (e^{ax+by}) = F(a, b) e^{ax+by}.$$

Operating both the sides by $\frac{1}{F(D, D')}$, we get

$$e^{ax+by} = F(a, b) \frac{1}{F(D, D')} e^{ax+by}$$

$$\Rightarrow \frac{1}{F(D, D')} e^{ax+by} = \frac{1}{F(a, b)} e^{ax+by}, \text{ provided } F(a, b) \neq 0.$$

Case II: The value of $\frac{1}{F(D, D')} \sin(ax + by)$ is obtained by putting

$$D^2 = -a^2, DD' = -ab \text{ and } D'^2 = -b^2,$$

provided the denominator is not zero. Similar is the rule for $\cos(ax + by)$.

Case III: $\frac{1}{F(D, D')} x^m y^n = [F(D, D')]^{-1} x^m y^n,$

which can be evaluated after expanding $[F(D, D')]^{-1}$ in ascending powers of D or D' .

Case IV: $\frac{1}{F(D, D')} (e^{ax+by} \cdot V) = e^{ax+by} \frac{1}{F(D+a, D'+b)} V.$

It follows from the fact that

$$F(D, D') \{e^{ax+by} \cdot V\} = e^{ax+by} F(D+a, D'+b) V.$$

Illustrative Examples

Example 24: Solve $r + 2s + t + 2p + 2q + z = 0$.

Solution: The given equation can be written as

$$(D^2 + 2DD' + D'^2 + 2D + 2D' + 1)z = 0$$

or $(D + D' + 1)^2 z = 0 \quad \text{or} \quad \{D - (-1)D' - (-1)\}^2 z = 0.$

There are repeated linear factors.

Hence the solution is

$$z = e^{-x} \phi_1(y - x) + x e^{-x} \phi_2(y - x).$$

Example 25: Solve $(D^2 - a^2 D'^2 + 2abD + 2abD')z = 0$.

Solution: The given equation can be written as

$$(D + aD')(D - aD' + 2ab)z = 0.$$

There are distinct linear factors. Hence the solution is

$$z = \phi_1(y - ax) + e^{-2abx} \phi_2(y + ax).$$

Example 26: Solve $(2D^4 - 3D^2 D' + D'^2)z = 0$.

Solution: The given equation can be written as

$$(2D^2 - D')(D^2 - D')z = 0.$$

Let $z = A e^{hx+ky}$ be the solution corresponding to

$$(D^2 - D')z = 0.$$

$$\therefore (D^2 - D')z = A h^2 e^{hx+ky} - A k e^{hx+ky}$$

$$= A (h^2 - k) e^{hx+ky} = 0$$

$$\Rightarrow h^2 - k = 0 \Rightarrow k = h^2.$$

Hence the general solution of $(D^2 - D')z = 0$ is

$$z = \Sigma A e^{hx + h^2 y}.$$

Similarly the general solution of $(2D^2 - D')z = 0$ is

$$z = \Sigma B e^{h'x + 2h^2 y}.$$

Hence the most general solution of the given equation is

$$z = \Sigma A e^{hx + h^2 y} + \Sigma B e^{h'x + 2h^2 y}.$$

Example 27: Solve $(D - D' - 1)(D - D' - 2)z = e^{2x-y} + x$.

Solution: C.F. = $e^x \phi_1(y + x) + e^{2x} \phi_2(y + x)$.

Now P.I. corresponding to e^{2x-y}

$$\begin{aligned} &= \frac{1}{(D - D' - 1)(D - D' - 2)} e^{2x-y} \\ &= \frac{1}{(2 + 1 - 1)(2 + 1 - 2)} e^{2x-y} = \frac{1}{2} e^{2x-y} \end{aligned}$$

and P.I. corresponding to x

$$\begin{aligned} &= \frac{1}{(D - D' - 1)(D - D' - 2)} x \\ &= \frac{1}{2} (1 - D + D')^{-1} (1 - \frac{1}{2} D + \frac{1}{2} D')^{-1} x \\ &= \frac{1}{2} (1 + D - D' \dots) (1 + \frac{1}{2} D - \frac{1}{2} D' \dots) x \\ &= \frac{1}{2} (1 + D + \frac{1}{2} D + \dots) x \\ &= \frac{1}{2} (1 + \frac{3}{2} D) x = \frac{1}{2} (x + \frac{3}{2}). \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = e^x \phi_1(y + x) + e^{2x} \phi_2(y + x) + \frac{1}{2} e^{2x-y} + \frac{x}{2} + \frac{3}{4}.$$

Example 28: Solve $(D^2 - DD' + D' - 1)z = \cos(x + 2y) + e^y$. (Avadh 2010, 11)

Solution: The given equation can be written as

$$(D - 1)(D - D' + 1)z = \cos(x + 2y) + e^y.$$

$\therefore$ C.F. = $e^x \phi_1(y) + e^{-x} \phi_2(y + x)$.

Now P.I. corresponding to $\cos(x + 2y)$

$$\begin{aligned} &= \frac{1}{D^2 - DD' + D' - 1} \cos(x + 2y) \\ &= \frac{1}{-1^2 - (-1 \cdot 2) + D' - 1} \cos(x + 2y) \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{D'} \cos(x+2y) = \frac{D'}{D'^2} \cos(x+2y) \\
 &= \frac{D'}{-2^2} \cos(x+2y) = -\frac{1}{4} \{-2 \sin(x+2y)\} \\
 &= \frac{1}{2} \sin(x+2y)
 \end{aligned}$$

and P.I. corresponding to e^y

$$\begin{aligned}
 &= \frac{1}{D^2 - DD' + D' - 1} e^y = \frac{1}{(D-1)(D-D'+1)} e^y \\
 &= \frac{1}{(0-1)(D-D'+1)} e^y = -e^y \frac{1}{D-(D'+1)+1} \\
 &= -e^y \frac{1}{D-D'} = -e^y \frac{1}{D} \left(1 - \frac{D'}{D}\right)^{-1} = -e^y \frac{1}{D} = -xe^y.
 \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = e^x \phi_1(y) + e^{-x} \phi_2(y+x) + \frac{1}{2} \sin(x+2y) - xe^y.$$

Example 29: Solve

$$(D^2 - D'^2 - 3D + 3D')z = xy + e^{x+2y}.$$

(Rohilkhand 2007; Lucknow 08; Purvanchal 2007)

Solution: The given equation can be written as

$$(D - D')(D + D' - 3)z = xy + e^{x+2y}.$$

$$\therefore \text{C.F.} = \phi_1(y+x) + e^{3x} \phi_2(y-x).$$

Now P.I. corresponding to xy

$$\begin{aligned}
 &= \frac{1}{(D-D')(D+D'-3)} xy \\
 &= -\frac{1}{3D} \left(1 - \frac{D'}{D}\right)^{-1} \left(1 - \frac{D}{3} - \frac{D'}{3}\right)^{-1} xy \\
 &= -\frac{1}{3D} \left(1 + \frac{D'}{D} + \dots\right) \left(1 + \frac{D}{3} + \frac{D'}{3} + \frac{2DD'}{9} + \dots\right) xy \\
 &= -\frac{1}{3D} \left(1 + \frac{D}{3} + \frac{D'}{3} + \frac{D'}{D} + \frac{D'}{3} + \frac{2DD'}{9} + \dots\right) xy \\
 &= -\frac{1}{3D} \left(xy + \frac{y}{3} + \frac{2}{3}x + \frac{1}{D}x + \frac{2}{9}\right) \\
 &= -\frac{1}{3} \left(\frac{1}{2}x^2y + \frac{1}{3}xy + \frac{1}{3}x^2 + \frac{1}{6}x^3 + \frac{2}{9}x\right)
 \end{aligned}$$

and P.I. corresponding to e^{x+2y}

$$\begin{aligned}
 &= \frac{1}{(D-D')(D+D'-3)} e^{x+2y} \\
 &= \frac{1}{(1-2)(D+D'-3)} e^{x+2y} = -\frac{1}{(D+D'-3)} e^{x+2y}
 \end{aligned}$$

$$\begin{aligned}
 &= -e^x \frac{1}{D' - 2} e^{2y}, \text{ putting } D = 1 \\
 &= -e^x \cdot e^{2y} \frac{1}{(D' + 2) - 2} = -e^{x+2y} \frac{1}{D'} \cdot 1 \\
 &= -ye^{x+2y}.
 \end{aligned}$$

Hence the general solution of the given equation is

$$\begin{aligned}
 z = \text{C.F.} + \text{P.I.} &= \phi_1(y + x) + e^{3x} \phi_2(y - x) \\
 &\quad - \frac{1}{3} \left(\frac{1}{2} x^2 y + \frac{1}{3} xy + \frac{1}{3} x^2 + \frac{1}{6} x^3 + \frac{2}{9} x \right) - ye^{x+2y}.
 \end{aligned}$$

Example 30: Solve $(D^2 - DD' - 2D'^2 + 2D + 2D')z = e^{2x+3y} + xy + \sin(2x + y)$.
(Avadh 2013)

Solution: The given equation can be written as

$$(D + D')(D - 2D' + 2)z = e^{2x+3y} + xy + \sin(2x + y).$$

$$\therefore \text{C.F.} = \phi_1(y - x) + e^{-2x} \phi_2(y + 2x).$$

Now P.I. corresponding to e^{2x+3y}

$$\begin{aligned}
 &= \frac{1}{(D + D')(D - 2D' + 2)} e^{2x+3y} \\
 &= \frac{1}{(2 + 3)(2 - 6 + 2)} e^{2x+3y} = -\frac{1}{10} e^{2x+3y},
 \end{aligned}$$

P.I. corresponding to xy

$$\begin{aligned}
 &= \frac{1}{(D + D')(D - 2D' + 2)} xy \\
 &= \frac{1}{2D} \left(1 + \frac{D'}{D} \right)^{-1} \left\{ 1 + \frac{1}{2}(D - 2D') \right\}^{-1} xy \\
 &= \frac{1}{2D} \left(1 - \frac{D'}{D} + \dots \right) \left\{ 1 - \frac{1}{2}(D - 2D') + \frac{1}{4}(D - 2D')^2 + \dots \right\} xy \\
 &= \frac{1}{2D} \left(1 - \frac{D'}{D} + \dots \right) \left(xy - \frac{1}{2} y + x - 1 \right) \\
 &= \frac{1}{2D} \left(xy - \frac{1}{2} y + x - 1 - \frac{1}{2} x^2 + \frac{1}{2} x \right) \\
 &= \frac{1}{2D} \left(xy - \frac{1}{2} y - \frac{1}{2} x^2 + \frac{3}{2} x - 1 \right) \\
 &= \frac{1}{2} \left(\frac{1}{2} x^2 y - \frac{1}{2} xy - \frac{1}{6} x^3 + \frac{3}{4} x^2 - x \right)
 \end{aligned}$$

and P.I. corresponding to $\sin(2x + y)$

$$\begin{aligned}
 &= \frac{1}{D^2 - DD' - 2D'^2 + 2D + 2D'} \sin(2x + y) \\
 &= \frac{1}{-2^2 - (-2 \cdot 1) - 2 \cdot (-1^2) + 2D + 2D'} \sin(2x + y) \\
 &= \frac{1}{2(D + D')} \sin(2x + y) = \frac{D - D'}{2(D^2 - D'^2)} \sin(2x + y)
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \frac{D - D'}{2^2 - (-1^2)} \sin(2x + y) = -\frac{1}{6} (D - D') \sin(2x + y) \\
 &= -\frac{1}{6} \{2 \cos(2x + y) - \cos(2x + y)\} = -\frac{1}{6} \cos(2x + y).
 \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.}$$

Example 31: Solve $(D - 3D' - 2)^2 z = 2e^{2x} \tan(y + 3x)$.

Solution: C.F. = $e^{2x} \phi_1(y + 3x) + xe^{2x} \phi_2(y + 3x)$.

$$\begin{aligned}
 \text{Now P.I.} &= \frac{1}{(D - 3D' - 2)^2} 2e^{2x} \tan(y + 3x) \\
 &= 2e^{2x} \frac{1}{\{(D + 2) - 3D' - 2\}^2} \tan(y + 3x) \\
 &= 2e^{2x} \frac{1}{(D - 3D')^2} \tan(y + 3x) \\
 &= 2e^{2x} \cdot \frac{x^2}{2!} \tan(y + 3x) = x^2 e^{2x} \tan(y + 3x).
 \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.}$$

Example 32: Solve $(D^3 - 3DD' + D + 1)z = e^{2x+3y}$. (Kanpur 2008)

Solution: Since $D^3 - 3DD' + D + 1$ cannot be resolved into linear factors in D and D' hence

$$\text{C.F.} = \Sigma A e^{hx+ky}$$

where
$$h^3 - 3hk + h + 1 = 0 \text{ i.e., } k = \frac{h^3 + h + 1}{3h}.$$

$$\begin{aligned}
 \text{Now P.I.} &= \frac{1}{D^3 - 3DD' + D + 1} e^{2x+3y} \\
 &= \frac{1}{2^3 - 3 \cdot 2 \cdot 3 + 2 + 1} e^{2x+3y} = -\frac{1}{7} e^{2x+3y}.
 \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = \Sigma A e^{hx+ky} - \frac{1}{7} e^{2x+3y},$$

where
$$h^3 - 3hk + h + 1 = 0.$$

Example 33: Solve $(D^2 - DD' - 2D)z = \sin(3x + 4y)$.

Solution: Since $D^2 - DD' - 2D$ cannot be resolved into linear factors in D and D' hence

$$\text{C.F.} = \Sigma A e^{hx+ky}, \text{ where } h^2 - hk - 2h = 0 \text{ i.e., } k = h - 2.$$

$$\begin{aligned}
 \text{Now P.I.} &= \frac{1}{D^2 - DD' - 2D} \sin(3x + 4y) \\
 &= \frac{1}{-9 - (-3 \cdot 4) - 2D} \sin(3x + 4y) \\
 &= \frac{1}{3 - 2D} \sin(3x + 4y) = \frac{3 + 2D}{9 - 4D^2} \sin(3x + 4y) \\
 &= \frac{3 + 2D}{9 - 4(-9)} \sin(3x + 4y) \\
 &= \frac{1}{45} \{3 \sin(3x + 4y) + 2 \cdot 3 \cos(3x + 4y)\} \\
 &= \frac{1}{15} \{\sin(3x + 4y) + 2 \cos(3x + 4y)\}.
 \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.}$$

Example 34: Solve $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6 \frac{\partial^2 z}{\partial y^2} = x^2 \sin(x + y)$.

Solution: The given equation can be written as

$$(D^2 + DD' - 6D'^2)z = x^2 \sin(x + y)$$

or $(D - 2D')(D + 3D')z = x^2 \sin(x + y)$.

$\therefore$ C.F. = $\phi_1(y + 2x) + \phi_2(y - 3x)$.

Now P.I. = $\frac{1}{D^2 + DD' - 6D'^2} x^2 \sin(x + y)$

$$= \text{imaginary part of } \frac{1}{D^2 + DD' - 6D'^2} x^2 e^{i(x+y)}$$

$$= \text{I.P. of } e^{i(x+y)} \frac{1}{(D+i)^2 + (D+i)(D'+i) - 6(D'+i)^2} x^2$$

$$= \text{I.P. of } e^{i(x+y)} \frac{1}{D^2 + 3iD + DD' - 11D'i - 6D'^2 + 4} x^2$$

$$= \text{I.P. of } e^{i(x+y)} \cdot \frac{1}{4} \left[1 + \left(\frac{D^2}{4} + \frac{3iD}{4} + \frac{DD'}{4} - \frac{11D'i}{4} - \frac{6D'^2}{4} \right) \right]^{-1} x^2$$

$$\begin{aligned}
 &= \text{I.P. of } e^{i(x+y)} \cdot \frac{1}{4} \left[1 - \frac{D^2}{4} - \frac{3iD}{4} - \frac{DD'}{4} + \frac{11D'i}{4} + \frac{6}{4} D'^2 \right. \\
 &\quad \left. + \frac{9i^2 D^2}{16} + \dots \right] x^2
 \end{aligned}$$

$$= \text{I.P. of } e^{i(x+y)} \cdot \frac{1}{4} \left[x^2 - \frac{1}{2} - \frac{3i}{2} x - \frac{9}{8} \right]$$

$$\begin{aligned}
 &= \text{I.P. of } \frac{1}{4} \{ \cos(x+y) + i \sin(x+y) \} \left\{ x^2 - \frac{13}{8} - \frac{3ix}{2} \right\} \\
 &= \frac{1}{4} \left(x^2 - \frac{13}{8} \right) \sin(x+y) - \frac{3}{8} x \cos(x+y).
 \end{aligned}$$

Hence the general solution of the given equation is

$$\begin{aligned}
 z = \text{C.F.} + \text{P.I.} &= \phi_1(y+2x) + \phi_2(y-3x) \\
 &\quad + \frac{1}{4} \left(x^2 - \frac{13}{8} \right) \sin(x+y) - \frac{3}{8} x \cos(x+y).
 \end{aligned}$$

Comprehensive Exercise 5

1. Solve $(D^2 - D'^2 + D - D')z = 0$.
2. Solve $DD'(D - 2D' - 3)z = 0$.
3. Solve $(D - 2D' - 1)(D - 2D'^2 - 1)z = 0$.
4. Solve $t + s + q = 0$. (Avadh 2007)
5. Solve $r - s + p = 1$. (Avadh 2010)
6. Solve $(D^2 - D')z = 2y - x^2$.
7. Solve $(D^2 + DD' + D' - 1)z = \sin(x+2y)$. (Rohilkhand 2008)
8. Solve $\frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x \partial y} + 4 \frac{\partial^2 z}{\partial y^2} + \frac{\partial z}{\partial x} - 2 \frac{\partial z}{\partial y} = e^{x+y}$.
9. Solve $(D^2 - DD' - 2D)z = \sin(3x+4y) - e^{2x+y}$. (Avadh 2010)
10. Solve $(D^2 - D')z = xe^{ax+a^2y}$.
11. Solve $(D^2 - D'^2 + D - D')z = e^{2x+3y}$.
12. Solve $(D^2 - 4DD' + D - 1)z = e^{3x-2y}$.
13. Solve $(D^2 - D' - 1)z = x^2y$.
14. Solve $(D^2 - DD' + D' - 1)z = 2 \cos(x+2y) - e^y$. (Avadh 2009)
15. Solve $\frac{\partial^2 z}{\partial x^2} - 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} - \frac{\partial z}{\partial x} + 2 \frac{\partial z}{\partial y} = (2+4x)e^{-y}$.

Answers 5

1. $z = \phi_1(y+x) + e^{-x} \phi_2(y-x)$
2. $z = \phi_1(x) + \phi_2(y) + e^{3x} \phi_3(y+2x)$

3. $z = e^x \phi_1(y + 2x) + \Sigma A e^{(2k^2 + 1)x + ky}$
4. $z = \phi_1(x) + e^{-x} \phi_2(y - x)$
5. $z = \phi_1(y) + e^{-x} \phi_2(y + x) + x$
6. $z = \Sigma A e^{hx + h^2 y + yx^2}$
7. $z = e^{-x} \phi_1(y) + e^x \phi_2(y - x) - \frac{1}{10} \{\cos(x + 2y) + 2 \sin(x + 2y)\}$
8. $z = \phi_1(y + 2x) + e^{-x} \phi_2(y + 2x) + \frac{1}{2} y e^{x+y}$
9. $z = \phi_1(y) + e^{2x} \phi_2(y + x) + \frac{1}{15} \{\sin(3x + 4y) + 2 \cos(3x + 4y)\} + \frac{1}{2} e^{2x+y}.$
10. $z = \Sigma A e^{hx + h^2 y} + \left(\frac{x^2}{4a} - \frac{x}{4a^2} \right) e^{ax + a^2 y}$
11. $z = \phi_1(y + x) + e^{-x} \phi_2(y - x) - \frac{1}{6} e^{2x+3y}$
12. $z = \Sigma A e^{hx + ky} + \frac{1}{35} e^{3x-2y}, \text{ where } k = \frac{1}{4h} (h^2 + h - 1)$
13. $z = \Sigma A e^{hx + (h^2 - 1)y + x^2 - x^2 y - 2y + 4}$
14. $z = e^x \phi_1(y) + e^{-x} \phi_2(y + x) + \sin(x + 2y) + x e^y$
15. $z = \phi_1(y + 2x) + e^x \phi_2(y + x) + e^{-y} (x + x^2)$

9.10 Equations Reducible to Linear Form with Constant Coefficients

A partial differential equation having variable coefficients can sometimes be reduced to an equation with constant coefficients by suitable substitutions. We shall discuss the reduction of an equation of the form

$$\left[A_0 x^n \frac{\partial^n z}{\partial x^n} + A_1 x^{n-1} y \frac{\partial^n z}{\partial x^{n-1} \partial y} + \dots + A_n y^n \frac{\partial^n z}{\partial y^n} + \dots \right] = f(x, y) \quad \dots(1)$$

into a linear equation with constant coefficients.

It should be noted that in the equation (1) the term $\frac{\partial^n z}{\partial x^r \partial y^{n-r}}$ is multiplied by the variable expression $x^r y^{n-r}$.

To transform the equation (1), we put $x = e^X, y = e^Y$ so that, $X = \log x$ and $Y = \log y$.

Then
$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial X} \cdot \frac{\partial X}{\partial x} = \frac{1}{x} \frac{\partial z}{\partial X}$$

or
$$x \frac{\partial z}{\partial x} = \frac{\partial z}{\partial X}$$

$\therefore x \frac{\partial}{\partial x} \equiv \frac{\partial}{\partial X} \equiv D$ (say). ...(2)

Now
$$x \frac{\partial}{\partial x} \left(x \frac{\partial z}{\partial x} \right) = x^2 \frac{\partial^2 z}{\partial x^2} + x \frac{\partial z}{\partial x}$$

$\Rightarrow x^2 \frac{\partial^2 z}{\partial x^2} = \left(x \frac{\partial}{\partial x} - 1 \right) x \frac{\partial z}{\partial x}$
 $= (D - 1) Dz$
 $= D(D - 1) z.$...(3)

In general
$$x^n \frac{\partial^n z}{\partial x^n} = D(D - 1)(D - 2) \dots (D - n + 1) z.$$
 ...(4)

Similarly differentiating w.r.t. y , we get

$$y \frac{\partial z}{\partial y} = \frac{\partial z}{\partial Y} = D' z \text{ i.e., } y \frac{\partial}{\partial y} \equiv \frac{\partial}{\partial Y} \equiv D',$$

$$y^2 \frac{\partial^2 z}{\partial y^2} = D'(D' - 1) z,$$

$$y^n \frac{\partial^n z}{\partial y^n} = D'(D' - 1) \dots (D' - n + 1) z.$$

Also
$$xy \frac{\partial^2 z}{\partial x \partial y} = DD' z$$

and
$$x^m y^n \frac{\partial^{m+n} z}{\partial x^m \partial y^n} = D(D - 1) \dots (D - m + 1) D'(D' - 1) \dots (D' - n + 1) z.$$

These substitutions reduce the equation (1) to an equation having constant coefficients and now it can easily be solved by the methods discussed for homogeneous and non-homogeneous linear equations with constant coefficients.

Illustrative Examples

Example 35: Solve $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = 0.$

Solution: Putting $x = e^X$, $y = e^Y$

and denoting $\frac{\partial}{\partial X}$ and $\frac{\partial}{\partial Y}$ by D and D' respectively, the given equation transforms to

$$[D(D - 1) + 2DD' + D'(D' - 1)] z = 0$$

$$\text{or} \quad [D^2 + 2DD' + D'^2 - D - D'] z = 0$$

$$\text{or} \quad (D + D')(D + D' - 1) z = 0.$$

Hence the general solution of the given equation is

$$\begin{aligned} z &= \phi_1(Y - X) + e^X \phi_2(Y - X) \\ &= \phi_1(\log y - \log x) + x \phi_2(\log y - \log x) \\ &= \phi_1\left(\log \frac{y}{x}\right) + x \phi_2\left(\log \frac{y}{x}\right) = f_1\left(\frac{y}{x}\right) + x f_2\left(\frac{y}{x}\right). \end{aligned}$$

Example 36: Solve $x^2 \frac{\partial^2 z}{\partial x^2} - 4xy \frac{\partial^2 z}{\partial x \partial y} + 4y^2 \frac{\partial^2 z}{\partial y^2} + 6y \frac{\partial z}{\partial y} = x^3 y^4$.

Solution: Putting $x = e^X, y = e^Y$

and denoting $\frac{\partial}{\partial X}$ and $\frac{\partial}{\partial Y}$ by D and D' respectively, the given equation transforms to

$$[D(D-1) - 4DD' + 4D'(D'-1) + 6D'] z = e^{3X+4Y}$$

$$\text{or} \quad (D^2 - D - 4DD' + 4D'^2 + 2D') z = e^{3X+4Y}.$$

$$\text{or} \quad (D - 2D')(D - 2D' - 1) z = e^{3X+4Y}.$$

$$\begin{aligned} \therefore \text{C.F.} &= \phi_1(Y + 2X) + e^X \phi_2(Y + 2X) \\ &= \phi_1(\log y + 2 \log x) + x \phi_2(\log y + 2 \log x) \\ &= \phi_1(\log(yx^2)) + x \phi_2(\log(yx^2)) \\ &= f_1(yx^2) + x f_2(yx^2). \end{aligned}$$

$$\begin{aligned} \text{Now P.I.} &= \frac{1}{(D - 2D')(D - 2D' - 1)} e^{3X+4Y} \\ &= \frac{1}{(3 - 2.4)(3 - 2.4 - 1)} e^{3X+4Y} = \frac{1}{30} x^3 y^4. \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = f_1(yx^2) + x f_2(yx^2) + \frac{1}{30} x^3 y^4.$$

Example 37: Solve $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = (x^2 + y^2)^{n/2}$.

Solution: Putting $x = e^X, y = e^Y$ and denoting $\frac{\partial}{\partial X}$ and $\frac{\partial}{\partial Y}$ by D and D' respectively,

the given equation transforms to

$$\{D(D-1) + 2DD' + D'(D'-1)\} z = (e^{2X} + e^{2Y})^{n/2}$$

$$\text{or} \quad \{D^2 + 2DD' + D'^2 - D - D'\} z = (e^{2X} + e^{2Y})^{n/2}$$

$$\text{or} \quad (D + D')(D + D' - 1) z = (e^{2X} + e^{2Y})^{n/2}.$$

$$\begin{aligned}
 \therefore \quad \text{C.F.} &= \phi_1(Y - X) + e^X \phi_2(Y - X) \\
 &= \phi_1(\log y - \log x) + x \phi_2(\log y - \log x) \\
 &= \phi_1\left(\log \frac{y}{x}\right) + x \phi_2\left(\log \frac{y}{x}\right) = f_1\left(\frac{y}{x}\right) + x f_2\left(\frac{y}{x}\right).
 \end{aligned}$$

$$\begin{aligned}
 \text{Now P.I.} &= \frac{1}{(D + D')(D + D' - 1)} (e^{2X} + e^{2Y})^{n/2} \\
 &= \frac{1}{(D + D')(D + D' - 1)} e^{nX} \{1 + e^{2(Y - X)}\}^{n/2} \\
 &= \frac{1}{(D + D')(D + D' - 1)} \left[e^{nX} + \frac{1}{2} n e^{(n-2)X + 2Y} \right. \\
 &\quad \left. + \frac{1}{2} n \left(\frac{1}{2} n - 1\right) e^{(n-4)X + 4Y} + \dots \right] \\
 &= \frac{e^{nX}}{n(n-1)} \left[1 + \frac{1}{2} n e^{2(Y - X)} + \dots \right] \\
 &= \frac{e^{nX} [1 + e^{2(Y - X)}]^{n/2}}{n(n-1)} = \frac{(e^{2X} + e^{2Y})^{n/2}}{n(n-1)} = \frac{(x^2 + y^2)^{n/2}}{n(n-1)}.
 \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = f_1\left(\frac{y}{x}\right) + x f_2\left(\frac{y}{x}\right) + \frac{(x^2 + y^2)^{n/2}}{n(n-1)}.$$

Example 38: Solve $yt - q = xy$.

(Avadh 2009)

Solution: The given equation can be written as

$$y \frac{\partial^2 z}{\partial y^2} - \frac{\partial z}{\partial y} = xy$$

$$\text{or} \quad y^2 \frac{\partial^2 z}{\partial y^2} - y \frac{\partial z}{\partial y} = xy^2.$$

Putting $x = e^X$, $y = e^Y$ and denoting $\frac{\partial}{\partial X}$ and $\frac{\partial}{\partial Y}$ by D and D' respectively the given equation transforms to

$$\{D'(D' - 1) - D'\} z = e^{X + 2Y}$$

$$\text{or} \quad D'(D' - 2) z = e^{X + 2Y}.$$

$$\begin{aligned}
 \therefore \quad \text{C.F.} &= \phi_1(X) + e^{2Y} \phi_2(X) \\
 &= \phi_1(\log x) + y^2 \phi_2(\log x) \\
 &= f_1(x) + y^2 f_2(x).
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } \text{P.I.} &= \frac{1}{D'(D'-2)} e^{X+2Y} \\
 &= \frac{1}{2(D'-2)} e^{X+2Y} \\
 &= \frac{1}{2} e^{X+2Y} \frac{1}{D'+2-2} 1 \\
 &= \frac{1}{2} e^{X+2Y} \frac{1}{D'} 1 = \frac{1}{2} Y e^{X+2Y} \\
 &= \frac{1}{2} xy^2 \log y.
 \end{aligned}$$

Hence the general solution of the given equation is

$$z = \text{C.F.} + \text{P.I.} = f_1(x) + y^2 f_2(x) + \frac{1}{2} xy^2 \log y.$$

Example 39: Solve $\frac{1}{x^2} \frac{\partial^2 z}{\partial x^2} - \frac{1}{x^3} \frac{\partial z}{\partial x} = \frac{1}{y^2} \frac{\partial^2 z}{\partial y^2} - \frac{1}{y^3} \frac{\partial z}{\partial y}$.

Solution: Put $\frac{1}{2} x^2 = X$ and $\frac{1}{2} y^2 = Y$,

so that $x dx = dX$ and $y dy = dY$.

Hence $\frac{\partial z}{\partial X} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial X} = \frac{1}{x} \frac{\partial z}{\partial x}$

and $\frac{\partial^2 z}{\partial X^2} = \frac{\partial}{\partial X} \left(\frac{\partial z}{\partial X} \right)$

$$\begin{aligned}
 &= \frac{\partial}{\partial x} \left(\frac{1}{x} \frac{\partial z}{\partial x} \right) \cdot \frac{\partial x}{\partial X} \\
 &= \frac{1}{x} \cdot \frac{\partial}{\partial x} \left(\frac{1}{x} \frac{\partial z}{\partial x} \right) \\
 &= -\frac{1}{x^3} \frac{\partial z}{\partial x} + \frac{1}{x^2} \frac{\partial^2 z}{\partial x^2}
 \end{aligned}$$

i.e., $\frac{1}{x^2} \frac{\partial^2 z}{\partial x^2} - \frac{1}{x^3} \frac{\partial z}{\partial x} = \frac{\partial^2 z}{\partial X^2}$.

Similarly $\frac{1}{y^2} \frac{\partial^2 z}{\partial y^2} - \frac{1}{y^3} \frac{\partial z}{\partial y} = \frac{\partial^2 z}{\partial Y^2}$.

Thus the given equation transforms to

$$\frac{\partial^2 z}{\partial X^2} = \frac{\partial^2 z}{\partial Y^2} \quad \text{or} \quad \frac{\partial^2 z}{\partial X^2} - \frac{\partial^2 z}{\partial Y^2} = 0$$

or $(D^2 - D'^2)z = 0$

where $D \equiv \frac{\partial}{\partial X}$, $D' \equiv \frac{\partial}{\partial Y}$

or $(D + D')(D - D')z = 0.$

Hence the solution is

$$\begin{aligned} z &= \phi_1(Y - X) + \phi_2(Y + X) \\ &= f_1(y^2 - x^2) + f_2(y^2 + x^2). \end{aligned}$$

Comprehensive Exercise 6

1. Solve $x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} - y \frac{\partial z}{\partial y} + x \frac{\partial z}{\partial x} = 0.$
2. Solve $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} + x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} - z = 0.$
3. Solve $x^2 r - 3xys + 2y^2 t + px + 2qy = x + 2y.$
4. Solve $x^2 \frac{\partial^2 z}{\partial x^2} - y^2 \frac{\partial^2 z}{\partial y^2} = x^2 y.$
5. Solve $(x^2 D^2 + 2xy DD' + y^2 D'^2)z = x^m y^n.$
6. Solve $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} - x \frac{\partial z}{\partial x} = \frac{x^3}{y^2}.$
7. Solve $x^2 r - y^2 t + px - qy = \log x.$

(Lucknow 2006)

Answers 6

1. $z = f_1(xy) + f_2\left(\frac{y}{x}\right)$
2. $z = x f_1\left(\frac{y}{x}\right) + \frac{1}{x} f_2\left(\frac{y}{x}\right)$
3. $z = f_1(xy) + f_2(x^2 y) + x + y$
4. $z = f_1(xy) + x f_2(y/x) + \frac{1}{2} x^2 y$
5. $z = f_1\left(\frac{y}{x}\right) + x f_2\left(\frac{y}{x}\right) + \frac{1}{(m+n)(m+n-1)} x^m y^n$
6. $z = f_1(y) + x^2 f_2(y/x^2) - \frac{1}{9} (x^3 / y^2)$
7. $z = f_1(y/x) + f_2(xy) + \frac{1}{6} (\log x)^3$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. Out of the following four P.D.E., the equation which is linear:

(a) $\frac{\partial^3 z}{\partial x^3} - 3 \frac{\partial^2 z}{\partial x^2} \frac{\partial z}{\partial y} + 8 \frac{\partial^2 z}{\partial y^2} = \sin x$

(b) $\frac{\partial^2 z}{\partial x^2} + \left(\frac{\partial z}{\partial y} \right)^2 + 9z = 0$

(c) $\frac{\partial^2 z}{\partial x^2} + 5 \frac{\partial^2 z}{\partial y^2} = 0$

(d) None of these.

(Rohilkhand 2003)

2. The A.E. of the equation $2r + 5s + 2t = 0$ is

(a) $2m^2 + 5m + 2 = 0$

(b) $2m^2 - 5m + 2 = 0$

(c) $2m^2 + 5m - 2 = 0$

(d) $2m^2 - 5m - 2 = 0$

3. The general solution of the differential equation $(D^2 - 2DD' + D'^2)z = 0$ is

(a) $z = c_1 e^x + c_2 e^y$

(b) $z = c e^{(x + c_2 y)}$

(c) $z = \phi_1(x + y) + \phi_2(y + x)$

(d) $z = \phi_1(y + x) + x \phi_2(y + x)$.

(Rohilkhand 2002)

4. The C.F. of the equation $(D^3 - 3DD'^2 + 2D'^3)z = (x + 2y)^{1/2}$ is

(a) $\phi_1(y + x) + \phi_2(y - 2x)$

(b) $\phi_1(y + x) + x \phi_2(y + x) + \phi_3(y + 2x)$

(c) $\phi_1(y + x) + x \phi_2(y + x) + \phi_3(y - 2x)$

(d) None of these.

5. The P.I. of the differential equation $(D^2 + 3DD' + 2D'^2)z = x + y$ is

(a) $\frac{(x + y)^3}{6}$

(b) $\frac{(x + y)^3}{12}$

(c) $\frac{(x + y)^3}{36}$

(d) None of these.

6. The C.F. of the equation $\log s = x + y$ is
- (a) $\phi_1(x) + \phi_2(y)$ (b) $\phi_1(y) + x\phi_2(y)$
 (c) $\phi_1(x) + x\phi_2(y)$ (d) None of these.
7. The C.F. of the equation $(D^2 + 2DD' + D'^2)z = 2\cos y - x\sin y$ is
- (a) $\phi_1(y+x) + \phi_2(y-x)$
 (b) $\phi_1(y+x) + x\phi_2(y+x)$
 (c) $\phi_1(y-x) + x\phi_2(y-x)$
 (d) None of these.
8. The solution of non-homogeneous equation $(D - mD' - k)z = 0$ is
- (a) $z = e^{kx} \phi(y + mx)$
 (b) $z = e^{kx} \phi(y - mx)$
 (c) $z = e^x \phi(y + mx)$
 (d) $z = e^x \phi(y - mx)$.
9. If $z = Ae^{hx + ky}$ be the solution of $(D - D'^2)z = 0$ then
- (a) $h = k$ (b) $h = k^2$
 (c) $h = k^3$ (d) None of these.
10. The C.F. of $(D^2 - DD' - 2D)z = \sin(3x + 4y)$ is $\Sigma Ae^{hx + ky}$ where
- (a) $h^2 - hk - 2h = 0$ (b) $h^2 + hk - 2h = 0$
 (c) $h^2 - hk + 2h = 0$ (d) None of these.
11. P.I. of the equation $(D^3 - 3DD' + D' + 1)z = e^{4x + 5y}$ is
- (a) $\frac{1}{4}e^{4x + 5y}$ (b) $\frac{1}{5}e^{4x + 5y}$
 (c) $\frac{1}{10}e^{4x + 5y}$ (d) None of these.
12. The equation
- $$A_0 x^n \frac{\partial^n z}{\partial x^n} + A_1 x^{n-1} y \frac{\partial^n z}{\partial x^{n-1} \partial y} + \dots + A_n y^n \frac{\partial^n z}{\partial y^n} + \dots = f(x, y)$$
- can be reduced into a linear equation with constant coefficients in the variables X and Y by substitutions
- (a) $x = \log X, y = \log Y$
 (b) $x = e^X, y = e^Y$
 (c) $x = e^{2X}, y = e^{2Y}$
 (d) None of these

Fill In The Blank(s)

Fill in the blanks “.....” so that the following statements are complete and correct.

1. In the homogeneous linear partial differential equation with constant coefficients all the partial derivatives appearing in the equation are of the order.
2. If u is the complementary function and z_1 a particular integral of a linear partial differential equation $F(D, D')z = f(x, y)$ then is a general solution of the equation.
3. A linear partial differential equation which is not homogeneous is called a linear equation.
4. The A.E. of a linear homogeneous n th order partial differential equation with constant coefficients will give in general roots.
5. The A.E. of $(D^2 + D'^2)z = 30(2x + y)$ is

True or False

Write ‘T’ for true and ‘F’ for false statement.

1. The A.E. of a linear homogeneous n th order partial differential equation with constant coefficients of the form $F(D, D')z = f(x, y)$ is obtained by putting $D = 1, D' = m$ in $F(D, D') = 0$.
2. The value of $\frac{1}{(bD - aD')} x^r \phi(ax + by) = \frac{x^{r+1}}{b(r+1)} \phi(ax + by)$.
3. The solution of $(D^2 - 2aDD' + a^2 D'^2)z = 0$ is

$$z = \phi_1(y + ax) + x \phi_2(y + ax).$$
4. In case of non-homogeneous linear partial differential equation with constant coefficients the value of $\frac{1}{F(D, D')} e^{(ax + by)} \cdot V = e^{ax + by} \frac{1}{(D - a, D' - b)} V$.
5. The C.F. of $(D^3 - 3DD' + D + 1)z = e^{2x + 3y}$ is $\sum A e^{hx + ky}$
 where $h^3 - 3hk + h + 1 = 0$.
6. In the equation $(D - mD')z = \phi(x, y)$ the value of

$$z = \frac{1}{D - mD'} \phi(x, y) = \int \phi(x, a - mx) dx, \text{ where } a = y + mx.$$

Answers

Multiple Choice Questions

- | | | |
|---------|---------|---------|
| 1. (c) | 2. (a) | 3. (d) |
| 4. (c) | 5. (c) | 6. (a) |
| 7. (c) | 8. (a) | 9. (b) |
| 10. (a) | 11. (c) | 12. (b) |

Fill in the Blank(s)

- | | | |
|---------|------------------|---------------------|
| 1. same | 3. $u + z_1$ | 3. non-homogeneous. |
| 4. n | 5. $m^2 + 1 = 0$ | |

True or False

- | | | |
|--------|--------|--------|
| 1. F | 2. T | 3. T |
| 4. F | 5. T | 6. T |



Chapter

10

Partial Differential Equations of Second Order with Variable Coefficients

10.1 Partial Differential Equations of Second Order with Variable Coefficients

A partial differential equation is said to be of second order if it contains at least one of the second order partial differential coefficients r , s and t but none of higher order. The differential coefficients p and q may also appear in the equation. Thus the general form of a second order partial differential equation is

$$F(x, y, z, p, q, r, s, t) = 0.$$

The complete solutions of these equations will contain two arbitrary functions.

In this chapter we deal with the second order partial differential equations with variable coefficients also.

Below we give some examples of equations that are readily solvable. It is to be noted that x and y , being independent, are constant with respect to each other in differentiation and integration.

Illustrative Examples

Example 1: Solve $s = \frac{x}{y} + a$.

Solution: The given equation can be written as $\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y} + a$.

Integrating w.r.t. 'x', we get $\frac{\partial z}{\partial y} = \frac{1}{y} \cdot \frac{x^2}{2} + ax + f(y)$.

Now integrating w.r.t. 'y' we get

$$z = \frac{x^2}{2} \log y + axy + \int f(y) dy + \psi(x)$$

or
$$z = \frac{1}{2} x^2 \log y + axy + \phi(y) + \psi(x).$$

Example 2: Solve $s = 2x + 2y$.

Solution: The given equation can be written as $\frac{\partial^2 z}{\partial x \partial y} = 2x + 2y$.

Integrating w.r.t. 'x', we get $\frac{\partial z}{\partial y} = x^2 + 2xy + f(y)$.

Now integrating w.r.t. 'y', we get

$$z = x^2 y + xy^2 + \int f(y) dy + \psi(x)$$

or
$$z = x^2 y + xy^2 + \phi(y) + \psi(x).$$

Example 3: Solve $t = \sin xy$.

Solution: The given equation can be written as $\frac{\partial^2 z}{\partial y^2} = \sin xy$.

Integrating w.r.t. 'x', we get $\frac{\partial z}{\partial y} = -\frac{1}{x} \cos xy + f(x)$.

Again integrating w.r.t. 'y', we get $z = -\frac{1}{x^2} \sin xy + yf(x) + \psi(x)$.

Example 4: Solve $ys + p = \cos(x + y) - y \sin(x + y)$.

Solution: The given equation can be written as

$$y \frac{\partial q}{\partial x} + \frac{\partial z}{\partial x} = \cos(x + y) - y \sin(x + y).$$

Integrating w.r.t. 'x', we get

$$yq + z = \sin(x + y) + y \cos(x + y) + f(y)$$

or
$$y \frac{\partial z}{\partial y} + z = \sin(x + y) + y \cos(x + y) + f(y).$$

Now integrating w.r.t. 'y', we get

$$yz = y \sin (x + y) + \int f (y) dy + \psi (x)$$

or $yz = y \sin (x + y) + \phi (y) + \psi (x).$

Example 5: Solve $t - xq = x^2$.

(Lucknow 2011)

Solution: The given equation can be written as

$$\frac{\partial q}{\partial y} - xq = x^2, \text{ which is linear in } q \text{ regarding } x \text{ as constant.}$$

$$\text{I.F.} = e^{-\int x dy} = e^{-xy}.$$

$\therefore$ solution is $q e^{-xy} = \int x^2 e^{-xy} dy + f(x) = -x e^{-xy} + f(x)$

or $q = \frac{\partial z}{\partial y} = -x + e^{xy} f(x).$

Integrating, $z = -xy + f(x) \int e^{xy} dy + \psi(x)$

or $z = -xy + \frac{1}{x} f(x) e^{xy} + \psi(x)$

or $z = -xy + \phi(x) e^{xy} + \psi(x).$

Example 6: Solve $yt - q = xy$.

(Avadh 2009)

Solution: The given equation can be written as

$$\frac{\partial q}{\partial y} - \frac{1}{y} q = x, \text{ which is linear in } q \text{ regarding } x \text{ as constant.}$$

$$\text{I.F.} = e^{\int (-1/y) dy} = e^{-\log y} = 1/y.$$

$\therefore$ $q \frac{1}{y} = \int x \cdot \frac{1}{y} dy + f(x) = x \log y + f(x)$

or $q = \frac{\partial z}{\partial y} = xy \log y + y f(x).$

Integrating, $z = x \int y \log y dy + f(x) \int y dy + \psi(x)$

or $z = x \left[\frac{1}{2} y^2 \log y - \int \frac{1}{y} \cdot \frac{y^2}{2} dy \right] + \frac{y^2}{2} f(x) + \psi(x)$

or $z = \frac{1}{2} xy^2 \log y - \frac{1}{4} x y^2 + \frac{1}{2} y^2 f(x) + \psi(x).$

Example 7: Solve $rx = (n-1)p$.

Solution: The given equation can be written as $\frac{\partial^2 z / \partial x^2}{\partial z / \partial x} = \frac{n-1}{x}.$

Integrating, $\log \frac{\partial z}{\partial x} = (n-1) \log x + \log f(y)$ or $\frac{\partial z}{\partial x} = x^{n-1} f(y).$

Again integrating w.r.t. 'x', we get

$$z = \frac{x^n}{n} f(y) + \psi(y) \quad \text{or} \quad z = x^n \phi(y) + \psi(y).$$

Example 8: Solve $p + r + s = 1$.

(Avadh 2006)

Solution: The given equation can be written as $\frac{\partial z}{\partial x} + \frac{\partial p}{\partial x} + \frac{\partial q}{\partial x} = 1$.

Integrating w.r.t. 'x', we get

$$z + p + q = x + f(y) \quad \text{or} \quad p + q = x + f(y) - z.$$

Lagrange's auxiliary equations for this are $\frac{dx}{1} = \frac{dy}{1} = \frac{dz}{x + f(y) - z}$.

The first two members give $x - y = a$.

From the last two members, we get

$$\frac{dz}{dy} + z = x + f(y) \quad \text{or} \quad \frac{dz}{dy} + z = a + y + f(y), \text{ which is linear in } z.$$

$$\text{I.F.} = e^{\int dy} = e^y.$$

$$\begin{aligned} \therefore z e^y &= \int \{a + y + f(y)\} e^y dy + b = a e^y + \int \{y + f(y)\} e^y dy + b \\ &= a e^y + \phi(y) + b \end{aligned}$$

$$\text{or} \quad z = a + e^{-y} \phi(y) + b e^{-y}$$

$$\text{or} \quad z = x - y + e^{-y} \phi(y) + e^{-y} \psi(x - y).$$

Solutions of equations under given conditions.

After finding the general solutions by the usual methods, the geometrical conditions given in the problem are used to find the arbitrary functions.

Example 9: Find the surface passing through the parabolas

$$z = 0, y^2 = 4ax \quad \text{and} \quad z = 1, y^2 = -4ax$$

and satisfying the equation

$$xr + 2p = 0.$$

Solution: The given equation can be written as

$$x \frac{\partial p}{\partial x} + 2p = 0 \quad \text{or} \quad x^2 \frac{\partial p}{\partial x} + 2px = 0.$$

Integrating w.r.t. 'x', we get $px^2 = f(y)$ or $p = \frac{\partial z}{\partial x} = \frac{1}{x^2} f(y)$.

Again integrating w.r.t. 'x', we get $z = -\frac{1}{x} f(y) + \phi(y)$ (1)

Now using the geometrical conditions of the problem we are to determine the values of $f(y)$ and $\phi(y)$.

Since the required surface is to pass through the parabola $z = 0, y^2 = 4ax$ hence putting

$$z = 0 \text{ and } x = y^2 / 4a \text{ in (1), we get } 0 = -\frac{4a}{y^2} f(y) + \phi(y). \quad \dots (2)$$

Again putting $z = 1$ and $x = -y^2 / 4a$ in (1), we get

$$1 = \frac{4a}{y^2} f(y) + \phi(y). \quad \dots(3)$$

Solving (2) and (3) for $f(y)$ and $\phi(y)$,

we have $\phi(y) = \frac{1}{2}$ and $f(y) = y^2 / 8a$.

Thus we have determined the arbitrary functions.

Putting the values of $f(y)$ and $\phi(y)$ in (1), the required surface is

$$z = -\frac{1}{x} \cdot \frac{y^2}{8a} + \frac{1}{2}$$

or $8axz - 4ax + y^2 = 0$.

Example 10: Find a surface passing through the two lines $z = x = 0$,
 $z - 1 = x - y = 0$, satisfying $r - 4s + 4t = 0$.

Solution: The given equation can be written as $(D^2 - 4DD' + 4D'^2)z = 0$. $\dots(1)$

A.E. is $m^2 - 4m + 4 = 0$

or $(m - 2)^2 = 0$.

$\therefore m = 2, 2$.

Hence the general solution of (1) is $z = \phi_1(y + 2x) + x\phi_2(y + 2x)$. $\dots(2)$

If the surface (2) passes through the lines $z = x = 0$ and $z - 1 = x - y = 0$, then we have

$$0 = \phi_1(y + 2x) \quad \dots(3)$$

and $1 = \phi_1(y + 2x) + x\phi_2(y + 2x)$. $\dots(4)$

From (3) and (4), we get

$$\phi_2(y + 2x) = \frac{1}{x} = \frac{3}{3x} = \frac{3}{2x + y} \quad [\because y - x = 0]$$

Putting the values of ϕ_1 and ϕ_2 in (2), the required surface is

$$z = x \cdot \frac{3}{2x + y} \text{ or } z(2x + y) = 3x.$$

Comprehensive Exercise 1

1. Solve $xy + p = 9x^2y^3$.
2. Solve $r = 2y^2$.
3. Solve $\log s = x + y$.
4. Solve $r = 6x$.

5. Solve $xr + 2p = 0$.
6. Solve $xr = p$.
7. Solve $xs + q = 4x + 2y + 2$.
8. Solve $t + s + q = 0$.
9. Solve $2yq + y^2t = 1$.
10. Find a surface satisfying $r + s = 0$ and touching the elliptic paraboloid $z = 4x^2 + y^2$ along its section by the plane $y = 2x + 1$.
11. Solve the equation $r + t = 2s$ and determine the arbitrary functions by the conditions that $bz = y^2$ when $x = 0$ and $az = x^2$ when $y = 0$.

Answers 1

1. $z = x^3 y^3 + f(y) \log x + \phi(y)$
2. $z = x^2 y^2 + x f(y) + \phi(y)$
3. $z = e^{x+y} + \phi(y) + \psi(x)$
4. $z = x^3 + x f(y) + \phi(y)$
5. $z = -\frac{1}{x} f(y) + \phi(y)$
6. $z = \frac{1}{2} x^2 f(y) + \phi(y)$
7. $z = 2x^2 y + xy^2 + 2x y + \phi(x) + \psi(y)$
8. $z e^x = \phi(x) + \psi(x - y)$
9. $yz = y \log y - f(x) + y \phi(x)$
10. $z + 4x^2 + y^2 - 8xy - 4y + 8x + 2 = 0$
11. $z = (y + x) \left(\frac{x}{a} + \frac{y}{b} \right)$

10.2 Classification of Linear Partial Differential Equation of Second Order

The linear partial differential equation of the second order in n independent variables $x_1, x_2, \dots, x_n$ can be written as

$$\sum_{i=1}^n \sum_{j=1}^n a_{ij} \frac{\partial^2 u}{\partial x_i \partial x_j} + \sum_{i=1}^n b_i \frac{\partial u}{\partial x_i} + c u = 0 \quad \dots(1)$$

where a_{ij} , b_i and c are constants or functions of $x_1, x_2, \dots, x_n$.

Let δ_i represent $\frac{\partial}{\partial x_i}$, for $i = 1, 2, \dots, n$

and $\delta_i \delta_j$ represent $\frac{\partial^2}{\partial x_i \partial x_j}$, for $i = 1, 2, 3, \dots, n$ and $j = 1, 2, 3, \dots, n$.

Consider the operator

$$\phi = \sum_{i=1}^n \sum_{j=1}^n a_{ij} \delta_i \delta_j \quad \dots(2)$$

for all real values of δ_i positive or negative.

Now at a point $x_1, x_2, \dots, x_n$ we call the linear partial differential equation given by (1) as :

- (i) **elliptic** if ϕ is positive for all real values of δ_i and it reduces to zero only when all δ_i 's are zero.
- (ii) **hyperbolic** if ϕ can be both positive or negative.
- (iii) **parabolic** if the determinant Δ vanishes, where

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix} = 0 .$$

If a_{ij} are functions of $x_1, x_2, \dots, x_n$ the same differential equation can be elliptic, hyperbolic and parabolic at different points.

If a_{ij} are constants, the equation will have the same nature throughout.

10.3 Classification of Linear Partial Differential Equation of Second Order in Two Independent Variables

Let us consider the equation of second order in two independent variables x and y

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + f\left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right) = 0, \quad \dots(1)$$

where A is positive.

Here $\phi = A \delta_1^2 + B \delta_1 \delta_2 + C \delta_2^2$.

The equation (1) is

- (i) **elliptic** if $B^2 - 4AC < 0$,
- (ii) **hyperbolic** if $B^2 - 4AC > 0$,
- (iii) **parabolic** if $B^2 - 4AC = 0$.

Note 1: If A, B, C are constants then the nature of the equation (1) will be the same in the whole region *i.e.*, for all values of x and y . The nature will depend on $B^2 - 4AC$.

The equation (1) will be elliptic if $B^2 - 4AC < 0$.

The equation (1) will be hyperbolic if $B^2 - 4AC > 0$.

The equation (1) will be parabolic if $B^2 - 4AC = 0$.

Note 2: If A, B, C are functions of x and y then the nature of equation (1) will not be same in the whole region *i.e.*, for all values of x and y .

The equation (1) will be elliptic in the region where $B^2 - 4AC < 0$.

The equation (1) will be hyperbolic in the region where $B^2 - 4AC > 0$.

The equation (1) will be parabolic in the region where $B^2 - 4AC = 0$.

Illustrative Examples

Example 11: Classify the operator

$$(i) \quad \frac{\partial^2 u}{\partial t^2} + \frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial x^2}$$

(Meerut 2011)

$$(ii) \quad \frac{\partial^2 u}{\partial t^2} - 4 \frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial x^2}$$

$$(iii) \quad \frac{\partial^2 u}{\partial t^2} + 4 \frac{\partial^2 u}{\partial x \partial t} + 4 \frac{\partial^2 u}{\partial x^2}.$$

Solution: (i) Here $A = 1, B = 1, C = 1$ and so $B^2 - 4AC = 1 - 4 = -3 < 0$.

Therefore, the given operator is elliptic.

(ii) Here $A = 1, B = -4, C = 1$ and so $B^2 - 4AC = 16 - 4 = 12 > 0$.

Therefore the given operator is hyperbolic.

(iii) Here $A = 1, B = 4, C = 4$ and so $B^2 - 4AC = 16 - 16 = 0$.

Therefore the given operator is parabolic.

Example 12: Classify the equation

$$(1 - x^2) \frac{\partial^2 z}{\partial x^2} - 2xy \frac{\partial^2 z}{\partial x \partial y} + (1 - y^2) \frac{\partial^2 z}{\partial y^2} + x \frac{\partial z}{\partial x} + 3x^2 y \frac{\partial z}{\partial y} - 2z = 0.$$

Solution: Consider the operator

$$\phi = A \delta_1^2 + B \delta_1 \delta_2 + C \delta_2^2, \quad \text{where } \delta_1 \equiv \frac{\partial}{\partial x}, \delta_2 \equiv \frac{\partial}{\partial y}.$$

Here $A = 1 - x^2, B = -2xy, C = 1 - y^2,$

and so $B^2 - 4AC = 4x^2y^2 - 4(1 - x^2)(1 - y^2) = 4(-1 + x^2 + y^2).$

Since A, B, C are functions of x and y , the given differential equation is hyperbolic in the region where $B^2 - 4AC > 0$ i.e., $x^2 + y^2 > 1$,
parabolic in the region where $B^2 - 4AC = 0$ i.e., at points on the circle $x^2 + y^2 = 1$,
and elliptic in the region where $B^2 - 4AC < 0$ i.e., $x^2 + y^2 < 1$.

Comprehensive Exercise 2

1. Classify the following equations :

$$(i) \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

(Laplace's equation) (Meerut 2007, 08, 11)

$$(ii) \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{1}{C^2} \frac{\partial^2 u}{\partial t^2}$$

(Wave equation) (Meerut 2007, 10)

$$(iii) \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{1}{C^2} \frac{\partial u}{\partial t}$$

(Heat equation) (Meerut 2007)

2. Find where the following operator is hyperbolic, parabolic and elliptic

$$(i) \frac{\partial^2 u}{\partial t^2} + t \frac{\partial^2 u}{\partial x \partial t} + x \frac{\partial^2 u}{\partial x^2}$$

$$(ii) x^2 \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} + u$$

$$(iii) t \frac{\partial^2 u}{\partial t^2} + 2 \frac{\partial^2 u}{\partial x \partial t} + x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x}$$

3. Show that the equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ is hyperbolic.

(Rohilkhand 2008, 10)

4. Classify the following as elliptic, parabolic or hyperbolic :

$$(i) \frac{\partial^2 z}{\partial x^2} = \frac{\partial z}{\partial y}$$

$$(ii) \frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial y^2}$$

$$(iii) \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0.$$

Answers 2

1. (i) elliptic. (ii) hyperbolic. (iii) parabolic.

2. hyperbolic if $t^2 > 4x$,
parabolic if $t^2 = 4x$ and elliptic if $t^2 < 4x$.

4. (i) Parabolic (ii) Hyperbolic (iii) Elliptic.

10.4 Canonical Forms (Method of Transformations)

Now we shall consider the equation of the type

$$Rr + Ss + Tt + F(x, y, z, p, q) = 0, \quad \dots(1)$$

where R, S, T are continuous functions of x and y possessing continuous partial derivatives of as high an order as necessary. We shall show that any equation of the type (1) can be reduced to one of the three canonical forms by a suitable change of the independent variables. Suppose we change the independent variables from x, y to u, v where

$$u = u(x, y), \quad v = v(x, y). \quad \dots(2)$$

Then, we have

$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x},$$

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y}.$$

$$\therefore \frac{\partial}{\partial x} \equiv \frac{\partial u}{\partial x} \cdot \frac{\partial}{\partial u} + \frac{\partial v}{\partial x} \cdot \frac{\partial}{\partial v}, \quad \frac{\partial}{\partial y} \equiv \frac{\partial u}{\partial y} \cdot \frac{\partial}{\partial u} + \frac{\partial v}{\partial y} \cdot \frac{\partial}{\partial v}.$$

Now

$$\begin{aligned} r &= \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \left(\frac{\partial u}{\partial x} \frac{\partial}{\partial u} + \frac{\partial v}{\partial x} \frac{\partial}{\partial v} \right) \left(\frac{\partial u}{\partial x} \frac{\partial z}{\partial u} + \frac{\partial v}{\partial x} \frac{\partial z}{\partial v} \right) \\ &= \frac{\partial^2 z}{\partial u^2} \left(\frac{\partial u}{\partial x} \right)^2 + 2 \frac{\partial^2 z}{\partial u \partial v} \cdot \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial^2 z}{\partial v^2} \left(\frac{\partial v}{\partial x} \right)^2 + \frac{\partial z}{\partial u} \frac{\partial^2 u}{\partial x^2} + \frac{\partial z}{\partial v} \frac{\partial^2 v}{\partial x^2}, \\ s &= \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \left(\frac{\partial u}{\partial x} \frac{\partial}{\partial u} + \frac{\partial v}{\partial x} \frac{\partial}{\partial v} \right) \left(\frac{\partial u}{\partial y} \frac{\partial z}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial z}{\partial v} \right) \\ &= \frac{\partial^2 z}{\partial u^2} \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial^2 z}{\partial u \partial v} \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right) \\ &\quad + \frac{\partial^2 z}{\partial v^2} \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} + \frac{\partial z}{\partial u} \frac{\partial^2 u}{\partial y \partial x} + \frac{\partial z}{\partial v} \frac{\partial^2 v}{\partial y \partial x} \end{aligned}$$

and

$$\begin{aligned} t &= \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \left(\frac{\partial u}{\partial y} \frac{\partial}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial}{\partial v} \right) \left(\frac{\partial u}{\partial y} \frac{\partial z}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial z}{\partial v} \right) \\ &= \frac{\partial^2 z}{\partial u^2} \left(\frac{\partial u}{\partial y} \right)^2 + 2 \frac{\partial^2 z}{\partial u \partial v} \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} + \frac{\partial^2 z}{\partial v^2} \left(\frac{\partial v}{\partial y} \right)^2 + \frac{\partial z}{\partial u} \frac{\partial^2 u}{\partial y^2} + \frac{\partial z}{\partial v} \frac{\partial^2 v}{\partial y^2}. \end{aligned}$$

Substituting these values of p, q, r, s and t in (1), it takes the form

$$A \frac{\partial^2 z}{\partial u^2} + 2B \frac{\partial^2 z}{\partial u \partial v} + C \frac{\partial^2 z}{\partial v^2} + F\left(u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}\right) = 0 \quad \dots(3)$$

where

$$A = R \left(\frac{\partial u}{\partial x} \right)^2 + S \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + T \left(\frac{\partial u}{\partial y} \right)^2, \quad \dots(4)$$

$$B = R \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial x} + \frac{1}{2} S \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right) + T \frac{\partial u}{\partial y} \frac{\partial v}{\partial y}, \quad \dots(5)$$

$$C = R \left(\frac{\partial v}{\partial x} \right)^2 + S \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} + T \left(\frac{\partial v}{\partial y} \right)^2 \quad \dots(6)$$

and the function F is the transformed form of the function f .

Now the problem is to determine u and v so that the equation (3) takes the simplest possible form. The procedure is simple when the discriminant $S^2 - 4RT$ of the quadratic equation

$$R\lambda^2 + S\lambda + T = 0 \quad \dots(7)$$

is everywhere either positive, negative or zero, and we shall discuss these three cases separately.

Case I: $S^2 - 4RT > 0$. If this condition is satisfied then the roots λ_1, λ_2 of the equation (7) are real and distinct. The coefficients of $\frac{\partial^2 z}{\partial u^2}$ and $\frac{\partial^2 z}{\partial v^2}$ in the equation (3) will vanish if we choose u and v such that

$$\frac{\partial u}{\partial x} = \lambda_1 \frac{\partial u}{\partial y}, \quad \dots(8)$$

$$\text{and} \quad \frac{\partial v}{\partial x} = \lambda_2 \frac{\partial v}{\partial y}. \quad \dots(9)$$

The differential equations (8) and (9) will determine the form of u and v as functions of x and y .

For this, from (8), Lagrange's auxiliary equations are

$$\frac{dx}{1} = \frac{dy}{-\lambda_1} = \frac{du}{0}.$$

The last member gives $du = 0$ i.e., $u = \text{constant}$.

The first two members give

$$\frac{dy}{dx} + \lambda_1 = 0. \quad \dots(10)$$

Let $f_1(x, y) = \text{constant}$ be the solution of the equation (10).

Then the solution of the equation (8) can be taken as

$$u = f_1(x, y). \quad \dots(11)$$

Similarly, if $f_2(x, y) = \text{constant}$ is a solution of

$$\frac{dy}{dx} + \lambda_2 = 0,$$

then the solution of the equation (9) can be taken as

$$v = f_2(x, y). \quad \dots(12)$$

Also it can be easily seen that, in general,

$$AC - B^2 = (4RT - S^2) \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right)^2,$$

so that when A and C are zero

$$B^2 = (S^2 - 4RT) \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right)^2. \quad \dots(13)$$

It follows that $B^2 > 0$ since $S^2 - 4RT > 0$ and hence we can divide both sides of the equation by it.

Thus making the substitutions defined by the equations (11) and (12), the equation (1) transforms to the form

$$\frac{\partial^2 z}{\partial u \partial v} = \phi \left(u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v} \right), \quad \dots(14)$$

which is the canonical form in this case.

Case II: $S^2 - 4RT = 0$. In this case the roots of the equation (7) are equal. We define the function u as in Case I and take v to be any function of x and y , which is independent of u . Then, we have, as before, $A = 0$.

Since $S^2 - 4RT = 0$,

hence from (13), $B^2 = 0$ i.e., $B = 0$.

On the other hand, in this case, $C \neq 0$, otherwise v would be a function of u .

Putting $A = 0$, $B = 0$ and dividing by C , we see that in this case the canonical form of the equation (1) is,

$$\frac{\partial^2 z}{\partial v^2} = \phi \left(u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v} \right). \quad \dots(15)$$

Case III: $S^2 - 4RT < 0$. Formally it is the same as Case I except that now the roots of the equation (7) are complex.

Proceeding as in Case I, we find that the equation (1) reduces to the form (14) but that the variables u, v are not real but are in fact complex conjugates.

To find a real canonical form let $u = \alpha + i\beta$, $v = \alpha - i\beta$

so that $\alpha = \frac{1}{2}(u + v)$, $\beta = \frac{1}{2}i(v - u)$.

Now
$$\begin{aligned} \frac{\partial z}{\partial u} &= \frac{\partial z}{\partial \alpha} \frac{\partial \alpha}{\partial u} + \frac{\partial z}{\partial \beta} \frac{\partial \beta}{\partial u} \\ &= \frac{1}{2} \left(\frac{\partial z}{\partial \alpha} - i \frac{\partial z}{\partial \beta} \right). \end{aligned}$$

Similarly
$$\frac{\partial z}{\partial v} = \frac{1}{2} \left(\frac{\partial z}{\partial \alpha} + i \frac{\partial z}{\partial \beta} \right).$$

$$\begin{aligned} \therefore \frac{\partial^2 z}{\partial u \partial v} &= \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial v} \right) \\ &= \frac{1}{4} \left(\frac{\partial}{\partial \alpha} - i \frac{\partial}{\partial \beta} \right) \left(\frac{\partial z}{\partial \alpha} + i \frac{\partial z}{\partial \beta} \right) \\ &= \frac{1}{4} \left(\frac{\partial^2 z}{\partial \alpha^2} + \frac{\partial^2 z}{\partial \beta^2} \right). \end{aligned}$$

Thus, transforming the independent variables u, v to α, β the desired canonical form is

$$\frac{\partial^2 z}{\partial \alpha^2} + \frac{\partial^2 z}{\partial \beta^2} = \Psi \left(\alpha, \beta, z, \frac{\partial z}{\partial \alpha}, \frac{\partial z}{\partial \beta} \right). \quad \dots(16)$$

Second order partial differential equations of the type (1) are classified by their canonical forms ; we say that an equation of this type is :

(i) Hyperbolic if $S^2 - 4RT > 0$,

(ii) Parabolic if $S^2 - 4RT = 0$,

(iii) Elliptic if $S^2 - 4RT < 0$.

(Rohilkhand 2009)

Illustrative Examples

Example 13: Reduce the equation

$$\frac{\partial^2 z}{\partial x^2} = x^2 \frac{\partial^2 z}{\partial y^2},$$

to canonical form.

(Lucknow 2007)

Solution: The given equation can be written as

$$r - x^2 t = 0. \quad \dots(1)$$

Comparing the equation (1) with

$$\begin{aligned} Rr + Ss + Tt + f(x, y, z, p, q) &= 0, \text{ we have} \\ R &= 1, S = 0, T = -x^2. \end{aligned}$$

The quadratic equation $R\lambda^2 + S\lambda + T = 0$ therefore becomes

$$\lambda^2 - x^2 = 0 \Rightarrow \lambda = x, -x \text{ (real and distinct roots).}$$

The equations $\frac{dy}{dx} + \lambda_1 = 0$ and $\frac{dy}{dx} + \lambda_2 = 0$ become

$$\frac{dy}{dx} + x = 0 \text{ and } \frac{dy}{dx} - x = 0.$$

These on integration give

$$y + \frac{1}{2}x^2 = \text{constant and } y - \frac{1}{2}x^2 = \text{constant},$$

so that to change the independent variables from x, y to u, v , we take

$$u = y + \frac{1}{2}x^2 \quad \text{and} \quad v = y - \frac{1}{2}x^2.$$

$$\therefore p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = x \frac{\partial z}{\partial u} - x \frac{\partial z}{\partial v} = x \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right),$$

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v},$$

$$\begin{aligned} r = \frac{\partial^2 z}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial x} \left\{ x \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \right\} \\ &= x \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) + 1 \cdot \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \\ &= x \left[\frac{\partial}{\partial u} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \frac{\partial u}{\partial x} + \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \cdot \frac{\partial v}{\partial x} \right] + \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \\ &= x^2 \left(\frac{\partial^2 z}{\partial u^2} - 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \right) + \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \end{aligned}$$

and

$$\begin{aligned} t = \frac{\partial^2 z}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \\ &= \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}. \end{aligned}$$

Substituting these values in (1), it reduces to

$$\frac{\partial^2 z}{\partial u \partial v} = \frac{1}{4x^2} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right)$$

or

$$\frac{\partial^2 z}{\partial u \partial v} = \frac{1}{4(u-v)} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right),$$

which is the required canonical form of the given equation.

Example 14: Reduce the equation

$$\frac{\partial^2 z}{\partial x^2} + 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 0$$

to canonical form and hence solve it.

(Lucknow 2010)

Solution: The given equation can be written as

$$r + 2s + t = 0. \quad \dots(1)$$

Comparing the equation (1) with

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0, \text{ we have}$$

$$R = 1, S = 2, T = 1.$$

The quadratic equation $R\lambda^2 + S\lambda + T = 0$ is therefore given by

$$\lambda^2 + 2\lambda + 1 = 0, \text{ or } (\lambda + 1)^2 = 0.$$

$\therefore \lambda = -1, -1.$ (equal roots).

The equation $\frac{dy}{dx} + \lambda = 0$ becomes $\frac{dy}{dx} - 1 = 0$,

which on integration gives $x - y = \text{constant}$.

To change the independent variables x, y to u, v we take

$$u = x - y.$$

We have to take v as some function of x and y independent of u .

Let $v = x + y$.

Then

$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v},$$

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = -\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v},$$

$$\begin{aligned} r = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) &= \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \\ &= \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}, \end{aligned}$$

$$s = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) = -\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2}$$

and

$$\begin{aligned} t = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) &= \left(-\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \\ &= \frac{\partial^2 z}{\partial u^2} - 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}. \end{aligned}$$

Substituting these values in (1), it reduces to

$$\frac{\partial^2 z}{\partial v^2} = 0$$

which is the required canonical form.

Integrating it w.r.t. v , we get

$$\frac{\partial z}{\partial v} = \phi_1(u).$$

Again integrating w.r.t. v , we get

$$z = v \phi_1(u) + \phi_2(u),$$

where ϕ_1 and ϕ_2 are arbitrary functions of u .

Hence the solution is

$$z = (x + y) \phi_1(x - y) + \phi_2(x - y).$$

Example 15: Reduce the equation

$$\frac{\partial^2 z}{\partial x^2} + x^2 \frac{\partial^2 z}{\partial y^2} = 0,$$

to canonical form.

(Lucknow 2011; Meerut 11)

Solution: The given equation can be written as

$$r + x^2 t = 0. \quad \dots(1)$$

Comparing the equation (1) with

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0,$$

we have

$$R = 1, S = 0, T = x^2.$$

The quadratic equation $R\lambda^2 + S\lambda + T = 0$ is therefore given by

$$\lambda^2 + x^2 = 0 \Rightarrow \lambda = ix, -ix. \text{ (Complex roots)}$$

The equations $\frac{dy}{dx} + \lambda_1 = 0$ and $\frac{dy}{dx} + \lambda_2 = 0$ become

$$\frac{dy}{dx} + ix = 0 \text{ and } \frac{dy}{dx} - ix = 0.$$

These on integration give

$$y + \frac{1}{2} ix^2 = \text{constant and } y - \frac{1}{2} ix^2 = \text{constant},$$

so that to change the independent variables from x, y to u, v , we take

$$u = y + \frac{1}{2} ix^2 = \alpha + i\beta \text{ (say)}$$

and

$$v = y - \frac{1}{2} ix^2 = \alpha - i\beta.$$

Then

$$\alpha = y, \quad \beta = \frac{1}{2} x^2.$$

Now we shall transform the independent variables x and y to α and β . We have

$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial x} + \frac{\partial z}{\partial \beta} \cdot \frac{\partial \beta}{\partial x} = x \frac{\partial z}{\partial \beta},$$

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial \alpha} \cdot \frac{\partial \alpha}{\partial y} + \frac{\partial z}{\partial \beta} \cdot \frac{\partial \beta}{\partial y} = \frac{\partial z}{\partial \alpha},$$

$$r = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial x} \left(x \frac{\partial z}{\partial \beta} \right)$$

$$= 1 \cdot \frac{\partial z}{\partial \beta} + x \cdot \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial \beta} \right)$$

$$= \frac{\partial z}{\partial \beta} + x \left[\frac{\partial}{\partial \alpha} \left(\frac{\partial z}{\partial \beta} \right) \frac{\partial \alpha}{\partial x} + \frac{\partial}{\partial \beta} \left(\frac{\partial z}{\partial \beta} \right) \cdot \frac{\partial \beta}{\partial x} \right]$$

$$= \frac{\partial z}{\partial \beta} + x^2 \frac{\partial^2 z}{\partial \beta^2}$$

and
$$t = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial \alpha} \right) = \frac{\partial^2 z}{\partial \alpha^2}.$$

Substituting these values in (1), it reduces to

$$\frac{\partial^2 z}{\partial \alpha^2} + \frac{\partial^2 z}{\partial \beta^2} = -\frac{1}{2\alpha} \frac{\partial z}{\partial \alpha},$$

which is the required canonical form of (1).

Example 16: Reduce the equation

$$(n-1)^2 \frac{\partial^2 z}{\partial x^2} - y^{2n} \frac{\partial^2 z}{\partial y^2} = ny^{2n-1} \frac{\partial z}{\partial y},$$

to canonical form and find its general solution.

(Lucknow 2009)

Solution: The given equation can be written as

$$(n-1)^2 r - y^{2n} t - ny^{2n-1} q = 0. \quad \dots(1)$$

Comparing the equation (1) with

$$Rr + Ss + Tt + f(x, y, z, p, q) = 0, \text{ we have}$$

$$R = (n-1)^2, \quad S = 0, \quad T = -y^{2n}.$$

The quadratic equation $R\lambda^2 + S\lambda + T = 0$ is therefore given by

$$(n-1)^2 \lambda^2 - y^{2n} = 0 \quad \text{or} \quad \lambda^2 = \frac{1}{(n-1)^2} y^{2n}.$$

$$\therefore \lambda = \frac{1}{(n-1)} y^n, -\frac{1}{(n-1)} y^n. \quad (\text{Real and distinct roots}).$$

The equations $\frac{dy}{dx} + \lambda_1 = 0$ and $\frac{dy}{dx} + \lambda_2 = 0$ become

$$\frac{dy}{dx} + \frac{1}{(n-1)} y^n = 0$$

and
$$\frac{dy}{dx} - \frac{1}{(n-1)} y^n = 0$$

or
$$(n-1) y^{-n} dy + dx = 0$$

and
$$(n-1) y^{-n} dy - dx = 0.$$

These on integration give

$$x - y^{-n+1} = \text{constant}$$

and
$$x + y^{-n+1} = \text{constant},$$

so that to change the independent variables from x, y to u, v , we take

$$u = x - y^{-n+1} \quad \text{and} \quad v = x + y^{-n+1}.$$

$\therefore$

$$\begin{aligned}
 p &= \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}, \\
 q &= \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} = (n-1) y^{-n} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right), \\
 r &= \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \\
 &= \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}
 \end{aligned}$$

and

$$\begin{aligned}
 t &= \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial y} \left[(n-1) y^{-n} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \right] \\
 &= -n(n-1) y^{-n-1} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) + (n-1) y^{-n} \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \\
 &= -n(n-1) y^{-n-1} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \\
 &\quad + (n-1) y^{-n} \left[\frac{\partial}{\partial u} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \frac{\partial u}{\partial y} + \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \frac{\partial v}{\partial y} \right] \\
 &= -n(n-1) y^{-n-1} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) + (n-1)^2 y^{-2n} \left[\frac{\partial^2 z}{\partial u^2} - 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \right].
 \end{aligned}$$

Substituting these values in (1), it reduces to

$$\frac{\partial^2 z}{\partial u \partial v} = 0, \text{ which is the required canonical form.}$$

 Integrating it, w.r.t. v , we get

$$\frac{\partial z}{\partial u} = \phi_1(u), \text{ where } \phi_1 \text{ is an arbitrary function.}$$

 Again integrating w.r.t. u , we get

$$z = \psi_1(u) + \psi_2(v),$$

 where ψ_1 and ψ_2 are arbitrary functions.

Hence the required general solution is

$$z = \psi_1(x - y^{-n+1}) + \psi_2(x + y^{-n+1}).$$

Comprehensive Exercise 3

1. Reduce the equation

$$y^2 \frac{\partial^2 z}{\partial x^2} - 2xy \frac{\partial^2 z}{\partial x \partial y} + x^2 \frac{\partial^2 z}{\partial y^2} = \frac{y^2}{x} \frac{\partial z}{\partial x} + \frac{x^2}{y} \frac{\partial z}{\partial y}$$

to canonical form and hence solve it.

(Meerut 2006, 07)

2. Reduce the equation

$$xyr - (x^2 - y^2)s - xyt + py - qx = 2(x^2 - y^2)$$

to canonical form and hence solve it.

3. Reduce the equation

$$x^2(y-1)r - x(y^2-1)s + y(y-1)t + xyp - q = 0$$

to canonical form and hence solve it.

4. Reduce the equation

$$x(xy-1)r - (x^2y^2-1)s + y(xy-1)t + (x-1)p + (y-1)q = 0$$

to canonical form and hence solve it.

5. Reduce the equation

$$x^2r - 2xy s + y^2t - xp + 3yq = 8y/x$$

to canonical form and hence solve it.

Answers 3

- $\frac{\partial^2 z}{\partial v^2} = 0$; $z = (x^2 - y^2)\phi_1(x^2 + y^2) + \phi_2(x^2 + y^2)$
- $\frac{\partial^2 z}{\partial u \partial v} = \frac{v^2 - 1}{(v^2 + 1)^2}$; $z = -xy + \psi_1(x^2 + y^2) + \psi_2(y/x)$
- $\frac{\partial^2 z}{\partial u \partial v} = 0$; $z = \phi_1(xy) + \phi_2(xe^y)$
- $\frac{\partial^2 z}{\partial u \partial v} = 0$; $z = \phi_1(xe^y) + \phi_2(ye^x)$
- $v \frac{\partial^2 z}{\partial v^2} + 2 \frac{\partial z}{\partial v} = 2$; $z = \frac{y}{x} + x^2 \psi_1(xy) + \phi_2(xy)$, where $\psi_1(xy) = -\frac{1}{xy} \phi_1(xy)$.

Objective Type Questions

Fill In The Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- The solution of the equation $s = 0$ is $z = \dots\dots\dots$
- The solution of the equation $r = 6x$ is $z = \dots\dots\dots$
- The solution of the equation $ar = xy$ is $az = \dots\dots\dots$

4. The equation $A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + f \left(x, y, u, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right) = 0$
is elliptic, hyperbolic or parabolic according as $B^2 - 4AC$

Answers

Fill in the Blank(s)

- | | |
|---|-----------------------------|
| 1. $\phi(y) + \psi(x)$ | 2. $x^3 + x f(y) + \phi(y)$ |
| 3. $\frac{1}{6} x^3 y + x f(y) + \phi(y)$ | 4. $<, >, = 0$ |



Chapter

11



Monge's Method

11.1 Introduction

It is only in special cases that a partial differential equation

$$F(x, y, z, p, q, r, s, t) = 0$$

of the second order can be integrated. The most important method of solution, due to Monge, is applicable to a wide class of such equations but by no means to all. Monge's method depends on establishing one or two intermediate integrals (first integrals) of the form

$$u = f(v)$$

where u and v are functions of x, y, z, p, q and f is some arbitrary function.

11.2 Monge's Method of Integrating

$$Rr + Ss + Tt = V;$$

where r, s, t have their usual meanings and R, S, T, V are functions of x, y, z, p and q .

(Avadh 2006, 12; Rohilkhand 07; Lucknow 09, 11)

The given equation is $Rr + Ss + Tt = V$.

...(1)

We have
$$dp = \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy = r dx + s dy$$

and
$$dq = \frac{\partial q}{\partial x} dx + \frac{\partial q}{\partial y} dy = s dx + t dy.$$

$\therefore r = \frac{dp - s dy}{dx}$ and $t = \frac{dq - s dx}{dy}.$

Substituting these values of r and t in (1), we get

$$R \left(\frac{dp - s dy}{dx} \right) + Ss + T \left(\frac{dq - s dx}{dy} \right) = V$$

or
$$R dy (dp - s dy) + Ss dx dy + T dx (dq - s dx) = V dx dy$$

or
$$(R dp dy + T dq dx - V dx dy) - s (R dy^2 - S dx dy + T dx^2) = 0.$$

If any relation between x, y, z, p, q makes each of the bracketed expressions vanish, this relation will satisfy the differential equation (1).

From
$$\begin{cases} R dy^2 - S dx dy + T dx^2 = 0 \\ R dp dy + T dq dx - V dx dy = 0 \end{cases} \quad \dots(2)$$

and
$$dz = p dx + q dy,$$

it may be possible to obtain either one or two relations between x, y, z, p, q called *intermediate integrals*, and therefrom to deduce the general solution of (1).

The equations (2) are called **Monge's Subsidiary Equations**.

The method of solution is explained in the following procedure:

Resolve the equation

$$R dy^2 - S dx dy + T dx^2 = 0 \quad \dots(3)$$

into the two equations

$$dy - m_1 dx = 0, \quad dy - m_2 dx = 0. \quad \dots(4)$$

Now from the first of the equations (4) and from the equation

$$R dp dy + T dq dx - V dx dy = 0 \quad \dots(5)$$

combining if necessary with $dz = p dx + q dy$, find two integrals $u_1 = a, v_1 = b$. Then

$$u_1 = f_1(v_1)$$

is an intermediate integral, f_1 being an arbitrary function.

In the same way, taking the second of the equations (4), find another pair of integrals, $u_2 = a, v_2 = b$. Then $u_2 = f_2(v_2)$,

where f_2 is an arbitrary function, is another intermediate integral.

To obtain the final integral, either of these intermediate integrals may be integrated and this must be done when $m_1 = m_2$. If m_1 and m_2 are unequal then solve the two intermediate integrals for p and q and substitute the values of p and q thus found in $dz = p dx + q dy$.

Integrating it obtain the complete integral of (1).

Illustrative Examples

Example 1: Solve $r = a^2 t$(1)

(Avadh 2007, 10, 11, 13; Lucknow 07, 08, 10; Rohilkhand 08; Purvanchal 10, 14)

Solution: We have

$$dp = \frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy = r dx + s dy$$

and
$$dq = \frac{\partial q}{\partial x} dx + \frac{\partial q}{\partial y} dy = s dx + t dy.$$

$\therefore r = \frac{dp - s dy}{dx} \text{ and } t = \frac{dq - s dx}{dy}.$

Putting these values of r and t in (1), we get

$$\frac{dp - s dy}{dx} = a^2 \frac{dq - s dx}{dy}$$

or
$$(dp dy - a^2 dq dx) - s (dy^2 - a^2 dx^2) = 0.$$

Monge's subsidiary equations are

$$dp dy - a^2 dq dx = 0 \quad \dots(2)$$

and
$$dy^2 - a^2 dx^2 = 0. \quad \dots(3)$$

The equation (3) resolves into the two equations:

$$dy - a dx = 0 \quad \dots(4)$$

and
$$dy + a dx = 0. \quad \dots(5)$$

From (2) and (4), we get

$$dp (a dx) - a^2 dq dx = 0 \quad \text{or} \quad dp - a dq = 0. \quad \dots(6)$$

From (4) and (6), we get $y - ax = a_1$ and $p - aq = b_1$.

$$\therefore p - aq = f_1 (y - ax) \quad \dots(7)$$

is an intermediate integral, f_1 being an arbitrary function.

Similarly from (2) and (5) we find another intermediate integral

$$p + aq = f_2 (y + ax), \quad \dots(8)$$

where f_2 is an arbitrary function.

Solving (7) and (8) for p and q , we have

$$p = \frac{1}{2} \{ f_1 (y - ax) + f_2 (y + ax) \}$$

and
$$q = \frac{1}{2a} \{ f_2 (y + ax) - f_1 (y - ax) \}.$$

Substituting these values of p and q in $dz = p dx + q dy$, we get

$$dz = \frac{1}{2} [f_1 (y - ax) + f_2 (y + ax)] dx + \frac{1}{2a} [f_2 (y + ax) - f_1 (y - ax)] dy$$

$$= \frac{1}{2a} f_2 (y + ax)(dy + a dx) - \frac{1}{2a} f_1 (y - ax)(dy - a dx).$$

Integrating,
$$z = \frac{1}{2a} \phi_2 (y + ax) - \frac{1}{2a} \phi_1 (y - ax)$$

or
$$z = F_2 (y + ax) + F_1 (y - ax),$$

which is the complete integral of the equation (1).

Example 2: Solve $r + (a + b)s + abt = xy$(1)

(Avadh 2006)

Solution: We have

$$dp = r dx + s dy \text{ and } dq = s dx + t dy.$$

$$\therefore r = \frac{dp - s dy}{dx} \text{ and } t = \frac{dq - s dx}{dy}.$$

Putting these values of r and t in (1), we get

$$\frac{dp - s dy}{dx} + (a + b)s + ab \frac{dq - s dx}{dy} = xy$$

or
$$\{dp dy + ab dq dx - xy dx dy\} - s \{dy^2 - (a + b) dx dy + ab dx^2\} = 0.$$

Monge's subsidiary equations are

$$dp dy + ab dq dx - xy dx dy = 0 \quad \dots(2)$$

and
$$dy^2 - (a + b) dx dy + ab dx^2 = 0. \quad \dots(3)$$

The equation (3) resolves into the two equations:

$$dy - b dx = 0 \quad \dots(4)$$

and
$$dy - a dx = 0. \quad \dots(5)$$

From (4), we get $y - bx = A$.

From (2) and (4), we have

$$dp + a dq - xy dx = 0 \text{ or } dp + a dq - x (A + bx) dx = 0.$$

Integrating,
$$p + aq - b \cdot \frac{1}{3} x^3 - A \cdot \frac{1}{2} x^2 = B.$$

$$\therefore p + aq - \frac{1}{3} b \cdot x^3 - \frac{1}{2} (y - bx) x^2 = f_1 (y - bx)$$

or
$$p + aq + \frac{1}{6} bx^3 - \frac{1}{2} x^2 y = f_1 (y - bx), \quad \dots(6)$$

is an intermediate integral, f_1 being an arbitrary function. Similarly from (2) and (5) we find another intermediate integral

$$p + bq + \frac{1}{6} ax^3 - \frac{1}{2} x^2 y = f_2 (y - ax), \quad \dots(7)$$

where f_2 is an arbitrary function.

Solving (6) and (7) for p and q , we have

$$p = \frac{1}{2} x^2 y - \frac{1}{6} (a + b) x^3 + \frac{1}{a - b} \{a f_2 (y - ax) - b f_1 (y - bx)\}$$

and
$$q = \frac{1}{6} x^3 + \frac{1}{a-b} \{f_1(y-bx) - f_2(y-ax)\}.$$

Substituting these values of p and q in $dz = p dx + q dy$, we get

$$\begin{aligned} dz = \frac{1}{2} x^2 y dx + \frac{1}{6} x^3 dy - \frac{1}{6} (a+b) x^3 dx \\ + \frac{1}{a-b} \{a f_2(y-ax) - b f_1(y-bx)\} dx \\ + \frac{1}{a-b} \{f_1(y-bx) - f_2(y-ax)\} dy \end{aligned}$$

or
$$\begin{aligned} dz = \frac{1}{6} (3x^2 y dx + x^3 dy) - \frac{1}{6} (a+b) x^3 dx \\ + \frac{1}{(b-a)} [f_2(y-ax)(dy - a dx) - f_1(y-bx)(dy - b dx)]. \end{aligned}$$

Integrating,

$$z = \frac{1}{6} x^3 y - \frac{1}{24} (a+b) x^4 + \frac{1}{b-a} [\phi_2(y-ax) - \phi_1(y-bx)]$$

or
$$z = \frac{1}{6} x^3 y - \frac{1}{24} (a+b) x^4 + \psi_2(y-ax) + \psi_1(y-bx),$$

which is the complete integral of the equation (1).

Example 3: Solve $q^2 r - 2 pqs + p^2 t = 0$(1)

Show that the integral represents a surface generated by straight lines which are parallel to a fixed plane. (Agra 2003; Kanpur 14)

Solution: We have $dp = r dx + s dy$ and $dq = s dx + t dy$.

$\therefore r = \frac{dp - s dy}{dx}$ and $t = \frac{dq - s dx}{dy}$.

Putting these values of r and t in (1), we get

$$q^2 \frac{dp - s dy}{dx} - 2 pqs + p^2 \frac{dq - s dx}{dy} = 0$$

or
$$(q^2 dp dy + p^2 dq dx) - s (q^2 dy^2 + 2 pq dx dy + p^2 dx^2) = 0.$$

Monge's subsidiary equations are

$$q^2 dp dy + p^2 dq dx = 0 \quad \dots(2)$$

and
$$q^2 dy^2 + 2 pq dx dy + p^2 dx^2 = 0 \quad \dots(3)$$

The equation (3) gives $(q dy + p dx)^2 = 0$

or
$$q dy + p dx = 0. \quad \dots(4)$$

$\therefore dz = p dx + q dy \Rightarrow dz = 0 \Rightarrow z = a. \quad \dots(5)$

From (2) and (4), we get

$$q dp - p dq = 0 \quad \text{or} \quad \frac{dp}{p} = \frac{dq}{q}$$

$\Rightarrow p = qb. \quad \dots(6)$

From (5) and (6), we get $p - q f(z) = 0$, ... (7)

as an intermediate integral, f being an arbitrary function. For the differential equation (7), Lagrange's auxiliary equations are

$$\frac{dx}{1} = \frac{dy}{-f(z)} = \frac{dz}{0}.$$

The last two members give $dz = 0$ i.e., $z = A$.

Taking the first two members, we get

$$dy + f(z) dx = 0 \quad \text{or} \quad dy + f(A) dx = 0.$$

$$\therefore y + x f(A) = B$$

$$\text{or} \quad y + x f(z) = \phi(z), \quad \dots (8)$$

which is the complete integral of the equation (1).

The integral of the given differential equation is the surface (8) which is the locus of the straight lines given by the intersection of planes

$$y + x f(c) = \phi(c) \quad \text{and} \quad z = c.$$

These lines are all parallel to the planes $z = 0$ as they lie on the plane $z = c$ for varying values of c .

Example 4: Solve $t - r \sec^4 y = 2q \tan y$ (1)

(Rohilkhand 2002, 04, 09; Lucknow 07; Purvanchal 10; Avadh 10, 14; Meerut 13B)

Solution: We have

$$dp = r dx + s dy \quad \text{and} \quad dq = s dx + t dy.$$

$$\therefore r = \frac{dp - s dy}{dx} \quad \text{and} \quad t = \frac{dq - s dx}{dy}.$$

Putting these values of r and t in (1), we get

$$\frac{dq - s dx}{dy} - \frac{dp - s dy}{dx} \sec^4 y = 2q \tan y$$

$$\text{or} \quad \{dq dx - dp dy \sec^4 y - 2q \tan y dx dy\} - s \{dx^2 - dy^2 \sec^4 y\} = 0.$$

Monge's subsidiary equations are

$$dq dx - dp dy \sec^4 y - 2q \tan y dx dy = 0. \quad \dots (2)$$

$$\text{and} \quad dx^2 - dy^2 \sec^4 y = 0. \quad \dots (3)$$

The equation (3) resolves into the two equations

$$dx - dy \sec^2 y = 0 \quad \dots (4)$$

$$\text{and} \quad dx + dy \sec^2 y = 0. \quad \dots (5)$$

Integrating (4), we get $x - \tan y = a$.

From (2) and (4), we get

$$dq - dp \sec^2 y - 2q \tan y dy = 0$$

$$\text{or} \quad dp - (\cos^2 y dq - 2q \sin y \cos y dy) = 0.$$

Integrating, we have $p - q \cos^2 y = b$.

$$\therefore p - q \cos^2 y = f_1(x - \tan y), \quad \dots(6)$$

which is an intermediate integral, f_1 being an arbitrary function.

Similarly from (2) and (5) we find another intermediate integral

$$p + q \cos^2 y = f_2(x + \tan y), \quad \dots(7)$$

where f_2 is an arbitrary function.

Solving (6) and (7) for p and q , we have

$$p = \frac{1}{2} \{ f_1(x - \tan y) + f_2(x + \tan y) \}$$

$$\text{and } q = \frac{1}{2} \sec^2 y \{ f_2(x + \tan y) - f_1(x - \tan y) \}.$$

Substituting these values of p and q in $dz = p dx + q dy$, we get

$$\begin{aligned} dz &= \frac{1}{2} \{ f_1(x - \tan y) + f_2(x + \tan y) \} dx \\ &\quad + \frac{1}{2} \sec^2 y \{ f_2(x + \tan y) - f_1(x - \tan y) \} dy \\ &= \frac{1}{2} f_1(x - \tan y) (dx - \sec^2 y dy) + \frac{1}{2} f_2(x + \tan y) (dx + \sec^2 y dy). \end{aligned}$$

Integrating, we get $z = \phi_1(x - \tan y) + \phi_2(x + \tan y)$,

which is the complete integral of the equation (1).

Example 5: Solve $r - t \cos^2 x + p \tan x = 0$(1)

(Lucknow 2006)

Solution: We have $dp = r dx + s dy$ and $dq = s dx + t dy$.

$$\therefore r = \frac{dp - s dy}{dx} \text{ and } t = \frac{dq - s dx}{dy}.$$

Putting these values of r and t in (1), we get

$$\frac{dp - s dy}{dx} - \frac{dq - s dx}{dy} \cos^2 x + p \tan x = 0$$

$$\text{or } (dp dy - dq dx \cos^2 x + p \tan x dx dy) - s (dy^2 - \cos^2 x dx^2) = 0.$$

Monge's subsidiary equations are

$$dp dy - dq dx \cos^2 x + p \tan x dx dy = 0 \quad \dots(2)$$

$$\text{and } dy^2 - \cos^2 x dx^2 = 0. \quad \dots(3)$$

The equation (3) resolves into the two equations

$$dy - \cos x dx = 0 \quad \dots(4)$$

$$\text{and } dy + \cos x dx = 0. \quad \dots(5)$$

Integrating (4), we get $y - \sin x = a$.

Also from (2) and (4), we get

$$dp - dq \cos x + p \tan x dx = 0$$

or $\sec x \, dp + p \sec x \tan x \, dx - dq = 0$.

Integrating, we get $p \sec x - q = b$.

$$\therefore p \sec x - q = f_1 (y - \sin x) \quad \dots(6)$$

is an intermediate integral, f_1 being an arbitrary function.

Similarly from (2) and (5) we find another intermediate integral

$$p \sec x + q = f_2 (y + \sin x), \quad \dots(7)$$

where f_2 is an arbitrary function.

Solving (6) and (7) for p and q , we have

$$p = \frac{1}{2} \cos x \{ f_1 (y - \sin x) + f_2 (y + \sin x) \}$$

$$\text{and } q = \frac{1}{2} \{ f_2 (y + \sin x) - f_1 (y - \sin x) \}.$$

Substituting these values of p and q in $dz = p \, dx + q \, dy$, we get

$$\begin{aligned} dz &= \frac{1}{2} \cos x \{ f_1 (y - \sin x) + f_2 (y + \sin x) \} \, dx \\ &\quad + \frac{1}{2} \{ f_2 (y + \sin x) - f_1 (y - \sin x) \} \, dy \end{aligned}$$

$$= -\frac{1}{2} f_1 (y - \sin x) (dy - \cos x \, dx) + \frac{1}{2} f_2 (y + \sin x) (dy + \cos x \, dx)$$

$$\text{or } dz = -\frac{1}{2} f_1 (y - \sin x) d(y - \sin x) + \frac{1}{2} f_2 (y + \sin x) d(y + \sin x).$$

Integrating, we get

$$z = \phi_1 (y - \sin x) + \phi_2 (y + \sin x),$$

which is the complete integral of the equation (1).

Example 6: Solve $pt - qs = q^3$ (1)

(Rohilkhand 2003, 14; Lucknow 07; Purvanchal 09, 11; Meerut 13)

Solution: We have $dq = s \, dx + t \, dy$.

$$\therefore t = \frac{dq - s \, dx}{dy}.$$

Putting this value of t in (1), we get

$$p \frac{dq - s \, dx}{dy} - qs = q^3 \quad \text{or} \quad (p \, dq - q^3 \, dy) - s (p \, dx + q \, dy) = 0.$$

Monge's subsidiary equations are

$$p \, dq - q^3 \, dy = 0, \quad \dots(2)$$

$$\text{and } p \, dx + q \, dy = 0. \quad \dots(3)$$

From (3), we have $dz = 0$. [$\because dz = p \, dx + q \, dy$]

$$\therefore z = a.$$

Again combining (3) with (2), we get

$$dq + q^2 \, dx = 0 \quad \text{or} \quad \frac{dq}{q^2} + dx = 0.$$

Integrating, we get $-\frac{1}{q} + x = b$.

$$\therefore -\frac{1}{q} + x = f(z), \text{ or } -\frac{1}{(\partial z / \partial y)} + x = f(z), \quad \dots(4)$$

which is an intermediate integral, f being an arbitrary function.

Now we shall integrate the equation (4) regarding x as constant.

Regarding x as constant, the equation (4) can be written as $\frac{dy}{dz} - x = -f(z)$.

Integrating w.r.t. z , treating x as constant, we get

$$y - xz = -\int f(z) dz + c = f_1(z) + f_2(x),$$

taking the constant of integration c as a function of x

$$\text{or } y = xz + f_1(z) + f_2(x),$$

which is the complete integral of the equation (1).

Example 7: Solve $y^2 r - 2ys + t = p + 6y$(1)

(Rohilkhand 2006)

Solution: We have $dp = r dx + s dy$ and $dq = s dx + t dy$.

$$\therefore r = \frac{dp - s dy}{dx} \text{ and } t = \frac{dq - s dx}{dy}.$$

Putting these values of r and t in (1), we get

$$y^2 \frac{dp - s dy}{dx} - 2ys + \frac{dq - s dx}{dy} = p + 6y$$

$$\text{or } \{y^2 dp dy + dq dx - (p + 6y) dx dy\} - s \{y^2 dy^2 + 2y dx dy + dx^2\} = 0.$$

Monge's subsidiary equations are

$$y^2 dp dy + dq dx - (p + 6y) dx dy = 0 \quad \dots(2)$$

$$\text{and } y^2 dy^2 + 2y dx dy + dx^2 = 0. \quad \dots(3)$$

$$\text{The equation (3) gives } (y dy + dx)^2 = 0 \text{ or } y dy + dx = 0. \quad \dots(4)$$

$$\text{Integrating, we get } y^2 + 2x = a. \quad \dots(5)$$

$$\text{From (2) and (4), we get } y dp - dq + p dy + 6y dy = 0$$

$$\text{or } (y dp + p dy) - dq + 6y dy = 0.$$

$$\text{Integrating, we get } py - q + 3y^2 = b. \quad \dots(6)$$

$$\text{From (5) and (6), we have } py - q + 3y^2 = f(y^2 + 2x) \quad \dots(7)$$

which is an intermediate integral, f being an arbitrary function.

Lagrange's subsidiary equations for the equation (7) are

$$\frac{dx}{y} = \frac{dy}{-1} = \frac{dz}{-3y^2 + f(y^2 + 2x)}.$$

Taking the first two members, we get $y \, dy + dx = 0$.

$$\therefore y^2 + 2x = A.$$

Again from the last two members, we get

$$dz = \{3y^2 - f(y^2 + 2x)\} \, dy$$

$$\text{or } dz = 3y^2 \, dy - f(A) \, dy.$$

$$\text{Integrating, } z = y^3 - y f(A) + B$$

$$\text{or } z = y^3 - y f(y^2 + 2x) + \phi(y^2 + 2x),$$

which is the complete integral of the equation (I).

Comprehensive Exercise 1

Solve the following equations by Monge's method :

1. $r = t$.
2. $(q + 1)s = (p + 1)t$.
3. $x^2r + 2xys + y^2t = 0$.
4. $x^2r - 2xst + q = 0$.
5. $r - 2s + t = \sin(2x + 3y)$.
6. $(r - t)xy - s(x^2 - y^2) = qx - py$.
7. $rx^2 - 3sxy + 2ty^2 + px + 2qy = x + 2y$.
8. $2xr - (x + 2y)s + yt = \frac{x + 2y}{x - 2y}(2p - q)$.
9. $(r - s)y + (s - t)x + q - p = 0$.
10. $r + ka^2t - 2as = 0$.
11. $x^{-2}r - y^{-2}t = x^{-3}p - y^{-3}q$.

(Purvanchal 2007)

(Kanpur 2014)

Answers 1

1. $z = f_1(x + y) + f_2(y - x)$
2. $z = \phi(x) + \psi(x + y + z)$
3. $z = y f_1(y/x) + f_2(y/x)$
4. $z = f_1(y + \log x) + x f_2(y + \log x)$
5. $z = f(x + y) + x \phi(x + y) - \sin(2x + 3y)$
6. $z = f_1(x^2 + y^2) + f_2(y/x)$

7. $z = x + y + f(x, y) + \phi(x^2, y)$
8. $z = f_1(x + 2, y) + f_2(x, y)$
9. $z = f_1(x + y) + f_2(x^2 - y^2)$
10. $z = f_1\{y + a(1 + t)x\} + f_2\{y + a(1 - t)x\}, t^2 = 1 - k$
11. $z = f_1(x^2 + y^2) + f_2(x^2 - y^2)$

11.3 Monge's Method of Integrating

$$Rr + Ss + Tt + U(rt - s^2) = V, \quad \dots(1)$$

where r, s, t have their usual meanings and R, S, T, U, V are functions of x, y, z, p and q .

We have $dp = r dx + s dy, dq = s dx + t dy$.

$$\therefore r = \frac{dp - s dy}{dx} \quad \text{and} \quad t = \frac{dq - s dx}{dy}.$$

Putting these values of r and t in (1), we get

$$R \cdot \frac{dp - s dy}{dx} + Ss + T \cdot \frac{dq - s dx}{dy} + U \left\{ \frac{(dp - s dy)(dq - s dx)}{dx dy} - s^2 \right\} = V$$

$$\text{or} \quad (R dp dy + T dq dx + U dp dq - V dx dy) - s(R dy^2 - S dx dy + T dx^2 + U dp dx + U dq dy) = 0.$$

$\therefore$ Monge's subsidiary equations are

$$L \equiv R dp dy + T dq dx + U dp dq - V dx dy = 0 \quad \dots(2)$$

$$\text{and} \quad M \equiv R dy^2 - S dx dy + T dx^2 + U dp dx + U dq dy = 0. \quad \dots(3)$$

Here the equation (3) cannot be resolved into two linear equations on account of the presence of the term

$$U dp dx + U dq dy.$$

However we try to resolve $M + \lambda L = 0$ into two linear equations, where λ is some multiplier to be determined.

$$\begin{aligned} \text{Now} \quad M + \lambda L &= R dy^2 + T dx^2 - (S + \lambda V) dx dy + U dp dx \\ &\quad + U dq dy + \lambda R dp dy + \lambda T dq dx + \lambda U dp dq. \quad \dots(4) \end{aligned}$$

Also let

$$M + \lambda L \equiv (\alpha dy + \beta dx + \gamma dp)(\alpha' dy + \beta' dx + \gamma' dq). \quad \dots(5)$$

Equating the coefficients of dy^2, dx^2 and $dp dq$ in (4) and (5), we have

$$R = \alpha\alpha', T = \beta\beta', \lambda U = \gamma\gamma'$$

If we choose $\alpha' = 1, \beta' = \frac{1}{k}, \gamma' = \frac{\lambda}{m}$ then $\alpha = R, \beta = kT, \gamma = mU$.

Now equating the coefficients of the remaining five terms in (4) and (5), we have

$$kT + \frac{R}{k} = -(S + \lambda V) \quad \dots(6)$$

$$\frac{kT \lambda}{m} = \lambda T, \quad U = \frac{mU}{k}, \quad \dots(7)$$

$$\frac{\lambda R}{m} = U, \lambda R = mU. \quad \dots(8)$$

From (7), $m = k$ and from (8), $m = \frac{\lambda R}{U}$.

Putting $k = \lambda R / U$ in (6), we have

$$\frac{\lambda R}{U} T + R \frac{U}{\lambda R} = -(S + \lambda V)$$

$$\text{or} \quad \lambda^2 (RT + UV) + \lambda US + U^2 = 0. \quad \dots(9)$$

Apart from the special case when $S^2 = 4(RT + UV)$, this equation will have two distinct roots λ_1, λ_2 .

For $\lambda = \lambda_1$, $m = k = R \lambda_1 / U$. Then from (5), $M + \lambda L = 0$ gives

$$\left(R dy + \lambda_1 \frac{RT}{U} dx + \lambda_1 R dp \right) \left(dy + \frac{U}{R \lambda_1} dx + \frac{U}{R} dq \right) = 0$$

$$\text{or} \quad (U dy + T \lambda_1 dx + U \lambda_1 dp) (R \lambda_1 dy + U dx + U \lambda_1 dq) = 0 \quad \dots(10)$$

Similarly for $\lambda = \lambda_2$, $M + \lambda L = 0$ gives

$$(U dy + T \lambda_2 dx + U \lambda_2 dp) (R \lambda_2 dy + U dx + U \lambda_2 dq) = 0. \quad \dots(11)$$

Now one factor of (10) is combined with one factor of (11) to give an intermediate integral and similarly other pair gives another intermediate integral. These cannot be obtained if we combine the first factor of (10) with the first factor of (11) or the second factor of (10) with the second factor of (11).

Thus the problem reduces to the solution of the pairs

$$\left. \begin{aligned} U dy + \lambda_1 T dx + \lambda_1 U dp &= 0 \\ \lambda_2 R dy + U dx + \lambda_2 U dq &= 0 \end{aligned} \right\} \quad \dots(12)$$

$$\text{and} \quad \left. \begin{aligned} U dy + \lambda_2 T dx + \lambda_2 U dp &= 0 \\ \lambda_1 R dy + U dx + \lambda_1 U dq &= 0 \end{aligned} \right\} \quad \dots(13)$$

From each of these pairs we shall derive two integrals of the form $u = a, v = b$. Let $u_1 = a_1, v_1 = b_1$ be the integrals obtained from the equations (12) and $u_2 = a_2, v_2 = b_2$ be the integrals obtained from the equations (13). Then the two intermediate integrals are

$$u_1 = f_1(v_1) \text{ and } u_2 = f_2(v_2),$$

which can often be solved to find the values of p and q as functions of x, y and z . Substituting these values of p and q in $dz = p dx + q dy$ and integrating it we obtain the solution of the original equation.

Note 1: If the two values of λ are equal, then it is possible to find only one intermediate integral.

Note 2: If it is not possible to solve the equations

$$u_1 = f_1(v_1) \text{ and } u_2 = f_2(v_2)$$

for p and q , then we may take one of the intermediate integrals say $u_1 = f_1(v_1)$ and one of the integrals from $u_2 = a_2$ and $v_2 = b_2$ to determine p and q .

Note 3: An integral of a more general form can be obtained by taking the arbitrary function occurring in the intermediate integral to be linear.

Let $u_1 = m v_1 + n$, where m and n are some constants.

Then integrating it by Lagrange's method we find the solution of the given equation.

Illustrative Examples

Example 8: Solve $r + 3s + t + (rt - s^2) = 1$(1)

Solution: Comparing the given equation (1) with

$$Rr + Ss + Tt + U(rt - s^2) = V,$$

we have $R = 1, S = 3, T = 1, U = 1, V = 1$.

The λ -quadratic equation is

$$\lambda^2 (UV + RT) + \lambda SU + U^2 = 0$$

or $2\lambda^2 + 3\lambda + 1 = 0$ or $(2\lambda + 1)(\lambda + 1) = 0$.

$\therefore \lambda_1 = -1, \lambda_2 = -\frac{1}{2}$.

One intermediate integral is given by the equations

$$U dy + \lambda_1 T dx + \lambda_1 U dp = 0 \quad \text{and} \quad \lambda_2 R dy + U dx + \lambda_2 U dq = 0$$

or $dy - dx - dp = 0$ and $-\frac{1}{2} dy + dx - \frac{1}{2} dq = 0$

or $dy - dx - dp = 0$ and $dy - 2 dx + dq = 0$.

Integrating, $y - x - p = a_1$ and $y - 2x + q = b_1$.

Hence one intermediate integral is

$$y - x - p = f_1'(y - 2x + q) = f_1'(\alpha), \quad \dots(2)$$

where $\alpha = y - 2x + q$.

Another intermediate integral is given by the equations

$$U dy + \lambda_2 T dx + \lambda_2 U dp = 0 \quad \text{and} \quad \lambda_1 R dy + U dx + \lambda_1 U dq = 0$$

or $dy - \frac{1}{2} dx - \frac{1}{2} dp = 0$ and $-dy + dx - dq = 0$.

Integrating, $2y - x - p = a_2$ and $-y + x - q = b_2$.

Hence another intermediate integral is

$$2y - x - p = f_2'(-y + x - q) = f_2'(\beta), \quad \dots(3)$$

where $\beta = -y + x - q$.

From the above relations, we have

$$x = -(\alpha + \beta) \text{ so that } dx = -(d\alpha + d\beta),$$

$$y = f_2'(\beta) - f_1'(\alpha) \text{ so that } dy = f_2''(\beta) d\beta - f_1''(\alpha) d\alpha$$

and $p = y - x - f_1'(\alpha), q = x - y - \beta$.

Putting these values in $dz = p dx + q dy$, we get

$$\begin{aligned} dz &= \{y - x - f_1'(\alpha)\} dx + \{x - y - \beta\} dy \\ &= -(x - y)(dx - dy) - f_1'(\alpha)\{-d\alpha - d\beta\} - \beta\{f_2''(\beta) d\beta - f_1''(\alpha) d\alpha\} \\ &= -(x - y)(dx - dy) + f_1'(\alpha) d\alpha + \{f_1'(\alpha) d\beta \\ &\quad + \beta f_1''(\alpha) d\alpha\} - \beta f_2''(\beta) d\beta. \end{aligned}$$

Integrating, $z = -\frac{1}{2}(x - y)^2 + f_1(\alpha) + \beta f_1'(\alpha) - \{\beta f_2'(\beta) - f_2(\beta)\}$

$$\begin{aligned} \text{or} \quad z &= -\frac{1}{2}(x-y)^2 + f_1(\alpha) + f_2(\beta) + \beta \{f_1'(\alpha) - f_2'(\beta)\} \\ &= -\frac{1}{2}(x-y)^2 + f_1(\alpha) + f_2(\beta) - \beta y, \end{aligned}$$

which is the required solution of the equation (1).

Example 9: Solve $2s + (rt - s^2) = 1$(1)

(Lucknow 2011; Purvanchal 11)

Solution: Comparing the given equation (1) with

$$Rr + Ss + Tt + U(rt - s^2) = V,$$

we have $R = 0, S = 2, T = 0, U = 1$ and $V = 1$.

The λ -quadratic equation is

$$\lambda^2(UV + RT) + \lambda SU + U^2 = 0$$

$$\text{or} \quad \lambda^2 + 2\lambda + 1 = 0.$$

$$\therefore \lambda_1 = -1, \lambda_2 = -1.$$

In this case we can find only one intermediate integral, which is given by the equations

$$U dy + \lambda_1 T dx + \lambda_1 U dp = 0$$

$$\text{and} \quad \lambda_2 R dy + U dx + \lambda_2 U dq = 0$$

$$\text{or} \quad dy - dp = 0 \quad \text{and} \quad dx - dq = 0. \quad \dots(2)$$

Integrating, $y - p = a$ and $x - q = b$.

$\therefore$ The intermediate integral is

$$y - p = f(x - q). \quad \dots(3)$$

From (2), $p = y - a$ and $q = x - b$.

Putting these values of p and q in $dz = p dx + q dy$,

$$\text{we get} \quad dz = (y - a) dx + (x - b) dy$$

$$\text{or} \quad dz = (y dx + x dy) - a dx - b dy.$$

Integrating, we get

$$z = xy - ax - by + c, \quad \text{which is the complete integral of (1).}$$

Alite:. To find the more general solution, let

$$y - p = m(x - q) + n, \quad \text{where } m \text{ and } n \text{ are constants}$$

$$\text{or} \quad p - mq = y - mx - n.$$

Lagrange's auxiliary equations are

$$\frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{y - mx - n}.$$

From the first two members, we have

$$dy + m dx = 0.$$

$$\therefore y + mx = a.$$

Again from the first and the last members, we have

$$dz = (y - mx - n) dx = (a - 2mx - n) dx.$$

Integrating, $z = ax - mx^2 - nx + b$

$$\text{or} \quad z = (y + mx)x - mx^2 - nx + b$$

$$\text{or} \quad z = xy - nx + \phi(y + mx), \quad \text{which is the required general solution of (1).}$$

Example 10: Solve $z(1+q^2)r - 2pqzs + z(1+p^2)t + z^2(rt-s^2) + (1+p^2+q^2) = 0$.
 $\dots(1)$

Solution: Comparing the equation (1) with

$$Rr + Ss + Tt + U(rt - s^2) = V,$$

we have $R = z(1+q^2), S = -2pqz, T = z(1+p^2), U = z^2$

and $V = -(1+p^2+q^2)$.

The λ -quadratic equation is

$$\lambda^2(UV + RT) + \lambda SU + U^2 = 0$$

or $\{-z^2(1+p^2+q^2) + z^2(1+p^2)(1+q^2)\}\lambda^2 - 2pqz^3\lambda + z^4 = 0$

or $p^2q^2\lambda^2 - 2pqz\lambda + z^2 = 0$ or $(pq\lambda - z)^2 = 0$.

$\therefore \lambda_1 = \lambda_2 = z/(pq)$.

The intermediate integral is given by the equations

$$U dy + \lambda_1 T dx + \lambda_1 U dp = 0 \quad \text{and} \quad \lambda_2 R dy + U dx + \lambda_2 U dq = 0$$

or $pq dy + (1+p^2) dx + z dp = 0$ and $pq dx + (1+q^2) dy + z dq = 0$

or $p(p dx + q dy) + dx + z dp = 0$ and $q(p dx + q dy) + dy + z dq = 0$

or $p dz + z dp + dx = 0$ and $q dz + z dq + dy = 0$.

Integrating, $zp + x = a$ and $zq + y = b$.
 $\dots(2)$

$\therefore$ The intermediate integral is

$$zp + x = f(zq + y). \quad \dots(3)$$

From (2), $p = \frac{a-x}{z}$ and $q = \frac{b-y}{z}$.

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = \frac{a-x}{z} dx + \frac{b-y}{z} dy$$

or $z dz = (a-x) dx + (b-y) dy$.

Integrating, we have $z^2 = -(a-x)^2 - (b-y)^2 + c$, which is the required complete integral of (1).

Aliter. To find the more general solution, let

$$z(p+x) = m(zq+y) + n, \text{ where } m \text{ and } n \text{ are constants}$$

or $z(p-mzq) = -x+my+n$.

Lagrange's auxiliary equations are

$$\frac{dx}{z} = \frac{dy}{-mz} = \frac{dz}{-x+my+n}.$$

Taking the first two members, we get $y+mx=a$.

Also each fraction = $\frac{x dx + y dy + z dz}{nz}$

$\therefore \frac{dx}{z} = \frac{x dx + y dy + z dz}{nz} \Rightarrow x dx + y dy + z dz - n dx = 0$.

Integrating, $x^2 + y^2 + z^2 - 2nx = b$

or $x^2 + y^2 + z^2 - 2nx = \phi(y+mx),$

which is the required general solution of (1).

Example 11: Solve $(rt - s^2) - s(\sin x + \sin y) = \sin x \sin y$ (1)

Solution: Comparing the equation (1) with

$$Rr + Ss + Tt + U(rt - s^2) = V, \text{ we have}$$

$$R = 0, S = -(\sin x + \sin y), T = 0, U = 1, V = \sin x \sin y.$$

The λ -quadratic equation is

$$\lambda^2 (UV + RT) + \lambda SU + U^2 = 0$$

$$\text{or} \quad \lambda^2 \sin x \sin y - \lambda (\sin x + \sin y) + 1 = 0$$

$$\text{or} \quad (\lambda \sin x - 1)(\lambda \sin y - 1) = 0.$$

$$\therefore \quad \lambda_1 = \operatorname{cosec} y \quad \text{and} \quad \lambda_2 = \operatorname{cosec} x.$$

One intermediate integral is given by the equations

$$U dy + \lambda_1 T dx + \lambda_1 U dp = 0$$

$$\text{and} \quad \lambda_2 R dy + U dx + \lambda_2 U dq = 0$$

$$\text{or} \quad dy + \operatorname{cosec} y dp = 0 \quad \text{and} \quad dx + \operatorname{cosec} x dq = 0$$

$$\text{or} \quad \sin y dy + dp = 0 \quad \text{and} \quad \sin x dx + dq = 0$$

Integrating, $p - \cos y = a$, $q - \cos x = b$.

$\therefore$ Intermediate integral is $p - \cos y = f(q - \cos x)$.

Let $p - \cos y = m(q - \cos x) + n$, where m and n are constants

$$\text{or} \quad p - mq = \cos y - m \cos x + n.$$

Lagrange's auxiliary equations are

$$\frac{dx}{1} = \frac{dy}{-m} = \frac{dz}{\cos y - m \cos x + n}.$$

The first two members give $y + mx = A$.

Again taking the first and the last members, we get

$$\begin{aligned} dz &= (\cos y - m \cos x + n) dx \\ &= \{\cos(A - mx) - m \cos x + n\} dx. \end{aligned}$$

$$\text{Integrating,} \quad z = -\frac{1}{m} \sin(A - mx) - m \sin x + nx + B$$

$$\text{or} \quad mz + \sin y + m^2 \sin x - mn x = m\phi(y + mx).$$

Example 12: Solve $qr + (p + x)s + yt + y(rt - s^2) = -q$ (1)

Solution: Comparing the equation (1) with

$$Rr + Ss + Tt + U(rt - s^2) = V,$$

$$\text{we have} \quad R = q, S = p + x, T = y, U = y, V = -q.$$

The λ -quadratic equation is

$$\lambda^2 (UV + RT) + \lambda SU + U^2 = 0$$

$$\text{or} \quad \lambda^2 (-yq + qy) + \lambda y(p + x) + y^2 = 0.$$

$$\therefore \quad \lambda_1 = -y / (p + x), \lambda_2 = \infty.$$

One intermediate integral is given by the equations

$$U dy + \lambda_1 T dx + \lambda_1 U dp = 0$$

$$\text{and} \quad \lambda_2 R dy + U dx + \lambda_2 U dq = 0$$

or $y \, dy - \frac{y^2}{p+x} \, dx - \frac{y^2}{p+x} \, dp = 0.$

and $q \, dy + 0 \cdot dx + y \, dq = 0$

or $-\frac{dy}{y^2} (p+x) + \frac{dp+dx}{y} = 0$ and $q \, dy + y \, dq = 0.$

Integrating, we get $(p+x)/y = a$ and $qy = b.$

Hence one intermediate integral is

$$qy = f\left(\frac{p+x}{y}\right).$$

Charpit's auxiliary equations are

$$\frac{dx}{-\frac{1}{y} f'\left(\frac{p+x}{y}\right)} = \frac{dy}{y} = \frac{dp}{\frac{1}{y} f'\left(\frac{p+x}{y}\right)}.$$

From the first and the last members, we have

$$dx + dp = 0. \quad \therefore \quad p + x = c.$$

Also from $dz = p \, dx + q \, dy$, we get

$$dz = (c-x) \, dx + (1/y) f(c/y) \, dy.$$

Integrating, $z = cx - \frac{1}{2} x^2 + \phi(c/y) + \psi(c)$, which is the required solution.

Comprehensive Exercise 2

1. Solve $r \, t - s^2 + a^2 = 0.$
2. Solve $rt - s^2 + 1 = 0.$
3. Solve $2pr + 2qt - 4pq(rt - s^2) = 1.$
4. Solve $3r + 4s + t + (rt - s^2) = 1.$
5. Solve $3s + (rt - s^2) = 2.$
6. Solve $r + t - (rt - s^2) = 1.$

(Rohilkhand 2011)

Answers 2

1. $z = -ax \, y + \frac{1}{2a} \{f_2(\beta) - f_1(\alpha)\} + \beta y$
2. $z = -xy - \frac{1}{2} \{f_1(\alpha) - f_2(\beta)\} + \beta y$
3. $3z = 3c \pm 2(b+x)^{3/2} \pm 2(a+y)^{3/2}$
4. $z = 2xy - \frac{1}{2} x^2 - \frac{3}{2} y^2 + ax + by + c$
5. $z = xy - f_1(\alpha) + f_2(\beta) + \beta y$
6. $z = \frac{1}{2} (x^2 + y^2) + ax + by + c$

Objective Type Questions

Multiple Choice Questions

- Monge's subsidiary equations for $r = t$ are
 - $dy^2 + dx^2 = 0, dp dy - dx dq = 0$
 - $dy^2 - dx^2 = 0, dp dy - dx dq = 0$
 - $dy^2 - dx^2 = 0, dp dy + dx dq = 0$
 - $dy^2 + dx^2 = 0, dp dy + dx dq = 0$
- The λ -quadratic equation for the equation $rt - s^2 + 1 = 0$ is
 - $\lambda^2 + 1 = 0$
 - $\lambda^2 - 1 = 0$
 - $\lambda^2 + 2 = 0$
 - $\lambda^2 - 2 = 0$

Fill In The Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- Monge's method is usually used to integrate the order partial differential equations.
- The equations $R dy^2 - S dx dy + T dx^2 = 0, R dp dy + T dq dx - V dx dy = 0$ are called in integrating the equation of the form $Rr + Ss + Tt = V$.
- We have $r = \frac{dp - \dots}{dx}$ and $t = \frac{\dots}{dy}$.
- In integrating the equation of the type $Rr + Ss + Tt + U(rt - s^2) = V$ by Monge's method the λ -quadratic equation is

Answers

Multiple Choice Questions

- (b)
- (b)

Fill in the Blank(s)

- second
- Monge's subsidiary equations
- $s dy, dq - s dx$
- $\lambda^2 (UV + RT) + \lambda SU + U^2 = 0$





Chapter

12

Series Solutions Of Differential Equations

12.1 Introduction

If a homogeneous linear differential equation has **constant coefficients**, it can be solved by algebraic methods, and its solutions are elementary functions known from calculus. However, if such an equation has **variable coefficients** (functions of x), it must usually be solved by other **methods**. **Legendre's equation**, the **hypergeometric equation**, and **Bessel's equation** are very important equations of this type, since these equations and their solutions play a basic role in applied mathematics.

In the present chapter we shall discuss the solutions of some linear differential equations of second order with variable coefficients in the form of series.

Consider the linear differential equation of the second order

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x) y = 0, \quad \dots(1)$$

where $P(x)$ and $Q(x)$ are functions of x .

A point $x = a$ may have a special nature with respect to the differential equation (1).

Without loss of generality, we can restrict our study to the nature of the origin with respect to (1), because the form of the differential equation (1) remains unchanged by shifting the origin. Hence to check the nature of a point a with respect to (1), it is convenient to change the differential equation by shifting the origin to the point a and then to determine the nature of the origin with respect to the transformed equation.

Ordinary and singular points: The point $x = 0$ (i.e., the origin) is called an **ordinary point** of the differential equation if $P(x)$ and $Q(x)$ do not become infinite in a neighbourhood of the origin and they can be expanded in the form of a power series.

The point $x = 0$ is called a **singular point** of the differential equation (1) if it is not an ordinary point of the differential equation. The singular points are of the two types:

- (i) Regular singular points, (ii) Irregular singular points.

The origin is a **regular singular point** if both $xP(x)$ and $x^2 Q(x)$ can be expanded in a power series of x in the neighbourhood of $x = 0$.

Otherwise $x = 0$ is called an **irregular singular point**.

If we consider the differential equation

$$P_0(x) \frac{d^2 y}{dx^2} + P_1(x) \frac{dy}{dx} + P_2(x) y = 0, \quad \dots(2)$$

then $x = a$ is called an **ordinary point** of (2) if $P_0(a) \neq 0$, otherwise it is a **singular point**.

A singular point $x = a$ of (2) is called *regular*, if when (2) is put in the form

$$\frac{d^2 y}{dx^2} + \frac{Q_1(x)}{x-a} \frac{dy}{dx} + \frac{Q_2(x)}{(x-a)^2} = 0,$$

then $Q_1(x)$ and $Q_2(x)$ possess derivatives of all orders in the neighbourhood of a .

12.2 Frobenius Method

The method of finding a solution in a series for the linear differential equation (1) near a regular singular point or near an ordinary point is called **Frobenius method**.

12.3 Solution Near an Ordinary Point

We begin by taking a trial solution of the given differential equation

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x) y = 0 \quad \dots(1)$$

in the form of the series

$$y = \sum_{n=0}^{\infty} c_n x^n. \quad \dots(2)$$

$$\therefore \frac{dy}{dx} = \sum n c_n x^{n-1}$$

and
$$\frac{d^2 y}{dx^2} = \sum n(n-1) c_n x^{n-2}. \quad \dots(3)$$

If $P(x)$ and $Q(x)$ are not polynomials in x , then they can be expanded in the form of power series given by

$$P(x) = \sum_{n=0}^{\infty} p_n x^n \quad \text{and} \quad Q(x) = \sum_{n=0}^{\infty} q_n x^n. \quad \dots(4)$$

Putting the respective values from (2), (3), (4) in (1) and then equating to zero the coefficients of various powers of x we determine the values of different coefficients of (2). Putting these values in (2) we get the general solution of (1) because in general it contains two arbitrary constants.

Illustrative Examples

Example 1 : Solve $\frac{d^2 y}{dx^2} - 2x^2 \frac{dy}{dx} + 4xy = x^2 + 2x + 2$ in powers of x .

(Avadh 2008, 14)

Solution : Here $x = 0$ is an ordinary point. Let a trial solution in the form of the series of the given differential equation be

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots + c_n x^n + \dots \quad \dots(1)$$

Differentiating (1), we get

$$\frac{dy}{dx} = c_1 + 2c_2 x + 3c_3 x^2 + \dots + n c_n x^{n-1} + \dots \quad \dots(2)$$

and
$$\frac{d^2 y}{dx^2} = 2c_2 + 6c_3 x + \dots + n(n-1) c_n x^{n-2} + \dots \quad \dots(3)$$

Putting these values in the given equation, we get

$$\begin{aligned} (2c_2 + 6c_3 x + \dots) - 2x^2 (c_1 + 2c_2 x + 3c_3 x^2 + \dots) \\ + 4x(c_0 + c_1 x + c_2 x^2 + c_3 x^3 + \dots) - x^2 - 2x - 2 = 0 \end{aligned}$$

or
$$\begin{aligned} (2c_2 - 2) + (6c_3 + 4c_0 - 2)c_1 x + (12c_4 + 2c_1 - 1)c_2 x^2 + 20c_5 x^3 + \dots \\ + \{(n+2)(n+1)c_{n+2} - 2(n-1)c_{n-1} + 4c_{n-1}\} x^n + \dots = 0, \end{aligned}$$

which is an identity in x . We can equate to zero the coefficients of various powers of x .

Equating to zero the coefficients of various powers of x , we get

$$\begin{aligned} 2c_2 - 2 = 0 \quad \text{i.e., } c_2 = 1 \\ 6c_3 + 4c_0 - 2 = 0 \quad \text{i.e., } c_3 = \frac{1}{3} - \frac{2}{3}c_0 \\ 12c_4 + 2c_1 - 1 = 0 \quad \text{i.e., } c_4 = \frac{1}{12} - \frac{1}{6}c_1. \end{aligned}$$

All other coefficients are given by the relation

$$c_{n+2} = \frac{2(n-3)}{(n+1)(n+2)} c_{n-1}, n \geq 3.$$

Hence the required complete solution in series is

$$y = c_0 \left(1 - \frac{2}{3} x^3 - \frac{2}{45} x^6 - \dots \right) + c_1 \left(x - \frac{1}{6} x^4 - \frac{1}{63} x^7 - \dots \right) + x^2 + \frac{1}{3} x^3 + \frac{1}{12} x^4 + \frac{1}{45} x^6 + \dots,$$

where c_0 and c_1 are arbitrary constants.

12.4 Solution Near a Regular Singular Point

We assume a trial series solution

$$y = x^m (c_0 + c_1 x + c_2 x^2 + \dots) = x^m \sum_{n=0}^{\infty} c_n x^n$$

where all the c 's are constants and $c_0 \neq 0$.

Here we have to determine m and c 's.

For this, we find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ and substitute their values in the given differential equation. Then the equation (1) reduces to an identity in x .

By equating to zero the coefficient of the lowest power of x in this identity, we get a quadratic equation in m . It is called the **indicial equation**. It will determine m .

Equating to zero the coefficients of other various powers of x , we can determine the values of the constants $c_1, c_2, \dots$ etc. in terms of c_0 .

The following cases arise according to the nature of the roots of the indicial equation :

- (i) The roots of the indicial equation *equal*.
- (ii) The roots of the indicial equation *unequal* and differing by a quantity not an integer.
- (iii) The roots of the indicial equation *unequal*, differing by an integer and making the coefficients of some powers of x in the series for y infinity.
- (iv) The roots of the indicial equation *unequal*, differing by an integer and making the coefficient of some power of x in the series for y indeterminate.

Case I: Roots of the indicial equation equal:

Let $m = \alpha$ be two equal roots of the indicial equation. Then putting $m = \alpha$ in y and in $\partial y / \partial m$, we can obtain the two independent solutions.

The second solution always consists of a numerical multiple of the product of the first solution and $\log x$ added to another series.

Illustrative Examples

Example 2: Solve $x \frac{d^2y}{dx^2} + \frac{dy}{dx} + xy = 0$. (Bessel's equation when $n = 0$). ... (1)

(Avadh 2008; Purvanchal 11)

Solution : Putting $y = x^m$ in the left hand side of the given equation (1), we get

$$x \cdot m(m-1)x^{m-2} + mx^{m-1} + x \cdot x^m$$

or $x^{m+1} + m^2 x^{m-1}.$

Obviously the common difference of the powers is $(m+1) - (m-1)$ i.e., 2.

$\therefore$ Let the series solution of the given equation (1) in ascending powers of x be

$$y = \sum_{r=0}^{\infty} c_r x^{m+2r} = c_0 x^m + c_1 x^{m+2} + c_2 x^{m+4} + \dots \quad \dots(2)$$

$$\therefore \frac{dy}{dx} = \sum_{r=0}^{\infty} c_r (m+2r) x^{m+2r-1}$$

and $\frac{d^2 y}{dx^2} = \sum_{r=0}^{\infty} c_r (m+2r)(m+2r-1) x^{m+2r-2}.$

Since (2) is a solution of (1) so substituting the values of $y, dy/dx$ and $d^2 y/dx^2$ in (1), we have

$$\sum_{r=0}^{\infty} c_r [(m+2r)(m+2r-1) x^{m+2r-1} + (m+2r) x^{m+2r-1} + x^{m+2r+1}] = 0$$

or $\sum_{r=0}^{\infty} c_r [x^{m+2r+1} + (m+2r)^2 x^{m+2r-1}] = 0,$

which is an identity in x . Equating to zero the coefficient of the lowest power of x i.e., of x^{m-1} , we have the indicial equation as $c_0 m^2 = 0$.

Now $c_0 \neq 0$, as it is the coefficient of the first term with which we start to write the series.

$$\therefore m = 0, 0.$$

Thus the indicial equation has equal roots.

Now equating to zero the coefficient of the general term i.e., of x^{m+2p+1} , we get

$$c_p + (m+2p+2)^2 c_{p+1} = 0.$$

$$\therefore c_{p+1} = -\frac{1}{(m+2p+2)^2} c_p. \quad \dots(3)$$

Putting $p = 0, 1, 2, \dots$ in (3), we get

$$c_1 = -\frac{1}{(m+2)^2} c_0, c_2 = -\frac{1}{(m+4)^2} c_1 = (-1)^2 \frac{1}{(m+2)^2 (m+4)^2} c_0,$$

$$c_3 = -\frac{1}{(m+6)^2} c_2 = (-1)^3 \frac{1}{(m+2)^2 (m+4)^2 (m+6)^2} c_0, \text{ etc.}$$

Substituting these values in (2), we get

$$y = c_0 x^m \left[1 - \frac{x^2}{(m+2)^2} + \frac{x^4}{(m+2)^2 (m+4)^2} - \dots \right] \quad \dots(4)$$

Putting $m = 0$ in (4), we get

$$y = c_0 \left[1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots \right] \quad \dots(5)$$

or $y = au$ (say),

which is one solution of the given equation (1).

Here
$$u = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots,$$

and a is an arbitrary constant which replaces c_0 .

Since the two values of m are equal so a second solution cannot be obtained from (4).

Differentiating (4), we get

$$\frac{dy}{dx} = c_0 \left[m x^{m-1} - \frac{(m+2)x^{m+1}}{(m+2)^2} + \frac{(m+4)x^{m+3}}{(m+2)^2(m+4)^2} - \dots \right]$$

and
$$\frac{d^2y}{dx^2} = c_0 \left[m(m-1)x^{m-2} - \frac{(m+2)(m+1)}{(m+2)^2} x^m + \frac{(m+4)(m+3)}{(m+2)^2(m+4)^2} x^{m+2} - \dots \right].$$

Substituting these values in the given equation (1), we have

$$\begin{aligned} \text{L.H.S.} &= x c_0 \left[m(m-1)x^{m-2} - \frac{(m+2)(m+1)}{(m+2)^2} x^m + \frac{(m+4)(m+3)}{(m+2)^2(m+4)^2} x^{m+2} - \dots \right] \\ &\quad + c_0 \left[m x^{m-1} - \frac{(m+2)x^{m+1}}{(m+2)^2} + \frac{(m+4)x^{m+3}}{(m+2)^2(m+4)^2} - \dots \right] \\ &\quad + x c_0 \left[x^m - \frac{x^{m+2}}{(m+2)^2} + \frac{x^{m+4}}{(m+2)^2(m+4)^2} - \dots \right] \\ &= c_0 m^2 x^{m-1}. \quad [\because \text{all other terms cancel}] \end{aligned}$$

$$\therefore \left[x \frac{d^2}{dx^2} + \frac{d}{dx} + x \right] y = c_0 m^2 x^{m-1}.$$

Differentiating both sides partially w.r.t. m , we get

$$\frac{\partial}{\partial m} \left[x \frac{d^2}{dx^2} + \frac{d}{dx} + x \right] y = \frac{\partial}{\partial m} (c_0 m^2 x^{m-1})$$

or
$$\left[x \frac{d^2}{dx^2} + \frac{d}{dx} + x \right] \left(\frac{\partial y}{\partial m} \right) = c_0 \cdot 2 m x^{m-1} + c_0 m^2 x^{m-1} \log x.$$

$$\left[\because \text{the operators } \left(x \frac{d^2}{dx^2} + \frac{d}{dx} + x \right) \text{ and } \frac{\partial}{\partial m} \text{ commute} \right]$$

Putting $m = 0$, we get

$$\left[x \frac{d^2}{dx^2} + \frac{d}{dx} + x \right] \left(\frac{\partial y}{\partial m} \right)_0 = 0.$$

Thus $\left(\frac{\partial y}{\partial m} \right)_{m=0}$ satisfies the equation (1). Hence it is also a solution of (1).

Differentiating (4) partially w.r.t. m , we get

$$\begin{aligned} \frac{\partial y}{\partial m} &= c_0 x^m \log x \left[1 - \frac{x^2}{(m+2)^2} + \frac{x^4}{(m+2)^2 (m+4)^2} - \dots \right] \\ &+ c_0 x^m \left[\frac{2x^2}{(m+2)^3} + \left\{ \frac{-2}{(m+2)^3 (m+4)^2} + \frac{-2}{(m+2)^2 (m+4)^3} \right\} x^4 + \dots \right] \end{aligned}$$

Putting $m = 0$, we obtain

$$\begin{aligned} \left(\frac{\partial y}{\partial m} \right)_{m=0} &= c_0 \log x \left[1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \dots \right] \\ &+ c_0 \left[\frac{x^2}{2^2} + \left\{ \frac{-2}{2^3 \cdot 4^2} + \frac{-2}{2^2 \cdot 4^3} \right\} x^4 + \dots \right] \\ &= bu \log x + b \left[\frac{x^2}{2^2} - \frac{3}{2^3 \cdot 4^2} x^4 + \dots \right] = bv, \text{ (say)} \end{aligned}$$

which is another independent solution of (1).

Here
$$v = u \log x + \left[\frac{x^2}{2^2} - \frac{3}{2^3 \cdot 4^2} x^4 + \dots \right]$$

and b is an arbitrary constant which replaces c_0 .

Hence the required general solution of (1) is

$$y = au + bv,$$

where a and b are arbitrary constants.

Note. The equation (1) is called the Bessel's equation of order zero.

Example 3 : Solve $x \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + 2y = 0$(1)

(Purvanchal 2010)

Solution : Putting $y = x^m$ in the left hand side of the given equation (1), we get

$$x \cdot m(m-1)x^{m-2} + (1+x)m x^{m-1} + 2x^m$$

or
$$(m+2)x^m + m^2 x^{m-1}.$$

Obviously the common difference of the powers is 1.

$\therefore$ Let the solution of (1) in ascending powers of x be

$$y = c_0 x^m + c_1 x^{m+1} + c_2 x^{m+2} + \dots = \sum_{r=0}^{\infty} c_r x^{m+r}. \quad \dots(2)$$

$$\therefore \frac{dy}{dx} = \sum_{r=0}^{\infty} c_r (m+r) x^{m+r-1}$$

$$\text{and } \frac{d^2 y}{dx^2} = \sum_{r=0}^{\infty} c_r (m+r)(m+r-1) x^{m+r-2}.$$

Putting these values in the equation (1), we get

$$\sum_{r=0}^{\infty} c_r [(m+r)(m+r-1) x^{m+r-1} + (1+x)(m+r) x^{m+r-1} + 2x^{m+r}] = 0$$

$$\text{or } \sum_{r=0}^{\infty} c_r [(m+r+2) x^{m+r} + (m+r)^2 x^{m+r-1}] = 0,$$

which is an identity in x . Equating to zero the coefficient of the lowest power of x i.e., of x^{m-1} , we have the indicial equation as $c_0 m^2 = 0$.

Now $c_0 \neq 0$, as it is the coefficient of the first term with which we start to write the series.

$$\therefore m = 0, 0.$$

Thus the indicial equation has equal roots.

Now equating to zero the coefficient of the general term i.e., of x^{m+p} , we get

$$c_p (m+p+2) + (m+p+1)^2 c_{p+1} = 0.$$

$$\therefore c_{p+1} = -\frac{(m+p+2)}{(m+p+1)^2} c_p \quad \dots(3)$$

Putting $p = 0, 1, 2, \dots$ in (3), we get

$$c_1 = -\frac{(m+2)}{(m+1)^2} c_0, \quad c_2 = -\frac{(m+3)}{(m+2)^2} c_1 = (-1)^2 \frac{(m+2)(m+3)}{(m+1)^2 (m+2)^2} c_0,$$

$$c_3 = -\frac{(m+4)}{(m+3)^2} c_2 = (-1)^3 \frac{(m+2)(m+3)(m+4)}{(m+1)^2 (m+2)^2 (m+3)^2} c_0, \text{ etc.}$$

Substituting these values in (2), we get

$$y = c_0 x^m \left[1 - \frac{(m+2)}{(m+1)^2} x + \frac{(m+2)(m+3)}{(m+1)^2 (m+2)^2} x^2 - \dots \right]. \quad \dots(4)$$

Putting $m = 0$ in (4), we get

$$y = c_0 \left[1 - 2x + \frac{3}{2!} x^2 - \frac{4}{3!} x^3 + \dots \right] \quad \dots(5)$$

$= au$ (say), which is one solution of (1).

Here $u = 1 - 2x + \frac{3}{2!} x^2 - \frac{4}{3!} x^3 + \dots$ and a is an arbitrary constant which replaces c_0 .

Proceeding as in Ex. 2, we find that the second solution is $\left(\frac{\partial y}{\partial m} \right)_{m=0}$.

Differentiating (4) partially w.r.t. m and then putting $m = 0$, we get

$$\left(\frac{\partial y}{\partial m}\right)_{m=0} = bu \log x + b \left[2 \left(2 - \frac{1}{2} \right) x - \frac{3}{2!} \left(-\frac{1}{3} + 2 + \frac{1}{2} \right) x^2 + \dots \right]$$

$= bv$ (say), which is another independent solution of (1).

Hence the required general solution of (1) is

$$y = au + bv, \text{ where } a \text{ and } b \text{ are arbitrary constants.}$$

Case II: Roots of the indicial equation unequal and not differing by an integer:

If the indicial equation has two unequal roots $m = \alpha$ and β which do not differ by an integer, then the two independent solutions can be obtained by putting $m = \alpha$ and β in the series for y .

Illustrative Examples

Example 4 : Solve in series the Bessel's equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2) y = 0 \quad \dots(1)$$

taking $2n$ as non-integral. (Avadh 2009, 10; Rohilkhand 10; Kanpur 09; Lucknow 08)

Solution : Putting $y = x^m$ in the left hand side of the given equation (1), we get

$$x^2 m(m-1) x^{m-2} + x m x^{m-1} + (x^2 - n^2) x^m$$

$$\text{or} \quad \{m(m-1) + m - n^2\} x^m + x^{m+2}.$$

Obviously the common difference of the powers is

$$(m+2) - m \text{ i.e., } 2.$$

Let the solution of (1) be

$$y = \sum_{r=0}^{\infty} c_r x^{m+2r} \quad \dots(2)$$

$$\therefore \frac{dy}{dx} = \sum_{r=0}^{\infty} c_r (m+2r) x^{m+2r-1}$$

$$\text{and} \quad \frac{d^2 y}{dx^2} = \sum_{r=0}^{\infty} c_r (m+2r)(m+2r-1) x^{m+2r-2}.$$

Substituting the values of y , dy/dx , $d^2 y/dx^2$ in (1), we get

$$\begin{aligned} \sum_{r=0}^{\infty} c_r [(m+2r)(m+2r-1) x^{m+2r} \\ + (m+2r) x^{m+2r} + (x^2 - n^2) x^{m+2r}] = 0 \end{aligned}$$

$$\text{or} \quad \sum_{r=0}^{\infty} c_r [\{(m+2r)^2 - n^2\} x^{m+2r} + x^{m+2r+2}] = 0$$

$$\text{or} \quad \sum_{r=0}^{\infty} c_r [(m+2r+n)(m+2r-n) x^{m+2r} + x^{m+2r+2}] = 0,$$

which is an identity in x . Equating to zero the coefficient of the lowest power of x i.e., of x^m , we have the indicial equation as

$$c_0 (m+n)(m-n) = 0.$$

Now $c_0 \neq 0$, as it is the coefficient of the first term with which we start to write the series.

$$\therefore m = n, -n.$$

Here the roots of the indicial equation are unequal and differ by $2n$ which is not an integer.

Now equating to zero the coefficient of the general term i.e., of x^{m+2p} , we get

$$c_p (m+2p+n)(m+2p-n) + c_{p-1} = 0.$$

$$\therefore c_p = -\frac{1}{(m+2p+n)(m+2p-n)} c_{p-1}.$$

Putting $p = 1, 2, \dots$, we get

$$c_1 = -\frac{1}{(m+2+n)(m+2-n)} c_0,$$

$$c_2 = -\frac{1}{(m+4+n)(m+4-n)} c_1$$

$$= (-1)^2 \frac{1}{(m+4+n)(m+4-n)(m+2+n)(m+2-n)} c_0, \text{ etc.}$$

Substituting these values in (2), we get

$$y = c_0 x^m \left[1 - \frac{1}{(m+2+n)(m+2-n)} x^2 + \frac{x^4}{(m+4+n)(m+4-n)(m+2+n)(m+2-n)} - \dots \right].$$

Putting $m = n$ and $-n$ successively, we get

$$y = c_0 x^n \left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4 \cdot (2n+2)(2n+4)} - \dots \right]$$

$$= c_0 x^n \left[1 - \frac{x^2}{2^2 \cdot 1!(n+1)} + \frac{x^4}{2^4 \cdot 2!(n+1)(n+2)} - \dots \right]$$

$$= au \text{ (say), which is one solution of (1)}$$

and

$$y = c_0 x^{-n} \left[1 - \frac{x^2}{2(-2n+2)} + \frac{x^4}{2 \cdot 4 \cdot (-2n+2)(-2n+4)} - \dots \right]$$

$$= c_0 x^{-n} \left[1 - \frac{x^2}{2^2 \cdot 1!(-n+1)} + \frac{x^4}{2^4 \cdot 2!(-n+1)(-n+2)} - \dots \right]$$

$$= bv \text{ (say), which is another independent solution of (1).}$$

Hence the required general solution of (1) is

$$y = au + bv, \text{ where } a \text{ and } b \text{ are arbitrary constants.}$$

Example 5: Solve in descending powers of x the Legendre's equation

$$(1-x^2)y'' - 2xy' + p(p+1)y = 0. \quad \dots(1)$$

Solution : Putting $y = x^m$ in the left hand side of the given equation (1), we get

$$(1-x^2)m(m-1)x^{m-2} - 2x \cdot mx^{m-1} + p(p+1)x^m$$

$$\text{or} \quad (-m^2 - m + p^2 + p)x^m + m(m-1)x^{m-2}.$$

Obviously the common difference of the powers is $m - (m-2)$ i.e., 2.

$\therefore$ Let the solution of (1) in a series in descending powers of x be

$$y = c_0 x^m + c_1 x^{m-2} + c_2 x^{m-4} + \dots = \sum_{r=0}^{\infty} c_r x^{m-2r}. \quad \dots(2)$$

$$\therefore y' = \sum_{r=0}^{\infty} c_r (m-2r) x^{m-2r-1}$$

$$\text{and} \quad y'' = \sum_{r=0}^{\infty} c_r (m-2r)(m-2r-1) x^{m-2r-2}.$$

Substituting the values of y, y', y'' in (1), we get

$$\sum_{r=0}^{\infty} c_r [(1-x^2)(m-2r)(m-2r-1)x^{m-2r-2} - 2x(m-2r)x^{m-2r-1} + p(p+1)x^{m-2r}] = 0$$

$$\text{or} \quad \sum_{r=0}^{\infty} c_r [- (m-2r)(m-2r-1) - 2(m-2r) + p(p+1)] x^{m-2r} + (m-2r)(m-2r-1)x^{m-2r-2} = 0$$

$$\text{or} \quad \sum_{r=0}^{\infty} c_r [\{p^2 - (m-2r)^2 + (p-m+2r)\} x^{m-2r} + (m-2r)(m-2r-1)x^{m-2r-2}] = 0$$

$$\text{or} \quad \sum_{r=0}^{\infty} c_r [(p-m+2r)(p+m-2r+1)x^{m-2r} + (m-2r)(m-2r-1)x^{m-2r-2}] = 0$$

which is an identity in x . Equating to zero the coefficient of the highest power of x i.e., of x^m , we have the indicial equation as

$$c_0 (p-m)(p+m+1) = 0.$$

Now $c_0 \neq 0$, as it is the coefficient of the first term with which we start to write the series.

$$\therefore m = p - (p+1).$$

Now equating to zero the coefficient of x^{m-2r} , we get

$$c_r (p-m+2r)(p+m-2r+1) + (m-2r+2)(m-2r+1)c_{r-1} = 0.$$

$$\therefore c_r = - \frac{(m-2r+2)(m-2r+1)}{(p-m+2r)(p+m-2r+1)} c_{r-1}.$$

Putting $r = 1, 2, \dots$, we get

$$\begin{aligned}
 c_1 &= -\frac{m(m-1)}{(p-m+2)(p+m-1)}c_0, \\
 c_2 &= -\frac{(m-2)(m-3)}{(p-m+4)(p+m-3)}c_1 \\
 &= (-1)^2 \frac{m(m-1)(m-2)(m-3)}{(p-m+2)(p-m+4)(p+m-1)(p+m-3)}c_0, \text{ etc.}
 \end{aligned}$$

Substituting these values in (2), we get

$$\begin{aligned}
 y &= c_0 \left[x^m - \frac{m(m-1)}{(p-m+2)(p+m-1)}x^{m-2} \right. \\
 &\quad \left. + \frac{m(m-1)(m-2)(m-3)}{(p-m+2)(p-m+4)(p+m-1)(p+m-3)}x^{m-4} - \dots \right]
 \end{aligned}$$

Putting

$m = p, - (p+1)$ successively, we get

$$\begin{aligned}
 y &= c_0 \left[x^p - \frac{p(p-1)}{2(2p-1)}x^{p-2} + \frac{p(p-1)(p-2)(p-3)}{2 \cdot 4 \cdot (2p-1)(2p-3)}x^{p-4} - \dots \right] \\
 &= au \text{ (say), which is one solution of (1)}
 \end{aligned}$$

and

$$\begin{aligned}
 y &= c_0 \left[x^{-p-1} + \frac{(p+1)(p+2)}{2(2p+3)}x^{-p-3} \right. \\
 &\quad \left. + \frac{(p+1)(p+2)(p+3)(p+4)}{2 \cdot 4 \cdot (2p+3)(2p+5)}x^{-p-5} + \dots \right]
 \end{aligned}$$

$= bv$ (say), which is another independent solution of (1).

Hence the required general solution of (1) is $y = au + bv$, where a and b are arbitrary constants.

Case III: Roots of the indicial equation differing by an integer and making the coefficients of some powers of x in the series for y infinity:

Let $m = \alpha$ and β be two roots of the indicial equation which differ by an integer and some of the coefficients of powers of x in the series for y become infinite for $m = \beta$.

In this case substitute $c(m - \beta)$ for c_0 .

It will give two independent solutions for $m = \beta$, namely the modified y and $\partial y / \partial m$ as in case I.

Hence, in all we find three solutions :

- (i) the solution obtained by putting $m = \alpha$ in y ,
- (ii) the solution obtained by putting $m = \beta$ in the modified y ,
- (iii) the solution obtained by putting $m = \beta$ in the partial differential coefficient w.r.t. m of the modified y i.e., in $\partial y / \partial m$.

But the solution (i) is a numerical multiple of (ii). Thus only two of these three solutions are independent.

Illustrative Examples

Example 6 : Obtain a general solution in series of powers of x of the equation (Bessel's equation of order one)

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - 1)y = 0. \quad \dots(1) \quad (\text{Avadh 2006})$$

Solution : Putting $y = x^m$ in the left hand side of the given equation (1), we get

$$x^2 \cdot m(m-1)x^{m-2} + x \cdot m x^{m-1} + (x^2 - 1)x^m$$

$$\text{or} \quad (m^2 - 1)x^m + x^{m+2}.$$

Obviously the common difference of the powers is 2.

$\therefore$ Let the solution of (1) in a series of ascending powers of x be

$$\begin{aligned} y &= c_0 x^m + c_1 x^{m+2} + c_2 x^{m+4} + \dots \\ &= \sum_{r=0}^{\infty} c_r x^{m+2r} \end{aligned} \quad \dots(2)$$

$$\therefore \quad \frac{dy}{dx} = \sum_{r=0}^{\infty} c_r (m+2r) x^{m+2r-1}$$

$$\text{and} \quad \frac{d^2 y}{dx^2} = \sum_{r=0}^{\infty} c_r (m+2r)(m+2r-1) x^{m+2r-2}.$$

Putting these values in the given equation (1), we get

$$\begin{aligned} \sum_{r=0}^{\infty} c_r [x^2 (m+2r)(m+2r-1) x^{m+2r-2} \\ + x \cdot (m+2r) x^{m+2r-1} + (x^2 - 1) x^{m+2r}] = 0 \end{aligned}$$

$$\text{or} \quad \sum_{r=0}^{\infty} c_r [\{(m+2r)^2 - 1\} x^{m+2r} + x^{m+2r+2}] = 0$$

$$\text{or} \quad \sum_{r=0}^{\infty} c_r [(m+2r+1)(m+2r-1) x^{m+2r} + x^{m+2r+2}] = 0$$

which is an identity in x . Equating to zero the coefficient of the lowest power of x i.e., of x^m , we have the indicial equation as

$$c_0 (m+1)(m-1) = 0.$$

Now $c_0 \neq 0$, as it is the coefficient of the first term with which we start to write the series.

$$\therefore \quad m = 1, -1.$$

Here the roots of the indicial equation are unequal and the difference of the roots is 2 which is an integer.

Now equating to zero the coefficient of the general term i.e., of x^{m+2p} , we get

$$c_p (m+2p+1)(m+2p-1) + c_{p-1} = 0$$

$$\text{or} \quad c_p = - \frac{1}{(m+2p+1)(m+2p-1)} c_{p-1}.$$

Putting $p = 1, 2, 3, \dots$, we get

$$c_1 = -\frac{1}{(m+3)(m+1)} c_0,$$

$$c_2 = -\frac{1}{(m+5)(m+3)} c_1 = (-1)^2 \frac{1}{(m+5)(m+3)^2(m+1)} c_0,$$

$$c_3 = -\frac{1}{(m+7)(m+5)} c_2 = (-1)^3 \frac{1}{(m+7)(m+5)^2(m+3)^2(m+1)} c_0, \text{ etc.}$$

Substituting these values in (2), we get

$$y = c_0 x^m \left[1 - \frac{x^2}{(m+1)(m+3)} + \frac{x^4}{(m+1)(m+3)^2(m+5)} - \frac{x^6}{(m+1)(m+3)^2(m+5)^2(m+7)} + \dots \right] \quad \dots(3)$$

Now if we put $m = -1$ in the above series, the coefficients of x^2 and onwards in the bracketed expression become infinite because of the factor $(m+1)$ in the denominator. To overcome this difficulty, put $c(m+1)$ for c_0 in (3).

$$\text{Thus } y = cx^m \left[(m+1) - \frac{1}{(m+3)} x^2 + \frac{1}{(m+3)^2(m+5)} x^4 - \frac{1}{(m+3)^2(m+5)^2(m+7)} x^6 + \dots \right] \quad \dots(4)$$

Putting $m = -1$ in (4), we get

$$y = cx^{-1} \left[-\frac{1}{2} x^2 + \frac{1}{2^2 \cdot 4} x^4 - \frac{1}{2^2 \cdot 4^2 \cdot 6} x^6 + \dots \right] \quad \dots(5)$$

$$= au \quad (\text{say}),$$

which is one solution of the equation (1).

Putting $m = 1$ in (3), we get a series which is a numerical multiple of the series in (5). So this solution is not independent of the solution obtained in (5).

Hence so far, by all means we have obtained only one series solution. As in case I, to find the second series solution, students may show that $(\partial y / \partial m)_{m=-1}$ also satisfies the given differential equation while y is given by (4). Thus $(\partial y / \partial m)_{m=-1}$ will give the second series solution.

Note that substituting (4) in the given differential equation (1) and on simplification,

$$\text{we have } x \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - 1) y = cx^m (m+1)^2 (m-1).$$

There is a factor $(m+1)^2$ present on the R.H.S. showing that y as well as $\partial y / \partial m$ satisfy the differential equation when $m = -1$.

Differentiating (4) partially w.r.t. m , we get

$$\frac{\partial y}{\partial m} = cx^m \log x \cdot \left[(m+1) - \frac{x^2}{m+3} + \frac{x^4}{(m+3)^2(m+5)} - \dots \right]$$

$$+ cx^m \left[1 + \frac{x^2}{(m+3)^2} - \left\{ \frac{2}{(m+3)^3(m+5)} + \frac{1}{(m+3)^2(m+5)^2} \right\} x^4 + \dots \right]$$

Putting $m = -1$ in this, we get

$$\left(\frac{\partial y}{\partial m} \right)_{m=-1} = cu \log x + c x^{-1} \left[1 + \frac{x^2}{4} - \frac{5}{64} x^4 + \dots \right] = bv \text{ (say),}$$

which is another independent solution of (1).

Hence the required general solution of the given equation (1) is $y = au + bv$, where a and b are arbitrary constants.

Case IV: Roots of the indicial equation differing by an integer and making a coefficient of the series for y indeterminate:

Let $m = \alpha$ and β (say $\alpha > \beta$) be two roots of the indicial equation which differ by an integer. If one of the coefficients of the series for y becomes indeterminate when $m = \beta$ the complete primitive is obtained by putting $m = \beta$ in y , which contains two arbitrary constants. The series obtained by putting $m = \alpha$ in y is merely a numerical multiple of one of the series contained in the first solution.

Illustrative Examples

Example 7 : Solve $(1 - x^2) \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} + y = 0$ (1) (Rohilkhand 2008)

Solution : Let the series solution of the given equation (1) be

$$y = c_0 x^m + c_1 x^{m+1} + c_2 x^{m+2} + \dots = \sum_{r=0}^{\infty} c_r x^{m+r} \quad \dots (2)$$

$$\therefore \frac{dy}{dx} = \sum_{r=0}^{\infty} c_r (m+r) x^{m+r-1}$$

$$\text{and } \frac{d^2 y}{dx^2} = \sum_{r=0}^{\infty} c_r (m+r)(m+r-1) x^{m+r-2}.$$

Putting these values in the given equation (1), we get

$$\sum_{r=0}^{\infty} c_r [(1 - x^2)(m+r)(m+r-1) x^{m+r-2} + 2x(m+r) x^{m+r-1} + x^{m+r}] = 0$$

$$\text{or } \sum_{r=0}^{\infty} c_r \{ -(m+r)(m+r-1) + 2(m+r) + 1 \} x^{m+r} + (m+r)(m+r-1) x^{m+r-2} = 0$$

$$\text{or } \sum_{r=0}^{\infty} c_r [-\{(m+r)(m+r-3) - 1\} x^{m+r} + (m+r)(m+r-1) x^{m+r-2}] = 0$$

which is an identity in x . Equating to zero the coefficient of the lowest power of x i.e., of x^{m-2} , we have the indicial equation as

$$c_0 m (m - 1) = 0.$$

Now $c_0 \neq 0$, as it is the coefficient of the first term with which we start to write the series.

$$\therefore m = 0, 1.$$

Now equating to zero the coefficient of the next higher power of x i.e., of x^{m-1} , we get

$$c_1 (m + 1) \cdot m = 0, \quad \dots(3)$$

which makes c_1 indeterminate ($0/0$) when $m = 0$.

But if $m = 0$, from (3) we have the identity

$$c \cdot 0 = 0,$$

which is true for every value of c_1 . Hence, in this case c_1 can be taken as an arbitrary constant.

Further equating to zero the coefficient of the general term i.e., of x^{m+p} , we get

$$-c_p \{(m+p)(m+p-3)-1\} + c_{p+2} (m+p+2)(m+p+1) = 0$$

$$\text{or} \quad c_{p+2} = \frac{(m+p)(m+p-3)-1}{(m+p+1)(m+p+2)} c_p.$$

Putting $p = 0, 1, 2, \dots$, we get

$$c_2 = \frac{m(m-3)-1}{(m+1)(m+2)} c_0, \quad c_3 = \frac{(m+1)(m-2)-1}{(m+2)(m+3)} c_1,$$

$$c_4 = \frac{(m+2)(m-1)-1}{(m+3)(m+4)} c_2 = \frac{\{(m+2)(m-1)-1\} \{m(m-3)-1\}}{(m+1)(m+2)(m+3)(m+4)} c_0,$$

$$c_5 = \frac{(m+3)m-1}{(m+4)(m+5)} c_3 = \frac{\{m(m+3)-1\} \{(m+1)(m-2)-1\}}{(m+2)(m+3)(m+4)(m+5)} c_1, \text{ etc.}$$

Substituting these values in (2), we get

$$y = c_0 x^m \left[1 + \frac{m(m-3)-1}{(m+1)(m+2)} x^2 + \frac{\{(m+2)(m-1)-1\} \{m(m-3)-1\}}{(m+1)(m+2)(m+3)(m+4)} x^4 + \dots \right] \\ + c_1 x^m \left[x + \frac{(m+1)(m-2)-1}{(m+2)(m+3)} x^3 + \frac{\{m(m+3)-1\} \{(m+1)(m-2)-1\}}{(m+2)(m+3)(m+4)(m+5)} x^5 + \dots \right].$$

Putting $m = 0$ in this, we get

$$y = c_0 \left[1 - \frac{1}{2} x^2 + \frac{1}{8} x^4 + \dots \right] + c_1 \left[x - \frac{1}{2} x^3 + \frac{1}{40} x^5 + \dots \right] \\ = a \left[1 - \frac{1}{2} x^2 + \frac{1}{8} x^4 + \dots \right] + b \left[x - \frac{1}{2} x^3 + \frac{1}{40} x^5 + \dots \right], \quad \dots(4)$$

where a and b are arbitrary constants.

This is the required complete primitive of the given equation (1) because it contains two arbitrary constants.

If we take $m = 1$, we get from (3) $c_1 = 0$ and hence $c_3, c_5, \dots$ all vanish. In this case we shall get a series which is merely a numerical multiple of the second series in the general solution (4).

12.5 Some Cases of Failure of the Method of Frobenius

It is not always necessary that every equation has a convergent series solution in ascending or descending powers of x . There may be cases in which the indicial equation may not have any root or may have one root, which may lead to a divergent series. These cases lead to the failure of the method of Frobenius.

Illustrative Examples

Example 8 : Transform the Bessel's equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2) y = 0 \quad \dots(1)$$

by the substitution $x = 1/z$.

Hence show that it has no integrals that are regular in descending powers of x .

Solution : We have $x = \frac{1}{z}$, so that $\frac{dx}{dz} = -\frac{1}{z^2}$.

Now
$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = -z^2 \frac{dy}{dz}$$

and
$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(-z^2 \frac{dy}{dz} \right) = \frac{d}{dz} \left(-z^2 \frac{dy}{dz} \right) \cdot \frac{dz}{dx} \\ &= \left(-z^2 \frac{d^2 y}{dz^2} - 2z \frac{dy}{dz} \right) (-z^2) = z^4 \frac{d^2 y}{dz^2} + 2z^3 \frac{dy}{dz}. \end{aligned}$$

Putting these values in the given equation (1), we get the transformed equation as

$$\frac{1}{z^2} \left(z^4 \frac{d^2 y}{dz^2} + 2z^3 \frac{dy}{dz} \right) + \frac{1}{z} \left(-z^2 \frac{dy}{dz} \right) + \left(\frac{1}{z^2} - n^2 \right) y = 0$$

or
$$z^4 \frac{d^2 y}{dz^2} + z^3 \frac{dy}{dz} + (1 - n^2 z^2) y = 0. \quad \dots(2)$$

Let the solution of (2) in ascending powers of z be

$$y = \sum_{r=0}^{\infty} c_r z^{m+r}. \quad \dots(3)$$

Then
$$\frac{dy}{dz} = \sum_{r=0}^{\infty} c_r (m+r) z^{m+r-1}$$

and
$$\frac{d^2 y}{dz^2} = \sum_{r=0}^{\infty} c_r (m+r)(m+r-1) z^{m+r-2}.$$

Putting these values in (2), we get

$$\sum_{r=0}^{\infty} c_r [(m+r)(m+r-1)z^{m+r+2} + (m+r)z^{m+r+2} + (1-n^2z^2)z^{m+r}] = 0$$

or
$$\sum_{r=0}^{\infty} c_r \{[(m+r)^2 - n^2]z^{m+r+2} + z^{m+r}\} = 0$$

which is an identity in z . Equating to zero the coefficient of the lowest power of z i.e., of z^m , we have the indicial equation as $c_0 = 0$, which has no roots as $c_0 \neq 0$, it being the coefficient of the first term with which we start to write the series.

Hence the differential equation (2) has no regular integrals in ascending powers of z i.e., the given equation (1) has no regular integrals in descending powers of x .

Illustrative Examples

Example 9 : Transform the equation

$$\frac{d^2 y}{dx^2} - y = 0, \quad \dots(1)$$

by the substitution $x = 1/z$ and show that it has no integrals that are regular in descending powers of x .

Solution : Proceeding exactly as in Ex. 8 after article 12.5, we get the transformed equation

as
$$z^4 \frac{d^2 y}{dz^2} + 2z^3 \frac{dy}{dz} - y = 0 \quad \dots(2)$$

Let the solution of (2) in ascending powers of z be

$$y = \sum_{r=0}^{\infty} c_r z^{m+r} \quad \dots(3)$$

$$\therefore \frac{dy}{dz} = \sum_{r=0}^{\infty} c_r (m+r) z^{m+r-1}$$

and
$$\frac{d^2 y}{dz^2} = \sum_{r=0}^{\infty} c_r (m+r)(m+r-1) z^{m+r-2}.$$

Putting these values in (2), we get

$$\sum_{r=0}^{\infty} c_r [(m+r)(m+r-1)z^{m+r+2} + 2(m+r)z^{m+r+2} - z^{m+r}] = 0$$

or
$$\sum_{r=0}^{\infty} c_r [(m+r)(m+r+1)z^{m+r+2} - z^{m+r}] = 0,$$

which is an identity in z . Equating to zero the coefficient of the lowest power of z i.e., of z^m , we have the indicial equation as $-c_0 = 0$, which has no roots as $c_0 \neq 0$, it being the coefficient of the first term with which we start to write the series.

Hence the differential equation (2) has no regular integrals in ascending powers of z i.e., the given equation (1) has no regular integrals in descending powers of x .

12.6 Method of Differentiation

Illustrative Examples

Example 10 : Solve $(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + m^2 y = 0$, where $x=0, y=0, dy/dx=m$.

Solution : The given equation can be put in the form

$$(1-x^2) y_2 - x y_1 + m^2 y = 0. \quad \dots(1)$$

Differentiating the equation (1) n times by Leibnitz's theorem w.r.t. x , we get

$$(1-x^2) y_{n+2} - (2n+1) x y_{n+1} + (m^2 - n^2) y_n = 0 \quad \dots(2)$$

Putting $x=0$ in (1) and (2), we get

$$(y_2)_0 = -m^2 (y)_0 = 0$$

and

$$(y_{n+2})_0 = (n^2 - m^2) (y_n)_0 \quad \dots(3)$$

Putting $n=2, 4, \dots$ in (3), we get

$$(y_4)_0 = (y_6)_0 = \dots = 0.$$

Putting $n=1, 3, 5, \dots$ in (3), we get

$$(y_3)_0 = (1^2 - m^2) (y_1)_0 = (1^2 - m^2) m$$

$$(y_5)_0 = (3^2 - m^2) (y_3)_0 = (3^2 - m^2) (1^2 - m^2) m, \text{ etc.}$$

By Maclaurin's Theorem, we have

$$y = (y)_0 + \frac{x}{1!} (y_1)_0 + \frac{x^2}{2!} (y_2)_0 + \frac{x^3}{3!} (y_3)_0 + \dots$$

Putting the values of $(y)_0, (y_1)_0, \dots$, we get

$$y = mx + (1^2 - m^2) m \frac{x^3}{3!} + (3^2 - m^2) (1^2 - m^2) m \frac{x^5}{5!} + \dots$$

12.7 The Particular Integral

If in a linear equation, right hand side is of the form x^m , then the usual method to determine the particular integral corresponding to x^m , is followed.

Illustrative Examples

Example 11 : Solve $x^4 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = x^{-1}$.

Solution : Substituting $y = x^m$ in the left hand side of the given equation, we have

$$x^4 \cdot m(m-1)x^{m-2} + x \cdot m x^{m-1} + x^m$$

or
$$m(m-1)x^{m+2} + (m+1)x^m$$

i.e., the common difference of the powers is $(m+2) - m = 2$.

$\therefore$ Let us assume
$$y = \sum_{r=0}^{\infty} A_r x^{m-2r} \quad \dots(1)$$

as the C.F. of the given equation i.e., the solution of the equation

$$x^4 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0. \quad \dots(2)$$

Then substituting the value of y from (1) in (2), we get

$$\sum_{r=0}^{\infty} [(m-2r)(m-2r-1) A_r x^{m-2r+2} + (m-2r+1) A_r x^{m-2r}] = 0$$

which is an identity.

$\therefore$ Equating to zero the coefficient of the highest power of x i.e., of x^{m+2} , we get

$$m(m-1) A_0 = 0. \quad \therefore m = 0, 1. \quad [\because A_0 \neq 0]$$

Now equating to zero the coefficient of the general term i.e., of x^{m-2r} , we get

$$(m-2r-2)(m-2r-3) A_{r+1} + (m-2r+1) A_r = 0$$

or
$$A_{r+1} = -\frac{(m-2r+1)}{(m-2r-3)(m-2r-2)} A_r. \quad \dots(3)$$

Case I: When $m = 0$, $A_{r+1} = \frac{2r-1}{(2r+2)(2r+3)} A_r. \quad \dots(4)$

Now putting $r = 0, 1, 2, 3, \dots$ in (4), we get

$$A_1 = -\frac{1}{2 \cdot 3} A_0 = -\frac{1}{3!} A_0, A_2 = \frac{1}{4 \cdot 5} A_1 = -\frac{1}{5!} A_0,$$

$$A_3 = \frac{3}{6 \cdot 7} A_2 = -\frac{1 \cdot 3}{7!} A_0, \text{ etc.}$$

$\therefore$
$$y = \sum_{r=0}^{\infty} A_r x^{m-2r} = \sum_{r=0}^{\infty} A_r x^{-2r} \quad [\because m = 0]$$

$$= A_0 + A_1 x^{-2} + A_2 x^{-4} + A_3 x^{-6} + \dots$$

$$= A \left[1 - \frac{1}{3!} x^{-2} - \frac{1}{5!} x^{-4} - \frac{1 \cdot 3}{7!} x^{-6} \dots \right]. \quad [\text{Taking } A_0 = A]$$

Case II: When $m = 1$, from (3), we have $A_{r+1} = \frac{r-1}{(r+1)(2r+1)} A_r. \quad \dots(5)$

Now putting $r = 0$, we have $A_1 = -A_0$, putting $r = 1$, we have $A_2 = 0$,

putting $r = 2, 3, \dots$, we have $A_3 = A_4 = \dots = 0$ (each).

$\therefore$
$$y = \sum_{r=0}^{\infty} A_r x^{m-2r} = \sum_{r=0}^{\infty} A_r x^{1-2r} \quad [\because m = 1]$$

$$= A_0 x + A_1 x^{-1} + A_2 x^{-3} + \dots$$

$$= A_0 \left(x - \frac{1}{x} \right) = B \left(x - \frac{1}{x} \right), \text{ taking } A_0 = B.$$

∴ the C.F. of the equation is

$$A \left(1 - \frac{1}{3!} x^{-2} - \frac{1}{5!} x^{-4} - \frac{1.3}{7!} x^{-6} \dots \right) + B \left(x - \frac{1}{x} \right).$$

To find P.I. To find the P.I., put $y = C_0 x^m$. Thus we must have

$$m(m-1)C_0 x^{m+2} = x^{-1} \Rightarrow m(m-1)C_0 = 1 \text{ and } m+2 = -1$$

$$\Rightarrow m = -3 \text{ and } C_0 = \frac{1}{12}.$$

$$\therefore \text{ Let } y = \sum_{r=0}^{\infty} C_r x^{m-2r} = \sum_{r=0}^{\infty} C_r x^{-3-2r}$$

be the P.I. of the given equation.

Substituting $y = \sum_{r=0}^{\infty} C_r x^{-3-2r}$ in (2) and proceeding as above, we get

$$C_{r+1} = \frac{2(r+1)}{(2r+5)(2r+6)} C_r.$$

Putting $r = 0, 1, 2, \dots$

$$C_1 = \frac{2}{5.6} C_0, C_2 = \frac{4}{7.8} C_1 = \frac{2.4}{5.6.7.8} C_0, \text{ etc.}$$

$$\therefore \text{ P.I.} = \sum_{r=0}^{\infty} C_r x^{-3-2r} = C_0 x^{-3} + C_1 x^{-5} + C_2 x^{-7} + \dots$$

$$= C_0 x^{-3} \left[1 + \frac{2}{5.6} x^{-2} + \frac{2.4}{5.6.7.8} x^{-4} + \dots \right]$$

$$= \frac{1}{12} x^{-3} \left[1 + \frac{2}{5.6} x^{-2} + \frac{2.4}{5.6.7.8} x^{-4} + \dots \right]$$

$$= 2x^{-3} \left[\frac{1}{4!} + \frac{2}{6!} x^{-2} + \frac{2.4}{8!} x^{-4} + \dots \right].$$

Hence the solution of the given equation is

$$y = A \left(1 - \frac{1}{3!} x^{-2} - \frac{1}{5!} x^{-4} - \frac{1.3}{7!} x^{-6} \dots \right) + B \left(x - \frac{1}{x} \right) \\ + 2x^{-3} \left(\frac{1}{4!} + \frac{2}{6!} x^{-2} + \frac{2.4}{8!} x^{-4} + \dots \right).$$

Comprehensive Exercise 1

Solve completely in series the following equations :

$$1. (1-x^2) y'' - 2xy' + p(p+1)y = 0.$$

2. $y'' + (x-1)^2 y' - 4(x-1)y = 0$ about $x = 1$.
3. $(x-x^2) \frac{d^2 y}{dx^2} + (1-5x) \frac{dy}{dx} - 4y = 0$.
(Agra 2008)
4. $(2x+x^3) \frac{d^2 y}{dx^2} - \frac{dy}{dx} - 6xy = 0$.
5. $2x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + (1-x^2)y = x^2$.
(Avadh 2006)
6. $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2-4)y = 0$.
(Purvanchal 2010)
7. $(2+x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (1+x)y = 0$.
8. $(x-x^2) \frac{d^2 y}{dx^2} + (1-x) \frac{dy}{dx} - y = 0$.
9. $9x(1-x) \frac{d^2 y}{dx^2} - 12 \frac{dy}{dx} + 4y = 0$.
10. $\frac{d^2 y}{dx^2} - y = x$.

Answers 1

1. $y = c_0 \left[1 - \frac{p(p+1)}{2!} x^2 + \frac{(p-2)p(p+1)(p+3)}{4!} x^4 - \dots \right]$
 $+ c_1 \left[x - \frac{(p-1)(p+2)}{3!} x^3 + \frac{(p-3)(p-1)(p+2)(p+4)}{5!} x^5 - \dots \right]$.
2. $y = a \left[1 + \frac{2}{3}(x-1)^3 + \frac{1}{45}(x-1)^6 - \frac{1}{1620}(x-1)^9 + \dots \right] + b \left[(x-1) + \frac{1}{4}(x-1)^4 \right]$.
3. $y = au + bv$, where $u = 1 + 2^2 x + 3^2 x^2 + 4^2 x^3 + \dots$
 and $v = u \log x - 2 [1.2x + 2.3x^2 + 3.4x^3 + \dots]$.
4. $y = a \left[1 + 3x^2 + \frac{3}{5}x^4 - \frac{1}{15}x^6 + \dots \right]$
 $+ bx^{3/2} \left[1 + \frac{3}{8}x^2 - \frac{1.3}{8.16}x^4 + \frac{1.3.5}{8.16.24}x^6 - \dots \right]$.
5. $y = au + bv + f(x)$, where
 $u = x \left[1 + \frac{x^2}{2.5} + \frac{x^4}{2.4.5.9} + \frac{x^6}{2.4.6.5.9.13} + \dots \right]$,
 $v = x^{1/2} \left[1 + \frac{x^2}{2.3} + \frac{x^4}{2.4.3.7} + \frac{x^6}{2.4.6.3.7.11} + \dots \right]$.

$$f(x) = \frac{1}{3}x^2 + \frac{1}{3} \cdot \frac{1}{3 \cdot 7}x^4 + \frac{1}{3} \cdot \frac{1}{3 \cdot 5 \cdot 7 \cdot 11}x^6 + \dots$$

6. $y = au + bv$, where

$$u = x^{-2} \left[-\frac{x^4}{2^2 \cdot 4} + \frac{x^6}{2^3 \cdot 4 \cdot 6} - \frac{x^8}{2^3 \cdot 4^2 \cdot 6 \cdot 8} + \dots \right],$$

$$v = u \log x + x^{-2} \left[1 + \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} + \dots \right].$$

7. $y = a \left[1 - \frac{1}{4}x^2 - \frac{1}{12}x^3 + \frac{5}{96}x^4 \dots \right] + b \left[x - \frac{1}{6}x^3 - \frac{1}{24}x^4 + \dots \right].$

8. $y = au + bv$,

where $u = 1 + x + \frac{2}{4}x^2 + \frac{2 \cdot 5}{4 \cdot 9}x^3 + \dots$, $v = u \log x + \left(-2x - x^2 - \frac{14}{27}x^3 - \dots \right).$

9. $y = au + bv$,

where $u = 1 + \frac{1}{3}x + \frac{1 \cdot 4}{3 \cdot 6}x^2 + \frac{1 \cdot 4 \cdot 7}{3 \cdot 6 \cdot 9}x^3 + \dots$,

$$v = x^{7/3} \left[1 + \frac{8}{10}x + \frac{8 \cdot 11}{10 \cdot 13}x^2 + \frac{8 \cdot 11 \cdot 14}{10 \cdot 13 \cdot 16}x^3 + \dots \right]$$

10. $y = a \left(1 + \frac{1}{2}x^2 + \frac{1}{24}x^4 + \dots \right) + b \left(x + \frac{1}{6}x^3 + \frac{1}{120}x^5 + \dots \right) + \left(\frac{1}{6}x^3 + \frac{1}{120}x^5 + \dots \right).$

Objective Type Questions

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. The point $x = 0$ is called an of the differential equation

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x) y = 0,$$

if $P(x)$ and $Q(x)$ do not become infinite in a neighbourhood of the origin and they can be expanded in the form of a power series.

2. The origin is a point if both $xP(x)$ and $x^2Q(x)$ can be expanded in a power series of x in the neighbourhood of $x = 0$.
3. The method of finding a solution in a series for the linear differential equation

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x) y = 0$$

near a regular singular point or near an ordinary point is called

4. In Question 1 if $x = 0$ is not an ordinary point, it is called a point.
5. The singular points are of two types :
..... points and points
6. The indicial equation of $x \frac{d^2 y}{dx^2} + \frac{dy}{dx} + xy = 0$ is ...
7. The indicial equation of $(2x + x^3) \frac{d^2 y}{dx^2} - \frac{dy}{dx} - 6xy = 0$ is ...

True or False

Write 'T' for true and 'F' for false statement.

1. $x = 0$ is an ordinary point of $(1 - x^2) y'' - 2xy' + p(p+1)y = 0$.
2. $x = 0$ is a singular point of $\frac{d^2 y}{dx^2} - 2x^2 \frac{dy}{dx} + 4xy = x^2 + 2x + 2$.

Answers

Fill in the Blank(s)

- | | |
|--|--------------------------|
| 1. ordinary point. | 2. regular singular. |
| 3. Frobenius method. | 4. singular. |
| 5. regular singular, irregular singular. | |
| 6. $c_0 m^2 = 0$. | 7. $c_0 m(2m - 3) = 0$. |

True or False

- | | |
|-------|-------|
| 1. T. | 2. F. |
|-------|-------|



Chapter

13



Legendre's Functions

13.1 Legendre's Equation.

(Avadh 2006, 08)

The differential equation of the form

$$(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0$$

is called **Legendre's differential equation (or Legendre's equation)**, where n is a constant.

This equation can also be written as $\frac{d}{dx} \left\{ (1 - x^2) \frac{dy}{dx} \right\} + n(n+1)y = 0$.

13.2 Solution of Legendre's Equation

(Avadh 2013; Bundelkhand 13)

The Legendre's equation is

$$(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0. \quad \dots(1)$$

It can be solved in a series of ascending or descending powers of x . The solution in descending powers of x is more important than the one in ascending powers.

Let us assume, $y = \sum_{r=0}^{\infty} a_r x^{k-r}, a_0 \neq 0$.

$$\therefore \frac{dy}{dx} = \sum_{r=0}^{\infty} a_r (k-r) x^{k-r-1}$$

$$\text{and} \quad \frac{d^2y}{dx^2} = \sum_{r=0}^{\infty} a_r (k-r)(k-r-1) x^{k-r-2}.$$

Substituting in (1), we have

$$(1-x^2) \sum_{r=0}^{\infty} a_r (k-r)(k-r-1) x^{k-r-2} - 2x \sum_{r=0}^{\infty} a_r (k-r) x^{k-r-1} + n(n+1) \sum_{r=0}^{\infty} a_r x^{k-r} = 0$$

$$\text{or} \quad \sum_{r=0}^{\infty} a_r [(k-r)(k-r-1) x^{k-r-2} + \{n(n+1) - (k-r)(k-r-1) - 2(k-r)\} x^{k-r}] = 0$$

$$\text{or} \quad \sum_{r=0}^{\infty} a_r [(k-r)(k-r-1) x^{k-r-2} + \{n(n+1) - (k-r)(k-r+1)\} x^{k-r}] = 0$$

$$\text{or} \quad \sum_{r=0}^{\infty} a_r [(k-r)(k-r-1) x^{k-r-2} + \{n^2 - (k-r)^2 + n - (k-r)\} x^{k-r}] = 0$$

$$\text{or} \quad \sum_{r=0}^{\infty} a_r [(k-r)(k-r-1) x^{k-r-2} + (n-k+r)(n+k-r+1) x^{k-r}] = 0.$$

...(2)

Now (2) being an identity, we can equate to zero the coefficients of various powers of x .

$\therefore$ Equating to zero the coefficient of the highest power of x , i.e. of x^k , we have

$$a_0 (n-k)(n+k+1) = 0.$$

Now $a_0 \neq 0$, as it is the coefficient of the first term with which we start to write the series.

$$\therefore \left. \begin{array}{l} k = n \\ \text{or} \quad k = -(n-1) \end{array} \right\} \quad \dots(3)$$

Equating to zero the coefficient of the next lower power of x i.e. of x^{k-1} , we have

$$a_1 (n-k+1)(n+k) = 0.$$

$\therefore a_1 = 0$, since neither $(n-k+1)$ nor $(n+k)$ is zero by virtue of (3).

Again equating to zero the coefficient of the general term i.e. of x^{k-r} , we have

$$a_{r-2} (k-r+2)(k-r+1) + (n-k+r)(n+k-r+1) a_r = 0.$$

$$\therefore a_r = - \frac{(k-r+2)(k-r+1)}{(n-k+r)(n+k-r+1)} a_{r-2}. \quad \dots(4)$$

Putting $r = 3, a_3 = -\frac{(k-1)(k-2)}{(n-k+3)(n+k-2)} a_1 = 0$, since $a_1 = 0$.

$\therefore$ we have, $a_1 = a_3 = a_5 = \dots = 0$ (each).

Now two cases arise :

Case I: When $k = n$, from (4), we have

$$a_r = -\frac{(n-r+2)(n-r+1)}{r \cdot (2n-r+1)} a_{r-2}.$$

Putting $r = 2, 4, \dots$ etc.

$$a_2 = -\frac{n(n-1)}{2(2n-1)} \cdot a_0,$$

$$a_4 = -\frac{(n-2)(n-3)}{4 \cdot (2n-3)} \cdot a_2 = \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4 \cdot (2n-1)(2n-3)} \cdot a_0, \text{ etc.}$$

$$\therefore y = a_0 x^n + a_2 x^{n-2} + a_4 x^{n-4} + \dots$$

$$\text{or } y = a_0 \left[x^n - \frac{n(n-1)}{2 \cdot (2n-1)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4 \cdot (2n-1)(2n-3)} x^{n-4} - \dots \right], \quad \dots(5)$$

which is one solution of Legendre's equation.

Case II: When $k = -(n+1)$, from (4), we have

$$a_r = \frac{(n+r-1)(n+r)}{r(2n+r+1)} a_{r-2}.$$

Putting $r = 2, 4, \dots$ etc.

$$a_2 = \frac{(n+1)(n+2)}{2(2n+3)} \cdot a_0$$

$$a_4 = \frac{(n+3)(n+4)}{4 \cdot (2n+5)} \cdot a_2 = \frac{(n+1)(n+2)(n+3)(n+4)}{2 \cdot 4 \cdot (2n+3)(2n+5)} \cdot a_0 \text{ etc.}$$

$$\therefore y = \sum_{r=0}^{\infty} a_r x^{-n-1-r} = a_0 x^{-n-1} + a_2 x^{-n-3} + a_4 x^{-n-5} + \dots$$

$$= a_0 \left[x^{-n-1} + \frac{(n+1)(n+2)}{2 \cdot (2n+3)} x^{-n-3} + \frac{(n+1)(n+2)(n+3)(n+4)}{2 \cdot 4 \cdot (2n+3)(2n+5)} x^{-n-5} + \dots \right] \quad \dots(6)$$

which is other solution of Legendre's equation.

13.3 Definition of $P_n(x)$ (Legendre's Polynomial) and $Q_n(x)$

The solution of Legendre's equation is called Legendre's function.

When n is a positive integer and $a_0 = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{n!}$,

the solution (5) of article 13.2 is denoted by $P_n(x)$ and is called **Legendre's function of the first kind**.

$$\therefore P_n(x) = \frac{1.3.5 \dots (2n-1)}{n!} \left[x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2.4.(2n-1)(2n-3)} x^{n-4} - \dots \right].$$

$P_n(x)$ is a terminating series and gives what are called **Legendre's Polynomials** for different values of n .

$$\text{We can write } P_n(x) = \sum_{r=0}^{(n/2)} (-1)^r \frac{(2n-2r)!}{2^n r! (n-2r)! (n-r)!} x^{n-2r}$$

(Avadh 2006; Rohilkhand 10)

where $(n/2) = \begin{cases} n/2, & \text{if } n \text{ is even} \\ (n-1)/2, & \text{if } n \text{ is odd} \end{cases}$

Again when n is a positive integer and $a_0 = \frac{n!}{1.3.5 \dots (2n+1)}$,

the solution (6) of 13.2 is denoted by $Q_n(x)$ and is called the **Legendre's function of the second kind**.

$$\therefore Q_n(x) = \frac{n!}{1.3.5 \dots (2n+1)} \left[x^{-n-1} + \frac{(n+1)(n+2)}{2(2n+3)} x^{-n-3} + \frac{(n+1)(n+2)(n+3)(n+4)}{2.4(2n+3)(2n+5)} x^{-n-5} + \dots \right].$$

$Q_n(x)$ is an infinite or non-terminating series as n is positive.

13.4 General Solution of Legendre's Equation

The most general solution of the Legendre's equation is

$$y = AP_n(x) + BQ_n(x),$$

where A and B are arbitrary constants.

13.5 $P_n(x)$ is the Coefficient of h^n in the Expansion of $(1 - 2xh + h^2)^{-1/2}$ in Ascending Powers of h

(Rohilkhand 2002; Avadh 07, 09; Lucknow 11; Purvanchal 10)

We have

$$\begin{aligned} (1 - 2xh + h^2)^{-1/2} &= \{1 - h(2x - h)\}^{-1/2} \\ &= 1 + \frac{1}{2} h(2x - h) + \frac{1.3}{2.4} h^2 (2x - h)^2 + \dots \\ &\quad + \frac{1.3 \dots (2n-3)}{2.4 \dots (2n-2)} h^{n-1} (2x - h)^{n-1} + \frac{1.3 \dots (2n-1)}{2.4 \dots (2n)} h^n (2x - h)^n + \dots \end{aligned}$$

∴ Coefficient of h^n

$$\begin{aligned}
 &= \frac{1.3 \dots (2n-1)}{2.4 \dots 2n} \cdot (2x)^n - \frac{1.3 \dots (2n-3)}{2.4 \dots (2n-2)} {}^{n-1}C_1 (2x)^{n-2} \\
 &\quad + \frac{1.3 \dots (2n-5)}{2.4 \dots (2n-4)} {}^{n-2}C_2 (2x)^{n-4} - \dots \\
 &= \frac{1.3 \dots (2n-1)}{2.4 \dots 2n} 2^n \left[x^n - \frac{2n}{2n-1} (n-1) \frac{x^{n-2}}{2^2} \right. \\
 &\quad \left. + \frac{2n(2n-2)}{(2n-1)(2n-3)} \cdot \frac{(n-2)(n-3)}{2!} \cdot \frac{x^{n-4}}{2^4} - \dots \right] \\
 &= \frac{1.3 \dots (2n-1)}{n!} \left[x^n - \frac{2n}{2n-1} \cdot (n-1) \cdot \frac{x^{n-2}}{2^2} \right. \\
 &\quad \left. + \frac{2n(2n-2)}{(2n-1)(2n-3)} \cdot \frac{(n-2)(n-3)}{2!} \cdot \frac{x^{n-4}}{2^4} - \dots \right] \\
 &= \frac{1.3 \dots (2n-1)}{n!} \left[x^n - \frac{n(n-1)}{2 \cdot (2n-1)} x^{n-2} \right. \\
 &\quad \left. + \frac{n(n-1)(n-2)(n-3)}{2.4 \cdot (2n-1)(2n-3)} x^{n-4} - \dots \right] \\
 &= P_n(x).
 \end{aligned}$$

Thus we can say that

$$\sum_{n=0}^{\infty} h^n P_n(x) = (1 - 2xh + h^2)^{-1/2}, \text{ where } P_0(x) = 1. \quad (\text{Meerut 2009})$$

Note. $(1 - 2xh + h^2)^{-1/2}$ is called the **generating function** of the Legendre polynomials.

13.6 Laplace's Definite Integrals For $P_n(x)$

(I) Laplace's First Integral for $P_n(x)$: When n is a positive integer,

$$P_n(x) = \frac{1}{\pi} \int_0^\pi [x \pm \sqrt{(x^2 - 1) \cos \phi}]^n d\phi.$$

(Rohilkhand 2004, 14; Meerut 06, 08; Agra 10; Lucknow 07; Avadh 14)

Proof: From integral calculus, we have

$$\int_0^\pi \frac{d\phi}{a \pm b \cos \phi} = \frac{\pi}{\sqrt{a^2 - b^2}}, \text{ where } a^2 > b^2.$$

Putting $a = 1 - hx$ and $b = h\sqrt{(x^2 - 1)}$ so that

$$\begin{aligned}
 a^2 - b^2 &= (1 - hx)^2 - h^2(x^2 - 1) \\
 &= 1 - 2xh + h^2,
 \end{aligned}$$

we have $\pi (1 - 2xh + h^2)^{-1/2} = \int_0^\pi [1 - hx \pm h\sqrt{(x^2 - 1) \cos \phi}]^{-1} d\phi$

$$\begin{aligned}
 &= \int_0^\pi [1 - h \{x \mp \sqrt{(x^2 - 1) \cos \phi}\}]^{-1} d\phi \\
 &= \int_0^\pi [1 - ht]^{-1} d\phi, \text{ where } t = x \mp \sqrt{(x^2 - 1) \cos \phi}
 \end{aligned}$$

or
$$\pi \sum_{n=0}^{\infty} h^n P_n(x) = \int_0^\pi (1 + ht + h^2 t^2 + \dots + h^n t^n + \dots) d\phi.$$

Equating the coefficients of h^n on both sides, we have

$$\pi P_n(x) = \int_0^\pi t^n d\phi = \int_0^\pi [x \mp \sqrt{(x^2 - 1) \cos \phi}]^n d\phi.$$

$\therefore P_n(x) = \frac{1}{\pi} \int_0^\pi [x \pm \sqrt{(x^2 - 1) \cos \phi}]^n d\phi.$

Note: Putting $x = \cos \theta$, we have

$$P_n(\cos \theta) = \frac{1}{\pi} \int_0^\pi (\cos \theta \pm i \sin \theta \cos \phi)^n d\phi.$$

(II) **Laplace's Second Integral for $P_n(x)$:** When n is a positive integer,

$$P_n(x) = \frac{1}{\pi} \int_0^\pi \frac{d\phi}{[x \pm \sqrt{(x^2 - 1) \cos \phi}]^{n+1}}.$$

(Meerut 2007; Avadh 08; Kanpur 07)

Proof: From integral calculus, we have

$$\int_0^\pi \frac{d\phi}{a \pm b \cos \phi} = \frac{\pi}{\sqrt{a^2 - b^2}}, \text{ where } a^2 > b^2.$$

Putting $a = xh - 1$ and $b = h\sqrt{(x^2 - 1)}$ so that $a^2 - b^2 = 1 - 2xh + h^2$,

we have
$$\pi (1 - 2xh + h^2)^{-1/2} = \int_0^\pi [-1 + xh \pm h\sqrt{(x^2 - 1) \cos \phi}]^{-1} d\phi$$

or
$$\frac{\pi}{h} \left[1 - 2x \frac{1}{h} + \frac{1}{h^2} \right]^{-1/2} = \int_0^\pi [h\{x \pm \sqrt{(x^2 - 1) \cos \phi}\} - 1]^{-1} d\phi$$

or
$$\begin{aligned}
 \frac{\pi}{h} \sum_{n=0}^{\infty} \frac{1}{h^n} P_n(x) &= \int_0^\pi (ht - 1)^{-1} d\phi, \text{ where } t = x \pm \sqrt{(x^2 - 1) \cos \phi} \\
 &= \int_0^\pi \frac{1}{ht} \left[1 - \frac{1}{ht} \right]^{-1} d\phi = \int_0^\pi \frac{1}{ht} \left[1 + \frac{1}{ht} + \frac{1}{h^2 t^2} + \dots + \frac{1}{h^n t^n} + \dots \right] d\phi \\
 &= \int_0^\pi \left[\frac{1}{ht} + \frac{1}{h^2 t^2} + \frac{1}{h^3 t^3} + \dots + \frac{1}{h^{n+1} t^{n+1}} + \dots \right] d\phi \\
 &= \sum_{n=0}^{\infty} \left[\frac{1}{h^{n+1}} \int_0^\pi \frac{1}{t^{n+1}} d\phi \right] \\
 &= \sum_{n=0}^{\infty} \left[\frac{1}{h^{n+1}} \int_0^\pi \frac{d\phi}{\{x \pm \sqrt{(x^2 - 1) \cos \phi}\}^{n+1}} \right].
 \end{aligned}$$

$\therefore$ Equating the coefficients of $\frac{1}{h^{n+1}}$, we have

$$\pi P_n(x) = \int_0^\pi \frac{d\phi}{\{x \pm \sqrt{(x^2 - 1) \cos \phi}\}^{n+1}}$$

or
$$P_n(x) = \frac{1}{\pi} \int_0^\pi \frac{d\phi}{\{x \pm \sqrt{(x^2 - 1) \cos \phi}\}^{n+1}}.$$

Note. Replacing n by $-(n+1)$ in Laplace's second integral, we have

$$\begin{aligned} P_{-(n+1)}(x) &= \frac{1}{\pi} \int_0^\pi \{x \pm \sqrt{(x^2 - 1) \cos \phi}\}^n d\phi \\ &= P_n(x), \text{ from Laplace's first integral.} \end{aligned}$$

Hence
$$P_{-(n+1)} = P_n.$$

Illustrative Examples

Example 1 : Show that (i) $P_n(1) = 1$, (Meerut 2007, 11; Rohilkhand 11; Kanpur 12; Purvanchal 10)

(ii) $P_n(-x) = (-1)^n P_n(x)$. (Meerut 2009; Agra 03, 10; Lucknow 10)

Hence deduce that $P_n(-1) = (-1)^n$. (Kanpur 2007, 12)

(iii) Prove that $P_n(x)$ is an even or odd function of x according as n is even or odd respectively.

Solution : (i) We know that

$$\sum_{n=0}^{\infty} h^n P_n(x) = (1 - 2xh + h^2)^{-1/2}.$$

Putting $x = 1$ on both sides, we have $\sum_{n=0}^{\infty} h^n P_n(1) = (1 - 2h + h^2)^{-1/2} = (1 - h)^{-1}$

$$= 1 + h + h^2 + \dots + h^n + \dots = \sum_{n=0}^{\infty} h^n.$$

Equating the coefficients of h^n on both sides, we have $P_n(1) = 1$.

(ii) We have, $(1 - 2xh + h^2)^{-1/2} = \sum_{n=0}^{\infty} h^n P_n(x)$ (1)

Replacing h by $-h$ on both sides of (1), we have

$$(1 + 2xh + h^2)^{-1/2} = \sum_{n=0}^{\infty} (-h)^n P_n(x) = \sum_{n=0}^{\infty} (-1)^n h^n P_n(x). \quad \dots (2)$$

Again replacing x by $-x$ on both sides of (1), we have

$$(1 + 2xh + h^2)^{-1/2} = \sum_{n=0}^{\infty} h^n P_n(-x). \quad \dots (3)$$

From (2) and (3), we have

$$\sum_{n=0}^{\infty} h^n P_n(-x) = \sum_{n=0}^{\infty} (-1)^n h^n P_n(x).$$

Equating the coefficients of h^n on both sides, we have

$$P_n(-x) = (-1)^n P_n(x).$$

Deduction: Putting $x = 1$, we have

$$P_n(-1) = (-1)^n P_n(1) = (-1)^n. \quad [\because P_n(1) = 1. \text{ See Ex. 1}]$$

(iii) From part (ii) of this question, we have

$$P_n(-x) = (-1)^n P_n(x). \quad \dots(4)$$

If n is even then $(-1)^n = 1$.

$\therefore$ From (4), if n is even, we have $P_n(-x) = P_n(x)$.

Hence $P_n(x)$ is an even function of x , if n is even.

Again if n is odd, then $(-1)^n = -1$.

$\therefore$ From (4), if n is odd, we have $P_n(-x) = -P_n(x)$.

Hence $P_n(x)$ is an odd function of x , if n is odd.

Example 2 : Prove that $P_n(0) = 0$, for n odd (Agra 2006)

and $P_n(0) = \frac{(-1)^{n/2} n!}{2^n \{(n/2)!\}^2}$, for n even.

Solution : We know that $\sum_{n=0}^{\infty} h^n P_n(x) = (1 - 2xh + h^2)^{-1/2}$(1)

Putting $x = 0$ on both sides of (1), we have

$$\begin{aligned} \sum_{n=0}^{\infty} h^n P_n(0) &= (1 + h^2)^{-1/2} = \{1 - (-h^2)\}^{-1/2} \\ &= 1 + \frac{1}{2} \cdot (-h^2) + \frac{1.3}{2.4} (-h^2)^2 + \frac{1.3.5}{2.4.6} (-h^2)^3 \\ &\quad + \dots + \frac{1.3.5 \dots (2r-1)}{2.4 \dots 2r} (-h^2)^r + \dots \quad \dots(2) \end{aligned}$$

We observe that all the powers of h on the R.H.S. of (2) are even. Therefore equating the coefficients of h^n on both sides of (2), we have

$$P_n(0) = 0, \text{ if } n \text{ is odd.}$$

Again equating the coefficients of h^{2m} on both sides of (2), we have

$$P_{2m}(0) = \frac{1.3.5 \dots (2m-1)}{2.4.6 \dots 2m} (-1)^m = (-1)^m \frac{(2m)!}{2^{2m} (m!)^2}$$

i.e. when $n = 2m$, we have $P_n(0) = \frac{(-1)^{n/2} n!}{2^n \{(n/2)!\}^2}$. Proved.

Example 3 : Show that $\frac{1 - z^2}{(1 - 2xz + z^2)^{3/2}} = \sum_{n=0}^{\infty} (2n+1) P_n(x) z^n$. (Agra 2003; Meerut 10)

Solution : We have $(1 - 2xz + z^2)^{-1/2} = \sum_{n=0}^{\infty} z^n P_n(x)$(1)

Differentiating w.r.t. z , we have

$$(x - z)(1 - 2xz + z^2)^{-3/2} = \sum_{n=0}^{\infty} n z^{n-1} P_n(x).$$

$\therefore 2(x - z)z(1 - 2xz + z^2)^{-3/2} = \sum_{n=0}^{\infty} 2n z^n P_n(x)$(2)

Adding (1) and (2), we have

$$\frac{1 - 2xz + z^2 + 2(x - z)z}{(1 - 2xz + z^2)^{3/2}} = \sum_{n=0}^{\infty} (2n+1) z^n P_n(x)$$

or
$$\frac{1 - z^2}{(1 - 2xz + z^2)^{3/2}} = \sum_{n=0}^{\infty} (2n+1) z^n P_n(x). \quad \text{Proved.}$$

Example 4 : Express $P_n(\cos \theta)$ as a series in cosines of multiples of θ .

Solution : We have $\sum_{n=0}^{\infty} h^n P_n(x) = (1 - 2xh + h^2)^{-1/2} \dots (1)$

Putting $x = \cos \theta$ in (1), we get

$$\begin{aligned} \sum_{n=0}^{\infty} h^n P_n(\cos \theta) &= (1 - 2h \cos \theta + h^2)^{-1/2} \\ &= \{1 - h(e^{i\theta} + e^{-i\theta}) + h^2\}^{-1/2} = (1 - he^{i\theta})^{-1/2} (1 - he^{-i\theta})^{-1/2} \\ &= \left\{ 1 + \frac{1}{2} he^{i\theta} + \frac{1.3}{2.4} h^2 e^{2i\theta} + \dots + \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots 2n} h^n e^{ni\theta} + \dots \right\} \\ &\quad \times \left\{ 1 + \frac{1}{2} he^{-i\theta} + \frac{1.3}{2.4} h^2 e^{-2i\theta} + \dots + \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots 2n} h^n e^{-ni\theta} + \dots \right\}. \end{aligned}$$

Equating the coefficients of h^n from both sides, we get

$$\begin{aligned} P_n(\cos \theta) &= \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots 2n} \left[(e^{ni\theta} + e^{-ni\theta}) + \frac{1}{2} \cdot \frac{2n}{2n-1} \{e^{(n-2)i\theta} + e^{-(n-2)i\theta}\} \right. \\ &\quad \left. + \frac{1.3}{2.4} \cdot \frac{2n(2n-2)}{(2n-1)(2n-3)} \{e^{(n-4)i\theta} + e^{-(n-4)i\theta}\} + \dots \right] \\ &= \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots 2n} \left[2 \cos n\theta + 2 \cdot \frac{1}{2} \cdot \frac{2n}{2n-1} \cos(n-2)\theta \right. \\ &\quad \left. + 2 \cdot \frac{1.3}{2.4} \cdot \frac{2n(2n-2)}{(2n-1)(2n-3)} \cos(n-4)\theta + \dots \right]. \end{aligned}$$

13.7 Orthogonal Properties of Legendre's Polynomials

(i) $\int_{-1}^{+1} P_m(x) P_n(x) dx = 0$ if $m \neq n$.
(Rohilkhand 2009; Agra 06, 08, 09, 10;
Meerut 09, 10, 11; Avadh 10; Kanpur 07, 08, 10;
Lucknow 10, 11; Purvanchal 10)

(ii) $\int_{-1}^{+1} [P_n(x)]^2 dx = \frac{2}{2n+1}$.
(Agra 2009, 10; Rohilkhand 09, 10;
Kanpur 10; Purvanchal 10)

Proof: (i) Legendre's equation may be written as

$$\frac{d}{dx} \left\{ (1-x^2) \frac{dy}{dx} \right\} + n(n+1)y = 0.$$

Since $P_n(x)$ is a solution of Legendre's equation, therefore

$$\frac{d}{dx} \left\{ (1-x^2) \frac{dP_n}{dx} \right\} + n(n+1)P_n = 0. \quad \dots(1)$$

$$\text{Similarly, } \frac{d}{dx} \left\{ (1-x^2) \frac{dP_m}{dx} \right\} + m(m+1)P_m = 0. \quad \dots(2)$$

Multiplying (1) by P_m and (2) by P_n and then subtracting, we have

$$P_m \frac{d}{dx} \left\{ (1-x^2) \frac{dP_n}{dx} \right\} - P_n \frac{d}{dx} \left\{ (1-x^2) \frac{dP_m}{dx} \right\} + \{n(n+1) - m(m+1)\}P_n P_m = 0.$$

Integrating between the limits -1 to 1 , we have

$$\int_{-1}^1 \left[P_m \frac{d}{dx} \left\{ (1-x^2) \frac{dP_n}{dx} \right\} \right] dx - \int_{-1}^1 \left[P_n \frac{d}{dx} \left\{ (1-x^2) \frac{dP_m}{dx} \right\} \right] dx$$

$$\text{Integrating by parts, we have} \quad + \{n(n+1) - m(m+1)\} \int_{-1}^1 P_m P_n dx = 0.$$

$$\begin{aligned} & \left[P_m (1-x^2) \frac{dP_n}{dx} \right]_{-1}^{+1} - \int_{-1}^{+1} \frac{dP_m}{dx} \left\{ (1-x^2) \frac{dP_n}{dx} \right\} dx \\ & - \left[P_n (1-x^2) \frac{dP_m}{dx} \right]_{-1}^{+1} + \int_{-1}^{+1} \frac{dP_n}{dx} \left\{ (1-x^2) \frac{dP_m}{dx} \right\} dx \\ & + [n(n+1) - m(m+1)] \int_{-1}^{+1} P_m P_n dx = 0. \end{aligned}$$

$$\therefore \{n(n+1) - m(m+1)\} \int_{-1}^{+1} P_m P_n dx = 0.$$

$$\text{Hence, } \int_{-1}^{+1} P_m(x) P_n(x) dx = 0 \text{ since } m \neq n.$$

$$(ii) \text{ We have } (1-2xh+h^2)^{-1/2} = \sum_{n=0}^{\infty} h^n P_n(x).$$

Squaring both sides, we have

$$(1-2xh+h^2)^{-1} = \sum_{n=0}^{\infty} h^{2n} \{P_n(x)\}^2 + \sum_{\substack{m,n=0 \\ m \neq n}}^{\infty} h^{m+n} P_m(x) P_n(x).$$

Integrating between the limits -1 to $+1$, we have

$$\begin{aligned} & \sum_{n=0}^{\infty} \int_{-1}^{+1} h^{2n} [P_n(x)]^2 dx + \sum_{\substack{m,n=0 \\ m \neq n}}^{\infty} \int_{-1}^{+1} h^{m+n} P_m(x) P_n(x) dx \\ & = \int_{-1}^{+1} \frac{dx}{(1-2xh+h^2)} \end{aligned}$$

$$\text{or } \sum_{n=0}^{\infty} \int_{-1}^{+1} h^{2n} [P_n(x)]^2 dx = \int_{-1}^{+1} \frac{dx}{(1-2xh+h^2)},$$

$$\begin{aligned} & \text{since other integrals on the L.H.S. are zero by (i) as } m \neq n \\ & = -\frac{1}{2h} \left[\log(1-2xh+h^2) \right]_{-1}^{+1} = -\frac{1}{2h} \{ \log(1-h)^2 - \log(1+h)^2 \} \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2h} \left[\log \left\{ \frac{1+h}{1-h} \right\}^2 \right] = \frac{1}{h} \log \left\{ \frac{1+h}{1-h} \right\} = \frac{2}{h} \left\{ h + \frac{h^3}{3} + \frac{h^5}{5} + \dots \right\} \\
 &= 2 \left\{ 1 + \frac{h^2}{3} + \frac{h^4}{5} + \dots + \frac{h^{2n}}{2n+1} + \dots \right\} = \sum_{n=0}^{\infty} \frac{2h^{2n}}{2n+1}.
 \end{aligned}$$

Equating the coefficients of h^{2n} , we have

$$\int_{-1}^{+1} [P_n(x)]^2 dx = \frac{2}{2n+1}.$$

Remark : Making use of the Kronecker delta, the above results (i) and (ii) can be written in compact form as

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn},$$

(Avadh 2008; Lucknow 09)

where Kronecker delta, δ_{mn} , is defined as

$$\delta_{mn} = \begin{cases} 0, & \text{if } m \neq n \\ 1, & \text{if } m = n. \end{cases}$$

13.8 Recurrence Formulae

$$(I) \quad (2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1}.$$

(Meerut 2006, 08; Agra 02, 06;

Rohilkhand 06; Avadh 06, 13; Lucknow 09, 11; Purvanchal 09, 11)

Proof: We have $(1-2xh+h^2)^{-1/2} = \sum_{n=0}^{\infty} h^n P_n(x)$.

Differentiating both sides w.r.t. 'h', we have

$$-\frac{1}{2}(1-2xh+h^2)^{-3/2}(-2x+2h) = \sum_{n=0}^{\infty} nh^{n-1} P_n(x)$$

$$\text{or} \quad (x-h)(1-2xh+h^2)^{-1/2} = (1-2xh+h^2) \sum_{n=0}^{\infty} nh^{n-1} P_n(x)$$

$$\text{or} \quad (x-h) \sum_{n=0}^{\infty} h^n P_n(x) = (1-2xh+h^2) \sum_{n=0}^{\infty} nh^{n-1} P_n(x)$$

$$\begin{aligned}
 \text{or} \quad & (x-h)[P_0(x) + hP_1(x) + \dots + h^{n-1}P_{n-1}(x) + h^n P_n(x) + \dots] \\
 &= (1-2xh+h^2)[P_1(x) + 2hP_2(x) + \dots + (n-1)h^{n-2}P_{n-1}(x) \\
 &\quad + nh^{n-1}P_n(x) + (n+1)h^n P_{n+1}(x) + \dots] \quad \dots (1)
 \end{aligned}$$

Equating the coefficients of h^n on both sides of (1), we have

$$xP_n(x) - P_{n-1}(x) = (n+1)P_{n+1}(x) - 2xn P_n(x) + (n-1)P_{n-1}(x)$$

$$\text{or} \quad (2n+1)xP_n(x) = (n+1)P_{n+1}(x) + nP_{n-1}(x).$$

In short $(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1}$.

Note. Equating the coefficients of h^{n-1} on both sides of (1), we get

$$xP_{n-1}(x) - P_{n-2}(x) = nP_n(x) - 2x(n-1)P_{n-1}(x) + (n-2)P_{n-2}(x)$$

or $nP_n = (2n-1)xP_{n-1} - (n-1)P_{n-2}$.

(II) $nP_n = xP_n' - P_{n-1}'$, where dashes denote differentiation w.r.t. 'x'.

(Agra 2003; Avadh 07; Lucknow 06; Kanpur 08, 11, 12)

Proof: We have $(1 - 2xh + h^2)^{-1/2} = \sum_{n=0}^{\infty} h^n P_n(x)$... (1)

Differentiating (1) w.r.t. 'h', we have

$$(x-h)(1-2xh+h^2)^{-3/2} = \sum_{n=0}^{\infty} nh^{n-1}P_n(x). \quad \dots(2)$$

Again differentiating (1) w.r.t. 'x', we have

$$h(1-2xh+h^2)^{-3/2} = \sum_{n=0}^{\infty} h^n P_n'(x)$$

or $h(x-h)(1-2xh+h^2)^{-3/2} = (x-h) \sum_{n=0}^{\infty} h^n P_n'(x). \quad \dots(3)$

From (2) and (3), we have

$$h \sum_{n=0}^{\infty} nh^{n-1}P_n(x) = (x-h) \sum_{n=0}^{\infty} h^n P_n'(x)$$

or $h[h^0 P_1(x) + 2hP_2(x) + \dots + nh^{n-1}P_n(x) + \dots]$
 $= (x-h)[P_0'(x) + hP_1'(x) + \dots + h^{n-1}P_{n-1}'(x) + h^n P_n'(x) + \dots].$

Equating the coefficients of h^n on both sides, we have

$$nP_n(x) = xP_n'(x) - P_{n-1}'(x).$$

In short $nP_n = xP_n' - P_{n-1}'$.

(III) $(2n+1)P_n = P_{n+1}' - P_{n-1}'$. (Lucknow 2008; Avadh 12)

Proof: From recurrence formula I, we have

$$(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1}.$$

Differentiating w.r.t. x, we have

$$(2n+1)xP_n' + (2n+1)P_n = (n+1)P_{n+1}' + nP_{n-1}'. \quad \dots(1)$$

From recurrence formula II, we have

$$xP_n' = nP_n + P_{n-1}'. \quad \dots(2)$$

Eliminating xP_n' from (1) and (2), we have

$$(2n+1)(nP_n + P_{n-1}') + (2n+1)P_n = (n+1)P_{n+1}' + nP_{n-1}'$$

or $(2n+1)(n+1)P_n = (n+1)P_{n+1}' + nP_{n-1}' - (2n+1)P_{n-1}'$

or $(2n+1)(n+1)P_n = (n+1)P_{n+1}' - (n+1)P_{n-1}'$.

$\therefore (2n+1)P_n = P_{n+1}' - P_{n-1}'$.

(IV) $(n+1)P_n = P_{n+1}' - xP_n'$. (Rohilkhand 2006, 11; Kanpur 10; Purvanchal 14)

Proof: Writing recurrence formulae II and III, we have

$$nP_n = xP_n' - P_{n-1}' \quad \dots(1)$$

$$\text{and} \quad (2n+1)P_n = P'_{n+1} - P'_{n-1}. \quad \dots(2)$$

Subtracting (1) from (2), we have

$$(n+1)P_n = P'_{n+1} - xP'_n.$$

$$(V) \quad (1-x^2)P'_n = n(P_{n-1} - xP_n). \quad (\text{Roholkhand 2006, 11; Kanpur 10})$$

Proof: Replacing n by $(n-1)$ in recurrence formula IV, we have

$$nP_{n-1} = P'_n - xP'_{n-1}. \quad \dots(1)$$

Writing recurrence formula II, we have

$$nP_n = xP'_n - P'_{n-1}. \quad \dots(2)$$

Multiplying (2) by x and then subtracting from (1), we have

$$n(P_{n-1} - xP_n) = (1-x^2)P'_n$$

$$\text{i.e.} \quad (1-x^2)P'_n = n(P_{n-1} - xP_n).$$

Aliter: From Laplace's first integral, we have

$$P_n(x) = \frac{1}{\pi} \int_0^\pi \{x + \sqrt{(x^2-1)} \cos \phi\}^n d\phi.$$

Replacing n by $(n-1)$, we have

$$P_{n-1}(x) = \frac{1}{\pi} \int_0^\pi \{x + \sqrt{(x^2-1)} \cos \phi\}^{n-1} d\phi.$$

$$\begin{aligned} \therefore P_{n-1} - xP_n &= \frac{1}{\pi} \int_0^\pi [\{x + \sqrt{(x^2-1)} \cos \phi\}^{n-1} - x\{x + \sqrt{(x^2-1)} \cos \phi\}^n] d\phi \\ &= \frac{1}{\pi} \int_0^\pi \{x + \sqrt{(x^2-1)} \cos \phi\}^{n-1} [1 - x\{x + \sqrt{(x^2-1)} \cos \phi\}] d\phi \\ &= \frac{1}{\pi} \int_0^\pi \{x + \sqrt{(x^2-1)} \cos \phi\}^{n-1} [(1-x^2) - x\sqrt{(x^2-1)} \cos \phi] d\phi \\ &= -\frac{(x^2-1)}{\pi} \int_0^\pi \{x + \sqrt{(x^2-1)} \cos \phi\}^{n-1} \left\{1 + \frac{x}{\sqrt{(x^2-1)}} \cos \phi\right\} d\phi \\ &= -\frac{(x^2-1)}{\pi} \int_0^\pi \left[\{x + \sqrt{(x^2-1)} \cos \phi\}^{n-1} \times \frac{d}{dx} \{x + \sqrt{(x^2-1)} \cos \phi\}\right] d\phi \\ &= -\frac{(x^2-1)}{\pi} \int_0^\pi \left\{\frac{1}{n} \frac{d}{dx} \{x + \sqrt{(x^2-1)} \cos \phi\}^n\right\} d\phi \\ &= \frac{(1-x^2)}{\pi n} \frac{d}{dx} \int_0^\pi \{x + \sqrt{(x^2-1)} \cos \phi\}^n d\phi \\ &= \frac{(1-x^2)}{\pi n} \frac{d}{dx} \{\pi P_n(x)\} \quad [\text{from Laplace's first integral}] \\ &= \frac{(1-x^2)}{n} \cdot P'_n(x). \end{aligned}$$

$$\text{Hence } (1-x^2)P'_n = n(P_{n-1} - xP_n).$$

$$(VI) \quad (1-x^2)P'_n = (n+1)(xP_n - P_{n+1}) \quad (\text{Agra 2011})$$

Proof: Writing recurrence formula I, we have

$$(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1} \quad \dots(1)$$

which may be written as

$$(n+1)xP_n + nP_n = (n+1)P_{n+1} + nP_{n-1}$$

$$\text{or} \quad (n+1)(xP_n - P_{n+1}) = n(P_{n-1} - xP_n).$$

Writing recurrence formula V, we have

$$(1-x^2)P_n' = n(P_{n-1} - xP_n). \quad \dots(2)$$

From (1) and (2), we have

$$(1-x^2)P_n' = (n+1)(xP_n - P_{n+1}).$$

Aliter: From Laplace's second integral, we have

$$P_n(x) = \frac{1}{\pi} \int_0^\pi \frac{d\phi}{[x + \sqrt{(x^2-1)\cos\phi}]^{n+1}}.$$

$$\begin{aligned} \therefore xP_n - P_{n+1} &= \frac{x}{\pi} \int_0^\pi \frac{d\phi}{[x + \sqrt{(x^2-1)\cos\phi}]^{n+1}} - \frac{1}{\pi} \int_0^\pi \frac{d\phi}{[x + \sqrt{(x^2-1)\cos\phi}]^{n+2}} \\ &= \frac{1}{\pi} \int_0^\pi \{x + \sqrt{(x^2-1)\cos\phi}\}^{-n-2} [x\{x + \sqrt{(x^2-1)\cos\phi}\} - 1] d\phi \\ &= \frac{1}{\pi} \int_0^\pi \{x + \sqrt{(x^2-1)\cos\phi}\}^{-n-2} \{(x^2-1) + x\sqrt{(x^2-1)\cos\phi}\} d\phi \\ &= \frac{(x^2-1)}{\pi} \int_0^\pi \{x + \sqrt{(x^2-1)\cos\phi}\}^{-n-2} \left\{1 + \frac{x}{\sqrt{(x^2-1)}} \cos\phi\right\} d\phi \\ &= \frac{(x^2-1)}{\pi} \int_0^\pi \left[\{x + \sqrt{(x^2-1)\cos\phi}\}^{-n-2} \cdot \frac{d}{dx} \{x + \sqrt{(x^2-1)\cos\phi}\}\right] d\phi \\ &= -\frac{(x^2-1)}{\pi} \int_0^\pi \left[\frac{1}{n+1} \frac{d}{dx} \{x + \sqrt{(x^2-1)\cos\phi}\}^{-n-1}\right] d\phi \\ &= \frac{(1-x^2)}{\pi(n+1)} \frac{d}{dx} \int_0^\pi \{x + \sqrt{(x^2-1)\cos\phi}\}^{-n-1} d\phi \\ &= \frac{(1-x^2)}{\pi(n+1)} \frac{d}{dx} \{\pi P_n(x)\} = \frac{(1-x^2)P_n'}{n+1}. \end{aligned}$$

$$\text{Hence } (1-x^2)P_n' = (n+1)(xP_n - P_{n+1}).$$

13.9 Beltrami's Result

To prove that $(2n+1)(x^2-1)P_n' = n(n+1)(P_{n+1} - P_{n-1})$.

Proof: From recurrence formulae V and VI, we have

$$(1-x^2)P_n' = n(P_{n-1} - xP_n) \quad \dots(1)$$

$$\text{and} \quad (1-x^2)P_n' = (n+1)(xP_n - P_{n+1}). \quad \dots(2)$$

Multiplying (1) by $n+1$, (2) by n and adding, we get

$$[(n+1)+n](1-x^2)P_n' = n(n+1)(P_{n-1} - P_{n+1})$$

$$\text{or} \quad (2n+1)(x^2-1)P_n' = n(n+1)(P_{n+1} - P_{n-1}).$$

This result is known as **Beltrami's result**.

13.10 Christoffel's Expansion

To prove that $P_n' = (2n-1)P_{n-1} + (2n-5)P_{n-3} + (2n-9)P_{n-5} + \dots$, the last term of the series being $3P_1$ or P_0 according as n is even or odd.

Proof. From Recurrence formula III, we have

$$P_{n+1}' = (2n+1)P_n + P_{n-1}'. \quad \dots(A)$$

$$\text{Replacing } n \text{ by } (n-1), \text{ we have } P_n' = (2n-1)P_{n-1} + P_{n-2}'. \quad \dots(1)$$

Replacing n by $(n-2)$, $(n-4)$, ... in (i), we have

$$P_{n-2}' = (2n-5)P_{n-3} + P_{n-4}' \quad \dots(2)$$

$$P_{n-4}' = (2n-9)P_{n-5} + P_{n-6}' \quad \dots(3)$$

$$\dots \dots \dots \dots \dots \dots \dots$$

$$P_2' = 3P_1 + P_0', \text{ when } n \text{ is even.}$$

Adding (1), (2), (3), etc., we have when n is even

$$\begin{aligned} P_n' &= (2n-1)P_{n-1} + (2n-5)P_{n-3} + (2n-9)P_{n-5} + \dots + 3P_1 + P_0' \\ &= (2n-1)P_{n-1} + (2n-5)P_{n-3} + (2n-9)P_{n-5} + \dots + 3P_1 \quad [\because P_0' = 0]. \end{aligned}$$

Again when n is odd, the last of the above relations is

$$P_3' = 5P_2 + P_1' = 5P_2 + P_0. \quad [\because P_1' = 1 = P_0]$$

Adding as before, we have when n is odd

$$P_n' = (2n-1)P_{n-1} + (2n-5)P_{n-3} + \dots + 5P_2 + P_0.$$

$$\text{Hence} \quad P_n' = (2n-1)P_{n-1} + (2n-5)P_{n-3} + (2n-9)P_{n-5} + \dots$$

the last term of the series being $3P_1$ or P_0 according as n is even or odd.

13.11 Christoffel's Summation Formula

$$\text{To prove that } \sum_{r=0}^n (2r+1)P_r(x)P_r(y) = (n+1) \frac{P_{n+1}(x)P_n(y) - P_{n+1}(y)P_n(x)}{(x-y)}.$$

Proof: From Recurrence formula I, we have

$$(2r+1)xP_r(x) = (r+1)P_{r+1}(x) + rP_{r-1}(x) \quad \dots(1)$$

$$\text{and} \quad (2r+1)yP_r(y) = (r+1)P_{r+1}(y) + rP_{r-1}(y). \quad \dots(2)$$

Multiplying (1) by $P_r(y)$ and (2) by $P_r(x)$ and then subtracting, we have

$$\begin{aligned} (2r+1)(x-y)P_r(x)P_r(y) &= (r+1)[P_{r+1}(x)P_r(y) - P_{r+1}(y)P_r(x)] \\ &\quad - r[P_{r-1}(y)P_r(x) - P_{r-1}(x)P_r(y)]. \quad \dots(3) \end{aligned}$$

$$\text{Now} \quad (x-y)P_0(x)P_0(y) = P_1(x)P_0(y) - P_1(y)P_0(x) \quad \dots(A_0)$$

$$[\because P_0(x) = 1 = P_0(y), \text{ and } P_1(x) = x, P_1(y) = y]$$

Again putting $r = 1, 2, 3, \dots, (n-1), n$ in (3), we have

$$3(x-y) P_1(x) P_1(y) = 2 [P_2(x) P_1(y) - P_2(y) P_1(x)] \\ - 1 \cdot [P_0(y) P_1(x) - P_0(x) P_1(y)] \quad \dots (A_1)$$

$$5(x-y) P_2(x) P_2(y) = 3 [P_3(x) P_2(y) - P_3(y) P_2(x)] \\ - 2 \cdot [P_1(y) P_2(x) - P_1(x) P_2(y)] \quad \dots (A_2)$$

$$\dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots \dots$$

$$(2n-1)(x-y) P_{n-1}(x) P_{n-1}(y) \\ = n \cdot [P_n(x) \cdot P_{n-1}(y) - P_n(y) P_{n-1}(x)] \\ - (n-1) [P_{n-2}(y) P_{n-1}(x) - P_{n-2}(x) P_{n-1}(y)] \quad \dots (A_{n-1})$$

$$(2n+1)(x-y) P_n(x) P_n(y) \\ = (n+1) [P_{n+1}(x) P_n(y) - P_{n+1}(y) P_n(x)] \\ - n [P_{n-1}(y) P_n(x) - P_{n-1}(x) P_n(y)] \quad \dots (A_n)$$

Adding $(A_0), (A_1), (A_2), \dots, (A_{n-1})$ and (A_n) , we have

$$(x-y) \sum_{r=0}^n (2r+1) P_r(x) P_r(y) \\ = (n+1) [P_{n+1}(x) P_n(y) - P_{n+1}(y) P_n(x)].$$

Hence $\sum_{r=0}^n (2r+1) P_r(x) P_r(y) = (n+1) \frac{P_{n+1}(x) P_n(y) - P_{n+1}(y) P_n(x)}{(x-y)}.$

This is Christoffel's Summation Formula.

Illustrative Examples

Example 5 : Show that $xP_n'(x) = nP_n(x) + (2n-3)P_{n-2}(x) + (2n-7)P_{n-4}(x) + \dots$ and hence deduce the following :

(i) $\int_{-1}^1 x P_n P_n' dx = \frac{2n}{2n+1} \cdot$ (ii) $\int_{-1}^1 x P_n P_m' dx = \text{either } 0, 2 \text{ or } \frac{2n}{2n+1}.$

Solution : From recurrence formula II, we have

$$xP_n' = nP_n + P_{n-1}' \quad \dots (1)$$

Also from recurrence formula III, we have

$$P_{n+1}' = (2n+1) P_n + P_{n-1}' \quad \dots (2)$$

In (2) replacing n by $(n-2), (n-4), (n-6)$ and so on, we get

$$P_{n-1}' = (2n-3) P_{n-2} + P_{n-3}' \quad \dots (3)$$

$$P_{n-3}' = (2n-7) P_{n-4} + P_{n-5}' \quad \dots (4)$$

$$P_{n-5}' = (2n-11) P_{n-6} + P_{n-7}' \quad \dots (5)$$

$$\dots \dots \dots \dots \dots$$

Adding (1), (3), (4), (5) etc., we get

$$xP_n' = nP_n + (2n-3) P_{n-2} + (2n-7) P_{n-4} + \dots \quad \dots (6)$$

Deduction (i): Multiplying both sides of (6) by P_n and then integrating w.r.t. 'x' between the limits -1 to 1 , we get

$$\int_{-1}^1 x P_n P_n' dx = n \int_{-1}^1 P_n^2 dx, \text{ the other integrals being zero as}$$

$$\int_{-1}^1 P_m P_n dx = 0, \text{ if } m \neq n$$

$$= n \cdot \frac{2}{2n+1} = \frac{2n}{2n+1}.$$

Deduction (ii): From (6), we have

$$\begin{aligned} x P_m' &= m P_m + (2m-3) P_{m-2} + (2m-7) P_{m-4} + \dots \\ \Rightarrow \int_{-1}^1 x P_n P_m' dx &= m \int_{-1}^1 P_n P_m dx + (2m-3) \int_{-1}^1 P_n P_{m-2} dx \\ &\quad + (2m-7) \int_{-1}^1 P_n P_{m-4} dx + \dots \end{aligned}$$

Now if m and n are distinct positive integers, then

$$\int_{-1}^1 x P_n P_m' dx = 0, \text{ since all the integrals on the R.H.S. vanish.}$$

Again if $n = m$, then $\int_{-1}^1 x P_n P_m' dx = n \int_{-1}^1 P_n^2 dx$, the other integrals on the R.H.S.

$$\text{being zero} \quad = n \cdot \frac{2}{2n+1} = \frac{2n}{2n+1}$$

$$\text{If } n = m-2, \text{ then } \int_{-1}^1 x P_n P_m' dx = 0 + (2m-3) \int_{-1}^1 P_n^2 dx + 0 + \dots$$

$$= \{2(n+2)-3\} \cdot \frac{2}{2n+1} = 2.$$

$$\text{If } n = m-4, \text{ then } \int_{-1}^1 x P_n P_m' dx = 0 + 0 + (2m-7) \int_{-1}^1 P_n^2 dx + 0 + \dots$$

$$= \{2(n+4)-7\} \frac{2}{2n+1} = 2.$$

In the same way, if $n = (m-6), (m-8)$ etc., the value of the integral is 2.

Example 6 : Prove that $P_0^2(x) + 3P_1^2(x) + 5P_2^2(x) + \dots + (2n+1)P_n^2(x)$

$$\begin{aligned} &= (n+1)[P_n(x)P_{n+1}'(x) - P_{n+1}(x)P_n'(x)] \\ &= (n+1)^2 P_n^2(x) + (1-x^2)\{P_n'(x)\}^2. \end{aligned}$$

Solution : From Christoffel's summation formula, we have

$$(x-y) \sum_{r=0}^n (2r+1) P_r(x) P_r(y)$$

$$= (n+1)[P_{n+1}(x)P_n(y) - P_{n+1}(y)P_n(x)].$$

Putting $y = x+h$, where h is a small quantity, we have

$$-h \sum_{r=0}^n (2r+1) P_r(x) P_r(x+h) = (n+1)[P_{n+1}(x)P_n(x+h) - P_{n+1}(x+h)P_n(x)].$$

Expanding by Taylor's theorem, we have

$$\begin{aligned}
 & -h \sum_{r=0}^n (2r+1)P_r(x)\{P_r(x) + hP_r'(x) + \dots\} \\
 & = (n+1) \left[P_{n+1}(x) \left\{ P_n(x) + hP_n'(x) + \frac{h^2}{2!} P_n''(x) + \dots \right\} \right. \\
 & \quad \left. - \left\{ P_{n+1}(x) + hP_{n+1}'(x) + \frac{h^2}{2!} P_{n+1}''(x) + \dots \right\} P_n(x) \right] \\
 & = -h(n+1)[\{P_n(x)P_{n+1}'(x) - P_n'(x)P_{n+1}(x)\} + h(\dots) + \dots]
 \end{aligned}$$

or

$$\begin{aligned}
 & \sum_{r=0}^n (2r+1)P_r(x)\{P_r(x) + hP_r'(x) + \dots\} \\
 & = (n+1)[P_n(x)P_{n+1}'(x) - P_n'(x)P_{n+1}(x) + h(\dots) + \dots].
 \end{aligned}$$

Taking limit as $h \rightarrow 0$, we have

$$\sum_{r=0}^n (2r+1)P_r^2(x) = (n+1)[P_n(x)P_{n+1}'(x) - P_n'(x)P_{n+1}(x)]$$

or

$$\begin{aligned}
 & P_0^2(x) + 3P_1^2(x) + 5P_2^2(x) + \dots + (2n+1)P_n^2(x) \\
 & = (n+1)[P_n(x)P_{n+1}'(x) - P_{n+1}(x)P_n'(x)]. \quad \text{Proved.}
 \end{aligned}$$

Again

$$\begin{aligned}
 & (n+1)^2 P_n^2(x) + (1-x^2)\{P_n'(x)\}^2 \\
 & = (n+1)P_n(x)[(n+1)P_n(x)] + P_n'(x)[(1-x^2)P_n'(x)] \\
 & = (n+1)P_n(x)[P_{n+1}'(x) - xP_n'(x)] + P_n'(x)[(n+1)\{xP_n(x) - P_{n+1}(x)\}], \\
 & \quad \text{from recurrence formula IV and VI} \\
 & = (n+1)[P_n(x)P_{n+1}'(x) - P_{n+1}(x)P_n'(x)].
 \end{aligned}$$

Example 7 : Prove that $\int_{-1}^{+1} x^2 P_{n+1} P_{n-1} dx = \frac{2n(n+1)}{(2n-1)(2n+1)(2n+3)}.$

Solution : From Recurrence formula I, we have (Rohilkhand 2011; Purvanchal 10)

$$(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1}.$$

Replacing n by $(n-1)$ and $(n+1)$ successively, we have

$$(2n-1)xP_{n-1} = nP_n + (n-1)P_{n-2} \quad \dots(1)$$

and

$$(2n+3)xP_{n+1} = (n+2)P_{n+2} + (n+1)P_n. \quad \dots(2)$$

Multiplying (1) and (2), we get

$$\begin{aligned}
 (2n-1)(2n+3)x^2 P_{n+1} P_{n-1} & = n(n+1)P_n^2 + n(n+2)P_n P_{n+2} \\
 & \quad + (n-1)(n+2)P_{n-2} P_{n+2} + (n-1)(n+1)P_{n-2} P_n.
 \end{aligned}$$

Integrating between the limits -1 to $+1$, we have

$$(2n-1)(2n+3) \int_{-1}^{+1} x^2 P_{n+1} P_{n-1} dx = n(n+1) \int_{-1}^{+1} P_n^2 dx,$$

all other integrals being zero by article 13.7 (i)

$$= n(n+1) \frac{2}{(2n+1)}. \quad [\text{By article 13.7}]$$

$$\therefore \int_{-1}^{+1} x^2 P_{n+1} P_{n-1} dx = \frac{2n(n+1)}{(2n-1)(2n+1)(2n+3)}.$$

Example 8 : Prove that $\int_{-1}^{+1} (P_n')^2 dx = n(n+1)$.

Solution : From Christoffel's expansion, we have

$$P_n' = (2n-1)P_{n-1} + (2n-5)P_{n-3} + (2n-9)P_{n-5} + \dots \quad \dots(1)$$

the last term is P_0 i.e. 1 or P_1 i.e. $3x$ according as n is odd or even.

When n is odd, let m be the number of terms on the R.H.S. of (1).

Then the last coeff. $1 = (2n-1) + (m-1) \cdot (-4)$;

$$\therefore m = \frac{n+1}{2}.$$

Again **when n is even,** let m' be the number of terms on the R.H.S. of (1).

Then the last coeff. $3 = (2n-1) + (m'-1)(-4)$.

$$\therefore m' = n/2.$$

$$\begin{aligned} \text{Now } (P_n')^2 &= (2n-1)^2 P_{n-1}^2 + (2n-5)^2 P_{n-3}^2 + \dots \\ &\quad + 2(2n-1)(2n-5)P_{n-1} P_{n-3} + \dots \end{aligned}$$

$$\begin{aligned} \therefore \int_{-1}^{+1} (P_n')^2 dx &= (2n-1)^2 \int_{-1}^{+1} P_{n-1}^2 dx + (2n-5)^2 \int_{-1}^{+1} P_{n-3}^2 dx + \dots, \\ &\quad \text{the other integrals are zero by article 13.7 (i)} \\ &= (2n-1)^2 \frac{2}{2(n-1)+1} + (2n-5)^2 \cdot \frac{2}{2(n-3)+1} \\ &\quad + (2n-9)^2 \frac{2}{2(n-5)+1} + \dots \\ &= 2 [(2n-1) + (2n-5) + (2n-9) + \dots], \quad \dots(2) \end{aligned}$$

the last term of the A.P. on the R.H.S. of (2) being 1 or 3 according as n is odd or even.

Case I: When n is even, the number of terms in the arithmetic progression on the R.H.S. of (2) is $n/2$.

$$\begin{aligned} \therefore \int_{-1}^{+1} (P_n')^2 dx &= 2 \cdot \frac{n/2}{2} \cdot [\text{first term} + \text{last term}], \text{ summing the A.P.} \\ &= \frac{n}{2} [(2n-1) + 3] = n(n+1). \end{aligned}$$

Case II: When n is odd, the number of terms in the A.P. on the R.H.S. of (2) is $(n+1)/2$.

$$\begin{aligned} \therefore \int_{-1}^{+1} (P_n')^2 dx &= 2 \cdot \frac{(n+1)/2}{2} \cdot [\text{first term} + \text{last term}] \\ &= \frac{n+1}{2} \cdot [(2n-1) + 1] = n(n+1). \end{aligned}$$

$$\text{Hence } \int_{-1}^{+1} (P_n')^2 dx = n(n+1).$$

Proved.

13.12 Rodrigue's Formula

To prove that $P_n(x) = \frac{1}{n!2^n} \cdot \frac{d^n}{dx^n} (x^2 - 1)^n$.

(Meerut 2002, 07, 09, 11, 13; Avadh 11; Rohilkhand 02, 04, 07, 11; Lucknow 06, 08, 10; Agra 03, 11; Kanpur 08; Purvanchal 09, 11, 14)

Proof: Let $y = (x^2 - 1)^n$.

Differentiating, $\frac{dy}{dx} = n(x^2 - 1)^{n-1} \cdot 2x$.

$$\therefore (x^2 - 1) \frac{dy}{dx} = 2nxy.$$

Differentiating both sides $(n+1)$ times by Leibnitz's Theorem, we have

$$\begin{aligned} (x^2 - 1) \frac{d^{n+2}y}{dx^{n+2}} + (n+1) \cdot \frac{d^{n+1}y}{dx^{n+1}} \cdot 2x + \frac{(n+1)n}{2!} \cdot \frac{d^n y}{dx^n} \cdot 2 \\ = 2n \left[x \cdot \frac{d^{n+1}y}{dx^{n+1}} + (n+1) \frac{d^n y}{dx^n} \cdot 1 \right] \end{aligned}$$

$$\text{or} \quad (x^2 - 1) \frac{d^{n+2}y}{dx^{n+2}} + 2x \frac{d^{n+1}y}{dx^{n+1}} - n(n+1) \frac{d^n y}{dx^n} = 0$$

$$\text{or} \quad (1 - x^2) \frac{d^{n+2}y}{dx^{n+2}} - 2x \frac{d^{n+1}y}{dx^{n+1}} + n(n+1) \frac{d^n y}{dx^n} = 0.$$

Putting $\frac{d^n y}{dx^n} = z$, the above equation becomes

$$(1 - x^2) \frac{d^2 z}{dx^2} - 2x \frac{dz}{dx} + n(n+1)z = 0,$$

which is Legendre's equation.

Hence its solution is $z = cP_n(x)$, where c is a constant.

$$\therefore \frac{d^n y}{dx^n} = c P_n(x). \quad \dots(1)$$

Putting $x = 1$, we have $c = \left(\frac{d^n y}{dx^n} \right)_{x=1}$, since $P_n(1) = 1$.

$$\text{Now} \quad y = (x^2 - 1)^n = (x+1)^n \cdot (x-1)^n.$$

Differentiating n times by Leibnitz's theorem, we have

$$\begin{aligned} \frac{d^n y}{dx^n} &= (x-1)^n \cdot \frac{d^n}{dx^n} (x+1)^n + n \cdot \left\{ \frac{d^{n-1}}{dx^{n-1}} (x+1)^n \right\} \cdot n(x-1)^{n-1} + \dots \\ &\quad + n \left\{ \frac{d}{dx} (x+1)^n \right\} \frac{d^{n-1}}{dx^{n-1}} (x-1)^n + (x+1)^n \cdot \frac{d^n}{dx^n} (x-1)^n \end{aligned}$$

$$\begin{aligned}
 &= (x-1)^n n! + n \cdot \frac{n!}{1!} (x+1)n(x-1)^{n-1} + \dots \\
 &\quad + n \cdot n(x+1)^{n-1} \frac{n!}{1!} (x-1) + (x+1)^n \cdot n!
 \end{aligned}$$

Putting $x = 1, \left(\frac{d^n y}{dx^n} \right)_{x=1} = (1+1)^n \cdot n! = 2^n \cdot n! = c.$

$\therefore$ From (1), we have $P_n(x) = \frac{1}{c} \frac{d^n y}{dx^n}$

or $P_n(x) = \frac{1}{2^n \cdot n!} \frac{d^n}{dx^n} (x^2 - 1)^n.$

This is Rodrigue's formula.

Corollary: Show that $\int_0^1 x^m P_n(x) dx = \frac{m(m-1)(m-2)\dots(m-n+2)}{(m+n+1)(m+n-1)\dots(m-n+3)},$

when $m > n-1$ and n is a positive integer.

By Rodrigue's formula, we have

$$P_n(x) = \frac{1}{2^n \cdot n!} \frac{d^n}{dx^n} (x^2 - 1)^n.$$

$$\begin{aligned}
 \therefore \int_0^1 x^m P_n(x) dx &= \frac{1}{2^n \cdot n!} \int_0^1 x^m \frac{d^n}{dx^n} (x^2 - 1)^n dx \\
 &= \frac{1}{2^n \cdot n!} \left\{ x^m \frac{d^{n-1}}{dx^{n-1}} (x^2 - 1)^n \right\}_0^1 - m \int_0^1 x^{m-1} \frac{d^{n-1}}{dx^{n-1}} (x^2 - 1)^n dx, \\
 &\quad \text{integrating by parts} \\
 0 &= \frac{(-1)^m}{2^n \cdot n!} \int_0^1 x^{m-1} \frac{d^{n-1}}{dx^{n-1}} (x^2 - 1)^n dx.
 \end{aligned}$$

[The first term vanishes in the given limits.]

Again integrating the R.H.S. by parts, we get

$$\int_0^1 x^m P_n(x) dx = \frac{(-1)^2 m(m-1)}{2^n \cdot n!} \int_0^1 x^{m-2} \frac{d^{n-2}}{dx^{n-2}} (x^2 - 1)^n dx.$$

Proceeding in the same way, we get

$$\begin{aligned}
 \int_0^1 x^m P_n(x) dx &= \frac{(-1)^n \cdot m(m-1)\dots(m-n+1)}{2^n \cdot n!} \int_0^1 x^{m-n} (x^2 - 1)^n dx \\
 &= \frac{m(m-1)\dots(m-n+1)}{2^n \cdot n!} \int_0^1 x^{m-n} (1-x^2)^n dx \\
 &= \frac{m(m-1)\dots(m-n+1)}{2^{n+1} \cdot n!} \int_0^1 (x^2)^{(m-n-1)/2} (1-x^2)^n 2x dx,
 \end{aligned}$$

$$\begin{aligned}
&= \frac{m(m-1)\dots(m-n+1)}{2^{n+1} \cdot n!} \int_0^1 z^{(m-n-1)/2} (1-z)^n dz, \\
&\quad \text{putting } x^2 = z, 2x dx = dz \\
&= \frac{m(m-1)\dots(m-n+1)}{2^{n+1} \cdot n!} \cdot \mathbf{B} \left[\frac{m-n-1}{2} + 1, n+1 \right] \\
&= \frac{m(m-1)\dots(m-n+1)}{2^{n+1} \cdot n!} \cdot \mathbf{B} \left[\frac{m-n+1}{2}, n+1 \right] \\
&= \frac{m(m-1)\dots(m-n+1)}{2^{n+1} \cdot n!} \cdot \frac{\Gamma\left(\frac{m-n+1}{2}\right) \Gamma(n+1)}{\Gamma\left(\frac{m-n+3}{2}\right)} \left[\because \mathbf{B}(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} \right] \\
&= \frac{m(m-1)\dots(m-n+1)}{2^{n+1} \cdot n!} \cdot \frac{\Gamma\left(\frac{m-n+1}{2}\right) n!}{\left(\frac{m-n+1}{2}\right) \left(\frac{m-n-1}{2}\right) \dots \left(\frac{m-n+3}{2}\right) \left(\frac{m-n+1}{2}\right)} \\
&= \frac{m(m-1)\dots(m-n+1)}{2^{n+1} \cdot n!} \cdot \frac{2^{n+1} \cdot n!}{(m+n+1)(m+n-1)\dots(m-n+3)(m-n+1)} \\
&= \frac{m(m-1)(m-2)\dots(m-n+2)}{(m+n+1)(m+n-1)\dots(m-n+3)} \dots(1)
\end{aligned}$$

Case I: When n is even.

$$\int_0^1 x^m P_n(x) dx = \frac{m(m-2)(m-4)\dots(m-n+2)}{(m+n+1)(m+n-1)\dots(m+1)} \dots(2)$$

Since for $n=2$, the last factor i.e., $(m-n+3)$ in the Dr. is $(m+1)$ and its preceding terms are $(m-1)$, $(m-3)$ etc., which cancel out with the corresponding factors in the Nr.

Case II: When n is odd.

$$\int_0^1 x^m P_n(x) dx = \frac{(m-1)(m-3)\dots(m-n+2)}{(m+n+1)(m+n-1)\dots(m+2)} \dots(3)$$

Since for $n=1$, the last factor i.e., $(m-n+3)$ in the Dr. is $(m+2)$ and its preceding terms are m , $(m-2)$ etc. which cancel out with the corresponding factors in the Nr.

Illustrative Examples

Example 9 : Prove that

$$(i) \int_{-1}^{+1} P_n(x) dx = 0, n \neq 0$$

(Kanpur 2012; Rohilkhand 14)

$$\text{and } (ii) \int_{-1}^{+1} P_0(x) dx = 2.$$

Solution : (i) From Rodrigue's formula, we have

$$P_n(x) = \frac{1}{2^n n!} \cdot \frac{d^n}{dx^n} (x^2 - 1)^n.$$

$$\begin{aligned} \therefore \int_{-1}^{+1} P_n(x) dx &= \frac{1}{2^n n!} \int_{-1}^{+1} \frac{d^n}{dx^n} (x^2 - 1)^n dx \\ &= \frac{1}{2^n n!} \left[\frac{d^{n-1}}{dx^{n-1}} (x^2 - 1)^n \right]_{-1}^{+1} \quad \dots(1) \end{aligned}$$

$$\begin{aligned} \text{Now } \frac{d^{n-1}}{dx^{n-1}} (x^2 - 1)^n &= \frac{d^{n-1}}{dx^{n-1}} \{ (x+1)^n (x-1)^n \} \\ &= (x+1)^n \frac{d^{n-1}}{dx^{n-1}} (x-1)^n + (n-1)n(x+1)^{n-1} \frac{d^{n-2}}{dx^{n-2}} (x-1)^n \\ &\quad + \dots + (x-1)^n \frac{d^{n-1}}{dx^{n-1}} (x+1)^n, \end{aligned}$$

by Leibnitz's theorem of differential calculus

$$\begin{aligned} &= (x+1)^n \frac{n!}{1!} (x-1) + n(n-1)(x+1)^{n-1} \frac{n!}{2!} (x-1)^2 + \dots \\ &\quad \dots + (x-1)^n n!(x+1) \\ &\left[\because D^r (x+a)^n = \frac{n!}{(n-r)!} (x+a)^{n-r} \right] \end{aligned}$$

$$= 0 \quad \text{when } x = -1 \quad \text{or } 1,$$

since each term contains both $x+1$ and $x-1$ as factors.

$$\therefore \text{ from (1), } \int_{-1}^{+1} P_n(x) dx = 0.$$

(ii) We know that $P_0(x) = 1$.

$$\begin{aligned} \therefore \int_{-1}^{+1} P_0(x) dx &= \int_{-1}^{+1} dx \\ &= [x]_{-1}^{+1} = 2. \end{aligned}$$

Example 10 : Evaluate $\int_{-1}^1 x^4 P_6(x) dx$.

(Meerut 2006, 13B)

Solution : We know that $P_n(x)$ is a polynomial in x of degree n . Therefore we can express x^4 as a linear combination of P_0, P_1, P_2, P_3 and P_4 .

$$\text{Let } x^4 = c_0 P_0 + c_1 P_1 + c_2 P_2 + c_3 P_3 + c_4 P_4,$$

where c_0, c_1, c_2, c_3, c_4 are some constants.

$$\begin{aligned} \text{Then } I &= \int_{-1}^1 x^4 P_6(x) dx \\ &= \int_{-1}^1 (c_0 P_0 + c_1 P_1 + c_2 P_2 + c_3 P_3 + c_4 P_4) P_6 dx \end{aligned}$$

$$\begin{aligned}
 &= c_0 \int_{-1}^1 P_0 P_6 dx + c_1 \int_{-1}^1 P_1 P_6 dx + c_2 \int_{-1}^1 P_2 P_6 dx \\
 &\quad + c_3 \int_{-1}^1 P_3 P_6 dx + c_4 \int_{-1}^1 P_4 P_6 dx \\
 &= 0, \text{ since } \int_{-1}^1 P_m P_n dx = 0 \text{ if } m \neq n.
 \end{aligned}$$

Comprehensive Exercise 1

1. Show that

$$\begin{aligned}
 P_0(x) &= 1, P_1(x) = x, P_2(x) = (3x^2 - 1)/2, P_3(x) = (5x^3 - 3x)/2, \\
 P_4(x) &= (35x^4 - 30x^2 + 3)/8. \quad (\text{Kanpur 2008; Lucknow 08; Avadh 14})
 \end{aligned}$$

2. Express $P(x) = x^4 + 2x^3 + 2x^2 - x - 3$ in terms of Legendre's polynomials.

3. (i) Prove that $P_n' - P_{n-2}' = (2n-1)P_{n-1}$.

[Hint : Replace n by $(n-1)$ in recurrence formula (III)].

(ii) Prove that $x P_9' = P_8' + 9P_9$.

(Meerut 2008)

[Hint : Put $n = 9$ in recurrence formula II].

4. Prove that $(1 - 2xz + z^2)^{-1/2}$ is a solution of the equation

$$z \frac{\partial^2(zv)}{\partial z^2} + \frac{\partial}{\partial x} \left\{ (1-x^2) \frac{\partial v}{\partial x} \right\} = 0.$$

5. (i) Prove that $\frac{1+z}{z \sqrt{(1-2xz+z^2)}} - \frac{1}{z} = \sum_{n=0}^{\infty} [P_n(x) + P_{n+1}(x)] z^n$.

(ii) Prove that

$$P'_{n+1} + P'_n = P_0 + 3P_1 + 5P_2 + \dots + (2n+1)P_n = \sum_{r=0}^n (2r+1)P_r(x).$$

6. Prove that (i) $\int P_n(x) dx = [P_{n+1}(x) - P_{n-1}(x)] / (2n+1) + C$,

and (ii) $\int_x^1 P_n(x) dx = [P_{n-1}(x) - P_{n+1}(x)] / (2n+1)$.

[Hint : Integrate recurrence formula III].

7. (i) Prove that $\int_{-1}^{+1} (x^2 - 1)P_{n+1} P_n' dx = \frac{2n(n+1)}{(2n+1)(2n+3)}$.

(Agra 2007)

(ii) Show that $\int_{-1}^1 x P_n(x) P_{n-1}(x) dx = 2n / (4n^2 - 1)$.

(Rohilkhand 2004;
Purvanchal 07)

[Hint: Using recurrence formula I, we have

$$\begin{aligned}
 \int_{-1}^1 x P_n(x) P_{n-1}(x) dx &= \frac{n+1}{2n+1} \int_{-1}^1 P_{n+1}(x) P_{n-1}(x) dx \\
 &\quad + \frac{n}{2n+1} \int_{-1}^1 [P_{n-1}(x)]^2 dx.
 \end{aligned}$$

8. (i) Prove that $\int_{-1}^1 \frac{P_n(x)}{\sqrt{(1-2xh+h^2)}} dx = \frac{2h^n}{2n+1}.$

(ii) Prove that $\int_{-1}^{+1} (1-x^2)(P_n')^2 dx = \frac{2n(n+1)}{2n+1}.$

(Agra 2010)

9. (i) Prove that $\int_{-1}^{+1} (1-x^2)P_m' P_n' dx = 0$

where m and n are distinct positive integers.

(ii) Prove that $\int_{-1}^1 (1-x^2)P_m' P_n' dx = \frac{2n(n+1)}{2n+1} \delta_{mn},$

where δ_{mn} is Kronecker delta.

(Lucknow 2006)

10. Prove that (i) $\int_{-1}^1 x^m P_n(x) dx = 0$

where m and n are integers and $m < n$.

and (ii) $\int_{-1}^1 x^n P_n(x) dx = \frac{2^{n+1} \cdot (n!)^2}{(2n+1)!}.$

11. Prove that (i) $P_n'(1) = \frac{1}{2} n(n+1)$

(Lucknow 2008)

and (ii) $P_n'(-1) = (-1)^{n-1} \cdot \frac{1}{2} n(n+1).$

12. Prove that

$$P_n\left(-\frac{1}{2}\right) = P_0\left(-\frac{1}{2}\right) P_{2n}\left(\frac{1}{2}\right) + P_1\left(-\frac{1}{2}\right) P_{2n-1}\left(\frac{1}{2}\right) + \dots + P_{2n}\left(-\frac{1}{2}\right) P_0\left(\frac{1}{2}\right).$$

13. Prove that $1 + \frac{1}{2} P_1(\cos \theta) + \frac{1}{3} P_2(\cos \theta) + \frac{1}{4} P_3(\cos \theta) + \dots = \log \frac{1 + \sin \frac{1}{2} \theta}{\sin \frac{1}{2} \theta}.$

14. Show that $\int_0^\pi P_n(\cos \theta) \cos n\theta d\theta = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} \pi.$ (Avadh 2012)

15. Show that $\int_0^\pi P_n(\cos \theta) \cos n\theta d\theta = B\left(n + \frac{1}{2}, \frac{1}{2}\right).$ (Avadh 2012)

16. Prove that if n is odd

$$\int_0^1 P_n(x) dx = \frac{(-1)^{(n-1)/2} (n-1)!}{2^n \cdot \{(n+1)/2\}! \{(n-1)/2\}!}.$$

(Avadh 2008)

17. Prove that

$$\int_{-1}^1 x^2 P_n^2(x) dx = \frac{1}{8(2n-1)} + \frac{3}{4(2n+1)} + \frac{1}{8(2n+3)}.$$

18. Show that all the roots of $P_n(x) = 0$ are real and lie between -1 and 1 .

19. Show that all the roots of $P_n(x) = 0$ are distinct.

Answers 1

$$2. \quad P(x) = \frac{8}{35} P_4(x) + \frac{4}{5} P_3(x) + \frac{40}{21} P_2(x) + \frac{1}{5} P_1(x) - \frac{224}{105} P_0(x).$$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The value of the integral $\int_{-1}^1 [P_n(x)]^2 dx$ is
 - 1
 - 0
 - $\frac{2}{2n+1}$
 - $2n^2$.

(Rohilkhand 2011)
- The value of $xP_n'(x) - P_{n-1}'(x)$ is
 - $nP_n(x)$
 - n^2
 - $P_{n-1}'(x)$
 - None of these.
- The value of $P_{n+1}'(x) - P_{n-1}'(x)$ is
 - $P_n(x)$
 - $(2n+1)P_n(x)$
 - $P_n'(x)$
 - None of these.
- The value of $n(P_{n-1} - xP_n)$ is
 - $(1-x^2)P_n(x)$
 - $(1-x^2)P_n'(x)$
 - $P_{n-1}'(x)$
 - None of these.
- The value of $(n+1)(xP_n - P_{n+1})$ is
 - $(1-x^2)P_n'$
 - xP_n
 - $(n+1)P_n'$
 - None of these.
- The value of $P_n(1)$ is
 - 0
 - 1
 - 1
 - None of these.

(Rohilkhand 2010, 11; Agra 10)
- The value of $P_n(-1)$ is
 - 1
 - 0
 - 1
 - $(-1)^n$.

(Rohilkhand 2008, 09; Agra 08)

8. If $P_n(x)$ is a solution of Legendre's equation, then value of $P_0(x)$ will be
 (a) 1 (b) 0
 (c) -1 (d) x . (Rohilkhand 2004)
9. The value of $\frac{1}{2^n \cdot n!} \frac{d^n}{dx^n} (x^2 - 1)^n$ is
 (a) $P_n'(x)$ (b) $Q_n(x)$
 (c) $P_{n-1}(x)$ (d) $P_n(x)$. (Agra 2006, 10)

Fill In The Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- The differential equation $(1 - x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0$ is called ...
- The polynomial solution $\frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{n!} \left[x^n - \frac{n(n-1)}{2 \cdot (2n-1)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2 \cdot 4 \cdot (2n-1)(2n-3)} x^{n-4} - \dots \right]$ of Legendre's equation is called Legendre's function of ...
- ... is called the generating function of Legendre's polynomials.
- If $m \neq n$, $\int_{-1}^1 P_m(x) P_n(x) dx = \dots$
- Laplace's first integral for $P_n(x)$ gives $P_n(x) = \dots$
- $Q_n(x)$ is called Legendre's function of ...
- The general solution of Legendre's equation is given by ...

True or False

Write 'T' for true and 'F' for false statement.

- Laplace's second integral for $P_n(x)$ gives
$$P_n(x) = \frac{1}{\pi} \int_0^\pi \frac{d\phi}{[x \pm \sqrt{(x^2 - 1) \cos \phi}]^{n+1}}.$$
- $P_n(x)$ is the coefficient of h^{n+1} in the expansion of $(1 - 2xh + h^2)^{-1/2}$ in ascending powers of h .
- $\delta_{mn} = \begin{cases} 0, & \text{if } m \neq n \\ 1, & \text{if } m = n. \end{cases}$
- $(2n+1)(x^2 - 1)P_n' = n(n-1)(P_{n+1} - P_{n-1})$ is known as Beltrami's result.
- $(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1}$.

Answers

Multiple Choice Questions

- | | | | | |
|---------|---------|---------|---------|---------|
| 1. (c). | 2. (a). | 3. (b). | 4. (b). | 5. (a). |
| 6. (b). | 7. (d). | 8. (a). | 9. (d). | |

Fill in the Blank(s)

- | | |
|--|-----------------|
| 1. Legendre's equation. | 2. first kind. |
| 3. $(1 - 2xh + h^2)^{-1/2}$. | 4. 0. |
| 5. $\frac{1}{\pi} \int_0^\pi [x \pm \sqrt{(x^2 - 1) \cos \phi}]^n d\phi$. | 6. second kind. |
| 7. $AP_n(x) + BQ_n(x)$. | |

True or False

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. T. | 2. F. | 3. T. | 4. F. | 5. T. |
|-------|-------|-------|-------|-------|



Chapter

14



Bessel's Functions

14.1 Bessel's Equation

(Avadh 2007, 10; Lucknow 07)

The differential equation of the form

$$\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + \left(1 - \frac{n^2}{x^2}\right) y = 0$$

is called **Bessel's differential equation** (or **Bessel's equation**).

14.2 Solution of Bessel's Differential Equation

The Bessel's differential equation is

$$\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + \left(1 - \frac{n^2}{x^2}\right) y = 0. \quad \dots(1)$$

We can integrate (1) in a series of ascending powers of x . Assume that its series solution is

$$y = \sum_{r=0}^{\infty} a_r x^{k+r}. \quad \therefore \quad \frac{dy}{dx} = \sum_{r=0}^{\infty} a_r (k+r) x^{k+r-1}$$

and
$$\frac{d^2 y}{dx^2} = \sum_{r=0}^{\infty} a_r (k+r)(k+r-1) x^{k+r-2}.$$

Substituting these values in (1), we have

$$\sum_{r=0}^{\infty} a_r \left[(k+r)(k+r-1) x^{k+r-2} + \frac{1}{x} (k+r) x^{k+r-1} + \left(1 - \frac{n^2}{x^2} \right) x^{k+r} \right] = 0$$

or
$$\sum_{r=0}^{\infty} a_r [(k+r)^2 - n^2] x^{k+r-2} + x^{k+r} = 0. \quad \dots(2)$$

Since the relation (2) is an identity, the coefficients of various powers of x must be zero.

$\therefore$ Equating to zero the coefficient of the lowest power of x , i.e. of x^{k-2} in (2), we have

$$a_0 (k^2 - n^2) = 0.$$

Now $a_0 \neq 0$ as it is the coefficient of the first term with which we begin to write the series.

$\therefore k^2 - n^2 = 0$ or $k = \pm n. \quad \dots(3)$

Now equating to zero the coefficient of x^{k-1} in (2), we have

$$a_1 \{(k+1)^2 - n^2\} = 0.$$

But $(k+1)^2 - n^2 \neq 0$ for $k = \pm n$ given by (3); $\therefore a_1 = 0.$

Again equating to zero the coefficient of the general term i.e. of x^{k+r} in (2), we have

$$a_{r+2} \{(k+r+2)^2 - n^2\} + a_r = 0$$

or
$$a_{r+2} (k+r+n+2)(k+r-n+2) = -a_r.$$

$\therefore a_{r+2} = -\frac{a_r}{(k+r+n+2)(k+r-n+2)}. \quad \dots(4)$

Putting $r=1$ in (4), we have $a_3 = \frac{a_1}{(k+n+3)(k-n+3)} = 0$, since $a_1 = 0.$

Similarly putting $r=3, 5, 7$ etc. in (4), we have $a_1 = a_3 = a_5 = \dots = 0$ (each).

Now two cases arise.

Case I: When $k = n$, from (4), we have $a_{r+2} = -\frac{a_r}{(2n+r+2)(r+2)}.$

Putting $r=0, 2, 4$ etc., we have $a_2 = -\frac{a_0}{(2n+2)(2)} = -\frac{a_0}{2^2 \cdot 1!(n+1)}$

$$a_4 = -\frac{a_2}{(2n+4)(4)} = -\frac{a_2}{2^2 \cdot 2 \cdot (n+2)} = \frac{a_0}{2^4 \cdot 2!(n+1)(n+2)}, \text{ etc.}$$

$\therefore y = a_0 \left[x^n - \frac{x^{n+2}}{2^2 \cdot 1!(n+1)} + \frac{x^{n+4}}{2^4 \cdot 2!(n+1)(n+2)} - \dots \right]$

$$= a_0 x^n \left[1 + (-1) \frac{x^2}{2^2 \cdot 1!(n+1)} + (-1)^2 \frac{x^4}{2^4 \cdot 2!(n+1)(n+2)} + \dots \right].$$

If $a_0 = \frac{1}{2^n \Gamma(n+1)}$, this solution is called $J_n(x)$, known as **Bessel's function of the first**

kind of order n .

$$\begin{aligned}
 \therefore J_n(x) &= \frac{x^n}{2^n \Gamma(n+1)} \left[1 + (-1) \frac{x^2}{2^2 \cdot 1! (n+1)} + (-1)^2 \frac{x^4}{2^4 \cdot 2! (n+1)(n+2)} + \dots \right] \\
 &= \frac{x^n}{2^n \Gamma(n+1)} \sum_{r=0}^{\infty} (-1)^r \frac{x^{2r}}{2^{2r} r! (n+1)(n+2) \dots (n+r)} \\
 &= \sum_{r=0}^{\infty} (-1)^r \left(\frac{x}{2} \right)^{n+2r} \frac{1}{r! \Gamma(n+r+1)}.
 \end{aligned}$$

Case II: When $k = -n$.

The series solution is obtained by replacing n by $-n$ in the value of J_n .

$$\therefore J_{-n}(x) = \sum_{r=0}^{\infty} (-1)^r \left(\frac{x}{2} \right)^{-n+2r} \frac{1}{r! \Gamma(-n+r+1)}.$$

When n is not an integer $J_{-n}(x)$ is distinct from $J_n(x)$.

Note. The solutions of Bessel's equation are called Bessel's functions.

14.3 General Solution of Bessel's Equation

(Lucknow 2007)

When n is not an integer, the most **general solution** of Bessel's equation is

$$y = A J_{-n}(x) + B J_n(x),$$

where A, B are two arbitrary constants.

14.4 Integration of Bessel's Equation in Series for $n = 0$

(Avadh 2008)

$$\text{The differential equation } \frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + y = 0 \quad \dots(1)$$

is known as Bessel's equation for $n = 0$.

Let us assume that its solution be $y = \sum_{r=0}^{\infty} a_r x^{k+r}$.

$$\therefore \frac{dy}{dx} = \sum_{r=0}^{\infty} a_r (k+r) x^{k+r-1} \text{ and } \frac{d^2 y}{dx^2} = \sum_{r=0}^{\infty} a_r (k+r)(k+r-1) x^{k+r-2}.$$

Substituting these values in (1), we get

$$\sum_{r=0}^{\infty} a_r \left[(k+r)(k+r-1) x^{k+r-2} + \frac{1}{x} (k+r) x^{k+r-1} + x^{k+r} \right] = 0$$

$$\text{or } \sum_{r=0}^{\infty} a_r [(k+r)^2 x^{k+r-2} + x^{k+r}] = 0, \quad \dots(2)$$

which is an identity. Equating to zero the coefficient of the lowest power of x i.e., of x^{k-2} , we have $a_0 k^2 = 0$.

Since $a_0 \neq 0$ for the same reason, as in 14.2, therefore $k^2 = 0$. $\therefore k = 0$.

Now equating to zero the coefficient of the next lowest power of x , i.e., of x^{k-1} , we have

$$a_1 (k+1)^2 = 0.$$

Since $k+1 \neq 0$ by virtue of (3), we have $a_1 = 0$.

Again equating to zero the coefficient of the general term, i.e., of x^{k+r} , we have

$$a_{r+2}(k+r+2)^2 + a_r = 0. \quad \therefore a_{r+2} = -\frac{a_r}{(k+r+2)^2}.$$

When $k=0$, we have $a_{r+2} = -\frac{a_r}{(r+2)^2}$.

Putting $r=1, 3, 5$, etc., we have $a_1 = a_3 = a_5 = \dots = 0$ (each).

Again putting $r=0, 2, 4$, etc., we have

$$a_2 = -\frac{a_0}{2^2}, a_4 = -\frac{a_2}{4^2} = \frac{a_0}{2^2 \cdot 4^2} \text{ etc.}$$

Since $y = \sum_{r=0}^{\infty} a_r x^r$, when $k=0$

$$\therefore y = a_0 \left(1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots \right).$$

If $a_0 = 1$, this solution is denoted by $J_0(x)$.

$$\therefore J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots$$

where $J_0(x)$ is called Bessel's function of zeroth order.

14.5 Definition of $J_0(x)$

$J_0(x)$ is the solution of Bessel's equation for $n=0$ i.e., of

$$\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + y = 0.$$

which is equal to 1 for $x=0$.

Illustrative Examples

Example 1: Show that when n is a positive integer

- (i) $J_{-n}(x) = (-1)^n J_n(x)$; (Meerut 2006, 08, 11; Kanpur 07, 08; Agra 06, 11; Rohilkhand 14; Purvanchal 10)
- (ii) $J_n(-x) = (-1)^n J_n(x)$ for +ive or -ive integers; (Meerut 2006, 11; Purvanchal 10)
- and (iii) $J_n(x)$ is an even or an odd function of x according as n is even or odd respectively.

Solution: (i) We have

$$J_{-n}(x) = \sum_{r=0}^{\infty} (-1)^r \left(\frac{x}{2}\right)^{-n+2r} \cdot \frac{1}{r! \Gamma(-n+r+1)}.$$

If p is an integer, then $\Gamma(-p)$ is infinity for $p \geq 0$. Therefore we get terms in J_{-n} equal to zero till $-n+r+1 < 1$ i.e., $r < n$. Hence we can write

$$\begin{aligned}
 J_{-n}(x) &= \sum_{r=n}^{\infty} \frac{(-1)^r}{r! \Gamma(-n+r+1)} \left(\frac{x}{2}\right)^{-n+2r} \\
 &= \sum_{s=0}^{\infty} \frac{(-1)^{n+s}}{(n+s)! \Gamma(s+1)} \left(\frac{x}{2}\right)^{n+2s}, \quad \text{putting } r = n+s \\
 &= (-1)^n \sum_{s=0}^{\infty} \frac{(-1)^s}{\Gamma(n+s+1) \cdot s!} \left(\frac{x}{2}\right)^{n+2s} = (-1)^n J_n(x).
 \end{aligned}$$

(ii) We know that $J_n(x) = \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$.

Case I: Let n be a +ive integer. Replacing x by $-x$, we have

$$\begin{aligned}
 J_n(-x) &= \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma(n+r+1)} \left(-\frac{x}{2}\right)^{n+2r} \\
 &= (-1)^n \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \quad [\because (-1)^{2r} = 1] \\
 &= (-1)^n J_n(x).
 \end{aligned}$$

Case II: Let n be a -ive integer, say $n = -m$ where m is a +ive integer.

Then $J_n(x) = J_{-m}(x) = (-1)^m J_m(x)$, as proved in part (i) of this question.

Replacing x by $-x$, we have

$$\begin{aligned}
 J_n(-x) &= (-1)^m J_m(-x) = (-1)^m (-1)^m J_m(x) \quad [\text{as proved in case I}] \\
 &= (-1)^m J_{-m}(x), \text{ by part (i) of this question} \\
 &= (-1)^{2m} (-1)^{-m} \cdot J_{-m}(x) = (-1)^{-m} J_{-m}(x) \quad [\because (-1)^{2m} = 1] \\
 &= (-1)^n J_n(x).
 \end{aligned}$$

Hence $J_n(-x) = (-1)^n J_n(x)$ for +ive or -ive integers. **Proved**

(iii) We have $J_n(-x) = (-1)^n J_n(x)$(1)

When n is even, $(-1)^n = 1$, $\therefore$ from (1), $J_n(-x) = J_n(x)$.

$\therefore J_n(x)$ is an even function of x if n is even. When n is odd, $(-1)^n = -1$,

$\therefore$ from (1), $J_n(-x) = -J_n(x)$.

$\therefore J_n(x)$ is an odd function of x , if n is odd.

Example 2: Prove that $J_n(x) = (-2)^n x^n \frac{d^n}{d(x^2)^n} J_0(x)$.

Solution: $J_0(x)$ is the solution of Bessel's equation of zeroeth order

$$\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + y = 0. \quad \dots(1)$$

Let us change the independent variable from x to X , by the relation $x^2 = X$, so that

$$\frac{dy}{dx} = \frac{dy}{dX} \cdot \frac{dX}{dx} = 2x \frac{dy}{dX} = 2\sqrt{X} \frac{dy}{dX}$$

and

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left(2 \sqrt{X} \frac{dy}{dX} \right) = \frac{d}{dX} \left(2 \sqrt{X} \frac{dy}{dX} \right) \cdot \frac{dX}{dx} \\ &= \left(2 \sqrt{X} \frac{d^2 y}{dX^2} + \frac{1}{\sqrt{X}} \frac{dy}{dX} \right) \cdot 2 \sqrt{X} = 4X \frac{d^2 y}{dX^2} + 2 \frac{dy}{dX}. \end{aligned}$$

Putting these values in (1), we get

$$\left(4X \frac{d^2 y}{dX^2} + 2 \frac{dy}{dX} \right) + \frac{1}{\sqrt{X}} 2 \sqrt{X} \frac{dy}{dX} + y = 0$$

or

$$4X \frac{d^2 y}{dX^2} + 4 \frac{dy}{dX} + y = 0. \quad \dots(2)$$

Now differentiating (2) n times w.r.t. X , by Leibnitz's theorem, we get

$$4 \left[X \frac{d^{n+2} y}{dX^{n+2}} + n \cdot 1 \cdot \frac{d^{n+1} y}{dX^{n+1}} \right] + 4 \frac{d^{n+1} y}{dX^{n+1}} + \frac{d^n y}{dX^n} = 0$$

or

$$4X \frac{d^{n+2} y}{dX^{n+2}} + 4(n+1) \frac{d^{n+1} y}{dX^{n+1}} + \frac{d^n y}{dX^n} = 0. \quad \dots(3)$$

Let $Y = \frac{d^n y}{dX^n} = \frac{d^n}{dX^n} J_0(x).$... (4)

Using this substitution the equation (3) reduces to

$$4X \frac{d^2 Y}{dX^2} + 4(n+1) \frac{dY}{dX} + Y = 0. \quad \dots(5)$$

Again, $J_n(x)$ is the solution of the Bessel's equation

$$\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + \left(1 - \frac{n^2}{x^2} \right) y = 0. \quad \dots(6)$$

Let $y = x^n Z$ or $J_n(x) = x^n Z$... (7)

so that $\frac{dy}{dx} = x^n \frac{dZ}{dx} + nx^{n-1} Z$ and $\frac{d^2 y}{dx^2} = x^n \frac{d^2 Z}{dx^2} + 2nx^{n-1} \frac{dZ}{dx} + n(n-1)x^{n-2} Z.$

Putting these values in (6), we get

$$x^n \frac{d^2 Z}{dx^2} + 2nx^{n-1} \frac{dZ}{dx} + n(n-1)x^{n-2} Z + \frac{1}{x} \left(x^n \frac{dZ}{dx} + nx^{n-1} Z \right) + \left(1 - \frac{n^2}{x^2} \right) x^n Z = 0$$

or

$$x^n \frac{d^2 Z}{dx^2} + (2n+1)x^{n-1} \frac{dZ}{dx} + x^n Z = 0$$

or

$$\frac{d^2 Z}{dx^2} + (2n+1) \frac{1}{x} \frac{dZ}{dx} + Z = 0. \quad \dots(8)$$

Changing the independent variable from x to X , by the relation $x^2 = X$, the equation (8) becomes

$$\left(4X \frac{d^2 Z}{dX^2} + 2 \frac{dZ}{dX} \right) + \frac{(2n+1)}{\sqrt{X}} 2 \sqrt{X} \frac{dZ}{dX} + Z = 0$$

or

$$4X \frac{d^2 Z}{dX^2} + 4(n+1) \frac{dZ}{dX} + Z = 0. \quad \dots(9)$$

Comparing (5) and (9), we get $Z = cY$, where c is a constant to be determined.

$$\therefore x^n Z = cx^n Y \quad \text{or} \quad y = cx^n Y.$$

$$\text{Hence} \quad J_n(x) = cx^n \cdot \frac{d^n}{d(x^2)^n} J_0(x). \quad \dots(10)$$

Now it remains to find the value of c .

$$\text{We have} \quad J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots = \sum_{r=0}^{\infty} \frac{(-1)^r x^{2r}}{(2^2 r!)^2}.$$

$$\therefore \frac{d^n}{d(x^2)^n} J_0(x) = \frac{d^n}{d(x^2)^n} \sum_{r=0}^{\infty} \frac{(-1)^r x^{2r}}{(2^2 r!)^2} = \frac{d^n}{d(x^2)^n} \sum_{r=0}^{\infty} \frac{(-1)^{n+r} (x^2)^{n+r}}{\{2^{n+r} (n+r)!\}^2}$$

[$\because$ On differentiation all those terms on R.H.S. will vanish in which the power of x^2 is less than n]

$$= \sum_{r=0}^{\infty} \frac{(-1)^{n+r} (n+r)(n+r-1)\dots(r+1)(x^2)^r}{\{2^{n+r} (n+r)!\}^2}$$

$$= \sum_{r=0}^{\infty} \frac{(-1)^{n+r} (n+r)! x^{2r}}{r! 2^{2n+2r} \{(n+r)!\}^2}$$

$$= (-1)^n \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! 2^{2n+2r} (n+r)!} x^{2r}.$$

$$\begin{aligned} \therefore J_n(x) &= cx^n (-1)^n \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! 2^{2n+2r} (n+r)!} x^{2r} \\ &= \frac{c(-1)^n}{2^n} \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} = \frac{c}{(-2)^n} \cdot J_n(x). \end{aligned}$$

$$\therefore c = (-2)^n.$$

$$\text{Putting this value of } c \text{ in (10), we get } J_n(x) = (-2)^n x^n \frac{d^n}{d(x^2)^n} J_0(x).$$

14.6 Recurrence Formulae

$$(I) \quad x J_n'(x) = n \cdot J_n(x) - x \cdot J_{n+1}(x).$$

(Meerut 2006, 07, 09, 11; Lucknow 07, 11; Rohilkhand 04, 14)

$$\text{Proof: We know that } J_n(x) = \sum_{r=0}^{\infty} (-1)^r \left(\frac{x}{2}\right)^{n+2r} \frac{1}{r! \Gamma(n+r+1)}.$$

Differentiating w.r.t. x , we have

$$J_n'(x) = \sum_{r=0}^{\infty} (-1)^r \frac{(n+2r)}{r! \Gamma(n+r+1)} \frac{1}{2} \left(\frac{x}{2}\right)^{n+2r-1}$$

$$\begin{aligned} \therefore x J_n'(x) &= \sum_{r=0}^{\infty} (-1)^r \frac{(n+2r)}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} = \sum_{r=0}^{\infty} (-1)^r \frac{n}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\ &\quad + \sum_{r=0}^{\infty} (-1)^r \frac{2r}{r! \Gamma(n+r+1)} \cdot \frac{x}{2} \left(\frac{x}{2}\right)^{n+2r-1} \end{aligned}$$

$$\begin{aligned}
 &= n J_n(x) + x \sum_{r=1}^{\infty} (-1)^r \frac{1}{(r-1)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\
 &= n J_n(x) - x \sum_{s=0}^{\infty} (-1)^s \frac{1}{s! \Gamma(n+1+s+1)} \left(\frac{x}{2}\right)^{n+1+2s},
 \end{aligned}$$

putting $r-1=s$

$$= n J_n(x) - x J_{n+1}(x).$$

Hence $x J_n'(x) = n J_n(x) - x J_{n+1}(x)$.

Multiplying both sides by x^{-n-1} , we have

$$x^{-n} J_n' = n x^{-n-1} J_n - x^{-n} J_{n+1} \quad \text{or} \quad -n x^{-n-1} J_n + x^{-n} J_n' = -x^{-n} J_{n+1}$$

or $\frac{d}{dx} (x^{-n} J_n) = -x^{-n} J_{n+1}$, which is another form of recurrence formula (I).

(II) $x J_n'(x) = -n J_n(x) + x J_{n-1}(x)$. (Meerut 2008, 13, 10; Agra 10)

Proof: As in I, we have

$$\begin{aligned}
 x J_n'(x) &= \sum_{r=0}^{\infty} (-1)^r \frac{(n+2r)}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\
 &= \sum_{r=0}^{\infty} (-1)^r \frac{(2n+2r-n)}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\
 &= -n \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\
 &\quad + \sum_{r=0}^{\infty} (-1)^r \frac{(2n+2r)}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\
 &= -n J_n(x) + \sum_{r=0}^{\infty} (-1)^r \frac{2(n+r)}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \left(\frac{x}{2}\right) \\
 &= -n J_n(x) + x \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma(n+r)} \left(\frac{x}{2}\right)^{n+2r-1} \\
 &= -n J_n(x) + x \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma\{(n-1)+r+1\}} \left(\frac{x}{2}\right)^{(n-1)+2r} \\
 &= -n J_n(x) + x J_{n-1}(x).
 \end{aligned}$$

Hence $x J_n'(x) = -n J_n(x) + x J_{n-1}(x)$

Multiplying both sides by x^{n-1} , we have

$$x^n J_n' = -n x^{n-1} J_n + x^n J_{n-1} \quad \text{or} \quad n x^{n-1} J_n + x^n J_n' = x^n J_{n-1}$$

or $\frac{d}{dx} (x^n J_n) = x^n J_{n-1}$, which is another form of recurrence formula (II).

(III) $2 J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$. (Rohilkhand 2002, 07; Avadh 07, 10; Kanpur 07)

Proof: Recurrence formulae I and II, are

$$x J_n'(x) = n J_n(x) - x J_{n+1}(x) \quad \text{and} \quad x J_n'(x) = -n J_n(x) + x J_{n-1}(x).$$

Adding, we have $2x J_n'(x) = x [J_{n-1}(x) - J_{n+1}(x)]$.

Hence $2 J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$.

Aliter. $J_n(x) = \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$.

$$\begin{aligned} \therefore 2 J_n'(x) &= \sum_{r=0}^{\infty} (-1)^r \frac{2}{r! \Gamma(n+r+1)} (n+2r) \frac{1}{2} \left(\frac{x}{2}\right)^{n+2r-1} \\ &= \sum_{r=0}^{\infty} (-1)^r \frac{[(n+r)+r]}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\ &= \sum_{r=0}^{\infty} (-1)^r \frac{(n+r)}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} + \sum_{r=0}^{\infty} (-1)^r \frac{r}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\ &= \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma(n+r)} \left(\frac{x}{2}\right)^{n+2r-1} \\ &\quad - \sum_{r=1}^{\infty} (-1)^{r-1} \frac{1}{(r-1)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \\ &= \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma[(n-1)+r+1]} \left(\frac{x}{2}\right)^{(n-1)+2r} \\ &\quad - \sum_{s=0}^{\infty} (-1)^s \frac{1}{s! \Gamma[(n+1)+s+1]} \left(\frac{x}{2}\right)^{(n+1)+2s}, \text{ putting } r-1=s \\ &= J_{n-1}(x) - J_{n+1}(x). \end{aligned}$$

Hence $2 J_n'(x) = J_{n-1}(x) - J_{n+1}(x)$.

(IV) $2n J_n(x) = x [J_{n-1}(x) + J_{n+1}(x)]$ (Avadh 2009; Kanpur 11)

Proof: Writing Recurrence formulae I and II, we have

$$x J_n'(x) = n J_n(x) - x J_{n+1}(x) \quad \text{and} \quad x J_n'(x) = -n J_n(x) + x J_{n-1}(x).$$

Subtracting, we have

$$0 = 2n J_n(x) - x [J_{n+1}(x) + J_{n-1}(x)].$$

Hence $2n J_n(x) = x [J_{n+1}(x) + J_{n-1}(x)]$

Aliter: $J_n(x) = \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$.

$$\begin{aligned} \therefore 2n J_n(x) &= \sum_{r=0}^{\infty} (-1)^r \frac{(2n+2r-2r)}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\ &= \sum_{r=0}^{\infty} (-1)^r \frac{2(n+r)}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\ &\quad - \sum_{r=0}^{\infty} (-1)^r \frac{2r}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \\ &= x \sum_{r=0}^{\infty} (-1)^r \frac{1}{r! \Gamma[(n-1)+r+1]} \left(\frac{x}{2}\right)^{(n-1)+2r} \\ &\quad + x \sum_{r=1}^{\infty} (-1)^{r-1} \frac{1}{(r-1)! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r-1} \end{aligned}$$

$$= x J_{n-1}(x) + x \sum_{s=0}^{\infty} (-1)^s \frac{1}{s! \Gamma[(n+1)+s+1]} \left(\frac{x}{2}\right)^{(n+1)+2s},$$

putting $r-1=s$

$$= x J_{n-1}(x) + x J_{n+1}(x).$$

$$\text{Hence } 2n J_n(x) = x [J_{n-1}(x) + J_{n+1}(x)]$$

$$(V) \quad \frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x). \quad (\text{Avadh 2006, 10; Agra 08; Lucknow 06, 08; Meerut 13B; Rohilkhand 14})$$

Proof: $\frac{d}{dx} [x^{-n} J_n(x)] = -n x^{-n-1} J_n(x) + x^{-n} J_n'(x)$

$$= x^{-n-1} [-n J_n(x) + x J_n'(x)]$$

$$= x^{-n-1} [-n J_n(x) + \{n J_n(x) - x J_{n+1}(x)\}], \text{ from recurrence formula I}$$

$$= x^{-n-1} [-x J_{n+1}(x)] = -x^{-n} J_{n+1}(x).$$

$$\text{Hence } \frac{d}{dx} [x^{-n} J_n(x)] = -x^{-n} J_{n+1}(x).$$

Aliter: We have $\frac{d}{dx} [x^{-n} J_n(x)] = \frac{d}{dx} \left[x^{-n} \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \right]$

$$= \frac{d}{dx} \left[\sum_{r=0}^{\infty} \frac{(-1)^r}{r! \Gamma(n+r+1)} \cdot \left(\frac{1}{2}\right)^n \cdot \left(\frac{x}{2}\right)^{2r} \right]$$

$$= \sum_{r=0}^{\infty} \left[\frac{(-1)^r}{r! \Gamma(n+r+1)} \left(\frac{1}{2}\right)^n \cdot 2r \left(\frac{x}{2}\right)^{2r-1} \cdot \frac{1}{2} \right]$$

$$= x^{-n} \sum_{r=1}^{\infty} \frac{(-1)^r}{(r-1)! \Gamma(n+r+1)} \cdot \left(\frac{x}{2}\right)^{n-1+2r}$$

$$= x^{-n} \sum_{s=0}^{\infty} \frac{(-1)^{s+1}}{s! \Gamma(n+s+2)} \cdot \left(\frac{x}{2}\right)^{n+2s+1}, \text{ putting } r-1=s$$

$$= -x^{-n} \sum_{s=0}^{\infty} \frac{(-1)^s}{s! \Gamma(n+1+s+1)} \left(\frac{x}{2}\right)^{n+1+2s} = -x^{-n} J_{n+1}(x).$$

$$(VI) \quad \frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x). \quad (\text{Agra 2003; Avadh 09, 11; Kanpur 10; Purvanchal 10; Bundelkhand 13})$$

Proof: $\frac{d}{dx} [x^n J_n(x)] = n x^{n-1} J_n(x) + x^n J_n'(x) = x^{n-1} [n J_n(x) + x J_n'(x)]$

$$= x^{n-1} [n J_n(x) + \{n J_n(x) - x J_{n-1}(x)\}], \text{ from recurrence formula II}$$

$$= x^{n-1} [x J_{n-1}(x)] = x^n J_{n-1}(x).$$

$$\text{Hence } \frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x).$$

Aliter: We have

$$\frac{d}{dx} [x^n J_n(x)] = \frac{d}{dx} \left[x^n \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r} \right]$$

$$\begin{aligned}
&= \frac{d}{dx} \left[\sum_{r=0}^{\infty} \frac{(-1)^r}{r! \Gamma(n+r+1)} \cdot 2^n \cdot \left(\frac{x}{2}\right)^{2(n+r)} \right] \\
&= \sum_{r=0}^{\infty} \left[\frac{(-1)^r \cdot 2^n \cdot 2(n+r)}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{2n+2r-1} \cdot \frac{1}{2} \right] \\
&= x^n \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \Gamma(n+r)} \cdot \left(\frac{x}{2}\right)^{n+2r-1} \\
&= x^n \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \Gamma(n-1+r+1)} \left(\frac{x}{2}\right)^{n-1+2r} \\
&= x^n \cdot J_{n-1}(x).
\end{aligned}$$

Illustrative Examples

Example 3: Prove that $\frac{d}{dx} [J_n^2 + J_{n+1}^2] = 2 \left(\frac{n}{x} J_n^2 - \frac{n+1}{x} J_{n+1}^2 \right)$.

or $J_n J_n' + J_{n+1} J_{n+1}' = \frac{1}{x} [n J_n^2 - (n+1) J_{n+1}^2].$

Solution: We have $\frac{d}{dx} (J_n^2 + J_{n+1}^2) = 2 J_n J_n' + 2 J_{n+1} J_{n+1}'.$... (1)

From recurrence formula I, we have $J_n' = \frac{n}{x} J_n - J_{n+1}.$... (2)

Also from recurrence formula II, we have $J_n' = -\frac{n}{x} J_n + J_{n-1}.$

Replacing n by $n+1$ in this result, we have

$$J_{n+1}' = -\frac{n+1}{x} J_{n+1} + J_n. \quad \dots (3)$$

Substituting the values of J_n' and J_{n+1}' from (2) and (3) in (1), we have

$$\begin{aligned}
\frac{d}{dx} (J_n^2 + J_{n+1}^2) &= 2 J_n \left(\frac{n}{x} J_n - J_{n+1} \right) + 2 J_{n+1} \left(-\frac{n+1}{x} J_{n+1} + J_n \right) \\
&= 2 \left(\frac{n}{x} J_n^2 - \frac{n+1}{x} J_{n+1}^2 \right).
\end{aligned}$$

Example 4: Prove that

(i) $J_{1/2}(x) = \sqrt{\left(\frac{2}{\pi x}\right)} \cdot \sin x$

(Avadh 2007, 12, 13; Lucknow 08, 10; Meerut 09; Kanpur 09; Rohilkhand 10; Purvanchal 08)

(ii) $J_{-1/2}(x) = \sqrt{\left(\frac{2}{\pi x}\right)} \cdot \cos x.$

(Avadh 2007, 11; Purvanchal 10)

(iii) $[J_{1/2}(x)]^2 + [J_{-1/2}(x)]^2 = 2 / \pi x.$

(Avadh 2007, 08, 14; Purvanchal 09, 14)

$$(iv) \quad J_{-3/2}(x) = -\sqrt{\left(\frac{2}{\pi x}\right)} \left(\frac{1}{x} \cos x + \sin x\right). \quad (\text{Avadh 2008})$$

$$(v) \quad J_{-5/2}(x) = \sqrt{\left(\frac{2}{\pi x}\right)} \cdot \left[\left(\frac{3-x^2}{x^2}\right) \cos x + \frac{3}{x} \sin x\right].$$

Solution: We know that

$$J_n(x) = \frac{x^n}{2^n \Gamma(n+1)} \left[1 - \frac{x^2}{2 \cdot (2n+2)} + \frac{x^4}{2 \cdot 4 \cdot (2n+2)(2n+4)} - \dots \right] \quad \dots(1)$$

(i) Putting $n = \frac{1}{2}$ in (1), we have

$$\begin{aligned} J_{1/2}(x) &= \frac{x^{1/2}}{2^{1/2} \Gamma(3/2)} \left[1 - \frac{x^2}{2 \cdot 3} + \frac{x^4}{2 \cdot 4 \cdot 3 \cdot 5} - \dots \right] \\ &= \sqrt{\left(\frac{2}{\pi x}\right)} \left[x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right] = \sqrt{\left(\frac{2}{\pi x}\right)} \cdot \sin x \quad \dots(2) \end{aligned}$$

(ii) Again putting $n = -\frac{1}{2}$ in (1), we have

$$\begin{aligned} J_{-1/2}(x) &= \frac{x^{-1/2}}{2^{-1/2} \Gamma\left(\frac{1}{2}\right)} \left[1 - \frac{x^2}{2 \cdot 1} + \frac{x^4}{2 \cdot 4 \cdot 1 \cdot 3} - \dots \right] \\ &= \sqrt{\left(\frac{2}{\pi x}\right)} \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right] = \sqrt{\left(\frac{2}{\pi x}\right)} \cos x. \quad \dots(3) \end{aligned}$$

(iii) Squaring and adding (2) and (3), we have

$$[J_{1/2}(x)]^2 + [J_{-1/2}(x)]^2 = \frac{2}{\pi x}.$$

(iv) From recurrence formula IV, we have

$$2n J_n(x) = x [J_{n-1}(x) + J_{n+1}(x)]$$

$$\text{or} \quad J_{n-1}(x) = (2n/x) J_n(x) - J_{n+1}(x). \quad \dots(4)$$

Putting $n = -\frac{1}{2}$ in (4), we have

$$J_{-3/2}(x) = -(1/x) J_{-1/2}(x) - J_{1/2}(x) \quad \dots(5)$$

Substituting for $J_{1/2}(x)$ and $J_{-1/2}(x)$ from (2) and (3) in (5), we have

$$\begin{aligned} J_{-3/2}(x) &= -\frac{1}{x} \cdot \sqrt{\left(\frac{2}{\pi x}\right)} \cos x - \sqrt{\left(\frac{2}{\pi x}\right)} \sin x \\ &= -\sqrt{\left(\frac{2}{\pi x}\right)} \left(\frac{1}{x} \cos x + \sin x\right). \quad \dots(6) \end{aligned}$$

Aliter: We know that

$$J_n(x) = \frac{x^n}{2^n \Gamma(n+1)} \left[1 - \frac{x^2}{2 \cdot (2n+2)} + \frac{x^4}{2 \cdot 4 \cdot (2n+2)(2n+4)} - \dots \right]$$

$$= \frac{(n+1)x^n}{2^n \Gamma(n+2)} \left[1 - \frac{x^2}{2 \cdot (2n+2)} + \frac{x^4}{2 \cdot 4 \cdot (2n+2)(2n+4)} - \dots \right],$$

multiplying the numerator and the denominator by $n+1$.

Now putting $n = -3/2$, we have

$$\begin{aligned} J_{-3/2}(x) &= \frac{\left(-\frac{1}{2}\right)x^{-3/2}}{2^{-3/2} \Gamma\left(\frac{1}{2}\right)} \left[1 - \frac{x^2}{2 \cdot (-1)} + \frac{x^4}{2 \cdot 4 \cdot (-1)(+1)} - \frac{x^6}{2 \cdot 4 \cdot 6 \cdot (-1)(1)(3)} + \dots \right] \\ &= -\sqrt{\left(\frac{2}{\pi x}\right)} \cdot \frac{1}{x} \left[1 + \frac{x^2}{2} - \frac{x^4}{2 \cdot 4} + \frac{x^6}{2 \cdot 3 \cdot 4 \cdot 6} - \dots \right] \\ &= -\sqrt{\left(\frac{2}{\pi x}\right)} \cdot \frac{1}{x} \left[1 + \frac{1}{2!}x^2 - \frac{3}{4!}x^4 + \frac{5}{6!}x^6 - \dots \right] \\ &= -\sqrt{\left(\frac{2}{\pi x}\right)} \cdot \frac{1}{x} \left[1 + \frac{2-1}{2!}x^2 - \frac{4-1}{4!}x^4 + \frac{6-1}{6!}x^6 - \dots \right] \\ &= -\sqrt{\left(\frac{2}{\pi x}\right)} \cdot \left[\left(\frac{1}{x} - \frac{1}{2!}x + \frac{1}{4!}x^3 - \dots\right) + \left(\frac{2}{2!}x - \frac{4}{4!}x^3 + \frac{6}{6!}x^5 - \dots\right) \right] \\ &= -\sqrt{\left(\frac{2}{\pi x}\right)} \left[\frac{1}{x} \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) + \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right) \right] \\ &= -\sqrt{\left(\frac{2}{\pi x}\right)} \left(\frac{1}{x} \cos x + \sin x \right). \end{aligned}$$

(v) Putting $n = -3/2$ in (4), we have $J_{-5/2}(x) = (-3/x) J_{-3/2}(x) - J_{-1/2}(x)$

$$= \left(-\frac{3}{x}\right) \cdot \left[-\sqrt{\left(\frac{2}{\pi x}\right)} \cdot \left(\frac{1}{x} \cos x + \sin x\right) \right] - \sqrt{\left(\frac{2}{\pi x}\right)} \cos x,$$

substituting for $J_{-3/2}(x)$ and $J_{-1/2}(x)$ from (6) and (3)

$$\begin{aligned} &= \sqrt{\left(\frac{2}{\pi x}\right)} \cdot \left[\frac{3}{x} \left(\frac{1}{x} \cos x + \sin x \right) - \cos x \right] \\ &= \sqrt{\left(\frac{2}{\pi x}\right)} \left[\left(\frac{3-x^2}{x^2} \right) \cos x + \frac{3}{x} \sin x \right]. \end{aligned}$$

Aliter: We know that

$$\begin{aligned} J_n(x) &= \frac{x^n}{2^n \Gamma(n+1)} \left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4(2n+2)(2n+4)} - \dots \right] \\ &= \frac{(n+2)(n+1)x^n}{2^n \Gamma(n+3)} \left[1 - \frac{x^2}{2 \cdot (2n+2)} + \frac{x^4}{2 \cdot 4 \cdot (2n+2)(2n+4)} - \dots \right], \end{aligned}$$

multiplying the numerator and the denominator by $(n+2)(n+1)$.

Now putting $n = -5/2$, we have

$$\begin{aligned}
 J_{-5/2}(x) &= \frac{(-1/2)(-3/2)x^{-5/2}}{2^{-5/2} \Gamma(1/2)} \cdot \left[1 - \frac{x^2}{2 \cdot (-3)} + \frac{x^4}{2 \cdot 4 \cdot (-3) \cdot (-1)} \right. \\
 &\quad \left. - \frac{x^6}{2 \cdot 4 \cdot 6 \cdot (-3) \cdot (-1) \cdot (1)} + \dots \right] \\
 &= \sqrt{\left(\frac{2}{\pi x}\right)} \cdot \frac{3}{x^2} \left[1 + \frac{x^2}{3!} + \frac{x^4}{4!} - \frac{5x^6}{6!} + \dots \right] \\
 &= \sqrt{\left(\frac{2}{\pi x}\right)} \left[\frac{3}{x^2} + \frac{1}{2!} + \frac{3x^2}{4!} - \frac{15x}{6!} + \dots \right] \\
 &= \sqrt{\left(\frac{2}{\pi x}\right)} \left[\frac{3}{x^2} - \frac{3-4}{2!} + \frac{3+0}{4!} x^2 - \frac{3+12}{6!} x^4 + \dots \right] \\
 &= \sqrt{\left(\frac{2}{\pi x}\right)} \left[\left(\frac{3}{x^2} - \frac{3}{2!} + \frac{3}{4!} x^2 - \frac{3}{6!} x^4 - \dots \right) + \left(\frac{4}{2!} + \frac{0}{4!} x^2 - \frac{12}{6!} x^4 - \dots \right) \right] \\
 &= \sqrt{\left(\frac{2}{\pi x}\right)} \left[\frac{3}{x^2} \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} - \dots \right) \right. \\
 &\quad \left. - \frac{2(1-3)}{2!} + \frac{4(3-3)}{4!} x^2 - \frac{6(5-3)}{6!} x^4 - \dots \right] \\
 &= \sqrt{\left(\frac{2}{\pi x}\right)} \left[\frac{3}{x^2} \cos x - \frac{2!}{2!} + \frac{4!}{4!} x^2 - \frac{6!}{6!} x^4 + \dots \right. \\
 &\quad \left. + 3 \left(\frac{2}{2!} - \frac{4}{4!} x^2 + \frac{6}{6!} x^4 - \dots \right) \right] \\
 &= \sqrt{\left(\frac{2}{\pi x}\right)} \left[\frac{3}{x^2} \cos x - \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} \dots \right) + \frac{3}{x} \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} \dots \right) \right] \\
 &= \sqrt{\left(\frac{2}{\pi x}\right)} \left[\frac{3}{x^2} \cos x - \cos x + \frac{3}{x} \sin x \right] = \sqrt{\left(\frac{2}{\pi x}\right)} \left[\frac{3-x^2}{x^2} \cos x + \frac{3}{x} \sin x \right].
 \end{aligned}$$

14.7 Generating Function for $J_n(x)$

Prove that when n is a positive integer, $J_n(x)$ is the coefficient of z^n in the expansion of $e^{x[z - (1/z)]/2}$ in ascending and descending powers of z . Also prove that $J_n(x)$ is the coefficient of z^{-n} multiplied by $(-1)^n$ in the expansion of the above expression.

(Rohilkhand 2002, 06; Lucknow 06, 09, 11; Meerut 08; Avadh 06, 08)

Proof: $e^{x[z - (1/z)]/2} = e^{xz/2} \cdot e^{-x/2z}$

$$\begin{aligned}
 &= \left[1 + \frac{xz}{2} + \frac{1}{2!} \left(\frac{xz}{2} \right)^2 + \dots + \frac{1}{n!} \left(\frac{xz}{2} \right)^n + \frac{1}{(n+1)!} \left(\frac{xz}{2} \right)^{n+1} + \frac{1}{(n+2)!} \left(\frac{xz}{2} \right)^{n+2} + \dots \right] \\
 &\times \left[1 - \frac{x}{2z} + \frac{1}{2!} \left(\frac{x}{2z} \right)^2 - \dots + \frac{(-1)^n}{n!} \left(\frac{x}{2z} \right)^n + \frac{(-1)^{n+1}}{(n+1)!} \left(\frac{x}{2z} \right)^{n+1} + \frac{(-1)^{n+2}}{(n+2)!} \left(\frac{x}{2z} \right)^{n+2} + \dots \right]
 \end{aligned}$$

Coefficient of z^n in this product

$$\begin{aligned}
&= \frac{1}{n!} \left(\frac{x}{2}\right)^n - \frac{1}{(n+1)!} \left(\frac{x}{2}\right) \left(\frac{x}{2}\right)^{n+1} + \frac{1}{(n+2)!} \cdot \frac{1}{2!} \left(\frac{x}{2}\right)^2 \left(\frac{x}{2}\right)^{n+2} - \dots \\
&= \frac{1}{n!} \left(\frac{x}{2}\right)^n - \frac{1}{(n+1)!} \left(\frac{x}{2}\right)^{n+2} + \frac{1}{2!(n+2)!} \left(\frac{x}{2}\right)^{n+4} - \dots \\
&= \frac{(-1)^0}{\Gamma(n+1)} \left(\frac{x}{2}\right)^n + \frac{(-1)}{1!\Gamma(n+2)} \cdot \left(\frac{x}{2}\right)^{n+2} + \frac{(-1)^2}{2!\Gamma(n+3)} \cdot \left(\frac{x}{2}\right)^{n+4} + \dots \\
&= \sum_{r=0}^{\infty} (-1)^r \frac{1}{r!\Gamma(n+r+1)} \cdot \left(\frac{x}{2}\right)^{n+2r} = J_n(x).
\end{aligned}$$

Similarly, the coefficient of z^{-n} in the above product

$$\begin{aligned}
&= \frac{(-1)^n}{n!} \left(\frac{x}{2}\right)^n + \frac{(-1)^{n+1}}{(n+1)!} \cdot \frac{x}{2} \cdot \left(\frac{x}{2}\right)^{n+1} + \frac{(-1)^{n+2}}{(n+2)!} \cdot \frac{1}{2!} \cdot \left(\frac{x}{2}\right)^2 \left(\frac{x}{2}\right)^{n+2} + \dots \\
&= (-1)^n \left[\frac{(-1)^0}{\Gamma(n+1)} \left(\frac{x}{2}\right)^n + \frac{(-1)}{\Gamma(n+2)} \left(\frac{x}{2}\right)^{n+2} + \frac{(-1)^2}{2!\Gamma(n+3)} \left(\frac{x}{2}\right)^{n+4} + \dots \right] \\
&= (-1)^n J_n(x).
\end{aligned}$$

Note. In the above product the term independent of z is

$$1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots = J_0(x).$$

$$\begin{aligned}
\text{Hence } e^{x(z-1/z)/2} &= J_0 + \left(z - \frac{1}{z}\right) J_1 + \left(z^2 + \frac{1}{z^2}\right) J_2 + \dots + \left[z^n + (-1)^n \frac{1}{z^n}\right] J_n + \dots \\
&= \sum_{-\infty}^{\infty} z^n J_n(x),
\end{aligned}$$

since $J_{-n}(x) = (-1)^n J_n(x)$. See Example 1, after article 14.5.

14.8 Orthogonal Property of Bessel's Functions

$$\int_0^1 x J_n(\alpha x) J_n(\beta x) dx = \begin{cases} 0 & , \alpha \neq \beta \\ \frac{1}{2} [J_{n+1}(\alpha)]^2 & , \alpha = \beta, \end{cases}$$

where α, β are the roots of $J_n(x) = 0$.

(Lucknow 2008, 11; Meerut 10; Agra 11)

Proof: The solutions of the equations

$$x^2 u'' + xu' + (\alpha^2 x^2 - n^2) u = 0 \quad \dots(1)$$

$$\text{and } x^2 v'' + xv' + (\beta^2 x^2 - n^2) v = 0 \quad \dots(2)$$

are $u = J_n(\alpha x)$ and $v = J_n(\beta x)$ respectively.

Multiplying (1) by v/x and (2) by u/x and subtracting, we get

$$x(u''v - uv'') + (u'v - uv') + (\alpha^2 - \beta^2)xuv = 0$$

$$\text{or } \frac{d}{dx} \{x(u'v - uv')\} = (\beta^2 - \alpha^2)xuv. \quad \dots(3)$$

Integrating both sides of (3) from 0 to 1, we get

$$(\beta^2 - \alpha^2) \int_0^1 xuv \, dx = [x(u'v - uv')]_0^1 = (u'v - uv')_{x=1} \quad \dots(4)$$

Now

$$u = J_n(\alpha x)$$

$$\Rightarrow u' = \frac{d}{dx} [J_n(\alpha x)] = \frac{d}{d(\alpha x)} [J_n(\alpha x)] \cdot \frac{d(\alpha x)}{dx} = \alpha J_n'(\alpha x)$$

Similarly, $v = J_n(\beta x) \Rightarrow v' = \beta J_n'(\beta x)$.

Substituting these values in (4), we get

$$\int_0^1 x J_n(\alpha x) J_n(\beta x) \, dx = \frac{\alpha J_n'(\alpha) J_n(\beta) - \beta J_n(\alpha) J_n'(\beta)}{\beta^2 - \alpha^2} \quad \dots(5)$$

If α and β are distinct roots of $J_n(x) = 0$, then $J_n(\alpha) = J_n(\beta) = 0$ and thus (5) reduces to

$$\int_0^1 x J_n(\alpha x) J_n(\beta x) \, dx = 0 \quad \dots(6)$$

This is known as the **orthogonality relation of Bessel's functions**.

If $\beta = \alpha$, then the right hand side of (5) is of the form $0/0$. Its value can be found by considering α as a root of $J_n(x) = 0$ and β as a variable approaching α . Then, from (5) we have

$$\lim_{\beta \rightarrow \alpha} \int_0^1 x J_n(\alpha x) J_n(\beta x) \, dx = \lim_{\beta \rightarrow \alpha} \frac{\alpha J_n'(\alpha) J_n(\beta)}{\beta^2 - \alpha^2}$$

or

$$\begin{aligned} \int_0^1 x J_n^2(\alpha x) \, dx &= \lim_{\beta \rightarrow \alpha} \frac{\alpha J_n'(\alpha) J_n'(\beta)}{2\beta} \\ &= \frac{1}{2} \{J_n'(\alpha)\}^2, \text{ by L'Hospital's rule} \\ &= \frac{1}{2} [J_{n+1}(\alpha)]^2, \text{ by recurrence formula.} \end{aligned} \quad \dots(7)$$

Corollary: If the interval be from 0 to a , then

$$\int_0^a x J_n^2(\alpha x) \, dx = \frac{a^2}{2} J_{n+1}^2(a\alpha), \quad \dots(8)$$

where α 's are the roots of $J_n(ax) = 0$.

14.9 Some Trigonometric Expansions Involving Bessel's Functions

Here we give some trigonometric expansions involving Bessel's functions.

- (i) $\cos(x \sin \theta) = J_0 + 2 \cos 2\theta J_2 + 2 \cos 4\theta J_4 + \dots$
- (ii) $\sin(x \sin \theta) = 2 \sin \theta J_1 + 2 \sin 3\theta J_3 + \dots$
- (iii) $\cos(x \cos \theta) = J_0 - 2 \cos 2\theta J_2 + 2 \cos 4\theta J_4 - \dots$
- (iv) $\sin(x \cos \theta) = 2 \cos \theta J_1 - 2 \cos 3\theta J_3 + 2 \cos 5\theta J_5 - \dots$

$$(v) \quad \cos x = J_0 - 2 J_2 + 2 J_4 - 2 J_6 + \dots$$

$$(vi) \quad \sin x = 2 J_1 - 2 J_3 + 2 J_5 - 2 J_7 + \dots$$

(Kanpur 2011)

(Kanpur 2011)

Proof: We know that

$$e^{x(z-1/z)/2} = J_0 + \left(z - \frac{1}{z}\right) J_1 + \left(z^2 + \frac{1}{z^2}\right) J_2 + \left(z^3 - \frac{1}{z^3}\right) J_3 + \dots \quad \dots(1)$$

Putting $z = e^{i\theta}$, so that $1/z = e^{-i\theta}$, $z - 1/z = e^{i\theta} - e^{-i\theta} = 2i \sin \theta$, $z^p = e^{ip\theta}$,

$$1/z^p = e^{-ip\theta}, z^p + 1/z^p = e^{ip\theta} + e^{-ip\theta} = 2 \cos p\theta,$$

$$z^p - 1/z^p = e^{ip\theta} - e^{-ip\theta} = 2i \sin p\theta$$

in (1), we have $e^{i(x \sin \theta)} = J_0 + (2i \sin \theta) J_1 + (2 \cos 2\theta) J_2$

$$+ (2i \sin 3\theta) J_3 + (2 \cos 4\theta) J_4 + \dots$$

or $\cos(x \sin \theta) + i \sin(x \sin \theta) = (J_0 + 2 \cos 2\theta J_2 + 2 \cos 4\theta J_4 + \dots)$

$$+ i(2 \sin \theta J_1 + 2 \sin 3\theta J_3 + \dots) \quad \dots(2)$$

(i) Equating the real parts on both sides of (2), we have

$$\cos(x \sin \theta) = J_0 + 2 \cos 2\theta J_2 + 2 \cos 4\theta J_4 + \dots$$

(ii) Equating the imaginary parts on both sides of (2), we have

$$\sin(x \sin \theta) = 2 \sin \theta J_1 + 2 \sin 3\theta J_3 + \dots$$

(iii) Replacing θ by $\pi/2 - \theta$ in (i), we have

$$\cos(x \cos \theta) = J_0 - 2 \cos 2\theta J_2 + 2 \cos 4\theta J_4 - \dots$$

(iv) Replacing θ by $\pi/2 - \theta$ in (ii), we have

$$\sin(x \cos \theta) = 2 \cos \theta J_1 - 2 \cos 3\theta J_3 + 2 \cos 5\theta J_5 - \dots$$

(v) Putting $\theta = \pi/2$ in (i), we have

$$\cos x = J_0 - 2 J_2 + 2 J_4 - 2 J_6 + \dots$$

(vi) Putting $\theta = \pi/2$ in (ii), we have

$$\sin x = 2 J_1 - 2 J_3 + 2 J_5 - 2 J_7 + \dots$$

Illustrative Examples

Example 5: Using generating function, prove that

$$J_n(x+y) = \sum_{r=-\infty}^{\infty} J_r(x) J_{n-r}(y).$$

Solution: We know that

$$\exp \left[\frac{1}{2} (x+y) \left(z - \frac{1}{z} \right) \right] = \sum_{n=-\infty}^{\infty} J_n(x+y) z^n$$

$$\text{or} \quad \exp \left[\frac{1}{2} x \left(z - \frac{1}{z} \right) \right] \cdot \exp \left[\frac{1}{2} y \left(z - \frac{1}{z} \right) \right] = \sum_{n=-\infty}^{\infty} J_n(x+y) z^n$$

$$\text{or} \quad \sum_{r=-\infty}^{\infty} J_r(x) z^r \cdot \sum_{k=-\infty}^{\infty} J_k(y) z^k = \sum_{n=-\infty}^{\infty} J_n(x+y) z^n$$

Equating the coefficients of z^n on both the sides, we get

$$\sum_{r=-\infty}^{\infty} J_r(x) J_{n-r}(y) = J_n(x+y).$$

Example 6: Show that when n is integral

$$(a) \quad \pi J_n = \int_0^\pi \cos(n\theta - x \sin \theta) d\theta.$$

(Purvanchal 2007, 08; Avadh 12)

$$(b) \quad (i) \quad \pi J_0 = \int_0^\pi \cos(x \cos \phi) d\phi$$

(Avadh 2012)

$$(ii) \quad \pi J_0 = \int_0^\pi \cos(x \sin \phi) d\phi$$

(Agra 2006; Purvanchal 09)

$$\text{and hence deduce that } J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots = \sum_{r=0}^{\infty} \frac{(-1)^r x^{2r}}{(2^r r!)^2}.$$

Solution: From article 14.9, we have

$$\cos(x \sin \theta) = J_0 + 2 J_2 \cos 2\theta + \dots + 2 J_{2m} \cos 2m\theta + \dots \quad \dots (1)$$

$$\text{and } \sin(x \sin \theta) = 2 \sin \theta \cdot J_1 + 2 \sin 3\theta \cdot J_3 + \dots + 2 J_{2m+1} \sin(2m+1)\theta + \dots \quad \dots (2)$$

(a) Multiplying both sides of (1) by $\cos 2m\theta$ and then integrating between the limits 0 to π ,

$$\begin{aligned} & \int_0^\pi \cos(x \sin \theta) \cos 2m\theta d\theta \\ &= J_0 \int_0^\pi \cos 2m\theta d\theta + 2 J_2 \int_0^\pi \cos 2\theta \cos 2m\theta d\theta + \dots \\ & \quad + 2 J_{2m} \int_0^\pi \cos^2 2m\theta d\theta + \dots \\ &= 0 + \dots + J_{2m} \int_0^\pi (1 + \cos 4m\theta) d\theta + 0 + \dots = \pi J_{2m}. \end{aligned}$$

Similarly we can prove that

$$\int_0^\pi \cos(x \sin \theta) \cos(2m+1)\theta d\theta = 0.$$

Again multiplying both sides of (2) by $\sin(2m+1)\theta$ and then integrating between the limits 0 to π , we have

$$\begin{aligned} & \int_0^\pi \sin(x \sin \theta) \sin(2m+1)\theta d\theta \\ &= 2 J_1 \int_0^\pi \sin \theta \cdot \sin(2m+1)\theta d\theta + 2 J_3 \int_0^\pi \sin 3\theta \sin(2m+1)\theta d\theta + \dots \\ & \quad + 2 J_{2m+1} \int_0^\pi \sin^2(2m+1)\theta d\theta + \dots \\ &= 0 + 0 + \dots + J_{2m+1} \int_0^\pi \{1 - \cos 2(2m+1)\theta\} d\theta + \dots \\ &= J_{2m+1} [\theta]_0^\pi = \pi J_{2m+1}. \end{aligned}$$

$$\text{Similarly, } \int_0^\pi \sin(x \sin \theta) \sin 2m\theta d\theta = 0.$$

$$\text{Therefore, } \int_0^\pi \cos(2m\theta - x \sin \theta) d\theta$$

$$= \int_0^\pi \cos 2m\theta \cdot \cos(x \sin \theta) d\theta + \int_0^\pi \sin 2m\theta \cdot \sin(x \sin \theta) d\theta = \pi J_{2m}.$$

$$\text{Also } \int_0^\pi \cos[(2m+1)\theta - x \sin \theta] d\theta$$

$$\begin{aligned} &= \int_0^\pi \cos(2m+1)\theta \cdot \cos(x \sin \theta) d\theta + \int_0^\pi \sin(2m+1)\theta \sin(x \sin \theta) d\theta \\ &= \pi J_{2m+1}. \end{aligned}$$

Hence for all positive integral values of n , we have

$$\int_0^\pi \cos(n\theta - x \sin \theta) d\theta = \pi J_n.$$

If n is negative, say $n = -m$, where m is +ive, then

$$\begin{aligned} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta &= \int_0^\pi \cos(-m\theta - x \sin \theta) d\theta \\ &= - \int_\pi^0 \cos\{-m(\pi - \phi) - x \sin(\pi - \phi)\} d\phi, \text{ putting } \theta = \pi - \phi \\ &= \int_0^\pi \cos\{-m\pi + (m\phi - x \sin \phi)\} d\phi \\ &= \int_0^\pi \{\cos m\pi \cdot \cos(m\phi - x \sin \phi) + \sin m\pi \sin(m\phi - x \sin \phi)\} d\phi \\ &= (-1)^m \int_0^\pi \cos(m\phi - x \sin \phi) d\phi \\ &= (-1)^m \pi J_m(x), \text{ since we have proved the result for positive integers} \\ &= \pi J_{-m}(x), \text{ since } J_{-m}(x) = (-1)^m J_m(x) \\ &= \pi J_n(x). \end{aligned}$$

Hence for all integral values of n ,

$$\int_0^\pi \cos(n\theta - x \sin \theta) d\theta = \pi J_n. \quad \text{Proved}$$

(b) Putting $\theta = \pi/2 + \phi$ in the value of $\cos(x \sin \theta)$ from (1), we have

$$\cos(x \cos \phi) = J_0 - 2J_2 \cos 2\phi + 2J_4 \cos 4\phi - \dots$$

$$\begin{aligned} \therefore \int_0^\pi \cos(x \cos \phi) d\phi &= J_0 \int_0^\pi d\phi - 2J_2 \int_0^\pi \cos 2\phi d\phi + \dots \\ &= \pi J_0. \end{aligned} \quad \text{Proved}$$

From (1) we have $\cos(x \sin \phi) = J_0 + 2J_2 \cos 2\phi + 2J_4 \cos 4\phi + \dots$

$$\begin{aligned} \therefore \int_0^\pi \cos(x \sin \phi) d\phi &= J_0 \int_0^\pi d\phi + 2J_2 \int_0^\pi \cos 2\phi d\phi + \dots \\ &= \pi J_0. \end{aligned} \quad \text{Proved}$$

Deduction: We have proved that $J_0(x) = \frac{1}{\pi} \int_0^\pi \cos(x \cos \phi) d\phi$

$$= \frac{1}{\pi} \int_0^\pi \left(1 - \frac{x^2 \cos^2 \phi}{2!} + \frac{x^4 \cos^4 \phi}{4!} - \frac{x^6 \cos^6 \phi}{6!} + \dots \right) d\phi. \quad \dots(1)$$

But from definite integrals, we have

$$\begin{aligned} \int_0^\pi \cos^{2r} \phi d\phi &= 2 \int_0^{\pi/2} \cos^{2r} \phi d\phi \\ &= 2 \cdot \frac{(2r-1)(2r-3)\dots 5.3.1}{2r(2r-2)\dots 6.4.2} \cdot \frac{\pi}{2} = \frac{1.3.5\dots(2r-1)}{2.4.6\dots(2r)} \pi. \end{aligned}$$

$\therefore$ from (1), we have

$$J_0(x) = \frac{1}{\pi} \left[\pi - \frac{x^2}{2!} \cdot \frac{1}{2} \pi + \frac{x^4}{4!} \cdot \frac{1.3}{2.4} \pi - \frac{x^6}{6!} \cdot \frac{1.3.5}{2.4.6} \pi + \dots \right]$$

$$\begin{aligned}
 &= 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots \\
 &= 1 - \frac{x^2}{2^2} + \frac{x^4}{2^4 \cdot (2!)^2} - \frac{x^6}{2^6 \cdot (3!)^2} + \dots = \sum_{r=0}^{\infty} \frac{(-1)^r x^{2r}}{(2^r \cdot r!)^2}.
 \end{aligned}$$

Comprehensive Exercise 1

1. (i) Prove that $\lim_{x \rightarrow 0} \frac{J_n(x)}{x^n} = \frac{1}{2^n \cdot \Gamma(n+1)}$; $n > -1$. (Kanpur 2007; Purvanchal 11)
 (ii) Prove that $J_{n+3} + J_{n+5} = (2/x)(n+4)J_{n+4}$. (Kanpur 2009; Rohilkhand 11)
 (iii) Prove that $4J_n''(x) = J_{n-2}(x) - 2J_n(x) + J_{n+2}(x)$.
2. Prove that (i) $J_0' = -J_1$ (Kanpur 2008, 12; Lucknow 08; Agra 10; Rohilkhand 10, 11; Agra 2007)
 (ii) $J_2 = J_0'' - x^{-1}J_0'$ and (iii) $J_2 - J_0 = 2J_0''$. (Agra 2007)
3. Establish the differential formula
 $x^2 J_n''(x) = (n^2 - n - x^2)J_n(x) + xJ_{n+1}(x)$, ($n = 0, 1, 2, \dots$).
4. (i) Prove that $J_{n-1} = \frac{2}{x}[nJ_n - (n+2)J_{n+2} + (n+4)J_{n+4} - \dots]$
 and hence deduce that
 $\frac{1}{2}xJ_n = (n+1)J_{n+1} - (n+3)J_{n+3} + (n+5)J_{n+5} - \dots$
 (Avadh 2008; Purvanchal 14)
 (ii) Prove that $J_n' = \frac{2}{x}\left[\frac{n}{2}J_n - (n+2)J_{n+2} + (n+4)J_{n+4} - \dots\right]$.
5. (i) Prove that $J_0^2 + 2(J_1^2 + J_2^2 + J_3^2 + \dots) = 1$. (Agra 2010, 11)
 Deduce that $|J_0(x)| \leq 1, |J_n(x)| \leq 2^{-1/2}, n \geq 1$. (Agra 2009)
 (ii) Prove that $\frac{d}{dx}(xJ_nJ_{n+1}) = x(J_n^2 - J_{n+1}^2)$. (Purvanchal 2011)
6. (i) Prove that $x = 2J_0J_1 + 6J_1J_2 + \dots + 2(2n+1)J_nJ_{n+1} + \dots$
 (ii) Prove that $\int J_{n+1}(x)dx = \int J_{n-1}(x)dx - 2J_n(x) + C$.
7. Prove that (i) $J_{3/2}(x) = \sqrt{\left(\frac{2}{\pi x}\right)}\left[\frac{1}{x}\sin x - \cos x\right]$ (Meerut 2007, 09; Agra 03; Rohilkhand 08, 11; Kanpur 11; Purvanchal 11)
 (ii) $J_{5/2}(x) = \sqrt{\left(\frac{2}{\pi x}\right)}\left[\left(\frac{3-x^2}{x^2}\right)\sin x - \frac{3}{x}\cos x\right]$.
8. Show that
 (i) $\int_0^x x^n J_{n-1}(x)dx = x^n J_n(x)$ and (ii) $\int_0^x x^{n+1} J_n(x)dx = x^{n+1} J_{n+1}(x)$.
9. If $n > -1$, show that $\int_0^x x^{-n} J_{n+1}(x)dx = \frac{1}{2^n \Gamma(n+1)} - x^{-n} J_n(x)$. (Purvanchal 2011)

10. Prove that $\int_0^{\pi/2} \sqrt{\pi x} J_{1/2}(2x) dx = 1$.
11. Prove that $\int_0^{\pi} e^{-ax} J_0(bx) dx = \frac{1}{\sqrt{(a^2 + b^2)}}, a > 0$. (Agra 2002; Lucknow 10)
12. From the recurrence formula $2J_n' = J_{n-1} - J_{n+1}$
deduce the result $2^r J_n^r = J_{n-r} - rJ_{n-r+2} + \frac{r(r-1)}{2!} J_{n-r+4} + \dots + (1)^r J_{n+r}$.
13. Prove that for $n > -\frac{1}{2}$, $J_n(x) = \frac{(x/2)^n}{\sqrt{\pi} \sqrt{n + \frac{1}{2}}} \int_{-1}^1 (1-t^2)^{n-\frac{1}{2}} e^{ixt} dt$.
14. Prove that $\int_0^x t \{J_n(t)\}^2 dt = \frac{x^2}{2} \{J_n^2(x) - J_{n-1}(x) J_{n+1}(x)\}$.
15. (i) Express $\int J_3(x) dx$ in terms of J_0 and J_1 .
(ii) Express $\int x^{-3} J_4(x) dx$ in terms of J_0 and J_1 . (Lucknow 2011)

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- $n J_n(x) - x J_{n+1}(x) =$
 - $x J_n(x)$
 - $(n+1) J_n'(x)$
 - $x J_n'(x)$
 - None of these.
- $-n J_n(x) + x J_{n-1}(x) = \dots$
 - $J_{n-1}'(x)$
 - $x J_n'(x)$
 - $(n-1) J_n(x)$
 - None of these.
- $J_{n-1}(x) - J_{n+1}(x) = \dots$
 - $2 J_n'(x)$
 - $2 J_n(x)$
 - $n J_n(x)$
 - None of these.
- $x [J_{n-1}(x) + J_{n+1}(x)] = \dots$
 - $2 J_n(x)$
 - $2 J_n'(x)$
 - $2n J_n(x)$
 - None of these. (Agra 2006)
- The value of $J_n(x)$ is
 - $\sum_{r=0}^{\infty} (-1)^r \left(\frac{x}{2}\right)^{n+2r} \frac{1}{r! \Gamma(n+r+1)}$
 - $\sum_{r=0}^{\infty} (-1)^r \left(\frac{x}{2}\right)^{n+2r} \frac{1}{r! \Gamma(n+r)}$

$$(c) \sum_{r=0}^{\infty} (-1)^r \left(\frac{x}{2}\right)^{n+r} \frac{1}{r! \Gamma(n+r+1)}$$

(d) None of these.

(Rohilkhand 2004)

Fill In The Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. The differential equation $\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + \left(1 - \frac{n^2}{x^2}\right) y = 0$ is called
2. The most general solution of Bessel's equation is
3. When n is a positive integer, $J_n(x)$ is the coefficient of z^n in the expansion of
4. $J_{-n}(x) = (-1)^n \dots$
5. $\frac{d}{dx} [x^{-n} J_n(x)] = \dots$
6. $J_0(x)$ is the solution of the differential equation
7. $\frac{d}{dx} [x^n J_n(x)] = \dots$

True or False

Write 'T' for true and 'F' for false statement.

1. $J_0(x)$ is called Bessel's function of zeroeth order.
2. The most general solution of Bessel's equation is $y = A J_n(x) + B J_{-n}(x)$.
3. $J_n(x)$ is called Bessel's function of the first kind of order n .
4. $J_{-n}(x)$ is called Bessel's function of second kind of order $-n$.

Answers 1

Multiple Choice Questions

1. (c).
2. (b).
3. (a).
4. (c).
5. (a)

Fill in the Blank(s)

1. Bessel's equation.
2. $A J_n(x) + B J_{-n}(x)$.
3. $e^{x[z - (1/z)]/2}$.
4. $J_n(x)$.
5. $-x^{-n} J_{n+1}(x)$.
6. $\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + y = 0$.
7. $x^n J_{n-1}(x)$.

True or False

1. T.
2. T.
3. T.
4. F.



SECTION

B



INTEGRAL TRANSFORMS

Chapters

A large, light gray arrow pointing horizontally to the right, positioned below the 'Chapters' header.

1. The Laplace Transform

2. The Inverse Laplace Transform

3. Applications of Laplace Transform

4. Fourier Transforms



5. Finite Fourier Transforms

**6. Applications of Fourier Transforms in
Initial and Boundary Value Problems**

7. Fourier Series

Chapter

1



The Laplace Transform

Laplace transform or Laplace transformation reduces the problem of solving a differential equation to an algebraic problem. It is a method for solving linear differential equations arising in Physics and Engineering.

1.1 Integral Transform

Let $K(p, t)$ be a function of two variables p and t , where p is a parameter (may be real or complex) independent of t . The function $f(p)$ defined by the integral (assumed to be convergent)

$$f(p) = \int_{-\infty}^{\infty} K(p, t) F(t) dt$$

is called the **integral transform** of the function $F(t)$ and is denoted by $T\{F(t)\}$. The function $K(p, t)$ is called the **kernel** of the transformation.

Remark: Some authors use the letter 's' in place of 'p'.

1.2 Laplace Transformation

If the kernel $K(p, t)$ is defined as

$$K(p, t) = \begin{cases} 0 & \text{for } t < 0 \\ e^{-pt} & \text{for } t \geq 0 \end{cases}$$

then
$$f(p) = \int_0^{\infty} e^{-pt} F(t) dt. \quad \dots(1)$$

The function $f(p)$ defined by the integral (1) is called the **Laplace transform** of the function $F(t)$ and is also denoted by

$$L\{F(t)\} \quad \text{or} \quad \bar{F}(p).$$

Thus *Laplace transform* is a function of a new variable (or parameter) p given by (1).

Note: The Laplace transform of $F(t)$ is said to exist if the integral (1) converges for some values of p , otherwise it does not exist.

1.3 Linearity Property of Laplace Transformation

(Bundelkhand 2014)

A transformation T is said to be *linear* if for every pair of functions $F_1(t)$ and $F_2(t)$ and for every pair of constants a_1 and a_2 , we have

$$T\{a_1 F_1(t) + a_2 F_2(t)\} = a_1 T\{F_1(t)\} + a_2 T\{F_2(t)\}.$$

Theorem: The Laplace transformation is a linear transformation, i.e.

$$L\{a_1 F_1(t) + a_2 F_2(t)\} = a_1 L\{F_1(t)\} + a_2 L\{F_2(t)\}$$

where a_1, a_2 are constants.

(Avadh 2010)

Proof: We have

$$L\{F(t)\} = \int_0^{\infty} e^{-pt} F(t) dt.$$

$$\begin{aligned} \therefore L\{a_1 F_1(t) + a_2 F_2(t)\} &= \int_0^{\infty} e^{-pt} \{a_1 F_1(t) + a_2 F_2(t)\} dt \\ &= a_1 \int_0^{\infty} e^{-pt} F_1(t) dt + a_2 \int_0^{\infty} e^{-pt} F_2(t) dt \\ &= a_1 L\{F_1(t)\} + a_2 L\{F_2(t)\}. \end{aligned}$$

1.4 Piecewise (or Sectionally) Continuous Function

A function $F(t)$ is said to be piecewise (or sectionally) continuous on a closed interval $a \leq t \leq b$, if it is defined on that interval and is such that the interval can be subdivided into a finite number of intervals, in each of which $F(t)$ is continuous and has finite right and left hand limits.

1.5 Functions of Exponential Order

A function $F(t)$ is said to be of exponential order α as $t \rightarrow \infty$ if there exists a positive constant (real) M , a number α and a finite number t_0 such that $|F(t)| < Me^{\alpha t}$ or $|e^{-\alpha t} F(t)| < M$, for all $t \geq t_0$.

If a function $F(t)$ is of exponential order α , it is also of β , $\beta > \alpha$.

1.6 A Function of Class A

A function which is piecewise (or sectionally) continuous on every finite interval in the range $t \geq 0$ and is of exponential order as $t \rightarrow \infty$ is known as 'a function of class A'.

1.7 Existence of Laplace Transform

Theorem: If $F(t)$ is a function which is piecewise continuous on every finite interval in the range $t \geq 0$ and satisfies $|F(t)| \leq Me^{at}$ for all $t \geq 0$ and for some constants a and M , then the Laplace transform of $F(t)$ exists for all $p > a$.

Proof: We have

$$\begin{aligned} L\{F(t)\} &= \int_0^{\infty} e^{-pt} F(t) dt \\ &= \int_0^{t_0} e^{-pt} F(t) dt + \int_{t_0}^{\infty} e^{-pt} F(t) dt \end{aligned} \quad \dots(1)$$

The integral $\int_0^{t_0} e^{-pt} F(t) dt$ exists since $F(t)$ is piecewise continuous on every finite interval $0 \leq t \leq t_0$.

$$\text{Now} \quad \left| \int_{t_0}^{\infty} e^{-pt} F(t) dt \right| \leq \int_{t_0}^{\infty} |e^{-pt} F(t)| dt \leq \int_{t_0}^{\infty} e^{-pt} M e^{at} dt,$$

$$\text{since } |F(t)| \leq M e^{at}$$

$$= \int_{t_0}^{\infty} e^{-(p-a)t} M dt = \frac{M e^{-(p-a)t_0}}{p-a}, \quad p > a.$$

$$\therefore \quad \left| \int_{t_0}^{\infty} e^{-pt} F(t) dt \right| \leq \frac{M e^{-(p-a)t_0}}{p-a}, \quad p > a.$$

But $\frac{M e^{-(p-a)t_0}}{p-a}$ can be made as small as we please by taking t_0 sufficiently large.

Thus from (1), we conclude that $L\{F(t)\}$ exists for all $p > a$.

Note 1: The above theorem of existence of Laplace transform can also be stated as :

"If $F(t)$ is a function which is piece-wise continuous on every finite interval in the range $t \geq 0$ and is of exponential order a as $t \rightarrow \infty$, the Laplace transform of $F(t)$ exists for all $p > a$ ".

Or

"If $F(t)$ is a function of class A , the Laplace transform of $F(t)$ exists for $p > a$ ".

Note 2: Conditions in the theorem are sufficient but not necessary for the existence of Laplace transform. If these conditions are satisfied, the Laplace transform must exist. If these conditions are not satisfied, the Laplace transform may or may not exist.

We can show this by the following example.

Example: Consider the function $F(t) = 1/\sqrt{t}$.

Here $F(t) \rightarrow \infty$ as $t \rightarrow 0$, from the right.

Thus the function $F(t)$ is not piece-wise continuous on every finite interval in the range $t \geq 0$.

But $F(t)$ is integrable from 0 to any positive value t_0 .

Also $|F(t)| < Me^{at}$ for all $t > 1$ with $M = 1$ and $a = 0$.

$$\begin{aligned} \text{Now } L\{F(t)\} &= \int_0^{\infty} e^{-pt} \cdot F(t) dt \\ &= \int_0^{\infty} e^{-pt} \cdot \frac{1}{\sqrt{t}} dt, \text{ which converges for } p > 0 \\ &= \frac{2}{\sqrt{p}} \int_0^{\infty} e^{-x^2} dx, \text{ putting } \sqrt{pt} = x \text{ so that } \frac{dt}{\sqrt{t}} = \frac{2}{\sqrt{p}} dx \\ &= \frac{2}{\sqrt{p}} \cdot \frac{\sqrt{\pi}}{2}, \text{ since } \int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2} \\ &= \sqrt{\pi/p}, \quad p > 0. \end{aligned}$$

Thus $L\{1/\sqrt{t}\}$ exists for $p > 0$ even if $1/\sqrt{t}$ is not piecewise continuous in the range $t \geq 0$.

1.8 Laplace Transforms of Some Elementary Functions

(i) Laplace transform of the function $F(t) = 1$.

Solution: We have $L\{F(t)\} = \int_0^{\infty} e^{-pt} F(t) dt$.

$$\therefore L\{1\} = \int_0^{\infty} e^{-pt} \cdot 1 dt = \left[-\frac{e^{-pt}}{p} \right]_0^{\infty} = \frac{1}{p}, \quad p > 0.$$

Here the condition $p > 0$ is necessary, since the integral is convergent for $p > 0$ and divergent for $p \leq 0$.

(ii) Laplace transform of the function $F(t) = t^n$, n is any real number greater than -1 .

Solution: We have,

$$L\{F(t)\} = \int_0^{\infty} e^{-pt} F(t) dt.$$

$$\therefore L\{t^n\} = \int_0^{\infty} e^{-pt} t^n dt = \int_0^{\infty} e^{-pt} t^{(n+1)-1} dt \quad \dots(1)$$

Now from Eulerian integral of the second kind known as 'Gamma function', we have

$$\int_0^{\infty} e^{-ax} x^{m-1} dx = \frac{\Gamma(m)}{a^m}, \text{ if } a > 0 \text{ and } m > 0.$$

$$\therefore \text{ from (1), } L\{t^n\} = \frac{\Gamma(n+1)}{p^{n+1}}, \text{ if } p > 0 \text{ and } n+1 > 0 \text{ i.e., } n > -1.$$

Thus if n is any real number greater than -1 , we have

$$L\{t^n\} = \frac{\Gamma(n+1)}{p^{n+1}}, p > 0.$$

Here the condition $p > 0$ is necessary for the convergence of the integral (1).

(iii) Laplace transform of the function $F(t) = t^n$, n is a positive integer.

(Gorakhpur 2008)

Solution: We have

$$L\{F(t)\} = \int_0^{\infty} e^{-pt} F(t) dt$$

$$\begin{aligned} \therefore L\{t^n\} &= \int_0^{\infty} e^{-pt} t^n dt = \int_0^{\infty} e^{-pt} t^{(n+1)-1} dt \\ &= \frac{\Gamma(n+1)}{p^{n+1}}, \quad \text{if } p > 0 \end{aligned}$$

$$\left[\because \int_0^{\infty} e^{-ax} x^{m-1} dx = \frac{\Gamma(m)}{a^m}, \right.$$

$$\left. \text{if } a > 0 \text{ and } m > 0. \text{ Here } n+1 > 0, n \text{ being a positive integer.} \right]$$

$$= \frac{n!}{p^{n+1}}, p > 0. \quad [\because \Gamma(n+1) = n!, n \text{ being a positive integer}]$$

Thus if n is any positive integer, we have $L\{t^n\} = \frac{n!}{p^{n+1}}, p > 0$.

Here the condition $p > 0$ is necessary for the convergence of the integral defining the Laplace transform of t^n .

(iv) Laplace transform of the function $F(t) = e^{at}$.

(Rohilkhand 2009)

Solution: Here

$$L\{e^{at}\} = \int_0^{\infty} e^{-pt} \cdot e^{at} dt = \int_0^{\infty} e^{-(p-a)t} dt$$

$$= \left[-\frac{e^{-(p-a)t}}{p-a} \right]_0^\infty, p \neq a$$

$$= \frac{1}{p-a}, p > a.$$

Here the condition $p > a$ is necessary, since the integral is convergent for $p > a$ and divergent for $p \leq a$.

(v) Laplace transform of the function $F(t) = \sin at$.

(Avadh 2010, 11)

Solution: $L\{\sin at\} = \int_0^\infty e^{-pt} \sin at \, dt$

$$= \left[\frac{e^{-pt}}{p^2 + a^2} (-p \sin at - a \cos at) \right]_0^\infty$$

$$\left[\because \int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) \right]$$

$$= \frac{a}{p^2 + a^2}, p > 0.$$

Here the condition $p > 0$ is necessary for the convergence of the integral (1).

(vi) Laplace transform of the function $F(t) = \cos at$.

(Rohilkhand 2006, 07, 10)

Solution: $L\{\cos at\} = \int_0^\infty e^{-pt} \cos at \, dt$

$$= \left[\frac{e^{-pt}}{p^2 + a^2} (-p \cos at + a \sin at) \right]_0^\infty$$

$$\left[\because \int e^{ax} \cos bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) \right]$$

$$= \frac{p}{p^2 + a^2}, p > 0.$$

(vii) Laplace transform of the function $F(t) = \cosh at$.

Solution: $L\{\cosh at\} = L\left\{\frac{1}{2}(e^{at} + e^{-at})\right\}$

$$= \frac{1}{2} L\{e^{at}\} + \frac{1}{2} L\{e^{-at}\}$$

$$= \frac{1}{2} \cdot \frac{1}{p-a} + \frac{1}{2} \cdot \frac{1}{p+a}, p > a \text{ and } p > -a$$

$$= \frac{p}{p^2 - a^2}, p > |a|.$$

(viii) Laplace transform of the function $F(t) = \sinh at$.

Solution: $L\{\sinh at\} = L\left\{\frac{1}{2}(e^{at} - e^{-at})\right\}$

$$= \frac{1}{2}[L\{e^{at}\} - L\{e^{-at}\}]$$

$$= \frac{1}{2}\left[\frac{1}{p-a} - \frac{1}{p+a}\right], p > a \text{ and } p > -a$$

$$= \frac{a}{p^2 - a^2}, p > |a|.$$

1.9 Laplace Transforms of Some Elementary Functions

Laplace Transforms of some elementary functions obtained in 1.8 are given here in the form of a table.

Laplace Transforms of Some Elementary Functions		
	$F(t)$	$L\{F(t)\}$
1.	1	$\frac{1}{p}, p > 0$
2.	t^n (n is a positive integer)	$\frac{n!}{p^{n+1}}, p > 0$
3.	t^a ($a > -1$)	$\frac{\Gamma(a+1)}{p^{a+1}}, p > 0$
4.	e^{at}	$\frac{1}{p-a}, p > a$
5.	$\sin at$	$\frac{a}{p^2 + a^2}, p > 0$
6.	$\cos at$	$\frac{p}{p^2 + a^2}, p > 0$
7.	$\sinh at$	$\frac{a}{p^2 - a^2}, p > a $
8.	$\cosh at$	$\frac{p}{p^2 - a^2}, p > a $

If we know the Laplace transforms in the above table then nearly all the transforms can be obtained by using the general theorems which we shall consider later on.

Illustrative Examples

Example 1: Find $L \{(t^2 + 1)^2\}$.

Solution: We have

$$\begin{aligned} L \{(t^2 + 1)^2\} &= L \{t^4 + 2t^2 + 1\} = L \{t^4\} + 2L \{t^2\} + L \{1\} \\ &= \frac{4!}{p^5} + 2 \cdot \frac{2!}{p^3} + \frac{1}{p} = \frac{24 + 4p^2 + p^4}{p^5}, p > 0. \end{aligned}$$

Example 2: Find the L.T. of the function

$$F(t) = (\sin t - \cos t)^2. \quad (\text{Gorakhpur 2006, 10})$$

Solution: We have $L \{(\sin t - \cos t)^2\} = L \{\sin^2 t + \cos^2 t - 2 \sin t \cos t\}$

$$\begin{aligned} &= L \{1\} - L \{\sin 2t\} = \frac{1}{p} - \frac{2}{p^2 + 2^2}, p > 0 \\ &= \frac{p^2 - 2p + 4}{p(p^2 + 4)}, p > 0. \end{aligned}$$

Example 3: (i) Evaluate $L \{4 \cos^2 2t\}$.

Solution: We have $L \{4 \cos^2 2t\}$

$$\begin{aligned} &= L \{2(1 + \cos 4t)\} = 2 [L \{1\} + L \{\cos 4t\}] \\ &= 2 \left[\frac{1}{p} + \frac{p}{p^2 + 4^2} \right], p > 0 \\ &= \frac{4(p^2 + 8)}{p(p^2 + 16)}, p > 0. \end{aligned}$$

Example 3: (ii) Evaluate $L \{\sin^2 at\}$.

(Rohilkhand 2014)

Solution: We have

$$\begin{aligned} L \{\sin^2 at\} &= L \left\{ \frac{1}{2} (1 - \cos 2at) \right\} = \frac{1}{2} [L \{1\} - L \{\cos 2at\}] \\ &= \frac{1}{2} \left[\frac{1}{p} - \frac{p}{p^2 + (2a)^2} \right], p > 0 \\ &= \frac{2a^2}{p(p^2 + 4a^2)}, p > 0. \end{aligned}$$

Example 4: Find $L \{3t^4 - 2t^3 + 4e^{-3t} - 2 \sin 5t + 3 \cos 2t\}$.

Solution: We have

$$\begin{aligned} &L \{3t^4 - 2t^3 + 4e^{-3t} - 2 \sin 5t + 3 \cos 2t\} \\ &= 3L \{t^4\} - 2L \{t^3\} + 4L \{e^{-3t}\} - 2L \{\sin 5t\} + 3L \{\cos 2t\} \end{aligned}$$

$$\begin{aligned}
 &= 3 \cdot \frac{4!}{p^5} - 2 \cdot \frac{3!}{p^4} + 4 \cdot \frac{1}{p+3} - 2 \cdot \frac{5}{p^2+5^2} + 3 \cdot \frac{p}{p^2+2^2}, \\
 &\quad p > 0 \text{ and } p > -3 \text{ i.e. } p > 0 \\
 &= \frac{72}{p^5} - \frac{12}{p^4} + \frac{4}{p+3} - \frac{10}{p^2+25} + \frac{3p}{p^2+4}, p > 0.
 \end{aligned}$$

Example 5: Find $L\{F(t)\}$ if $F(t) = \begin{cases} (t-1)^2, & t > 1 \\ 0, & 0 < t < 1. \end{cases}$

Solution: We have

$$\begin{aligned}
 L\{F(t)\} &= \int_0^\infty F(t) e^{-pt} dt = \int_0^1 0 \cdot e^{-pt} dt + \int_1^\infty (t-1)^2 e^{-pt} dt \\
 &= \int_1^\infty (t-1)^2 e^{-pt} dt = \int_0^\infty x^2 e^{-p(x+1)} dx, \text{ putting } t-1 = x, \\
 &\quad \text{so that } dt = dx \text{ and changing the limits} \\
 &= \int_0^\infty e^{-p} e^{-px} x^2 dx = e^{-p} \int_0^\infty e^{-px} x^{(3-1)} dx = e^{-p} \cdot \frac{\Gamma(3)}{p^3}, p > 0 \\
 &\quad \left[\because \int_0^\infty e^{-ax} x^{m-1} dx = \frac{\Gamma(m)}{a^m}, a > 0, m > 0 \right] \\
 &= e^{-p} \cdot \frac{2!}{p^3} = \frac{2e^{-p}}{p^3}, p > 0.
 \end{aligned}$$

Example 6: Show that the function e^{t^2} is not of exponential order as $t \rightarrow \infty$.

Solution: We have

$$\begin{aligned}
 \lim_{t \rightarrow \infty} \{e^{-at} F(t)\} &= \lim_{t \rightarrow \infty} \{e^{-at} e^{t^2}\} = \lim_{t \rightarrow \infty} e^{t(t-a)} \\
 &= \infty \text{ for all values of } a.
 \end{aligned}$$

Hence whatever be the value of a , we cannot find a number M such that

$$e^{t^2} < M e^{at}.$$

$\therefore$ the given function is not of exponential order as $t \rightarrow \infty$.

Example 7: Find $L\{\sin \sqrt{t}\}$.

(Purvanchal 2007; Avadh 12)

Solution: We have

$$\begin{aligned}
 L\{\sin \sqrt{t}\} &= L\left\{\sqrt{t} - \frac{(\sqrt{t})^3}{3!} + \frac{(\sqrt{t})^5}{5!} - \frac{(\sqrt{t})^7}{7!} + \dots\right\} \\
 &= L\left\{t^{1/2} - \frac{t^{3/2}}{3!} + \frac{t^{5/2}}{5!} - \frac{t^{7/2}}{7!} + \dots\right\} \\
 &= L\{t^{1/2}\} - \frac{1}{3!} L\{t^{3/2}\} + \frac{1}{5!} L\{t^{5/2}\} - \frac{1}{7!} L\{t^{7/2}\} + \dots \\
 &= \frac{\Gamma(\frac{3}{2})}{p^{3/2}} - \frac{1}{3!} \cdot \frac{\Gamma(\frac{5}{2})}{p^{5/2}} + \frac{1}{5!} \cdot \frac{\Gamma(\frac{7}{2})}{p^{7/2}} - \frac{1}{7!} \cdot \frac{\Gamma(\frac{9}{2})}{p^{9/2}} + \dots
 \end{aligned}$$

$$\begin{aligned}
&= \frac{\frac{1}{2}\sqrt{\pi}}{p^{3/2}} - \frac{1}{1 \cdot 2 \cdot 3} \cdot \frac{\frac{3}{2} \cdot \frac{1}{2}\sqrt{\pi}}{p^{5/2}} + \frac{1}{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} \cdot \frac{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2}\sqrt{\pi}}{p^{7/2}} - \dots \\
&= \frac{\sqrt{\pi}}{2p^{3/2}} \left[1 - \frac{1}{1!} \left(\frac{1}{4p} \right) + \frac{1}{2!} \left(\frac{1}{4p} \right)^2 - \frac{1}{3!} \left(\frac{1}{4p} \right)^3 + \dots \right] \\
&= \frac{\sqrt{\pi}}{2p^{3/2}} \cdot e^{-1/4p}.
\end{aligned}$$

Example 8: Show that the Laplace transform of the function

$$F(t) = t^n, -1 < n < 0,$$

exists, although it is not a function of class A.

Solution: Here $F(t) \rightarrow \infty$ as $t \rightarrow 0$ from the right i.e., the function is not piecewise continuous on every finite interval in the range $t \geq 0$.

$$\begin{aligned}
\text{We have } \lim_{t \rightarrow \infty} \{e^{-at} F(t)\} &= \lim_{t \rightarrow \infty} \left(\frac{t^n}{e^{at}} \right) \\
&= \lim_{t \rightarrow \infty} \frac{1}{t^{-n} e^{at}} = \lim_{t \rightarrow \infty} \frac{1}{t^m e^{at}}, \quad \text{where } 0 < m < 1 \\
&= 0, \text{ if } a > 0.
\end{aligned}$$

$\therefore F(t) = t^n$ is of exponential order.

Since $F(t) = t^n$ is not sectionally continuous over every finite interval in the range $t \geq 0$, hence it is not a function of class A. But t^n is integrable from 0 to any positive number t_0 .

$$\begin{aligned}
\text{Now } L\{F(t)\} &= \int_0^\infty e^{-pt} F(t) dt = \int_0^\infty e^{-pt} t^n dt = \int_0^\infty e^{-pt} t^{(n+1)-1} dt \\
&= \frac{\Gamma(n+1)}{p^{(n+1)}}, \text{ if } p > 0 \text{ and } n+1 > 0 \text{ i.e., } n > -1.
\end{aligned}$$

Hence the Laplace transform of $t^n, -1 < n < 0$, exists, although it is not a function of class A.

Comprehensive Exercise 1

Find the L.T. of the following functions.

- $2t^3 - 6t + 8$
 - $\sin t \cos t$
 - $\cosh^2 2t$
 - $6 \sin 2t - 5 \cos 2t$
- $3 \cosh 5t - 4 \sin 5t$
 - $7e^{2t} + 9e^{-2t} + 5 \cos t + 7t^3 + 5 \sin 3t + 2$
 - $2e^{3t} - e^{-3t}$
 - $F(t) = (e^{at} - 1) / a$

(Rohilkhand 2006)

3. (i) Find $L\{F(t)\}$, if $F(t) = \begin{cases} e^t, & 0 < t \leq 1 \\ 0, & t > 1 \end{cases}$
- (ii) Find $L\{F(t)\}$, where $F(t) = \begin{cases} 0, & 0 < t < 2 \\ 4, & t > 2 \end{cases}$
4. (i) Find the Laplace transform of the following function :
- $$f(x) = \begin{cases} x/a, & 0 < x < a \\ 1, & x > a. \end{cases}$$
- (ii) Find $L\{F(t)\}$, where $F(t) = \begin{cases} 0, & 0 < t < 1 \\ t, & 1 < t < 2 \\ 0, & t > 2. \end{cases}$ (Lucknow 2010)
5. (i) Find Laplace Transform of the function $F(t)$, where
- $$F(t) = \begin{cases} \sin t, & 0 < t < \pi \\ 0, & t > \pi. \end{cases}$$
- (Lucknow 2007)
- (ii) Find $L\{F(t)\}$, if $F(t) = \begin{cases} e^t, & 0 < t < 5 \\ 3, & t > 5. \end{cases}$
6. (i) Find $L\{F(t)\}$, if $F(t) = \begin{cases} 1, & 0 < t < 2 \\ t, & t > 2. \end{cases}$
- (ii) Find $L\{F(t)\}$ if $F(t) = \begin{cases} t, & 0 < t < 4 \\ 5, & t > 4. \end{cases}$
7. (i) Prove that $L\{H(t)\} = \frac{2(1 - e^{-\pi p})}{p^2 + 4}$, where $H(t) = \begin{cases} \sin 2t, & 0 < t < \pi \\ 0, & t > \pi. \end{cases}$
- (ii) Prove that $L\left\{\frac{1}{\sqrt{\pi t}}\right\} = \frac{1}{\sqrt{p}}$. (Avadh 2014)
8. Show that $L\left(\frac{\cos \sqrt{t}}{\sqrt{t}}\right) = \sqrt{\left(\frac{\pi}{p}\right)} e^{-1/4p}$. (Purvanchal 2007; Kashi 14)
9. Show that t^2 is of exponential order 3.

Answers 1

1. (i) $12/p^4 - 6/p^2 + 8/p$, $p > 0$ (ii) $\frac{1}{(p^2 + 4)}$, $p > 0$
- (iii) $\frac{p^2 - 8}{p(p^2 - 16)}$, $p > 4$ (iv) $\frac{12 - 5p}{p^2 + 4}$, $p > 0$
2. (i) $\frac{3p - 20}{p^2 - 25}$, $p > 5$

$$(ii) \frac{16p-4}{p^2-4} + \frac{5p}{p^2+1} + \frac{42+2p^3}{p^4} + \frac{15}{p^2+9}, p > 2$$

$$(iii) \frac{p+9}{p^2-9}, p > 3$$

$$(iv) \frac{1}{p(p-a)}, p > 0 \text{ if } a < 0 \text{ and } p > a \text{ if } a > 0$$

$$3. (i) \frac{1}{(p-1)} [1 - e^{-(p-1)}], p \neq 1 \quad (ii) \frac{4}{p} e^{-2p}, p > 0$$

$$4. (i) \frac{1}{a p^2} (1 - e^{-ap}), p > 0$$

$$(ii) -\left(\frac{2}{p} + \frac{1}{p^2}\right) e^{-2p} + \left(\frac{1}{p} + \frac{1}{p^2}\right) e^{-p}, p \neq 0.$$

$$5. (i) \frac{e^{-p\pi} + 1}{p^2 + 1} \quad (ii) \frac{1 - e^{-5(p-1)}}{p-1} + \frac{3}{p} e^{-5p}, p > 0$$

$$6. (i) \frac{1}{p} [1 + e^{-2p}] + \frac{1}{p^2} e^{-2p}, p > 0 \quad (ii) \frac{1 + (p-1) e^{-4p}}{p^2}, p > 0$$

1.10 First Translation or Shifting Theorem

If $L\{F(t)\} = f(p)$ when $p > \alpha$, then $L\{e^{at} F(t)\} = f(p-a)$, $p > \alpha + a$; i.e., if $f(p)$ is the Laplace transform of $F(t)$ then $f(p-a)$ is the Laplace transform of $e^{at} F(t)$.

(Gorakhpur 2008, 11)

Proof. By definition, we have

$$f(p) = L\{F(t)\} = \int_0^\infty e^{-pt} F(t) dt, \quad \dots(1)$$

the integral (1) being given to be convergent if $p > \alpha$.

$$\begin{aligned} \therefore f(p-a) &= \int_0^\infty e^{-(p-a)t} F(t) dt \quad \dots(2) \\ &= \int_0^\infty e^{-pt} \cdot e^{at} F(t) dt = L\{e^{at} F(t)\}. \end{aligned}$$

Obviously if the integral (1) converges when $p > \alpha$, the integral (2) converges when $p-a > \alpha$ i.e., $p > \alpha + a$.

1.11 Second Translation or Shifting Theorem

If $L\{F(t)\} = f(p)$ and $G(t) = \begin{cases} F(t-a), & t > a \\ 0, & t < a, \end{cases}$ then $L\{G(t)\} = e^{-ap} f(p)$.

(Avadh 2007; Rohilkhand 07; Lucknow 09)

Proof: By definition, we have

$$\begin{aligned}
 L\{G(t)\} &= \int_0^\infty e^{-pt} G(t) dt = \int_0^a e^{-pt} \cdot 0 dt + \int_a^\infty e^{-pt} \cdot F(t-a) dt \\
 &= \int_a^\infty e^{-pt} F(t-a) dt = \int_0^\infty e^{-p(a+x)} \cdot F(x) dx, \\
 &\quad \text{putting } t-a=x \text{ so that } dt=dx \\
 &= e^{-pa} \int_0^\infty e^{-px} F(x) dx = e^{-pa} \int_0^\infty e^{-pt} F(t) dt, \\
 &\quad \text{by a property of definite integrals} \\
 &= e^{-pa} L\{F(t)\} = e^{-pa} f(p).
 \end{aligned}$$

1.12 Change of Scale Property

If $L\{F(t)\} = f(p)$, then $L\{F(at)\} = \frac{1}{a} f\left(\frac{p}{a}\right)$.

(Avadh 2006, 11; Kanpur 07; 10; Lucknow 07; Rohilkhand 08, 11)

Proof: By definition, we have

$$\begin{aligned}
 L\{F(at)\} &= \int_0^\infty e^{-pt} F(at) dt \\
 &= \frac{1}{a} \int_0^\infty e^{-p(x/a)} F(x) dx, \text{ putting } at=x, \text{ so that } dt=(1/a) dx \\
 &= \frac{1}{a} \int_0^\infty e^{-(p/a)t} F(t) dt \quad \left[\because \int_a^b f(x) dx = \int_a^b f(t) dt \right] \\
 &= \frac{1}{a} f\left(\frac{p}{a}\right), \text{ since } f(p) = \int_0^\infty e^{-pt} F(t) dt.
 \end{aligned}$$

Illustrative Examples

Example 9: Find (i) $L\{e^{-t} \sin^2 t\}$

(Rohilkhand 2004)

(ii) $L\{e^t \sin^2 t\}$.

(Kanpur 2012)

Solution: We have $L\{\sin^2 t\} = L\left\{\frac{1}{2}(1 - \cos 2t)\right\}$

$$= \frac{1}{2} \left[\frac{1}{p} - \frac{p}{p^2 + 2^2} \right] = \frac{2}{p(p^2 + 4)} = f(p), \text{ say.}$$

$\therefore$ From first shifting theorem, we have

$$(i) \quad L\{e^{-t} \sin^2 t\} = f(p+1) = \frac{2}{(p+1)\{(p+1)^2 + 4\}} = \frac{2}{(p+1)(p^2 + 2p + 5)}.$$

$$(ii) \quad L\{e^t \sin^2 t\} = f(p-1) = \frac{2}{(p-1)\{(p-1)^2 + 4\}} = \frac{2}{(p-1)(p^2 - 2p + 5)}.$$

Example 10: Find $L \{e^{-t} (3 \sinh 2t - 5 \cosh 2t)\}$.

Solution: We have $L \{3 \sinh 2t - 5 \cosh 2t\}$

$$= 3 \cdot \frac{2}{p^2 - 2^2} - 5 \cdot \frac{p}{p^2 - 2^2} = \frac{6 - 5p}{p^2 - 4} = f(p), \text{ say.}$$

$\therefore$ From first shifting theorem, we have

$$\begin{aligned} L \{e^{-t} (3 \sinh 2t - 5 \cosh 2t)\} &= f(p+1) \\ &= \frac{6 - 5(p+1)}{(p+1)^2 - 4} = \frac{1 - 5p}{p^2 + 2p - 3}. \end{aligned}$$

Example 11: Using first shifting theorem evaluate

$$L \{e^{6t} (t+2)^2\}.$$

Solution: We have, $L \{(t+2)^2\} = L \{t^2 + 4t + 4\}$

$$= \frac{2!}{p^3} + 4 \cdot \frac{1!}{p^2} + \frac{4}{p} = \frac{2 + 4p + 4p^2}{p^3} = f(p), \text{ say.}$$

$\therefore$ From first shifting theorem, we have

$$\begin{aligned} L \{e^{6t} (t+2)^2\} &= f(p-6) \\ &= \frac{2 + 4(p-6) + 4(p-6)^2}{(p-6)^3} = \frac{4p^2 - 44p + 122}{(p-6)^3}. \end{aligned}$$

Example 12: Given $L \{F(t)\} = \frac{p^2 - p + 1}{(2p+1)^2 (p-1)}$,

applying the change of scale property show that

$$L \{F(2t)\} = \frac{p^2 - 2p + 4}{4(p+1)^2 (p-2)}.$$

Solution: We have

$$L \{F(t)\} = \frac{p^2 - p + 1}{(2p+1)^2 (p-1)} = f(p), \text{ say.}$$

$\therefore$ by the change of scale property, we have

$$\begin{aligned} L \{F(2t)\} &= \frac{1}{2} f(p/2) \\ &= \frac{1}{2} \cdot \frac{(p/2)^2 - (p/2) + 1}{[2 \cdot (p/2) + 1]^2 [(p/2) - 1]} = \frac{p^2 - 2p + 4}{4(p+1)^2 (p-2)}. \end{aligned}$$

Example 13: Find $L \{F(t)\}$, where $F(t) = \begin{cases} \cos(t - \frac{2}{3}\pi), & t > 2\pi/3 \\ 0, & t < 2\pi/3. \end{cases}$

Solution: Let $\phi(t) = \cos t$.

Then
$$F(t) = \begin{cases} \phi(t - 2\pi/3), & t > 2\pi/3 \\ 0, & t < 2\pi/3. \end{cases}$$

We have $L\{\phi(t)\} = L\{\cos t\} = \frac{p}{p^2 + 1} = f(p)$, say.

$\therefore$ From second shifting theorem, we have

$$L\{F(t)\} = e^{(-2\pi/3)p} \cdot f(p) = e^{-2\pi p/3} \cdot \frac{p}{p^2 + 1}.$$

Aliter: We have $L\{F(t)\} = \int_0^\infty e^{-pt} F(t) dt$

$$\begin{aligned} &= \int_0^{2\pi/3} e^{-pt} \cdot 0 dt + \int_{2\pi/3}^\infty e^{-pt} \cdot \cos\left(t - \frac{2\pi}{3}\right) dt \\ &= \int_{2\pi/3}^\infty e^{-pt} \cdot \cos\left(t - \frac{2\pi}{3}\right) dt \\ &= \int_0^\infty e^{-p[x+(2\pi/3)]} \cos x dx, \text{ putting } t - 2\pi/3 = x \text{ and } dt = dx \\ &= e^{-p(2\pi/3)} \int_0^\infty e^{-px} \cos x dx \\ &= e^{-p(2\pi/3)} \int_0^\infty e^{-pt} \cos t dt \\ &= e^{-p(2\pi/3)} L\{\cos t\} = e^{-2\pi p/3} [p / (p^2 + 1)], p > 0. \end{aligned}$$

Comprehensive Exercise 2

Find the Laplace transform of the following functions.

- (i) $t^3 e^{-3t}$ (Kanpur 2008) (ii) $e^{-at} t^{n-1} / (n-1)!$
 (iii) $t^n e^{at}$. (Avadh 2010)
- (i) $e^{3t} \sin 4t$. (Lucknow 2010) (ii) $e^{3t} \cos 5t$.
- (i) $e^t \cos^2 t$. (ii) $e^{-4t} \cosh 2t$.
- (i) $e^{-t} (3 \sin 2t - 5 \cosh 2t)$. (ii) $e^{-2t} (3 \cos 6t - 5 \sin 6t)$.
- $(t+3)^2 \cdot e^t$.
- If $L\{F(t)\} = f(p)$, find $L\{F(t) \cos \omega t\}$.
- Applying change of scale property, find
 (i) $L\{\sinh 3t\}$ and (ii) $L\{\cos 5t\}$.
- Find $L\{G(t)\}$, where $G(t) = \begin{cases} e^{t-a}, & t > a \\ 0, & t < a. \end{cases}$
- Find $L\{F(t)\}$, where $F(t) = \begin{cases} \sin(t - \pi/3), & t > \pi/3 \\ 0, & t < \pi/3. \end{cases}$
- If $L\{F(t)\} = (1/p) e^{-1/p}$, prove that $L\{e^{-t} F(3t)\} = \frac{e^{-3/(p+1)}}{p+1}$.

Answers 2

1. (i) $6/(p+3)^4$. (ii) $1/(p+a)^n$
 (iii) $n!/(p-a)^{n+1}$.
2. (i) $4/(p^2 - 6p + 25)$. (ii) $(p-3)/(p^2 - 6p + 34)$.
3. (i) $\frac{(p^2 - 2p + 3)}{(p-1)(p^2 - 2p + 5)}$. (ii) $\frac{p+4}{p^2 + 8p + 12}$.
4. (i) $\frac{6}{p^2 + 2p + 5} - \frac{5(p+1)}{p^2 + 2p - 3}$. (ii) $\frac{3p-24}{p^2 + 4p + 40}$.
5. $\frac{9p^2 - 12p + 5}{(p-1)^3}$. 6. $\frac{1}{2} [f(p-i\omega) + f(p+i\omega)]$.
7. (i) $\frac{3}{p^2 - 9}$. (ii) $\frac{p}{p^2 + 25}, p > 0$.
8. $\frac{e^{-ap}}{p-1}, p > 1$. 9. $\frac{e^{-\pi p/3}}{p^2 + 1}, p > 0$.

1.13 Laplace Transform of the Derivative of $F(t)$

Theorem: Let $F(t)$ be continuous for all $t \geq 0$ and be of exponential order a as $t \rightarrow \infty$ and if $F'(t)$ is of class A , then Laplace transform of the derivative $F'(t)$ exists when $p > a$, and

$$L\{F'(t)\} = pL\{F(t)\} - F(0).$$

Proof:

Case I: In case $F'(t)$ is continuous for all $t \geq 0$, then

$$L\{F'(t)\} = \int_0^\infty e^{-pt} F'(t) dt \quad \dots(1)$$

$$= [e^{-pt} F(t)]_0^\infty + p \int_0^\infty e^{-pt} F(t) dt \quad [\text{Integrating by parts}]$$

$$= \lim_{t \rightarrow \infty} e^{-pt} F(t) - F(0) + pL\{F(t)\}. \quad \dots(2)$$

Now $|F(t)| \leq Me^{at}$ for all $t \geq 0$ and for some constants a and M .

We have $|e^{-pt} F(t)| = e^{-pt} |F(t)| \leq e^{-pt} Me^{at}$

$$= Me^{-(p-a)t} \rightarrow 0 \text{ as } t \rightarrow \infty \text{ if } p > a.$$

$$\therefore \lim_{t \rightarrow \infty} e^{-pt} F(t) = 0 \text{ for } p > a.$$

Therefore from (2) we conclude that $L\{F'(t)\}$ exists and

$$L\{F'(t)\} = pL\{F(t)\} - F(0).$$

Case II: In case $F'(t)$ is merely piece-wise continuous, the integral (1) may be broken as the sum of integrals in different ranges from 0 to ∞ such that $F'(t)$ is continuous in each of such parts.

Then proceeding as in case I, we shall have

$$L\{F'(t)\} = pL\{F(t)\} - F(0).$$

Note 1: If $F(t)$ fails to be continuous at $t = 0$ but

$$\lim_{t \rightarrow 0} F(t) = F(0 + 0) \text{ exists,} \quad [F(0 + 0) \text{ is not equal to } F(0),$$

which may or may not exist]

$$\text{then} \quad L\{F'(t)\} = pL\{F(t)\} - F(0 + 0).$$

Note 2: If $F(t)$ fails to be continuous at $t = a$, then

$$L\{F'(t)\} = pL\{F(t)\} - F(0) - e^{-ap} [F(a + 0) - F(a - 0)]$$

where $F(a + 0)$ and $F(a - 0)$ are the limits of F at $t = a$, as t approaches a from the right and from the left respectively.

The quantity $F(a + 0) - F(a - 0)$ is called the jump at the discontinuity $t = a$.

$$\begin{aligned} \text{Proof:} \quad L\{F'(t)\} &= \int_0^\infty e^{-pt} F'(t) dt = \int_0^a e^{-pt} F'(t) dt + \int_a^\infty e^{-pt} F'(t) dt \\ &= [e^{-pt} F(t)]_0^a + p \int_0^a e^{-pt} F(t) dt + [e^{-pt} F(t)]_a^\infty \\ &\quad + p \int_a^\infty e^{-pt} F(t) dt \\ &= e^{-ap} F(a - 0) - F(0) + \lim_{t \rightarrow \infty} e^{-pt} F(t) \\ &\quad - e^{-ap} F(a + 0) + p \int_0^\infty e^{-pt} F(t) dt \\ &= pL\{F(t)\} - F(0) - e^{-ap} [F(a + 0) - F(a - 0)]. \\ &\quad \left[\because \lim_{t \rightarrow \infty} e^{-pt} F(t) = 0, \text{ as shows in the theorem} \right] \end{aligned}$$

Note 3: For more than one discontinuity of the function $F(t)$, appropriate modification can be made.

1.14 Laplace Transform of the n th Order Derivative of $F(t)$

Theorem: Let $F(t)$ and its derivatives $F'(t), F''(t), \dots, F^{n-1}(t)$ be continuous functions for all $t \geq 0$ and be of exponential orders as $t \rightarrow \infty$ and if $F^n(t)$ is of class A, then Laplace transform of $F^n(t)$ exists when $p > a$, and is given by

$$L\{F^n(t)\} = p^n L\{F(t)\} - p^{n-1} F(0) - p^{n-2} F'(0) - \dots - F^{n-1}(0).$$

Proof: From the theorem of 1.13, we have

$$L\{F'(t)\} = pL\{F(t)\} - F(0) \quad \dots(1)$$

Applying the result (1) to the second order derivative $F''(t)$, we have

$$\begin{aligned} L\{F''(t)\} &= pL\{F'(t)\} - F'(0) \\ &= p[pL\{F(t)\} - F(0)] - F'(0) \\ &= p^2 L\{F(t)\} - pF(0) - F'(0). \end{aligned} \quad \dots(2)$$

(Gorakhpur 2006, 10)

Again applying (1) to the third order derivative $F'''(t)$, we have

$$\begin{aligned} L\{F'''(t)\} &= pL\{F''(t)\} - F''(0) \\ &= p[p^2 L\{F(t)\} - pF(0) - F'(0)] - F''(0) \\ &= p^3 L\{F(t)\} - p^2 F(0) - pF'(0) - F''(0). \end{aligned}$$

Proceeding similarly, we have

$$L\{F^n(t)\} = p^n L\{F(t)\} - p^{n-1} F(0) - p^{n-2} F'(0) - \dots - F^{n-1}(0)$$

$$\text{or } L\{F^n(t)\} = p^n L\{F(t)\} - \sum_{r=0}^{n-1} p^{n-1-r} F^{(r)}(0).$$

1.15 Initial-Value Theorem

Let $F(t)$ be continuous for all $t \geq 0$ and be of exponential order as $t \rightarrow \infty$ and if $F'(t)$ is of class A ,

$$\text{then } \lim_{t \rightarrow 0} F(t) = \lim_{p \rightarrow \infty} p L\{F(t)\}.$$

Proof: By the theorem of 1.13, we have

$$L\{F'(t)\} = \int_0^\infty e^{-pt} F'(t) dt = pL\{F(t)\} - F(0) \quad \dots(1)$$

Since $F'(t)$ is sectionally continuous and of exponential order,

$$\therefore \lim_{p \rightarrow \infty} \int_0^\infty e^{-pt} F'(t) dt = 0.$$

Taking limit as $p \rightarrow \infty$ in (1), we have

$$0 = \lim_{p \rightarrow \infty} pL\{F(t)\} - F(0) \quad \text{or} \quad F(0) = \lim_{p \rightarrow \infty} pL\{F(t)\}$$

$$\text{or } \lim_{t \rightarrow 0} F(t) = \lim_{p \rightarrow \infty} pL\{F(t)\}.$$

Note: If $F(t)$ fails to be continuous at $t = 0$, but $\lim_{t \rightarrow 0} F(t)$ exists, the result still holds

by using Note (1), 1.13 theorem.

1.16 Final-Value Theorem

Theorem: Let $F(t)$ be continuous for all $t \geq 0$ and be of exponential order as $t \rightarrow \infty$ and if $F'(t)$ is of class A, then

$$\lim_{t \rightarrow \infty} F(t) = \lim_{p \rightarrow 0} p L\{F(t)\}.$$

Proof. By the theorem of 1.13, we have

$$\begin{aligned} L\{F'(t)\} &= \int_0^{\infty} e^{-pt} F'(t) dt \\ &= pL\{F(t)\} - F(0). \end{aligned} \quad \dots(1)$$

Taking limit as $p \rightarrow 0$ in (1), we have

$$\lim_{p \rightarrow 0} \int_0^{\infty} e^{-pt} F'(t) dt = \lim_{p \rightarrow 0} pL\{F(t)\} - F(0)$$

$$\text{or} \quad \int_0^{\infty} F'(t) dt = \lim_{p \rightarrow 0} pL\{F(t) - F(0)\}$$

$$\text{or} \quad [F(t)]_0^{\infty} = \lim_{p \rightarrow 0} pL\{F(t)\} - F(0)$$

$$\text{or} \quad \lim_{t \rightarrow \infty} F(t) - F(0) = \lim_{p \rightarrow 0} pL\{F(t)\} - F(0)$$

$$\text{or} \quad \lim_{t \rightarrow \infty} F(t) = \lim_{p \rightarrow 0} pL\{F(t)\}.$$

Note: If $F(t)$ fails to be continuous at $t = 0$, but $\lim_{t \rightarrow 0} F(t)$ exists, the result still holds by using Note (1), 1.13 theorem.

1.17 Laplace Transform of Integrals

Theorem: If $F(t)$ is piecewise continuous and satisfies $|F(t)| \leq Me^{at}$ for all $t \geq 0$ for some constants a and M , then

$$L\left\{\int_0^t F(x) dx\right\} = \frac{1}{p} L\{F(t)\}, \quad (p > 0, p > a) \quad (\text{Lucknow 2010})$$

Proof: Let $F(t)$ be piece-wise continuous such that $|F(t)| \leq Me^{at}$,
for some constants a and M .

If (1) holds for some negative value of a then it also holds for positive value of a . Therefore suppose that a is positive.

$$\text{Let} \quad G(t) = \int_0^t F(x) dx.$$

Then $G(t)$ is continuous because the integral of an integrable function is continuous.

Also $|G(t)| \leq \int_0^t |F(x)| dx \leq \int_0^t M e^{ax} dx.$

$$\therefore |G(t)| \leq \frac{M}{a} (e^{at} - 1), a > 0. \quad \dots(2)$$

Further $G'(t) = F(t)$, except for points at which $F(t)$ is discontinuous.

Therefore $G'(t)$ is piece-wise continuous on each finite interval.

Hence from the theorem of 1.13, we have

$$L\{G'(t)\} = pL\{G(t)\} - G(0) = pL\{G(t)\} \quad [\text{Since } G(0) = 0 \text{ from (2)}]$$

$$\begin{aligned} \therefore L\{G(t)\} &= \frac{1}{p} L\{G'(t)\} \\ L\left\{\int_0^t F(x) dx\right\} &= \frac{1}{p} L\{F(t)\} \end{aligned}$$

1.18 Multiplication By t

Theorem: If $F(t)$ is a function of class A and if $L\{F(t)\} = f(p)$, then $L\{tF(t)\} = -f'(p)$.

(Rohilkhand 2009, Gorakhpur 05, 09)

Proof: We have $f(p) = L\{F(t)\} = \int_0^\infty e^{-pt} F(t) dt.$

$$\begin{aligned} \therefore f'(p) &= \frac{d}{dp} \int_0^\infty e^{-pt} F(t) dt \\ &= \int_0^\infty \frac{\partial}{\partial p} \{e^{-pt} F(t)\} dt, \end{aligned}$$

by Leibnitz's rule for differentiating under the sign of integral

$$\begin{aligned} &= - \int_0^\infty t e^{-pt} F(t) dt \\ &= - \int_0^\infty e^{-pt} \{t F(t)\} dt = -L\{t F(t)\}. \end{aligned}$$

Thus $L\{t F(t)\} = -f'(p)$.

1.19 Multiplication by t^n

Theorem: If $F(t)$ is a function of class A and if $L\{F(t)\} = f(p)$ then

$$L\{t^n F(t)\} = (-1)^n \frac{d^n}{dp^n} f(p), \text{ where } n = 1, 2, 3, \dots$$

(Rohilkhand 2000; Lucknow 06, 09, 11; Kanpur 08;
Avadh 10; Agra 01; Gorakhpur 11)

Proof: We shall prove this theorem by mathematical induction. By 1.18, we have

$$L\{t F(t)\} = (-1)^1 \frac{d}{dp} f(p)$$

i.e. the theorem is true for $n = 1$.

Now assume that the theorem is true for a particular value of n say m . Then, we have

$$L \{ t^m F(t) \} = (-1)^m \frac{d^m}{dp^m} f(p)$$

$$\text{or} \quad \int_0^\infty e^{-pt} t^m F(t) dt = (-1)^m \frac{d^m}{dp^m} f(p).$$

Now differentiating both sides w. r. t. p , we have

$$\frac{d}{dp} \int_0^\infty e^{-pt} t^m F(t) dt = (-1)^m \frac{d^{m+1}}{dp^{m+1}} f(p)$$

$$\text{or} \quad \int_0^\infty \frac{\partial}{\partial p} \{ e^{-pt} t^m F(t) \} dt = (-1)^m \frac{d^{m+1}}{dp^{m+1}} f(p),$$

by Leibnitz's rule for differentiating under the sign of integral

$$\text{or} \quad - \int_0^\infty e^{-pt} \cdot t^{m+1} F(t) dt = (-1)^m \frac{d^{m+1}}{dp^{m+1}} f(p)$$

$$\text{or} \quad \int_0^\infty e^{-pt} \cdot \{ t^{m+1} F(t) \} dt = (-1)^{m+1} \frac{d^{m+1}}{dp^{m+1}} f(p)$$

$$\text{or} \quad L \{ t^{m+1} F(t) \} = (-1)^{m+1} \frac{d^{m+1}}{dp^{m+1}} f(p)$$

which shows that if the theorem is true for any particular value on n , it is true for the next value of n . But we have already seen that the theorem is true for $n = 1$. Hence it is true for $n = 1 + 1 = 2$ and $n = 2 + 1 = 3$, etc.

Therefore the theorem is true for every positive integral value of n .

1.20 Division by t

Theorem: If $L \{ F(t) \} = f(p)$, then $L \left\{ \frac{1}{t} F(t) \right\} = \int_p^\infty f(x) dx$

provided $\lim_{t \rightarrow 0} \left\{ \frac{1}{t} F(t) \right\}$ exists.

Proof: Let $G(t) = \frac{1}{t} F(t)$ i.e., $F(t) = t G(t)$.

$$\begin{aligned} \therefore L \{ F(t) \} &= L \{ t G(t) \} \\ &= - \frac{d}{dp} L \{ G(t) \}, \text{ by the theorem of 1.18} \end{aligned}$$

or
$$f(p) = -\frac{d}{dp} L\{G(t)\}.$$

Now integrating both sides with respect to p from p to ∞ , we have

$$-\left[L\{G(t)\} \right]_p^\infty = \int_p^\infty f(p) dp$$

or
$$-\lim_{p \rightarrow \infty} L\{G(t)\} + L\{G(t)\} = \int_p^\infty f(p) dp$$

[Note that $L\{G(t)\}$ is a function of p]

or
$$0 + L\{G(t)\} = \int_p^\infty f(p) dp$$

$$\left[\because \lim_{p \rightarrow \infty} L\{G(t)\} = \lim_{p \rightarrow \infty} \int_0^\infty e^{-pt} G(t) dt = 0 \right]$$

or
$$L\left\{\frac{1}{t} F(t)\right\} = \int_p^\infty f(x) dx$$

Illustrative Examples

Example 14: Evaluate

(i) $L\{t\}$

(ii) $L\{e^{at}\}$

(Rohilkhand 2011)

(iii) $L\{-a \sin at\}$

Solution: We have

$$L\{F'(t)\} = pL\{F(t)\} - F(0) \quad \dots(1)$$

(i) Here let $F(t) = t$, then $F'(t) = 1$ and $F(0) = 0$.

$\therefore$ from (1), we have $L\{1\} = pL\{t\} - 0$

or
$$L\{t\} = \frac{1}{p} L\{1\} = \frac{1}{p} \cdot \frac{1}{p} = \frac{1}{p^2}, p > 0.$$

(ii) Here let $F(t) = e^{at}$, then $F'(t) = a e^{at}$ and $F(0) = 1$.

$\therefore$ from (1), we have

$$L\{a e^{at}\} = pL\{e^{at}\} - 1$$

or
$$a L\{e^{at}\} = pL\{e^{at}\} - 1$$

or
$$(p - a) L\{e^{at}\} = 1.$$

$\therefore$
$$L\{e^{at}\} = \frac{1}{p - a}.$$

(iii) Here let $F(t) = -a \sin at$, then $F'(t) = -a^2 \cos at$ and $F''(t) = a^3 \sin at$

so that $F'(0) = -a^2$ and $F(0) = 0$.

$\therefore$ from
$$L\{F''(t)\} = p^2 L\{F(t)\} - pF(0) - F'(0),$$

we have
$$L\{a^3 \sin at\} = p^2 L\{-a \sin at\} - 0 - (-a^2)$$

or
$$(p^2 + a^2) L\{-a \sin at\} = -a^2.$$

$$\therefore L\{-a \sin at\} = -\frac{a^2}{p^2 + a^2}.$$

Example 15: Show that $L\{t^2 \cos at\} = \frac{2p(p^2 - 3a^2)}{(p^2 + a^2)^3}, p > 0$.

(Rohilkhand 2011; Kanpur 07)

Solution: Since $L\{\cos at\} = p/(p^2 + a^2), p > 0$,

$$\begin{aligned}\therefore L\{t^2 \cos at\} &= (-1)^2 \frac{d^2}{dp^2} \left\{ \frac{p}{(p^2 + a^2)} \right\} = \frac{d}{dp} \left(\frac{-p^2 + a^2}{(p^2 + a^2)^2} \right) \\ &= \frac{2p(p^2 - 3a^2)}{(p^2 + a^2)^3}.\end{aligned}$$

Example 16: Find $L\{\sin at - at \cos at\}$.

Solution: Let $F(t) = \sin at - at \cos at$, then $F'(t) = a^2 t \sin at$ and $F(0) = 0$.

$\therefore$ From $L\{F'(t)\} = pL\{F(t)\} - F(0)$, we have

$$L\{a^2 t \sin at\} = pL\{\sin at - at \cos at\} - 0.$$

$$\begin{aligned}\therefore L\{\sin at - at \cos at\} &= \frac{1}{p} L\{a^2 t \sin at\} \\ &= \frac{a^2}{p} L\{t \sin at\} = -\frac{a^2}{p} \frac{d}{dp} L\{\sin at\} \\ &= -\frac{a^2}{p} \cdot \frac{d}{dp} \left(\frac{a}{p^2 + a^2} \right) = \frac{2a^3}{(p^2 + a^2)^2}.\end{aligned}$$

Aliter. $L\{\sin at - at \cos at\} = L\{\sin at\} - a L\{t \cos at\}$

$$\begin{aligned}&= \frac{a}{p^2 + a^2} - a \cdot (-1) \frac{d}{dp} [L\{\cos at\}] \\ &= \frac{a}{p^2 + a^2} + a \frac{d}{dp} \left(\frac{p}{p^2 + a^2} \right) \\ &= \frac{a}{p^2 + a^2} + \frac{a(a^2 - p^2)}{(p^2 + a^2)^2} = \frac{2a^3}{(p^2 + a^2)^2}.\end{aligned}$$

Example 17: Show that

$$L\{(t^2 - 3t + 2) \sin 3t\} = \frac{6p^4 - 18p^3 + 126p^2 - 162p + 432}{(p^2 + 9)^3}.$$

Solution: We have $L\{(t^2 - 3t + 2) \sin 3t\}$

$$\begin{aligned}&= L\{t^2 \sin 3t\} - 3L\{t \sin 3t\} + 2L\{\sin 3t\} \\ &= (-1)^2 \frac{d^2}{dp^2} L\{\sin 3t\} - 3 \left[-\frac{d}{dp} L\{\sin 3t\} \right] + 2L\{\sin 3t\}\end{aligned}$$

$$\begin{aligned}
 &= \frac{d^2}{dp^2} \left(\frac{3}{p^2 + 9} \right) + 3 \frac{d}{dp} \left(\frac{3}{p^2 + 9} \right) + 2 \cdot \frac{3}{p^2 + 9} \\
 &= \frac{18p^2 - 54}{(p^2 + 9)^3} + 3 \cdot \left\{ \frac{-6p}{(p^2 + 9)^2} \right\} + \frac{6}{p^2 + 9} \\
 &= \frac{6p^4 - 18p^3 + 126p^2 - 162p + 432}{(p^2 + 9)^3}.
 \end{aligned}$$

Example 18: Find $L\{t e^{-t} \sin t\}$.

(Lucknow 2007)

Solution: Since $L\{\sin t\} = \frac{1}{p^2 + 1}$

$$\therefore L\{t \sin t\} = -\frac{d}{dp} \left\{ \frac{1}{p^2 + 1} \right\} = \frac{2p}{(p^2 + 1)^2}$$

$$\therefore L\{t e^{-t} \sin t\} = \frac{2(p+1)}{[(p+1)^2 + 1]^2} = \frac{2p+2}{(p^2 + 2p + 2)^2}.$$

Example 19: Prove that $L\left\{\frac{\sin t}{t}\right\} = \tan^{-1} \frac{1}{p}$ and hence find $L\left\{\frac{\sin at}{t}\right\}$. Does the Laplace

transform of $\frac{\cos at}{t}$ exist?

(Lucknow 2009; Gorakhpur 05; Rohilkhand 10; Avadh 13)

Solution: Let $F(t) = \sin t$.

$$\text{Now } \lim_{t \rightarrow 0} \frac{F(t)}{t} = \lim_{t \rightarrow 0} \frac{\sin t}{t} = 1.$$

We have $L\{\sin t\} = \frac{1}{p^2 + 1} = f(p)$, say.

$$\begin{aligned}
 \therefore L\left\{\frac{\sin t}{t}\right\} &= \int_p^\infty f(x) dx = \int_p^\infty \frac{dx}{x^2 + 1} = (\tan^{-1} x)_p^\infty = \frac{\pi}{2} - \tan^{-1} p \\
 &= \cot^{-1} p = \tan^{-1} (1/p).
 \end{aligned}$$

$$\text{Now } L\left\{\frac{\sin at}{t}\right\} = aL\left\{\frac{\sin at}{at}\right\} = a \cdot \frac{1}{a} \tan^{-1} \frac{1}{(p/a)},$$

$$\text{since } L\{F(at)\} = \frac{1}{p} f\left(\frac{p}{a}\right) = \tan^{-1}(a/p).$$

Again, since $L\{\cos at\} = \frac{p}{p^2 + a^2} = f(p)$, we have

$$\begin{aligned}
 L\left\{\frac{\sin at}{t}\right\} &= \int_p^\infty \frac{x}{x^2 + a^2} dx = \left[\frac{1}{2} \log(x^2 + a^2) \right]_p^\infty \\
 &= \frac{1}{2} \lim_{x \rightarrow \infty} \log(x^2 + a^2) - \frac{1}{2} \log(p^2 + a^2),
 \end{aligned}$$

which does not exist since $\lim_{x \rightarrow \infty} \log(x^2 + a^2)$ is infinite.

Hence $L\left\{\frac{\cos at}{t}\right\}$ does not exist.

Example 20: If $L\{F(t), t \rightarrow p\} = f(p)$,

show that $L\left\{\int_0^t \frac{F(u)}{u} du, t \rightarrow p\right\} = \frac{1}{p} \int_p^\infty f(y) dy$.

Hence show that $L\left\{\int_0^t \frac{\sin u}{u} du, t \rightarrow p\right\} = \frac{\cot^{-1} p}{p}$.

Solution: From 1.17, we have

$$L\left\{\int_0^t F(u) du\right\} = \frac{1}{p} f(p) \quad \dots(1)$$

where $f(p) = L\{F(t)\}$.

Let $G(t) = \frac{F(t)}{t}$.

Then $L\{G(t)\} = L\left\{\frac{F(t)}{t}\right\} = \int_p^\infty f(y) dy$ [from article 1.20]
 $= g(p)$, say.

$\therefore$ From (1), we have

$$L\left\{\int_0^t G(u) du\right\} = \frac{1}{p} g(p)$$

or $L\left\{\int_0^t F(u) du\right\} = \frac{1}{p} \int_p^\infty f(y) dy \quad \dots(2)$

Deduction. Let $F(t) = \sin t$

so that $f(p) = L\{\sin t\} = \frac{1}{p^2 + 1}$.

$\therefore$ From (2), we have

$$\begin{aligned} L\left\{\int_0^t \frac{\sin u}{u} du\right\} &= \frac{1}{p} \int_p^\infty \frac{dy}{y^2 + 1} = \frac{1}{p} [\tan^{-1} y]_p^\infty \\ &= \frac{1}{p} \left(\frac{\pi}{2} - \tan^{-1} p\right) = \frac{1}{p} \cot^{-1} p. \end{aligned}$$

Comprehensive Exercise 3

Find the Laplace transforms of the following functions.

1. (i) $t \cos at$. (Lucknow 2008) (ii) $t \cosh 3t$.

2. (i) $t^2 \sin at$. (Lucknow 2006) (ii) $t^2 e^{2t}$.

3. (i) Show that $L\{t^3 \cos t\} = \frac{6p^4 - 36p^2 + 6}{(p^2 + 1)^4}$.

(ii) Show that $L\{t^n \cdot e^{at}\} = \frac{n!}{(p-a)^{n+1}}, p > a$.

4. (i) Find $L\{t(3 \sin 2t - 2 \cos 2t)\}$.

(ii) Find L.T. of $f(t) = \sin \alpha t + t \cos \alpha t$.

5. Prove that $L\{t e^{at} \sin at\} = \frac{2a(p-a)}{(p^2 - 2ap + 2a^2)^2}$.

(Kanpur 2010)

6. Given $L\{\sin \sqrt{t}\} = \frac{\sqrt{\pi}}{2p^{3/2}} e^{-1/4p}$, show that

$$L\left\{\frac{\cos \sqrt{t}}{\sqrt{t}}\right\} = \sqrt{\left(\frac{\pi}{p}\right)} e^{-1/4p}.$$

7. Find $L\left\{\frac{\sinh t}{t}\right\}$.

8. Find $L\left\{\int_0^t \frac{\sin x}{x} dx\right\}$.

9. Prove that if $L\{F(t)\} = f(p)$, then

$$\int_0^\infty \frac{F(t)}{t} dt = \int_0^\infty f(x) dx \text{ provided that the integral converges.}$$

10. Find the Laplace transform of $F(t)$ defined as

$$F(t) = \begin{cases} t+1, & 0 \leq t \leq 2 \\ 3, & t > 2. \end{cases}$$

Also determine $L\{F'(t)\}$.

Answers 3

1. (i) $\frac{p^2 - a^2}{(p^2 + a^2)^2}, p > 0$.

(ii) $\frac{p^2 + 9}{(p^2 - 9)^2}, p > 0$.

2. (i) $\frac{2a(3p^2 - a^2)}{(p^2 + a^2)^3}, p > 0$.

(ii) $\frac{2}{(p-2)^3}$.

4. (i) $\frac{8+12p-2p^2}{(p^2+4)^2}$.

(ii) $\frac{(\alpha+1)p^2 + (\alpha-1)\alpha^2}{(p^2 + \alpha^2)^2}$.

7. $\frac{1}{2} \log \frac{p+1}{p-1}$.

8. $(1/p) \cot^{-1} p$.

10. $\frac{1}{p^2} [1 + p - e^{-2p}], \frac{1 - e^{-2p}}{p}$.

1.21 Evaluation of Integrals

If $L\{F(t)\} = f(p)$ i.e., $\int_0^\infty e^{-pt} F(t) dt = f(p)$, taking limit as $p \rightarrow 0$, we have

$$\int_0^{\infty} F(t) dt = f(0)$$

assuming that the integral is convergent.

Illustrative Examples

Example 21: Evaluate $\int_0^{\infty} \frac{(e^{-at} - e^{-bt})}{t} dt$.

Solution: Let $F(t) = e^{-at} - e^{-bt}$.

Then $L\{F(t)\} = L\{e^{-at}\} - L\{e^{-bt}\} = \frac{1}{p+a} - \frac{1}{p+b} = f(p)$, say.

$$\begin{aligned} \therefore L\left\{\frac{F(t)}{t}\right\} &= \int_p^{\infty} f(x) dx = \int_p^{\infty} \left(\frac{1}{x+a} - \frac{1}{x+b}\right) dx = \left[\log \frac{(x+a)}{(x+b)}\right]_p^{\infty} \\ &= \lim_{x \rightarrow \infty} \log \frac{x+a}{x+b} - \log \frac{p+a}{p+b} \\ &= \lim_{x \rightarrow \infty} \log \frac{1+(a/x)}{1+(b/x)} - \log \frac{p+a}{p+b} \\ &= 0 - \log \frac{p+a}{p+b} = \log \frac{p+b}{p+a}. \end{aligned}$$

Thus $L\left\{\frac{F(t)}{t}\right\} = \int_0^{\infty} e^{-pt} \cdot \frac{e^{-at} - e^{-bt}}{t} dt = \log \frac{p+b}{p+a}$

$\therefore$ Taking limit as $p \rightarrow 0$, we have

$$\int_0^{\infty} \frac{e^{-at} - e^{-bt}}{t} dt = \log \frac{b}{a}.$$

Example 22: Prove that $\int_0^{\infty} t^3 e^{-t} \sin t dt = 0$.

(Gorakhpur 2006)

Solution: Let $F(t) = t^3 \sin t$.

$$\therefore L\{F(t)\} = L\{t^3 \sin t\} = (-1)^3 \frac{d^3}{dp^3} L\{\sin t\}$$

$$= -\frac{d^3}{dp^3} \left\{ \frac{1}{p^2+1} \right\} = -\frac{d^2}{dp^2} \left\{ -\frac{2p}{(p^2+1)^2} \right\} = \frac{d}{dp} \left\{ \frac{2-6p^2}{(p^2+1)^3} \right\}$$

or $\int_0^{\infty} e^{-pt} \cdot t^3 \sin t dt = \frac{24(p^2-1)p}{(p^2+1)^4}.$

Taking limit as $p \rightarrow 1$, we have

$$\int_0^{\infty} t^3 e^{-t} \sin t dt = 0.$$

Comprehensive Exercise 4

1. Show that $\int_0^\infty \frac{\sin t}{t} dt = \frac{\pi}{2}$. (Purvanchal 2007; Meerut 13, 13B; Rohilkhand 10, 14)
2. Evaluate $\int_0^\infty \frac{e^{-t} - e^{-3t}}{t} dt$. (Rohilkhand 2010)
3. Show that $\int_0^\infty t e^{-3t} \sin t dt = \frac{3}{50}$.
4. Show that $\int_0^\infty t e^{-2t} \cos t dt = \frac{3}{25}$.
5. Prove that $\int_0^\infty \frac{e^{-x} \sin x}{x} dx = \frac{\pi}{4}$. (Rohilkhand 2009, 11)
6. Evaluate $\int_0^\infty t e^{-3t} \cos 4t dt$. (Gorakhpur 2007, 09)

Answers 4

3. $\log 3$.

6. $\frac{5}{169}$.

1.22 Periodic Functions

Fundamental Theorem: Let $F(t)$ be a periodic function with period $T > 0$, that is $F(u+T) = F(u)$, $F(u+2T) = F(u)$, etc., then

$$L\{F(t)\} = \frac{\int_0^T e^{-pt} F(t) dt}{1 - e^{-pT}}.$$

Proof: We have

$$\begin{aligned} L\{F(t)\} &= \int_0^\infty e^{-pt} F(t) dt \\ &= \int_0^T e^{-pt} F(t) dt + \int_T^{2T} e^{-pt} F(t) dt + \int_{2T}^{3T} e^{-pt} F(t) dt + \dots \\ &= \int_0^T e^{-pt} F(t) dt + \int_0^T e^{-p(u+T)} F(u+T) du \\ &\quad + \int_0^T e^{-p(u+2T)} F(u+2T) du + \dots, \end{aligned}$$

putting $t = u + T$, $t = u + 2T$, etc. in 2nd, 3rd, ... integrals respectively

$$\begin{aligned} &= \int_0^T e^{-pu} F(u) du + e^{-pT} \int_0^T e^{-pu} F(u) du \\ &\quad + e^{-2pT} \int_0^T e^{-pu} F(u) du + \dots \end{aligned}$$

$$\begin{aligned}
 &= (1 + e^{-pT} + e^{-2pT} + \dots) \int_0^T e^{-pu} F(u) du \\
 &= \frac{1}{1 - e^{-pT}} \int_0^T e^{-pu} F(u) du = \frac{\int_0^T e^{-pt} F(t) dt}{1 - e^{-pT}}.
 \end{aligned}$$

1.23 Some Special Functions

1. **The Sine and Cosine Integrals:** The sine and cosine integrals, denoted by $S_i(t)$ and $C_i(t)$ respectively are defined by

$$S_i(t) = \int_0^t \frac{\sin u}{u} du,$$

and

$$C_i(t) = \int_t^\infty \frac{\cos u}{u} du.$$

2. **The Error Function:** The error function denoted by $\text{erf}(t)$, is defined by

$$\text{erf}(t) = \frac{2}{\sqrt{\pi}} \int_0^t e^{-u^2} du.$$

3. **The Gamma Function:** If $n > 0$, the gamma function is defined by

$$\Gamma(n) = \int_0^\infty u^{n-1} e^{-u} du.$$

4. **The Unit Step Function (also called Heaviside's Unit function):**

The unit step function, denoted by $H(t - a)$, is defined by $H(t - a) = \begin{cases} 0, & t < a \\ 1, & t \geq a. \end{cases}$

5. **The Bessel Function:** Bessel function of order n is defined by

$$\begin{aligned}
 J_n(t) &= \frac{t^n}{2^n \Gamma(n+1)} \left[1 - \frac{t^2}{2 \cdot (2n+2)} + \frac{t^4}{2 \cdot 4 \cdot (2n+2)(2n+4)} - \dots \right] \\
 &= \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \Gamma(n+r+1)} \cdot \left(\frac{t}{2}\right)^{n+2r}
 \end{aligned}$$

The Bessel function of order zero i.e., $J_0(t)$ is given by

$$J_0(t) = 1 - \frac{t^2}{2^2} + \frac{t^4}{2^2 \cdot 4^2} - \frac{t^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots$$

Illustrative Examples

Example 23: Show that

$$(i) \quad L\{\sinh at \cos at\} = \frac{a(p^2 - 2a^2)}{p^4 + 4a^4}$$

$$\text{and } (ii) \quad L\{\sinh at \sin at\} = \frac{2a^2 p}{p^4 + 4a^4}.$$

Solution: We have $L \{ \sinh at \} = \frac{a}{p^2 - a^2} = f(p)$, say.

$$\therefore L \{ e^{iat} \sinh at \} = f(p - ia)$$

$$\begin{aligned} &= \frac{a}{(p - ia)^2 - a^2} = \frac{a}{(p^2 - 2a^2) - 2iap} \\ &= \frac{a \{ (p^2 - 2a^2) + 2iap \}}{(p^2 - 2a^2)^2 - (2ipa)^2} \end{aligned}$$

or
$$L \{ \sinh at (\cos at + i \sin at) \} = \frac{a (p^2 - 2a^2) + 2ia^2 p}{p^4 + 4a^4}$$

or
$$L \{ \sinh at \cos at \} + i L \{ \sinh at \sin at \} = \frac{a (p^2 - 2a^2)}{p^4 + 4a^4} + i \frac{2a^2 p}{p^4 + 4a^4}.$$

Hence
$$L \{ \sinh at \cos at \} = \frac{a (p^2 - 2a^2)}{p^4 + 4a^4}$$

and
$$L \{ \sinh at \sin at \} = \frac{2a^2 p}{p^4 + 4a^4}.$$

Example 24: Show that $L \{ (1 + t e^{-t})^3 \} = \frac{1}{p} + \frac{3}{(p+1)^2} + \frac{6}{(p+2)^3} + \frac{6}{(p+3)^4}.$

Solution: We have

$$\begin{aligned} L \{ (1 + t e^{-t})^3 \} &= L \{ 1 + 3t e^{-t} + 3t^2 e^{-2t} + t^3 e^{-3t} \} \\ &= L \{ 1 \} + 3(-1) \frac{d}{dp} L \{ e^{-t} \} + 3(-1)^2 \frac{d^2}{dp^2} L \{ e^{-2t} \} + (-1)^3 \frac{d^3}{dp^3} L \{ e^{-3t} \} \\ &= \frac{1}{p} + 3 \cdot \frac{1!}{(p+1)^2} + 3 \cdot \frac{2!}{(p+2)^3} + \frac{3!}{(p+3)^4} \\ &= \frac{1}{p} + \frac{3}{(p+1)^2} + \frac{6}{(p+2)^3} + \frac{6}{(p+3)^4}. \end{aligned}$$

Aliter:

$$\begin{aligned} L \{ (1 + t e^{-t})^3 \} &= \int_0^\infty (1 + t e^{-t})^3 \cdot e^{-pt} dt \\ &= \int_0^\infty [e^{-pt} + 3t e^{-(p+1)t} + 3t^2 e^{-(p+2)t} + t^3 e^{-(p+3)t}] dt \\ &= \int_0^\infty e^{-pt} dt + 3 \int_0^\infty t e^{-(p+1)t} dt + 3 \int_0^\infty t^2 e^{-(p+2)t} dt \\ &\quad + \int_0^\infty t^3 e^{-(p+3)t} dt \\ &= \int_0^\infty e^{-pt} t^{1-1} dt + 3 \int_0^\infty e^{-(p+1)t} t^{2-1} dt \\ &\quad + 3 \int_0^\infty e^{-(p+2)t} t^{3-1} dt + \int_0^\infty e^{-(p+3)t} t^{4-1} dt \end{aligned}$$

$$= \frac{1}{p} + \frac{3}{(p+1)^2} + \frac{6}{(p+2)^3} + \frac{6}{(p+3)^4}, p > 0.$$

$$\left[\because \int_0^\infty e^{-at} t^{n-1} dt = \frac{\Gamma(n)}{a^n}, \text{ if } a > 0 \text{ and } n > 0 \right]$$

Example 25: Prove that $L\{J_0(t)\} = \frac{1}{\sqrt{1+p^2}}$

and hence deduce that

(Avadh 2007)

$$(i) \quad L\{J_0(at)\} = \frac{1}{\sqrt{p^2 + a^2}}$$

$$(ii) \quad L\{t J_0(at)\} = \frac{p}{(p^2 + a^2)^{3/2}}$$

(Rohilkhand 2002)

$$(iii) \quad L\{e^{-at} J_0(at)\} = \frac{1}{\sqrt{p^2 + 2ap + 2a^2}}$$

$$(iv) \quad \int_0^\infty J_0(t) dt = 1.$$

Solution: We know that

$$J_0(t) = 1 - \frac{t^2}{2^2} + \frac{t^4}{2^2 \cdot 4^2} - \frac{t^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots$$

$$\therefore L\{J_0(t)\} = L\{1\} - \frac{1}{2^2} L\{t^2\} + \frac{1}{2^2 \cdot 4^2} L\{t^4\} - \frac{1}{2^2 \cdot 4^2 \cdot 6^2} L\{t^6\} + \dots$$

$$= \frac{1}{p} - \frac{1}{2^2} \cdot \frac{2!}{p^3} + \frac{1}{2^2 \cdot 4^2} \cdot \frac{4!}{p^5} - \frac{1}{2^2 \cdot 4^2 \cdot 6^2} \cdot \frac{6!}{p^7} + \dots$$

$$= \frac{1}{p} \left[1 - \frac{1}{2} \left(\frac{1}{p^2} \right) + \frac{1 \cdot 3}{2 \cdot 4} \left(\frac{1}{p^2} \right)^2 - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \left(\frac{1}{p^2} \right)^3 + \dots \right]$$

$$= \frac{1}{p} \left(1 + \frac{1}{p^2} \right)^{-1/2} = \frac{1}{\sqrt{1+p^2}}.$$

Deductions:

$$(i) \text{ Since } L\{F(at)\} = \frac{1}{a} f\left(\frac{p}{a}\right) \text{ where } f(p) = L\{F(t)\},$$

$$\therefore L\{J_0(at)\} = \frac{1}{a} \cdot \frac{1}{\sqrt{1 + (p/a)^2}} = \frac{1}{\sqrt{p^2 + a^2}}.$$

$$(ii) \quad L\{t J_0(at)\} = -\frac{d}{dp} L\{J_0(at)\} = -\frac{d}{dp} \left[\frac{1}{\sqrt{p^2 + a^2}} \right] = \frac{p}{(p^2 + a^2)^{3/2}}$$

$$(iii) \text{ Since } L\{e^{-at} F(t)\} = f(p+a) \text{ where } f(p) = L\{F(t)\},$$

$$\therefore L\{e^{-at} J_0(at)\} = \frac{1}{\sqrt{(p+a)^2 + a^2}} \quad \left[\because L\{J_0(at)\} = \frac{1}{\sqrt{p^2 + a^2}} \right]$$

$$= \frac{1}{\sqrt{p^2 + 2ap + 2a^2}}.$$

(iv) We have $L\{J_0(t)\} = \int_0^\infty e^{-pt} J_0(t) dt = \frac{1}{\sqrt{1+p^2}}$.

$\therefore$ Putting $p=0$, we have $\int_0^\infty J_0(t) dt = 1$.

Example 26: Prove that $L\{J_1(t)\} = 1 - \frac{p}{\sqrt{(p^2+1)}}$ where $J_1(t)$ is the Bessel function of order one (Kanpur 2009)

and hence deduce that $L\{t J_1(t)\} = \frac{1}{(p^2+1)^{3/2}}$.

Solution: We know that $J_0'(t) = -J_1(t)$.

$\therefore$ From $L\{F'(t)\} = pL\{F(t)\} - F(0)$, we have

$$\begin{aligned} L\{J_1(t)\} &= L\{-J_0'(t)\} = -L\{J_0'(t)\} = -[pL\{J_0(t)\} - J_0(0)] \\ &= -\left[p \cdot \frac{1}{\sqrt{(p^2+1)}} - 1\right] \\ &= -\left[\because \text{from Ex. 26, } L\{J_0(t)\} = \frac{1}{\sqrt{(1+p^2)}} \text{ and } J_0(0) = 1\right] \\ &= 1 - \frac{p}{\sqrt{(p^2+1)}}. \end{aligned}$$

Aliter: We have

$$J_n(t) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \Gamma(n+r+1)} \cdot \left(\frac{t}{2}\right)^{n+2r}.$$

$$\therefore J_1(t) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r!(r+1)!} \cdot \left(\frac{t}{2}\right)^{2r+1} = \frac{t}{2} - \frac{1}{2^2} \frac{t^3}{4} + \frac{1}{2^2 \cdot 4^2} \frac{t^5}{6} - \dots$$

$$\begin{aligned} \therefore L\{J_1(t)\} &= \frac{1}{2} \cdot L\{t\} - \frac{1}{2^2 \cdot 4} L\{t^3\} + \frac{1}{2^2 \cdot 4^2 \cdot 6} L\{t^5\} - \dots \\ &= \frac{1}{2} \cdot \frac{1!}{p^2} - \frac{1}{2^2 \cdot 4} \cdot \frac{3!}{p^4} + \frac{1}{2^2 \cdot 4^2 \cdot 6} \cdot \frac{5!}{p^6} - \dots \\ &= 1 - \left[1 - \frac{1}{2} \left(\frac{1}{p^2}\right) + \frac{1 \cdot 3}{2 \cdot 4} \left(\frac{1}{p^2}\right)^2 - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \left(\frac{1}{p^2}\right)^3 + \dots\right] \\ &= 1 - \left(1 + \frac{1}{p^2}\right)^{-1/2} = 1 - \frac{p}{\sqrt{(p^2+1)}} \end{aligned}$$

Deduction: $L\{t J_1(t)\} = -\frac{d}{dp} L\{J_1(t)\}$

$$= -\frac{d}{dp} \left(1 - \frac{p}{\sqrt{(p^2+1)}}\right) = \frac{1}{(p^2+1)^{3/2}}.$$

Example 27: Find $L\{\operatorname{erf} \sqrt{t}\}$ and hence prove that

$$L\{t \cdot \operatorname{erf}(2\sqrt{t})\} = \frac{3p+8}{p^2(p+4)^{3/2}}.$$

(Avadh 2012)

Solution: We know that

$$\begin{aligned}\operatorname{erf}(\sqrt{t}) &= \frac{2}{\sqrt{\pi}} \int_0^{\sqrt{t}} e^{-u^2} du \\ &= \frac{2}{\sqrt{\pi}} \int_0^{\sqrt{t}} \left(1 - u^2 + \frac{u^4}{2!} - \frac{u^6}{3!} + \dots\right) du \\ &= \frac{2}{\sqrt{\pi}} \left[u - \frac{u^3}{3} + \frac{u^5}{5(2!)} - \frac{u^7}{7(3!)} + \dots \right]_0^{\sqrt{t}} \\ &= \frac{2}{\sqrt{\pi}} \left[t^{1/2} - \frac{t^{3/2}}{3} + \frac{t^{5/2}}{5(2!)} - \frac{t^{7/2}}{7(3!)} + \dots \right]\end{aligned}$$

$$\begin{aligned}\therefore L\{\operatorname{erf} \sqrt{t}\} &= \frac{2}{\sqrt{\pi}} \left[L\{t^{1/2}\} - \frac{1}{3} L\{t^{3/2}\} - \frac{1}{5(2!)} L\{t^{5/2}\} - \frac{1}{7(3!)} L\{t^{7/2}\} + \dots \right] \\ &= \frac{2}{\sqrt{\pi}} \left[\frac{\Gamma(\frac{3}{2})}{p^{3/2}} - \left(\frac{1}{3}\right) \frac{\Gamma(\frac{5}{2})}{p^{5/2}} + \frac{1}{5(2!)} \cdot \frac{\Gamma(\frac{7}{2})}{p^{7/2}} - \frac{1}{7(3!)} \cdot \frac{\Gamma(\frac{9}{2})}{p^{9/2}} + \dots \right] \\ &= \frac{2}{\sqrt{\pi}} \left[\frac{\sqrt{\pi}}{2} \cdot \frac{1}{p^{3/2}} - \frac{\sqrt{\pi}}{2} \cdot \frac{1}{2} \cdot \frac{1}{p^{5/2}} + \frac{\sqrt{\pi}}{2} \cdot \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{1}{p^{7/2}} - \frac{\sqrt{\pi}}{2} \cdot \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \cdot \frac{1}{p^{9/2}} + \dots \right] \\ &= \frac{1}{p^{3/2}} \left[1 - \frac{1}{2} \cdot \frac{1}{p} + \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{1}{p^2} - \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \cdot \frac{1}{p^3} + \dots \right] = \frac{1}{p^{3/2}} \left(1 + \frac{1}{p} \right)^{-1/2} \\ &= \frac{1}{p \sqrt{p+1}}.\end{aligned}$$

(Rohilkhand 2004)

Since $L\{F(at)\} = \frac{1}{a} f\left(\frac{p}{a}\right)$, where $f(p) = L\{F(t)\}$,

$$\begin{aligned}\therefore L\{\operatorname{erf}(2\sqrt{t})\} &= L\{\operatorname{erf} \sqrt{4t}\} \\ &= \frac{1}{4} \cdot \frac{1}{\frac{p}{4} \sqrt{\left(\frac{p}{4} + 1\right)}} = \frac{2}{p \sqrt{p+4}}.\end{aligned}$$

$$\text{Hence } L\{t \cdot \operatorname{erf}(2\sqrt{t})\} = -\frac{d}{dp} L\{\operatorname{erf}(2\sqrt{t})\} = -\frac{d}{dp} \left[\frac{2}{p \sqrt{p+4}} \right] = \frac{3p+8}{p^2(p+4)^{3/2}}.$$

Example 28: Find the Laplace transform of $S_i(t)$.

(Rohilkhand 2009)

Solution: We know that

$$S_i(t) = \int_0^t \frac{\sin u}{u} du = \int_0^t \left(1 - \frac{u^2}{3!} + \frac{u^4}{5!} - \frac{u^6}{7!} + \dots\right) du$$

$$= t - \frac{t^3}{3(3!)} + \frac{t^5}{5(5!)} - \frac{t^7}{7(7!)} + \dots$$

$$\begin{aligned} \therefore L\{S_i(t)\} &= L\{t\} - \frac{1}{3(3!)} L\{t^3\} + \frac{1}{5(5!)} L\{t^5\} - \frac{1}{7(7!)} L\{t^7\} + \dots \\ &= \frac{1!}{p^2} - \frac{1}{3(3!)} \cdot \frac{3!}{p^4} + \frac{1}{5(5!)} \cdot \frac{5!}{p^6} - \frac{1}{7(7!)} \cdot \frac{7!}{p^8} + \dots \\ &= \frac{1}{p} \left[\frac{1}{p} - \frac{1}{3} \cdot \frac{1}{p^3} + \frac{1}{5} \cdot \frac{1}{p^5} - \frac{1}{7} \cdot \frac{1}{p^7} + \dots \right] \\ &= \frac{1}{p} \tan^{-1} \frac{1}{p}, \text{ by Gregory's series.} \end{aligned}$$

Example 29: Find $L\{C_i(t)\}$.

(Rohilkhand 2003, 08)

Solution: $L\{C_i(t)\} = L\left\{\int_t^\infty \frac{\cos u}{u} du\right\}.$

Let $F(t) = \int_t^\infty \frac{\cos u}{u} du = -\int_\infty^t \frac{\cos u}{u} du$

so that $F'(t) = -\frac{\cos t}{t}$ or $t F'(t) = -\cos t.$

$\therefore L\{t F'(t)\} = L\{-\cos t\}$

or $-\frac{d}{dp} L\{F'(t)\} = -\frac{p}{p^2 + 1}$

or $\frac{d}{dp} [p f(p) - F(0)] = \frac{p}{p^2 + 1}$ where $f(p) = L\{F(t)\}$

or $\frac{d}{dp} [p f(p)] = \frac{p}{p^2 + 1}$, since $F(0)$ is constant.

Integrating, $p f(p) = \frac{1}{2} \log(p^2 + 1) + C$ (constant) ...(1)

But from the final-value theorem 1.16,

$$\lim_{p \rightarrow 0} p f(p) = \lim_{t \rightarrow \infty} F(t) = 0.$$

$\therefore$ from (1) as $p \rightarrow 0$ we have $0 = 0 + C$ or $C = 0.$

$\therefore$ from (1), $p f(p) = \frac{1}{2} \log(p^2 + 1)$

or $f(p) = L\{F(t)\} = L\{C_i(t)\} = \frac{\log(p^2 + 1)}{2p}.$

Example 30: If $F(t) = \begin{cases} 3t, & 0 < t < 2 \\ 6, & 2 < t < 4, \end{cases}$

find $L\{F(t)\}$ where $F(t)$ has period 4.

Solution: Here $F(t)$ is a periodic function with period $T = 4.$

∴ from 1.22 theorem, we have

$$\begin{aligned}
 L\{F(t)\} &= \frac{\int_0^T e^{-pt} F(t) dt}{1 - e^{-pT}} = \frac{\int_0^2 3t e^{-pt} dt + \int_2^4 6 \cdot e^{-pt} dt}{1 - e^{-4p}} \\
 &= \frac{1}{1 - e^{-4p}} \left[\left(\frac{e^{-pt}}{-p} \right) \cdot 3t - \left(\frac{e^{-pt}}{p^2} \right) \cdot 3 \right]_0^2 + \frac{1}{1 - e^{-4p}} \left[\frac{6e^{-pt}}{-p} \right]_2^4 \\
 &= \frac{1}{1 - e^{-4p}} \left[-\frac{6}{p} e^{-2p} - \frac{3e^{-2p}}{p^2} + \frac{3}{p^2} - \frac{6}{p} e^{-4p} + \frac{6}{p} e^{-2p} \right] \\
 &= \frac{3 - 3e^{-2p} - 6pe^{-4p}}{p^2 (1 - e^{-4p})}.
 \end{aligned}$$

Comprehensive Exercise 5

- Given $L\left\{2\sqrt{\left(\frac{t}{\pi}\right)}\right\} = \frac{1}{p^{3/2}}$, show that $\frac{1}{p^{1/2}} = L\left\{\frac{1}{\sqrt{(\pi t)}}\right\}$.
- Find (i) $L\{F(t)\}$ and (ii) $L\{F'(t)\}$, for the function given by

$$F(t) = \begin{cases} 2t, & 0 \leq t \leq 1 \\ t, & t > 1. \end{cases}$$
- Show that $L\{(5e^{2t} - 3)^2\} = \frac{25}{p-4} - \frac{30}{p-2} + \frac{9}{p}$, $p > 4$.
- Show that $L\left\{\frac{\sin^2 t}{t}\right\} = \frac{1}{4} \log\left(\frac{p^2 + 4}{p^2}\right)$. (Rohilkhand 2010; Kashi 14)
- Show that $\int_0^\infty \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$. (Rohilkhand 2003)
- Show that $\int_0^\infty \frac{\cos 6t - \cos 4t}{t} dt = \log\left(\frac{2}{3}\right)$.
- Prove that $L\{J_0(a\sqrt{t})\} = (1/p) e^{-(a^2/4p)}$.
- If $F(t) = t^2$, $0 < t < 2$ and $F(t+2) = F(t)$, find $L\{F(t)\}$.
 [Hint. Here $F(t)$ is a periodic function with period $T = 2$]
- Compute $L\{F(t)\}$, if $F(t) = \begin{cases} \sin t, & 0 < t < \pi \\ 0, & \pi < t < 2\pi, \end{cases}$ where $F(t)$ has period 2π .
- Find the Laplace transform of the Heaviside's unit step function $H(t-a)$.
- Find Laplace transform of $\int_0^t \left(\frac{1 - e^{-2x}}{x}\right) dx$.

12. If $E(t) = \int_t^\infty \frac{e^{-u}}{u} du$, show that $L\{E(t)\} = \frac{\log(p+1)}{p}$.

13. Evaluate $\int_0^\infty e^{-t} \operatorname{erf} \sqrt{t} dt$.

Answers 5

2. (i) $2/p^2 - (1/p + 1/p^2) e^{-p}$

(ii) $\frac{2}{p} - \frac{e^{-p}}{p}$

8. $\frac{-(4p^2 + 4p + 2)e^{-2p} + 2}{p^3(1 - e^{-2p})}$

9. $\frac{1}{(p^2 + 1)(1 - e^{-p\pi})}$

10. $\frac{e^{-ap}}{p}$

11. $\frac{1}{p} \log\left(1 + \frac{2}{p}\right)$

12. $1/\sqrt{2}$

124 Table of Laplace Transform Theorems

Laplace Transform Theorems			
No.	Operation	$F(t)$	$L\{F(t)\} = f(p)$
1.	Linearity property	$a_1 F_1(t) + a_2 F_2(t)$	$a_1 L\{F_1(t)\} + a_2 L\{F_2(t)\}$
2.	First translation or shifting theorem	$e^{at} F(t)$	$f(p - a)$
3.	Second translation or shifting theorem	$G(t) = \begin{cases} F(t - a), & t > a \\ 0, & t < a \end{cases}$	$e^{-ap} f(p)$
4.	Change of scale property	$F(at)$	$\frac{1}{a} f\left(\frac{p}{a}\right)$
5.	Differentiation theorems	$F'(t)$	$pf(p) - F(0)$
		$F^n(t)$	$p^n f(p) - \sum_{r=0}^{n-1} p^{n-1-r} F^r(0)$

6.	Multiplication theorems	$t F(t)$	$-f'(p)$
		$t^n F(t)$	$(-1)^n \frac{d^n}{dp^n} f(p)$
7.	Division theorem	$\frac{1}{t} F(t)$	$\int_p^\infty f(x) dx$
8.	Integral theorem	$\int_0^t F(x) dx$	$\frac{1}{p} f(p)$
9.	Initial value theorem	$\lim_{t \rightarrow 0} F(t) = \lim_{p \rightarrow \infty} p L\{F(t)\}$	
10.	Final-value theorem	$\lim_{t \rightarrow \infty} F(t) = \lim_{p \rightarrow 0} p L\{F(t)\}$	
11.	Fundamental theorem for Periodic Functions	$L\{F(t)\} = \frac{\int_0^T e^{-pt} F(t) dt}{1 - e^{-pT}},$ $F(t)$ is periodic function of period T	

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The Laplace transform of $F(t)$ exists for all $p > a$, if
 - $F(t)$ is a continuous function
 - $F(t)$ is a differentiable function
 - $F(t)$ is a function of class A
 - None of these.

(Rohilkhand 2002)
- The Laplace transform of $F(t) = 1$ is
 - p
 - $\frac{1}{p}$, $p > 0$
 - 1
 - $\frac{1}{p-1}$, $p > 1$.

(Rohilkhand 2008)
- If $L\{F(t)\} = f(p)$ and $G(t) = \begin{cases} F(t-a), & t > a \\ 0, & t < a \end{cases}$ then $L\{G(t)\}$ is
 - $f(p-a)$
 - $e^{-ap} f(p)$
 - $\frac{1}{a} f\left(\frac{p}{a}\right)$
 - $f(ap)$.

(Rohilkhand 2003)

4. If $L\{F(t)\} = f(p)$ and $F'(t)$ is of class A , then $L\{F'(t)\}$ is
 (a) $pf(p) - F(0)$ (b) $p^2 f(p)$
 (c) $f(p) - F(0)$ (d) None of these. (Rohilkhand 2003)
5. The Laplace transform of $\cosh at$ is
 (a) $\frac{p}{p^2 - a^2}$ (b) $\frac{p}{p^2 + a^2}$
 (c) $\frac{1}{p^2 - a^2}$ (d) $\frac{1}{p^2 + a^2}$. (Rohilkhand 2003)
6. If $L\{F(t)\} = f(p)$ then $L\{tF(t)\}$ is
 (a) $f'(p)$ (b) $-f'(p)$
 (c) $\int_p^\infty f(x) dx$ (d) $\frac{1}{p} f(p)$. (Rohilkhand 2002)
7. The value of Laplace transform of $\frac{\sin t}{t}$ is
 (a) $\tan^{-1} \frac{1}{p}$ (b) $\tan^{-1} \frac{2}{p}$
 (c) $\tan^{-1} p$ (d) None of these.
8. If $L\{F(t)\} = f(p)$ then $L\{F(at)\}$ is
 (a) $f\left(\frac{p}{a}\right)$ (b) $\frac{1}{a} f(p)$
 (c) $\frac{1}{a} f\left(\frac{p}{a}\right)$ (d) None of these.
9. The Laplace transform of e^{t^2} is
 (a) $\frac{1}{p-2}$ (b) $\frac{1}{(p-2)^2}$
 (c) $\frac{1}{p^2}$ (d) not existing. (Rohilkhand 2003)
10. If $L\left\{\frac{\sin t}{t}\right\} = \tan^{-1} \frac{1}{p}$, then $L\left\{\frac{\sin 3t}{t}\right\}$ is
 (a) $\frac{1}{3} \tan^{-1} \left(\frac{3}{p}\right)$ (b) $\tan^{-1} \left(\frac{3}{p}\right)$
 (c) $\tan^{-1} (3p)$ (d) $3 \tan^{-1} (3p)$.
11. The value of $L\{t \sin at\}$ is
 (a) $\frac{a}{(p^2 + a^2)}$ (b) $\frac{2ap}{(p^2 + a^2)}$
 (c) $\frac{2ap}{(p^2 + a^2)^2}$ (d) None of these.

12. The value of $\int_0^\infty J_0(t) dt$ is
- (a) 1 (b) 0
(c) -1 (d) None of these.
13. The value of $\int_0^\infty \frac{\sin t}{t} dt$ is
- (a) 0 (b) $\frac{\pi}{2}$
(c) 1 (d) None of these.
14. If $L\{\operatorname{erf} \sqrt{t}\} = \frac{1}{p\sqrt{p+1}}$ then $L\{e^{3t} \operatorname{erf} \sqrt{t}\}$ is
- (a) $\frac{1}{(p-3)\sqrt{p-2}}$ (b) $\frac{1}{(p-3)\sqrt{p+2}}$
(c) $\frac{3}{(p-3)\sqrt{p-2}}$ (d) None of these.

Fill In The Blanks

Fill in the blanks “.....” so that the following statements are complete and correct.

- If the integral $f(p) = \int_{-\infty}^\infty K(p, t) F(t) dt$ is convergent, then $f(p)$ is called the of the function $F(t)$ and the function $K(p, t)$ is called the of the transformation.
- The integral $\int_0^\infty e^{-pt} F(t) dt$ is called the transform of the function $F(t)$.
- The Laplace transform of the function $F(t) = \cos at$ is
- If $f(p)$ is the Laplace transform of $F(t)$ then $f(p-a)$ is the Laplace transform of
- If $F(t)$ is a function of class A and if $L\{F(t)\} = f(p)$, then $L\{t^n F(t)\} = \dots$ where $n = 1, 2, 3, \dots$.
- The value of $L\{J_0(t)\}$ is
- The value of integral $\int_0^\infty \frac{\sin^2 t}{t^2} dt$ is
- If $F(t)$ is a periodic function with period $T > 0$ then $F(u+T) = \dots$

True or False

Write ‘T’ for true and ‘F’ for false statement.

- The Laplace transformation is a linear transformation.
- The function e^{t^2} is not of exponential order as $t \rightarrow \infty$.

3. If the Laplace transform of a function $F(t)$ exists then it must be of class A .
4. The Laplace transform of the function $F(t) = t^n$, n being any real number greater than -1 is $\frac{\Gamma n}{p^{n+1}}$, $p > 0$.
5. If $F(t)$ is a function of class A and if $L\{F(t)\} = f(p)$, then

$$L\{t F(t)\} = f'(p).$$
6. The value of $L\{e^{at}\} = \frac{1}{p+a}$.
7. The value of $L\{\sin \lambda t \cdot \cos \lambda t\} = \frac{\lambda}{p^2 + 4\lambda^2}$, $p > 0$.
8. The value of $L\{t^3 e^{-3t}\} = \frac{6}{(p+3)^4}$.

Answers

Multiple Choice Questions

- | | | |
|---------|---------|---------|
| 1. (c) | 2. (c) | 3. (b) |
| 4. (a) | 5. (a) | 6. (b) |
| 7. (a) | 8. (c) | 9. (d) |
| 10. (b) | 11. (c) | 12. (a) |
| 13. (b) | 14. (a) | |

Fill in the Blank(s)

- | | |
|------------------------------------|-----------------------------|
| 1. integral transform ; kernel. | 2. Laplace. |
| 3. $\frac{p}{p^2 + a^2}$, $p > 0$ | 4. $e^{at} F(t)$ |
| 5. $(-1)^n \frac{d^n}{dp^n} f(p)$ | 6. $\frac{1}{\sqrt{1+p^2}}$ |
| 7. $\frac{\pi}{2}$ | 8. $F(u)$ |

True or False

- | | |
|--------|--------|
| 1. T | 2. T |
| 3. F | 4. F |
| 5. F | 6. F |
| 7. T | 8. T |



Chapter

2



The Inverse Laplace Transform

2.1 Inverse Laplace Transform

If $f(p)$ is the Laplace Transform of a function $F(t)$, i.e. $L\{F(t)\} = f(p)$, then $F(t)$ is called the *inverse Laplace transform* of the function $f(p)$ and is written as $F(t) = L^{-1}\{f(p)\}$.

L^{-1} is called the **inverse Laplace transformation operator**.

2.2 Null Function

If $N(t)$ is a function of t such that $\int_0^t N(t) dt = 0$ for all $t > 0$ then $N(t)$ is called a **null function**.

2.3 Lerch's Theorem

If $F_1(t)$ and $F_2(t)$ are two functions having the same Laplace transform $f(p)$, then $F_1(t) - F_2(t) = N(t)$ where $N(t)$ is a null function for all $t > 0$.

From this it follows that, an inverse Laplace transform is unique except for the addition of a null function.

If we restrict ourselves to functions $F(t)$ which are sectionally continuous in every finite interval $0 \leq t \leq t_0$ and of exponential order for $t > t_0$, then the inverse Laplace transform of $f(p)$ i.e., $L^{-1}\{f(p)\} = F(t)$ is unique.

2.4 Linearity Property

Theorem: Let $f_1(p)$ and $f_2(p)$ be the Laplace transforms of functions $F_1(t)$ and $F_2(t)$ respectively and c_1, c_2 be two constants, then $L^{-1}\{c_1 f_1(p) + c_2 f_2(p)\}$

$$= c_1 L^{-1}\{f_1(p)\} + c_2 L^{-1}\{f_2(p)\} = c_1 F_1(t) + c_2 F_2(t). \text{ (Avadh 2014)}$$

Proof: We have

$$\begin{aligned} L\{c_1 F_1(t) + c_2 F_2(t)\} &= c_1 L\{F_1(t)\} + c_2 L\{F_2(t)\} \\ &= c_1 f_1(p) + c_2 f_2(p). \end{aligned}$$

$$\begin{aligned} \therefore L^{-1}\{c_1 f_1(p) + c_2 f_2(p)\} &= c_1 F_1(t) + c_2 F_2(t) \\ &= c_1 L^{-1}\{f_1(p)\} + c_2 L^{-1}\{f_2(p)\}. \end{aligned}$$

2.5 Inverse L.T. of Some Special Functions

(Rohilkhand 2006)

(i) Inverse L.T. of $1/p, p > 0$.

$$\therefore L\{1\} = 1/p, p > 0, \quad \therefore L^{-1}\{1/p\} = 1, p > 0.$$

(ii) Inverse L.T. of $1/p^{n+1}, p > 0, n > -1$.

$$\therefore L\{t^n\} = \frac{\Gamma(n+1)}{p^{n+1}}, p > 0, n > -1$$

$$\therefore L^{-1}\{1/p^{n+1}\} = t^n / \Gamma(n+1), n > -1.$$

If n is a positive integer, then $\Gamma(n+1) = n!$.

$$\therefore L^{-1}\{1/p^{n+1}\} = t^n / n!, p > 0, n \text{ is a positive integer.}$$

(iii) Inverse L.T. of $1/(p-a), p > a$.

$$\therefore L\{e^{at}\} = 1/(p-a), p > a, \quad \therefore L^{-1}\{1/(p-a)\} = e^{at}, p > a.$$

(iv) Inverse L.T. of $1/(p^2 + a^2), p > 0$.

$$\therefore L\{\sin at\} = \frac{a}{p^2 + a^2}, p > 0, \quad \therefore L^{-1}\left\{\frac{p}{p^2 + a^2}\right\} = \frac{1}{a} \sin at$$

(v) Inverse L.T. of $p/(p^2 + a^2), p > 0$.

$$\therefore L\{\cos at\} = \frac{p}{p^2 + a^2}, \quad \therefore L^{-1}\left\{\frac{p}{p^2 + a^2}\right\} = \cos at.$$

(vi) Inverse L.T. of $1/(p^2 - a^2), p > |a|$.

$$\therefore L\{\sinh at\} = \frac{a}{p^2 - a^2}, \quad \therefore L^{-1}\left\{\frac{1}{p^2 - a^2}\right\} = \frac{1}{a} \sinh at.$$

(vii) Inverse L.T. of $p/(p^2 - a^2)$, $p > |a|$.

$$\therefore L\{\cosh at\} = \frac{p}{p^2 - a^2}, \quad \therefore L^{-1}\left\{\frac{p}{p^2 - a^2}\right\} = \cosh at.$$

2.6 Table of Some Inverse Laplace Transforms

	$f(p)$	$L^{-1}\{f(p)\} = F(t)$
1.	$\frac{1}{p}$	1
2.	$\frac{1}{p^{n+1}}$, n is a positive integer	$\frac{t^n}{n!}$
3.	$\frac{1}{p-a}$	e^{at}
4.	$\frac{1}{p^2 + a^2}$	$\frac{1}{a} \sin at$
5.	$\frac{p}{p^2 + a^2}$	$\cos at$
6.	$\frac{1}{p^2 - a^2}$	$\frac{1}{a} \sinh at$
7.	$\frac{p}{p^2 - a^2}$	$\cosh at$
8.	$\frac{1}{p^{n+1}}$, $n > -1$	$\frac{t^n}{\Gamma(n+1)}$

If we know the inverse L.T. given in the above table then nearly all the inverse Laplace transforms can be obtained by using the general theorems, which we shall give later on.

Illustrative Examples

Example 1: Find (i) $L^{-1}\left\{\frac{1}{p^4}\right\}$ (ii) $L^{-1}\left\{\frac{1}{p^2 + 4}\right\}$.

(Meerut 2013B)

Solution: (i) $L^{-1}\left\{\frac{1}{p^4}\right\} = \frac{t^{4-1}}{3!} = \frac{t^3}{6}$

(ii) $L^{-1}\left\{\frac{1}{p^2 + 4}\right\} = L^{-1}\left\{\frac{1}{p^2 + 2^2}\right\} = \frac{1}{2} \sin 2t$

Example 2: Prove that

$$L^{-1} \left\{ \frac{5}{p^2} + \left(\frac{\sqrt{p-1}}{p} \right)^2 - \frac{7}{3p+2} \right\} = 1 + 6t - 4\sqrt{\frac{t}{\pi}} - \frac{7}{3}e^{-2t/3}.$$

Solution: We have
$$\begin{aligned} L^{-1} \left\{ \frac{5}{p^2} + \left(\frac{\sqrt{p-1}}{p} \right)^2 - \frac{7}{3p+2} \right\} \\ &= L^{-1} \left\{ \frac{5}{p^2} + \frac{p-2\sqrt{p+1}}{p^2} - \frac{7}{3} \cdot \frac{1}{p+(2/3)} \right\} \\ &= 6 L^{-1} \left\{ \frac{1}{p^2} \right\} + L^{-1} \left\{ \frac{1}{p} \right\} - 2 L^{-1} \left\{ \frac{1}{p^{3/2}} \right\} - \frac{7}{3} L^{-1} \left\{ \frac{1}{p+(2/3)} \right\} \\ &= 6 \frac{t^{2-1}}{1!} + 1 - 2 \cdot \frac{t^{3/2-1}}{\Gamma(3/2)} - \frac{7}{3} e^{-2t/3} \\ &= 6t + 1 - 4\sqrt{t/\pi} - (7/3)e^{-2t/3}. \end{aligned}$$

Example 3: Show that

$$L^{-1} \left\{ \frac{1}{p} \sin \frac{1}{p} \right\} = t - \frac{t^3}{(3!)^2} + \frac{t^5}{(5!)^2} - \frac{t^7}{(7!)^2} + \dots \quad (\text{Avadh 2014})$$

Solution:
$$\begin{aligned} L^{-1} \left\{ \frac{1}{p} \sin \frac{1}{p} \right\} &= L^{-1} \left\{ \frac{1}{p} \left(\frac{1}{p} - \frac{(1/p)^3}{3!} + \frac{(1/p)^5}{5!} - \frac{(1/p)^7}{7!} + \dots \right) \right\} \\ &= L^{-1} \left\{ \frac{1}{p^2} \right\} - \frac{1}{3!} L^{-1} \left\{ \frac{1}{p^4} \right\} + \frac{1}{5!} L^{-1} \left\{ \frac{1}{p^6} \right\} - \frac{1}{7!} L^{-1} \left\{ \frac{1}{p^8} \right\} + \dots \\ &= t - \frac{t^3}{(3!)^2} + \frac{t^5}{(5!)^2} - \frac{t^7}{(7!)^2} + \dots \end{aligned}$$

Comprehensive Exercise 1

Find the inverse Laplace transforms of each of the following functions.

1. $4/(p-2).$

2. $\frac{p}{p^2+2} + \frac{6p}{p^2-16} + \frac{3}{p-3}.$

3. $\frac{2p+1}{p(p+1)}.$

(Purvanchal 2014; Kashi 14, 14B)

4. $\frac{2p-5}{p^2-9}.$

(Meerut 2013)

5. $\frac{3p-2}{p^{5/2}} - \frac{7}{3p+2}.$

6. $\frac{6}{2p-3} - \frac{3+4p}{9p^2-16} + \frac{8-6p}{16p^2+9}.$

7. $\frac{3}{p^2-3} + \frac{3p+2}{p^3} - \frac{3p-27}{p^2+9} + \frac{6-30\sqrt{p}}{p^4}$.
8. $\frac{3(p^2-1)^2}{2p^5} + \frac{4p-18}{9-p^2} + \frac{(p+1)(2-\sqrt{p})}{p^{5/2}}$.
9. Show that $L^{-1} \left\{ \frac{1}{p} \cos \frac{1}{p} \right\} = 1 - \frac{t^2}{(2!)^2} + \frac{t^4}{(4!)^2} - \frac{t^6}{(6!)^2} + \dots$

Answers 1

1. $4e^{2t}$.
2. $\cos \sqrt{2}t + 6 \cosh 4t + 3e^{3t}$.
3. $e^{-t} + 1$.
4. $2 \cosh 3t - (5/3) \sinh 3t$.
5. $6 \sqrt{\left(\frac{t}{\pi}\right)} - \frac{8}{3}t \sqrt{\left(\frac{t}{\pi}\right)} - \frac{7}{3}e^{-2t/3}$.
6. $3e^{3t/2} - \frac{1}{4} \sinh \frac{4t}{3} - \frac{4}{9} \cosh \frac{4t}{3} + \frac{2}{3} \sin \frac{3t}{4} - \frac{3}{8} \cos \frac{3t}{4}$.
7. $\sqrt{3} \sinh(\sqrt{3}t) + 3t + t^2 - 3 \cos 3t + 9 \sin 3t + t^3 - 16t^2 \sqrt{(t/\pi)}$.
8. $\frac{1}{2} - \frac{3}{2}t^2 + \frac{1}{16}t^4 - 4 \cosh 3t + 6 \sinh 3t + 4 \sqrt{(t/\pi)} + \frac{8}{3}t \sqrt{(t/\pi)} - t$.

2.7 First Translation or Shifting Theorem

If $L^{-1} \{f(p)\} = F(t)$, then $L^{-1} \{f(p-a)\} = e^{at} F(t) = e^{at} L^{-1} \{f(p)\}$.

Proof: We have $f(p) = \int_0^\infty e^{-pt} F(t) dt$.

$$\therefore f(p-a) = \int_0^\infty e^{-(p-a)t} F(t) dt = \int_0^\infty e^{-pt} \cdot \{e^{at} \cdot F(t)\} dt = L \{e^{at} F(t)\}.$$

$$\therefore L^{-1} \{f(p-a)\} = e^{at} F(t) = e^{at} L^{-1} \{f(p)\}.$$

2.8 Second Translation or Shifting Theorem

If $L^{-1} \{f(p)\} = F(t)$, then $L^{-1} \{e^{-ap} f(p)\} = G(t)$, where $G(t) = \begin{cases} F(t-a), & t > a \\ 0, & t < a. \end{cases}$

Proof: We have $f(p) = \int_0^\infty e^{-pt} F(t) dt$.

$$\therefore e^{-ap} f(p) = \int_0^\infty e^{-p(t+a)} F(t) dt$$

$$\begin{aligned}
 &= \int_0^{\infty} e^{-px} F(x-a) dx, \text{ putting } t+a=x, \text{ so that } dt=dx \\
 &= \int_0^a e^{-px} \cdot 0 dx + \int_a^{\infty} e^{-px} F(x-a) dx \\
 &= \int_0^a e^{-pt} \cdot 0 dt + \int_a^{\infty} e^{-pt} F(t-a) dt \\
 &= \int_0^{\infty} e^{-pt} G(t) dt = L\{G(t)\},
 \end{aligned}$$

where

$$G(t) = \begin{cases} F(t-a), & t > a \\ 0, & t < a. \end{cases}$$

Note: In terms of Heaviside's unit step function $H(t-a)$ this theorem can be stated as :

If $L^{-1}\{f(p)\} = F(t),$

then $L^{-1}\{e^{-ap} f(p)\} = F(t-a) \cdot H(t-a).$

2.9 Change of Scale Property

Theorem: If $L^{-1}\{f(p)\} = F(t)$, then $L^{-1}\{f(ap)\} = (1/a) F(t/a).$

Proof: We have, $f(p) = \int_0^{\infty} e^{-pt} F(t) dt.$

$$\begin{aligned}
 \therefore f(ap) &= \int_0^{\infty} e^{-apt} F(t) dt = \frac{1}{a} \int_0^{\infty} e^{-px} F\left(\frac{x}{a}\right) dx. \\
 &\quad \text{putting } at = x, \text{ so that } dt = (1/a) dx \\
 &= \frac{1}{a} \int_0^{\infty} e^{-pt} F\left(\frac{t}{a}\right) dt = \frac{1}{a} L\left\{F\left(\frac{t}{a}\right)\right\} = L\left\{\frac{1}{a} F\left(\frac{t}{a}\right)\right\}.
 \end{aligned}$$

Hence $L^{-1}\{f(ap)\} = (1/a) F(t/a).$

2.10 Use of Partial Fractions

If $f(p)$ is of the form $\frac{g(p)}{h(p)}$, where g and h are polynomials in p , then break $f(p)$ into partial fractions and manipulate term by term.

Illustrative Examples

Example 4: Evaluate $L^{-1}\left\{\frac{3p+7}{p^2-2p-3}\right\}.$

(Bundelkhand 2013)

Solution: $L^{-1}\left\{\frac{3p+7}{p^2-2p-3}\right\} = L^{-1}\left\{\frac{3(p-1)+10}{(p-1)^2-4}\right\}$

$$\begin{aligned}
 &= L^{-1} \left\{ \frac{3(p-1)}{(p-1)^2 - 4} + \frac{10}{(p-1)^2 - 4} \right\} \\
 &= 3 L^{-1} \left\{ \frac{p-1}{(p-1)^2 - 4} \right\} + 10 L^{-1} \left\{ \frac{1}{(p-1)^2 - 4} \right\} \\
 &= 3e^t L^{-1} \left\{ \frac{p}{p^2 - 2^2} \right\} + 10 e^t L^{-1} \left\{ \frac{1}{p^2 - 2^2} \right\} \\
 &= 3e^t \cosh 2t + 5e^t \sinh 2t = 4e^{3t} - e^{-t}.
 \end{aligned}$$

Example 5: If $L^{-1} \left\{ \frac{e^{-1/p}}{p^{1/2}} \right\} = \frac{\cos 2\sqrt{t}}{\sqrt{\pi t}}$, find $L^{-1} \left\{ \frac{e^{-a/p}}{p^{1/2}} \right\}$, where $a > 0$.

Solution: Since $L^{-1} \left\{ \frac{e^{-1/p}}{p^{1/2}} \right\} = \frac{\cos 2\sqrt{t}}{\sqrt{\pi t}}$,

$$\therefore L^{-1} \left\{ \frac{e^{-1/pk}}{(pk)^{1/2}} \right\} = \frac{1}{k} \frac{\cos 2\sqrt{(t/k)}}{\sqrt{\pi(t/k)}} \quad \text{or} \quad L^{-1} \left\{ \frac{e^{-1/pk}}{p^{1/2}} \right\} = \frac{\cos 2\sqrt{(t/k)}}{\sqrt{\pi t}}.$$

Taking $k = 1/a$, we have $L^{-1} \left\{ \frac{e^{-a/p}}{p^{1/2}} \right\} = \frac{\cos 2\sqrt{at}}{\sqrt{\pi t}}$.

Example 6: Find a function $F(t)$ for which

$$F(t) = L^{-1} \left\{ \frac{3}{p} - \frac{4e^{-p}}{p^2} + \frac{4e^{-3p}}{p^2} \right\}.$$

Solution: $F(t) = L^{-1} \left\{ \frac{3}{p} - \frac{4e^{-p}}{p^2} + \frac{4e^{-3p}}{p^2} \right\}$

$$= 3 L^{-1} \left\{ \frac{1}{p} \right\} - 4 L^{-1} \left\{ \frac{e^{-p}}{p^2} \right\} + 4 L^{-1} \left\{ \frac{e^{-3p}}{p^2} \right\}. \quad \dots(1)$$

Now $L^{-1} \{1/p\} = 1$, $L^{-1} \{1/p^2\} = t$.

$$\therefore L^{-1} \{e^{-p}/p^2\} = (t-1)H(t-1),$$

$$L^{-1} \{e^{-3p}/p^2\} = (t-3)H(t-3).$$

Putting in (1), we have

$$F(t) = 3 - 4(t-1)H(t-1) + 4(t-3)H(t-3).$$

Example 7: Find the inverse Laplace transform of the following function.

$$\frac{p+1}{p(p^2+4p+8)}.$$

Solution: $L^{-1} \left\{ \frac{p+1}{p(p^2+4p+8)} \right\} = L^{-1} \left\{ \frac{1}{8p} - \frac{p-4}{8(p^2+4p+8)} \right\},$

resolving into partial fractions

$$\begin{aligned}
&= \frac{1}{8} L^{-1} \left\{ \frac{1}{p} \right\} - \frac{1}{8} L^{-1} \left\{ \frac{(p+2)-6}{(p+2)^2+4} \right\} \\
&= \frac{1}{8} - \frac{1}{8} e^{-2t} L^{-1} \left\{ \frac{p-6}{p^2+4} \right\} \\
&= \frac{1}{8} - \frac{1}{8} e^{-2t} \left[L^{-1} \left\{ \frac{p}{p^2+2^2} \right\} - 6 L^{-1} \left\{ \frac{1}{p^2+2^2} \right\} \right] \\
&= \frac{1}{8} [1 - e^{-2t} (\cos 2t - 3 \sin 2t)].
\end{aligned}$$

Example 8: Prove that $L^{-1} \left\{ \frac{p}{p^4 + p^2 + 1} \right\} = \frac{2}{\sqrt{3}} \sinh \frac{t}{2} \cdot \sin \frac{1}{2} \cdot \sqrt{3}t$.

(Kanpur 2010; Meerut 13B; Kashi 14; Rohilkhand 14)

Solution: We have $L^{-1} \left\{ \frac{p}{p^4 + p^2 + 1} \right\} = L^{-1} \left\{ \frac{p}{(p^2+1)^2 - p^2} \right\}$

$$\begin{aligned}
&= L^{-1} \left\{ \frac{p}{(p^2 - p + 1)(p^2 + p + 1)} \right\} \\
&= L^{-1} \left\{ \frac{1}{2} \frac{(p^2 + p + 1) - (p^2 - p + 1)}{(p^2 - p + 1)(p^2 + p + 1)} \right\} \\
&= L^{-1} \left\{ \frac{1}{2(p^2 - p + 1)} - \frac{1}{2(p^2 + p + 1)} \right\} \\
&= \frac{1}{2} L^{-1} \left\{ \frac{1}{(p - \frac{1}{2})^2 + \frac{3}{4}} \right\} - \frac{1}{2} L^{-1} \left\{ \frac{1}{(p + \frac{1}{2})^2 + \frac{3}{4}} \right\} \\
&= \frac{1}{2} e^{t/2} L^{-1} \left[\frac{1}{p^2 + (\frac{1}{2}\sqrt{3})^2} \right] - \frac{1}{2} e^{-t/2} L^{-1} \left[\frac{1}{p^2 + (\frac{1}{2}\sqrt{3})^2} \right] \\
&= \frac{1}{2} e^{t/2} \cdot \frac{2}{\sqrt{3}} \sin \left(\sqrt{3} \frac{t}{2} \right) - \frac{1}{2} e^{-t/2} \cdot \frac{2}{\sqrt{3}} \sin \left(\sqrt{3} \frac{t}{2} \right) \\
&= \frac{1}{\sqrt{3}} (e^{t/2} - e^{-t/2}) \sin \left(\sqrt{3} \frac{t}{2} \right) = \frac{2}{\sqrt{3}} \sinh \frac{t}{2} \sin \left(\sqrt{3} \frac{t}{2} \right).
\end{aligned}$$

Example 9: Evaluate (i) $L^{-1} \left\{ \frac{1}{(p^2+4)(p+1)^2} \right\}$,

(Meerut 2013)

(ii) $L^{-1} \left\{ \frac{3p^3 - 3p^2 - 40p + 36}{(p^2 - 4)^2} \right\}.$

Solution: (i) We have $L^{-1} \left\{ \frac{1}{(p^2 + 4)(p + 1)^2} \right\}$

$$= L^{-1} \left\{ \frac{2}{25(p + 1)} + \frac{1}{5(p + 1)^2} - \frac{2p + 3}{25(p^2 + 4)} \right\}$$

$$= \frac{1}{25} \left[2 L^{-1} \left\{ \frac{1}{p + 1} \right\} + 5 L^{-1} \left\{ \frac{1}{(p + 1)^2} \right\} \right]$$

$$- 2 L^{-1} \left\{ \frac{p}{p^2 + 4} \right\} - 3 L^{-1} \left\{ \frac{1}{p^2 + 4} \right\} \Bigg]$$

$$= \frac{1}{25} [2e^{-t} + 5e^{-t} L^{-1} \{1/p^2\} - 2 \cos 2t - (3/2) \sin 2t]$$

$$= \frac{1}{25} [e^{-t} (2 + 5t) - 2 \cos 2t - (3/2) \sin 2t].$$

(ii) $L^{-1} \left\{ \frac{3p^3 - 3p^2 - 40p + 36}{(p^2 - 4)^2} \right\}$

$$= L^{-1} \left\{ \frac{3p^3 - 3p^2 - 40p + 36}{(p - 2)^2 (p + 2)^2} \right\}$$

$$= L^{-1} \left\{ -\frac{2}{(p - 2)^2} + \frac{3}{p + 2} + \frac{5}{(p + 2)^2} \right\}$$

$$= -2 L^{-1} \left\{ \frac{1}{(p - 2)^2} \right\} + 3 L^{-1} \left\{ \frac{1}{p + 2} \right\} + 5 L^{-1} \left\{ \frac{1}{(p + 2)^2} \right\}$$

$$= -2e^{2t} L^{-1} \{1/p^2\} + 3e^{-2t} + 5e^{-2t} L^{-1} \{1/p^2\}$$

$$= -2e^{2t} \cdot t + 3e^{-2t} + 5e^{-2t} \cdot t$$

$$= (5t + 3)e^{-2t} - 2te^{2t}.$$

Example 10: Show that $L^{-1} \left\{ \frac{p^2}{p^4 + 4a^4} \right\} = \frac{1}{2a} (\cosh at \sin at + \sinh at \cos at).$

Solution: We have

$$L^{-1} \left\{ \frac{p^2}{p^4 + 4a^4} \right\} = L^{-1} \left\{ \frac{p^2}{(p^2 + 2a^2)^2 - 4a^2 p^2} \right\}$$

$$= L^{-1} \left\{ \frac{p^2}{(p^2 - 2ap + 2a^2)(p^2 + 2ap + 2a^2)} \right\}$$

$$= L^{-1} \left\{ \frac{1}{4a} \cdot \frac{p(p^2 + 2ap + 2a^2) - p(p^2 - 2ap + 2a^2)}{(p^2 + 2ap + 2a^2)(p^2 - 2ap + 2a^2)} \right\}$$

$$\begin{aligned}
 &= L^{-1} \left\{ \frac{p}{4a(p^2 - 2ap + 2a^2)} - \frac{p}{4a(p^2 + 2ap + 2a^2)} \right\} \\
 &= \frac{1}{4a} L^{-1} \left\{ \frac{(p-a)+a}{(p-a)^2 + a^2} \right\} - \frac{1}{4a} L^{-1} \left\{ \frac{(p+a)-a}{(p+a)^2 + a^2} \right\} \\
 &= \frac{1}{4a} e^{at} L^{-1} \left\{ \frac{p+a}{p^2 + a^2} \right\} - \frac{1}{4a} e^{-at} L^{-1} \left\{ \frac{p-a}{p^2 + a^2} \right\} \\
 &= \frac{1}{4a} e^{at} \left[L^{-1} \left\{ \frac{p}{p^2 + a^2} \right\} + a L^{-1} \left\{ \frac{1}{p^2 + a^2} \right\} \right] \\
 &\quad - \frac{1}{4a} e^{-at} \left[L^{-1} \left\{ \frac{p}{p^2 + a^2} \right\} - a L^{-1} \left\{ \frac{1}{p^2 + a^2} \right\} \right] \\
 &= \frac{1}{4a} e^{at} (\cos at + \sin at) - \frac{1}{4a} e^{-at} (\cos at - \sin at) \\
 &= \frac{1}{2a} \left[\left(\frac{e^{at} + e^{-at}}{2} \right) \sin at + \left(\frac{e^{at} - e^{-at}}{2} \right) \cos at \right] \\
 &= \frac{1}{2a} (\cosh at \sin at + \sinh at \cos at).
 \end{aligned}$$

Example 11: Find $L^{-1} \left\{ \frac{(p+1)e^{-\pi p}}{p^2 + p + 1} \right\}$.

Solution: We have $L^{-1} \left\{ \frac{p+1}{p^2 + p + 1} \right\} = L^{-1} \left\{ \frac{(p + \frac{1}{2}) + \frac{1}{2}}{(p + \frac{1}{2})^2 + \frac{3}{4}} \right\}$

$$\begin{aligned}
 &= e^{-t/2} L^{-1} \left\{ \frac{p + \frac{1}{2}}{p^2 + \frac{3}{4}} \right\} \\
 &= e^{-t/2} L^{-1} \left\{ \frac{p}{p^2 + (\sqrt{3}/2)^2} \right\} + \frac{1}{2} e^{-t/2} L^{-1} \left\{ \frac{1}{p^2 + (\sqrt{3}/2)^2} \right\} \\
 &= e^{-t/2} \cos(\sqrt{3}t/2) + \frac{1}{2} e^{-t/2} \cdot (2/\sqrt{3}) \sin(\sqrt{3}t/2) \\
 &= \frac{e^{-t/2}}{\sqrt{3}} [\sqrt{3} \cos(\sqrt{3}t/2) + \sin(\sqrt{3}t/2)].
 \end{aligned}$$

$$\begin{aligned}
 \therefore L^{-1} \left\{ \frac{(p+1)e^{-\pi p}}{p^2 + p + 1} \right\} &= \begin{cases} \frac{e^{-(t-\pi)/2}}{\sqrt{3}} \left[\sqrt{3} \cos \frac{\sqrt{3}}{2}(t-\pi) + \sin \frac{\sqrt{3}}{2}(t-\pi) \right], & t > \pi \\ 0, & t < \pi \end{cases} \\
 &= \frac{e^{-(t-\pi)/2}}{\sqrt{3}} \left[\sqrt{3} \cos \frac{\sqrt{3}}{2}(t-\pi) + \sin \frac{\sqrt{3}}{2}(t-\pi) \right] H(t-\pi).
 \end{aligned}$$

Comprehensive Exercise 2

Evaluate the following :

$$1. \quad (i) \ L^{-1} \left\{ \frac{1}{p^2 - 6p + 10} \right\}. \quad (ii) \ L^{-1} \left\{ \frac{p-1}{(p+3)(p^2 + 2p + 2)} \right\}.$$

$$2. \quad (i) \ L^{-1} \left\{ \frac{1}{(p+a)^n} \right\} \quad (ii) \ L^{-1} \left\{ \frac{p}{(p+1)^{5/2}} \right\}.$$

$$3. \quad (i) \ L^{-1} \left\{ \frac{p}{(p+1)^5} \right\} \text{ (Lucknow 2011)} \quad (ii) \ L^{-1} \left\{ \frac{3p+2}{4p^2 + 12p + 9} \right\}.$$

$$4. \quad (i) \ L^{-1} \left\{ \frac{p^2 - 2p + 3}{(p-1)^2 (p+1)} \right\} \quad (ii) \ L^{-1} \left\{ \frac{1}{(p+2)(p-1)^2} \right\}.$$

(Gorakhpur 2005)

$$5. \quad \text{If } L^{-1} \left\{ \frac{p}{(p^2 + 1)^2} \right\} = \frac{1}{2} t \sin t, \text{ find } L^{-1} \left\{ \frac{32p}{(16p^2 + 1)^2} \right\}.$$

$$6. \quad \text{Find (i) } L^{-1} \left\{ \frac{e^{-5p}}{(p-2)^4} \right\} \quad (ii) \ L^{-1} \left\{ \frac{e^{-4p}}{(p-3)^4} \right\}.$$

$$7. \quad \text{Find } L^{-1} \left\{ \frac{e^{4-3p}}{(p+4)^{5/2}} \right\}.$$

$$8. \quad \text{Find the inverse L.T. of } e^{-3p} / p^3.$$

(Lucknow 2009)

9. Prove the following :

$$(i) \ L^{-1} \left\{ \frac{e^{-p\pi}}{p^2 + 1} \right\} = -\sin t. H(t - \pi)$$

(Kanpur 2008)

$$(ii) \ L^{-1} \left\{ \frac{p e^{-2p\pi/3}}{p^2 + 9} \right\} = \cos 3(t - 2\pi/3) \cdot H(t - 2\pi/3).$$

10. Prove the following :

$$(i) \ L^{-1} \left\{ \frac{p e^{-ap}}{p^2 - \omega^2} \right\} = \cos h \omega(t - a) \cdot H(t - a), a > 0$$

$$(ii) \ L^{-1} \left\{ \frac{3(1 + e^{-p\pi})}{p^2 + 9} \right\} = -\sin 3t \cdot H(t - \pi) + \sin 3t.$$

11. Evaluate

$$(i) \ L^{-1} \left\{ \frac{p+2}{p^2 - 2p + 5} \right\}$$

(Kanpur 2008; Lucknow 10; Avadh 13)

$$(ii) \quad L^{-1} \left\{ \frac{p+8}{p^2+8p+5} \right\}.$$

$$12. \quad \text{Prove that } L^{-1} \left\{ \frac{4p+5}{(p-1)^2(p+2)} \right\} = 3te^t + \frac{1}{3}e^t - \frac{1}{3}e^{-2t}. \quad (\text{Purvanchal 2007})$$

$$13. \quad \text{Evaluate (i) } L^{-1} \left\{ \frac{4p+5}{(p-4)^2(p+3)} \right\}$$

$$(ii) \quad L^{-1} \left\{ \frac{5p^2-15p-11}{(p+1)(p-2)^3} \right\}. \quad (\text{Gorakhpur 2009})$$

$$14. \quad \text{Prove that } L^{-1} \left\{ \frac{2p+1}{(p+2)^2(p-1)^2} \right\} = \frac{1}{3}t(e^t - e^{-2t}). \quad (\text{Gorakhpur 2007})$$

$$15. \quad \text{Prove that } L^{-1} \left\{ \frac{p}{(p^2-2p+2)(p^2+2p+2)} \right\} = \frac{1}{2} \sin t \sinh t.$$

Answers 2

$$1. \quad (i) e^{3t} \sin t$$

$$(ii) -\frac{4}{5}e^{-3t} + \frac{1}{5}e^{-t}(4 \cos t - 3 \sin t)$$

$$2. \quad (i) e^{-at} \cdot \frac{t^{n-1}}{(n-1)!}$$

$$(ii) \frac{2}{3}e^{-t} \sqrt{\left(\frac{t}{\pi}\right)}(3-2t)$$

$$3. \quad (i) \frac{1}{24}e^{-t}(4t^3 - t^4)$$

$$(ii) \frac{1}{8}e^{-3t/2}(6-5t)$$

$$4. \quad (i) \left(t - \frac{1}{2}\right)e^t + \frac{3}{2}e^{-t}$$

$$(ii) \frac{1}{9}[(3t-1)e^t + e^{-2t}]$$

$$5. \quad \frac{1}{4}t \sin\left(\frac{1}{4}t\right)$$

$$6. \quad (i) \frac{1}{9}(t-5)^3 e^{2(t-5)} \cdot H(t-5)$$

$$(ii) \frac{1}{6}(t-4)^3 e^{3(t-4)} \cdot H(t-4)$$

$$7. \quad \frac{4}{3}(1/\sqrt{\pi}) \cdot (t-3)^{3/2} e^{-4(t-4)} \cdot H(t-3)$$

$$8. \quad \frac{1}{2}(t-3)^3 \cdot H(t-3)$$

$$9. \quad (i) e^t \left[\cos 2t + \frac{3}{2} \sin 2t \right]$$

$$(ii) e^{-4t} [\cosh(\sqrt{11}t) + (4/\sqrt{11}) \sinh(\sqrt{11}t)]$$

$$13. \quad (i) -\frac{1}{7}e^{-3t} + \frac{1}{7}e^{4t} + 3te^{4t}$$

$$(ii) \frac{1}{3}(-e^{-t} + e^{2t}) + 4te^{2t} - \frac{7}{2}t^2 e^{2t}$$

2.11 Inverse Laplace Transform of Derivatives

Theorem: If $L^{-1} \{f(p)\} = F(t)$, then

$$\begin{aligned} L^{-1} \{f^{(n)}(p)\} &= L^{-1} \left[\frac{d^n}{dp^n} f(p) \right] \\ &= (-1)^n t^n F(t) \\ &= (-1)^n t^n L^{-1} \{f(p)\}, \quad n = 1, 2, 3, \dots \end{aligned}$$

Proof: Since, we have

$$\begin{aligned} L \{t^n F(t)\} &= (-1)^n \frac{d^n}{dp^n} f(p) \\ &= (-1)^n f^{(n)}(p), \end{aligned}$$

$$\begin{aligned} \therefore L^{-1} \{f^{(n)}(p)\} &= L^{-1} \left[\frac{d^n}{dp^n} f(p) \right] = (-1)^n t^n F(t) \\ &= (-1)^n t^n L^{-1} \{f(p)\}. \end{aligned}$$

2.12 Inverse Laplace Transform of Integrals

Theorem: If $L^{-1} \{f(p)\} = F(t)$, then $L^{-1} \left[\int_p^\infty f(x) dx \right] = \frac{F(t)}{t}$.

Proof: Since, we have

$$L \left[\frac{1}{t} F(t) \right] = \int_p^\infty f(x) dx, \text{ provided } \lim_{t \rightarrow 0} \left[\frac{F(t)}{t} \right] \text{ exists.}$$

$$\therefore L^{-1} \left[\int_p^\infty f(x) dx \right] = \frac{F(t)}{t}.$$

2.13 Multiplication by Powers of p

Theorem: If $L^{-1} \{f(p)\} = F(t)$ and $F(0) = 0$, then $L^{-1} \{p f(p)\} = F'(t)$.

Proof: We have

$$L \{F'(t)\} = p L \{F(t)\} - F(0) = p f(p). \quad [\because F(0) = 0]$$

$$\therefore L^{-1} \{p f(p)\} = F'(t).$$

Note 1: If $F(0) \neq 0$, then

$$L^{-1} \{p f(p) - F(0)\} = F'(t)$$

$$\text{or} \quad L^{-1} \{p f(p)\} = F'(t) + F(0) \delta(t)$$

where $\delta(t)$ is the Dirac delta function or unit impulse function.

Note 2: Generalizations to $L^{-1} \{p^n f(p)\}$ are possible, for $n = 2, 3, \dots$ i.e., in general $L^{-1} \{p^n f(p)\} = F^n(t)$ if $F(0) = 0$.

2.14 Division by Powers of p

Theorem I: If $F(t)$ is sectionally continuous and of exponential order a and such that $\lim_{t \rightarrow 0} \frac{F(t)}{t}$ exists then for $p > a$, $L^{-1} \left[\frac{f(p)}{p} \right] = \int_0^t F(x) dx$.

Proof: Let $G(t) = \int_0^t F(x) dx$.

Then $G'(t) = F(t)$ and $G(0) = 0$.

$$\therefore L\{G'(t)\} = p L\{G(t)\} - G(0) = p L\{G(t)\}$$

$$\text{or } L\{F(t)\} = f(p) = p L\{G(t)\}.$$

$$\therefore L\{G(t)\} = \frac{f(p)}{p}. \text{ Hence } L^{-1} \left[\frac{f(p)}{p} \right] = G(t) = \int_0^t F(x) dx.$$

Theorem II: $L^{-1} \left[\frac{f(p)}{p^2} \right] = \int_0^t \int_0^y F(x) dx dy$.

Proof: Let $G(t) = \int_0^t \int_0^y F(x) dx dy$.

Then $G'(t) = \int_0^t F(x) dx$, $G''(t) = F(t)$. Also $G(0) = G'(0) = 0$.

$$\text{Now } L\{G''(t)\} = p^2 L\{G(t)\} - p G(0) - G'(0) = p^2 L\{G(t)\}$$

$$\text{or } L\{F(t)\} = f(p) = p^2 L\{G(t)\}. \quad \therefore L\{G(t)\} = \frac{f(p)}{p^2}$$

$$\text{or } L^{-1} \left[\frac{f(p)}{p^2} \right] = G(t) = \int_0^t \int_0^y F(x) dx dy.$$

This may also be written as $L^{-1} \left[\frac{f(p)}{p^2} \right] = \int_0^t \int_0^t F(t) dt^2$.

$$\text{In general } L^{-1} \left[\frac{f(p)}{p^n} \right] = \int_0^t \int_0^t \dots \int_0^t F(t) dt^n.$$

Illustrative Examples

Example 12: Find $L^{-1} \left\{ \frac{p}{(p^2 + a^2)^2} \right\}$.

(Purvanchal 2010)

Solution: $L^{-1} \left\{ \frac{p}{(p^2 + a^2)^2} \right\} = L^{-1} \left\{ -\frac{1}{2} \cdot \frac{d}{dp} \left(\frac{1}{p^2 + a^2} \right) \right\} = -\frac{1}{2} L^{-1} \left\{ \frac{d}{dp} \left(\frac{1}{p^2 + a^2} \right) \right\}$

$$= -\frac{1}{2} t \cdot (-1)^1 L^{-1} \left\{ \frac{1}{p^2 + a^2} \right\} = \frac{t}{2a} \sin at.$$

Example 13: Find $L^{-1} \left\{ \frac{p^2}{(p^2 + 4)^2} \right\}$.

Solution: Let $f(p) = \frac{p}{(p^2 + 4)^2}$.

$$\therefore L^{-1} \{f(p)\} = L^{-1} \left\{ \frac{p}{(p^2 + 4)^2} \right\} = \frac{t}{4} \sin 2t = F(t)$$

[See Ex. 12, here $a = 2$]

and $F(0) = 0$.

$$\therefore L^{-1} \{p f(p)\} = F'(t)$$

or $L^{-1} \left\{ \frac{p^2}{(p^2 + 4)^2} \right\} = \frac{d}{dt} \left(\frac{1}{4} t \sin 2t \right) = \frac{1}{4} (\sin 2t + 2t \cos 2t).$

Example 14: Find $L^{-1} \left\{ \frac{p+2}{p^2(p+3)} \right\}$.

Solution: We have $L^{-1} \left\{ \frac{p+2}{p^2(p+3)} \right\} = L^{-1} \left\{ \frac{(p+3)-1}{p^2(p+3)} \right\}$
 $= L^{-1} \left\{ \frac{1}{p^2} - \frac{1}{p^2(p+3)} \right\} = L^{-1} \left\{ \frac{1}{p^2} \right\} - L^{-1} \left\{ \frac{1}{p^2(p+3)} \right\} \quad \dots(1)$

Now $L^{-1} \{1/p^2\} = t/1! = t$.

Also $L^{-1} \left\{ \frac{1}{p+3} \right\} = e^{-3t} = F(t), \text{ say.}$

$$\therefore \text{ we have } L^{-1} \left\{ \frac{1}{p(p+3)} \right\} = \int_0^t F(x) dx. \quad (\text{by theorem I of article 2.14})$$

$$= \int_0^t e^{-3x} dx = \frac{1}{3} (1 - e^{-3t}) = F_1(t) \text{ say,}$$

$$\therefore L^{-1} \left\{ \frac{1}{p^2(p+3)} \right\} = \int_0^t F_1(x) dx = \frac{1}{3} \int_0^t (1 - e^{-3x}) dx$$

$$= \frac{1}{3} t + \frac{1}{9} (e^{-3t} - 1).$$

$\therefore$ From (1), we have

$$L^{-1} \left\{ \frac{p+2}{p^2(p+3)} \right\} = t - \frac{1}{3} t - \frac{1}{9} (e^{-3t} - 1) = \frac{2}{3} t - \frac{1}{9} e^{-3t} + \frac{1}{9}.$$

Example 15: Find (i) $L^{-1} \left\{ \log \left(1 + \frac{1}{p^2} \right) \right\}$ (Agra 2001)

$$(ii) \quad L^{-1} \left\{ \frac{1}{p} \log \left(1 + \frac{1}{p^2} \right) \right\}.$$

Solution: (i) Let

$$\begin{aligned} f(p) &= \log \left(1 + \frac{1}{p^2} \right) = -\log \left(\frac{p^2}{p^2 + 1} \right) \\ &= -2 \log p + \log (p^2 + 1). \end{aligned}$$

$$\therefore f'(p) = -\frac{2}{p} + \frac{2p}{p^2 + 1}.$$

$$\therefore L^{-1} \{ f'(p) \} = -2 + 2 \cos t$$

$$\text{or} \quad -t L^{-1} \{ f(p) \} = -2 (1 - \cos t)$$

$$\text{or} \quad L^{-1} \{ \log (1 + 1/p^2) \} = 2 (1 - \cos t)/t.$$

(ii) From part (i), we have

$$F(t) = L^{-1} \{ \log (1 + 1/p^2) \} = 2 (1 - \cos t)/t.$$

$$\begin{aligned} \therefore L^{-1} \left\{ \frac{1}{p} \log \left(1 + \frac{1}{p^2} \right) \right\} &= L^{-1} \left\{ \frac{1}{p} f(p) \right\} \\ &= \int_0^t F(x) dx \\ &= \int_0^t \frac{2}{x} (1 - \cos x) dx. \end{aligned}$$

Comprehensive Exercise 3

Evaluate the following :

$$1. \quad (i) \quad L^{-1} \left\{ \frac{1}{(p-a)^3} \right\}$$

$$(ii) \quad L^{-1} \left\{ \frac{1}{(p+a)^3} \right\}.$$

(Rohilkhand 2010)

$$2. \quad (i) \quad L^{-1} \left\{ \frac{p}{(p^2 - a^2)^2} \right\}$$

$$(ii) \quad L^{-1} \left\{ \frac{p+1}{(p^2 + 2p + 2)^2} \right\}.$$

$$3. \quad L^{-1} \left\{ \log \left(1 - \frac{1}{p^2} \right) \right\}$$

$$4. \quad L^{-1} \left\{ \frac{1}{p^3 (p+1)} \right\}.$$

$$5. \quad L^{-1} \left\{ \frac{1}{p^3 (p^2 + 1)} \right\}.$$

$$6. \quad L^{-1} \left\{ \frac{1}{p (p+1)^3} \right\}.$$

$$7. \quad (i) \quad L^{-1} \left\{ \log \frac{p+2}{p+1} \right\}$$

$$(ii) \quad L^{-1} \left\{ \frac{1}{p} \log \frac{p+2}{p+1} \right\}.$$

8. (i) $L^{-1} \left\{ \log \frac{p+3}{p+2} \right\}$ (Kanpur 2007)
 (ii) $L^{-1} \left\{ \frac{1}{p} \log \frac{p+3}{p+2} \right\}$.
9. If $L^{-1} \left\{ \frac{p}{(p^2+1)^2} \right\} = \frac{1}{2} t \sin t$, find $L^{-1} \left\{ \frac{1}{(p^2+1)^2} \right\}$. (Kanpur 2011)
10. $L^{-1} \left\{ \tan^{-1} \frac{1}{p} \right\}$.

Answers 3

- | | |
|--|--|
| 1. (i) $\frac{1}{2} t^2 e^{at}$ | (ii) $\frac{1}{2} t^2 e^{-at}$. |
| 2. (i) $\frac{1}{2} \left(\frac{t}{a} \right) \sinh at$ | (ii) $\frac{1}{2} t e^{-t} \sin t$. |
| 3. $2(1 - \cosh t)/t$. | 4. $1 - t + \frac{1}{2} t^2 - e^{-t}$. |
| 5. $\frac{1}{2} t^2 + \cos t - 1$. | 6. $1 - e^{-t} (1 + t + \frac{1}{2} t^2)$. |
| 7. (i) $(e^{-t} - e^{-2t})/t$ | (ii) $\int_0^t \frac{1}{x} (e^{-x} - e^{-2x}) dx$. |
| 8. (i) $(e^{-2t} - e^{-3t})/t$ | (ii) $\int_0^t \frac{1}{x} (e^{-2x} - e^{-3x}) dx$. |
| 9. $\frac{1}{2} (\sin t - t \cos t)$. | 10. $\frac{\sin t}{t}$. |

2.15 Convolution

Let $F(t)$ and $G(t)$ be two functions of class A, then the convolution of the two functions $F(t)$ and $G(t)$ denoted by $F * G$ is defined by the relation

$$F * G = \int_0^t F(x) G(t-x) dx.$$

This relation $F * G$ is also called the **resultant** or **falling** of F and G .

Properties of Convolution:

- (i) $F * G$ is commutative i.e., $F * G = G * F$.

Proof: $F * G = \int_0^t F(x) G(t-x) dx$

$$\begin{aligned}
 &= -\int_0^t F(t-y) G(y) dy, \text{ putting } t-x=y \text{ so that } dx=-dy \\
 &= \int_0^t G(y) F(t-y) dy = G * F.
 \end{aligned}$$

(ii) $F * G$ is associative i.e., $(F * G) * H = F * (G * H)$.

(ii) $F * G$ is distributive w.r.t. addition, i.e., $F * (G + H) = F * G + F * H$.

Proof: $F * (G + H) = \int_0^t F(x) \cdot [G(t-x) + H(t-x)] dx$

$$\begin{aligned}
 &= \int_0^t F(x) G(t-x) dx + \int_0^t F(x) H(t-x) dx \\
 &= F * G + F * H.
 \end{aligned}$$

2.16 Convolution Theorem (Convolution Property)

Let $F(t)$ and $G(t)$ be two functions of class A and let $L^{-1}\{f(p)\} = F(t)$ and $L^{-1}\{g(p)\} = G(t)$, then

$$L^{-1}\{f(p)g(p)\} = \int_0^t F(x) G(t-x) dx = F * G.$$

(Rohilkhand 2003, 07; Avadh 06, 09, 14; Lucknow 06, 10, 11; Kanpur 09)

Proof: Here we shall prove that the Laplace transform of the convolution of two functions is equal to the product of their Laplace transforms

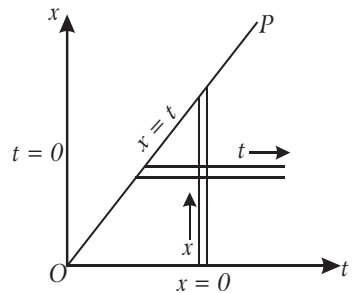
i.e.,
$$L\left[\int_0^t F(x) G(t-x) dx\right] = f(p) g(p)$$

from which the required result follows immediately.

We have,
$$\begin{aligned}
 L\left\{\int_0^t F(x) G(t-x) dx\right\} &= \int_{t=0}^{\infty} e^{-pt} \left[\int_0^t F(x) G(t-x) dx\right] dt \\
 &= \int_{t=0}^{\infty} \left[\int_{x=0}^t e^{-pt} F(x) G(t-x) dx\right] dt, \quad \dots(1)
 \end{aligned}$$

the integration being performed first with respect to x and then with respect to t .

In the double integral (1), the region of integration is the area in the tx -plane lying below the infinite line OP whose equation is $x = t$ and above the line Ot i.e., t -axis whose equation is $x = 0$. Here t is taken along the line Ot and x is taken along the perpendicular line Ox . The integration is first performed with respect to x regarding t as constant and so the strip is taken parallel to Ox .



If we change the order of integration, then the strip is taken parallel to Ot so that the limits of t are from x to ∞ and those of x are from 0 to ∞ .

$\therefore$ changing the order of integration in the double integral (1), we have

$$\begin{aligned} L \left\{ \int_0^t F(x) G(t-x) dx \right\} &= \int_{x=0}^{\infty} \left[\int_{t=x}^{\infty} e^{-pt} F(x) G(t-x) dt \right] dx \\ &= \int_{x=0}^{\infty} e^{-px} F(x) \left[\int_{t=x}^{\infty} e^{-pt} G(t-x) dt \right] dx \\ &= \int_{x=0}^{\infty} e^{-px} F(x) \left[\int_{t=x}^{\infty} e^{-p(t-x)} G(t-x) dt \right] dx \\ &= \int_{x=0}^{\infty} e^{-px} F(x) \left[\int_{z=0}^{\infty} e^{-pz} G(z) dz \right] dx, \end{aligned}$$

putting $t-x=z$, so that $dt=dz$; when $t=x$,
we have $z=0$ and when $t \rightarrow \infty$, $z \rightarrow \infty$

$$\begin{aligned} &= \left[\int_{x=0}^{\infty} e^{-px} F(x) dx \right] \left[\int_{z=0}^{\infty} e^{-pz} G(z) dz \right] \\ &= \left[\int_{t=0}^{\infty} e^{-pt} F(t) dt \right] \left[\int_{t=0}^{\infty} e^{-pt} G(t) dt \right] \\ &\quad \left[\because \int_a^b f(x) dx = \int_a^b f(t) dt = \int_a^b f(z) dz \right] \\ &= L\{F(t)\} \cdot L\{G(t)\} = f(p) g(p). \end{aligned}$$

Thus
$$L \left\{ \int_0^t F(x) G(t-x) dx \right\} = f(p) g(p).$$

Hence
$$L^{-1}\{f(p) g(p)\} = \int_0^t F(x) G(t-x) dx = F(t) * G(t).$$

2.17 Heaviside's Expansion Theorem or Formula

Let $F(p)$ and $G(p)$ be two polynomials in p where $F(p)$ has degree less than that of $G(p)$. If $G(p)$ has n distinct zeros α_r , $r=1, 2, \dots, n$, i.e. $G(p) = (p-\alpha_1)(p-\alpha_2)\dots(p-\alpha_n)$, then

$$L^{-1} \left[\frac{F(p)}{G(p)} \right] = \sum_{r=1}^n \frac{F(\alpha_r)}{G'(\alpha_r)} e^{\alpha_r t}.$$

Proof: Since $F(p)$ is a polynomial of degree less than that of $G(p)$ and $G(p)$ has n distinct zeros α_r , $r=1, 2, \dots, n$

$$\begin{aligned} \therefore \frac{F(p)}{G(p)} &= \frac{F(p)}{(p-\alpha_1)(p-\alpha_2)\dots(p-\alpha_n)} \\ &= \frac{A_1}{p-\alpha_1} + \frac{A_2}{p-\alpha_2} + \dots + \frac{A_r}{p-\alpha_r} + \dots + \frac{A_n}{p-\alpha_n}. \end{aligned}$$

Multiplying both sides by $(p-\alpha_r)$ and taking limits as $p \rightarrow \alpha_r$, we have

$$\begin{aligned}
 A_r &= \lim_{p \rightarrow \alpha_r} \frac{F(p) \cdot (p - \alpha_r)}{G(p)} \\
 &= F(\alpha_r) \cdot \lim_{p \rightarrow \alpha_r} \frac{(p - \alpha_r)}{G(p)} \quad \left[\text{Form } \frac{0}{0} \right] \\
 &= F(\alpha_r) \cdot \lim_{p \rightarrow \alpha_r} \frac{1}{G'(p)} \quad [\text{by L' Hospital's rule}] \\
 &= F(\alpha_r) \cdot \frac{1}{G'(\alpha_r)}.
 \end{aligned}$$

$$\begin{aligned}
 \therefore \frac{F(p)}{G(p)} &= \frac{F(\alpha_1)}{G'(\alpha_1)} \cdot \frac{1}{(p - \alpha_1)} + \frac{F(\alpha_2)}{G'(\alpha_2)} \cdot \frac{1}{(p - \alpha_2)} + \dots \\
 &\quad + \frac{F(\alpha_r)}{G'(\alpha_r)} \cdot \frac{1}{(p - \alpha_r)} + \dots + \frac{F(\alpha_n)}{G'(\alpha_n)} \cdot \frac{1}{p - \alpha_n} + \dots
 \end{aligned}$$

$$\begin{aligned}
 \text{Hence } L^{-1} \left[\frac{F(p)}{G(p)} \right] &= \frac{F(\alpha_1)}{G'(\alpha_1)} \cdot L^{-1} \left[\frac{1}{p - \alpha_1} \right] + \frac{F(\alpha_2)}{G'(\alpha_2)} \cdot L^{-1} \left[\frac{1}{(p - \alpha_2)} \right] + \dots \\
 &\quad + \frac{F(\alpha_r)}{G'(\alpha_r)} \cdot L^{-1} \left[\frac{1}{p - \alpha_r} \right] + \dots + \frac{F(\alpha_n)}{G'(\alpha_n)} \cdot L^{-1} \left[\frac{1}{p - \alpha_n} \right] \\
 &= \frac{F(\alpha_1)}{G'(\alpha_1)} \cdot e^{\alpha_1 t} + \dots + \frac{F(\alpha_r)}{G'(\alpha_r)} \cdot e^{\alpha_r t} + \dots + \frac{F(\alpha_n)}{G'(\alpha_n)} \cdot e^{\alpha_n t} \\
 &= \sum_{r=1}^n \frac{F(\alpha_r)}{G'(\alpha_r)} e^{\alpha_r t}.
 \end{aligned}$$

2.18 The Beta Function

If $m > 0, n > 0$, the Beta function is defined as

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx.$$

Illustrative Examples

Example 16: Use the convolution theorem to find $L^{-1} \left\{ \frac{p^2}{(p^2 + a^2)^2} \right\}$.

Solution: We have $L^{-1} \left\{ \frac{p}{p^2 + a^2} \right\} = \cos at$.

$\therefore$ By the convolution theorem, we have

$$L^{-1} \left\{ \frac{p^2}{(p^2 + a^2)^2} \right\} = L^{-1} \left\{ \frac{p}{p^2 + a^2} \cdot \frac{p}{p^2 + a^2} \right\}$$

$$\begin{aligned}
&= \int_0^t \cos ax \cos a(t-x) dx \\
&= \int_0^t \cos ax (\cos at \cos ax + \sin at \sin ax) dx \\
&= \cos at \int_0^t \cos^2 ax dx + \sin at \int_0^t \cos ax \sin ax dx \\
&= \frac{1}{2} \cos at \int_0^t (1 + \cos 2ax) dx + \frac{1}{2} \sin at \int_0^t \sin 2ax dx \\
&= \frac{1}{2} \cos at \left[x + \frac{1}{2a} \sin 2ax \right]_0^t + \frac{1}{2} \sin at \cdot \left[-\frac{1}{2a} \cos 2ax \right]_0^t \\
&= \frac{1}{2} \cos at \left[t + \frac{1}{2a} \sin 2at \right] + \frac{1}{4a} \sin at (1 - \cos 2at) \\
&= \frac{1}{2} t \cos at + \frac{1}{4a} \sin at + \frac{1}{4a} (\sin 2at \cos at - \sin at \cos 2at) \\
&= \frac{1}{2} t \cos at + \frac{1}{4a} [\sin at + \sin (2at - at)] = \frac{1}{2a} [at \cos at + \sin at].
\end{aligned}$$

Example 17: Find $L^{-1} \left\{ \frac{1}{\sqrt{p \cdot (p-a)}} \right\}$, by the convolution integral and deduce the value of

$$L^{-1} \left\{ \frac{1}{p \sqrt{(p+a)}} \right\}.$$

Solution: We have $L^{-1} \left\{ \frac{1}{\sqrt{p}} \right\} = L^{-1} \left\{ \frac{1}{p^{1/2}} \right\} = \frac{t^{1/2-1}}{\Gamma(\frac{1}{2})} = \frac{1}{\sqrt{\pi} \sqrt{t}} = F_1(t)$, say

and $L^{-1} \left\{ \frac{1}{p-a} \right\} = e^{at} = F_2(t)$, say.

$\therefore$ By the convolution theorem, we have

$$\begin{aligned}
L^{-1} \left\{ \frac{1}{\sqrt{p \cdot (p-a)}} \right\} &= F_1(t) * F_2(t) = \int_0^t \frac{1}{\sqrt{\pi} \sqrt{x}} \cdot e^{a(t-x)} dx \\
&= \frac{e^{at}}{\sqrt{\pi}} \int_0^{\sqrt{at}} \frac{\sqrt{a}}{u} e^{-u^2} \cdot \frac{2u}{a} du, \text{ putting } ax = u^2, \text{ so that } dx = \frac{2u du}{a} \\
&= \frac{e^{at}}{\sqrt{a}} \cdot \frac{2}{\sqrt{\pi}} \int_0^{\sqrt{at}} e^{-u^2} du = \frac{e^{at}}{\sqrt{a}} \operatorname{erf}(\sqrt{at}).
\end{aligned}$$

Deduction. $L^{-1} \left\{ \frac{1}{p \sqrt{(p+a)}} \right\} = L^{-1} \left\{ \frac{1}{(p+a-a) \sqrt{(p+a)}} \right\}$

$$= e^{-at} L^{-1} \left\{ \frac{1}{(p-a) \sqrt{p}} \right\} = e^{-at} \cdot \frac{e^{at}}{\sqrt{a}} \operatorname{erf}(\sqrt{at}) = \frac{1}{\sqrt{a}} \operatorname{erf}(\sqrt{at}).$$

Example 18: Apply convolution theorem to prove that

$$\mathbf{B}(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}, m > 0, n > 0.$$

(Gorakhpur 2008; Agra 02)

Hence deduce that

$$\int_0^{\pi/2} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta = \frac{1}{2} \mathbf{B}(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{2 \Gamma(m+n)}.$$

Solution: Consider the function

$$F(t) = \int_0^t x^{m-1} \cdot (t-x)^{n-1} dx.$$

We have, $F(t) = \int_0^t F_1(x) \cdot F_2(t-x) dx$, where $F_1(t) = t^{m-1}$ and $F_2(t) = t^{n-1}$

$$= F_1 * F_2.$$

$$\begin{aligned} \therefore L\{F(t)\} &= L\{F_1 * F_2\} \\ &= L\{F_1(t)\} \cdot L\{F_2(t)\}, \text{ by convolution theorem} \\ &= L\{t^{m-1}\} \cdot L\{t^{n-1}\} = \frac{\Gamma(m)}{p^m} \cdot \frac{\Gamma(n)}{p^n} = \frac{\Gamma(m)\Gamma(n)}{p^{m+n}}. \end{aligned}$$

$$\begin{aligned} \therefore F(t) &= \int_0^t x^{m-1} \cdot (t-x)^{n-1} dx = L^{-1} \left\{ \frac{\Gamma(m)\Gamma(n)}{p^{m+n}} \right\} \\ &= \Gamma(m) \cdot \Gamma(n) \cdot L^{-1} \left\{ \frac{1}{p^{m+n}} \right\} = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)} t^{m+n-1}. \end{aligned}$$

Taking $t=1$, we have

$$\begin{aligned} \mathbf{B}(m, n) &= \int_0^1 x^{m-1} (1-x)^{n-1} dx \\ &= \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}. \end{aligned} \quad \dots(1)$$

Deduction: Taking $x = \sin^2 \theta$, so that $dx = 2 \sin \theta \cos \theta d\theta$. From (1), we have

$$2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}.$$

$$\text{Hence} \quad \int_0^{\pi/2} \sin^{2m-1} \theta \cdot \cos^{2n-1} \theta d\theta = \frac{\Gamma(m)\Gamma(n)}{2\Gamma(m+n)} = \frac{1}{2} \mathbf{B}(m, n)$$

Example 19: Using Heaviside's expansion formula find $L^{-1} \left\{ \frac{3p+1}{(p-1)(p^2+1)} \right\}$

(Gorakhpur 2008)

Solution: Here $F(p) = 3p+1$

and $G(p) = (p-1)(p^2+1) = (p-1)(p-i)(p+i).$

$\therefore G(p)$ has 3 distinct zeros $\alpha_1 = 1, \alpha_2 = i$ and $\alpha_3 = -i$.

Also $G'(p) = 3p^2 - 2p + 1$.

$\therefore$ By the Heaviside's expansion formula, we have

$$L^{-1} \left\{ \frac{3p+1}{(p-1)(p^2+1)} \right\} = \frac{F(1)}{G'(1)} e^t + \frac{F(i)}{G'(i)} e^{it} + \frac{F(-i)}{G'(-i)} e^{-it}$$

$$\begin{aligned}
&= \frac{4e^t}{2} + \frac{(3i+1)}{-(2+2i)} e^{it} + \frac{(-3i+1)}{(-2+2i)} e^{-it} \\
&= 2e^t - \frac{(3i+1)(1-i)}{2(1+i)(1-i)} e^{it} + \frac{(3i-1)(1+i)}{2(1-i)(1+i)} e^{-it} \\
&= 2e^t - \frac{1}{2}(i+2)e^{it} + \frac{1}{2}(i-2)e^{-it} \\
&= 2e^t - \frac{1}{2}i(e^{it} - e^{-it}) - (e^{it} + e^{-it}) \\
&= 2e^t - \frac{1}{2}i \cdot 2i \sin t - 2 \cos t = 2e^t + \sin t - 2 \cos t.
\end{aligned}$$

Example 20: Using Heaviside's expansion formula find

$$L^{-1} \left\{ \frac{2p^2 + 5p - 4}{p^3 + p^2 - 2p} \right\}. \quad (\text{Avadh 2010})$$

Solution: Here $F(p) = 2p^2 + 5p - 4$

and $G(p) = p^3 + p^2 - 2p = p(p-1)(p+2)$

$$G'(p) = 3p^2 + 2p - 2.$$

$G(p)$ has 3 distinct zeros $\alpha_1 = 0, \alpha_2 = 1$ and $\alpha_3 = -2$.

$\therefore$ By the Heaviside's expansion formula, we have

$$\begin{aligned}
L^{-1} \left\{ \frac{2p^2 + 5p - 4}{p^3 + p^2 - 2p} \right\} &= \frac{F(0)}{G'(0)} e^{0t} + \frac{F(1)}{G'(1)} e^t + \frac{F(-2)}{G'(-2)} e^{-2t} \\
&= 2 + e^t - e^{-2t}.
\end{aligned}$$

Comprehensive Exercise 4

Use the convolution theorem to find:

$$1. \quad (i) \quad L^{-1} \left\{ \frac{1}{(p-1)(p+2)} \right\} \quad (ii) \quad L^{-1} \left\{ \frac{1}{(p+1)(p-2)} \right\}$$

$$(iii) \quad L^{-1} \left\{ \frac{1}{(p-1)(p+3)} \right\}.$$

(Kanpur 2012)

$$2. \quad (i) \quad L^{-1} \left\{ \frac{p}{(p^2 + a^2)^2} \right\} \quad (ii) \quad L^{-1} \left\{ \frac{1}{p(p^2 + 4)^2} \right\}.$$

(Lucknow 2009, 11)

$$\left[\text{Hint. (ii)} \quad L^{-1} \left\{ \frac{1}{p(p^2 + 4)^2} \right\} = L^{-1} \left\{ \frac{1}{p^2} \cdot \frac{p}{(p^2 + 4)^2} \right\} \right].$$

$$\text{Take } f_1(p) = \frac{1}{p^2}, \quad f_2(p) = \frac{p}{(p^2 + 4)^2}.$$

3. (i) $L^{-1} \left\{ \frac{1}{(p-2)(p^2+1)} \right\}$

(ii) $L^{-1} \left\{ \frac{1}{(p^2+4)(p+2)} \right\}.$

4. (i) $L^{-1} \left\{ \frac{1}{p^2(p+1)^2} \right\}$

(Gorakhpur 2006)

(ii) $L^{-1} \left\{ \frac{1}{(p+2)^2(p-2)} \right\}.$

(Kanpur 2011)

5. Find $L^{-1} \left\{ \frac{p^2}{(p^2+4)^2} \right\}.$

(Kanpur 2010)

[Hint. Proceed as in Ex. 16. Here $a = 2$].

6. Show that $L^{-1} \left\{ \frac{1}{p\sqrt{p+4}} \right\} = \frac{1}{2} \operatorname{erf} (2\sqrt{t}).$

Using Heaviside's expansion formula, find

7. $L^{-1} \left\{ \frac{2p^2 - 6p + 5}{p^3 - 6p^2 + 11p - 6} \right\}.$

8. $L^{-1} \left\{ \frac{p^2 - 6}{p^3 + 4p^2 + 3p} \right\}.$

9. (i) $L^{-1} \left\{ \frac{19p + 37}{(p+1)(p-2)(p+3)} \right\}$

(ii) $L^{-1} \left\{ \frac{p+5}{(p+1)(p^2+1)} \right\}.$

Answers 4

1. (i) $\frac{1}{3}(e^t - e^{-2t})$

(ii) $\frac{1}{3}(e^{2t} - e^{-t})$

(iii) $\frac{1}{4}(e^t - e^{-3t})$

2. (i) $(1/2a)t \sin at$

(ii) $\frac{1}{16}(1 - t \sin 2t - \cos 2t)$

3. (i) $\frac{1}{5}(e^{2t} - 2 \sin t - \cos t)$

(ii) $\frac{1}{8}(e^{-2t} + \sin 2t - \cos 2t)$

4. (i) $(t+2)e^{-t} + t - 2$

(ii) $\frac{1}{16}[e^{2t} - (4t+1)e^{2t}]$

5. $\frac{1}{4}(2t \cos 2t + \sin 2t).$

7. $\frac{1}{2}e^t - e^{2t} + \frac{5}{2}e^{3t}$

8. $-2 + \frac{5}{2}e^{-t} + \frac{1}{2}e^{-3t}$

9. (i) $-3e^{-t} + 5e^{2t} - 2e^{-3t}$

(ii) $2e^{-t} - 2 \cos t + 3 \sin t$

Illustrative Examples

Example 21: Find $L^{-1} \left\{ \frac{e^{-\sqrt[4]{p}}}{p} \right\}$ and hence deduce that $L^{-1} \left\{ \frac{e^{-x\sqrt[4]{p}}}{p} \right\} = \operatorname{erfc} \left(\frac{x}{2\sqrt[4]{t}} \right)$.

Solution: Let $f(p) = e^{-\sqrt[4]{p}}$.

$$\begin{aligned} \therefore F(t) &= L^{-1} \{ e^{-\sqrt[4]{p}} \} \\ &= L^{-1} \left\{ 1 - \sqrt[4]{p} + \frac{p}{2!} - \frac{p^{3/2}}{3!} + \frac{p^2}{4!} - \frac{p^{5/2}}{5!} + \dots \right\} \\ &= L^{-1} \{1\} - L^{-1} \{p^{1/2}\} + \frac{1}{2!} L^{-1} \{p\} - \frac{1}{3!} L^{-1} \{p^{3/2}\} \\ &\quad + \frac{1}{4!} L^{-1} \{p^2\} - \frac{1}{5!} L^{-1} \{p^{5/2}\} + \dots \quad \dots(1) \end{aligned}$$

$$\begin{aligned} \text{Now } L^{-1} \{p^{n+(1/2)}\} &= L^{-1} \left\{ \frac{1}{p^{-n-(1/2)}} \right\} \\ &= \frac{t^{-n-(3/2)}}{\Gamma(-n-\frac{1}{2})}, \quad \text{for } n=0,1,2,\dots \\ &= \frac{(-1)^{n+1}}{\sqrt{\pi}} \left(\frac{1}{2} \right) \left(\frac{3}{2} \right) \left(\frac{5}{2} \right) \dots \left(\frac{2n+1}{2} \right) t^{-n-(3/2)}. \\ &\quad \left[\because \Gamma(-n-\frac{1}{2}) = (-1)^{n+1} \left(\frac{2}{1} \right) \left(\frac{2}{3} \right) \left(\frac{2}{5} \right) \dots \left(\frac{2}{2n+1} \right) \sqrt{\pi} \right] \end{aligned}$$

Also $L^{-1} \{p^n\} = 0$, if n is a +ve integer.

$\therefore$ from (1), we have

$$\begin{aligned} F(t) &= -\frac{(-1)t^{-3/2}}{\sqrt{\pi}} \cdot \frac{1}{2} - \frac{1}{3!} \cdot \frac{(-1)^2}{\sqrt{\pi}} \cdot \left(\frac{1}{2} \right) \left(\frac{3}{2} \right) t^{-5/2} \\ &\quad - \frac{1}{5!} \frac{(-1)^3}{\sqrt{\pi}} \left(\frac{1}{2} \right) \left(\frac{3}{2} \right) \left(\frac{5}{2} \right) t^{-7/2} + \dots \\ &= \frac{1}{2\sqrt{\pi} t^{3/2}} \left[1 - \frac{1}{4t} + \frac{\left(\frac{1}{4t} \right)^2}{2!} - \frac{\left(\frac{1}{4t} \right)^3}{3!} + \dots \right] \\ &= \frac{1}{2\sqrt{\pi} t^{3/2}} e^{-1/4t}. \end{aligned}$$

Since $L^{-1} \left\{ \frac{f(p)}{p} \right\} = \int_0^t F(x) dx$, where $F(t) = L^{-1} \{f(p)\}$, we have

$$\begin{aligned}
 L^{-1} \left\{ \frac{e^{-\sqrt{p}}}{p} \right\} &= \int_0^t \frac{1}{2\sqrt{(\pi)} x^{3/2}} e^{-1/(4x)} dx \\
 &= -\frac{2}{\sqrt{\pi}} \int_{\infty}^{1/(2\sqrt{t})} e^{-y^2} dy, \text{ putting } x = \frac{1}{4y^2} \text{ so that } dx = -\frac{dy}{2y^3} \\
 &= \frac{2}{\pi} \int_{1/(2\sqrt{t})}^{\infty} e^{-y^2} dy = \operatorname{erfc} \left(\frac{1}{2\sqrt{t}} \right).
 \end{aligned}$$

$$\begin{aligned}
 \left[\because \text{Complementary Error Function } \operatorname{erfc}(t) = 1 - \operatorname{erf}(t) = 1 - \frac{2}{\sqrt{\pi}} \int_0^t e^{-x^2} dx \right. \\
 \left. = \frac{2}{\sqrt{\pi}} \left\{ \int_0^{\infty} e^{-x^2} dx + \int_t^0 e^{-x^2} dx \right\} = \frac{2}{\sqrt{\pi}} \int_t^{\infty} e^{-x^2} dx \right]
 \end{aligned}$$

Deduction: We have $L^{-1} \left\{ \frac{e^{-\sqrt{p}}}{p} \right\} = \operatorname{erfc} \left(\frac{1}{2\sqrt{t}} \right)$.

$\therefore L^{-1} \left\{ \frac{e^{-\sqrt{(x^2 p)}}}{x^2 p} \right\} = \frac{1}{x^2} \operatorname{erfc} \left(\frac{1}{2\sqrt{(t/x^2)}} \right)$, by change of scale property

or $L^{-1} \left\{ \frac{e^{-x\sqrt{p}}}{p} \right\} = \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right)$.

Example 22: (i) Prove that $L^{-1} \left\{ \frac{1}{p^3 + 1} \right\} = \frac{t^2}{2!} - \frac{t^5}{5!} + \frac{t^8}{8!} - \frac{t^{11}}{11!} + \dots$

(ii) Applying Heaviside's Expansion formula, prove that

$$L^{-1} \left\{ \frac{1}{p^3 + 1} \right\} = \frac{1}{3} [e^{-t} - e^{t/2} \{ \cos(\frac{1}{2}\sqrt{3}t) - \sqrt{3} \sin(\frac{1}{2}\sqrt{3}t) \}].$$

Solution: (i) We have, $\frac{1}{p^3 + 1} = \frac{1}{p^3} (1 + 1/p^3)^{-1}$

$$= \frac{1}{p^3} \left[1 - \frac{1}{p^3} + \frac{1}{p^6} - \frac{1}{p^9} + \frac{1}{p^{12}} - \dots \right]$$

$$= \frac{1}{p^3} - \frac{1}{p^6} + \frac{1}{p^9} - \frac{1}{p^{12}} + \dots$$

$\therefore L^{-1} \left\{ \frac{1}{p^3 + 1} \right\} = \frac{t^2}{2!} - \frac{t^5}{5!} + \frac{t^8}{8!} - \frac{t^{11}}{11!} + \dots$

(ii) Here $F(p) = 1$ and $G(p) = p^3 + 1 = (p + 1)(p^2 - p + 1)$, $G'(p) = 3p^2$.

$$G(p) \text{ has 3 distinct zeros } -1, \frac{1}{2}(1 + \sqrt{3}i) \text{ and } \frac{1}{2}(1 - \sqrt{3}i).$$

$\therefore$ By the Heaviside's expansion formula, we have

$$\begin{aligned}
 L^{-1} \left\{ \frac{1}{p^3 + 1} \right\} &= \frac{F(-1)}{G'(-1)} e^{-t} + \frac{F\left\{\frac{1}{2}(1 + \sqrt{3}i)\right\}}{G'\left\{\frac{1}{2}(1 + \sqrt{3}i)\right\}} e^{\frac{1}{2}(1 + \sqrt{3}i)t} \\
 &\quad + \frac{F\left\{\frac{1}{2}(1 - \sqrt{3}i)\right\}}{G'\left\{\frac{1}{2}(1 - \sqrt{3}i)\right\}} e^{\frac{1}{2}(1 - \sqrt{3}i)t} \\
 &= \frac{1}{3} e^{-t} + \frac{1}{\frac{3}{4}(1 + \sqrt{3}i)^2} e^{\frac{1}{2}(1 + \sqrt{3}i)t} + \frac{1}{\frac{3}{4}(1 - \sqrt{3}i)^2} e^{\frac{1}{2}(1 - \sqrt{3}i)t} \\
 &= \frac{1}{3} e^{-t} + \frac{2(\sqrt{3}i + 1)}{3(\sqrt{3}i - 1)(\sqrt{3}i + 1)} e^{\frac{1}{2}(1 + \sqrt{3}i)t} \\
 &\quad + \frac{2(\sqrt{3}i - 1)}{3(\sqrt{3}i + 1)(\sqrt{3}i - 1)} e^{\frac{1}{2}(1 - \sqrt{3}i)t} \\
 &= \frac{1}{3} e^{-t} - \frac{1}{6} e^{t/2} [(\sqrt{3}i + 1) e^{\sqrt{3}it/2} - (\sqrt{3}i - 1) e^{-\sqrt{3}it/2}] \\
 &= \frac{1}{3} e^{-t} - \frac{1}{6} e^{t/2} [\sqrt{3}i (e^{\sqrt{3}it/2} - e^{-\sqrt{3}it/2}) + (e^{\sqrt{3}it/2} + e^{-\sqrt{3}it/2})] \\
 &= \frac{1}{3} [e^{-t} - e^{t/2} \{-\sqrt{3} \sin(\frac{1}{2}\sqrt{3}t) + \cos(\frac{1}{2}\sqrt{3}t)\}].
 \end{aligned}$$

Example 23: Show that $\int_0^\infty \cos x^2 dx = \frac{1}{2} \sqrt{\pi/2}$ (Avadh 2008)

Solution: Let $F(t) = \int_0^\infty \cos tx^2 dx$.

$$\begin{aligned}
 \therefore L\{F(t)\} &= \int_0^\infty e^{-pt} F(t) dt = \int_0^\infty e^{-pt} \left[\int_0^\infty \cos tx^2 dx \right] dt \\
 &= \int_0^\infty \left[\int_0^\infty e^{-pt} \cos tx^2 dt \right] dx = \int_0^\infty L\{\cos tx^2\} dx = \int_0^\infty \frac{p}{p^2 + x^4} dx \\
 &= \frac{1}{2\sqrt{p}} \int_0^{\pi/2} \frac{d\theta}{\sqrt{\tan \theta}}, \text{ putting } x = \sqrt{p \tan \theta} \text{ so that} \\
 &\quad dx = \frac{p \sec^2 \theta d\theta}{2\sqrt{p \tan \theta}} \\
 &= \frac{1}{2\sqrt{p}} \int_0^{\pi/2} \sin^{-1/2} \theta \cos^{1/2} \theta d\theta \\
 &= \frac{1}{2\sqrt{p}} \frac{\Gamma(\frac{1}{4}) \Gamma(\frac{3}{4})}{2 \Gamma(1)} = \frac{1}{2\sqrt{p}} \frac{\Gamma(\frac{1}{4}) \Gamma(1 - \frac{1}{4})}{2} = \frac{1}{4\sqrt{p}} \cdot \frac{\pi}{\sin \frac{1}{4}\pi} = \frac{\pi}{2\sqrt{(2p)}}.
 \end{aligned}$$

$$\left[\because \Gamma(p) \Gamma(1-p) = \frac{\pi}{\sin p\pi}, 0 < p < 1 \right]$$

$$\therefore F(t) = \frac{\pi}{2\sqrt{2}} L^{-1} \left\{ \frac{1}{p^{1/2}} \right\} = \frac{\pi}{2\sqrt{2}} \frac{t^{(1/2)-1}}{\Gamma(\frac{1}{2})} = \frac{1}{2} \sqrt{\left(\frac{\pi}{2t}\right)}$$

$$\text{or } \int_0^\infty \cos tx^2 dx = \frac{1}{2} \sqrt{\left(\frac{\pi}{2t}\right)}.$$

Now taking $t = 1$, we have

$$\int_0^{\infty} \cos x^2 dx = \frac{1}{2} \sqrt{\pi/2}$$

Example 24: Prove that $\int_0^{\infty} e^{-x^2} dx = \frac{1}{2} \sqrt{\pi}$.

(Meerut 2013B)

Solution: Let $F(t) = \int_0^{\infty} e^{-tx^2} dx$.

$\therefore$ Proceeding as in Ex. 23,

$$L\{F(t)\} = \int_0^{\infty} L\{e^{-tx^2}\} dx = \int_0^{\infty} \frac{dx}{p+x^2} = \left(\frac{1}{\sqrt{p}} \tan^{-1} \frac{x}{\sqrt{p}} \right)_0^{\infty} = \frac{\pi}{2\sqrt{p}}$$

or
$$F(t) = \frac{\pi}{2} L^{-1} \left(\frac{1}{\sqrt{p}} \right) = \frac{\pi}{2} \cdot \frac{1}{\sqrt{(\pi t)}} = \frac{1}{2} \sqrt{\left(\frac{\pi}{t} \right)}$$

or
$$\int_0^{\infty} e^{-t x^2} dx = \frac{1}{2} \sqrt{\left(\frac{\pi}{t} \right)}.$$

Taking $t = 1$, we have $\int_0^{\infty} e^{-x^2} dx = \frac{1}{2} \sqrt{\pi}$.

Example 25: Prove that $L^{-1} \left\{ \frac{8}{(p^2+1)^3} \right\} = (3-t^2) \sin t - 3t \cos t$.

Solution: We have $L^{-1} \left\{ \frac{1}{p^2+1} \right\} = \sin t$.

$\therefore$ by the convolution theorem, we have

$$\begin{aligned} L^{-1} \left[\frac{1}{(p^2+1)} \cdot \frac{1}{(p^2+1)} \right] &= \int_0^t \sin x \sin(t-x) dx \\ &= \int_0^t \sin x (\sin t \cos x - \cos t \sin x) dx \\ &= \sin t \int_0^t \sin x \cos x dx - \cos t \int_0^t \sin^2 x dx \\ &= \frac{1}{2} \sin t \int_0^t \sin 2x dx - \frac{1}{2} \cos t \cdot \int_0^t (1 - \cos 2x) dx \\ &= \frac{1}{2} \sin t \cdot \frac{1}{2} (1 - \cos 2t) - \frac{1}{2} \cos t \cdot \left(t - \frac{1}{2} \sin 2t \right) \\ &= \frac{1}{2} \sin t \cdot \sin^2 t - \frac{1}{2} t \cos t + \frac{1}{2} \cos t \sin t \cos t \end{aligned}$$

or
$$L^{-1} \left[\frac{1}{(p^2+1)^2} \right] = \frac{1}{2} \sin t - \frac{t}{2} \cos t.$$

$\therefore$
$$L^{-1} \left[\frac{1}{(p^2+1)^3} \right] = 8 L^{-1} \left[\frac{1}{(p^2+1)^2} \cdot \frac{1}{(p^2+1)} \right]$$

$$= 8 \int_0^t \left(\frac{1}{2} \sin x - \frac{x}{2} \cos x \right) \sin(t-x) dx, \text{ by the convolution theorem}$$

$$= 4 \int_0^t (\sin x - x \cos x) (\sin t \cos x - \cos t \sin x) dx$$

$$= 4 \sin t \int_0^t (\sin x \cos x - x \cos^2 x) dx - 4 \cos t \int_0^t (\sin^2 x - x \sin x \cos x) dx$$

$$\begin{aligned}
&= 2 \sin t \int_0^t \{ \sin 2x - x (1 + \cos 2x) \} dx - 2 \cos t \int_0^t \{ (1 - \cos 2x) - x \sin 2x \} dx \\
&= 2 \sin t \left[-\frac{t^2}{2} + \frac{1 - \cos 2t}{2} - \frac{t}{2} \sin 2t + \frac{1 - \cos 2t}{4} \right] \\
&\quad - 2 \cos t \left[t - \frac{1}{2} \sin 2t + \frac{t \cos 2t}{2} - \frac{\sin 2t}{4} \right] \\
&= -t^2 \sin t + \frac{3}{2} \sin t - \frac{3}{2} \sin t \cos 2t - t \sin t \sin 2t \\
&\quad - 2t \cos t + \frac{3}{2} \sin 2t \cos t - t \cos t \cos 2t \\
&= -t^2 \sin t + \frac{3}{2} \sin t + \frac{3}{2} (-\sin t \cos 2t + \sin 2t \cos t) \\
&\quad - t (\cos 2t \cos t + \sin t \sin 2t) - 2t \cos t \\
&= (3 - t^2) \sin t - 3t \cos t.
\end{aligned}$$

Comprehensive Exercise 5

- Find $L^{-1} \left\{ \frac{1}{(p-1)^5 (p+2)} \right\}$.
(Kanpur 2009, 11)
- Prove that $L^{-1} \left\{ \frac{1}{p \sqrt{(p^2 + a^2)}} \right\} = \int_0^t J_0(ax) dx$.
- Apply Heaviside's expansion formula, to prove that

$$L^{-1} \left\{ \frac{1}{p^3 - 1} \right\} = \frac{1}{3} [e^t - e^{-1/2} \{ \cos(\frac{1}{2} \sqrt{3}t) + \sqrt{3} \sin(\frac{1}{2} \sqrt{3}t) \}]$$
- Show that $\int_0^\infty \sin x^2 dx = \frac{1}{2} \sqrt{(\pi/2)}$.
- Apply the convolution theorem to show that

$$\int_0^t \sin u \cos(t-u) du = \frac{t}{2} \sin t.$$

(Gorakhpur 2007, 11)
- Use Laplace transform to prove that $\int_0^\infty \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}$.
- Prove that $L^{-1} \left\{ \tan^{-1} \frac{2}{p^2} \right\} = \frac{2}{t} \sin t \sinh t$.
(Agra 2002)
- Show that $1 * 1 * 1 * \dots * 1$ (n times) $= t^{n-1} / (n-1)!$, where $n = 1, 2, 3, \dots$
(Rohilkhand 2002)

9. Find $L^{-1} \left[\frac{1}{(p^2 + a^2)^{3/2}} \right]$.
10. Prove that $\int_0^t \int_0^t \dots \int_0^t F(t) dt^n = \int_0^t \frac{(t-u)^{n-1}}{(n-1)!} F(u) du$.

Answers 5

1. $\frac{1}{72} e^t \left(t^4 - \frac{4}{3} t^3 + \frac{4}{3} t^2 - \frac{8}{9} t + \frac{8}{27} \right) - \frac{1}{243} e^{-2t}$
9. $\frac{t}{a} J_1(at)$

2.19 Table of Inverse Laplace Transform Theorems

Inverse Laplace Transform Theorems			
No.	Operation	$f(p)$	$L^{-1} \{f(p)\} = F(t)$
1.	Linearity property	$a_1 f_1(p) + a_2 f_2(p)$	$a_1 L^{-1} \{f_1(p)\} + a_2 L^{-1} \{f_2(p)\}$
2.	First translation or shifting theorem	$f(p-a)$	$e^{at} L^{-1} \{f(p)\}$
3.	Second translation or shifting theorem	$e^{-ap} f(p)$	$G(t) = \begin{cases} F(t-a), & t > a \\ 0, & t < a \end{cases}$
4.	Change of scale property	$f(ap)$	$\frac{1}{a} F\left(\frac{t}{a}\right)$
5.	Differentiation theorem	$f^n(p)$	$(-1)^n t^n F(t)$
6.	Integral theorem	$\int_p^\infty f(x) dx$	$\frac{1}{t} F(t)$
7.	Multiplication theorems	$p f(p)$	$F'(t)$
		$p^n f(p)$	$F^n(t)$

8.	Division theorems	$\frac{f(p)}{p}$ $\frac{f(p)}{p^n}$	$\int_0^t F(x) dx$ $\int_0^t \int_0^t \dots \int_0^t F(t) (dt)^n$
9.	Convolution theorem	$f(p) \cdot g(p)$	$F * G = \int_0^t F(x) \cdot G(t-x) dx$
10.	Heaviside's expansion theorem	$\frac{F(p)}{G(p)},$ degree $F(p) <$ degree $G(p)$	$\sum_{r=1}^n \frac{F(\alpha_r)}{G'(\alpha_r)} e^{\alpha_r t}$ where $\alpha_r, r = 1, 2, \dots, n$, are roots of $G(p) = 0$ and are all distinct.

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

- The value of $L^{-1} \left(\frac{1}{p^4} \right)$ is

(a) $\frac{t^3}{3!}$ (b) $\frac{t^4}{4!}$

(c) $\frac{t^5}{5!}$ (d) None of these.
- The value of $L^{-1} \left\{ \frac{1}{\sqrt{p}} \right\}$ is

(a) $\sqrt{\left(\frac{\pi}{t} \right)}$ (b) $\sqrt{\left(\frac{t}{\pi} \right)}$

(c) $\frac{1}{\sqrt{(\pi t)}}$ (d) $\sqrt{(\pi t)}$.
- The value of $L^{-1} \left\{ \frac{1}{p^{5/2}} \right\}$ is

(a) $\frac{4}{3} t^{3/2}$ (b) $\frac{4}{3} t^{5/2} [$

(c) $\frac{4}{3} t \sqrt{\left(\frac{t}{\pi} \right)}$ (d) $\frac{4}{3} t \sqrt{\left(\frac{\pi}{t} \right)}.$

4. If $L^{-1} \{ f(p) \} = F(t)$ then $L^{-1} \{ f(p-a) \}$ is
 (a) $e^{at} F(t)$ (b) $\frac{1}{a} F\left(\frac{t}{a}\right)$
 (c) $(-1)^n t^n F(t)$ (d) $\int_0^t F(u) du$ (Rohilkhand 2002)
5. If $L^{-1} \{ f(p) \} = F(t)$ then $L^{-1} \{ f(ap) \}$ is
 (a) $aF(at)$ (b) $F(at)$
 (c) $F\left(\frac{t}{a}\right)$ (d) $\frac{1}{a} F\left(\frac{t}{a}\right)$. (Rohilkhand 2003, 10)
6. If $L^{-1} \{ f(p) \} = F(t)$ then $L^{-1} \left\{ \frac{f(p)}{p} \right\}$ is
 (a) $(-1)^n t^n F(t)$ (b) $F'(t)$
 (c) $\int_0^t F(u) du$ (d) $\frac{F(t)}{t}$. (Rohilkhand 2002)
7. Inverse Laplace transform of $\frac{1}{p^2 + a^2}$ is
 (a) $\sin at$ (b) $\cos at$
 (c) $\frac{1}{a} \sin at$ (d) $\sinh at$. (Rohilkhand 2003)
8. The value of $L^{-1} \left\{ \frac{p}{2p^2 - 8} \right\}$ is
 (a) $\cosh 2t$ (b) $\sinh 2t$
 (c) $\frac{1}{2} \sinh 2t$ (d) $\frac{1}{2} \cosh 2t$.
9. If $L^{-1} \{ f(p) \} = F(t)$ and $F(0) = 0$, then $L^{-1} \{ p f(p) \}$ is
 (a) $(-1)^n t^n F(t)$ (b) $F'(t)$
 (c) $\frac{F(t)}{t}$ (d) $\int_0^t F(u) du$.
10. The value of $\int_0^\infty e^{-x^2} dx$ is
 (a) $\frac{1}{2} \sqrt{\pi}$ (b) $\frac{1}{2} \sqrt{\left(\frac{\pi}{2}\right)}$
 (c) $\sqrt{\pi}$ (d) None of these.
11. The value of $L^{-1} \left\{ \frac{1}{(p-1)^3} \right\}$ is
 (a) $t^2 e^t$ (b) $\frac{t^2 e^t}{2}$
 (c) $\frac{te^t}{2}$ (d) None of these. (Rohilkhand 2004)

Fill In The Blank(s)

Fill in the blanks “.....” so that the following statements are complete and correct.

1. If $f(p)$ is the Laplace transform of a function $F(t)$, then $F(t)$ is called the Laplace transform of the function $f(p)$.
2. Inverse Laplace transform of $\frac{1}{p-a}$, $p > a$ is
3. If $L^{-1}\{f(p)\} = F(t)$ then $L^{-1}\{e^{-ap} f(p)\} = G(t)$, where $G(t) = \dots\dots$
4. Let $F(t)$ and $G(t)$ be two functions of class A and let

$$L^{-1}\{f(p)\} = F(t)$$

 and $L^{-1}\{g(p)\} = G(t)$, then

$$L^{-1}\{f(p)g(p)\} = \dots\dots$$
5. The value of $\int_0^\infty \cos x^2 dx = \dots\dots$

True or False

Write ‘T’ for true and ‘F’ for false statement.

1. The value of $L^{-1}\left\{\frac{4}{p+4}\right\}$ is $3e^{-4t}$.
2. The value of $L^{-1}\left\{\frac{1}{(p+1)^2}\right\}$ is $t^2 e^{-t}$.
3. The value of $L^{-1}\left\{\frac{1}{p^2+8p+16}\right\}$ is te^{-4t} .
4. The convolution of two functions F and G obeys the commutative law.
5. The value of $L^{-1}\left\{\frac{1}{(p-1)(p+2)}\right\}$ is $\frac{1}{3}(e^{-t} - e^{2t})$.

Answers

Multiple Choice Questions

- | | | |
|---------|---------|--------|
| 1. (a) | 2. (c) | 3. (c) |
| 4. (a) | 5. (d) | 6. (c) |
| 7. (c) | 8. (d) | 9. (b) |
| 10. (a) | 11. (b) | |

Fill in the Blank(s)

1. inverse
2. e^{at}
3. $F(t-a), t > a$ and $0, t < a$
4. $\int_0^t F(x) G(t-x) dx$
5. $\frac{1}{2} \sqrt{\left(\frac{\pi}{2}\right)}$

True or False

1. T
2. F
3. T
4. T
5. F



Chapter

3



Applications of Laplace Transform

(To Solutions of Differential Equations and Integral Equations)

3.1 Solution of an Ordinary Differential Equation with Constant Coefficients

The Laplace transform is very useful in solving ordinary linear differential equations with constant coefficients.

Let us consider a linear differential equation with constant coefficients

$$\frac{d^n y}{dt^n} + C_1 \frac{d^{n-1} y}{dt^{n-1}} + \dots + C_{n-1} \frac{dy}{dt} + C_n y = F(t) \quad \dots(1)$$

where $F(t)$ is a function of the independent variable t .

$$\text{Let } y(0) = A_0, y'(0) = A_1, \dots, y^{(n-1)}(0) = A_{n-1} \quad \dots(2)$$

be the given initial or boundary conditions where $A_0, A_1, A_2, \dots, A_{n-1}$ are constants. On taking the Laplace transform of both sides of equation (1) and using conditions (2) we obtain an algebraic equation known as “*subsidiary equation*” from which $\bar{y}(p) = L\{y(t)\}$ is determined. The required solution is then obtained by finding the inverse Laplace transform of $\bar{y}(p)$.

Illustrative Examples

Example 1: Solve $y''(t) + y(t) = t$ with $y'(0) = 1, y(\pi) = 0$.

(Kanpur 08; Lucknow 10)

Solution: Taking Laplace transform of both sides of the given equation, we have

$$L\{y''\} + L\{y\} = L\{t\}$$

$$\text{or } p^2 L\{y\} - p y(0) - y'(0) + L\{y\} = 1/p^2$$

$$\text{or } (p^2 + 1) L\{y\} = Ap + 1 + 1/p^2 = Ap + (p^2 + 1)/p^2, \text{ where } y(0) = A$$

$$\text{or } L\{y\} = A \cdot \frac{p}{p^2 + 1} + \frac{1}{p^2}.$$

$$\therefore y = y(t) = AL^{-1}\left\{\frac{p}{p^2 + 1}\right\} + L^{-1}\left\{\frac{1}{p^2}\right\} = A \cos t + t.$$

But $y(\pi) = 0$, therefore $0 = A \cos \pi + \pi$ or $A = \pi$.

$\therefore y = \pi \cos t + t$, which is the required solution.

Example 2: Solve $(D^2 + m^2)x = a \cos nt, t > 0$, x, Dx equal to x_0 and x_1 , when $t = 0, n \neq m$.

Solution: (a) Taking Laplace transform of both sides of the given equation, we have

$$L\{x''\} + m^2 L\{x\} = a L\{\cos nt\}$$

$$\text{or } p^2 L\{x\} - px(0) - x'(0) + m^2 L\{x\} = \frac{ap}{p^2 + n^2}$$

$$\text{or } (p^2 + m^2) L\{x\} = px_0 + x_1 + \frac{ap}{p^2 + n^2}$$

$$\begin{aligned} \text{or } L\{x\} &= x_0 \cdot \frac{p}{p^2 + m^2} + x_1 \cdot \frac{1}{p^2 + m^2} + \frac{a \cdot p}{(p^2 + m^2)(p^2 + n^2)} \\ &= x_0 \cdot \frac{p}{p^2 + m^2} + x_1 \cdot \frac{1}{p^2 + m^2} + \frac{ap}{m^2 - n^2} \left[\frac{(p^2 + m^2) - (p^2 + n^2)}{(p^2 + m^2)(p^2 + n^2)} \right] \\ &= x_0 \cdot \frac{p}{p^2 + m^2} + x_1 \cdot \frac{1}{p^2 + m^2} + \frac{a}{(m^2 - n^2)} \cdot \left(\frac{p}{p^2 + n^2} - \frac{p}{p^2 + m^2} \right). \end{aligned}$$

Taking the inverse transform, we have

$$\begin{aligned} x &= x_0 \cdot L^{-1}\left\{\frac{p}{p^2 + m^2}\right\} + x_1 \cdot L^{-1}\left\{\frac{1}{p^2 + m^2}\right\} \\ &\quad + \frac{a}{(m^2 - n^2)} \cdot \left[L^{-1}\left\{\frac{p}{p^2 + n^2}\right\} - L^{-1}\left\{\frac{p}{p^2 + m^2}\right\} \right] \end{aligned}$$

$$\text{or } x = x_0 \cos mt + \frac{x_1}{m} \sin mt + \frac{a}{(m^2 - n^2)} (\cos nt - \cos mt),$$

which is the required solution.

Example 3: Solve $(D^2 + 9)y = \cos 2t$ if $y(0) = 1, y(\pi/2) = -1$.

(Rohilkhand 2000, 02, 04; Agra 01; Gorakhpur 07, 10, 11; Kashi 14)

Solution: Taking the Laplace transform of both sides of the given equation, we have

$$L\{y''\} + 9L\{y\} = L\{\cos 2t\}$$

$$\text{or } p^2 L\{y\} - p y(0) - y'(0) + 9L\{y\} = p/(p^2 + 4)$$

$$\text{or } (p^2 + 9)L\{y\} - p - A = p/(p^2 + 4), \text{ where } y'(0) = A$$

$$\begin{aligned} \text{or } L\{y\} &= \frac{p+A}{p^2+9} + \frac{p}{(p^2+9)(p^2+4)} \\ &= \frac{p}{p^2+9} + \frac{A}{p^2+9} + \frac{p}{5(p^2+4)} - \frac{p}{5(p^2+9)}. \end{aligned}$$

$$\begin{aligned} \therefore y &= L^{-1}\left\{\frac{p}{p^2+9}\right\} + AL^{-1}\left\{\frac{1}{p^2+9}\right\} + \frac{1}{5}L^{-1}\left\{\frac{p}{p^2+4}\right\} - \frac{1}{5}L^{-1}\left\{\frac{p}{p^2+9}\right\} \\ &= \cos 3t + \frac{1}{3}A \sin 3t + \frac{1}{5}\cos 2t - \frac{1}{5}\cos 3t \\ &= \frac{4}{5}\cos 3t + \frac{1}{3}A \sin 3t + \frac{1}{5}\cos 2t. \end{aligned}$$

But $y(\pi/2) = -1$.

$$\therefore -1 = \frac{4}{5}\cos \frac{3}{2}\pi + \frac{1}{3}A \sin \frac{3}{2}\pi + \frac{1}{5}\cos \pi$$

$$\text{or } -1 = -\frac{1}{3}A - \frac{1}{5} \quad \text{or } A = 12/5.$$

Hence the required solution is $y = \frac{4}{5}\cos 3t + \frac{4}{5}\sin 3t + \frac{1}{5}\cos 2t$.

Example 4: Solve $(D+1)^2 y = t$ given that $y = -3$, when $t = 0$ and $y = -1$, when $t = 1$.

(Rohilkhand 2006)

Solution: The given equation can be written as $(D^2 + 2D + 1)y = t$.

$$\therefore L\{y''\} + 2L\{y'\} + L\{y\} = L\{t\}$$

$$\text{or } p^2 L\{y\} - py(0) - y'(0) + 2[pL\{y\} - y(0)] + L\{y\} = 1/p^2$$

$$\text{or } (p^2 + 2p + 1)L\{y\} - p(-3) - A - 2(-3) = 1/p^2 \text{ where } y'(0) = A$$

$$\text{or } (p+1)^2 L\{y\} = (1/p^2) - 3p - 6 + A$$

$$\begin{aligned} \text{or } L\{y\} &= \frac{1}{p^2(p+1)^2} - \frac{3p+6}{(p+1)^2} + \frac{A}{(p+1)^2} \\ &= \frac{1}{p^2(p+1)^2} - \frac{3(p+1)+3}{(p+1)^2} + \frac{A}{(p+1)^2} \\ &= -\frac{2}{p} + \frac{1}{p^2} + \frac{2}{p+1} + \frac{1}{(p+1)^2} - \frac{3}{(p+1)} - \frac{3}{(p+1)^2} + \frac{A}{(p+1)^2} \\ &= -\frac{2}{p} + \frac{1}{p^2} - \frac{1}{(p+1)} + \frac{A-2}{(p+1)^2}. \end{aligned}$$

$$\begin{aligned}\therefore y &= -2L^{-1}\left\{\frac{1}{p}\right\} + L^{-1}\left\{\frac{1}{p^2}\right\} - L^{-1}\left\{\frac{1}{p+1}\right\} + (A-2)L^{-1}\left\{\frac{1}{(p+1)^2}\right\} \\ &= -2.1 + t - e^{-t} + (A-2)e^{-t}L^{-1}\{1/p^2\} = -2 + t - e^{-t} + (A-2)te^{-t}.\end{aligned}$$

Now, since $y = -1$, when $t = 1$,

$$\therefore -1 = -2 + 1 - e^{-1} + (A-2)e^{-1} \text{ or } A = 3.$$

Hence the complete solution is $y = -2 + t - e^{-t} + te^{-t}$.

Example 5: Solve $(D^2 + 1)y = t \cos 2t$, $y = 0, \frac{dy}{dt} = 0$ when $t = 0$. (Avadh 2006, 07; Meerut 13B)

Solution: Taking the Laplace transform of both the sides of the given equation, we have

$$L\{y''\} + L\{y\} = L\{t \cos 2t\}$$

$$\text{or } p^2 L\{y\} - p y(0) - y'(0) + L\{y\} = -\frac{d}{dp}(L\{\cos 2t\})$$

$$\text{or } (p^2 + 1)L\{y\} = -\frac{d}{dp}\left(\frac{p}{p^2 + 4}\right) = -\frac{1}{p^2 + 4} + \frac{2p^2}{(p^2 + 4)^2}$$

$$\text{or } L\{y\} = \frac{p^2 - 4}{(p^2 + 1)(p^2 + 4)^2} = -\frac{5}{9(p^2 + 1)} + \frac{5}{9(p^2 + 4)} + \frac{8}{3(p^2 + 4)^2}$$

$$\begin{aligned}\therefore y &= -\frac{5}{9}L^{-1}\left\{\frac{1}{p^2 + 1}\right\} + \frac{5}{9}L^{-1}\left\{\frac{1}{p^2 + 4}\right\} + \frac{8}{3}L^{-1}\left\{\frac{1}{(p^2 + 4)^2}\right\} \\ &= -\frac{5}{9}\sin t + \frac{5}{18}\sin 2t + \frac{8}{3}\int_0^t \frac{1}{2}\sin 2x \cdot \frac{1}{2}\sin 2(t-x) dx, \\ &\quad \left[\text{By the convolution theorem since } L^{-1}\left\{\frac{1}{p^2 + 4}\right\} = \frac{1}{2}\sin 2t \right]\end{aligned}$$

$$= -\frac{5}{9}\sin t + \frac{5}{18}\sin 2t + \frac{1}{3}\int_0^t \{\cos(2t - 4x) - \cos 2t\} dx$$

$$= -\frac{5}{9}\sin t + \frac{5}{18}\sin 2t + \frac{1}{3}\left[-\frac{1}{4}\sin(2t - 4x) - x \cos 2t\right]_0^t$$

$$= -\frac{5}{9}\sin t + \frac{5}{18}\sin 2t + \frac{1}{12}\sin 2t - \frac{1}{3}t \cos 2t + \frac{1}{12}\sin 2t$$

$$\text{or } y = -\frac{5}{9}\sin t + \frac{4}{9}\sin 2t - \frac{1}{3}t \cos 2t, \text{ which is the required solution.}$$

Example 6: Solve $(D^3 - D^2 - D + 1)y = 8te^{-t}$ if $y = D^2 y = 0, Dy = 1$ when $t = 0$.

Solution: Taking the Laplace transform of both sides of the given equation, we have

$$L\{y'''\} - L\{y''\} - L\{y'\} + L\{y\} = 8L\{te^{-t}\}$$

$$\begin{aligned}\text{or } p^3 L\{y\} - p^2 y(0) - py'(0) - y''(0) - [p^2 L\{y\} - p y(0) - y'(0)] \\ - [pL\{y\} - y(0)] + L\{y\} = -8\frac{d}{dp}[L\{e^{-t}\}]\end{aligned}$$

$$\text{or} \quad (p^3 - p^2 - p + 1) L\{y\} - p + 1 = -8 \frac{d}{dp} \left(\frac{1}{p+1} \right)$$

$$\text{or} \quad (p-1)^2 (p+1) L\{y\} = p-1 + \frac{8}{(p+1)^2}$$

$$\begin{aligned} \text{or} \quad L\{y\} &= \frac{1}{(p-1)(p+1)} + \frac{8}{(p-1)^2 (p+1)^3} \\ &= \frac{1}{2} \left(\frac{1}{p-1} - \frac{1}{p+1} \right) - \frac{3}{2(p-1)} + \frac{1}{(p-1)^2} + \frac{3}{2(p+1)} \\ &\quad + \frac{2}{(p+1)^2} + \frac{2}{(p+1)^3} \\ &= -\frac{1}{p-1} + \frac{1}{p+1} + \frac{1}{(p-1)^2} + \frac{2}{(p+1)^2} + \frac{2}{(p+1)^3}. \end{aligned}$$

$$\begin{aligned} \therefore y &= -L^{-1} \left\{ \frac{1}{p-1} \right\} + L^{-1} \left\{ \frac{1}{p+1} \right\} + L^{-1} \left\{ \frac{1}{(p-1)^2} \right\} \\ &\quad + 2 L^{-1} \left\{ \frac{1}{(p+1)^2} \right\} + 2 L^{-1} \left\{ \frac{1}{(p+1)^3} \right\} \\ &= -e^t + e^{-t} + e^t L^{-1} \left\{ \frac{1}{p^2} \right\} + 2 e^{-t} L^{-1} \left\{ \frac{1}{p^2} \right\} + 2 e^{-t} L^{-1} \left\{ \frac{1}{p^3} \right\} \\ &= -e^t + e^{-t} + e^t t + 2e^{-t} t + 2e^{-t} (t^2/2!) \\ &= (1 + 2t + t^2) e^{-t} - (1-t) e^t, \text{ which is the required solution.} \end{aligned}$$

Example 7: Solve $(D^4 + 2D^2 + 1)y = 0$, where $y(0) = 0$, $y'(0) = 1$, $y''(0) = 2$ and $y'''(0) = -3$.

Solution: Taking the Laplace transform of both sides of the given equation, we have

$$L\{y^{iv}\} + 2L\{y''\} + L\{y\} = 0$$

$$\begin{aligned} \text{or} \quad p^4 L\{y\} - p^3 y(0) - p^2 y'(0) - p y''(0) - y'''(0) \\ + 2[p^2 L\{y\} - p y(0) - y'(0)] + L\{y\} = 0 \end{aligned}$$

$$\text{or} \quad (p^4 + 2p^2 + 1) L\{y\} - p^2 - 2p + 3 + 2(-1) = 0$$

$$\text{or} \quad (p^2 + 1)^2 L\{y\} = p^2 + 2p - 1$$

$$\text{or} \quad L\{y\} = \frac{p^2 + 2p - 1}{(p^2 + 1)^2} = \frac{1}{p^2 + 1} + \frac{2p - 2}{(p^2 + 1)^2} = \frac{1}{p^2 + 1} + \frac{2p}{(p^2 + 1)^2} - \frac{2}{(p^2 + 1)^2}.$$

$$\therefore y = L^{-1} \left\{ \frac{1}{p^2 + 1} \right\} + L^{-1} \left\{ \frac{2p}{(p^2 + 1)^2} \right\} - 2 L^{-1} \left\{ \frac{1}{(p^2 + 1)^2} \right\}. \quad \dots(1)$$

$$\text{Now } L^{-1} \left\{ \frac{2p}{(p^2 + 1)^2} \right\} = -L^{-1} \left\{ \frac{d}{dp} \left(\frac{1}{p^2 + 1} \right) \right\} = -(-t) L^{-1} \left\{ \frac{1}{p^2 + 1} \right\} = t \sin t$$

$$\text{and } L^{-1} \left\{ \frac{1}{(p^2 + 1)^2} \right\} = L^{-1} \left\{ \frac{1}{p^2 + 1} \cdot \frac{1}{p^2 + 1} \right\} = F(t) * F(t),$$

$$\text{where } F(t) = L^{-1} \left\{ \frac{1}{p^2 + 1} \right\} = \sin t$$

$$= \int_0^t \sin x \cdot \sin(t - x) dx = \frac{1}{2} \int_0^t [\cos(2x - t) - \cos t] dx$$

$$= \frac{1}{2} \left[\frac{1}{2} \sin(2x - t) - x \cos t \right]_0^t = \frac{1}{2} (\sin t - t \cos t).$$

Putting in (1), the required solution is

$$y = \sin t + t \sin t - (\sin t - t \cos t) \quad \text{or} \quad y = t (\sin t + \cos t).$$

Example 8: Find the general solution of the differential equation

$$x''(t) + k^2 x(t) = F(t); \quad x(0) = A, \quad x'(0) = B.$$

Solution: Taking the Laplace transform of both sides of the given equation, we have

$$L\{x''\} + k^2 L\{x\} = L\{F(t)\}$$

$$\text{or } p^2 L\{x\} - px(0) - x'(0) + k^2 L\{x\} = L\{F(t)\}$$

$$\text{or } (p^2 + k^2) L\{x\} = pA + B + f(p), \text{ where } L\{F(t)\} = f(p)$$

$$\text{or } L\{x\} = A \cdot \frac{p}{p^2 + k^2} + B \cdot \frac{1}{p^2 + k^2} + \frac{1}{p^2 + k^2} f(p).$$

$$\therefore x = AL^{-1} \left\{ \frac{p}{p^2 + k^2} \right\} + BL^{-1} \left\{ \frac{1}{p^2 + k^2} \right\} + L^{-1} \left\{ \frac{1}{p^2 + k^2} \cdot f(p) \right\}.$$

$$= A \cos kt + (B/k) \sin kt + \{(1/k) \sin kt\} * F(t)$$

$$x(t) = A \cos kt + (B/k) \sin kt + (1/k) \int_0^t \sin k(t - x) \cdot F(x) dx.$$

[By the convolution theorem]

This is the required general solution.

Example 9: Solve $\frac{d^2 x}{dt^2} + x = F(t)$, if $x = x' = 0$ for $t = 0$.

Solution: Taking the Laplace transform of both sides of the given equation, we have

$$L\{x''\} + L\{x\} = L\{F(t)\}$$

$$\text{or } p^2 L\{x\} - px(0) - x'(0) + L\{x\} = f(p), \quad \text{where } L\{F(t)\} = f(p)$$

$$\text{or } L\{x\} = \frac{1}{p^2 + 1} f(p).$$

$$\therefore x = L^{-1} \left\{ \frac{1}{p^2 + 1} f(p) \right\} = (\sin t) * F(t)$$

or
$$x = \int_0^t \sin(t-x) \cdot F(x) dx, \quad (\text{by the convolution theorem})$$

which is the required solution.

Comprehensive Exercise 1

Solve the following problems by means of the Laplace transform :

1. (i) $\frac{dy}{dt} + y = 1$, given that $y = 2$ when $t = 0$.
 (ii) $\frac{d^2 y}{dt^2} + y = 0$, under the conditions that $y = 1, \frac{dy}{dt} = 0$ when $t = 0$.
 (Rohilkhand 2002)
2. (i) $(D^2 + 1)y = 6 \cos 2t$, if $y = 3, Dy = 1$, when $t = 0$.
 (Gorakhpur 2006, 09; Rohilkhand 07; Purvanchal 07)
 (ii) $(D^2 + D)x = 2$, when $x(0) = 3, x'(0) = 1$.
 (Rohilkhand 2011; Kanpur 07)
3. (i) $(D^2 + m^2)x = a \sin nt, t > 0$, where x, Dx equal to x_0 and x_1 , when
 $t = 0, n \neq m$.
 (ii) $(D + 2)^2 y = 4e^{-2t}, y(0) = -1$ and $y'(0) = 4$.
4. (i) $(D^2 - D - 6)y = 2, t > 0$, if $y = 1, Dy = 0$, when $t = 0$.
 (ii) $(D^2 + 6D + 9)y = \sin x$, where $y(0) = 1, y'(0) = 0$.
5. $(D^2 + 9)y = 18t$, if $y(0) = 0, y(\pi/2) = 0$.
6. $(D^2 + 6D + 25)y = 208 e^{3t}, t > 0$, if $y = 1, Dy = 0$ when $t = 0$.
7. $(D^2 - 4D + 5)y = 125t^2$, if $y = 0 = Dy$, when $t = 0$.
8. $(D^2 + 2D + 1)y = 3t e^{-t}, t > 0$, subject to the conditions, $y = 4, Dy = 2$ when $t = 0$.
9. $(D^2 - 3D + 2)y = 1 - e^{2t}, y = 1, Dy = 0$ when $t = 0$.
 (Avadh 08; Gorakhpur 05)
10. $(D^2 + 1)y = \sin t \sin 2t, t > 0$, if $y = 1, Dy = 0$ when $t = 0$.
11. $(D^3 - D)y = 2 \cos t, y = 3, Dy = 2, D^2 y = 1$, when $t = 0$.
12. $(D^3 + D)y = e^{2t}, y(0) = y'(0) = y''(0) = 0$.
13. $(D^3 - 2D^2 + 5D)y = 0$, if $y(0) = 0, y'(0) = 1, y(\pi/8) = 1$.
14. $(D^2 + D)y = t^2 + 2t$, where $y(0) = 4, y'(0) = -2$.

Answers 1

1. (i) $y = e^{-t} + 1$ (ii) $y = \cos t$
2. (i) $y = 5 \cos t + \sin t - 2 \cos 2t$ (ii) $y = 2 + 2t + e^{-t}$
3. (i) $x = x_0 \cos mt + \frac{x_1}{m} \sin mt + \frac{a}{m^2 - n^2} \left(\sin nt - \frac{n}{m} \sin mt \right)$
 (ii) $y = e^{-2t} (2t^2 + 2t - 1)$
4. (i) $y = -\frac{1}{3} + \frac{4}{5} e^{-2t} + \frac{8}{15} e^{3t}$
 (ii) $y = \frac{1}{50} [(53 + 155x) e^{-3x} - (3 \cos x - 4 \sin x)]$
5. $y = \pi \sin 3t + 2t$
6. $y = 4e^{3t} - \frac{3}{4} e^{-3t} (4 \cos 4t + 7 \sin 4t)$
7. $y = 25t^2 + 40t + 22 + 2e^{2t} (2 \sin t - \cos t)$
8. $y = \frac{1}{2} e^{-t} \cdot t^3 + 4e^{-t} + 6te^{-t}$ 9. $y = \frac{1}{2} + \frac{1}{2} e^{2t} - te^{2t}$
10. $y = \frac{15}{16} \cos t + \frac{1}{4} t \sin t + \frac{1}{16} \cos 3t$
11. $y = 3 \sinh t - \sin t + \cosh t + 2$
12. $y = -\frac{1}{2} + \frac{1}{10} e^{2t} + \frac{2}{5} \cos t - \frac{1}{5} \sin t$
13. $y = 1 + e^t (\sin 2t - \cos 2t)$ 14. $y = \frac{1}{3} t^3 + 2(1 + e^{-t})$

3.2 Solution of Ordinary Differential Equations with Variable Coefficients

The Laplace transform can be used in solving some ordinary differential equations with variable coefficients. The method is found useful in case of the equations having the terms of the form $t^m y^n(t)$ whose Laplace transform is $(-1)^m \frac{d^m}{dp^m} L\{y^n(t)\}$.

Illustrative Examples

Example 10: Solve $ty'' + y' + 4ty = 0$ if $y(0) = 3, y'(0) = 0$. (Purvanchal 2010)

Solution: Taking the Laplace transform of both sides of the given equation, we have

$$L\{ty''\} + L\{y'\} + 4L\{ty\} = 0$$

$$\text{or} \quad -\frac{d}{dp} L\{y''\} + L\{y'\} + 4(-1) \frac{d}{dp} L\{y\} = 0$$

$$\text{or} \quad -\frac{d}{dp} [p^2 \bar{y} - p y(0) - y'(0)] + [p \bar{y} - y(0)] - 4 \frac{d}{dp} \bar{y} = 0,$$

where $\bar{y} = L\{y\}$

$$\text{or} \quad -\frac{d}{dp} (p^2 \bar{y} - 3p) + (p \bar{y} - 3) - 4 \frac{d \bar{y}}{dp} = 0$$

$$\text{or} \quad -(p^2 + 4) \frac{d \bar{y}}{dp} - p \bar{y} = 0 \quad \text{or} \quad \frac{d \bar{y}}{\bar{y}} + \frac{p}{p^2 + 4} dp = 0.$$

$$\text{Integrating, } \log \bar{y} + \frac{1}{2} \log (p^2 + 4) = \log C_1 \quad \text{or} \quad \bar{y} = \frac{C_1}{\sqrt{(p^2 + 4)}}.$$

$$\therefore \quad y = L^{-1}\{\bar{y}\} = C_1 L^{-1}\left\{\frac{1}{\sqrt{(p^2 + 4)}}\right\}$$

$$\text{or} \quad y = C_1 J_0(2t) \quad (\text{See Ex. 25, after article 1.23 of Chapter 1})$$

$$\text{Since } y(0) = 3, \text{ therefore } 3 = C_1 J_0(0) = C_1. \quad [\because J_0(0) = 1]$$

Hence $y = 3 J_0(2t)$, which is the required solution.

Example 11: Solve $[t D^2 + (1 - 2t) D - 2] y = 0$ if $y(0) = 1, y'(0) = 2$.

Solution: The given equation can be written as $ty'' + y' - 2ty' - 2y = 0$.

$$\therefore \quad L\{t y''\} + L\{y'\} - 2L\{t y'\} - 2L\{y\} = 0$$

$$\text{or} \quad -\frac{d}{dp} L\{y''\} + L\{y'\} + 2 \frac{d}{dp} L\{y'\} - 2L\{y\} = 0$$

$$\text{or} \quad -\frac{d}{dp} [p^2 \bar{y} - p y(0) - y'(0)] + [p \bar{y} - y(0)] + 2 \frac{d}{dp} [p \bar{y} - y(0)] - 2 \bar{y} = 0, \text{ where } \bar{y} = L\{y\}$$

$$\text{or} \quad -\frac{d}{dp} (p^2 \bar{y} - p - 2) + (p \bar{y} - 1) + 2 \frac{d}{dp} (p \bar{y} - 1) - 2 \bar{y} = 0$$

$$\text{or} \quad -(p^2 - 2p) \frac{d \bar{y}}{dp} - p \bar{y} = 0 \quad \text{or} \quad \frac{d \bar{y}}{\bar{y}} + \frac{1}{p-2} dp = 0.$$

$$\text{Integrating, } \log \bar{y} + \log (p-2) = \log C_1 \quad \text{or} \quad \bar{y} = C_1 / (p-2).$$

$$\therefore \quad y = L^{-1}\{\bar{y}\} = C_1 L^{-1}\{1/(p-2)\} = C_1 e^{2t}.$$

$$\text{But } y(0) = 1.$$

$$\therefore \quad 1 = C_1.$$

Hence $y = e^{2t}$, which is the required solution.

Example 12: Solve $y'' - ty' + y = 1$, if $y(0) = 1, y'(0) = 2$.

Solution: Taking the Laplace transform of both sides of the given equation, we have

$$L\{y''\} - L\{t y'\} + L\{y\} = L\{1\}$$

$$\begin{aligned}
 \text{or} \quad & p^2 \bar{y} - p y'(0) - y''(0) + \frac{d}{dp} [L\{y'\}] + \bar{y} = \frac{1}{p} \\
 \text{or} \quad & p^2 \bar{y} - p - 2 + \frac{d}{dp} [p \bar{y} - y(0)] + \bar{y} = \frac{1}{p} \\
 \text{or} \quad & p^2 \bar{y} - p - 2 + \frac{d}{dp} (p \bar{y} - 1) + \bar{y} = \frac{1}{p} \\
 \text{or} \quad & p \frac{d\bar{y}}{dp} + (p^2 + 2) \bar{y} = p + 2 + \frac{1}{p} \\
 \text{or} \quad & \frac{d\bar{y}}{dp} + \left(p + \frac{2}{p}\right) \bar{y} = 1 + \frac{2}{p} + \frac{1}{p^2} \quad \dots(1)
 \end{aligned}$$

which is a linear differential equation in $\bar{y}$.

$$\therefore \text{I.F.} = e^{\int (p + 2/p) dp} = e^{p^2/2 + 2 \log p} = p^2 e^{p^2/2}.$$

$\therefore$ solution of (1) is given by

$$p^2 e^{p^2/2} \bar{y} = c_1 + \int \left(1 + \frac{2}{p} + \frac{2}{p^2}\right) p^2 e^{p^2/2} dp = c_1 + \int (p^2 + 2p + 1) e^{p^2/2} dp$$

$$= c_1 + \int (p^2 + 1) e^{p^2/2} dp + 2 \int p e^{p^2/2} dp$$

$$= c_1 + \int (2v + 1) e^v \cdot \frac{dv}{\sqrt{2v}} + 2 \int \sqrt{2v} e^v \frac{dv}{\sqrt{2v}},$$

$$\text{putting } \frac{p^2}{2} = v \text{ so that } p dp = dv \text{ or } dp = dv / \sqrt{2v}$$

$$= c_1 + \int \sqrt{2v} e^v dv + \int \frac{e^v}{\sqrt{2v}} dv + 2 \int e^v dv$$

$$= c_1 + \sqrt{2v} e^v + 2e^v = c_1 + p e^{p^2/2} + 2e^{p^2/2}$$

$$\text{or} \quad \bar{y} = L\{y\} = \frac{c_1}{p^2} e^{-p^2/2} + \frac{1}{p} + \frac{2}{p^2}$$

$$= \frac{c_1}{p^2} \left(1 - \frac{p^2}{2} + \frac{p^4}{4 \cdot 2!} - \dots\right) + \frac{1}{p} + \frac{2}{p^2}$$

$$= \frac{(2 + c_1)}{p^2} - \frac{c_1}{2} + \frac{c_1}{8} p^2 - \dots + \frac{1}{p}.$$

$$\begin{aligned}
 \therefore y &= (2 + c_1) L^{-1}\{1/p^2\} - \frac{1}{2} c_1 L^{-1}\{1\} + \frac{1}{8} c_1 L^{-1}\{p^2\} + \dots + L^{-1}\{1/p\} \\
 &= (2 + c_1) t + 1, \quad \text{since } L^{-1}\{p^n\} = 0, n = 0, 1, 2, \dots
 \end{aligned}$$

But given $y'(0) = 2$. Therefore,

$$2 = 2 + c_1 \quad \text{or} \quad c_1 = 0.$$

Hence $y = 2t + 1$, which is the required solution.

Comprehensive Exercise 2

Solve the following problems by means of the Laplace transform:

1. $y'' + t y' - y = 0$, if $y(0) = 0$, $y'(0) = 1$. (Agra 2002; Purvanchal 11)
2. $[t D^2 + (t-1) D - 1] y = 0$, if $y(0) = 5$, $y(\infty) = 0$.
3. $y''(t) + at y'(t) - 2a y(t) = 1$, $y(0) = y'(0) = 0$, $a > 0$.
4. $ty'' + 2y' + t y = 0$, $y(0+) = 1$, $y(\pi) = 0$.
5. Solve $ty''(t) + (2t+3)y'(t) + (t+3)y(t) = ae^{-t}$ if $y(t)$ and its derivatives have transforms.

Answers 2

1. $y = t$
2. $y = 5e^{-t}$
3. $y = \frac{1}{2}t^2$
4. $y = \frac{\sin t}{t}$
5. $y = \left(A + \frac{at}{3}\right)e^{-t}$

3.3 Solution of Simultaneous Ordinary Differential Equations

The Laplace transform can also be used in solving two or more simultaneous ordinary differential equations.

Illustrative Examples

Example 13: Solve $Dx + Dy = t$; $D^2x - y = e^{-t}$ if $x(0) = 3$, $x'(0) = -2$, $y(0) = 0$.

(Rohilkhand 2010)

Solution: Taking the Laplace transform of both sides of the two equations, we have

$$L\{x'\} + L\{y'\} = L\{t\} \quad \text{and} \quad L\{x''\} - L\{y\} = L\{e^{-t}\}$$

$$\text{or} \quad p\bar{x} - x(0) + p\bar{y} - y(0) = 1/p^2$$

$$\text{and} \quad p^2\bar{x} - px(0) - x'(0) - \bar{y} = 1/(p+1)$$

$$\text{or} \quad p\bar{x} + p\bar{y} = 3 + 1/p^2 \quad \text{and} \quad p^2\bar{x} - \bar{y} = 3p - 2 + 1/(p+1).$$

Solving for $\bar{x}$ and $\bar{y}$, we have

$$\bar{x} = \frac{3p^2 + 1}{p^3(1+p^2)} + \frac{3p}{1+p^2} - \frac{2}{1+p^2} - \frac{1}{(p+1)(p^2+1)}$$

$$\begin{aligned}
 &= \frac{1}{p^3} \left(1 + 2p^2 - \frac{2p^4}{1+p^2} \right) + \frac{3p}{1+p^2} - \frac{2}{1+p^2} + \frac{1}{2(p+1)} - \frac{p}{2(p^2+1)} + \frac{1}{2(p^2+1)} \\
 &= \frac{2}{p} + \frac{1}{p^3} + \frac{1}{2(p+1)} + \frac{p}{2(1+p^2)} - \frac{3}{2(p^2+1)}
 \end{aligned}$$

$$\begin{aligned}
 \text{and } \bar{y} &= \frac{1}{p(p+1)(p^2+1)} + \frac{2}{p^2+1} \\
 &= \frac{1}{p} - \frac{1}{2(p+1)} - \frac{p}{2(p^2+1)} - \frac{1}{2(p^2+1)} + \frac{2}{p^2+1} \\
 &= \frac{1}{p} - \frac{1}{2(p+1)} - \frac{p}{2(p^2+1)} + \frac{3}{2(p^2+1)}.
 \end{aligned}$$

$$\begin{aligned}
 \therefore x &= 2L^{-1} \left\{ \frac{1}{p} \right\} + L^{-1} \left\{ \frac{1}{p^3} \right\} + \frac{1}{2} L^{-1} \left\{ \frac{1}{p+1} \right\} \\
 &\quad + \frac{1}{2} L^{-1} \left\{ \frac{p}{p^2+1} \right\} - \frac{3}{2} L^{-1} \left\{ \frac{1}{p^2+1} \right\} \\
 &= 2 + \frac{1}{2} t^2 + \frac{1}{2} e^{-t} + \frac{1}{2} \cos t - \frac{3}{2} \sin t
 \end{aligned}$$

$$\begin{aligned}
 \text{and } y &= L^{-1} \left\{ \frac{1}{p} \right\} - \frac{1}{2} L^{-1} \left\{ \frac{1}{p+1} \right\} - \frac{1}{2} L^{-1} \left\{ \frac{p}{p^2+1} \right\} + \frac{3}{2} L^{-1} \left\{ \frac{1}{p^2+1} \right\} \\
 &= 1 - \frac{1}{2} e^{-t} - \frac{1}{2} \cos t + \frac{3}{2} \sin t.
 \end{aligned}$$

Example 14: Solve $Dx + 2D^2 y = e^{-t}$, $(D+2)x - y = 1$ if $x(0) = y(0) = y'(0) = 0$.

Solution: Taking the Laplace transform of both sides of the given equations, we have

$$L\{x'\} + 2L\{y''\} = L\{e^{-t}\}$$

$$\text{and } L\{x'\} + 2L\{x\} - L\{y\} = L\{1\}$$

$$\text{or } p\bar{x} - x(0) + 2[p^2\bar{y} - py(0) - y'(0)] = \frac{1}{p+1},$$

$$\text{where } \bar{x} = L\{x\} \text{ and } \bar{y} = L\{y\}$$

$$\text{and } p\bar{x} - x(0) + 2\bar{x} - \bar{y} = \frac{1}{p}$$

$$\text{or } p\bar{x} + 2p^2\bar{y} = \frac{1}{p+1} \quad \text{and} \quad (p+2)\bar{x} - \bar{y} = \frac{1}{p}.$$

Solving for $\bar{x}$ and $\bar{y}$, we have

$$\begin{aligned}
 \bar{x} &= \frac{1}{p(p+1)(2p^2+4p+1)} + \frac{2}{(2p^2+4p+1)} \\
 &= \frac{1}{2p(p+1) \left(p + \frac{2-\sqrt{2}}{2} \right) \left(p + \frac{2+\sqrt{2}}{2} \right)} + \frac{1}{\left(p + \frac{2-\sqrt{2}}{2} \right) \left(p + \frac{2+\sqrt{2}}{2} \right)}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{p} + \frac{1}{p+1} - \frac{1}{(2-\sqrt{2})\left(p+\frac{2-\sqrt{2}}{2}\right)} - \frac{1}{(2+\sqrt{2})\left(p+\frac{2+\sqrt{2}}{2}\right)} \\
 &\quad + \frac{1}{\sqrt{2}\left(p+\frac{2-\sqrt{2}}{2}\right)} - \frac{1}{\sqrt{2}\left(p+\frac{2+\sqrt{2}}{2}\right)} \\
 &= \frac{1}{p} + \frac{1}{p+1} - \frac{1}{\left(p+\frac{2-\sqrt{2}}{2}\right)} - \frac{1}{\left(p+\frac{2+\sqrt{2}}{2}\right)}
 \end{aligned}$$

$$\begin{aligned}
 \text{and } \bar{y} &= \frac{1}{p(p+1)(2p^2+4p+1)} \\
 &= \frac{1}{p} + \frac{1}{p+1} - \frac{1}{(2-\sqrt{2})\left(p+\frac{2-\sqrt{2}}{2}\right)} - \frac{1}{(2+\sqrt{2})\left(p+\frac{2+\sqrt{2}}{2}\right)} \\
 &= \frac{1}{p} + \frac{1}{p+1} - \left(\frac{2+\sqrt{2}}{2}\right) \frac{1}{\left(p+\frac{2-\sqrt{2}}{2}\right)} - \left(\frac{2-\sqrt{2}}{2}\right) \frac{1}{\left(p+\frac{2+\sqrt{2}}{2}\right)}
 \end{aligned}$$

$$\begin{aligned}
 \therefore x &= L^{-1} \left\{ \frac{1}{p} \right\} + L^{-1} \left\{ \frac{1}{p+1} \right\} - L^{-1} \left\{ \frac{1}{p+a} \right\} - L^{-1} \left\{ \frac{1}{p+b} \right\} \\
 &= 1 + e^{-t} - e^{-at} - e^{-bt}
 \end{aligned}$$

$$\begin{aligned}
 \text{and } y &= L^{-1} \left\{ \frac{1}{p} \right\} + L^{-1} \left\{ \frac{1}{p+1} \right\} - b L^{-1} \left\{ \frac{1}{p+a} \right\} - a L^{-1} \left\{ \frac{1}{p+b} \right\} \\
 &= 1 + e^{-t} - b e^{-at} - a e^{-bt}, \text{ where } a = \frac{2-\sqrt{2}}{2} \text{ and } b = \frac{2+\sqrt{2}}{2}.
 \end{aligned}$$

Example 15: Solve $(D^2 - 3)x - 4y = 0$, $x + (D^2 + 1)y = 0$, $t > 0$

if $x = y = D y = 0$, $D x = 2$ when $t = 0$.

Solution: Taking the Laplace transform of both sides of the two equations, we have

$$L\{x'\} - 3L\{x\} - 4L\{y\} = 0 \text{ and } L\{x\} + L\{y''\} + L\{y\} = 0$$

$$\text{or } p^2 \bar{x} - p x(0) - x'(0) - 3\bar{x} - 4\bar{y} = 0, \text{ where } \bar{x} = L\{x\}$$

$$\text{and } \bar{y} = L\{y\}$$

$$\text{and } \bar{x} + p^2 \bar{y} - p y(0) - y'(0) + \bar{y} = 0$$

$$\text{or } (p^2 - 3)\bar{x} - 4\bar{y} = 2 \text{ and } \bar{x} + (p^2 + 1)\bar{y} = 0.$$

Solving for $\bar{x}$ and $\bar{y}$, we have

$$\bar{x} = \frac{2(p^2 + 1)}{(p^2 - 1)^2} = \frac{1}{(p-1)^2} + \frac{1}{(p+1)^2}$$

and

$$\bar{y} = \frac{-2}{(p+1)^2 (p-1)^2} = \frac{1}{2} \left[-\frac{1}{p+1} + \frac{1}{p-1} - \frac{1}{(p+1)^2} - \frac{1}{(p-1)^2} \right].$$

 $\therefore$

$$x = L^{-1} \left\{ \frac{1}{(p-1)^2} \right\} + L^{-1} \left\{ \frac{1}{(p+1)^2} \right\} \\ = e^t L^{-1} \{1/p^2\} + e^{-t} L^{-1} \{1/p^2\} = (e^t + e^{-t}) t$$

and

$$y = \frac{1}{2} \left[-L^{-1} \left\{ \frac{1}{p+1} \right\} + L^{-1} \left\{ \frac{1}{p-1} \right\} - L^{-1} \left\{ \frac{1}{(p+1)^2} \right\} - L^{-1} \left\{ \frac{1}{(p-1)^2} \right\} \right] \\ = \frac{1}{2} [-e^{-t} + e^t - t e^{-t} - t e^t] = \frac{1}{2} (1-t) e^t - \frac{1}{2} (1+t) e^{-t}.$$

Comprehensive Exercise 3

Solve the following problems by means of the Laplace transform:

1. $(D-2)x + 3y = 0$, $2x + (D-1)y = 0$, if $x(0) = 8$ and $y(0) = 3$.
2. $(D^2 + 2)x - Dy = 1$, $Dx + (D^2 + 2)y = 0$, if $x = 0 = Dx = y = Dy$, when $t = 0$.
3. $(D^2 - 1)x + 5Dy = t$, $-2Dx + (D^2 - 4)y = -2$,
if $x = 0 = Dx = y = Dy$ when $t = 0$.
4. Applying Laplace transform solve the equations :

$$\frac{dx}{dt} + \frac{dy}{dt} = t \quad \text{and} \quad \frac{d^2x}{dt^2} - y = e^{-t},$$

given that $x(0) = 0 = y(0)$ and $\frac{dx}{dt} = 0$ when $t = 0$. (Rohilkhand 2007)

5. Solve $(D-2)x - (D+1)y = 6e^{3t}$, $(2D-3)x + (D-3)y = 6e^{3t}$, if $x = 3$, $y = 0$ when $t = 0$.
6. Solve $(D-2)x - (D-2)y = \sin t$, $(D^2 + 1)x + 2Dy = 0$,
if $x = 0 = x'(0) = y(0)$.

Answers 3

1. $x = 5e^{-t} + 3e^{4t}$, $y = 5e^{-t} - 2e^{4t}$.
2. $x = \frac{1}{2} - \frac{1}{6}(2 \cos t + \cos 2t)$, $y = \frac{1}{6}(\sin 2t - 2 \sin t)$.
3. $x = -t + 5 \sin t - 2 \sin 2t$, $y = 1 - 2 \cos t + \cos 2t$.
4. $x = \frac{1}{2}t^2 - 1 + \frac{1}{2}e^{-t} + \frac{1}{2} \cos t + \frac{1}{2} \sin t$, $y = 1 - \frac{1}{2}e^{-t} - \frac{1}{2} \cos t - \frac{1}{2} \sin t$.

$$5. \quad x = e^t + 2te^t + 2e^{3t}, \quad y = e^t - te^t - 2e^{3t}.$$

$$6. \quad x = \frac{1}{9}(1+3t) + e^{-t} + \frac{4}{45}e^{2t} - \frac{1}{5}(\cos t + 2\sin t), \quad y = \frac{1}{9}[(1+3t)e^{-t} - e^{2t}].$$

3.4 Solution of Partial Differential Equations

Laplace transform is also useful in solving partial differential equations when the boundary conditions are given.

Laplace transforms of some partial derivatives:

Theorem: If $y(x, t)$ is a function of x and t , then

$$(a) \quad L \left\{ \frac{\partial y}{\partial t} \right\} = p \bar{y}(x, p) - y(x, 0)$$

$$(b) \quad L \left\{ \frac{\partial^2 y}{\partial t^2} \right\} = p^2 \bar{y}(x, p) - p y(x, 0) - y_t(x, 0)$$

$$(c) \quad L \left\{ \frac{\partial y}{\partial x} \right\} = \frac{d \bar{y}}{dx} \quad \text{and} \quad (d) \quad L \left\{ \frac{\partial^2 y}{\partial x^2} \right\} = \frac{d^2 \bar{y}}{dx^2} \quad \text{where } L \{ y(x, t) \} = \bar{y}(x, p).$$

Proof: (a)
$$L \left\{ \frac{\partial y}{\partial t} \right\} = \int_0^\infty e^{-pt} \frac{\partial y}{\partial t} dt = \lim_{s \rightarrow \infty} \int_0^s e^{-pt} \frac{\partial y}{\partial t} dt$$

$$= \lim_{s \rightarrow \infty} \left[\left\{ e^{-pt} y(x, t) \right\}_0^s + p \int_0^s e^{-pt} y(x, t) dt \right]$$

$$= p \int_0^\infty e^{-pt} y(x, t) dt - y(x, 0) = p \bar{y}(x, p) - y(x, 0)$$

(b) Let $V = \frac{\partial y}{\partial t}$.

$$\therefore \quad L \left\{ \frac{\partial^2 y}{\partial t^2} \right\} = L \left\{ \frac{\partial V}{\partial t} \right\} = p \cdot L \{ V \} - V(x, 0)$$

$$= p [pL \{ y \} - y(x, 0)] - y_t(x, 0) \quad \left[\because V = \frac{\partial y}{\partial t} = y_t \right]$$

$$= p^2 \bar{y}(x, p) - p y(x, 0) - y_t(x, 0).$$

(c)
$$L \left\{ \frac{\partial y}{\partial x} \right\} = \int_0^\infty e^{-pt} \frac{\partial y}{\partial x} dt = \frac{d}{dx} \int_0^\infty e^{-pt} y dt = \frac{d \bar{y}}{dx}.$$

(d)
$$L \left\{ \frac{\partial^2 y}{\partial x^2} \right\} = L \left\{ \frac{\partial U}{\partial x} \right\}, \text{ where } U = \frac{\partial y}{\partial x}$$

$$= \frac{d}{dx} L \{U\} = \frac{d}{dx} L \left\{ \frac{\partial y}{\partial x} \right\} = \frac{d}{dx} \left(\frac{d \bar{y}}{dx} \right) = \frac{d^2 \bar{y}}{dx^2}.$$

Illustrative Examples

Example 16: Solve $\frac{\partial y}{\partial t} = 2 \frac{\partial^2 y}{\partial x^2}$ where $y(0, t) = 0 = y(5, t)$ and $y(x, 0) = 10 \sin 4\pi x$.

(Meerut 2008)

Solution: Taking the Laplace transform of both sides of given equation, we have

$$L \left\{ \frac{\partial y}{\partial t} \right\} = 2L \left\{ \frac{\partial^2 y}{\partial x^2} \right\} \quad \text{or} \quad p \bar{y} - y(x, 0) = 2 \frac{d^2 \bar{y}}{dx^2}$$

or $\frac{d^2 \bar{y}}{dx^2} - \frac{p}{2} \bar{y} = -5 \sin 4\pi x$ whose general solution is

$$\bar{y} = C_1 e^{\sqrt{(p/2)} x} + C_2 e^{-\sqrt{(p/2)} x} - \frac{5 \sin 4\pi x}{-(4\pi)^2 - p/2}$$

or $\bar{y} = C_1 e^{\sqrt{(p/2)} x} + C_2 e^{-\sqrt{(p/2)} x} + \frac{10}{32\pi^2 + p} \sin 4\pi x \quad \dots(1)$

But $y(0, t) = 0 = y(5, t)$. Therefore, $\bar{y}(0, p) = 0$, $\bar{y}(5, p) = 0$.

$\therefore$ from (1), we have $0 = C_1 + C_2$

and $0 = C_1 e^{5\sqrt{(p/2)}} + C_2 e^{-5\sqrt{(p/2)}} + \frac{10}{32\pi^2 + p} \sin 20\pi$

$$= C_1 e^{5\sqrt{(p/2)}} + C_2 e^{-5\sqrt{(p/2)}} + 0.$$

Solving $C_1 = 0 = C_2$.

$\therefore$ from (1), we have $\bar{y} = \frac{10}{32\pi^2 + p} \sin 4\pi x$.

$\therefore y = L^{-1} \left\{ \frac{10}{32\pi^2 + p} \sin 4\pi x \right\} \quad \text{or} \quad y = 10 e^{-32\pi^2 t} \sin 4\pi x,$

which is the required solution.

Example 17: Find the bounded solution of $\frac{\partial y}{\partial t} = \frac{\partial^2 y}{\partial x^2}$, $x > 0, t > 0$,

where $y(0, t) = 1, y(x, 0) = 0$.

(Meerut 2007)

Solution: Taking the Laplace transform of both the sides of the given equation, we have

$$L \left\{ \frac{\partial y}{\partial t} \right\} = L \left\{ \frac{\partial^2 y}{\partial x^2} \right\} \quad \text{or} \quad p \bar{y}(x, p) - y(x, 0) = \frac{d^2 \bar{y}}{dx^2}$$

or $\frac{d^2 \bar{y}}{dx^2} - p \bar{y} = 0$

whose general solution is $\bar{y} = c_1 e^{\sqrt{p}x} + c_2 e^{-\sqrt{p}x}$.

But $y(x, t)$ must be bounded as $x \rightarrow \infty$.

$\therefore \bar{y}(x, p) = L\{y(x, t)\}$ must also be bounded as $x \rightarrow \infty$

$\therefore c_1 = 0$ so that $\bar{y} = c_2 e^{-\sqrt{p}x}$, if $\sqrt{p} > 0$(1)

But $y(0, t) = 1$. Therefore,

$$L\{y(0, t)\} = L\{1\} \quad \text{or} \quad \bar{y}(0, p) = 1/p \quad \text{...(2)}$$

Thus, from (1) and (2) we have $1/p = c_2$.

$$\therefore \bar{y} = (1/p) e^{-\sqrt{p}x}$$

$$\text{or} \quad y = L^{-1} \left\{ \frac{1}{p e^{\sqrt{p}x}} \right\} = \operatorname{erfc} \left(\frac{x}{2\sqrt{t}} \right), \text{ which is the required solution.}$$

[See Ex. 21, after article 2.18 of Chapter 2]

Example 18: Solve $\frac{\partial^2 y}{\partial x^2} - \frac{\partial^2 y}{\partial t^2} = xt$ where $y = 0 = \frac{\partial y}{\partial t}$ at $t = 0$ and $y(0, t) = 0$.

Solution: Taking the Laplace transform of both the sides of the given equation, we have

$$L \left\{ \frac{\partial^2 y}{\partial x^2} \right\} - L \left\{ \frac{\partial^2 y}{\partial t^2} \right\} = L \{xt\}$$

$$\text{or} \quad \frac{d^2 \bar{y}}{dx^2} - [p^2 \bar{y}(x, p) - p y(x, 0) - y_t(x, 0)] = x L\{t\}$$

$$\text{or} \quad \frac{d^2 \bar{y}}{dx^2} - p^2 \bar{y} = x/p^2 \quad \text{whose general solution is}$$

$$\bar{y} = c_1 e^{px} + c_2 e^{-px} - x/p^4.$$

Since $\bar{y} = 0$ for all values of x , therefore, $c_1 = 0$, otherwise

$$\bar{y} = \infty, \text{ as } x \rightarrow \infty.$$

$$\therefore \bar{y} = c_2 e^{-px} - x/p^4.$$

Again $\bar{y} = 0$ when $x = 0$; $\therefore c_2 = 0$.

$$\bar{y} = -x/p^4 \quad \text{or} \quad y = -L^{-1}\{x/p^4\} = -\frac{1}{6}xt^3,$$

which is the required solution.

Comprehensive Exercise 4

Solve the following problems by means of the Laplace transform:

1. $\frac{\partial y}{\partial x} = 2 \frac{\partial y}{\partial t} + y$, $y(x, 0) = 6e^{-3x}$ which is bounded for $x > 0, t > 0$.
2. $\frac{\partial y}{\partial t} = 3 \frac{\partial^2 y}{\partial x^2}$, where $y\left(\frac{\pi}{2}, t\right) = 0, \left(\frac{\partial y}{\partial x}\right)_{x=0} = 0$ and $y(x, 0) = 30 \cos 5x$.

$$3. \quad \frac{\partial y}{\partial t} = \frac{\partial^2 y}{\partial x^2}, y(x, 0) = 3 \sin 2\pi x, y(0, t) = 0 = y(1, t), 0 < x < 1, t > 0.$$

(Meerut 2006)

$$4. \quad \frac{\partial y}{\partial t} = 2 \frac{\partial^2 y}{\partial x^2}, y(0, t) = 0, y(5, t) = 0, y(x, 0) = 10 \sin 4\pi x - 5 \sin 6\pi x.$$

(Meerut 2008)

$$5. \quad \frac{\partial u}{\partial x} - \frac{\partial u}{\partial t} = 1 - e^{-t}, 0 < x < 1, t > 0, u(x, 0) = x.$$

(Meerut 2007)

Answers 4

$$1. \quad y(x, t) = 6 e^{-2t-3x}.$$

$$2. \quad y = 30 e^{-75t} \cos 5x.$$

$$3. \quad y(x, t) = 3 e^{-4\pi^2 t} \sin 2\pi x.$$

$$4. \quad y(x, t) = -5 e^{-72\pi^2 t} \sin 6\pi x + 10 e^{-32\pi^2 t} \sin 4\pi x.$$

$$5. \quad u(x, t) = (x + 1) - e^{-t}.$$

3.5 Integral Equations

1. **Integral equation:** An equation of the form

$$F(t) = y(t) + \int_a^b K(u, t) F(u) du$$

is called an “*Integral equation*” where $y(t)$ and $K(u, t)$ are known, a and b are either constants or functions of t .

Here the function $F(t)$ which appears under the sign of integration is to be determined.

2. **Abel's integral equation:** An equation of the form

$$G(t) = \int_0^t \frac{F(u) du}{(t-u)^n}$$

is called *Abel's integral equation* where $F(u)$ is unknown and $G(t)$ is known and n is a constant between 0 and 1, i.e., $0 < n < 1$.

3. **Integral equation of Convolution type:** An integral equation of the form

$$F(t) = y(t) + \int_0^t K(t-u) F(u) du$$

which may also be expressed as

$$F(t) = y(t) + K(t) * F(t)$$

is called an *integral equation of convolution type*.

4. **Integro-differential equation:** An integral equation in which various derivatives of the unknown function $F(t)$ can also be present is called an *Integro-differential equation*.

For example, $F'(t) = F(t) + y(t) + \int_0^t \sin(t-u) F(u) du.$

3.6 Application of Laplace Transform to Integral Equations

Laplace transform may be used to solve various integral equations.

Example 19: Solve the integral equation

$$F(t) = e^{-t} - 2 \int_0^t \cos(t-u) F(u) du.$$

(Rohilkhand 2011)

Solution: The given integral equation may be expressed as

$$F(t) = e^{-t} - 2 F(t) * \cos t.$$

Taking the Laplace transform of both the sides, we have

$$\begin{aligned} L\{F(t)\} &= L\{e^{-t}\} - 2 L\{F(t) * \cos t\} \\ &= \frac{1}{p+1} - 2 \cdot L\{F(t)\} \cdot L\{\cos t\} \\ &= \frac{1}{p+1} - \frac{2p}{p^2+1} \cdot L\{F(t)\} \end{aligned}$$

$$\text{or} \quad \left(1 + \frac{2p}{p^2+1}\right) L\{F(t)\} = \frac{1}{p+1}$$

$$\text{or} \quad L\{F(t)\} = \frac{(p^2+1)}{(p+1)^3}.$$

$$\begin{aligned} \therefore F(t) &= L^{-1} \left\{ \frac{p^2+1}{(p+1)^3} \right\} = L^{-1} \left\{ \frac{(p+1-1)^2+1}{(p+1)^3} \right\} \\ &= e^{-t} L^{-1} \left\{ \frac{(p-1)^2+1}{p^3} \right\} = e^{-t} L^{-1} \left\{ \frac{1}{p} - \frac{2}{p^2} + \frac{2}{p^3} \right\} \\ &= e^{-t} \left\{ 1 - 2 \cdot \frac{t}{\Gamma(2)} + 2 \cdot \frac{t^3}{\Gamma(3)} \right\} \\ &= e^{-t} (1 - 2t + t^2) = e^{-t} (1-t)^2. \end{aligned}$$

Example 20: Solve the integral equation

$$F(t) = 1 + \int_0^t F(u) \cdot \sin(t-u) du$$

and verify your solution.

Solution: The given integral equation may be expressed as

$$F(t) = 1 + F(t) * \sin t.$$

Taking the Laplace transform of both the sides, we have

$$\begin{aligned} L\{F(t)\} &= L\{1\} + L\{F(t) * \sin t\} \\ &= L\{1\} + L\{F(t)\} \cdot L\{\sin t\} \\ &= \frac{1}{p} + L\{F(t)\} \cdot \frac{1}{p^2+1} \end{aligned}$$

or
$$\left\{ 1 - \frac{1}{p^2 + 1} \right\} L \{ F(t) \} = \frac{1}{p} \quad \text{or} \quad L \{ F(t) \} = \frac{p^2 + 1}{p^3}.$$

$\therefore$
$$F(t) = L^{-1} \left\{ \frac{p^2 + 1}{p^3} \right\} = L^{-1} \left\{ \frac{1}{p} \right\} + L^{-1} \left\{ \frac{1}{p^3} \right\}$$
$$= 1 + \frac{t^2}{\Gamma(3)} = 1 + \frac{t^2}{2}.$$

Verification: We have $F(t) = 1 + \frac{t^2}{2}.$

Putting in the R.H.S. of the given equation, we have

$$\begin{aligned} \text{R.H.S.} &= 1 + \int_0^t \left(1 + \frac{u^2}{2} \right) \cdot \sin(t-u) du \\ &= 1 + \left[\left(1 + \frac{u^2}{2} \right) \cos(t-u) \right]_0^t - \int_0^t u \cdot \cos(t-u) du \\ &= 1 + 1 + \frac{t^2}{2} - \cos t - [-u \cdot \sin(t-u)]_0^t - \int_0^t \sin(t-u) du \\ &= 2 + \frac{t^2}{2} - \cos t - [\cos(t-u)]_0^t = 2 + \frac{t^2}{2} - \cos t - (1 - \cos t) \\ &= 1 + \frac{t^2}{2} = F(t) = \text{L.H.S.} \end{aligned}$$

Example 21: Solve the following equation for $F(t)$ with the condition that $F(0) = 0,$

$$F'(t) = \sin t + \int_0^t F(t-u) \cos u du.$$

Solution: The given equation may be expressed as

$$F'(t) = \sin t + F(t) * \cos(t).$$

Taking the Laplace transform of both the sides, we have

$$L \{ F'(t) \} = L \{ \sin t \} + L \{ F(t) * \cos t \}$$

or
$$pL \{ F(t) \} - F(0) = \frac{1}{p^2 + 1} + L \{ F(t) \} \cdot L \{ \cos t \}$$

or
$$pL \{ F(t) \} = \frac{1}{p^2 + 1} + L \{ F(t) \} \cdot \frac{p}{p^2 + 1}$$

or
$$\left(p - \frac{p}{p^2 + 1} \right) L \{ F(t) \} = \frac{1}{p^2 + 1} \quad \text{or} \quad L \{ F(t) \} = \frac{1}{p^3}.$$

$\therefore$
$$F(t) = L^{-1} \left\{ \frac{1}{p^3} \right\} = \frac{t^2}{\Gamma(3)} = \frac{t^2}{2}.$$

Example 22: Solve the integral equation $\int_0^t F(u) F(t-u) du = 16 \sin 4t.$

Solution: The given integral equation may be expressed as

$$F(t) * F(t) = 16 \sin 4t.$$

Taking the Laplace transform of both the sides, we have

$$L \{ F(t) * F(t) \} = 16 L \{ \sin 4t \}$$

$$\text{or} \quad L \{ F(t) \} \cdot L \{ F(t) \} = 16 \cdot \frac{4}{p^2 + 4^2}$$

$$\text{or} \quad L \{ F(t) \} = \pm \frac{8}{\sqrt{(p^2 + 4^2)}}.$$

$$\therefore F(t) = \pm 8 L^{-1} \left\{ \frac{1}{\sqrt{(p^2 + 4^2)}} \right\} = \pm 8 J_0(4t).$$

(See Q. 2 of Exercise 5 Chapter 2)

Example 23: Solve the integral equation $\int_0^t \frac{F(u) du}{(t-u)^{1/3}} = t(1+t).$

Solution: The given integral equation may be expressed as

$$\int_0^t F(u) \cdot (t-u)^{-1/3} du = t + t^2$$

$$\text{or} \quad F(t) * t^{-1/3} = t + t^2.$$

Taking the Laplace transform of both the sides, we have

$$L \{ F(t) * t^{-1/3} \} = L \{ t + t^2 \}$$

$$\text{or} \quad L \{ F(t) \} \cdot L \{ t^{-1/3} \} = L \{ t \} + L \{ t^2 \}$$

$$\text{or} \quad L \{ F(t) \} \cdot \frac{\Gamma(-\frac{1}{3} + 1)}{p^{-1/3+1}} = \frac{\Gamma(2)}{p^2} + \frac{\Gamma(3)}{p^3}$$

$$\text{or} \quad L \{ F(t) \} \cdot \frac{\Gamma(2/3)}{p^{2/3}} = \frac{1}{p^2} + \frac{2}{p^3}$$

$$\text{or} \quad L \{ F(t) \} = \frac{1}{\Gamma(2/3)} \cdot \left[\frac{1}{p^{4/3}} + \frac{2}{p^{7/3}} \right].$$

$$\therefore F(t) = \frac{1}{\Gamma(2/3)} \cdot \left[L^{-1} \left\{ \frac{1}{p^{4/3}} \right\} + 2 L^{-1} \left\{ \frac{1}{p^{7/3}} \right\} \right]$$

$$= \frac{1}{\Gamma(2/3)} \cdot \left[\frac{t^{(4/3)-1}}{\Gamma(4/3)} + 2 \cdot \frac{t^{(7/3)-1}}{\Gamma(7/3)} \right]$$

$$= \frac{1}{\Gamma(2/3)} \cdot \left[\frac{t^{1/3}}{\frac{1}{3} \cdot \Gamma(1/3)} + 2 \cdot \frac{t^{4/3}}{\frac{4}{3} \cdot \frac{1}{3} \cdot \Gamma(1/3)} \right]$$

$$= \frac{3 t^{1/3}}{\Gamma(1 - \frac{1}{3}) \Gamma(\frac{1}{3})} \left[1 + \frac{3}{2} t \right]$$

$$= \frac{3 t^{1/3}}{(\pi / \sin \frac{1}{3} \pi)} \left[1 + \frac{3}{2} t \right] \quad \left[\because \Gamma(n) \Gamma(1-n) = \frac{\pi}{\sin n\pi} \right]$$

$$= \frac{3 \sqrt{3}}{4 \pi} \cdot t^{1/3} (2 + 3t).$$

Example 24: Solve $F'(t) = t + \int_0^t F(t-u) \cos u \, du$, $F(0) = 4$.

Solution: The given equation may be expressed as

$$F'(t) = t + F(t) * \cos t.$$

Taking Laplace transform of both the sides, we have

$$L\{F'(t)\} = L\{t\} + L\{F(t) * \cos t\}$$

$$\text{or} \quad pL\{F(t)\} - F(0) = \frac{1}{p^2} + L\{F(t)\} \cdot L\{\cos t\}$$

$$\text{or} \quad pL\{F(t)\} - 4 = \frac{1}{p^2} + L\{F(t)\} \cdot \frac{p}{p^2 + 1}$$

$$\text{or} \quad L\{F(t)\} \cdot \frac{p^3}{p^2 + 1} = \frac{1}{p^2} + 4$$

$$\begin{aligned} \text{or} \quad L\{F(t)\} &= \frac{p^2 + 1}{p^5} + \frac{4(p^2 + 1)}{p^3} \\ &= \frac{1}{p^3} + \frac{1}{p^5} + \frac{4}{p} + \frac{4}{p^3} = \frac{4}{p} + \frac{5}{p^3} + \frac{1}{p^5}. \end{aligned}$$

$$\begin{aligned} \therefore F(t) &= L^{-1} \left\{ \frac{4}{p} + \frac{5}{p^3} + \frac{1}{p^5} \right\} = 4 + \frac{5t^2}{\Gamma(3)} + \frac{t^4}{\Gamma(5)} \\ &= 4 + \frac{5t^2}{2} + \frac{t^4}{4!} = 4 + \frac{5}{2}t^2 + \frac{t^4}{24}. \end{aligned}$$

Example 25: Express

$$2F''(t) - 3F'(t) - 2F(t) = 4e^{-t} + 2\cos t, \quad F(0) = 4, \quad F'(0) = -1$$

into an integral equation.

Solution: We have

$$2F''(t) - 3F'(t) - 2F(t) = 4e^{-t} + 2\cos t, \quad \dots(1)$$

$$F(0) = 4, \quad F'(0) = -1 \quad \dots(2)$$

$$\text{Method 1: Let } F''(t) = G(t). \quad \dots(3)$$

Integrating (3), we get

$$F'(t) = \int_0^t G(u) \, du + C_1. \quad \dots(4)$$

For $t = 0$, (4) gives

$$-1 = F'(0) = 0 + C_1 \quad \text{or} \quad C_1 = -1.$$

$$\therefore F'(t) = \int_0^t G(u) \, du - 1. \quad \dots(5)$$

Again integrating, we get

$$F(t) = \int_0^t (t-u) G(u) \, du - t + C_2.$$

For $t = 0$, this gives

$$4 = F(0) = 0 - 0 + C_2 \quad \text{or} \quad C_2 = 4.$$

$$\therefore F(t) = \int_0^t (t-u) G(u) du - t + 4 \quad \dots(6)$$

Putting the values of $F''(t)$, $F'(t)$ and $F(t)$ from (3), (5) and (6) in (1), we get

$$\begin{aligned} 2G(t) - 3 \int_0^t G(u) du + 3 - 2 \int_0^t (t-u) G(u) du + 2t - 8 \\ = 4e^{-t} + 2 \cos t \end{aligned}$$

$$\text{or} \quad 2G(t) + \int_0^t (-2t + 2u - 3) G(u) du = 4e^{-t} + 2 \cos t - 2t + 5,$$

which is the required integral equation.

Method 2: Integrating (1) between the limits 0 to t , we get

$$2[F'(u)]_0^t - 3[F'(u)]_0^t - 2 \int_0^t F(u) du = -4[e^{-u}]_0^t + 2[\sin u]_0^t$$

$$\begin{aligned} \text{or} \quad 2F'(t) - 2F'(0) - 3F(t) + 3F(0) - 2 \int_0^t F(u) du \\ = 4(1 - e^{-t}) + 2 \sin t. \end{aligned}$$

$$\text{or} \quad 2F'(t) - 3F(t) - 2 \int_0^t F(u) du = -4e^{-t} + 2 \sin t - 10. \quad [\text{Using (2)}]$$

Again integrating, between the limits 0 to t , we get

$$\begin{aligned} 2[F(u)]_0^t - 3 \int_0^t F(u) du - 2 \int_0^t (t-u) F(u) du \\ = 4[e^{-u}]_0^t - 2[\cos u]_0^t - 10t. \end{aligned}$$

$$\begin{aligned} \text{or} \quad 2[F(t) - F(0)] + \int_0^t (-2t + 2u - 3) F(u) du \\ = 4(e^{-t} - 1) - 2(\cos t - 1) - 10t. \end{aligned}$$

$$\text{or} \quad 2F(t) + \int_0^t (-2t + 2u - 3) F(u) du = 4e^{-t} - 2 \cos t - 10t + 6$$

which is the required integral equation.

Example 26: Express the given differential equation

$$F''(t) + aF(t) = 0, F(0) = 0 = F(1)$$

as an integral equation.

Solution: We have

$$F''(t) + aF(t) = 0 \quad \dots(1)$$

$$F(0) = 0 = F(1). \quad \dots(2)$$

$$\text{Method 1: Let } F''(t) = G(t). \quad \dots(3)$$

Integrating (3) between the limits 0 to t , we get

$$F'(t) = \int_0^t G(u) du + C_1 \quad \dots(4)$$

Again integrating between the limits 0 to t , we get

$$F(t) = \int_0^t (t-u) G(u) du + C_1 t + C_2. \quad \dots(5)$$

For $t = 0$, (5) gives

$$0 = F(0) = 0 + C_2 \quad \text{or} \quad C_2 = 0.$$

Putting this value in (5), we get

$$F(t) = \int_0^t (t-u) G(u) du + C_1 t \quad \dots(6)$$

For $t = 1$, (6) gives

$$0 = F(1) = \int_0^1 (1-u) G(u) du + C_1$$

or
$$C_1 = - \int_0^1 (1-u) G(u) du.$$

Putting this value of C_1 in (6), we get

$$\begin{aligned} F(t) &= \int_0^t (t-u) G(u) du - \int_0^1 t(1-u) G(u) du \\ &= \int_0^t (t-u) G(u) du + \int_0^1 (ut-t) G(u) du \\ &= \int_0^t (t-u) G(u) du + \int_0^t (ut-t) G(u) du + \int_t^1 (ut-t) G(u) du \\ &= \int_0^t (t-1)u G(u) du + \int_t^1 t(u-1) G(u) du \end{aligned}$$

or
$$F(t) = \int_0^1 K(t, u) G(u) du, \text{ where } K(t, u) = \begin{cases} (t-1)u, & \text{if } u < t \\ t(u-1), & \text{if } u > t. \end{cases}$$

Hence from (1), we get

$$G(t) + a \int_0^t K(t, u) G(u) du = 0, \text{ where } K(t, u) = \begin{cases} (t-1)u, & u < t \\ (u-1)t, & u > t. \end{cases}$$

This is the required integral equation of the given differential equation.

Method 2: Integrating (1) between the limits 0 to t , we get

$$\int_0^t F''(u) du + a \int_0^t F(u) du = 0$$

or
$$[F'(u)]_0^t + a \int_0^t F(u) du = 0$$

or
$$F'(t) - F'(0) + a \int_0^t F(u) du = 0.$$

Again integrating between the same limits, we get

$$[F(u)]_0^t - F'(0)[u]_0^t + a \int_0^t (t-u) F(u) du = 0$$

or
$$F(t) - F(0) - tF'(0) + a \int_0^t (t-u) F(u) du = 0 \quad \dots(1)$$

or
$$F(t) - tF'(0) + a \int_0^t (t-u) F(u) du = 0$$

For $t = 1$, this gives

$$F(1) - F'(0) + a \int_0^1 (1-u) F(u) du = 0 \quad [\text{Using (2)}]$$

$$\text{or} \quad 0 - F'(0) + a \int_0^1 (1-u) F(u) du = 0$$

$$\text{or} \quad F'(0) = a \int_0^1 (1-u) F(u) du.$$

Putting this value in (I), we get

$$F(t) - ta \int_0^1 (1-u) F(u) du + a \int_0^t (t-u) F(u) du = 0$$

$$\text{or} \quad F(t) + a \int_0^1 t(u-1) F(u) du + a \int_0^t (t-u) F(u) du = 0$$

$$\begin{aligned} \text{or} \quad F(t) + a \int_0^t t(u-1) F(u) du + a \int_t^1 t(u-1) F(u) du \\ + a \int_0^t (t-u) F(u) du = 0 \end{aligned}$$

$$\text{or} \quad F(t) + a \int_0^t u(t-1) F(u) du + a \int_t^1 t(u-1) F(u) du = 0$$

$$\text{or} \quad F(t) + a \int_0^1 K(u, t) F(u) du = 0, \text{ where } K(u, t) = \begin{cases} u(t-1), & u < t \\ t(u-1), & u > t. \end{cases}$$

Example 27: Convert the given integral equation

$$F(t) - \int_0^t (t-u) \sec t \cdot F(u) du = t$$

into differential equation and associated conditions.

Solution: The given integral equation is

$$F(t) - \int_0^t (t-u) \sec t \cdot F(u) du = t \quad \dots(1)$$

We know that,

$$\frac{d}{dt} \int_{a(t)}^{b(t)} K(u, t) du = \int_{a(t)}^{b(t)} \frac{\partial K}{\partial t} du + K(b, t) \frac{db}{dt} - K(a, t) \frac{da}{dt}. \quad \dots(2)$$

Differentiating (1) using the fact (2), we get

$$F'(t) - \int_0^t [\sec t + (t-u) \sec t \tan t] F(u) du = 1$$

$$\text{or} \quad F'(t) - \int_0^t \sec t \cdot F(u) du - \tan t \int_0^t (t-u) \sec t F(u) du = 1$$

$$\text{or} \quad F'(t) - \int_0^t \sec t \cdot F(u) du + \tan t \cdot [t - F(t)] = 1, [\text{Using (1)}] \quad \dots(3)$$

Again differentiating (3), we get

$$\begin{aligned} F''(t) + \sec^2 t \cdot [t - F(t)] + \tan t \cdot [1 - F'(t)] \\ - \sec t \cdot \tan t \int_0^t F(u) du - \sec t \cdot F(t) = 0. \end{aligned}$$

$$\begin{aligned} \text{or} \quad F''(t) + \sec^2 t \cdot [t - F(t)] + \tan t \cdot [1 - F'(t)] \\ - \sec t \cdot F(t) + \tan t \cdot [1 - F'(t) - t \tan t + \tan t F(t)] = 0, \end{aligned}$$

[Using (2)]

$$\begin{aligned} \text{or} \quad F'' - F' \cdot [\tan t + \tan t] + [-\sec^2 t + \tan^2 t - \sec t] F \\ + [t \sec^2 t + \tan t - t \cdot \tan^2 t + \tan t] = 0 \end{aligned}$$

or $F''(t) - 2 \tan t \cdot F'(t) - (1 + \sec t) F(t) + (t + 2 \tan t) = 0. \quad \dots(4)$

Putting $t = 0$ in (1) and (3) respectively, we get

$$F(0) = 0, F'(0) = 1. \quad \dots(5)$$

Hence, (4) is the required differential equation and its associated conditions are given by (5).

Comprehensive Exercise 5

1. Solve the integral equation

$$F(t) = a \sin t - 2 \int_0^t F(u) \cdot \cos(t-u) du.$$

2. Solve $y(t) = t - 1 + \int_0^t y(\Gamma) \cdot \sin(t-\Gamma) d\Gamma.$

3. Show that the solution of the integral equation

$$F(t) = 4t - 3 \int_0^t F(u) \sin(t-u) du$$

is $F(t) = t + \frac{3}{2} \sin 2t.$

4. Solve the integral equation $F(t) = 1 + 2 \int_0^t F(t-u) \cdot e^{-2u} du.$

5. Solve $y(t) = t + \frac{1}{6} \int_0^t (t-u)^3 \cdot y(u) du.$

6. Solve the integral equation $F(t) = \frac{1}{2} t^2 - \int_0^t (t-u) F(u) du. \quad (\text{Agra 2003})$

7. Solve $2F(t) = 2 - t + \int_0^t F(t-u) F(u) du.$

8. Solve $\int_0^t \frac{F(u) du}{\sqrt[3]{t-u}} = 1 + t + t^2.$

9. Convert the given differential equation

$$F''(t) + 2F'(t) - 8F(t) = 5t^2 - 3t, F(0) = -2, F'(0) = 3$$

into an integral equation.

10. Convert the given integral equation

$$F(t) = t^2 - 3t + 4 - 3 \int_0^t (t-u)^2 F(u) du$$

into differential equation and associated conditions.

Answers 5

1. $F(t) = at e^{-t}$

2. $y(t) = -1 + t - \frac{1}{2} t^2 + \frac{1}{6} t^3$

4. $F(t) = 1 + 2t$

5. $y(t) = \frac{1}{2} (\sinh t + \sin t)$

6. $F(t) = 1 - \cos t$

7. $F(t) = 1, -1$

8. $F(t) = \frac{1}{\pi} \left[t^{-1/2} + 2t^{1/2} + \frac{8}{3}t^{3/2} \right]$

9. $G(t) + \int_0^t [2 - 8t + 8u] G(u) du - 5t^2 - 21t + 22 = 0$

or $F(t) + \int_0^t (-8t + 8u + 2) F(u) du = \frac{5t^4}{12} - \frac{t^3}{2} - t - 2$

10. $F'''(t) + 6F(t) = 0$ with $F(0) = 4, F'(0) = -3, F''(0) = 2$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. The solution of
- $(D+1)y=0, t>0$
- , given that
- $y=y_0$
- when
- $t=0$
- is

(a) $y_0 e^{-t}$

(b) $y_0 e^t$

(c) $2y_0 e^{-t}$

(d) $2y_0 e^t$

2. If
- $y(x, t)$
- is a function of
- x
- and
- t
- , then
- $L \left\{ \frac{\partial y}{\partial x} \right\}$
- is

(a) $\frac{\partial \bar{y}}{\partial x}$

(b) $\frac{d\bar{y}}{dx}$

(c) $\frac{\partial^2 \bar{y}}{\partial x^2}$

(d) $\frac{d^2 \bar{y}}{dx^2}$

where $L\{y(x, t)\} = \bar{y}(x, p)$.

3. An integral equation of the form

$$F(t) = y(t) + \int_0^t K(t-u) F(u) du$$
 is called

(a) an integral equation of convolution type

(b) Abel's integral equation

(c) integro-differential equation

(d) Cauchy's integral equation

Fill In The Blank(s)

Fill in the Blanks '.....' so that the following statements are complete and correct.

- On taking the Laplace transform of both sides of the given differential equation and using the given conditions, we obtain an algebraic equation known as from which $\bar{y}(p) = L\{y(t)\}$ is determined.
- The solution of the given differential equation is obtained by finding the transform of $\bar{y}(p)$, where $\bar{y}(p) = L\{y(x, t)\}$.

3. If $y(x, t)$ is a function of x and t , then $L \left\{ \frac{\partial y}{\partial t} \right\} = \dots\dots$
4. If $y(x, t)$ is a function of x and t , then $L \left\{ \frac{\partial^2 y}{\partial x^2} \right\} = \dots\dots$
5. An equation of the form $F(t) = y(t) + \int_a^b K(u, t) F(u) du$ is called $\dots\dots$.

equation where $y(t)$ and $K(u, t)$ are known, a and b are either constants or functions of t .

True or False

Write 'T' for true and 'F' for false statement.

1. If $y(x, t)$ is a function of x and t then $L \left\{ \frac{\partial^2 y}{\partial t^2} \right\} = \bar{y}(x, p) - py(x, 0) - y_t(x, 0)$ where $L \{y(x, t)\} = \bar{y}(x, p)$.
2. The differential equation $\frac{\partial y}{\partial t} = 2 \frac{\partial^2 y}{\partial x^2}$ with the condition $y(x, 0) = 10 \sin 4\pi x$ is takes the form $\frac{d^2 \bar{y}}{dx^2} - \frac{p}{2} \bar{y} = -5 \sin 4\pi x$, where $L \{y(x, t)\} = \bar{y}(x, p)$.
3. An equation of the form $G(t) = \int_0^t \frac{F(u)}{(t-u)^n} du$ is called Abel's integral equation where $F(u)$ is unknown and $G(t)$ is known and n is a constant between 0 and 1, i.e., $0 < n < 1$.
4. An integral equation in which various derivatives of the unknown function $F(t)$ can also be present is called an Integro transform equation.

Answers

Multiple Choice Questions

1. (a) 2. (b) 3. (a)

Fill in the Blank(s)

1. subsidiary equation 2. inverse Laplace
3. $p \bar{y}(x, p) - y(x, 0)$, where $L \{y(x, t)\} = \bar{y}(x, p)$
4. $\frac{d^2 \bar{y}}{dx^2}$, where $L \{y(x, t)\} = \bar{y}(x, p)$ 5. an integral

True or False

1. F 2. T 3. T 4. F



Chapter

4



Fourier Transforms

4.1 Dirichlet's Conditions

- A function $f(x)$ is said to satisfy Dirichlet conditions in the interval (a, b) , if
- (i) $f(x)$ is defined and is single-valued except possibly at a finite number of points in the interval (a, b) , and
 - (ii) $f(x)$ and $f'(x)$ are piecewise continuous in the interval (a, b) .

These conditions play an important role in the study of Fourier series and Fourier Transforms.

4.2 Fourier Series

(Meerut 2013, 13B; Rohilkhand 14)

If $f(x)$ is a periodic function with period $2l$ i.e. $f(x + 2l) = f(x)$ and satisfies Dirichlet conditions in the interval $(-l, l)$, then at every point of continuity we have

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right] \quad \dots(1)$$

where
$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx \quad \dots(2)$$

and
$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx. \quad \dots(3)$$

The series (1) with coefficients a_n and b_n given by (2) and (3) respectively is called the *Fourier series* of $f(x)$, and the coefficients a_n and b_n are called the *Fourier coefficients* corresponding to $f(x)$.

At a point of discontinuity

$$f(x) = \frac{1}{2} [f(x+0) + f(x-0)].$$

If the function $f(x)$ defined in the interval $(-l, l)$ be an even function of x i.e. if $f(-x) = f(x)$, then

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx$$

and
$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx = 0.$$

Therefore in this case we get *Fourier cosine series*.

Again if $f(x)$ be an odd function of x i.e. if $f(-x) = -f(x)$, then

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx = 0$$

and
$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$$

and thus in this case we get *Fourier sine series*.

Note: If $f(x)$ is a function of period $2l$ but is defined only in $(0, l)$, we can extend it to $(-l, 0)$ so as to be an even or an odd function of x in the interval $(-l, l)$

4.3 Fourier's Integral Formula

Let $f(x)$ be a function satisfying Dirichlet conditions in every finite interval $-l \leq x \leq l$ and defined as $\frac{1}{2} [f(x+0) + f(x-0)]$ at every point of discontinuity.

Further let $\int_{-\infty}^{\infty} |f(x)| dx$ converge i.e. $f(x)$ is absolutely integrable in $-\infty < x < \infty$,

then by Fourier's integral formula

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(v) \left[\int_{-\infty}^{\infty} \cos w(x-v) dw \right] dv \quad \dots(1)$$

or
$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dw \int_{-\infty}^{\infty} f(v) \cos w(x-v) dv.$$

The representation (1) of $f(x)$ is known as Fourier's integral formula.

The proof is out of scope of this book.

Another form: We have

$$0 = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(v) \left[\int_{-\infty}^{\infty} \sin w(x-v) dw \right] dv \quad \dots(2)$$

Multiplying both sides of (2) by i and then adding to (1), we get

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(v) \left[\int_{-\infty}^{\infty} e^{iw(x-v)} dw \right] dv$$

or
$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iwx} dw \int_{-\infty}^{\infty} f(v) e^{-iwx} dv.$$

4.4 Fourier Transform or Complex Fourier Transform

(Purvanchal 2014)

Let $f(x)$ be a function defined on $(-\infty, \infty)$ and be piecewise continuous in each finite partial interval and absolutely integrable in $(-\infty, \infty)$, then

$$F\{f(x)\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} f(x) dx$$

is called the *Fourier Transform* of $f(x)$ and is denoted by $F\{f(x)\}$ or $\tilde{f}(p)$.

The function $f(x)$ is called the *inverse Fourier transform* of $\tilde{f}(p)$ i.e.,

$$f(x) = F^{-1}\{\tilde{f}(p)\}.$$

4.5 Inversion Theorem for Complex Fourier Transform

If $\tilde{f}(p)$ is the Fourier transform of $f(x)$ and if $f(x)$ satisfies the Dirichlet conditions in every finite interval $(-l, l)$ and further if $\int_{-\infty}^{\infty} |f(x)| dx$ is convergent, then at every point of continuity of $f(x)$,

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(p) e^{-ipx} dp.$$

Proof: From Fourier integral formula, we have

$$\begin{aligned} f(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(v) \left[\int_{-\infty}^{\infty} e^{iw(x-v)} dw \right] dv \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iwx} dw \int_{-\infty}^{\infty} f(v) e^{-iwx} dv \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ipx} dp \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{ipx} dx, \end{aligned}$$

putting $w = -p$ so that $dw = -dp$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ipx} \tilde{f}(p) dp.$$

Note: Some authors also define the Fourier transform in the following forms :

$$(1) \quad \tilde{f}(p) = \int_{-\infty}^{\infty} e^{-ipx} f(x) dx$$

$$\text{and } f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(p) \cdot e^{ipx} dp.$$

$$(2) \quad \tilde{f}(p) = \int_{-\infty}^{\infty} e^{ipx} f(x) dx$$

$$\text{and } f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ipx} \tilde{f}(p) dp.$$

$$(3) \quad \tilde{f}(p) = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{-ipx} f(x) dx$$

$$\text{and } f(x) = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \tilde{f}(p) \cdot e^{ipx} dp.$$

4.6 Fourier Sine Transform

The infinite Fourier sine transform of $f(x)$, $0 < x < \infty$, is denoted by $F_s \{f(x)\}$ or $\tilde{f}_s(p)$, and is defined as

$$F_s \{f(x)\} = \tilde{f}_s(p) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} f(x) \sin px \, dx.$$

The function $f(x)$ is called the *inverse Fourier sine transform* of $\tilde{f}_s(p)$

$$\text{i.e.,} \quad f(x) = F_s^{-1} \{ \tilde{f}_s(p) \}.$$

Note: Some authors also define

$$\tilde{f}_s(p) = \int_0^{\infty} f(x) \sin px \, dx.$$

4.7 Inversion Formula for Fourier Sine Transform

If $\tilde{f}_s(p)$ is the Fourier sine transform of the function $f(x)$ which satisfies the Dirichlet conditions in every finite interval $(0, l)$ and is such that $\int_0^{\infty} |f(x)| \, dx$ exists, then

$$f(x) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \tilde{f}_s(p) \sin px \, dp$$

at every point of continuity of $f(x)$.

This is an inversion formula for infinite Fourier sine transform.

Proof: From Fourier integral formula, we have

$$f(x) = \frac{1}{\pi} \int_0^{\infty} dw \int_{-\infty}^{\infty} f(v) \cos w(x-v) \, dv$$

$$\begin{aligned}
 &= \frac{1}{\pi} \int_0^{\infty} dp \int_{-\infty}^{\infty} \{ f(v) \cos px \cos pv + f(v) \sin px \sin pv \} dv, \text{ where } w = p \\
 &= \frac{1}{\pi} \int_0^{\infty} \cos px dp \int_{-\infty}^{\infty} f(v) \cos pv dv + \frac{1}{\pi} \int_0^{\infty} \sin px dp \int_{-\infty}^{\infty} f(v) \sin pv dv
 \end{aligned}$$

$$\begin{aligned}
 \text{or } f(x) &= \frac{1}{\pi} \int_0^{\infty} \cos px dp \int_{-\infty}^{\infty} f(x) \cos px dx \\
 &\quad + \frac{1}{\pi} \int_0^{\infty} \sin px dp \int_{-\infty}^{\infty} f(x) \sin px dx \quad \dots(1)
 \end{aligned}$$

Now define $f(x)$ in $(-\infty, 0)$ such that $f(x)$ is an odd function of x in $(-\infty, \infty)$. Then obviously $f(x) \cos px$ is an odd function of x and $f(x) \sin px$ is an even function of x in $(-\infty, \infty)$.

$\therefore$ From (1), we have

$$f(x) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \sin px dp \times \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} f(x) \sin px dx$$

$$\text{or } f(x) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \tilde{f}_s(p) \sin px dp.$$

Note: According to the authors who define

$$\tilde{f}_s(p) = \int_0^{\infty} f(x) \sin px dx,$$

$$\text{we have } f(x) = \frac{2}{\pi} \int_0^{\infty} \tilde{f}_s(p) \sin px dp.$$

4.8 Fourier Cosine Transform

The infinite Fourier cosine transform of $f(x)$, $0 < x < \infty$, is denoted by $F_c \{f(x)\}$ or $\tilde{f}_c(p)$, and is defined as

$$F_c \{f(x)\} = \tilde{f}_c(p) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} f(x) \cos px dx.$$

The function $f(x)$ is called the inverse Fourier Cosine transform of $\tilde{f}_c(p)$

$$\text{i.e., } f(x) = F_c^{-1} \{ \tilde{f}_c(p) \}$$

Note: Some authors also define $\tilde{f}_c(p) = \int_0^{\infty} f(x) \cos px dx$.

4.9 Inversion Formula for Fourier Cosine Transform

If $\tilde{f}_c(p)$ is the Fourier cosine transform of the function $f(x)$ which satisfies the Dirichlet conditions in every finite interval $(0, l)$ and is such that $\int_0^{\infty} |f(x)| dx$ exists, then

$$f(x) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \tilde{f}_c(p) \cos px \, dp$$

at every point of continuity of $f(x)$.

This is an inversion formula for infinite Fourier cosine transform.

Proof: Proceeding as in article 4.7, we have

$$\begin{aligned} f(x) &= \frac{1}{\pi} \int_0^{\infty} \cos px \, dp \int_{-\infty}^{\infty} f(x) \cos px \, dx \\ &\quad + \frac{1}{\pi} \int_0^{\infty} \sin px \, dp \int_{-\infty}^{\infty} f(x) \sin px \, dx. \end{aligned} \quad \dots(1)$$

Now define $f(x)$ in $(-\infty, 0)$ such that $f(x)$ is an even function of x in $(-\infty, \infty)$. Then obviously $f(x) \cos px$ is an even function of x and $f(x) \sin px$ is an odd function of x in $(-\infty, \infty)$.

$\therefore$ from (1), we have

$$f(x) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \cos px \, dp \times \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} f(x) \cos px \, dx$$

or
$$f(x) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \tilde{f}_c(p) \cos px \, dp.$$

Note: According to the authors who define

$$\tilde{f}_c(p) = \int_0^{\infty} f(x) \cos px \, dx,$$

we have
$$f(x) = \frac{2}{\pi} \int_0^{\infty} \tilde{f}_c(p) \cos px \, dx.$$

4.10 Linearity Property of Fourier Transform

If $\tilde{f}(p)$ and $\tilde{g}(p)$ are Fourier transforms of $f(x)$ and $g(x)$ respectively, then

$$F\{a f(x) + b g(x)\} = a \tilde{f}(p) + b \tilde{g}(p),$$

where a and b are constants.

Proof: We have

$$F\{f(x)\} = \tilde{f}(p) = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipx} f(x) \, dx$$

and
$$F\{g(x)\} = \tilde{g}(p) = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipx} g(x) \, dx.$$

$$\begin{aligned} \therefore F\{a f(x) + b g(x)\} &= \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipx} \{af(x) + bg(x)\} \, dx \\ &= \frac{a}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipx} f(x) \, dx + \frac{b}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipx} g(x) \, dx \\ &= a \tilde{f}(p) + b \tilde{g}(p). \end{aligned}$$

4.11 Change of Scale Property

Theorem 1: (For Complex Fourier Transform). If $\tilde{f}(p)$ is the complex Fourier transform of $f(x)$, the complex Fourier transform of $f(ax)$ is $\frac{1}{a} \tilde{f}\left(\frac{p}{a}\right)$.

Proof. We have

$$\tilde{f}(p) = F\{f(x)\} = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipx} f(x) dx. \quad \dots(1)$$

Now

$$\begin{aligned} F\{f(ax)\} &= \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipx} f(ax) dx \\ &= \frac{1}{a} \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ip(t/a)} f(t) dt, \end{aligned}$$

putting $ax = t$ so that $dx = \frac{1}{a} dt$

$$\begin{aligned} &= \frac{1}{a} \cdot \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{i(p/a)t} f(t) dt \\ &= \frac{1}{a} \tilde{f}\left(\frac{p}{a}\right), \text{ from (1).} \end{aligned}$$

Theorem 2: (For Fourier Sine Transform). If $\tilde{f}_s(p)$ is the Fourier sine transform of $f(x)$, then the Fourier sine transform of $f(ax)$ is $\frac{1}{a} \tilde{f}_s\left(\frac{p}{a}\right)$.

Proof: We have $\tilde{f}_s(p) = F_s\{f(x)\}$

$$= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} f(x) \sin px dx \quad \dots(1)$$

$$\text{Now } F_s\{f(ax)\} = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} f(ax) \sin px dx$$

$$\begin{aligned} &= \frac{1}{a} \cdot \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} f(t) \cdot \sin\left(\frac{p}{a}t\right) dt, \text{ putting } ax = t \text{ so that } dx = (1/a) dt \\ &= \frac{1}{a} \tilde{f}_s\left(\frac{p}{a}\right), \text{ from (1).} \end{aligned}$$

Theorem 3: (For Fourier Cosine Transform). If $\tilde{f}_c(p)$ is the Fourier Cosine Transform of $f(x)$, then the Fourier Cosine transform of $f(ax)$ is $\frac{1}{a} \tilde{f}_c\left(\frac{p}{a}\right)$.

Proof: We have

$$\begin{aligned} \tilde{f}_c(p) &= F_c\{f(x)\} \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} f(x) \cos px dx \quad \dots(1) \end{aligned}$$

Now

$$\begin{aligned}
 F_c \{f(ax)\} &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(ax) \cos px \, dx \\
 &= \frac{1}{a} \cdot \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(t) \cos\left(\frac{p}{a}t\right) dt, \\
 &\qquad\qquad\qquad \text{putting } ax = t \text{ so that } dx = (1/a) dt \\
 &= \frac{1}{a} \tilde{f}_c\left(\frac{p}{a}\right), \text{ from (1).}
 \end{aligned}$$

4.12 Shifting Property

If $\tilde{f}(p)$ is the complex Fourier transform of $f(x)$, then the complex Fourier transform of $f(x-a)$ is $e^{ipa} \tilde{f}(p)$. (Purvanchal 2014)

Proof: We have

$$\tilde{f}(p) = F\{f(x)\} = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipx} f(x) \, dx \quad \dots(1)$$

Now

$$\begin{aligned}
 F\{f(x-a)\} &= \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipx} f(x-a) \, dx = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ip(a+t)} f(t) \, dt, \\
 &\qquad\qquad\qquad \text{putting } x-a = t \text{ so that } dx = dt \\
 &= e^{ipa} \cdot \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipt} f(t) \, dt \\
 &= e^{ipa} \tilde{f}(p), \text{ from (1).}
 \end{aligned}$$

4.13 Modulation Theorem

If $\tilde{f}(p)$ is the Complex Fourier transform of $f(x)$, then the Fourier transform of $f(x) \cos ax$ is

$$\frac{1}{2} [\tilde{f}(p-a) + \tilde{f}(p+a)].$$

Proof. We have

$$\tilde{f}(p) = F\{f(x)\} = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipx} f(x) \, dx \quad \dots(1)$$

Now

$$\begin{aligned}
 F\{f(x) \cos ax\} &= \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipx} \cdot f(x) \cdot \frac{e^{iax} + e^{-iax}}{2} \, dx \\
 &= \frac{1}{2} \left[\frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{i(p+a)x} f(x) \, dx + \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{i(p-a)x} f(x) \, dx \right] \\
 &= \frac{1}{2} [\tilde{f}(p+a) + \tilde{f}(p-a)].
 \end{aligned}$$

4.14 Important Results

Theorem: If $\tilde{f}_s(p)$ and $\tilde{f}_c(p)$ are Fourier sine and cosine transforms of $f(x)$ respectively, then

$$(i) \quad F_s \{ f(x) \cos ax \} = \frac{1}{2} [\tilde{f}_s(p+a) + \tilde{f}_s(p-a)]$$

$$(ii) \quad F_c \{ f(x) \sin ax \} = \frac{1}{2} [\tilde{f}_s(p+a) - \tilde{f}_s(p-a)]$$

$$(iii) \quad F_s \{ f(x) \sin ax \} = \frac{1}{2} [\tilde{f}_c(p-a) - \tilde{f}_c(p+a)].$$

Proof: (i) We have

$$\begin{aligned} F_s \{ f(x) \cos ax \} &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \cos ax \sin px \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{1}{2} \int_0^\infty f(x) \cdot [\sin(p+a)x + \sin(p-a)x] \, dx \\ &= \frac{1}{2} \left[\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \cdot \sin(p+a)x \, dx \right. \\ &\quad \left. + \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \cdot \sin(p-a)x \, dx \right] \\ &= \frac{1}{2} [\tilde{f}_s(p+a) + \tilde{f}_s(p-a)]. \end{aligned}$$

(ii) We have

$$\begin{aligned} F_c \{ f(x) \sin ax \} &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \sin ax \cos px \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{1}{2} \int_0^\infty f(x) [\sin(p+a)x - \sin(p-a)x] \, dx \\ &= \frac{1}{2} \left[\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \sin(p+a)x \, dx \right. \\ &\quad \left. - \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \cdot \sin(p-a)x \, dx \right] \\ &= \frac{1}{2} [\tilde{f}_s(p+a) - \tilde{f}_s(p-a)]. \end{aligned}$$

(iii) We have

$$\begin{aligned} F_s \{ f(x) \sin ax \} &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \sin ax \sin px \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{1}{2} \int_0^\infty f(x) [\cos(p-a)x - \cos(p+a)x] \, dx \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \left[\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \cos(p-a)x \, dx \right. \\
&\quad \left. - \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \cos(p+a)x \, dx \right] \\
&= \frac{1}{2} [\tilde{f}_c(p-a) - \tilde{f}_c(p+a)].
\end{aligned}$$

4.15 Theorem

If $\phi(p)$ is the Fourier sine transform of $f(x)$ for $p > 0$, then

$$F_s \{f(x)\} = -\phi(-p) \quad \text{for } p < 0.$$

Proof: We have

$$\begin{aligned}
F_s \{f(x)\} &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \sin px \, dx \\
&= \phi(p), \quad \text{for } p > 0.
\end{aligned} \tag{1}$$

For $p < 0$, let $p = -s$ where $s > 0$.

$$\begin{aligned}
\therefore F_s \{f(x)\} &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \sin(-sx) \, dx \\
&= -\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \sin sx \, dx = -\phi(s) \\
&= -\phi(-p), \quad \text{for } p < 0.
\end{aligned}$$

Hence in general

$$F_s \{f(x)\} = \begin{cases} \phi(|p|), & p > 0 \\ -\phi(|p|), & p < 0 \end{cases}$$

or $F_s \{f(x)\} = \phi(|p|) \cdot \text{Sgn } p,$

where the symbol $\text{Sgn } p = \begin{cases} +1, & p > 0 \\ -1, & p < 0. \end{cases}$

4.16 Multiple Fourier Transforms

Let $f(x, y)$ be a function of two variables x and y . Regarding $f(x, y)$, temporarily, as a function of x , its Fourier transform is

$$\tilde{f}(p, y) = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f(x, y) e^{ipx} \, dx.$$

Now regarding $\tilde{f}(p, y)$ as a function of y , its Fourier transform is

$$\tilde{F}(p, q) = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \tilde{f}(p, y) e^{ipy} \, dy$$

or
$$\tilde{F}(p, q) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{i(px + qy)} dx dy$$

which is Fourier transform of $f(x, y)$.

Inversion formula: Using inversion formula for Fourier transforms, we have

$$f(x, y) = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \tilde{f}(p, y) e^{-ipx} dp$$

and
$$\tilde{f}(p, y) = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \tilde{F}(p, q) e^{-iqy} dq.$$

Hence
$$f(x, y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{F}(p, q) e^{-i(px + qy)} dp dq,$$

which is the inversion formula for the Fourier transform of $f(x, y)$.

4.17 Convolution

The function

$$H(x) = F * G = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} F(u) \cdot G(x - u) du$$

is called the **convolution** or **Falting** of two integrable functions F and G over the interval $(-\infty, \infty)$.

Note: Some authors also define

$$F * G = \int_{-\infty}^{\infty} F(u) \cdot G(x - u) du.$$

4.18 The Convolution or Falting Theorem for Fourier Transforms

If $F\{f(x)\}$ and $F\{g(x)\}$ are the Fourier transforms of the functions $f(x)$ and $g(x)$ respectively, then the Fourier transform of the convolution of $f(x)$ and $g(x)$ is the product of their Fourier transforms

i.e.
$$F\{f(x) * g(x)\} = F\{f(x)\} \cdot F\{g(x)\}.$$

Proof: We have $F\{f(x) * g(x)\}$

$$\begin{aligned} &= F\left\{\frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f(u) \cdot g(x - u) du\right\} \\ &= \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f(u) g(x - u) du \right] e^{ipx} dx \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(u) \left[\int_{-\infty}^{\infty} g(x - u) e^{ipx} dx \right] du \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(u) \left[\int_{-\infty}^{\infty} g(y) e^{ip(u+y)} dy \right] du, \\
 &\quad \text{putting } x - u = y \text{ so that } dx = dy, \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(u) \left[\int_{-\infty}^{\infty} e^{ipu} g(y) e^{ipy} dy \right] du \\
 &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f(u) \left[e^{ipu} \int_{-\infty}^{\infty} g(y) e^{ipy} dy \right] du \\
 &= \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f(u) \left[e^{ipu} \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} g(x) e^{ipx} dx \right] du \\
 &= \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f(u) [e^{ipu} F\{g(x)\}] du \\
 &= \left[\frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f(u) e^{ipu} du \right] F\{g(x)\} \\
 &= \left[\frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f(x) e^{ipx} dx \right] F\{g(x)\} \\
 &= F\{f(x)\} \cdot F\{g(x)\}.
 \end{aligned}$$

4.19 Relationship Between Fourier and Laplace Transforms

Let us consider the function

$$f(t) = \begin{cases} e^{-x t} g(t), & t > 0 \\ 0, & t < 0. \end{cases} \quad \dots(1)$$

$\therefore$ The Fourier transform of $f(t)$ is given by

$$\begin{aligned}
 F\{f(t)\} &= \int_{-\infty}^{\infty} e^{ipt} \cdot f(t) dt, \quad (\text{Taking non-symmetrical form of F. T.}) \\
 &= \int_{-\infty}^0 0 \cdot e^{ipt} dt + \int_0^{\infty} e^{-xt} g(t) \cdot e^{ipt} dt \\
 &= \int_0^{\infty} e^{(ip-x)t} g(t) dt = \int_0^{\infty} e^{-st} g(t) dt, \quad \text{putting } x - ip = s \\
 &= L\{g(t)\}.
 \end{aligned}$$

Hence the Fourier transform of the function $f(t)$ defined by (1) is the Laplace transform of the function $g(t)$.

4.20 Fourier Transform of The Derivatives of A Function

(a) The Fourier transform of $f'(x)$, the derivative of $f(x)$ is $-ip \tilde{f}(p)$, where $\tilde{f}(p)$ is the Fourier transform of $f(x)$.

Proof: By definition

$$\begin{aligned}
 F\{f'(x)\} &= \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f'(x) \cdot e^{ipx} dx \\
 &= \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \cdot e^{ipx} dx \\
 &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{f(x+h)}{h} \cdot e^{ipx} dx - \lim_{h \rightarrow 0} \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{f(x)}{h} \cdot e^{ipx} dx \\
 &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{f(x+h)}{h} \cdot e^{ip(x+h)} e^{-iph} dx - \lim_{h \rightarrow 0} \frac{\tilde{f}(p)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{e^{-iph}}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{1}{h} \cdot f(y) e^{ipy} dy - \lim_{h \rightarrow 0} \frac{\tilde{f}(p)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{e^{-iph} \tilde{f}(p)}{h} - \lim_{h \rightarrow 0} \frac{\tilde{f}(p)}{h} \\
 &= \tilde{f}(p) \cdot \lim_{h \rightarrow 0} \frac{e^{-iph} - 1}{h} = (-ip) \tilde{f}(p).
 \end{aligned}$$

(b) The Fourier transform of $f^n(x)$, the n th derivative of $f(x)$ is $(-ip)^n$ times the Fourier transform of $f(x)$ provided that the first $(n-1)$ derivatives of $f(x)$ vanish as $x \rightarrow \pm \infty$.

Proof: By definition

$$F\{f^n(x)\} = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f^n(x) \cdot e^{ipx} dx.$$

Integrating by parts, we have

$$\begin{aligned}
 F\{f^n(x)\} &= \frac{1}{\sqrt{(2\pi)}} [f^{n-1}(x) \cdot e^{ipx}]_{-\infty}^{\infty} - \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f^{n-1}(x) ip e^{ipx} dx \\
 &= \frac{(-ip)}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f^{n-1}(x) e^{ipx} dx, \text{ since } \lim_{x \rightarrow \pm \infty} f^{n-1}(x) = 0.
 \end{aligned}$$

Repeating the same process of integration by parts $(n-1)$ times more, we have

$$F\{f^n(x)\} = (-ip)^n \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f(x) e^{ipx} dx$$

or

$$\tilde{f}^n(p) = (-ip)^n \tilde{f}(p).$$

(c) The Fourier cosine and sine transforms of the derivatives of $f(x)$ are given by

$$\tilde{f}_c^{2n}(p) = -\sqrt{\left(\frac{2}{\pi}\right)^{n-1}} \sum_{r=0}^{n-1} (-1)^r \alpha_{2n-2r-1} p^{2r} + (-1)^n p^{2n} \tilde{f}_c(p);$$

$$\tilde{f}_c^{2n+1}(p) = -\sqrt{\left(\frac{2}{\pi}\right)} \sum_{r=1}^n (-1)^r \alpha_{2n-2r} p^{2r} + (-1)^n p^{2n+1} \tilde{f}_s(p);$$

$$\tilde{f}_s^{2n}(p) = -\sqrt{\left(\frac{2}{\pi}\right)} \sum_{r=1}^n (-1)^r \alpha_{2n-2r} p^{2r-1} + (-1)^{n+1} p^{2n} \tilde{f}_c(p);$$

and

$$\tilde{f}_c^{2n+1}(p) = -\sqrt{\left(\frac{2}{\pi}\right)} \sum_{r=1}^n (-1)^r \alpha_{2n-2r-1} p^{2r-1} + (-1)^{n+1} p^{2n+1} \tilde{f}_c(p);$$

provided that first $(n-1)$ derivatives of $f(x)$ vanish as $x \rightarrow \infty$ and

$$\frac{d^{n-1}f}{dx^{n-1}} \rightarrow \alpha_{n-1} \text{ etc. as } x \rightarrow 0.$$

Proof: By definition, we have

$$\tilde{f}_c^n(p) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f^n(x) \cos px \, dx \quad \dots(1)$$

and

$$\tilde{f}_s^n(p) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f^n(x) \sin px \, dx. \quad \dots(2)$$

Integrating R.H.S. of (1) by parts, we have

$$\tilde{f}_c^n(p) = \sqrt{\left(\frac{2}{\pi}\right)} [f^{n-1}(x) \cos px]_0^\infty + p \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f^{n-1}(x) \sin px \, dx$$

or

$$\tilde{f}_c^n(p) = -\sqrt{\left(\frac{2}{\pi}\right)} \alpha_{n-1} + p \tilde{f}_s^{n-1}(p). \quad \dots(3)$$

Similarly integrating R.H.S. of (2), we have

$$\tilde{f}_s^n(p) = -p \tilde{f}_c^{n-1}(p). \quad \dots(4)$$

From (3) and (4), we have

$$\tilde{f}_c^n(p) = -\sqrt{\left(\frac{2}{\pi}\right)} \alpha_{n-1} - p^2 \tilde{f}_n^{n-2}(p). \quad \dots(5)$$

By repeated application of these rules $\tilde{f}_c^n(p)$ is obtained as a sum of α 's and $\tilde{f}_c'(p)$ or $\tilde{f}_c(p)$.

It is clear that $\tilde{f}_c'(p)$ will occur when n is odd and in that case it may be replaced by

$$-\sqrt{\left(\frac{2}{\pi}\right)} \alpha_0 + p \tilde{f}_s(p).$$

Thus, we have

$$\tilde{f}_c^{2n}(p) = -\sqrt{\left(\frac{2}{\pi}\right)} \sum_{r=0}^{n-1} (-1)^r \alpha_{2n-2r-1} p^{2r} + (-1)^n p^{2n} \tilde{f}_c(p)$$

and
$$\widetilde{f}_c^{2n+1}(p) = -\sqrt{\left(\frac{2}{\pi}\right)} \sum_{r=0}^n (-1)^r \alpha_{2n-2r} p^{2r} + (-1)^n p^{2n+1} \widetilde{f}_s(p)$$

Also from (3) and (4), we have

$$\widetilde{f}_s^n(p) = -p \left[-\sqrt{\left(\frac{2}{\pi}\right)} \alpha_{n-2} + p \widetilde{f}_s^{n-2}(p) \right]$$

or
$$\widetilde{f}_s^n(p) = p \sqrt{\left(\frac{2}{\pi}\right)} \alpha_{n-2} - p^2 \widetilde{f}_s^{n-2}(p).$$

By repeated application of these results $\widetilde{f}_s^n(p)$ is obtained as a sum of α 's and $\widetilde{f}_s'(p)$ or $\widetilde{f}_s(p)$.

It is clear that $\widetilde{f}_s'(p)$ will occur when n is odd and it may be replaced by $-p \widetilde{f}_c(p)$.

Thus, we have

$$\widetilde{f}_s^{2n}(p) = -\sqrt{\left(\frac{2}{\pi}\right)} \sum_{r=1}^n (-1)^r \alpha_{2n-2r} p^{2r-1} + (-1)^{n+1} p^{2n} \widetilde{f}_s(p)$$

and
$$\widetilde{f}_c^{2n+1}(p) = -\sqrt{\left(\frac{2}{\pi}\right)} \sum_{r=1}^n (-1)^r \alpha_{2n-2r+1} p^{2r-1} + (-1)^{n+1} p^{2n+1} \widetilde{f}_c(p).$$

Note. The infinite sine and cosine transforms can be applied when the range of the variable selected for exclusion is 0 to ∞ .

Illustrative Examples

Example 1: Find the Fourier complex transform of $f(x)$, if

$$f(x) = \begin{cases} e^{i\omega x} & a < x < b \\ 0 & x < a, x > b. \end{cases} \quad (\text{Meerut 2013B; Avadh 14})$$

Solution: We have

$$\begin{aligned} F\{f(x)\} &= \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} e^{ipx} f(x) dx \\ &= \frac{1}{\sqrt{(2\pi)}} \left[\int_{-\infty}^a 0 \cdot e^{ipx} dx + \int_a^b e^{ipx} \cdot e^{i\omega x} dx + \int_b^{\infty} 0 \cdot e^{ipx} dx \right] \\ &= \frac{1}{\sqrt{(2\pi)}} \int_a^b e^{i(p+\omega)x} dx = \frac{1}{\sqrt{(2\pi)}} \cdot \left[\frac{e^{i(p+\omega)x}}{i(p+\omega)} \right]_a^b \\ &= \frac{1}{\sqrt{(2\pi)}} \left[\frac{e^{i(p+\omega)a} - e^{i(p+\omega)b}}{p+\omega} \right]. \end{aligned}$$

Example 2: Find the Fourier transform of $F(x)$ defined by

$$F(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a. \end{cases}$$

(Kanpur 2008; Meerut 13; Bundelkhand 13; Avadh 13)

and hence evaluate

$$(a) \int_{-\infty}^{\infty} \frac{\sin pa \cos px}{p} dp, \quad \text{and} \quad (b) \int_0^{\infty} \frac{\sin p}{p} dp.$$

Solution: We have

$$\begin{aligned} \tilde{F}(p) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} F(x) dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a e^{ipx} dx = \frac{1}{\sqrt{2\pi}} \left[\frac{e^{ipx}}{ip} \right]_{-a}^a \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{e^{ipa}}{ip} - \frac{e^{-ipa}}{ip} \right) \\ &= \frac{2i \sin pa}{ip \sqrt{2\pi}} = \frac{2 \sin pa}{p \sqrt{2\pi}}, p \neq 0. \end{aligned}$$

For $p = 0$, $\tilde{F}(p) = 2a / \sqrt{2\pi}$.

(a) We know that if

$$\tilde{F}(p) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(x) e^{ipx} dx$$

$$\text{then} \quad F(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{F}(p) e^{-ipx} dp.$$

$$\therefore \quad \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{2 \sin pa}{p \sqrt{2\pi}} \cdot e^{-ipx} dp = \begin{cases} 1, & |x| < a \\ 0, & |x| > a. \end{cases}$$

$$\begin{aligned} \text{But L.H.S.} &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin pa \cos px}{p} dp - \frac{i}{\pi} \int_{-\infty}^{\infty} \frac{\sin pa \sin px}{p} dp \\ &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin pa \cos px}{p} dp, \end{aligned}$$

since the integrand in the other integral is an odd function of p .

$$\therefore \quad \int_{-\infty}^{\infty} \frac{\sin pa \cos px}{p} dp = \begin{cases} \pi, & |x| < a \\ 0, & |x| > a. \end{cases}$$

(b) If $x = 0$ and $a = 1$ in (a), then

$$\int_{-\infty}^{\infty} \frac{\sin p}{p} dp = \pi$$

$$\text{or} \quad 2 \int_0^{\infty} \frac{\sin p}{p} dp = \pi \quad \text{or} \quad \int_0^{\infty} \frac{\sin p}{p} dp = \frac{\pi}{2}. \quad \dots(1)$$

Note: Putting $x = 0$ in (a) and on simplification, we get

$$\int_0^{\infty} \frac{\sin ap}{p} dp = \frac{\pi}{2}. \quad \dots(2)$$

The results (1) and (2) can be used as standard formulae.

Example 3: Find Fourier sine and cosine transforms of e^{-x} and using the inversion formulae recover the original functions, in both the cases. (Kanpur 2009)

Solution: Let $f(x) = e^{-x}$.

$$\begin{aligned} \text{Then } \tilde{f}_s(p) &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} f(x) \sin px \, dx = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} e^{-x} \sin px \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \left[\frac{e^{-x}}{1+p^2} (-\sin px - p \cos px) \right]_0^{\infty} \\ &= \frac{p}{1+p^2} \sqrt{\left(\frac{2}{\pi}\right)} \end{aligned}$$

$$\begin{aligned} \text{and } \tilde{f}_c(p) &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} f(x) \cos px \, dx = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} e^{-x} \cos px \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \left[\frac{e^{-x}}{1+p^2} (-\cos px + p \sin px) \right]_0^{\infty} \\ &= \frac{1}{1+p^2} \sqrt{\left(\frac{2}{\pi}\right)}. \end{aligned}$$

Applying inversion to the sine transform, we have

$$\begin{aligned} f(x) &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \tilde{f}_s(p) \cdot \sin px \, dp \\ &= \frac{2}{\pi} \int_0^{\infty} \frac{p \sin px}{1+p^2} dp \quad \dots(1) \end{aligned}$$

and applying inversion to the cosine transform, we have

$$\begin{aligned} f(x) &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \tilde{f}_c(p) \cos px \, dp \\ &= \frac{2}{\pi} \int_0^{\infty} \frac{\cos px}{1+p^2} dp. \quad \dots(2) \end{aligned}$$

Now from Fourier integral theorem, we have

$$f(x) = \frac{1}{\pi} \int_0^{\infty} dp \int_{-\infty}^{\infty} f(v) \cos p(x-v) \, dv$$

$$\begin{aligned} \text{or } f(x) &= \frac{1}{\pi} \int_0^{\infty} \cos px \, dp \int_{-\infty}^{\infty} f(v) \cos pv \, dv \\ &\quad + \frac{1}{\pi} \int_0^{\infty} \sin px \, dp \int_{-\infty}^{\infty} f(v) \sin pv \, dv. \quad \dots(3) \end{aligned}$$

Case I: Defining $f(x)$ in $(-\infty, 0)$ such that $f(x)$ is an even function of x , from (3), we have

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \cos px \, dp \int_0^{\infty} f(v) \cos pv \, dv.$$

Taking $f(x) = e^{-x}$, we have

$$\begin{aligned} e^{-x} &= \frac{2}{\pi} \int_0^{\infty} \cos px \, dp \int_0^{\infty} e^{-v} \cos pv \, dv \\ &= \frac{2}{\pi} \int_0^{\infty} \cos px \left[\frac{e^{-v}}{1+p^2} (-\cos pv + p \sin pv) \right]_0^{\infty} dp \\ &= \frac{2}{\pi} \int_0^{\infty} \frac{\cos px}{1+p^2} dp. \end{aligned}$$

$$\therefore \int_0^{\infty} \frac{\cos px}{1+p^2} dp = \frac{\pi}{2} e^{-x}.$$

$$\therefore \text{from (2) we have } f(x) = \frac{2}{\pi} \cdot \frac{\pi}{2} e^{-x} = e^{-x}.$$

Case II: Again defining $f(x)$ in $(-\infty, 0)$ such that $f(x)$ is an odd function of x , from (2), we have

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \sin px \, dp \int_0^{\infty} f(v) \sin pv \, dv.$$

Taking $f(x) = e^{-x}$ and simplifying, we have

$$\int_0^{\infty} \frac{p \sin px}{1+p^2} dp = \frac{\pi}{2} e^{-x}.$$

$$\therefore \text{from (1), } f(x) = \frac{2}{\pi} \cdot \frac{\pi}{2} e^{-x} = e^{-x}.$$

Example 4: Find Fourier cosine transform of $f(x) = \frac{1}{1+x^2}$ and hence find Fourier sine

transform of $F(x) = \frac{x}{1+x^2}$. (Kanpur 2007, 10; Meerut 13; Rohilkhand 14)

Solution: We have

$$\begin{aligned} \tilde{f}_c(p) &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} f(x) \cos px \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \frac{\cos px}{1+x^2} dx. \end{aligned}$$

Differentiating both sides w.r.t. p , we have

$$\frac{d}{dp} \tilde{f}_c(p) = -\sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \frac{x \sin px}{1+x^2} dx$$

$$\begin{aligned}
&= -\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{(x^2 + 1 - 1) \sin px}{x(1 + x^2)} dx \\
&= -\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\sin px}{x} dx + \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\sin px}{x(1 + x^2)} dx \\
&= -\sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{\pi}{2} + \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\sin px}{x(1 + x^2)} dx. \quad \left[\because \int_0^\infty \frac{\sin px}{x} dx = \frac{\pi}{2} \right]
\end{aligned}$$

Differentiating again w.r.t. p , we have

$$\frac{d^2}{dp^2} \tilde{f}_c(p) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\cos px}{1 + x^2} dx = \tilde{f}_c(p)$$

or $(D^2 - 1) \tilde{f}_c(p) = 0$

whose general solution is

$$\tilde{f}_c(p) = Ae^p + Be^{-p}. \quad \dots(1)$$

Now when $p = 0$,

$$\begin{aligned}
\tilde{f}_c(p) &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{dx}{1 + x^2} = \sqrt{\left(\frac{2}{\pi}\right)} [\tan^{-1} x]_0^\infty \\
&= \frac{\pi}{2} \sqrt{\left(\frac{2}{\pi}\right)} = \sqrt{\left(\frac{\pi}{2}\right)}
\end{aligned}$$

and $\frac{d}{dp} \tilde{f}_c(p) = -\sqrt{\left(\frac{\pi}{2}\right)}.$

$\therefore$ from (1), we have $\sqrt{\left(\frac{\pi}{2}\right)} = A + B$

and $-\sqrt{\left(\frac{\pi}{2}\right)} = A - B.$

Solving, $A = 0, B = \sqrt{\left(\frac{\pi}{2}\right)}.$

$\therefore$ from (1), we have $\tilde{f}_c(p) = \sqrt{\left(\frac{\pi}{2}\right)} e^{-p}.$

Second part: We have

$$\tilde{f}_c(p) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\cos px}{1 + x^2} dx = \sqrt{\left(\frac{\pi}{2}\right)} \cdot e^{-p}.$$

Now differentiating both sides w.r.t. p , we have

$$-\int_0^\infty \frac{x \sin px}{1 + x^2} dx = -\frac{\pi}{2} \cdot e^{-p}.$$

$\therefore \tilde{F}_s(p) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{x}{1 + x^2} \sin px dx = \sqrt{\left(\frac{\pi}{2}\right)} e^{-p}.$

Example 5: Find the sine and cosine transforms of $x^n e^{-ax}$.

Solution: Let $f(x) = x^n e^{-ax}$.

$$\begin{aligned} \therefore \quad \widetilde{f}_s(p) &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \sin px \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty x^n e^{-ax} \sin px \, dx \end{aligned} \quad \dots(1)$$

$$\begin{aligned} \text{We have } \int_0^\infty e^{-ax} \sin px \, dx &= \left[\frac{e^{-ax}}{a^2 + p^2} (-a \sin px - p \cos px) \right]_0^\infty \\ &= \frac{p}{a^2 + p^2} = \frac{1}{2i} \left(\frac{1}{a - ip} - \frac{1}{a + ip} \right). \end{aligned}$$

Differentiating both sides w.r.t. a, n times, we have

$$\begin{aligned} &(-1)^n \int_0^\infty x^n e^{-ax} \sin px \, dx \\ &= \frac{1}{2i} \left[\frac{d^n}{da^n} (a - ip)^{-1} - \frac{d^n}{da^n} (a + ip)^{-1} \right] \\ &= \frac{1}{2i} (-1)^n (n!) [(a - ip)^{-(n+1)} - (a + ip)^{-(n+1)}] \\ &= \frac{1}{2i} (-1)^n (n!) [2i r^{-(n+1)} \sin(n+1)\theta], \text{ putting } a = r \cos \theta, p = r \sin \theta \\ &= (-1)^n n! (1/r)^{n+1} \sin(n+1)\theta. \end{aligned}$$

$$\begin{aligned} \therefore \quad \int_0^\infty x^n e^{-ax} \sin px \, dx &= (n!) \cdot [1 / (a^2 + p^2)^{(n+1)/2}] \sin \{(n+1) \tan^{-1}(p/a)\}. \\ &[\because r = (a^2 + p^2)^{1/2} \text{ and } \theta = \tan^{-1}(p/a)] \end{aligned}$$

Hence, from (1),

$$\widetilde{f}_s(p) = \sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{n! \sin \{(n+1) \tan^{-1}(p/a)\}}{(a^2 + p^2)^{(n+1)/2}}$$

$$\begin{aligned} \text{Also } \widetilde{f}_c(p) &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \cos px \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty x^n e^{-ax} \cos px \, dx. \end{aligned} \quad \dots(2)$$

$$\begin{aligned} \text{We have, } \int_0^\infty e^{-ax} \cos px \, dx &= \left[\frac{e^{-ax}}{a^2 + p^2} (-a \cos px + p \sin px) \right]_0^\infty \\ &= \frac{a}{a^2 + p^2} \\ &= \frac{1}{2} \left(\frac{1}{a - ip} + \frac{1}{a + ip} \right). \end{aligned}$$

Differentiating both sides w.r.t. a , n times, we have

$$\begin{aligned}
 & (-1)^n \int_0^\infty x^n e^{-ax} \cos px \, dx \\
 &= \frac{1}{2} (-1)^n (n!) [(a - ip)^{-(n+1)} + (a + ip)^{-(n+1)}] \\
 &= (-1)^n (n!) (1/r)^{n+1} \cos (n+1) \theta, \\
 &\quad \text{putting } a = r \cos \theta, p = r \sin \theta \text{ and on simplification.}
 \end{aligned}$$

$$\therefore \int_0^\infty x^n e^{-ax} \cos px \, dx = (n!) \cdot \frac{\cos \{(n+1) \tan^{-1} (p/a)\}}{(a^2 + p^2)^{(n+1)/2}}.$$

Hence from (2), we have

$$\tilde{f}_c(p) = \sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{n! \cos \{(n+1) \tan^{-1} (p/a)\}}{(a^2 + p^2)^{(n+1)/2}}.$$

Alternative method to evaluate the integrals

$$\int_0^\infty e^{-ax} x^n \sin px \, dx \text{ and } \int_0^\infty e^{-ax} x^n \cos px \, dx.$$

$$\begin{aligned}
 \text{We have } & \int_0^\infty e^{-ax} x^n \cos px \, dx + i \int_0^\infty e^{-ax} x^n \sin px \, dx \\
 &= \int_0^\infty e^{-ax} x^n (\cos px + i \sin px) \, dx \\
 &= \int_0^\infty e^{-ax} x^n e^{ipx} \, dx = \int_0^\infty e^{-(a - ip)x} x^{(n+1)-1} \, dx \\
 &= \frac{\Gamma(n+1)}{(a - ip)^{n+1}} \quad \left[\because \int_0^\infty e^{-az} z^{n-1} \, dz = \frac{\Gamma(n)}{a^n} \right] \\
 &= \frac{\Gamma(n+1)}{r^{n+1} (\cos \theta - i \sin \theta)^{n+1}}, \text{ putting } a = r \cos \theta \text{ and } p = r \sin \theta \\
 &= \frac{n!}{r^{n+1}} (\cos \theta + i \sin \theta)^{n+1} \quad \left[\because \frac{1}{\cos \theta - i \sin \theta} = \cos \theta + i \sin \theta \right] \\
 &= \frac{n!}{r^{n+1}} [\cos (n+1) \theta + i \sin (n+1) \theta]. \quad \dots(1)
 \end{aligned}$$

Equating real and imaginary parts on both sides of (1), we have

$$\int_0^\infty e^{-ax} x^n \cos px \, dx = \frac{n!}{r^{n+1}} \cos (n+1) \theta$$

$$\text{and } \int_0^\infty e^{-ax} x^n \sin px \, dx = \frac{n!}{r^{n+1}} \sin (n+1) \theta,$$

$$\text{where } r^2 = a^2 + p^2 \text{ i.e., } r = (a^2 + p^2)^{1/2} \text{ and } \theta = \tan^{-1} (p/a).$$

Hence

$$\begin{aligned} & \int_0^{\infty} e^{-ax} x^n \cos px \, dx \\ &= \frac{n!}{(a^2 + p^2)^{(n+1)/2}} \cos \{(n+1) \tan^{-1} (p/a)\} \end{aligned}$$

and

$$\begin{aligned} & \int_0^{\infty} e^{-ax} x^n \sin px \, dx \\ &= \frac{n!}{(a^2 + p^2)^{(n+1)/2}} \sin \{(n+1) \tan^{-1} (p/a)\}. \end{aligned}$$

Example 6: Find the sine transform of

$$\frac{e^{ax} + e^{-ax}}{e^{\pi x} - e^{-\pi x}}.$$

(Meerut 2013B; Purvanchal 14; Kanpur 14)

Solution: If $f(x) = \frac{e^{ax} + e^{-ax}}{e^{\pi x} - e^{-\pi x}}$, then we have

$$\begin{aligned} \tilde{f}_s(p) &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} f(x) \sin px \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \frac{e^{ax} + e^{-ax}}{e^{\pi x} - e^{-\pi x}} \sin px \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \frac{e^{ax} + e^{-ax}}{e^{\pi x} - e^{-\pi x}} \cdot \frac{e^{ipx} - e^{-ipx}}{2i} \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \left[\frac{1}{2i} \int_0^{\infty} \frac{e^{(a+ip)x} - e^{-(a+ip)x}}{e^{\pi x} - e^{-\pi x}} \, dx \right. \\ &\quad \left. - \frac{1}{2i} \int_0^{\infty} \frac{e^{(a-ip)x} - e^{-(a-ip)x}}{e^{\pi x} - e^{-\pi x}} \, dx \right] \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \left[\frac{1}{2i} \cdot \frac{1}{2} \tan \frac{a+ip}{2} - \frac{1}{2i} \cdot \frac{1}{2} \tan \frac{a-ip}{2} \right] \\ &\quad \left[\because \text{From definite integrals, } \int_0^{\infty} \frac{e^{az} - e^{-az}}{e^{\pi z} - e^{-\pi z}} \, dz = \frac{1}{2} \tan \frac{a}{2} \right] \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \left[\frac{1}{4i} \frac{\sin \frac{a+ip}{2}}{\cos \frac{a+ip}{2}} - \frac{1}{4i} \frac{\sin \frac{a-ip}{2}}{\cos \frac{a-ip}{2}} \right] \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \frac{\sin \frac{a+ip}{2} \cos \frac{a-ip}{2} - \sin \frac{a-ip}{2} \cos \frac{a+ip}{2}}{4i \cos \frac{a+ip}{2} \cos \frac{a-ip}{2}} \end{aligned}$$

$$\begin{aligned}
 &= \sqrt{\left(\frac{2}{\pi}\right)} \frac{\sin a + \sin ip - (\sin a - \sin ip)}{2 \cdot 2i [\cos ip + \cos a]} \\
 &= \frac{\sin ip}{\sqrt{(2\pi) i [\cos ip + \cos a]}} = \frac{\sinh p}{\sqrt{(2\pi) (\cosh p + \cos a)}} \\
 &= \frac{e^p - e^{-p}}{\sqrt{(2\pi) (e^p + e^{-p} + 2 \cos a)}}.
 \end{aligned}$$

Example 7: Find the Fourier sine transform of

$$f(x) = \frac{1}{x(a^2 + x^2)}.$$

Solution: We have

$$\tilde{f}_s(p) = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{1}{x(a^2 + x^2)} \sin px \, dx. \quad \dots(1)$$

$$\text{Let} \quad I = \int_0^\infty \frac{1}{x(a^2 + x^2)} \sin px \, dx. \quad \dots(2)$$

$$\begin{aligned}
 \text{Then} \quad \frac{dI}{dp} &= \frac{d}{dp} \int_0^\infty \frac{\sin px}{x(a^2 + x^2)} \, dx \\
 &= \int_0^\infty \left[\frac{\partial}{\partial p} \left\{ \frac{\sin px}{x(a^2 + x^2)} \right\} \right] dx \\
 &= \int_0^\infty \frac{\cos px}{a^2 + x^2} \, dx. \quad \dots(3)
 \end{aligned}$$

$$\begin{aligned}
 \therefore \quad \frac{d^2 I}{dp^2} &= - \int_0^\infty \frac{x \sin px}{a^2 + x^2} \, dx \\
 &= - \int_0^\infty \frac{x^2 \sin px}{x(a^2 + x^2)} \, dx = - \int_0^\infty \frac{(x^2 + a^2) - a^2}{x(a^2 + x^2)} \sin px \, dx \\
 &= - \int_0^\infty \frac{\sin px}{x} \, dx + a^2 \int_0^\infty \frac{\sin px}{x(a^2 + x^2)} \, dx \\
 &= -\frac{\pi}{2} + a^2 I. \quad \left[\because \int_0^\infty \frac{\sin px}{x} \, dx = \frac{\pi}{2} \right].
 \end{aligned}$$

$$\therefore \quad \frac{d^2 I}{dp^2} - a^2 I = -\frac{\pi}{2}$$

$$\text{or} \quad (D^2 - a^2) I = -\frac{\pi}{2}, \text{ where } D \equiv \frac{d}{dp}.$$

The solution of the above differential equation is

$$I = Ae^{-ap} + Be^{ap} + \frac{\pi}{2a^2}. \quad \dots(4)$$

$$\therefore \frac{dI}{dp} = -Aae^{-ap} + Bae^{ap}. \quad \dots(5)$$

Now from (2), when $p=0$, we have $I=0$ and from (3), when $p=0$, we have

$$\frac{dI}{dp} = \int_0^\infty \frac{1}{a^2 + x^2} dx = \frac{1}{a} \left[\tan^{-1} \frac{x}{a} \right]_0^\infty = \frac{\pi}{2a}.$$

So putting $p=0$ in (4) and (5), we get

$$A + B = -\frac{\pi}{2a^2} \quad \dots(6)$$

and $a(-A + B) = \frac{\pi}{2a}$ i.e., $-A + B = \frac{\pi}{2a^2}$ (7)

Solving (6) and (7), we get $B=0$, $A = -\frac{\pi}{2a^2}$.

Putting the values of A and B in (4), we get

$$I = \int_0^\infty \frac{\sin px}{x(a^2 + x^2)} dx = -\frac{\pi}{2a^2} e^{-ap} + \frac{\pi}{2a^2} = \frac{\pi}{2a^2} (1 - e^{-ap}).$$

Now putting the value of I in (1), we get

$$\tilde{f}_s(p) = \sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{\pi}{2a^2} (1 - e^{-ap}) = \frac{1}{a^2} \sqrt{\left(\frac{\pi}{2}\right)} \cdot (1 - e^{-ap}).$$

Example 8: Find the Fourier cosine transform of e^{-x^2} .

Solution: We have

$$\tilde{F}_c \{e^{-x^2}\} = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty e^{-x^2} \cos px dx = I \quad \dots(1)$$

Differentiating w.r.t. ' p ' we have

$$\begin{aligned} \frac{dI}{dp} &= -\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty x e^{-x^2} \sin px dx \\ &= \frac{1}{2} \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty (-2 x e^{-x^2}) \cdot \sin px dx \\ &= \frac{1}{2} \sqrt{\left(\frac{2}{\pi}\right)} \left[(e^{-x^2} \sin px)_0^\infty - p \int_0^\infty e^{-x^2} \cos px dx \right] \\ &\quad \text{(Integrating by parts taking } \sin px \text{ as first function)} \\ &= -\frac{p}{2} I. \quad \therefore \quad \frac{dI}{I} = -\frac{p}{2} dp. \end{aligned}$$

Integrating, we have

$$\log I = -\frac{p^2}{4} + \log A \quad \text{or} \quad I = A e^{-p^2/4}. \quad \dots(2)$$

But when $p=0$, from (1) $I = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty e^{-x^2} dx = \frac{1}{\sqrt{2}}$.

$\therefore$ from (2), $A = 1/\sqrt{2}$.

Hence $I = F_c \{e^{-x^2}\} = (1/\sqrt{2}) e^{-p^2/4}$.

Example 9: Use the sine inversion formula to obtain $f(x)$ if $\tilde{f}_s(p) = \frac{p}{1+p^2}$.

Solution: Using Fourier sine inversion formula, we have

$$\begin{aligned} f(x) &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{p}{1+p^2} \cdot \sin px \, dp \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{p^2}{p(1+p^2)} \cdot \sin px \, dp = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{(p^2+1)-1}{p(1+p^2)} \cdot \sin px \, dp \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\sin px}{p} \, dp - \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\sin px}{p(1+p^2)} \, dp \end{aligned}$$

$$\text{or} \quad f(x) = \sqrt{\left(\frac{\pi}{2}\right)} - \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\sin px}{p(1+p^2)} \, dp \quad \dots(1)$$

$$\left[\because \int_0^\infty \frac{\sin px}{p} \, dp = \frac{\pi}{2} \right]$$

$$\therefore \quad \frac{df}{dx} = -\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\cos px}{1+p^2} \, dp \quad \dots(2)$$

$$\text{and} \quad \frac{d^2 f}{dx^2} = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{p \sin px}{1+p^2} \, dp$$

$$\text{or} \quad \frac{d^2 f}{dx^2} - f = 0$$

$$\text{whose solution is } f = A e^x + B e^{-x}. \quad \dots(3)$$

$$\therefore \quad \frac{df}{dx} = A e^x - B e^{-x}. \quad \dots(4)$$

Now when $x=0$, $f = \sqrt{\left(\frac{\pi}{2}\right)}$, from (1)

$$\text{and} \quad \frac{df}{dx} = -\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{dp}{1+p^2} = -\sqrt{\left(\frac{\pi}{2}\right)}, \text{ from (2).}$$

$$\therefore \text{ from (3) and (4), } \sqrt{\left(\frac{\pi}{2}\right)} = A + B \text{ and } -\sqrt{\left(\frac{\pi}{2}\right)} = A - B.$$

Solving, $A=0$, $B = \sqrt{\left(\frac{\pi}{2}\right)}$. Hence $f(x) = \sqrt{\left(\frac{\pi}{2}\right)} e^{-x}$.

Comprehensive Exercise 1

1. (i) Find the Fourier transform of $f(x)$, if

$$f(x) = \begin{cases} \frac{\sqrt{2\pi}}{2\varepsilon}, & |x| \leq \varepsilon \\ 0, & |x| > \varepsilon. \end{cases}$$

- (ii) Find the Fourier transform of

$$f(x) = \begin{cases} x, & |x| \leq a \\ 0, & |x| > a. \end{cases} \quad (\text{Kanpur 2008})$$

2. (i) Show that the Fourier transform of $f(x) = e^{-x^2/2}$ is $e^{-p^2/2}$. (Kanpur 2011)

- (ii) Find the Fourier transform of the function

$$f(x) = \begin{cases} 1 + \frac{x}{a}, & \text{for } -a < x < 0 \\ 1 - \frac{x}{a}, & \text{for } 0 < x < a \\ 0, & \text{other wise.} \end{cases}$$

3. (i) Find the Fourier transform of $F(x) = \begin{cases} 1 - x^2, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases}$

(Kanpur 2010; Purvanchal 14)

and hence evaluate $\int_0^\infty \left(\frac{x \cos x - \sin x}{x^3} \right) \cos \frac{x}{2} dx$.

- (ii) Find the cosine transform of the function $f(x)$, if

$$f(x) = \begin{cases} \cos x, & 0 < x < a \\ 0, & x > a. \end{cases}$$

4. (i) Find the Fourier sine and cosine transform of $f(x)$, if

$$f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 < x < 2 \\ 0, & x > 2. \end{cases} \quad (\text{Kanpur 2007, 11, 14})$$

- (ii) Find the Fourier sine and cosine transform of the function

$$f(x) = x^{m-1}.$$

$$\begin{aligned} & \left[\text{Hint. } \int_0^\infty x^{m-1} \cos px \, dx - i \int_0^\infty x^{m-1} \sin px \, dx \right. \\ & \left. = \int_0^\infty e^{ipx} x^{m-1} \, dx = \frac{\Gamma(m)}{(ip)^m} = \frac{\Gamma(m)}{p^m} \left(\cos \frac{m\pi}{2} - i \sin \frac{m\pi}{2} \right) \right] \end{aligned}$$

5. (i) Find the Fourier sine transform of $x/(1+x^2)$.

(ii) Find the cosine transform of $\frac{e^{ax} + e^{-ax}}{e^{\pi x} + e^{-\pi x}}$.

(Purvanchal 2014; Kanpur 14)

6. Find the sine transform of $\frac{1}{e^{\pi x} - e^{-\pi x}}$ and deduce that

$$F_s (\operatorname{cosech} \pi x) = \frac{1}{\sqrt{(2\pi)}} \tanh (p/2).$$

7. Find the Fourier sine transform of $f(x)$, if

$$f(x) = \begin{cases} 0, & 0 < x < a \\ x, & a \leq x \leq b \\ 0, & x > b. \end{cases}$$

8. Find $f(x)$ if its cosine transform is $1/(1+p^2)$.

9. Find $f(x)$ if $\tilde{f}_c(p) = \frac{1}{2} \pi e^{-p}$.

10. Find $f(x)$ if its sine transform is $\pi/2$.

11. Find $f(x)$ if (i) its sine transform is e^{-ap} ,

(Kanpur 2012)

(ii) its cosine transform is e^{-ap} .

12. Find the inverse Fourier transform of $\tilde{f}(p) = e^{-|p|y}$.

13. Find $f(x)$ if $\tilde{f}_s(p) = p^n e^{-ap}$.

14. Find $f(x)$ if its cosine transform is

$$\tilde{f}_c(p) = \begin{cases} \frac{1}{\sqrt{(2\pi)}} \left(a - \frac{p}{2}\right), & \text{if } p < 2a \\ 0, & \text{if } p \geq 2a \end{cases}$$

15. Find $f(x)$ if $\tilde{f}_s(p) = \frac{e^{-ap}}{p}$. Hence deduce $F_s^{-1}\{1/p\}$.

Answers 1

1. (ii) $\frac{\sin p\epsilon}{p\epsilon}$

(ii) $-\frac{i}{p^2} \cdot \sqrt{\left(\frac{2}{\pi}\right)} (ap \cos ap - \sin ap)$

2. (ii) $\frac{1}{ap^2} \cdot \sqrt{\frac{2}{\pi}} [1 - \cos pa]$

3. (i) $-2\sqrt{\left(\frac{2}{\pi}\right)} \cdot \left(\frac{p \cos p - \sin p}{p^3}\right); -\frac{3\pi}{16}$

(ii) $\frac{1}{\sqrt{(2\pi)}} \left[\frac{\sin(1+p)a}{1+p} + \frac{\sin(1-p)a}{1-p} \right]$

4. (i) $2\sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{\sin p}{p^2} (1 - \cos p); 2\sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{\cos p}{p^2} (1 - \cos p).$
 (ii) $\frac{\Gamma(m)}{p^m} \sqrt{\left(\frac{2}{\pi}\right)} \sin \frac{m\pi}{2}; \frac{\Gamma(m)}{p^m} \sqrt{\left(\frac{2}{\pi}\right)} \cos \frac{m\pi}{2}$
5. (i) $\sqrt{(\pi/2)} \cdot e^{-p}$ (ii) $\sqrt{\left(\frac{2}{\pi}\right)} \frac{\cos(a/2) \cdot (e^{p/2} + e^{-p/2})}{2 \cos a + e^p + e^{-p}}$
6. $\frac{1}{2\sqrt{(2\pi)}} \frac{e^p - 1}{e^p + 1}$
7. $\sqrt{\left(\frac{2}{\pi}\right)} \left[\frac{-b \cos pb + a \cos pa}{p} + \frac{\sin pb - \sin pa}{p^2} \right]$
8. $\sqrt{(\pi/2)} e^{-x}$
9. $\sqrt{\left(\frac{\pi}{2}\right)} \cdot \frac{1}{1+x^2}$ 10. $\sqrt{(\pi/2)} \cdot (1/x)$
11. $\sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{x}{a^2 + x^2}; \sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{a}{a^2 + x^2}$ 12. $\frac{y\sqrt{2}}{\sqrt{\pi(y^2 + x^2)}}$
13. $\sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{n! \sin\{(n+1) \tan^{-1}(x/a)\}}{(a^2 + x^2)^{(n+1)/2}}$ 14. $\pi^{-1} x^{-2} \sin^2 ax$
15. $\sqrt{(2/\pi)} \tan^{-1}(x/a); \sqrt{(\pi/2)}$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. If $\tilde{f}(p)$ is the complex Fourier transform of $f(x)$, then the complex Fourier transform of $f(ax)$ is
 (a) $\frac{1}{a} \tilde{f}\left(\frac{p}{a}\right)$ (b) $\tilde{f}\left(\frac{p}{a}\right)$
 (c) $\frac{1}{a} \tilde{f}(p)$ (d) None of these.
2. If $\tilde{f}_s(p)$ and $\tilde{f}_c(p)$ are infinite Fourier sine and cosine transforms of $f(x)$ respectively, then $F_c\{f(x) \sin ax\}$ is
 (a) $[\tilde{f}_s(p+a) - \tilde{f}_s(p-a)]$ (b) $\frac{1}{2} [\tilde{f}_s(p+a) - \tilde{f}_s(p-a)]$
 (c) $2 [\tilde{f}_s(p+a) - \tilde{f}_s(p-a)]$ (d) $\frac{1}{2} [\tilde{f}_s(p-a) - \tilde{f}_s(p+a)]$

3. The infinite Fourier sine transform of e^{-x} is
- (a) $\frac{1}{1+p^2} \sqrt{\left(\frac{2}{\pi}\right)}$ (b) $\frac{p}{1+p^2} \sqrt{\left(\frac{2}{\pi}\right)}$
 (c) $\frac{1}{p} \sqrt{\left(\frac{2}{\pi}\right)}$ (d) $\frac{1}{p^2} \sqrt{\left(\frac{2}{\pi}\right)}$
4. What is $f(x)$ if its infinite Fourier sine transform is $\frac{e^{-ap}}{p}$?
- (a) $\sqrt{\left(\frac{2}{\pi}\right)} \tan^{-1} \frac{x}{a}$ (b) $\sqrt{\left(\frac{2}{\pi}\right)} \tan^{-1} \frac{a}{x}$
 (c) $\sqrt{\left(\frac{2}{\pi}\right)} \tan^{-1} \frac{2x}{a}$ (d) $\sqrt{\left(\frac{2}{\pi}\right)} \tan^{-1} \frac{2a}{x}$.

Fill In The Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- If the function $f(x)$ is defined on $(-\infty, \infty)$ and is piecewise continuous in each finite partial interval and absolutely integrable in $(-\infty, \infty)$, then the integral $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ipx} f(x) dx$ is called the transform of $f(x)$.
- The infinite Fourier sine transform $\tilde{f}_s(p)$ of $f(x)$ in $0 < x < \infty$, is
- If $\tilde{f}(p)$ is the complex Fourier transform of $f(x)$, then the complex Fourier transform of $f(x-a)$ is
- If $\tilde{f}_s(p)$ and $\tilde{f}_c(p)$ are infinite Fourier sine and cosine transforms of $f(x)$ respectively, then $F_s \{f(x) \cos ax\} = \dots\dots\dots$.
- The Fourier transform of $F(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| > a \end{cases}$ is

True or False

Write 'T' for true and 'F' for false statement.

- The Fourier transform is a linear transformation.
- If $\tilde{f}_c(p)$ is the infinite Fourier cosine transform of $f(x)$, then the infinite Fourier cosine transform of $f(ax)$ is $\tilde{f}_c\left(\frac{p}{a}\right)$.
- $\sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{x}{a^2 + x^2}$ is the inverse Fourier sine transform of e^{-ap} .
- If $\tilde{f}_s(p)$ and $\tilde{f}_c(p)$ are infinite Fourier sine and cosine transforms of $f(x)$ respectively, then $F_s \{f(x) \sin ax\} = \frac{1}{2} [\tilde{f}_c(p-a) + \tilde{f}_c(p+a)]$.

5. The Fourier transform of $f'(x)$, the derivative of $f(x)$ is $(-ip)\tilde{f}(p)$, where $\tilde{f}(p)$ is the Fourier transform of $f(x)$.

6. The Fourier transform of the function

$$f(t) = \begin{cases} e^{-xt}g(t), & t > 0 \\ 0, & t < 0 \end{cases} \text{ is the Laplace transform of the function } g(t).$$

Answers

Multiple Choice Questions

1. (a) 2. (b) 3. (b) 4. (a)

Fill in the Blank(s)

1. Fourier 2. $\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty f(x) \sin px \, dx$ 3. $e^{ipa} \tilde{f}(p)$
 4. $\frac{1}{2} [\tilde{f}_s(p+a) + \tilde{f}_s(p-a)]$ 5. $\frac{2 \sin pa}{p \sqrt{(2\pi)}}, p \neq 0$

True or False

1. T 2. F 3. T 4. F 5. T
 6. T



Chapter

5



Finite Fourier Transforms

5.1 Finite Fourier Sine Transforms

Let $f(x)$ denote a function that is sectionally continuous over some finite interval $(0, l)$ of the variable x . The finite Fourier sine transform of $f(x)$ on this interval is defined as

$$\tilde{f}_s(p) = \int_0^l f(x) \sin \frac{p\pi x}{l} dx,$$

where p is an integer.

By the proper choice of the origin and the unit of length, if the end points of the interval become $x = 0$ and $x = \pi$, then

$$\tilde{f}_s(p) = \int_0^\pi f(x) \sin px \, dx.$$

Note that $\tilde{f}_s(p)$ is always zero when $p = 0$.

The function $f(x)$ is called the inverse finite Fourier sine transform of $\tilde{f}_s(p)$.

i.e.,
$$f(x) = F_s^{-1} \{ \tilde{f}_s(p) \}.$$

5.2 Inversion Formula For Sine Transform

If $\tilde{f}_s(p)$ is the finite Fourier sine transform of $f(x)$ over the interval $(0, l)$ then the inversion formula for sine transform is given by

$$f(x) = \frac{2}{l} \sum_{p=1}^{\infty} \tilde{f}_s(p) \sin \frac{p\pi x}{l} \quad \text{or} \quad f(x) = \frac{2}{\pi} \sum_{p=1}^{\infty} \tilde{f}_s(p) \sin px$$

if $(0, \pi)$ is the interval considered for $\tilde{f}_s(p)$.

Proof: Defining $f(x)$ in the interval $(-l, 0)$ such that $f(x)$ is an odd function of x in $(-l, l)$, by Fourier series, we have

$$f(x) = \sum_{p=1}^{\infty} b_p \cdot \sin \frac{p\pi x}{l}$$

where
$$b_p = \frac{2}{l} \int_0^l f(x) \sin \frac{p\pi x}{l} dx = \frac{2}{l} \tilde{f}_s(p),$$

$\tilde{f}_s(p)$ is the finite Fourier sine transform of $f(x)$.

Hence
$$f(x) = \frac{2}{l} \sum_{p=1}^{\infty} \tilde{f}_s(p) \sin \frac{p\pi x}{l} \quad \text{or} \quad f(x) = \frac{2}{\pi} \sum_{p=1}^{\infty} \tilde{f}_s(p) \sin px$$

where $f(x)$ is an odd function of x in the interval $(-\pi, \pi)$ such that

$$\tilde{f}_s(p) = \int_0^{\pi} f(x) \sin px \, dx$$

5.3 Finite Fourier Cosine Transforms

Let $f(x)$ denote a function that is sectionally continuous over some finite interval $(0, l)$ of the variable x . The finite Fourier cosine transform of $f(x)$ on this interval is defined as

$$\tilde{f}_c(p) = \int_0^l f(x) \cos \frac{p\pi x}{l} dx$$

where p is an integer. On the interval $(0, \pi)$, $\tilde{f}_c(p)$ is defined as

$$\tilde{f}_c(p) = \int_0^{\pi} f(x) \cos px \, dx.$$

The function $f(x)$ is called the inverse finite Fourier cosine transform of $\tilde{f}_c(p)$ i.e.,

$$f(x) = F_c^{-1} \{ \tilde{f}_c(p) \}.$$

5.4 Inversion Formula for Cosine Transform

If $\tilde{f}_c(p)$ is the finite Fourier cosine transform of $f(x)$ over the interval $(0, l)$ then the inversion formula for cosine transform is given by

$$f(x) = \frac{1}{l} \tilde{f}_c(0) + \frac{2}{l} \sum_{p=1}^{\infty} \tilde{f}_c(p) \cos \frac{p\pi x}{l},$$

where
$$\tilde{f}_c(0) = \int_0^l f(x) dx.$$

If π is taken as the upper limit for the finite Fourier cosine transform then the inversion formula is given by

$$f(x) = \frac{1}{\pi} \tilde{f}_c(0) + \frac{2}{\pi} \sum_{p=1}^{\infty} \tilde{f}_c(p) \cos px,$$

where
$$\tilde{f}_c(0) = \int_0^{\pi} f(x) dx.$$

Proof: Defining $f(x)$ in the interval $(-l, 0)$ such that $f(x)$ is an even function of x in $(-l, l)$, by the Fourier series, we have

$$f(x) = \frac{a_0}{2} + \sum_{p=1}^{\infty} a_p \cos \frac{p\pi x}{l},$$

where
$$a_p = \frac{2}{l} \int_0^l f(x) \cos \frac{p\pi x}{l} dx = \frac{2}{l} \tilde{f}_c(p).$$

$\therefore$
$$a_0 = \frac{2}{l} \int_0^l f(x) dx = \frac{2}{l} \tilde{f}_c(0).$$

$\therefore$
$$f(x) = \frac{1}{l} \tilde{f}_c(0) + \frac{2}{l} \sum_{p=1}^{\infty} \tilde{f}_c(p) \cos \frac{p\pi x}{l}.$$

If $f(x)$ is an even function of x in the interval $(-\pi, \pi)$, then

$$f(x) = \frac{1}{\pi} \tilde{f}_c(0) + \frac{2}{\pi} \sum_{p=1}^{\infty} \tilde{f}_c(p) \cos px$$

where
$$\tilde{f}_c(p) = \int_0^{\pi} f(x) \cos px dx$$

and
$$\tilde{f}_c(0) = \int_0^{\pi} f(x) dx.$$

Note: The upper limit for Fourier sine or cosine transforms will be taken as $x = \pi$, if not given in the problem.

5.5 Multiple Finite Fourier Transforms

Let $f(x, y)$ be a function of two variables x and y , defined in the square $0 \leq x \leq \pi$ and $0 \leq y \leq \pi$.

Considering it, temporarily, as a function of x , the finite sine transform is given by

$$\tilde{f}_s(p, y) = \int_0^{\pi} f(x, y) \sin px dx$$

and now the finite sine transform of $\tilde{f}_s(p, y)$ which is a function of y is given by

$$\tilde{F}_s(p, q) = \int_0^\pi \tilde{f}_s(p, y) \sin qy \, dy.$$

$$\therefore \tilde{F}_s(p, q) = \int_0^\pi \int_0^\pi f(x, y) \sin px \sin qy \, dx \, dy,$$

which is the double finite sine transform of $f(x, y)$ in the region $0 \leq x \leq \pi, 0 \leq y \leq \pi$.

Similarly

$$\tilde{f}_c(p, q) = \int_0^\pi \int_0^\pi f(x, y) \cos px \cos qy \, dx \, dy,$$

which is the double finite cosine transform of $f(x, y)$ in the region $0 \leq x \leq \pi, 0 \leq y \leq \pi$.

Using inversion formula, we have

$$\tilde{f}_s(p, y) = \frac{2}{\pi} \sum_{q=1}^{\infty} \tilde{F}_s(p, q) \sin qy$$

and

$$f(x, y) = \frac{2}{\pi} \sum_{p=1}^{\infty} \tilde{f}_s(p, y) \sin px.$$

Hence

$$f(x, y) = \frac{4}{\pi^2} \sum_{p=1}^{\infty} \sum_{q=1}^{\infty} \tilde{F}_s(p, q) \sin px \sin qy.$$

Similarly we can find the inversion formula for double finite cosine transform of $f(x, y)$.

5.6 Operational Properties of Finite Fourier Sine Transform

Theorem 1: The finite Fourier sine transformation resolves the differential form $f''(x)$ into a linear algebraic form in the transform $\tilde{f}_s(p)$ and the boundary values $f(0)$ and $f(\pi)$ in the manner

$$F_s \{ f''(x) \} = -p^2 \tilde{f}_s(p) + p \{ f(0) - (-1)^p f(\pi) \}$$

whenever $f(x)$ and $f'(x)$ are continuous and $f''(x)$ is sectionally continuous, on the interval, $0 \leq x \leq \pi$.

Proof: We have

$$F_s \{ f''(x) \} = \int_0^\pi f''(x) \sin px \, dx$$

$$= [f'(x) \sin px]_0^\pi - p \int_0^\pi f'(x) \cos px \, dx$$

(Integrating by parts taking $f'(x)$ as second function)

$$= 0 - [pf(x) \cos px]_0^\pi - p^2 \int_0^\pi f(x) \sin px \, dx$$

$$= -p f(\pi) \cos p\pi + p f(0) - p^2 \tilde{f}_s(p).$$

Hence

$$F_s \{ f''(x) \} = -p^2 \tilde{f}_s(p) + p \{ f(0) - (-1)^p f(\pi) \}.$$

Note: If $f''(x)$ and $f'''(x)$ are continuous and $f^{(4)}(x)$ is sectionally continuous, on the interval, $0 \leq x \leq \pi$ then replacing $f(x)$ by $f''(x)$, in the result of the above theorem we have

$$\begin{aligned} F_s \{ f^{(4)}(x) \} &= -p^2 F_s \{ f''(x) \} + p [f''(0) - (-1)^p f''(\pi)] \\ &= -p^2 [-p^2 \tilde{f}_s(p) + p \{ f(0) - (-1)^p f(\pi) \}] + p \{ f''(0) - (-1)^p f''(\pi) \} \\ &= p^4 \tilde{f}_s(p) - p^3 [f(0) - (-1)^p f(\pi)] + p [f''(0) - (-1)^p f''(\pi)]. \end{aligned}$$

In the similar way finite Fourier sine transforms of other derivatives of even order can be found.

Theorem 2: If $\tilde{f}_s(p)$ is the sine transform of a sectionally continuous function $f(x)$, $0 \leq x \leq \pi$, then

$$\begin{aligned} F_s^{-1} \left\{ \frac{\tilde{f}_s(p)}{p^2} \right\} &= \frac{x}{\pi} \int_0^\pi \int_0^t f(r) dr dt - \int_0^x \int_0^t f(r) dr dt \\ &= \frac{x}{\pi} \int_0^\pi (\pi - r) f(r) dr - \int_0^\pi (x - r) f(r) dr. \end{aligned}$$

5.7 Operational Properties of Finite Fourier Cosine Transforms

Theorem 1: If $f(x)$ and $f'(x)$ are continuous and if $f''(x)$ is sectionally continuous, the finite Fourier cosine transformation resolves the differential form $f''(x)$ into an algebraic form in $\tilde{f}_c(p)$ and the boundary values $f'(0)$ and $f'(\pi)$, in the manner

$$F_c \{ f''(x) \} = -p^2 \tilde{f}_c(p) - f'(0) + (-1)^p f'(\pi).$$

Proof: We have

$$\begin{aligned} F_c \{ f''(x) \} &= \int_0^\pi f''(x) \cdot \cos px dx \\ &= [f'(x) \cos px]_0^\pi + p \int_0^\pi f'(x) \sin px dx \\ &\quad \text{(Integrating by parts taking } f''(x) \text{ as second function)} \\ &= f'(\pi) \cos p\pi - f'(0) + p [f(x) \sin px]_0^\pi - p^2 \int_0^\pi f(x) \cos px dx \\ &= (-1)^p f'(\pi) - f'(0) - p^2 \tilde{f}_c(p). \end{aligned}$$

$$\text{Hence } F_c \{ f''(x) \} = -p^2 \tilde{f}_c(p) - f'(0) + (-1)^p f'(\pi). \quad \dots(1)$$

Note: If $f''(x)$ and $f'''(x)$ are continuous and if $f^{(4)}(x)$ is sectionally continuous, then replacing $f(x)$ by $f''(x)$ in (1), we have

$$\begin{aligned} F_c \{ f^{(4)}(x) \} &= -p^2 F_c \{ f''(x) \} - f'''(0) + (-1)^p f'''(\pi) \\ &= -p^2 [-p^2 \tilde{f}_c(p) - f'(0) + (-1)^p f'(\pi)] - f'''(0) + (-1)^p f'''(\pi) \end{aligned}$$

$$= p^4 \tilde{f}_c(p) + p^2 [f'(0) - (-1)^p f'(\pi)] - f'''(0) + (-1)^p f'''(\pi).$$

Similarly finite Fourier cosine transforms of other derivatives of even order can be found.

Theorem 2: If $\tilde{f}_c(p)$ is the cosine transform of a sectionally continuous function $f(x)$, $0 \leq x \leq \pi$, then

$$F_c^{-1} \left\{ \frac{\tilde{f}_c(p)}{p^2} \right\} = \int_0^x \int_t^\pi f(r) dr dt + \frac{\tilde{f}_c(0)}{2\pi} (x - \pi)^2 + A,$$

where A is an arbitrary constant.

5.8 Combined Properties of Finite Fourier Sine and Cosine Transforms

When $f(x)$ is continuous and $f'(x)$ is sectionally continuous then

$$(a) F_s \{ f'(x) \} = -p F_c \{ f(x) \}, (p = 1, 2, 3, \dots)$$

$$\text{and } (b) F_c \{ f'(x) \} = p F_s \{ f(x) \} - f(0) + (-1)^p f(\pi), (p = 0, 1, 2, \dots).$$

Proof: (a) We have

$$\begin{aligned} F_s \{ f'(x) \} &= \int_0^\pi f'(x) \sin px dx \\ &= [f(x) \sin px]_0^\pi - p \int_0^\pi f(x) \cos px dx \\ &\quad \text{(Integrating by parts, taking } f'(x) \text{ as second function)} \\ &= -p F_c \{ f(x) \}, (p = 1, 2, 3, \dots). \end{aligned}$$

(b) We have

$$\begin{aligned} F_c \{ f'(x) \} &= \int_0^\pi f'(x) \cos px dx \\ &= [f(x) \cos px]_0^\pi + p \int_0^\pi f(x) \sin px dx \\ &\quad \text{[Integrating by parts]} \\ &= f(\pi) \cos p\pi - f(0) + p F_s \{ f(x) \}, (p = 0, 1, 2, \dots) \\ &= p F_s \{ f(x) \} - f(0) + (-1)^p f(\pi). \end{aligned}$$

Note: If $H(x)$ is any sectionally continuous function, the above properties may be written as

$$F_s \{ H(x) \} = -p F_c \left\{ \int_0^x H(r) dr \right\}, (p = 1, 2, \dots)$$

and

$$F_c \left\{ H(x) - \frac{1}{\pi} \tilde{H}_c(0) \right\} = p F_s \left\{ \int_0^\pi H(r) dr - \frac{x}{\pi} \tilde{H}_c(0) \right\},$$

$$(p = 0, 1, 2, \dots)$$

5.9 Convolution

Let $F(x)$ and $G(x)$ be two functions defined on the interval $-2\pi < x < 2\pi$, then the function

$$F(x) * G(x) = \int_{-\pi}^{\pi} F(x-y) G(y) dy$$

is called the convolution of $F(x)$ and $G(x)$ on the interval $-\pi < x < \pi$.

Illustrative Examples

Example 1: Find the finite Fourier sine and cosine transforms of the function $f(x) = 2x$, $0 < x < 4$. (Agra 2003)

Solution: We have

$$\begin{aligned} \tilde{f}_s(p) &= \int_0^l f(x) \sin(p\pi x/l) dx \\ &= \int_0^4 2x \sin(p\pi x/4) dx, \text{ as } l = 4 \\ &= \left[\frac{-2x \cos(p\pi x/4)}{p\pi/4} \right]_0^4 + 2 \int_0^4 \frac{\cos(p\pi x/4)}{p\pi/4} dx \\ &= -\frac{32}{p\pi} \cos p\pi + \frac{8}{p\pi} \left[\frac{\sin(p\pi x/4)}{p\pi/4} \right]_0^4 = -\frac{32}{p\pi} \cos p\pi. \end{aligned} \quad (\text{Given})$$

Also

$$\begin{aligned} \tilde{f}_c(p) &= \int_0^l f(x) \cos(p\pi x/l) dx = \int_0^4 2x \cos(p\pi x/4) dx, \text{ as } l = 4 \\ &= \left[\frac{2x \sin(p\pi x/4)}{p\pi/4} \right]_0^4 - 2 \int_0^4 \frac{\sin(p\pi x/4)}{p\pi/4} dx \\ &= \frac{8}{p\pi} \left[\frac{\cos(p\pi x/4)}{p\pi/4} \right]_0^4 = \frac{32}{p^2 \pi^2} (\cos p\pi - 1), \text{ if } p > 0 \end{aligned}$$

and if $p = 0$, then $\tilde{f}_c(p) = \int_0^4 2x \cdot 1 dx = 16$.

Example 2: Find the finite Fourier cosine transform of $f(x)$ if

$$(i) f(x) = \frac{\pi}{3} - x + \frac{x^2}{2\pi}, \quad (ii) f(x) = \sin nx.$$

Solution: (i) We have

$$\begin{aligned} \tilde{f}_c(p) &= \int_0^{\pi} f(x) \cos px dx. \quad \therefore \tilde{f}_c(p) = \int_0^{\pi} \left(\frac{\pi}{3} - x + \frac{x^2}{2\pi} \right) \cos px dx \\ &= \left[\left(\frac{\pi}{3} - x + \frac{x^2}{2\pi} \right) \frac{1}{p} \sin px \right]_0^{\pi} - \frac{1}{p} \int_0^{\pi} \left(-1 + \frac{x}{\pi} \right) \sin px dx \end{aligned}$$

$$\begin{aligned}
 &= -\frac{1}{p} \left[-\left(-1 + \frac{x}{\pi} \right) \frac{1}{p} \cos px \right]_0^{\pi} - \frac{1}{p^2} \int_0^{\pi} \frac{1}{\pi} \cos px \, dx \\
 &= \frac{1}{p^2} - \frac{1}{p^3 \pi} [\sin px]_0^{\pi} = \frac{1}{p^2}, \quad \text{if } p > 0
 \end{aligned}$$

and when $p=0$,

$$\tilde{f}_c(p) = \int_0^{\pi} \left(\frac{\pi}{3} - x + \frac{x^2}{2\pi} \right) dx = 0$$

$$\begin{aligned}
 \text{(ii)} \quad \tilde{f}_c(p) &= \int_0^{\pi} \sin nx \cos px \, dx \\
 &= \frac{1}{2} \int_0^{\pi} [\sin(n+p)x + \sin(n-p)x] \, dx \\
 &= \frac{1}{2} \left[-\frac{\cos(n+p)x}{n+p} - \frac{\cos(n-p)x}{n-p} \right]_0^{\pi}, \text{ if } p \neq n \\
 \therefore \text{ if } p \neq n, \tilde{f}_c(p) &= \frac{1}{2} \left[-\frac{\cos(n+p)\pi}{n+p} - \frac{\cos(n-p)\pi}{n-p} + \frac{1}{n+p} + \frac{1}{n-p} \right].
 \end{aligned}$$

If $n-p$ is even, then $n+p$ is also even and so

$$\tilde{f}_c(p) = \frac{1}{2} \left[-\frac{1}{n+p} - \frac{1}{n-p} + \frac{1}{n+p} + \frac{1}{n-p} \right] = 0.$$

If $n-p$ is odd, then $n+p$ is also odd and so

$$\tilde{f}_c(p) = \frac{1}{2} \left[\frac{2}{n+p} + \frac{2}{n-p} \right] = \frac{2n}{n^2 - p^2}.$$

If $p=n$,

$$\begin{aligned}
 \text{then} \quad \tilde{f}_c(p) &= \int_0^{\pi} \sin nx \cos nx \, dx = \frac{1}{2} \int_0^{\pi} \sin 2nx \, dx \\
 &= \frac{1}{2} \left[-\frac{\cos 2nx}{2n} \right]_0^{\pi} = 0.
 \end{aligned}$$

$$\therefore \tilde{f}_c(p) = 0 \quad \text{or} \quad \frac{2n}{n^2 - p^2} \text{ according as } n-p \text{ is even or odd.}$$

Example 3: Find the finite cosine transform of $f(x)$ if $f(x) = -\frac{\cos k(\pi-x)}{k \sin k\pi}$.

Solution: We have

$$\begin{aligned}
 \tilde{f}_c(p) &= -\int_0^{\pi} \frac{\cos \{k(\pi-x)\}}{k \sin k\pi} \cos px \, dx \\
 &= -\frac{1}{2k \sin k\pi} \int_0^{\pi} [\cos \{k(\pi-x) + px\} + \cos \{k(\pi-x) - px\}] \, dx
 \end{aligned}$$

$$\begin{aligned}
 &= -\frac{1}{2k \sin k\pi} \left[\frac{\sin(k\pi - kx + px)}{p-k} - \frac{\sin(k\pi - kx - px)}{p+k} \right]_0^\pi \\
 &= -\frac{1}{2k \sin k\pi} \left[\frac{\sin p\pi}{p-k} - \frac{\sin(-p\pi)}{p+k} - \frac{\sin k\pi}{p-k} + \frac{\sin k\pi}{p+k} \right] \\
 &= \frac{1}{2k} \left(\frac{1}{p-k} - \frac{1}{p+k} \right) = \frac{1}{p^2 - k^2}, k \neq 0, 1, 2, 3, \dots
 \end{aligned}$$

Example 4: Find finite Fourier sine transform of $f(x)$ if

$$f(x) = \frac{\pi \sin kx}{2k \sin^2 k\pi} - \frac{x \cos k(\pi - x)}{2k \sin k\pi}.$$

Solution: We have

$$\begin{aligned}
 \tilde{f}_s(p) &= \int_0^\pi f(x) \sin px \, dx \\
 &= \int_0^\pi \left\{ \frac{\pi \sin kx}{2k \sin^2 k\pi} - \frac{x \cos k(\pi - x)}{2k \sin k\pi} \right\} \sin px \, dx \\
 &= \frac{\pi}{2k \sin^2 k\pi} \int_0^\pi \sin kx \sin px \, dx \\
 &\quad - \frac{1}{2k \sin k\pi} \int_0^\pi x \cos k(\pi - x) \sin px \, dx \\
 &= \frac{\pi}{4k \sin^2 k\pi} \int_0^\pi [\cos(p-k)x - \cos(p+k)x] \, dx \\
 &\quad - \frac{1}{4k \sin k\pi} \int_0^\pi x [\sin(k\pi - kx + px) + \sin(px - k\pi + kx)] \, dx \\
 &= \frac{\pi}{4k \sin^2 k\pi} \left[\frac{\sin(p-k)x}{p-k} - \frac{\sin(p+k)x}{p+k} \right]_0^\pi \\
 &\quad - \frac{1}{4k \sin k\pi} \cdot \left[x \left\{ -\frac{\cos(k\pi - kx + px)}{(p-k)} - \frac{\cos(px - k\pi + kx)}{(p+k)} \right\} \right]_0^\pi \\
 &\quad + \frac{1}{4k \sin k\pi} \int_0^\pi 1 \cdot \left\{ -\frac{\cos(k\pi - kx + px)}{(p-k)} - \frac{\cos(px - k\pi + kx)}{(p+k)} \right\} dx \\
 &= \frac{\pi}{4k \sin^2 k\pi} \left[\frac{\sin(p-k)\pi}{p-k} - \frac{\sin(p+k)\pi}{p+k} \right] \\
 &\quad + \frac{\pi}{4k \sin k\pi} \left[\cos p\pi \cdot \left(\frac{1}{p-k} + \frac{1}{p+k} \right) \right] \\
 &\quad - \frac{1}{4k \sin k\pi} \left[\frac{\sin(k\pi - kx + px)}{(p-k)^2} + \frac{\sin(px - k\pi + kx)}{(p+k)^2} \right]_0^\pi
 \end{aligned}$$

$$\begin{aligned}
&= \frac{\pi}{4k \sin^2 k\pi} \cdot \left[\sin p\pi \cos k\pi \cdot \left(\frac{1}{p-k} - \frac{1}{p+k} \right) \right. \\
&\quad \left. - \cos p\pi \sin k\pi \cdot \left(\frac{1}{p-k} + \frac{1}{p+k} \right) \right] + \frac{\pi}{4k \sin k\pi} \cdot \frac{2p}{p^2 - k^2} \cos p\pi \\
&\quad + \frac{1}{4k \sin k\pi} \cdot \left[\sin k\pi \cdot \left\{ \frac{1}{(p-k)^2} - \frac{1}{(p+k)^2} \right\} \right] \\
&= \frac{1}{4k} \cdot \frac{(p+k)^2 - (p-k)^2}{(p^2 - k^2)^2} \\
&= \frac{p}{(p^2 - k^2)^2}, (|k| \neq 0, 1, 2, \dots).
\end{aligned}$$

Example 5: Find $f(x)$ if

$$\tilde{f}_c(p) = \frac{6 \left(\sin \frac{p\pi}{2} - \cos p\pi \right)}{(2p+1)\pi} \text{ for } p = 1, 2, 3, \dots \text{ and } \frac{2}{\pi} \text{ for } p = 0,$$

where $0 < x < 4$.

Solution: We have $f(x) = \frac{1}{l} \tilde{f}_c(0) + \frac{2}{l} \sum_{p=1}^{\infty} \tilde{f}_c(p) \cos \frac{p\pi x}{l}$

$$\begin{aligned}
&= \frac{1}{4} \cdot \frac{2}{\pi} + \frac{2}{4} \cdot \sum_{p=1}^{\infty} \frac{6 \left(\sin \frac{p\pi}{2} - \cos p\pi \right)}{(2p+1)\pi} \cos \frac{p\pi x}{4} \\
&= \frac{1}{2\pi} + \frac{3}{\pi} \sum_{p=1}^{\infty} \frac{\left(\sin \frac{p\pi}{2} - \cos p\pi \right)}{2p+1} \cos \left(\frac{p\pi x}{4} \right).
\end{aligned}$$

Example 6: Find $f(x)$ if its finite sine transform is given by

$$\tilde{f}_s(p) = \frac{1 - \cos p\pi}{p^2 \pi^2}, \text{ where } 0 < x < \pi.$$

(Kanpur 2008)

Solution: We have

$$\begin{aligned}
f(x) &= \frac{2}{\pi} \sum_{p=1}^{\infty} \tilde{f}_s(p) \sin px \\
&= \frac{2}{\pi} \sum_{p=1}^{\infty} \left(\frac{1 - \cos p\pi}{p^2 \pi^2} \right) \sin px \\
&= \frac{2}{\pi^3} \sum_{p=1}^{\infty} \left(\frac{1 - \cos p\pi}{p^2} \right) \sin px.
\end{aligned}$$

Comprehensive Exercise 1

1. (i) Find the finite Fourier sine and cosine transforms of $f(x) = 1$.
 (ii) Find the finite Fourier sine and cosine transforms of $f(x) = x$.
2. (i) Find the finite Fourier sine transforms of

$$\left(1 - \frac{x}{\pi}\right) \text{ and } \frac{x}{4\pi}.$$

- (ii) Find the finite Fourier cosine transform of

$$\left(1 - \frac{x}{\pi}\right) \text{ and } \frac{x}{4\pi}.$$

3. Find the finite Fourier sine transform of $f(x)$ if

$$(i) f(x) = \begin{cases} x, & 0 \leq x \leq \pi/2 \\ \pi - x, & \pi/2 \leq x \leq \pi. \end{cases}$$

$$(ii) f(x) = \begin{cases} -x, & x < c \\ \pi - x, & x > c, \text{ where } 0 \leq c \leq \pi. \end{cases}$$

4. (i) Find the finite Fourier cosine transform of $f(x)$ if

$$f(x) = \begin{cases} 1, & 0 < x < \pi/2 \\ -1, & \pi/2 < x < \pi. \end{cases}$$

(Kanpur 2012)

- (ii) Find the finite cosine transform of $(1 - x/\pi)^2$.

(Avadh 2013)

5. Find the finite Fourier sine transforms of

$$(i) x(\pi - x) \quad (ii) x(\pi^2 - x^2).$$

6. Find the finite Fourier sine transform of $f(x)$ if

$$(i) f(x) = \sin nx \quad (ii) f(x) = \frac{\pi}{3} - x + \frac{x^2}{2\pi}.$$

7. Find the finite sine transform of $f(x)$, if

$$(i) f(x) = \cos kx \quad (ii) f(x) = x^3 \quad (iii) f(x) = e^{cx}.$$

8. Find finite Fourier cosine transform of $f(x)$ if $f(x) = \frac{\cosh \{c(\pi - x)\}}{\sinh(\pi c)}$.

9. Find finite Fourier sine transform of $f(x)$, if

$$f(x) = \frac{\sin k(\pi - x)}{\sin(k\pi)}.$$

10. Find the finite Fourier sine and cosine transforms of

$$f(x) = x^2, \quad 0 < x < 4.$$

11. Find $f(x)$ if $\tilde{f}_c(p) = \frac{\cos(2p\pi/3)}{(2p+1)^2}$, if $0 < x < 1$.

12. When $f(x) = \sin mx$, where m is a positive integer, show that

$$\tilde{f}_s(p) = 0 \text{ if } p \neq m \text{ and } \tilde{f}_s(p) = \pi/2 \text{ if } p = m.$$

Answers 1

1. (i) $\frac{1}{p} [1 - (-1)^p]; 0$
 (ii) $\frac{\pi (-1)^{p+1}}{p}; \frac{(-1)^p - 1}{p^2}, \text{ if } p = 1, 2, 3, \dots \text{ and } \frac{\pi^2}{2}, \text{ if } p = 0.$
2. (i) $\frac{1}{p}; \frac{(-1)^{p+1}}{4p}$
 (ii) $\frac{1}{\pi p^2} [1 - (-1)^p]; \frac{1}{4\pi p^2} [(-1)^p - 1]$
3. (i) $(2/p^2) \sin(p\pi/2),$ (ii) $(\pi/p) \cos pc.$
4. (i) $(2/p) \sin(p\pi/2), p > 0 \text{ and } 0, \text{ if } p = 0.$
 (ii) $\frac{2}{\pi p^2}, \text{ if } p > 0 \text{ and } \frac{\pi}{3}, \text{ if } p = 0$
5. (i) $\frac{2}{p^3} [1 - (-1)^p],$ (ii) $\frac{6\pi}{p^3} (-1)^{p+1}$
6. (i) $0, \text{ if } p \neq n, \text{ and } \pi/2, \text{ if } p = n$
 (ii) $\frac{\pi}{6p} \{(-1)^{p+2}\} + \frac{1}{\pi p^3} \{(-1)^p - 1\}$
7. (i) $\frac{p}{p^2 - k^2} [1 - (-1)^p \cos k\pi],$ (ii) $\pi (-1)^p \left(\frac{6}{p^3} - \frac{\pi^2}{p} \right),$
 (iii) $\frac{p}{c^2 + p^2} [1 - (-1)^p e^{c\pi}]$
8. $\frac{c}{c^2 + p^2}$ 9. $\frac{p}{p^2 - k^2}, k \neq 0, 1, 2, \dots$
10. $-\frac{64}{p\pi} \cos p\pi + \frac{128}{p^3 \pi^3} (\cos p\pi - 1); \frac{128}{p^2 \pi^2} \cos p\pi, \text{ if } p > 0; \frac{64}{3}, \text{ if } p = 0$
11. $1 + 2 \sum_{p=1}^{\infty} \frac{\cos(2p\pi/3)}{(2p+1)^2} \cos p\pi x$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. If $f(x)$ is continuous and $f'(x)$ is sectionally continuous then $F_s \{f'(x)\}$ is
 (a) $-F_c \{f(x)\}$ (b) $F_c \{f(x)\}$
 (c) $-p F_c \{f(x)\}$ (d) $-p^2 F_c \{f(x)\}$

2. The finite Fourier sine transform of $f(x) = x$ is
- (a) $\frac{\pi}{p} (-1)^{p+1}$ (b) $\frac{\pi}{p} (-1)^p$
- (c) $\frac{\pi}{p^2} (-1)^{p+1}$ (d) $\frac{\pi}{p^2} (-1)^p$
3. What is $f(x)$ if its finite sine transform $\tilde{f}_s(p)$ is $\frac{2\pi(-1)^{p-1}}{p^3}$, where $0 < x < \pi$?
- (a) $\frac{2}{\pi} \sum_{p=1}^{\infty} \frac{(-1)^{p-1}}{p^3} \sin px$ (b) $4 \sum_{p=1}^{\infty} \frac{(-1)^p}{p^3} \sin px$
- (c) $4 \sum_{p=1}^{\infty} \frac{(-1)^{p-1}}{p^3} \sin px$ (d) $\frac{4}{\pi} \sum_{p=1}^{\infty} \frac{(-1)^{p-1}}{p^3} \sin px$
4. The finite Fourier sine transform of $\frac{x}{\pi}$ is
- (a) $\frac{(-1)^p}{p^2}$ (b) $\frac{(-1)^{p+1}}{p^2}$
- (c) $\frac{(-1)^{p-1}}{p}$ (d) $\frac{(-1)^{p+1}}{p}$

Fill In The Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

- The finite Fourier sine transform $\tilde{f}_s(p)$ of $f(x)$ on the interval $(0, \pi)$ is
- On the interval $(0, \pi)$, $\tilde{f}_c(p) = \int_0^\pi f(x) \cos px \, dx$, then the function $f(x)$ is called Fourier cosine transform of $\tilde{f}_c(p)$.
- If $F(x)$ and $G(x)$ are two functions defined on the interval $-\pi < x < \pi$, then the function $F(x) * G(x) = \int_{-\pi}^\pi F(x-y) G(y) \, dy$ is called the of $F(x)$ and $G(x)$ on the interval $-\pi < x < \pi$.
- The finite Fourier sine transform of the function $f(x) = 2x, 0 < x < 4$ is

True or False

Write 'T' for true and 'F' for false statement.

- If $\tilde{f}_s(p)$ is the finite Fourier sine transform of $f(x)$ on the interval $(0, \pi)$ then the inversion formula for sine transform is $f(x) = \frac{2}{\pi} \sum_{p=1}^{\infty} \tilde{f}_s(p) \sin px$
- If $f(x)$ is continuous and $f'(x)$ is sectionally continuous then $F_c\{f'(x)\} = p F_s\{f(x)\} - f(0) + f(\pi)$.

3. The finite Fourier cosine transform of the function $f(x) = 1$ is 0.
4. If the finite sine transform is given by $\tilde{f}_s(p) = \frac{1 - \cos p\pi}{p^2 \pi^2}$, where $0 < x < \pi$, then the value of $f(x)$ is $\frac{2}{\pi^3} \sum_{p=1}^{\infty} \left(\frac{1 - \cos p\pi}{p^2} \right) \sin px$.

Answers

Multiple Choice Questions

1. (c) 2. (a) 3. (c) 4. (d)

Fill in the Blank(s)

1. $\int_0^{\pi} f(x) \sin px \, dx$ 2. inverse finite 3. Convolution
4. $\frac{-32}{p\pi} \cos p\pi$

True or False

1. T 2. F 3. T 4. T



Chapter

6

Applications of Fourier Transforms in Initial and Boundary Value Problems

6.1 Application of Infinite Fourier Transforms

First we shall consider the problems in which one of the variables ranges from $-\infty$ to ∞ or 0 to ∞ . Such problems are solved by taking the infinite Fourier transform of both the sides and then taking the corresponding inverse Fourier transform of the solution of the differential equation thus obtained.

6.2 Choice of Infinite Sine or Cosine Transform

The choice of sine or cosine transform is decided by the form of the boundary conditions at the lower limit of the variable selected for exclusion.

If we want to remove a term $\frac{\partial^2 V}{\partial x^2}$ in a differential equation, applying sine transform,

we have

$$\begin{aligned} \int_0^\infty \sin px \cdot \frac{\partial^2 V}{\partial x^2} dx &= \left[\frac{\partial V}{\partial x} \sin px \right]_0^\infty - p \int_0^\infty \cos px \cdot \frac{\partial V}{\partial x} dx \\ &= -p \int_0^\infty \cos px \cdot \frac{\partial V}{\partial x} dx, \text{ if } \frac{\partial V}{\partial x} \rightarrow 0 \text{ as } x \rightarrow \infty \end{aligned}$$

$$\begin{aligned}
 &= -p \left[(V \cdot \cos px)_0^\infty + p \int_0^\infty V \sin px \, dx \right] \\
 &= p(V)_{x=0} - p^2 \tilde{V}_s \text{ where } \tilde{V}_s \text{ is the sine transform of } V \text{ and}
 \end{aligned}$$

assuming that $V \rightarrow 0$ as $x \rightarrow \infty$.

Again applying the cosine transform, we have

$$\begin{aligned}
 \int_0^\infty \cos px \cdot \frac{\partial^2 V}{\partial x^2} dx &= \left[\frac{\partial V}{\partial x} \cos px \right]_0^\infty + p \int_0^\infty \sin px \frac{\partial V}{\partial x} dx \\
 &= - \left(\frac{\partial V}{\partial x} \right)_{x=0} + p [V \sin px]_0^\infty - p^2 \int_0^\infty V \cos px \cdot dx \\
 &\quad \text{if } \frac{\partial V}{\partial x} \rightarrow 0 \text{ as } x \rightarrow \infty \\
 &= - \left(\frac{\partial V}{\partial x} \right)_{x=0} - p^2 \tilde{V}_c \text{ if } V \rightarrow 0 \text{ as } x \rightarrow \infty,
 \end{aligned}$$

where $\tilde{V}_c$ is the cosine transform of V .

Thus we see that for the exclusion of $\frac{\partial^2 V}{\partial x^2}$ from a differential equation we require

$(V)_{x=0}$ in sine transform

and $\left(\frac{\partial V}{\partial x} \right)_{x=0}$ in cosine transform.

Note 1: By the Fourier sine or cosine transform we cannot exclude a derivative of odd order from the given differential equation.

Note 2: When one of the variables in a differential equation ranges from $-\infty$ to ∞ then that variable can be excluded with the help of complex Fourier transform.

Illustrative Examples

Example 1: The temperature U in the semi-infinite rod $0 \leq x < \infty$ is determined by the differential equation

$$\frac{\partial U}{\partial t} = k \frac{\partial^2 U}{\partial x^2}$$

subject to the conditions

- (i) $U = 0$ when $t = 0, x \geq 0$
- (ii) $\frac{\partial U}{\partial x} = -\mu$ (a constant) when $x = 0$ and $t > 0$.

Making use of cosine transform, show that

$$U(x, t) = \frac{2\mu}{\pi} \int_0^\infty \frac{\cos px}{p^2} (1 - e^{-kp^2 t}) dp.$$

Solution: Since $\left(\frac{\partial U}{\partial x}\right)_{x=0}$ is given, so taking the Fourier cosine transform of both the

sides, we have

$$\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\partial U}{\partial t} \cos px \, dx = k \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\partial^2 U}{\partial x^2} \cos px \, dx$$

$$\text{or} \quad \sqrt{\left(\frac{2}{\pi}\right)} \frac{d}{dt} \int_0^\infty U \cos px \, dx = k \sqrt{\left(\frac{2}{\pi}\right)} \left[\frac{\partial U}{\partial x} \cos px \right]_0^\infty + kp \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\partial U}{\partial x} \sin px \, dx$$

$$\text{or} \quad \frac{d}{dt} (\tilde{U}_c) = -k \sqrt{\left(\frac{2}{\pi}\right)} \cdot \left(\frac{\partial U}{\partial x}\right)_{x=0} + kp \sqrt{\left(\frac{2}{\pi}\right)} (U \sin px)_0^\infty - kp^2 \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty U \cos px \, dx$$

if $\frac{\partial U}{\partial x} \rightarrow 0$ as $x \rightarrow \infty$

$$\text{or} \quad \frac{d\tilde{U}_c}{dt} = \sqrt{\left(\frac{2}{\pi}\right)} \cdot k\mu - kp^2 \tilde{U}_c \quad \text{if } U \rightarrow 0 \text{ as } x \rightarrow \infty$$

$$\text{or} \quad \frac{d}{dt} \tilde{U}_c + kp^2 \tilde{U}_c = \sqrt{\left(\frac{2}{\pi}\right)} k\mu,$$

which is a linear differential equation of first order.

$$\text{I.F.} = e^{\int kp^2 dt} = e^{kp^2 t}.$$

$\therefore$ its solution is

$$\begin{aligned} e^{kp^2 t} \cdot \tilde{U}_c &= A + \sqrt{\left(\frac{2}{\pi}\right)} k\mu \int e^{kp^2 t} dt \\ &= A + \sqrt{\left(\frac{2}{\pi}\right)} \frac{\mu}{p^2} e^{kp^2 t}. \end{aligned}$$

$$\text{But when } t=0, \tilde{U}_c=0. \quad [\because U=0 \text{ when } t=0]$$

$$\therefore 0 = A + \sqrt{\left(\frac{2}{\pi}\right)} \frac{\mu}{p^2}$$

$$\text{or} \quad A = -\frac{\mu}{p^2} \cdot \sqrt{\left(\frac{2}{\pi}\right)}.$$

$$\therefore e^{kp^2 t} \tilde{U}_c = \sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{\mu}{p^2} (-1 + e^{kp^2 t})$$

$$\text{or} \quad \tilde{U}_c = \sqrt{\left(\frac{2}{\pi}\right)} \cdot \frac{\mu}{p^2} (1 - e^{-kp^2 t}).$$

Applying the inverse Fourier cosine transform, we have

$$U_c = \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \tilde{U}_c \cos px \, dp$$

or

$$U_c = \frac{2\mu}{\pi} \int_0^\infty \frac{\cos px}{p^2} \cdot \left(1 - e^{-kp^2 t}\right) dp.$$

Example 2: Solve $\frac{\partial U}{\partial t} = 2 \frac{\partial^2 U}{\partial x^2}$

if $U(0, t) = 0, U(x, 0) = e^{-x}, x > 0, U(x, t)$ is bounded where $x > 0, t > 0$. (Meerut 2013B)

Solution: Since $U(0, t)$ is given, so taking the Fourier sine transform of both the sides, we have

$$\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\partial U}{\partial t} \sin px \, dx = 2 \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\partial^2 U}{\partial x^2} \sin px \, dx$$

or

$$\begin{aligned} \frac{d}{dt} \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty U \sin px \, dx &= 2 \sqrt{\left(\frac{2}{\pi}\right)} \cdot \left(\frac{\partial U}{\partial x} \sin px\right)_0^\infty \\ &\quad - 2p \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\partial U}{\partial x} \cos px \, dx \\ &\text{if } \frac{\partial U}{\partial x} \rightarrow 0 \text{ as } x \rightarrow \infty. \end{aligned}$$

or

$$\begin{aligned} \frac{d}{dt} \tilde{U}_s &= -2p \sqrt{\left(\frac{2}{\pi}\right)} (U \cos px)_0^\infty - 2p^2 \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty U \sin px \, dx \\ &\text{if } U \rightarrow 0 \text{ as } x \rightarrow \infty \end{aligned}$$

$$= 2p \sqrt{\left(\frac{2}{\pi}\right)} U(0, t) - 2p^2 \tilde{U}_s$$

or

$$\frac{d\tilde{U}_s}{dt} + 2p^2 \tilde{U}_s = 2p \sqrt{\left(\frac{2}{\pi}\right)} U(0, t) = 0$$

whose solution is $\tilde{U}_s = A e^{-2p^2 t}$... (1)

But $U(x, 0) = e^{-x}$.

$$\begin{aligned} \therefore \text{ when } t = 0, \tilde{U}_s &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty e^{-x} \sin px \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \left[\frac{e^{-x}}{1+p^2} (-\sin px - p \cos px) \right]_0^\infty \\ &= \frac{p}{1+p^2} \sqrt{\left(\frac{2}{\pi}\right)}. \end{aligned}$$

$\therefore$ from (1) when $t = 0$

$$\tilde{U}_s = \frac{p}{1+p^2} \sqrt{\left(\frac{2}{\pi}\right)} = A.$$

Hence
$$\tilde{U}_s = \sqrt{\left(\frac{2}{\pi}\right)} \frac{p}{1+p^2} e^{-2p^2 t}.$$

Now applying the inverse Fourier sine transform, we have

$$U(x, t) = \frac{2}{\pi} \int_0^\infty \frac{pe^{-2p^2 t} \sin px}{1+p^2} dp.$$

Example 3: Using the Fourier sine transform, solve the partial differential equation $\frac{\partial V}{\partial t} = k \frac{\partial^2 V}{\partial x^2}$ for $x > 0, t > 0$, under the boundary conditions $V = V_0$ when $x = 0, t > 0$, and the initial condition $V = 0$, when $t = 0, x > 0$.

Solution: Taking the Fourier sine transform of both the sides, we have

$$\sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\partial V}{\partial t} \sin px \, dx = k \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \frac{\partial^2 V}{\partial x^2} \sin px \, dx$$

or
$$\frac{d}{dt} \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty V \sin px \, dx = -kp^2 \tilde{V}_s + kp \sqrt{\left(\frac{2}{\pi}\right)} V(0, t)$$

[See Ex. 2.]

or
$$\frac{d\tilde{V}_s}{dt} + kp^2 \tilde{V}_s = kp \sqrt{\left(\frac{2}{\pi}\right)} \cdot V_0, \quad [\because V(0, t) = V_0]$$

which is linear differential equation of first order.

$$\therefore \text{I.F.} = e^{\int kp^2 dt} = e^{kp^2 t}.$$

Hence its solution is

$$\tilde{V}_s e^{kp^2 t} = c + \sqrt{\left(\frac{2}{\pi}\right)} \int kp V_0 \cdot e^{kp^2 t} dt$$

or
$$\tilde{V}_s e^{kp^2 t} = c + \sqrt{\left(\frac{2}{\pi}\right)} \frac{V_0}{p} e^{kp^2 t} \quad \dots(1)$$

But when $t = 0, \tilde{V}_s = 0. \quad [\because V = 0 \text{ when } t = 0]$

$\therefore$ from (1), we have

$$0 = c + \frac{V_0}{p} \cdot \sqrt{\left(\frac{2}{\pi}\right)} \quad \text{or} \quad c = -\frac{V_0}{p} \cdot \sqrt{\left(\frac{2}{\pi}\right)}.$$

$\therefore$ from (1), $\tilde{V}_s \cdot e^{kp^2 t} = \frac{V_0}{p} \sqrt{\left(\frac{2}{\pi}\right)} (e^{kp^2 t} - 1)$

or
$$\tilde{V}_s = \sqrt{\left(\frac{2}{\pi}\right)} \frac{V_0}{p} (1 - e^{-kp^2 t}).$$

Now applying the inverse Fourier transform, we have

$$V = \frac{2}{\pi} \cdot V_0 \int_0^\infty \frac{(1 - e^{-kp^2 t})}{p} \sin px \, dp$$

$$\text{or} \quad V = \frac{2V_0}{\pi} \left[\int_0^\infty \frac{\sin px}{p} dp - \int_0^\infty \frac{e^{-kp^2 t}}{p} \cdot \sin px dp \right]$$

$$= \frac{2V_0}{\pi} \left[\frac{\pi}{2} - \int_0^\infty \frac{e^{-kp^2 t}}{p} \sin px dp \right]$$

$$\text{or} \quad V(x, t) = V_0 \left[1 - \frac{2}{\pi} \int_0^\infty \frac{e^{-kp^2 t}}{p} \sin px dp \right].$$

Note: In terms of Error function or Complementary error function the solution may be written as

$$V(x, t) = V_0 \left[1 - \operatorname{erf} \frac{x}{2\sqrt{(kt)}} \right] = V_0 \operatorname{erf} c \left(\frac{x}{2\sqrt{(kt)}} \right) \text{ as shown below.}$$

We know that

$$\int_0^\infty e^{-\alpha^2 x^2} \cos 2\beta x dx = \frac{\sqrt{\pi}}{2\alpha} e^{-\beta^2/\alpha^2}.$$

Integrating both sides w.r.t. β from 0 to β , we have

$$\int_0^\infty e^{-\alpha^2 x^2} \left(\int_0^\beta \cos 2\beta x d\beta \right) dx = \frac{\sqrt{\pi}}{2\alpha} \int_0^\beta e^{-\beta^2/\alpha^2} d\beta$$

$$\text{or} \quad \int_0^\infty e^{-\alpha^2 x^2} \frac{1}{2x} \sin 2\beta x dx = \frac{\sqrt{\pi}}{2\alpha} \alpha \int_0^{\beta/\alpha} e^{-u^2} du,$$

putting $\frac{\beta}{\alpha} = u$ so that $d\beta = \alpha du$

$$\text{or} \quad \int_0^\infty e^{-\alpha^2 x^2} \cdot \frac{\sin 2\beta x}{x} dx = \frac{\pi}{2} \cdot \frac{2}{\sqrt{\pi}} \int_0^{\beta/\alpha} e^{-u^2} du$$

$$= \frac{\pi}{2} \operatorname{erf} (\beta/\alpha).$$

$$\therefore \quad \frac{2}{\pi} \int_0^\infty e^{-kp^2 t} \cdot \frac{\sin px}{p} dp$$

$$= \frac{2}{\pi} \int_0^\infty e^{-4ktu^2} \cdot \frac{\sin 2ux}{u} du, \text{ putting } p = 2u \text{ so that } dp = 2du$$

$$= \frac{2}{\pi} \cdot \frac{\pi}{2} \operatorname{erf} \left(\frac{x}{2\sqrt{(kt)}} \right) = \operatorname{erf} \left(\frac{x}{2\sqrt{(kt)}} \right).$$

Example 4: Use the method of Fourier transform to determine the displacement $y(x, t)$ of an infinite string, given that the string is initially at rest and that the initial displacement is $f(x)$, $-\infty < x < \infty$. Show that the solution can also be put in the form

$$y(x, t) = \frac{1}{2} [f(x+ct) + f(x-ct)]. \quad (\text{Meerut 2006})$$

Solution: Displacement of a string is governed by one dimensional wave equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \dots(1)$$

where $y(x, t)$ is the displacement at any time t .

$$-\infty < x < \infty, \quad t > 0$$

and
$$c^2 = \frac{T}{\rho}.$$

Taking the Fourier transform of both the sides of equation (1), we have

$$\frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{\partial^2 y}{\partial t^2} e^{ipx} dx = c^2 \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{\partial^2 y}{\partial x^2} e^{ipx} dx$$

or
$$\frac{d^2}{dt^2} \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} y e^{ipx} dx = c^2 (-ip)^2 \tilde{y}(p, t)$$

or
$$\frac{d^2 \tilde{y}(p, t)}{dt^2} = -c^2 p^2 \tilde{y}(p, t)$$

or
$$\frac{d^2 \tilde{y}(p, t)}{dt^2} + c^2 p^2 \tilde{y}(p, t) = 0$$

whose solution is

$$\tilde{y}(p, t) = A \cos cpt + B \sin cpt. \quad \dots(2)$$

Initially the string is at rest.

$\therefore \frac{\partial y}{\partial t} = 0, \text{ at } t = 0.$

$\therefore \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{\partial y}{\partial t} \cdot e^{ipx} dx = \frac{d}{dt} \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} y e^{ipx} dx$

$$= \frac{d \tilde{y}(p, t)}{dt} = 0, \text{ at } t = 0.$$

$\therefore$ from (2), we have $0 = Bcp$ or $B = 0$.

Also at $t = 0, y = f(x)$.

$\therefore$ at $t = 0, \tilde{y}(p, 0) = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f(u) e^{ipu} du = \tilde{f}(p).$

$\therefore$ from (2), we have $\tilde{f}(p) = A$.

Hence $\tilde{y}(p, t) = \tilde{f}(p) \cos cpt$.

Taking the inverse Fourier transform, we have

$$y(x, t) = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \tilde{f}(p) \cos cpt e^{-ipx} dp$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(u) e^{ipu} du \right] \cos cpt e^{-ipx} dp$$

$$\begin{aligned}
&= \frac{1}{4\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(u) e^{ipu} du \right] (e^{icpt} + e^{-icpt}) e^{-ipx} dp \\
&= \frac{1}{2} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} \left\{ \int_{-\infty}^{\infty} f(u) e^{-iau} du \right\} (e^{-i\alpha t} + e^{i\alpha t}) e^{i\alpha x} d\alpha \right], \\
&\quad \text{putting } p = -\alpha \text{ so that } dp = -d\alpha \\
&= \frac{1}{2} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} f(u) e^{-iau} \left\{ \int_{-\infty}^{\infty} e^{i\alpha(x+ct)} d\alpha \right\} du \right. \\
&\quad \left. + \frac{1}{2\pi} \int_{-\infty}^{\infty} f(u) e^{-iau} \left\{ \int_{-\infty}^{\infty} e^{i\alpha(x-ct)} d\alpha \right\} du \right] \\
&= \frac{1}{2} [f(x+ct) + f(x-ct)].
\end{aligned}$$

(From Fourier integral formula, see article 4.3)

Example 5: If the flow of heat is linear so that the variation of θ (temperature) with z and y may be neglected and if it is assumed that no heat is generated in the medium, then solve the differential equation $\frac{\partial \theta}{\partial t} = k \frac{\partial^2 \theta}{\partial x^2}$ (one dimensional heat equation) where $-\infty < x < \infty$ and $\theta = f(x)$ where $t = 0$, $f(x)$ being a given function of x .

Solution: Taking the Fourier transform of both the sides of the given equation, we have

$$\frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{\partial \theta}{\partial t} e^{ipx} dx = k \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{\partial^2 \theta}{\partial x^2} e^{ipx} dx$$

$$\text{or} \quad \frac{d}{dt} \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \theta \cdot e^{ipx} dx = k (-ip)^2 \tilde{\theta} \quad [\text{From article 4.20 (b)}]$$

where $\tilde{\theta} = \tilde{\theta}(p, t)$ is the Fourier transform of $\theta(x, t)$

$$\text{or} \quad \frac{d}{dt} \tilde{\theta} = -kp^2 \tilde{\theta}$$

whose solution is

$$\tilde{\theta} = A e^{-kp^2 t} \quad \dots(1)$$

Now at $t = 0$, $\theta = f(x)$.

$$\therefore \text{ at } t = 0, \tilde{\theta} = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f(x) e^{ipx} dx = \tilde{f}(p).$$

$$\therefore \text{ from (1), at } t = 0, \tilde{\theta} = \tilde{f}(p) = A;$$

$$\text{hence from (1), } \tilde{\theta} = \tilde{f}(p) e^{-kp^2 t}.$$

Taking the inverse Fourier transform, we have

$$\theta(x, t) = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \tilde{f}(p) e^{-kp^2 t} e^{-ipx} dp$$

$$\text{or} \quad \theta(x, t) = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \tilde{f}(p) e^{-kp^2 t - ipx} dp.$$

Example 6: Solve $\frac{\partial^4 V}{\partial x^4} + \frac{\partial^2 V}{\partial y^2} = 0$, $-\infty < x < \infty$, $y \geq 0$

satisfying the conditions

(i) V and its partial derivatives tend to zero as $x \rightarrow \pm \infty$ and

(ii) $V = f(x)$, $\frac{\partial V}{\partial y} = 0$ on $y = 0$.

Solution: Taking the Fourier transform of both the sides, we have

$$\frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{\partial^4 V}{\partial x^4} e^{ipx} dx + \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{\partial^2 V}{\partial y^2} e^{ipx} dx = 0$$

or

$$\frac{1}{\sqrt{(2\pi)}} \left(\frac{\partial^3 V}{\partial x^3} \cdot e^{ipx} \right)_{-\infty}^{\infty} - \frac{ip}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{\partial^3 V}{\partial x^3} e^{ipx} dx + \frac{d^2}{dy^2} \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} V \cdot e^{ipx} dx = 0.$$

or

$$-\frac{ip}{\sqrt{(2\pi)}} \left[\left(\frac{\partial^2 V}{\partial x^2} e^{ipx} \right)_{-\infty}^{\infty} - ip \int_{-\infty}^{\infty} \frac{\partial^2 V}{\partial x^2} e^{ipx} dx \right] + \frac{d^2 \tilde{V}}{dy^2} = 0,$$

since $\frac{\partial^3 V}{\partial x^3} \rightarrow 0$ as $x \rightarrow \pm \infty$

or

$$\frac{(ip)^2}{\sqrt{(2\pi)}} \left[\left(\frac{\partial V}{\partial x} e^{ipx} \right)_{-\infty}^{\infty} - ip \int_{-\infty}^{\infty} \frac{\partial V}{\partial x} e^{ipx} dx \right] + \frac{d^2 \tilde{V}}{dy^2} = 0,$$

since $\frac{\partial^2 V}{\partial x^2} \rightarrow 0$ as $x \rightarrow \pm \infty$

or

$$-\frac{(ip)^3}{\sqrt{(2\pi)}} \left[\left(V e^{ipx} \right)_{-\infty}^{\infty} - ip \int_{-\infty}^{\infty} V e^{ipx} dx \right] + \frac{d^2 \tilde{V}}{dy^2} = 0$$

or

$$\frac{(ip)^4}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} V e^{ipx} dx + \frac{d^2 \tilde{V}}{dy^2} = 0$$

or

$$p^4 \tilde{V} + \frac{d^2 \tilde{V}}{dy^2} = 0$$

whose solution is

$$\tilde{V} = A \cos p^2 y + B \sin p^2 y. \quad \dots(1)$$

Since on $y = 0$, $V = f(x)$ and $\frac{\partial V}{\partial y} = 0$,

∴ taking Fourier transform, we have

on
$$y = 0, \tilde{V} = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} f(x) e^{ipx} dx = \tilde{f}(p)$$

and
$$\frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \frac{\partial V}{\partial y} e^{ipx} dx = 0 \text{ i.e., } \frac{d\tilde{V}}{dy} = 0.$$

∴ from (1), we have

$$\tilde{f}(p) = A$$

and
$$\frac{d\tilde{V}}{dy} = 0 = Bp^2 \quad \text{or} \quad B = 0.$$

∴ the solution (1) reduces to

$$\tilde{V} = \tilde{f}(p) \cos p^2 y.$$

∴ applying inversion theorem for Fourier transform, we have

$$V = \frac{1}{\sqrt{(2\pi)}} \int_{-\infty}^{\infty} \tilde{f}(p) \cos(p^2 y) \cdot e^{-ipx} dp.$$

Example 7: Use a cosine transform to show that the steady temperature in the semi-infinite solid $y > 0$ when the temperature on the surface $y = 0$ is kept at unity over the strip $|x| < a$ and at zero outside the strip, is

$$\frac{1}{\pi} \left[\tan^{-1} \left(\frac{a+x}{y} \right) + \tan^{-1} \left(\frac{a-x}{y} \right) \right].$$

The result $\int_0^{\infty} e^{-sx} x^{-1} \sin rx dx = \tan^{-1} \frac{r}{s}, r > 0, s > 0$, may be assumed.

Solution: Here the steady temperature $U(x, y)$ in the semi-infinite solid is governed by two-dimensional Laplace equation

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} = 0 \quad \dots(1)$$

$$0 < y < \infty, -\infty < x < \infty$$

subject to the conditions

$$U = 1, y = 0 \quad -a < x < a$$

and
$$U = 0, y = 0 \quad \text{when } |x| \nless a.$$

Taking Fourier cosine transform of both sides of (1), we have

$$\sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \frac{\partial^2 U}{\partial x^2} \cos px dx + \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \frac{\partial^2 U}{\partial y^2} \cos px dx = 0$$

or
$$\sqrt{\left(\frac{2}{\pi}\right)} \left[\frac{\partial U}{\partial x} \cdot \cos px \right]_0^{\infty} + p \sqrt{\left(\frac{2}{\pi}\right)} \int_0^{\infty} \sin px dx + \frac{d^2 \tilde{U}_c}{dy^2} = 0$$

$$\left(\frac{\partial U}{\partial x} \rightarrow 0 \text{ as } x \rightarrow \infty \text{ and } \frac{\partial U}{\partial x} \rightarrow 0 \text{ as } x \rightarrow 0. \text{ By symmetry} \right)$$

$$\text{or} \quad p \sqrt{\left(\frac{2}{\pi}\right)} [U \sin px]_0^\infty - p^2 \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty U \cos px \, dx + \frac{d^2 \tilde{U}_c}{dy^2} = 0$$

$$\text{or} \quad \frac{d^2 \tilde{U}_c}{dy^2} - p^2 \tilde{U}_c = 0 \quad (\text{if } U \rightarrow 0 \text{ as } x \rightarrow \infty)$$

$$\text{whose solution is } \tilde{U}_c = A e^{py} + B e^{-py}. \quad \dots(2)$$

Now as $y \rightarrow \infty, \tilde{U}_c \rightarrow 0 \therefore A = 0$.

$$\therefore \quad \tilde{U}_c = B e^{-py}. \quad \dots(3)$$

Also when $y = 0$,

$$\begin{aligned} \tilde{U}_c &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^a 1 \cdot \cos px \, dx + \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty 0 \cdot \cos px \, dx \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \left(\frac{\sin px}{p} \right)_0^a \\ &= \sqrt{\left(\frac{2}{\pi}\right)} \frac{\sin pa}{p}. \end{aligned}$$

$$\therefore \quad \text{from (3), } \sqrt{\left(\frac{2}{\pi}\right)} \frac{\sin pa}{p} = B.$$

$$\text{Hence } \tilde{U}_c = \sqrt{\left(\frac{2}{\pi}\right)} \frac{\sin pa}{p} \cdot e^{-py}.$$

Taking the inverse Fourier cosine transform, we have

$$\begin{aligned} U(x, y) &= \sqrt{\left(\frac{2}{\pi}\right)} \int_0^\infty \sqrt{\left(\frac{2}{\pi}\right)} \frac{\sin pa}{p} e^{-py} \cos px \, dp \\ &= \frac{1}{\pi} \int_0^\infty \frac{e^{-py}}{p} \cdot [\sin(a+x)p + \sin(a-x)p] \, dp \\ &= \frac{1}{\pi} \int_0^\infty e^{-py} \cdot p^{-1} \sin(a+x)p \, dp + \frac{1}{\pi} \int_0^\infty e^{-py} \cdot p^{-1} \sin(a-x)p \, dp \\ &= \frac{1}{\pi} \left[\tan^{-1} \left(\frac{a+x}{y} \right) + \tan^{-1} \left(\frac{a-x}{y} \right) \right] \end{aligned}$$

assuming the result $\int_0^\infty e^{-sx} x^{-1} \sin rx \, dx = \tan^{-1} \frac{r}{s}, r > 0, s > 0$.

6.3 Application of Finite Fourier Transforms

In the problems in which the range of one of the variables is finite, finite Fourier transforms are applied.

6.4 Finite Fourier Transforms of Partial Derivatives

(a) The finite Fourier sine and cosine transforms of $\frac{\partial U}{\partial x}$ where U is a function of x and t for

$$0 < x < l, t > 0.$$

We have

$$\begin{aligned} F_s \left\{ \frac{\partial U}{\partial x} \right\} &= \int_0^l \frac{\partial U}{\partial x} \sin \frac{p\pi x}{l} dx \\ &= \left[U(x, t) \sin \frac{p\pi x}{l} \right]_0^l - \frac{p\pi}{l} \int_0^l U \cos \frac{p\pi x}{l} dx \quad (\text{Integrating by parts}) \end{aligned}$$

$$\text{or} \quad F_s \left\{ \frac{\partial U}{\partial x} \right\} = -\frac{p\pi}{l} F_c \{U\}$$

$$\begin{aligned} \text{and} \quad F_c \left\{ \frac{\partial U}{\partial x} \right\} &= -\int_0^l \frac{\partial U}{\partial x} \cos \frac{p\pi x}{l} dx \\ &= \left[U(x, t) \cos \frac{p\pi x}{l} \right]_0^l + \frac{p\pi}{l} \int_0^l U \sin \frac{p\pi x}{l} dx \end{aligned}$$

$$\text{or} \quad F_c \left\{ \frac{\partial U}{\partial x} \right\} = \frac{p\pi}{l} F_s \{U\} - \{U(0, t) - U(l, t) \cos p\pi\}.$$

(b) The finite Fourier sine and cosine transforms of $\frac{\partial^2 U}{\partial x^2}$ where U is a function of x and t , for

$$0 < x < l, t > 0.$$

$$\begin{aligned} F_s \left\{ \frac{\partial^2 U}{\partial x^2} \right\} &= \int_0^l \frac{\partial^2 U}{\partial x^2} \sin \frac{p\pi x}{l} dx = \left[\frac{\partial U}{\partial x} \sin \frac{p\pi x}{l} \right]_0^l - \frac{p\pi}{l} \int_0^l \frac{\partial U}{\partial x} \cos \frac{p\pi x}{l} dx \\ &= -\frac{p\pi}{l} \left[\left(U \cos \frac{p\pi x}{l} \right)_0^l + \frac{p\pi}{l} \int_0^l U \sin \frac{p\pi x}{l} dx \right] \\ &= -\frac{p\pi}{l} \left[\frac{p\pi}{l} F_s \{U\} - \{U(0, t) - U(l, t) \cos p\pi\} \right] \end{aligned}$$

$$\text{or} \quad F_s \left\{ \frac{\partial^2 U}{\partial x^2} \right\} = -\frac{p^2 \pi^2}{l} F_s \{U\} + \frac{p\pi}{l} \{U(0, t) - U(l, t) \cos p\pi\}$$

$$\begin{aligned} \text{and} \quad F_c \left\{ \frac{\partial^2 U}{\partial x^2} \right\} &= \int_0^l \frac{\partial^2 U}{\partial x^2} \cos \frac{p\pi x}{l} dx = \left[\frac{\partial U}{\partial x} \cos \frac{p\pi x}{l} \right]_0^l + \frac{p\pi}{l} \int_0^l \frac{\partial U}{\partial x} \sin \frac{p\pi x}{l} dx \\ &= \frac{p\pi}{l} F_s \left\{ \frac{\partial U}{\partial x} \right\} - \{U_x(0, t) - U_x(l, t) \cos p\pi\} \end{aligned}$$

$$\text{or} \quad F_c \left\{ \frac{\partial^2 U}{\partial x^2} \right\} = -\frac{p^2 \pi^2}{l^2} F_c \{U\} - \{U_x(0, t) - U_x(l, t) \cos p\pi\}$$

where U_x denotes the partial derivative w.r.t. 'x'.

6.5 Choice of Finite Sine or Cosine Transform

The choice of finite sine or cosine transform is decided by the form of the boundary conditions.

From the last article we see that for exclusion of $\frac{\partial^2 U}{\partial x^2}$ from a differential equation, we require

$$U(0, t) \text{ and } U(l, t) \text{ in finite sine transform}$$

$$\text{and} \quad U_x(0, t) \text{ and } U_x(l, t) \text{ in finite cosine transform.}$$

Illustrative Examples

Example 8: Use finite Fourier transforms to solve

$$\frac{\partial U}{\partial t} = \frac{\partial^2 U}{\partial x^2},$$

$U(0, t) = 0, U(4, t) = 0, U(x, 0) = 2x$, where $0 < x < 4, t > 0$. Give the physical interpretation of the problem. (Meerut 2013; Rohilkhand 14)

Solution: Taking the finite Fourier sine transform (with $l = 4$) of both sides of the given partial differential equation, we have

$$\int_0^4 \frac{\partial U}{\partial t} \sin \frac{p\pi x}{4} dx = \int_0^4 \frac{\partial^2 U}{\partial x^2} \sin \frac{p\pi x}{4} dx$$

$$\text{or} \quad \frac{d\tilde{U}_s}{dt} = -\frac{p^2 \pi^2}{4^2} \tilde{U}_s + \frac{p\pi}{4} [U(0, t) - U(4, t) \cos p\pi] \quad [\text{See article 6.4 (b)}]$$

where $\tilde{U}_s$ is finite Fourier sine transform of U

$$\text{or} \quad \frac{d\tilde{U}_s}{dt} = -\frac{p^2 \pi^2}{16} \tilde{U}_s$$

whose solution is $\tilde{U}_s = A e^{-p^2 \pi^2 t / 16}$ (1)

Since $U(x, 0) = 2x$, where $0 < x < 4$, taking finite Fourier sine transforms, we have

$$\text{at } t = 0, \tilde{U}_s = \int_0^4 2x \cdot \sin \frac{p\pi x}{4} dx$$

$$= \left[-2x \cdot \frac{4}{p\pi} \cos \frac{p\pi x}{4} + 2 \cdot \frac{4}{p\pi} \cdot \frac{4}{p\pi} \sin \frac{p\pi x}{4} \right]_0^4$$

$$= \frac{32 (-\cos p\pi)}{p\pi} = \frac{32 \{-(-1)^p\}}{p\pi}.$$

$$\therefore \text{ from (1), } \frac{32 (-1)^{p+1}}{p\pi} = A.$$

$$\text{Hence } \tilde{U}_s = \frac{32 (-1)^{p+1}}{p\pi} \cdot e^{-p^2 \pi^2 t/16}.$$

Taking the inverse finite Fourier sine transform, we have

$$\begin{aligned} U(x, t) &= \frac{2}{4} \sum_{p=1}^{\infty} \frac{32 (-1)^{p+1}}{p\pi} \cdot e^{-p^2 \pi^2 t/16} \cdot \sin \frac{p\pi x}{4} \\ &= \frac{16}{\pi} \sum_{p=1}^{\infty} \frac{(-1)^{p+1}}{p} \cdot e^{-p^2 \pi^2 t/16} \cdot \sin \frac{p\pi x}{4}, \end{aligned}$$

which is the required solution.

Physical Interpretation: Physically $U(x, t)$ represents the temperature at any point x at any time t in solid bounded by the planes $x = 0$ and $x = 4$ (or a bar on the x -axis with the ends $x = 0$ and $x = 4$, whose surface is insulated laterally). The conditions $U(0, t) = 0$ and $U(4, t) = 0$ imply that the ends are kept at zero temperature while $U(x, 0) = 2x$ implies that the initial temperature is a function of x .

Example 9: Use the finite cosine transform to solve

$$\frac{\partial V}{\partial t} = k \frac{\partial^2 V}{\partial x^2} \text{ (one dimensional heat equation)}$$

with the boundary conditions $\frac{\partial V}{\partial x} = 0$, when $x = 0$ and $x = \pi$, $t > 0$ and the initial condition

$V = f(x)$, when $t = 0$, $0 < x < \pi$.

(Kanpur 2014)

Solution: Taking the finite Fourier cosine transform (with $l = \pi$) of both the sides of the given equation, we have

$$\int_0^{\pi} \frac{\partial V}{\partial t} \cos px \, dx = k \int_0^{\pi} \frac{\partial^2 V}{\partial x^2} \cos px \, dx$$

$$\text{or } \frac{d\tilde{V}_c}{dt} = k [-p^2 \tilde{V}_c - \{V_x(0, t) - V_x(\pi, t) \cos p\pi\}],$$

where $\tilde{V}_c$ is finite Fourier cosine transform of V

[See article 6.4 (b)]

$$\text{or } \frac{d\tilde{V}_c}{dt} = -k p^2 \tilde{V}_c$$

whose solution is

$$\tilde{V}_c = A e^{-kp^2 t}. \quad \dots(1)$$

Now taking finite Fourier cosine transform of initial condition, we have

at $t = 0, \tilde{V}_c = \int_0^\pi f(y) \cos py \, dy.$

$\therefore$ from (1), $A = \int_0^\pi f(x) \cos px \, dx.$

Hence $\tilde{V}_c = e^{-kp^2t} \int_0^\pi f(y) \cos py \, dy.$

Taking the inverse finite Fourier cosine transform, we have

$$\begin{aligned} V(x, t) &= \frac{1}{\pi} \tilde{V}_c(0) + \frac{2}{\pi} \sum_{p=1}^{\infty} \tilde{V}_c(p) \cos px \\ &= \frac{1}{\pi} \int_0^\pi f(y) \, dy + \frac{2}{\pi} \sum_{p=1}^{\infty} \left[e^{-kp^2t} \cos px \cdot \int_0^\pi f(y) \cos py \, dy \right]. \end{aligned}$$

Example 10: Solve $\frac{\partial U}{\partial t} = \frac{\partial^2 U}{\partial x^2}, \quad 0 < x < 6, t > 0$

subject to the conditions

$$U(0, t) = 0, U(6, t) = 0, U(x, 0) = \begin{cases} 1, & 0 < x < 3 \\ 0, & 3 < x < 6 \end{cases}$$

and interpret physically.

(Kanpur 2011)

Solution: Taking the finite Fourier sine transform ($l = 6$) of both the sides of the given partial differential equation, we have

$$\int_0^6 \frac{\partial U}{\partial t} \cdot \sin \frac{p\pi x}{6} \, dx = \int_0^6 \frac{\partial^2 U}{\partial x^2} \sin \frac{p\pi x}{6} \, dx$$

or $\frac{d\tilde{U}_s}{dt} = -\frac{p^2\pi^2}{6^2} \tilde{U}_s + \frac{p\pi}{6} [U(0, t) - U(6, t) \cos p\pi]$

or $\frac{d\tilde{U}_s}{dt} = -\frac{p^2\pi^2}{36} \tilde{U}_s,$

whose solution is

$$\tilde{U}_s = A e^{-p^2\pi^2t/36}. \quad \dots(1)$$

Now at $t = 0, U = \begin{cases} 1, & 0 < x < 3 \\ 0, & 3 < x < 6. \end{cases}$

$$\begin{aligned} \therefore \text{ at } t = 0, \tilde{U}_s &= \int_0^6 U \cdot \sin \frac{p\pi x}{6} \, dx \\ &= \int_0^3 1 \cdot \sin \frac{p\pi x}{6} \, dx + \int_3^6 0 \cdot \sin \frac{p\pi x}{6} \, dx = \frac{6}{p\pi} \left(1 - \cos \frac{p\pi}{2} \right). \end{aligned}$$

$\therefore$ from (1), when $t = 0$, $\frac{6}{p\pi} \left(1 - \cos \frac{p\pi}{2}\right) = A$.

Hence $\tilde{U}_s = \frac{6}{p\pi} \left(1 - \cos \frac{p\pi}{2}\right) e^{-p^2 \pi^2 t / 36}$.

Taking the inverse Fourier sine transform, we have

$$\begin{aligned} U(x, t) &= \frac{2}{6} \cdot \sum_{p=1}^{\infty} \frac{6}{p\pi} \left(1 - \cos \frac{p\pi}{2}\right) e^{-p^2 \pi^2 t / 36} \sin \frac{p\pi x}{6} \\ &= \frac{2}{\pi} \sum_{p=1}^{\infty} \frac{1}{p} \left(1 - \cos \frac{p\pi}{2}\right) e^{-p^2 \pi^2 t / 36} \cdot \sin \frac{p\pi x}{6} \end{aligned}$$

Physical Interpretation: Physically $U(x, t)$ represents the temperature at any point x at any time t in a bar with the ends $x = 0$ and $x = 6$ kept at zero temperature which is insulated laterally. Initially the temperature in the half bar from $x = 0$ to $x = 3$ is constant equal to 1 unit while the half bar from $x = 3$ to $x = 6$ is at zero temperature.

Example 11: Solve the boundary value problem

$$\frac{\partial U}{\partial t} = \frac{\partial^2 U}{\partial x^2}, U(0, t) = 1, U(\pi, t) = 3, U(x, 0) = 1,$$

where $0 < x < \pi, t > 0$.

Give the physical interpretation of the problem.

Solution: Taking the finite Fourier sine transform ($l = \pi$) of both the sides of the given differential equation, we have

$$\int_0^{\pi} \frac{\partial U}{\partial t} \sin px \, dx = \int_0^{\pi} \frac{\partial^2 U}{\partial x^2} \cdot \sin px \, dx$$

$$\text{or} \quad \frac{d\tilde{U}_s}{dt} = -p^2 \tilde{U}_s + p[U(0, t) - U(\pi, t) \cos p\pi]$$

$$\text{or} \quad \frac{d\tilde{U}_s}{dt} = -p^2 \tilde{U}_s + p[1 - 3 \cos p\pi]$$

$$\text{or} \quad \frac{d\tilde{U}_s}{dt} + p^2 \tilde{U}_s = p(1 - 3 \cos p\pi) \quad \dots(1)$$

which is a linear diff. equation of first order.

$$\text{I.F.} = e^{\int p^2 dt} = e^{p^2 t}.$$

$\therefore$ solution of (1) is

$$\begin{aligned} \tilde{U}_s \cdot e^{p^2 t} &= A + \int p(1 - 3 \cos p\pi) e^{p^2 t} dt \\ &= A + \frac{(1 - 3 \cos p\pi)}{p} e^{p^2 t} \end{aligned}$$

or
$$\tilde{U}_s = Ae^{-p^2 t} + \frac{(1 - 3 \cos p\pi)}{p}. \quad \dots(2)$$

Now when $t = 0, U = 1$.

Taking finite Fourier sine transform

$$\tilde{U}_s = \int_0^\pi 1 \cdot \sin px \, dx = \frac{1 \cdot (1 - \cos p\pi)}{p}.$$

$\therefore$ At $t = 0$, from (2), $\frac{1 \cdot (1 - \cos p\pi)}{p} = A + \frac{1 - 3 \cos p\pi}{p}$

or
$$A = \frac{2 \cos p\pi}{p}.$$

Hence from (2),

$$\tilde{U}_s = \frac{2 \cos p\pi}{p} e^{-p^2 t} + \frac{(1 - 3 \cos p\pi)}{p}.$$

Taking the inverse infinite Fourier sine transform, we have

$$\begin{aligned} U(x, t) &= \frac{2}{\pi} \sum_{p=1}^{\infty} \frac{2 \cos p\pi}{p} \cdot e^{-p^2 t} \sin px \\ &\quad + \frac{2}{\pi} \sum_{p=1}^{\infty} \frac{(1 - 3 \cos p\pi)}{p} \cdot \sin px \end{aligned} \quad \dots(3)$$

Now
$$F_s\{1\} = \int_0^\pi 1 \cdot \sin px \, dx = \frac{1 - \cos p\pi}{p}$$

$$= \frac{1 - (-1)^p}{p}, (p = 1, 2, \dots)$$

and
$$F_s\{x\} = \int_0^\pi x \cdot \sin px \, dx$$

$$= \left(-\frac{x}{p} \cos px \right)_0^\pi + \int_0^\pi \frac{\cos px}{p} \, dx$$

$$= -\frac{\pi}{p} \cos p\pi = \frac{\pi}{p} (-1)^{p+1}.$$

$\therefore 1 = \frac{2}{\pi} \sum_{p=1}^{\infty} \frac{1 - (-1)^p}{p} \sin px$

and
$$x = \frac{2}{\pi} \sum_{p=1}^{\infty} \frac{\pi (-1)^{p+1}}{p} \sin px = 2 \sum_{p=1}^{\infty} \frac{(-1)^{p+1}}{p} \sin px.$$

$\therefore \frac{2}{\pi} \sum_{p=1}^{\infty} \frac{(1 - 3 \cos p\pi)}{p} \sin px = \frac{2}{\pi} \sum_{p=1}^{\infty} \frac{1 - 3(-1)^p}{p} \sin px$

$$\begin{aligned}
 &= \frac{2}{\pi} \sum_{p=1}^{\infty} \frac{1 - (-1)^p}{p} \sin px + \frac{4}{\pi} \sum_{p=1}^{\infty} \frac{(-1)^{p+1}}{p} \sin px \\
 &= 1 + \frac{2}{\pi} x.
 \end{aligned}$$

Hence from (3),

$$U(x, t) = \frac{4}{\pi} \sum_{p=1}^{\infty} \frac{\cos p\pi}{p} e^{-p^2 t} \sin px + 1 + \frac{2x}{\pi}.$$

Example 12: Using finite Fourier transform, find the solution of the wave equation

$$\frac{\partial^2 U}{\partial t^2} = 4 \frac{\partial^2 U}{\partial x^2},$$

subject to the conditions

$$U(0, t) = 0, \quad U(\pi, t) = 0, \quad U(x, 0) = (0 \cdot 1) \sin x + (0 \cdot 01) \sin 4x$$

and $U_t(x, 0) = 0$ for $0 < x < \pi, t > 0$. (Meerut 2007)

Solution: Taking the finite Fourier sine transform of both the sides of the given equation, we have

$$\int_0^{\pi} \frac{\partial^2 U}{\partial t^2} \sin px \, dx = 4 \int_0^{\pi} \frac{\partial^2 U}{\partial x^2} \sin px \, dx$$

$$\text{or} \quad \frac{d^2 \tilde{U}_s}{dt^2} = -4p^2 \tilde{U}_s + 4p[U(0, t) - U(\pi, t) \cos p\pi]$$

$$\text{or} \quad \frac{d^2 \tilde{U}_s}{dt^2} + 4p^2 \tilde{U}_s = 0$$

whose solution is

$$\tilde{U}_s = A \cos 2pt + B \sin 2pt. \quad \dots(1)$$

$$\text{Now at } t = 0, U = (0 \cdot 1) \sin x + (0 \cdot 01) \sin 4x$$

$$\text{and } U_t = \frac{\partial U}{\partial t} = 0.$$

Taking finite sine transforms, we have

$$\begin{aligned}
 \text{At } t = 0, \tilde{U}_s &= \int_0^{\pi} \{(0 \cdot 1) \sin x + (0 \cdot 01) \sin 4x\} \sin px \, dx \\
 &= (0 \cdot 1) \int_0^{\pi} \sin x \cdot \sin px \, dx + (0 \cdot 01) \int_0^{\pi} \sin 4x \sin px \, dx
 \end{aligned}$$

$$\text{and } \frac{d\tilde{U}_s}{dt} = 0.$$

$\therefore$ from (1), we have

$$(0 \cdot 1) \int_0^{\pi} \sin x \sin px \, dx + (0 \cdot 01) \int_0^{\pi} \sin 4x \sin px \, dx = A$$

or $\frac{d\tilde{U}_s}{dt} = 0 = 2Bp \quad \text{or} \quad B = 0.$

Hence from (1),

$$\tilde{U}_s = \left[(0 \cdot 1) \int_0^\pi \sin x \sin px \, dx + (0 \cdot 01) \int_0^\pi \sin 4x \sin px \, dx \right] \cos 2pt.$$

Taking the inverse finite Fourier sine transform, we have

$$\begin{aligned} U(x, t) &= \frac{2}{\pi} \sum_{p=1}^{\infty} \tilde{U}_s \cdot \sin px \\ &= \frac{2}{\pi} (0 \cdot 1) \sum_{p=1}^{\infty} \left\{ \int_0^\pi \sin \pi \sin px \, dx \right\} \cos 2pt \sin px \\ &\quad + \frac{2}{\pi} (0 \cdot 01) \sum_{p=1}^{\infty} \left\{ \int_0^\pi \sin 4x \sin px \, dx \right\} \cos 2pt \sin px \\ &= \frac{2}{\pi} (0 \cdot 1) \left[\int_0^\pi \sin x \cdot \sin x \, dx \right] \cos 2t \sin x \\ &\quad + \frac{2}{\pi} (0 \cdot 01) \left[\int_0^\pi \sin 4x \cdot \sin 4x \, dx \right] \cos 8t \sin 4x \end{aligned}$$

(Since the integrals in first summation are all zero if $p \neq 1$
and the integrals in second summation are all zero if $p \neq 4$)

or $U(x, t) = (0 \cdot 1) \cos 2t \sin x + (0 \cdot 01) \cos 8t \sin 4x.$

Comprehensive Exercise 1

1. Solve $\frac{\partial U}{\partial t} = \frac{\partial^2 U}{\partial x^2}$, $x > 0, t > 0$ subject to the conditions

(i) $U = 0$, when $x = 0, t > 0$ (ii) $U = \begin{cases} 1, & 0 < x < 1 \\ 0, & x \geq 1 \end{cases}$ when $t = 0$

(iii) $U(x, t)$ is bounded.

2. If the function $U(x, y)$ is determined by the differential equation

$$\frac{\partial U}{\partial x} = \frac{\partial^2 U}{\partial y^2} \text{ for } x \geq 0, -\infty < y < \infty \text{ and } U = f(y) \text{ when } x = 0,$$

show that $U(x, y) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(p) e^{-p^2 x - ipy} dp$

where $\tilde{f}(p)$ is the Fourier transform of $f(y)$.

[Hint: Replace $k \rightarrow 1, \theta \rightarrow U, t \rightarrow x$ and $x \rightarrow y$ in Ex. 5]

3. Use finite Fourier transform to solve

$$\frac{\partial V}{\partial t} = \frac{\partial^2 V}{\partial x^2}, \quad 0 < x < 6, t > 0, V_x(0, t) = 0, V_x(6, t) = 0, V(x, 0) = 2x. \quad (\text{Kanpur 2009})$$

4. Solve the boundary value problem $\frac{\partial^2 U}{\partial t^2} = 9 \frac{\partial^2 U}{\partial x^2}$ subject to the conditions

$$U(0, t) = 0, U(2, t) = 0, U(x, 0) = (0.05)x(2 - x)$$

and $U_t(x, 0) = 0$, where $0 < x < 2, t > 0$.

5. Find the steady temperature $V(x, y)$ in a long square bar of side π when one face is kept at constant and the other faces at zero temperature. Also $V(x, y)$ is bounded.

Or

Determine a function $V(x, y)$ which is harmonic in the open square, $0 < x < \pi, 0 < y < \pi$, takes a constant value V_0 on the edge $y = \pi$ and vanishes on the other edges of the square.

6. Determine the displacement $U(x, t)$ in a horizontal string stretched from origin to the point $(\pi, 0)$ when the motion is due to the weight of the string alone. The string may be taken to be initially at rest in the position $U = 0$.

[Hint: Here displacement is governed by

$$\frac{\partial^2 U}{\partial t^2} = c^2 \frac{\partial^2 U}{\partial x^2} + g, \quad 0 < x < \pi, t > 0$$

Conditions $U = 0 = \frac{\partial U}{\partial t}$, at $t = 0$ and $U = 0$, when $x = 0$ or π .]

Answers 1

$$1. \quad U(x, t) = \frac{2}{\pi} \int_0^\infty \frac{1 - \cos p}{p} e^{-p^2 t} \sin px \, dp$$

$$3. \quad V(x, t) = 6 + \frac{24}{\pi^2} \sum_{p=1}^\infty \left(\frac{\cos p\pi - 1}{p^2} \right) e^{-p^2 \pi^2 t/36} \cos \frac{p\pi x}{6}$$

$$4. \quad U(x, t) = \frac{(1 \cdot 6)}{\pi^3} \sum_{n=1}^\infty \frac{1}{(2n-1)^3} \cos \frac{3(2n-1)\pi t}{2} \sin \frac{(2n-1)\pi x}{2}$$

$$5. \quad V(x, y) = \frac{4V_0}{\pi} \sum_{n=0}^\infty \frac{\sinh(2n+1)y \sin(2n+1)x}{(2n+1) \sinh(2n+1)\pi}$$

$$6. \quad U(x, t) = \frac{2g}{\pi c^2} \sum_{p=1}^\infty \frac{1}{p^3} [1 - (-1)^p] (1 - \cos pat) \sin px.$$



Chapter

7



Fourier Series

7.1 Some Important Results

The following results are useful in Fourier Series :

- (i) $\sin n\pi = 0, \cos n\pi = (-1)^n$,
 $\sin\left(n + \frac{1}{2}\right)\pi = (-1)^n, \cos\left(n + \frac{1}{2}\right)\pi = 0$, where $n \in I$.
- (ii) $\int uv = uv_1 - u'v_3 + u''v_2 - u'''v_4 + \dots$,
where $u' = \frac{du}{dx}$, $u'' = \frac{d^2u}{dx^2}$, ... and $v_1 = \int v dx$, $v_2 = \int v_1 dx, \dots$.
- (iii) $\int_0^{2\pi} \sin nx dx = 0$. (iv) $\int_0^{2\pi} \cos nx dx = 0$.
- (v) $\int_0^{2\pi} \sin^2 nx dx = \pi$. (vi) $\int_0^{2\pi} \cos^2 nx dx = \pi$.
- (vii) $\int_0^{2\pi} \sin nx \cdot \sin mx dx = 0$. (viii) $\int_0^{2\pi} \cos nx \cdot \cos mx dx = 0$.
- (ix) $\int_0^{2\pi} \sin nx \cdot \cos mx dx = 0$. (x) $\int_0^{2\pi} \sin nx \cdot \cos nx dx = 0$.

7.2 Determination of Fourier Coefficients (Euler's Formulae)

The Fourier series is

$$f(x) = \frac{a_0}{2} + a_1 \cos x + a_2 \cos 2x + \dots + a_n \cos nx + \dots \\ + b_1 \sin x + b_2 \sin 2x + \dots + b_n \sin nx + \dots \quad \dots(1)$$

To find a_0 . Integrating both sides of (1) from $x = 0$ to $x = 2\pi$, we have

$$\int_0^{2\pi} f(x) dx = \frac{a_0}{2} \int_0^{2\pi} dx + a_1 \int_0^{2\pi} \cos x dx + a_2 \int_0^{2\pi} \cos 2x dx + \dots \\ + a_n \int_0^{2\pi} \cos nx dx + \dots + b_1 \int_0^{2\pi} \sin x dx + b_2 \int_0^{2\pi} \sin 2x dx \\ + \dots + b_n \int_0^{2\pi} \sin nx dx + \dots \\ = \frac{a_0}{2} \int_0^{2\pi} dx, \quad [\text{Other integrals vanish}]$$

or
$$\int_0^{2\pi} f(x) dx = \frac{a_0}{2} 2\pi.$$

$\therefore a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx. \quad \dots(2)$

To find a_n . Multiplying each side of (1) by $\cos nx$ and integrating from $x = 0$ to $x = 2\pi$, we have

$$\int_0^{2\pi} f(x) \cos nx dx = \frac{a_0}{2} \int_0^{2\pi} \cos nx dx + a_1 \int_0^{2\pi} \cos x \cos nx dx + \dots \\ + a_n \int_0^{2\pi} \cos^2 nx dx + \dots + b_1 \int_0^{2\pi} \sin x \cos nx dx + b_2 \int_0^{2\pi} \sin 2x \cos nx dx + \dots \\ = a_n \int_0^{2\pi} \cos^2 nx dx \quad [\text{Other integrals vanish}]$$

$\therefore a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx. \quad \dots(3)$

By taking $n = 1, 2, \dots$ we get the values of $a_1, a_2, \dots$

To find b_n . Multiplying each side of (1) by $\sin nx$ and integrating from $x = 0$ to $x = 2\pi$, we have

$$\int_0^{2\pi} f(x) \sin nx dx = \frac{a_0}{2} \int_0^{2\pi} \sin nx dx + a_1 \int_0^{2\pi} \cos x \sin nx dx + \dots \\ + a_n \int_0^{2\pi} \cos nx \sin nx dx + \dots \\ + b_1 \int_0^{2\pi} \sin x \sin nx dx + \dots + b_n \int_0^{2\pi} \sin^2 nx dx + \dots \\ = b_n \int_0^{2\pi} \sin^2 nx dx \quad [\text{Other integrals vanish}] \\ = b_n \pi.$$

$$\therefore b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx. \quad \dots(4)$$

By taking $n = 1, 2, \dots$ we get the values of $b_1, b_2, \dots$.

Note. To get similar formula of $a_0, \frac{1}{2}$ has been written with a_0 in Fourier series.

If we write the Fourier series as $f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$,

then the values of constants is given as :

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(x) \, dx; a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx \text{ and } b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx.$$

7.3 Fourier Series Expansion In The Interval $\alpha < x < \alpha + 2\pi$

The Fourier series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx. \quad \dots(1)$$

To find a_0 . Integrating both sides of (1) from $x = \alpha$ to $x = \alpha + 2\pi$, we have

$$\begin{aligned} \int_{\alpha}^{\alpha+2\pi} f(x) \, dx &= \frac{a_0}{2} \int_{\alpha}^{\alpha+2\pi} dx + \int_{\alpha}^{\alpha+2\pi} \left\{ \sum_{n=1}^{\infty} a_n \cos nx \right\} dx \\ &\quad + \int_{\alpha}^{\alpha+2\pi} \left\{ \sum_{n=1}^{\infty} b_n \sin nx \right\} dx \\ &= \frac{a_0}{2} \cdot 2\pi + 0 + 0. \end{aligned}$$

$$\therefore a_0 = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \, dx.$$

To find a_n . Multiplying both sides of (1) by $\cos nx$ and then integrating from $x = \alpha$ to $x = \alpha + 2\pi$, we have

$$\begin{aligned} \int_{\alpha}^{\alpha+2\pi} f(x) \cos nx \, dx &= \frac{a_0}{2} \int_{\alpha}^{\alpha+2\pi} \cos nx \, dx + \int_{\alpha}^{\alpha+2\pi} \left(\sum_{n=1}^{\infty} a_n \cos nx \right) \cos nx \, dx \\ &\quad + \int_{\alpha}^{\alpha+2\pi} \left(\sum_{n=1}^{\infty} b_n \sin nx \right) \cos nx \, dx \\ &= 0 + \pi a_n + 0. \end{aligned}$$

$$\therefore a_n = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \cos nx \, dx.$$

$$\text{Similarly } b_n = \frac{1}{\pi} \int_{\alpha}^{\alpha+2\pi} f(x) \sin nx \, dx.$$

Note 1. Putting $\alpha = 0$, the interval becomes $0 < x < 2\pi$ and

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) \, dx, a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx \, dx \text{ and } b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx.$$

Note 2. Putting $\alpha = -\pi$, the interval becomes $-\pi < x < \pi$ and

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx, a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx \text{ and } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx.$$

7.4 Fourier Series For Discontinuous Functions

Let the function $f(x)$ be defined by $f(x) = \begin{cases} f_1(x), & \alpha < x < x_0 \\ f_2(x), & x_0 < x < \alpha + 2\pi, \end{cases}$

in the interval $(\alpha, \alpha + 2\pi)$. Here x_0 is the point of discontinuity.

The Fourier series for $f(x)$ in such cases is obtained in the usual way. The values of a_0, a_n, b_n are given by

$$\begin{aligned} a_0 &= \frac{1}{\pi} \left[\int_{\alpha}^{x_0} f_1(x) dx + \int_{x_0}^{\alpha+2\pi} f_2(x) dx \right]; \\ a_n &= \frac{1}{\pi} \left[\int_{\alpha}^{x_0} f_1(x) \cos nx dx + \int_{x_0}^{\alpha+2\pi} f_2(x) \cos nx dx \right]; \\ b_n &= \frac{1}{\pi} \left[\int_{\alpha}^{x_0} f_1(x) \sin nx dx + \int_{x_0}^{\alpha+2\pi} f_2(x) \sin nx dx \right]. \end{aligned}$$

At the point of discontinuity, $x = \alpha$, the Fourier series gives the value of $f(x)$ as the arithmetic mean of left and right limits.

$$\therefore \quad \text{At } x = \alpha, f(x) = \frac{1}{2} [f(\alpha - 0) + f(\alpha + 0)].$$

Illustrative Examples

Example 1: Find a Fourier series to represent $x - x^2$ from $x = -\pi$ to $x = \pi$.

Deduce that $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$.

Solution: The Fourier series for $f(x)$ in $(-\pi, \pi)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx.$$

Here
$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} (x - x^2) dx = \frac{1}{2\pi} \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_{-\pi}^{\pi} = -\frac{\pi^2}{3};$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} (x - x^2) \cos nx dx \\ &= \frac{1}{\pi} \left[(x - x^2) \frac{\sin nx}{n} - (1 - 2x) \left(-\frac{\cos nx}{n^2} \right) + (-2) \left(-\frac{\sin nx}{n^3} \right) \right]_{-\pi}^{\pi} \\ &= \frac{-4(-1)^n}{n^2}; \quad [\because \cos n\pi = (-1)^n] \end{aligned}$$

and
$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} (x - x^2) \sin nx dx \\ &= \frac{1}{\pi} \left[(x - x^2) \left(-\frac{\cos nx}{n} \right) - (1 - 2x) \left(-\frac{\sin nx}{n^2} \right) + (-2) \left(\frac{\cos nx}{n^3} \right) \right]_{-\pi}^{\pi} \\ &= -2(-1)^n / n. \end{aligned}$$

∴ The required Fourier series is

$$x - x^2 = -\frac{\pi^2}{3} + 4 \left[\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \frac{\cos 4x}{4^2} + \dots \right] \\ + 2 \left[\frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \frac{\sin 4x}{4} + \dots \right] \dots (1)$$

Deduction: Putting $x = 0$ in (1), we get

$$0 = -\frac{\pi^2}{3} + 4 \left(\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \right) \quad \text{or} \quad \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}.$$

Example 2: Find the Fourier series expansion for $f(x)$, if

$$f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi. \end{cases} \quad \text{Deduce that } \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}.$$

Solution: The Fourier series of $f(x)$ in $(-\pi, \pi)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx.$$

$$\text{Then } a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \left[\int_{-\pi}^0 (-\pi) dx + \int_0^{\pi} x dx \right] \\ = \frac{1}{\pi} \left[-\pi(x)_{-\pi}^0 + \left(\frac{x^2}{2} \right)_0^{\pi} \right] = \frac{1}{\pi} \left[-(\pi^2 + \frac{\pi^2}{2}) \right] = -\frac{\pi}{2};$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \left[\int_{-\pi}^0 (-\pi) \cos nx dx + \int_0^{\pi} x \cos nx dx \right] \\ = \frac{1}{\pi} \left[-\pi \left(\frac{\sin nx}{n} \right)_{-\pi}^0 + \left(\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right)_0^{\pi} \right] \\ = \frac{1}{\pi} \left[0 + \frac{1}{n^2} \cos n\pi - \frac{1}{n^2} \right] = \frac{1}{\pi n^2} (\cos n\pi - 1);$$

$$\text{and } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \left[\int_{-\pi}^0 (-\pi) \sin nx dx + \int_0^{\pi} x \sin nx dx \right] \\ = \frac{1}{\pi} \left[\left(\frac{\pi \cos nx}{n} \right)_{-\pi}^0 + \left(-\frac{\cos nx}{n} + \frac{\sin nx}{n^2} \right)_0^{\pi} \right] \\ = \frac{1}{\pi} \left[\frac{\pi}{n} (0 - \cos n\pi) - \frac{\pi}{n} \cos n\pi \right] = \frac{1}{n} (1 - 2 \cos n\pi).$$

∴ The required Fourier series is

$$f(x) = -\frac{\pi}{4} - \frac{2}{\pi} \left(\cos x + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right) \\ + \left(3 \sin x - \frac{2 \sin 2x}{2} + \frac{3 \sin 3x}{3} - \frac{\sin 4x}{4} + \dots \right) \dots (1)$$

Deduction: Putting $x = 0$ in (1), we get

$$f(0) = -\frac{\pi}{4} - \frac{2}{\pi} \left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots \infty \right). \dots (2)$$

But $f(x)$ is discontinuous at $x = 0$, and we have $f(0-0) = -\pi$ and $f(0+0) = 0$.

$$\therefore f(0) = \frac{1}{2} [f(0-0) + f(0+0)] = -(\pi/2) \dots (3)$$

Hence from (2) and (3), we have

$$-\frac{\pi}{2} = -\frac{\pi}{4} - \frac{2}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right] \quad \text{or} \quad \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}.$$

Example 3: Obtain the Fourier series of $f(x) = \left(\frac{\pi-x}{2}\right)$ in the interval $(0, 2\pi)$ and hence deduce

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

Solution: The Fourier series for $f(x)$ in $(0, 2\pi)$ is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

Here

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{2\pi} \left(\frac{\pi-x}{2}\right) dx$$

$$= \frac{1}{2\pi} \left[\pi x - \frac{x^2}{2} \right]_0^{2\pi} = \frac{1}{2\pi} [2\pi^2 - 2\pi^2] = 0;$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx = \frac{1}{2\pi} \int_0^{2\pi} (\pi-x) \cos nx dx$$

$$= \frac{1}{2\pi} \left[(\pi-x) \frac{\sin nx}{n} - (-1) \left\{ \frac{-\cos nx}{n^2} \right\} \right]_0^{2\pi} = 0;$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx = \frac{1}{2\pi} \int_0^{2\pi} (\pi-x) \sin nx dx$$

$$= \frac{1}{2\pi} \left[-(\pi-x) \frac{\cos nx}{n} - \frac{\sin nx}{n^2} \right]_0^{2\pi} = \frac{1}{n}.$$

$$\therefore f(x) = \frac{\pi-x}{2} = 0 + 0 + \sum_{n=1}^{\infty} \frac{1}{n} \sin nx = \sin x + \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x + \dots \quad \dots(1)$$

Deduction: Putting $x = \frac{\pi}{2}$ in (1), we get $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$

Comprehensive Exercise 1

1. Find a series of sines and cosines of multiples of x which will represent $\frac{\pi}{2 \sinh \pi} e^x$ in the interval $-\pi < x < \pi$.

2. Find the Fourier series of the function defined as

$$f(x) = \begin{cases} x + \pi & ; \quad 0 \leq x \leq \pi \\ -x - \pi & ; \quad -\pi \leq x < 0 \end{cases} \quad \text{and} \quad f(x + 2\pi) = f(x).$$

3. Obtain the Fourier series for $f(x) = e^{-x}$ in the interval $0 < x < 2\pi$.

4. Find the Fourier series of the function

$$f(x) = \begin{cases} -1 & \text{for } -\pi < x < -\pi/2 \\ 0 & \text{for } -\pi/2 < x < \pi/2 \\ +1 & \text{for } \pi/2 < x < \pi. \end{cases}$$

Answers

1. $\frac{\pi}{2 \sinh \pi} e^x = \frac{1}{2} - \left(\frac{1}{2} \cos x - \frac{1}{2} \cos 2x + \frac{1}{10} \cos 3x - \frac{1}{17} \cos 4x + \dots \right) + \left(\frac{1}{2} \sin x - \frac{2}{5} \sin 2x + \frac{3}{10} \sin 3x - \frac{4}{17} \sin 4x + \dots \right)$
2. $f(x) = \frac{\pi}{2} - \frac{4}{\pi} \left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \dots \right) + 4 \left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \dots \right)$
3. $e^{-x} = \frac{1 - e^{-2\pi}}{\pi} \left\{ \frac{1}{2} + \left(\frac{1}{2} \cos x + \frac{1}{5} \cos 2x + \frac{1}{10} \cos 3x + \dots \right) + \left(\frac{1}{2} \sin x + \frac{2}{5} \sin 2x + \frac{3}{10} \sin 3x + \dots \right) \right\}$
4. $f(x) = \frac{1}{\pi} \left[2 \sin x - 2 \sin 2x + \frac{2}{3} \sin 3x + \dots \right]$

7.5 Change of Period

Generally in engineering problems, the period of the given function is not always 2π but is T or $2c$. To use the formula of a_n and b_n this period must be converted to the length 2π .

The independent variable x is also to be changed proportionally.

Let the function $f(x)$ be defined in the interval $(0, 2c)$.

$\therefore 2c$ is the interval for the variable x .

$\therefore 2\pi$ is the interval for the variable $= \frac{2\pi x}{2c} = \frac{\pi x}{c}$.

Putting $z = \frac{\pi x}{c}$ or $x = \frac{zc}{\pi}$, the function $f(x)$ of period $2c$ is transformed to the

function $f\left(\frac{cz}{\pi}\right)$ or $F(z)$ of period 2π .

Now $F(z)$ can be expanded in the Fourier series as

$$F(z) = f\left(\frac{cz}{\pi}\right) = \frac{a_0}{2} + a_1 \cos z + a_2 \cos 2z + \dots + b_1 \sin z + b_2 \sin 2z + \dots$$

$$\text{where } a_0 = \frac{1}{\pi} \int_0^{2\pi} F(z) dz = \frac{1}{\pi} \int_0^{2\pi} f\left(\frac{cz}{\pi}\right) dz$$

$$= \frac{1}{\pi} \int_0^{2c} f(x) \frac{\pi}{c} dx = \frac{1}{c} \int_0^{2c} f(x) dx; \quad \left[\text{Putting } z = \frac{\pi x}{c} \right]$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} F(z) \cos nz dz = \frac{1}{\pi} \int_0^{2\pi} f\left(\frac{cz}{\pi}\right) \cos nx dz$$

$$= \frac{1}{\pi} \int_0^{2c} f(x) \cos \frac{n\pi x}{c} \frac{\pi}{c} dx = \frac{1}{c} \int_0^{2c} f(x) \cos \frac{n\pi x}{c} dx. \quad \left[\text{Putting } z = \frac{\pi x}{c} \right]$$

Similarly, $b_n = \frac{1}{c} \int_0^{2c} f(x) \sin \frac{n\pi x}{c} dx$.

Illustrative Examples

Example 4: Obtain Fourier Series for the function $f(x) = \begin{cases} \pi x, & 0 \leq x \leq 1 \\ \pi(2-x), & 1 \leq x \leq 2. \end{cases}$

Deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

Solution: Here the function is defined in the interval $(0, 2)$. Let

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\pi x + b_n \sin n\pi x). \quad \dots(1)$$

Then, we have $a_0 = \frac{1}{c} \int_0^{2c} f(x) dx$, where $c = 1$

$$\begin{aligned} &= \frac{1}{1} \int_0^2 f(x) dx = \int_0^1 \pi x dx + \int_1^2 \pi(2-x) dx \\ &= \pi \left[\frac{x^2}{2} \right]_0^1 + \pi \left[2x - \frac{x^2}{2} \right]_1^2 = \frac{\pi}{2} + \frac{\pi}{2} = \pi; \\ a_n &= \frac{1}{c} \int_0^{2c} f(x) \cos \frac{n\pi x}{c} dx = \frac{1}{1} \int_0^2 f(x) \cos n\pi x dx \\ &= \int_0^1 \pi x \cos n\pi x dx + \int_1^2 \pi(2-x) \cos n\pi x dx \\ &= \pi \left[x \frac{\sin n\pi x}{n\pi} + \frac{\cos n\pi x}{(n\pi)^2} \right]_0^1 + \pi \left[-(2-x) \frac{\sin n\pi x}{n\pi} - \frac{\cos n\pi x}{(n\pi)^2} \right]_1^2 \\ &= \pi \left[\frac{\cos n\pi}{(n\pi)^2} - \frac{1}{(n\pi)^2} \right] + \pi \left[-\frac{1}{(n\pi)^2} + \frac{\cos n\pi}{(n\pi)^2} \right] = \frac{2}{\pi} \left[\frac{\cos n\pi}{n^2} - \frac{1}{n^2} \right] \\ &= \begin{cases} -\frac{4}{\pi n^2}, & \text{if } n \text{ is odd} \\ 0, & \text{if } n \text{ is even;} \end{cases} \end{aligned}$$

and
$$\begin{aligned} b_n &= \frac{1}{c} \int_0^{2c} f(x) \sin \frac{n\pi x}{c} dx = \frac{1}{1} \int_0^2 f(x) \sin n\pi x dx \\ &= \pi \int_0^1 x \sin n\pi x dx + \pi \int_1^2 (2-x) \sin n\pi x dx \\ &= \pi \left[x \left(-\frac{\cos n\pi x}{n\pi} \right) + \frac{\sin n\pi x}{(n\pi)^2} \right]_0^1 + \pi \left[(2-x) \left(-\frac{\cos n\pi x}{n\pi} \right) - \frac{\sin n\pi x}{(n\pi)^2} \right]_1^2 \\ &= -\frac{\cos n\pi}{n} + \frac{\cos n\pi}{n} = 0. \end{aligned}$$

Putting these values of a_0, a_n, b_n in (1), we get the required Fourier series as

$$f(x) = \frac{\pi}{2} - \frac{4}{\pi} \left[\frac{\cos \pi x}{1^2} + \frac{\cos 3\pi x}{3^2} + \frac{\cos 5\pi x}{5^2} + \dots \right] \quad \dots(2)$$

Deduction: Putting $x = 0$ in (2), we get

$$f(0) = \frac{\pi}{2} - \frac{4}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right] \quad \text{or} \quad 0 = \frac{\pi}{2} - \frac{4}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right]$$

or
$$\frac{\pi}{2} = \frac{4}{\pi} \left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right] \quad \text{or} \quad \frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

Example 5: Find the Fourier series corresponding to the function $f(x)$ defined in $(-2, 2)$ as follows

$$f(x) = \begin{cases} 2, & \text{if } -2 \leq x \leq 0 \\ x, & \text{if } 0 < x < 2. \end{cases}$$

Solution: Here the interval is $(-2, 2)$ and $l = 2$.

Let
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right].$$

Then
$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx = \frac{1}{2} \left[\int_{-2}^0 2 dx + \int_0^2 x dx \right]$$

$$= \frac{1}{2} \left[(2x)_{-2}^0 + \left(\frac{x^2}{2} \right)_0^2 \right] = \frac{1}{2} [4 + 2] = 3;$$

$$\begin{aligned} a_n &= \frac{1}{l} \int_{-l}^l f(x) \cos \left(\frac{n\pi x}{l} \right) dx \\ &= \frac{1}{2} \left[\int_{-2}^0 2 \cos \frac{n\pi x}{2} dx + \int_0^2 x \cos \frac{n\pi x}{2} dx \right] \\ &= \frac{1}{2} \left[\frac{4}{n\pi} \left(\sin \frac{n\pi x}{2} \right)_{-2}^0 + \left(x \frac{2}{n\pi} \sin \frac{n\pi x}{2} + \frac{4}{n^2 \pi^2} \cos \frac{n\pi x}{2} \right)_0^2 \right] \\ &= \frac{1}{2} \left[\frac{4}{n^2 \pi^2} \cos n\pi - \frac{4}{n^2 \pi^2} \right] = \frac{2}{n^2 \pi^2} [(-1)^n - 1] \\ &= \begin{cases} -\frac{4}{n^2 \pi^2}, & \text{when } n \text{ is odd} \\ 0, & \text{when } n \text{ is even;} \end{cases} \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx = \frac{1}{2} \int_{-2}^0 2 \sin \frac{n\pi x}{2} dx + \frac{1}{2} \int_0^2 x \sin \frac{n\pi x}{2} dx \\ &= \frac{1}{2} \left[2 \left(-\frac{2}{n\pi} \cos \frac{n\pi x}{2} \right) \right]_{-2}^0 + \frac{1}{2} \left[x \left(-\frac{2}{n\pi} \cos \frac{n\pi x}{2} \right) + (l) \frac{4}{n^2 \pi^2} \sin \frac{n\pi x}{2} \right]_0^2 \\ &= \frac{1}{2} \left[-\frac{4}{n\pi} + \frac{4}{n\pi} \cos n\pi \right] + \frac{1}{2} \left[-\frac{4}{n\pi} \cos n\pi + \frac{4}{n^2 \pi^2} \sin n\pi \right] = \frac{1}{2} \left[-\frac{4}{n\pi} \right] = -\frac{2}{n\pi}. \end{aligned}$$

$\therefore$ The required Fourier series is

$$\begin{aligned} f(x) &= \frac{a_0}{2} + a_1 \cos \frac{\pi x}{l} + a_2 \cos \frac{2\pi x}{l} + a_3 \cos \frac{3\pi x}{l} + \dots \\ &\quad + b_1 \sin \frac{\pi x}{l} + b_2 \sin \frac{2\pi x}{l} + b_3 \sin \frac{3\pi x}{l} + \dots \end{aligned}$$

$$= \frac{3}{2} - \frac{4}{\pi^2} \left\{ \frac{1}{l^2} \cos \frac{\pi x}{2} + \frac{1}{3^2} \cos \frac{3\pi x}{2} + \dots \right\} - \frac{2}{\pi} \left\{ \frac{1}{1} \sin \frac{\pi x}{2} + \frac{1}{2} \sin \frac{2\pi x}{2} + \frac{1}{3} \sin \frac{3\pi x}{2} + \dots \right\}.$$

Comprehensive Exercise 2

1. Prove that $\frac{l}{2} - x = \frac{l}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{2n\pi x}{l}, 0 < x < l$.
2. Obtain Fourier series of the function $F(x) = \begin{cases} 1 + \frac{2x}{l}, & -1 < x < 0 \\ 1 - \frac{2x}{l}, & 0 < x < l. \end{cases}$
3. Given $f(x) = \begin{cases} x+1 & \text{for } -1 < x < 0, \\ x-1 & \text{for } 0 < x < 1. \end{cases}$
Expand f into a Fourier series on $(-1, 1)$.
4. Obtain the Fourier series expansion of the periodic function of period 1
$$f(x) = \begin{cases} \frac{1}{2} + x, & -\frac{1}{2} < x \leq 0 \\ \frac{1}{2} - x, & 0 < x < \frac{1}{2}. \end{cases}$$
5. Given $f(x) = \begin{cases} 0, & 0 < x < c \\ 1, & c < x < 2c \end{cases}$; expand $f(x)$ in a Fourier series of period $2c$.

Answers 2

2. $F(x) = \frac{8}{\pi^2} \left[\frac{1}{l^2} \cos \frac{\pi x}{l} + \frac{1}{3^2} \cos \frac{3\pi x}{l} + \frac{1}{5^2} \cos \frac{5\pi x}{l} + \dots \right]$
3. $f(x) = -\frac{2}{\pi} \left\{ \frac{1}{1} \sin \pi x + \frac{1}{2} \sin 2\pi x + \frac{1}{3} \sin 3\pi x + \dots \right\}$
4. $f(x) = \frac{1}{4} + \frac{2}{\pi^2} \left[\frac{\cos 2\pi x}{l^2} + \frac{\cos 6\pi x}{3^2} + \frac{\cos 10\pi x}{5^2} + \dots \right]$
5. $f(x) = \frac{1}{2} - \frac{2}{\pi} \left(\frac{1}{1} \sin \frac{\pi x}{c} + \frac{1}{3} \sin \frac{3\pi x}{c} + \dots \right)$

7.6 Fourier Series Expansion of An Even or an Odd Function in $(-\pi, \pi)$

A function $f(x)$ is called an even (symmetric) function if $f(-x) = f(x)$.

The area under such a curve from $-\pi$ to π is double the area from 0 to π .

$$\therefore \int_{-\pi}^{\pi} f(x) dx = 2 \int_0^{\pi} f(x) dx.$$

A function $f(x)$ is called an odd (or skew symmetric) function if $f(-x) = -f(x)$.

Here the area under the curve from $-\pi$ to π is zero.

$$\therefore \int_{-\pi}^{\pi} f(x) dx = 0.$$

Expansion Of An Even Function

In $(-\pi, \pi)$:

Here

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} f(x) dx.$$

As $f(x)$ and $\cos nx$ are both even functions, therefore the product of $f(x) \cdot \cos nx$ is also an even function.

$$\begin{aligned} \therefore a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx \\ &= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx. \end{aligned}$$

As $\sin nx$ is an odd function so $f(x) \cdot \sin nx$ is also an odd function. We need not to calculate b_n . It saves our labour a lot.

$$\therefore b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = 0.$$

Thus the series of the even function will contain only cos terms.

Expansion Of An Odd Function In $(-\pi, \pi)$:

Here
$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = 0 ;$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = 0 \quad [\because f(x) \cdot \cos nx \text{ is an odd function}]$$

and
$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$$

$$[\because f(x) \cdot \sin nx \text{ is an even function}]$$

Thus the series of the odd function will contain only sin terms.

Illustrative Examples

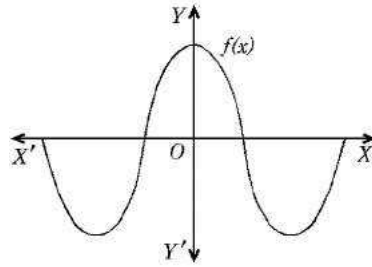
Example 6: Obtain Fourier's series for the expansion of $f(x) = x \sin x$ in the interval $-\pi < x < \pi$. Hence deduce that

$$\frac{\pi - 2}{4} = \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \dots$$

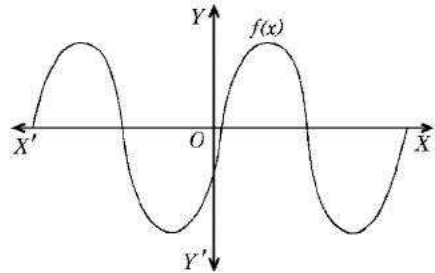
Solution: Here $f(x) = x \sin x$ being an even function of x . The Fourier series is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

Even Function



Odd Function



$$f(x) = \frac{1}{\pi} \int_0^{\pi} f(u) du + \frac{2}{\pi} \sum_{n=1}^{\infty} \cos nx \int_0^{\pi} f(u) \cos nu du,$$

where $f(u) = u \sin u$.

Now $\int_0^{\pi} u \sin u du = [-u \cos u + \sin u]_0^{\pi} = \pi$

and
$$\begin{aligned} \int_0^{\pi} u \sin u \cos nu du &= \frac{1}{2} \int_0^{\pi} u [\sin(n+1)u - \sin(n-1)u] du \\ &= \frac{1}{2} \left[\frac{-u \cos(n+1)u}{n+1} + \frac{\sin(n+1)u}{(n+1)^2} \right]_0^{\pi} - \frac{1}{2} \left[\frac{-u \cos(n-1)u}{n-1} + \frac{\sin(n-1)u}{(n-1)^2} \right]_0^{\pi} \\ &= \frac{\pi}{2} \left[\frac{\cos(n-1)\pi}{n-1} - \frac{\cos(n+1)\pi}{n+1} \right] = \frac{\pi}{2} \left[-\frac{\cos n\pi}{n-1} + \frac{\cos n\pi}{n+1} \right] \\ &= \frac{\pi \cos n\pi}{1-n^2}, \text{ when } n > 1. \end{aligned}$$

When $n = 1$, we have

$$\int_0^{\pi} u \sin u \cos u du = \frac{1}{2} \int_0^{\pi} u \sin 2u du = \frac{1}{2} \left[-\frac{u \cos 2u}{2} + \frac{\sin 2u}{4} \right]_0^{\pi} = -\frac{\pi}{4}.$$

$$\begin{aligned} \therefore x \sin x &= 1 + \frac{2}{\pi} \left(-\frac{\pi}{4} \cos x + \sum_{n=2}^{\infty} \frac{\pi \cos n\pi}{1-n^2} \cos nx \right) \\ &= 1 + 2 \left[-\frac{1}{4} \cos x - \frac{1}{1.3} \cos 2x + \frac{1}{2.4} \cos 3x - \frac{1}{3.5} \cos 4x + \dots \right] \end{aligned}$$

Deduction: Putting $x = \frac{1}{2} \pi$, we get

$$\frac{\pi}{2} = 1 + 2 \left\{ \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \dots \right\} \text{ or } \frac{\pi}{4} = \frac{1}{2} + \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \dots$$

or
$$\frac{\pi - 2}{4} = \frac{1}{1.3} - \frac{1}{3.5} + \frac{1}{5.7} - \dots$$

Example 7: Obtain Fourier Series of the function

$$f(x) = \begin{cases} x, & -\pi < x < 0 \\ -x, & 0 < x < \pi. \end{cases}$$

Solution: We have $f(x) = \begin{cases} x, & -\pi < x < 0 \\ -x, & 0 < x < \pi \end{cases}$ and $f(-x) = \begin{cases} -x, & \pi > x > 0 \\ x, & 0 > x > -\pi. \end{cases}$

Thus $f(x)$ is an even function in $(-\pi, \pi)$ and so $b_n = 0$.

The Fourier series in this case is $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$.

Here $a_0 = \frac{1}{\pi} \int_0^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} -x dx = -\frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} = -\frac{1}{\pi} \left[\frac{\pi^2}{2} \right] = -\frac{\pi}{2};$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} -x \cos nx dx = -\frac{2}{\pi} \left[x \frac{\sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^{\pi}$$

$$= -\frac{2}{\pi} \left[\frac{(-1)^n}{n^2} - \frac{1}{n^2} \right] = \frac{2}{\pi n^2} [1 - (-1)^n] = \begin{cases} 0, & n \text{ is even} \\ 4/\pi n^2, & n \text{ is odd} \end{cases}$$

$$\therefore \text{The required series is } f(x) = -\frac{\pi}{2} + \frac{4}{\pi} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right]$$

Example 8: A periodic function of period 4 is defined as

$f(x) = |x|, -2 < x < 2$. Obtain its Fourier series expansion.

Solution: We have

$$f(x) = \begin{cases} |x|, & -2 < x < 2 \\ x, & 0 < x < 2 \\ -x, & -2 < x < 0. \end{cases}$$

Here $f(x)$ is an even function in the interval $(-2, 2)$ and so $b_n = 0$.

The Fourier series in this case is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx.$$

$$\text{Here } a_0 = \frac{1}{c} \int_{-c}^c f(x) dx = \frac{1}{2} \int_0^2 x dx + \frac{1}{2} \int_{-2}^0 (-x) dx$$

$$= \frac{1}{2} \left[\frac{x^2}{2} \right]_0^2 + \frac{1}{2} \left[-\frac{x^2}{2} \right]_{-2}^0 = \frac{1}{4} (4 - 0) + \frac{1}{4} (0 + 4) = 2;$$

$$a_n = \frac{1}{c} \int_{-c}^c f(x) \cos \frac{n\pi x}{c} dx = \frac{1}{2} \int_0^2 x \cos \frac{n\pi x}{2} dx + \frac{1}{2} \int_{-2}^0 (-x) \cos \frac{n\pi x}{2} dx$$

$$= \frac{1}{2} \left[x \left(\frac{2}{n\pi} \sin \frac{n\pi x}{2} \right) - (1) \left(-\frac{4}{n^2 \pi^2} \cos \frac{n\pi x}{2} \right) \right]_0^2$$

$$+ \frac{1}{2} \left[(-x) \left(\frac{2}{n\pi} \sin \frac{n\pi x}{2} \right) - (-1) \left(-\frac{4}{n^2 \pi^2} \cos \frac{n\pi x}{2} \right) \right]_{-2}^0$$

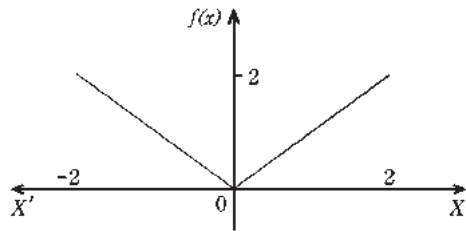
$$= \frac{1}{2} \left[0 + \frac{4}{n^2 \pi^2} (-1)^n - \frac{4}{n^2 \pi^2} \right] + \frac{1}{2} \left[0 - \frac{4}{n^2 \pi^2} + \frac{4}{n^2 \pi^2} (-1)^n \right]$$

$$= \frac{1}{2} \frac{4}{n^2 \pi^2} [(-1)^n - 1 - 1 + (-1)^n] = \frac{4}{n^2 \pi^2} [(-1)^n - 1]$$

$$= \begin{cases} -\frac{8}{n^2 \pi^2}, & \text{if } n \text{ is odd} \\ 0, & \text{if } n \text{ is even.} \end{cases}$$

$$\therefore \text{The required Fourier series is } f(x) = \frac{a_0}{2} + a_1 \cos \frac{\pi x}{c} + a_2 \cos \frac{2\pi x}{c} + \dots$$

$$= 1 - \frac{8}{\pi^2} \left[\frac{\cos \frac{\pi x}{2}}{1^2} + \frac{\cos \frac{3\pi x}{2}}{3^2} + \frac{\cos \frac{5\pi x}{2}}{5^2} + \dots \right].$$



Comprehensive Exercise 3

1. Obtain Fourier series for the function $f(x)$ given by

$$f(x) = \begin{cases} 1 + (2x/\pi), & -\pi \leq x \leq 0 \\ 1 - (2x/\pi), & 0 \leq x \leq \pi. \end{cases} \text{ Deduce that } \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}.$$

2. Find the Fourier series expansion of the periodic function of period 2π
 $f(x) = x^2$, $-\pi \leq x \leq \pi$.

Hence, find the sum of the series $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$

3. Obtain a Fourier expression for $f(x) = x^3$ for $-\pi < x < \pi$.

4. Find the Fourier Series expansion for the function

$$f(x) = x \cos x, -\pi < x < \pi.$$

Answers 3

1. $f(x) = \frac{8}{\pi^2} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right]$

2. $x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} (-1)^n \frac{\cos nx}{n^2}; \frac{\pi^2}{12}$

3. $x^3 = 2 \left[- \left(-\frac{\pi^2}{1} + \frac{6}{1^3} \right) \sin x + \left(-\frac{\pi^2}{2} + \frac{6}{2^3} \right) \sin 2x - \left(-\frac{\pi^2}{3} + \frac{6}{3^3} \right) \sin 3x \dots \right]$

4. $x \cos x = -\frac{1}{2} \sin x + \frac{4 \sin 2x}{2^2 - 1} - \frac{6 \sin 3x}{3^2 - 1} + \dots$

7.7 Half Range Fourier Cosine and Sine Series [Interval $(0, T)$]

For the interval $(0, T)$ the **Half Range Fourier Cosine Series** is

$$f(x) = \frac{a_0}{2} + a_1 \cos \frac{\pi x}{T} + a_2 \cos \frac{2\pi x}{T} + \dots + a_n \cos \frac{n\pi x}{T} + \dots$$

where $a_0 = \frac{2}{T} \int_0^T f(x) dx, a_n = \frac{2}{T} \int_0^T f(x) \cos \frac{n\pi x}{T} dx.$

Thus when $f(x)$ is an even function with period T , we have

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{T},$$

where $a_0 = \frac{2}{T} \int_0^T f(x) dx, a_n = \frac{2}{T} \int_0^T f(x) \cos \frac{n\pi x}{T} dx.$

In this case $b_n = 0.$

Again for the interval $(0, T)$ the **Half Range Fourier sine series** is

$$f(x) = b_1 \sin \frac{\pi x}{T} + b_2 \sin \frac{2\pi x}{T} + \dots + b_n \sin \frac{n\pi x}{T} + \dots,$$

where
$$b_n = \frac{2}{T} \int_0^T f(x) \sin \frac{n\pi x}{T} dx.$$

Thus when $f(x)$ is an odd function with period T , we have

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{T}, \text{ where } b_n = \frac{2}{T} \int_0^T f(x) \sin \frac{n\pi x}{T} dx.$$

In this case $a_0 = 0 = a_n$.

Remark: While expanding a function in $(0, T)$ as a series of sines or cosines, we only see if it is an odd or even function of period $2T$. It makes no matter if the function is odd or even or neither.

Illustrative Examples

Example 9: Find the Fourier half-range cosine series of the function

$$f(t) = \begin{cases} 2t, & 0 < t < 1 \\ 2(2-t), & 1 < t < 2. \end{cases}$$

Solution: The Fourier half range cosine series in interval $(0, T)$ is

$$f(t) = \frac{a_0}{2} + a_1 \cos \frac{\pi t}{T} + a_2 \cos \frac{2\pi t}{T} + a_3 \cos \frac{3\pi t}{T} + \dots \quad \dots(1)$$

Here $T = 2$. We have

$$\begin{aligned} a_0 &= \frac{2}{T} \int_0^T f(t) dt = \frac{2}{2} \int_0^1 2t dt + \frac{2}{2} \int_1^2 2(2-t) dt \\ &= \left[t^2 \right]_0^1 + \left[2 \left(2t - \frac{t^2}{2} \right) \right]_1^2 = 1 + \left[(4t - t^2) \right]_1^2 = 1 + (8 - 4 - 4 + 1) = 2, \\ a_n &= \frac{2}{T} \int_0^T f(t) \cos \frac{n\pi t}{T} dt = \frac{2}{2} \int_0^1 2t \cos \frac{n\pi t}{2} dt + \frac{2}{2} \int_1^2 2(2-t) \cos \frac{n\pi t}{2} dt \\ &= \left[2t \left(\frac{2}{n\pi} \sin \frac{n\pi t}{2} \right) - (2) \left(-\frac{4}{n^2 \pi^2} \cos \frac{n\pi t}{2} \right) \right]_0^1 \\ &\quad + \left[(4-2t) \left(\frac{2}{n\pi} \sin \frac{n\pi t}{2} \right) - (-2) \left(-\frac{4}{n^2 \pi^2} \cos \frac{n\pi t}{2} \right) \right]_1^2 \\ &= \left[\frac{4}{n\pi} \sin \frac{n\pi}{2} + \frac{8}{n^2 \pi^2} \cos \frac{n\pi}{2} - \frac{8}{n^2 \pi^2} \right] \\ &\quad + \left[0 - \frac{8}{n^2 \pi^2} \cos n\pi - \frac{4}{n\pi} \sin \frac{n\pi}{2} + \frac{8}{n^2 \pi^2} \cos \frac{n\pi}{2} \right] \\ &= \frac{8}{n^2 \pi^2} \cos \frac{n\pi}{2} - \frac{8}{n^2 \pi^2} - \frac{8}{n^2 \pi^2} \cos n\pi + \frac{8}{n^2 \pi^2} \frac{\cos n\pi}{2} \end{aligned}$$

$$= \frac{8}{n^2 \pi^2} \left[2 \cos \frac{n\pi}{2} - 1 - \cos n\pi \right]$$

$$\therefore f(x) = 1 + \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \left(2 \cos \frac{n\pi}{2} - 1 - \cos n\pi \right) \cos \frac{n\pi x}{2}$$

Example 10: (i) Express $f(x) = x$ as a half-range sine series in $0 < x < 2$.

(ii) Express $f(x) = x$ as a half-range cosine series in $0 < x < 2$.

Solution: (i) The Fourier sine series for $f(x)$ in $(0, 2)$ is

$$\begin{aligned} f(x) &= \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{2} \text{ where } b_n = \frac{2}{2} \int_0^2 f(x) \sin \frac{n\pi x}{2} dx = \int_0^2 x \sin \frac{n\pi x}{2} dx \\ &= \left[-\frac{2x}{n\pi} \cos \frac{n\pi x}{2} + \frac{4}{n^2 \pi^2} \sin \frac{n\pi x}{2} \right]_0^2 = -\frac{4(-1)^n}{n\pi}. \end{aligned}$$

$$\therefore b_1 = 4/\pi, b_2 = -4/2\pi, b_3 = 4/3\pi, b_4 = -4/4\pi, \text{ etc.}$$

Hence, the required half-range Fourier sine series for $f(x)$ in $(0, 2)$ is

$$f(x) = \frac{4}{\pi} \left[\sin \frac{\pi x}{2} - \frac{1}{2} \sin \frac{2\pi x}{2} + \frac{1}{3} \sin \frac{3\pi x}{2} - \frac{1}{4} \sin \frac{4\pi x}{2} + \dots \right].$$

(ii) The Fourier cosine series for $f(x)$ in $(0, 2)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{2}$$

$$\text{where } a_0 = \frac{2}{2} \int_0^2 f(x) dx = \int_0^2 x dx = 2 \quad \text{and} \quad a_n = \frac{2}{2} \int_0^2 f(x) \cos \frac{n\pi x}{2} dx$$

$$= \int_0^2 x \cos \frac{n\pi x}{2} dx = \left[\frac{2x}{n\pi} \sin \frac{n\pi x}{2} + \frac{4}{n^2 \pi^2} \cos \frac{n\pi x}{2} \right]_0^2 = \frac{4}{n^2 \pi^2} [(-1)^n - 1].$$

$$\therefore a_1 = -8/\pi^2, a_2 = 0, a_3 = -8/3^2 \pi^2, a_4 = 0, a_5 = -8/5^2 \pi^2, \text{ etc.}$$

Hence, the required half-range Fourier series for $f(x)$ in $(0, 2)$ is

$$f(x) = 1 - \frac{8}{\pi^2} \left[\frac{\cos \pi x/2}{1^2} + \frac{\cos 3\pi x/2}{3^2} + \frac{\cos 5\pi x/2}{5^2} + \dots \right]$$

Comprehensive Exercise 4

- Find Fourier half-range even expansion of the function,
 $f(x) = (-x/l) + 1, 0 \leq x \leq l$.
- Obtain the half-range sine series for the function $f(x) = x^2$ in the interval
 $0 < x < 3$.
- Find a series of sines of multiples of x which will represent $f(x)$ in the interval
 $(0, \pi)$ where

$$f(x) = \begin{cases} \pi/3 & 0 < x < \pi/3 \\ 0 & \pi/3 < x < 2\pi/3 \\ -\pi/3 & 2\pi/3 < x < \pi \end{cases}$$

4. Given $f(x) = \begin{cases} x, & \text{for } 0 \leq x \leq \pi/2, \\ \pi - x, & \text{for } \pi/2 \leq x \leq \pi. \end{cases}$

Express this function by a sine series and also by a cosine series.

5. Develop $\sin(\pi t/l)$ in half range cosine series in the range $0 < t < l$.

6. Expand $f(x) = e^x$ in a cosine series over $(0, l)$.

7. Expand $f(x) = \begin{cases} \frac{1}{4} - x, & \text{if } 0 < x < \frac{1}{2}, \\ x - \frac{3}{4}, & \text{if } \frac{1}{2} < x < 1. \end{cases}$

Answers 4

1. $f(x) = \frac{1}{2} + \frac{4}{\pi^2} \left[\frac{1}{1^2} \cos \frac{\pi x}{l} + \frac{1}{3^2} \cos \frac{3\pi x}{l} + \frac{1}{5^2} \cos \frac{5\pi x}{l} \dots \right]$

2. $f(x) = \frac{2}{3} \left[\frac{108}{\pi^3} \left(\frac{\sin x}{1^3} + \frac{\sin 3x}{3^3} + \frac{\sin 5x}{5^3} + \dots \right) \right. \\ \left. + \frac{27}{\pi} \left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \dots \right) - \frac{18}{\pi} \left(\frac{\sin 2x}{2} + \frac{\sin 4x}{4} + \dots \right) \right]$

3. $f(x) = \frac{1}{2} \left[\frac{1}{2} \sin 2x + \frac{1}{2} \sin 4x + \frac{1}{8} \sin 8x + \frac{1}{10} \sin 10x + \dots \right]$

4. $f(x) = \frac{\pi}{4} - \frac{8}{\pi} \left[\frac{1}{2^2} \cos 2x + \frac{1}{6^2} \cos 6x + \dots \right]$

5. $f(t) = \frac{2}{\pi} - \frac{4}{\pi} \left[\frac{1}{3} \cos \frac{2\pi t}{l} + \frac{1}{15} \cos \frac{4\pi t}{l} + \frac{1}{35} \cos \frac{6\pi t}{l} + \dots \right]$

6. $e^x = e - 1 + 2 \left[\frac{-e - 1}{\pi^2 + 1} \cos \pi x + \frac{e - 1}{4\pi^2 + 1} \cos 2\pi x + \frac{-e - 1}{9\pi^2 + 1} \cos 3\pi x + \dots \right]$

7. $f(x) = \left(\frac{1}{\pi} - \frac{4}{\pi^2} \right) \sin \pi x + \left(\frac{1}{3\pi} + \frac{4}{3^2\pi^2} \right) \sin 3\pi x + \left(\frac{1}{5\pi} - \frac{4}{5^2\pi^2} \right) \sin 5\pi x + \dots$

7.8 Parseval's Formula

If the Fourier series for $f(x)$ converges uniformly in $(-l, l)$, then

$$\int_{-l}^l [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right].$$

Proof: The Fourier series for $f(x)$ in $(-l, l)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l} \right), \quad \dots(1)$$

where $a_0 = \frac{1}{l} \int_{-l}^l f(x) dx$; $\dots(2)$ $a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx$ $\dots(3)$

$$\text{and} \quad b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx. \quad \dots(4)$$

Multiplying (1) by $f(x)$, we get

$$[f(x)]^2 = \frac{a_0}{2} f(x) + \sum_{n=1}^{\infty} a_n f(x) \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n f(x) \sin \frac{n\pi x}{l} \quad \dots(5)$$

Integrating (5) term by term from $-l$ to l , we get

$$\begin{aligned} \int_{-l}^l [f(x)]^2 dx &= \frac{a_0}{2} \int_{-l}^l f(x) dx + \sum_{n=1}^{\infty} a_n \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx \\ &\quad + \sum_{n=1}^{\infty} b_n \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx \quad \dots(6) \end{aligned}$$

Putting the values of a_0, a_n, b_n from (2), (3) and (4) in (6), we get

$$\int_{-l}^l [f(x)]^2 dx = \frac{la_0^2}{2} + \sum_{n=1}^{\infty} l a_n^2 + \sum_{n=1}^{\infty} l b_n^2$$

$$\text{or} \quad \int_{-l}^l [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right].$$

This is the **Parseval's formula**.

Corollary 1: If the Fourier series for $f(x)$ in the interval $0 < x < 2l$ converges uniformly, then $\int_0^{2l} [f(x)]^2 dx = l \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right]$.

Corollary 2: If the half range Fourier cosine series for $f(x)$ in the interval $0 < x < l$ converges uniformly, then $\int_{-l}^l [f(x)]^2 dx = \frac{l}{2} \left[\frac{a_0^2}{2} + \sum_{n=1}^{\infty} a_n^2 \right]$.

Corollary 3: If the half range Fourier sine series for $f(x)$ in the interval $0 < x < l$ converges uniformly, then $\int_0^l [f(x)]^2 dx = \frac{l}{2} \sum_{n=1}^{\infty} b_n^2$.

Corollary 4: Root Mean Square Value. The root mean square value or the effective value of the function $f(x)$ denoted by $[f(x)]_{rms}$ over an interval (a, b) is defined as

$$[f(x)]_{rms} = \left\{ \frac{\int_a^b [f(x)]^2 dx}{b-a} \right\}^{1/2}.$$

The root mean square value of a periodic function is frequently used in electric circuit theory and in the theory of mechanical vibrations.

Illustrative Examples

Example 12: Prove that for $0 < x < \pi$

$$(a) \quad x(\pi - x) = \frac{\pi^2}{6} - \left[\frac{\cos 2x}{1^2} + \frac{\cos 4x}{2^2} + \frac{\cos 6x}{3^2} + \dots \right].$$

$$(b) \quad x(\pi - x) = \frac{8}{\pi} \left[\frac{\sin x}{1^2} + \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} + \dots \right].$$

Deduce from (a) and (b) respectively that

$$(c) \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

$$(d) \sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945}.$$

Solution: Let $f(x) = x(\pi - x)$, $0 < x < \pi$.

(a) The half range cosine series for $f(x)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx.$$

Here
$$a_0 = \frac{2}{\pi} \int_0^{\pi} x(\pi - x) dx = \frac{2}{\pi} \left[\frac{\pi x^2}{2} - \frac{x^3}{3} \right]_0^{\pi} = \frac{2}{\pi} \left[\frac{\pi^3}{2} - \frac{\pi^3}{3} \right] = \frac{\pi^2}{3}$$

and
$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} x(\pi - x) \cos nx dx \\ &= \frac{2}{\pi} \left[(\pi x - x^2) \frac{\sin nx}{n} - (\pi - 2x) \left(\frac{-\cos nx}{n^2} \right) + (-2) \left(\frac{-\sin nx}{n^3} \right) \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[0 - \frac{\pi(-1)^n}{n^2} + 0 - \frac{\pi}{n^2} \right] = \frac{2}{\pi} \left(\frac{\pi}{n^2} \right) [-(-1)^n - 1] \\ &= \begin{cases} -4/n^2, & \text{when } n \text{ is even} \\ 0, & \text{when } n \text{ is odd.} \end{cases} \end{aligned}$$

$$\therefore x(\pi - x) = \frac{\pi^2}{6} - 4 \left[\frac{\cos 2x}{2^2} + \frac{\cos 4x}{4^2} + \frac{\cos 6x}{6^2} + \dots \right].$$

By Parseval's formula, we get

$$\frac{2}{\pi} \int_0^{\pi} x^2 (\pi - x)^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} a_n^2$$

or
$$\frac{2}{\pi} \int_0^{\pi} (\pi^2 x^2 - 2\pi x^3 + x^4) dx = \frac{1}{2} \left(\frac{\pi^4}{9} \right) + 16 \left[\frac{1}{2^4} + \frac{1}{4^4} + \frac{1}{6^4} + \dots \right]$$

or
$$\frac{2}{\pi} \left[\frac{\pi^2 x^3}{3} - \frac{2\pi x^4}{4} + \frac{x^5}{5} \right]_0^{\pi} = \frac{\pi^4}{18} + \left[\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \right]$$

or
$$\frac{2}{\pi} \left[\frac{x^5}{3} - \frac{2\pi^5}{4} + \frac{\pi^5}{5} \right] = \frac{\pi^4}{18} + \left[\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \right]$$

or
$$\frac{\pi^4}{15} = \frac{\pi^4}{18} + \sum_{n=1}^{\infty} \frac{1}{n^4} \quad \text{or} \quad \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$

(b) The half range sine series for $f(x)$ is $f(x) = \sum_{n=1}^{\infty} b_n \sin nx$.

Here
$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^{\pi} x(\pi - x) \sin nx dx \\ &= \frac{2}{\pi} \left[(\pi x - x^2) \left(\frac{-\cos nx}{n} \right) - (\pi - 2x) \left(\frac{-\sin nx}{n^2} \right) + (-2) \frac{\cos nx}{n^3} \right]_0^{\pi} \\ &= \frac{2}{\pi} \left[-2 \frac{(-1)^n}{n^3} + \frac{2}{n^3} \right] = \frac{4}{\pi n^3} [-(-1)^n + 1] = \begin{cases} \frac{8}{n^3 \pi}, & \text{when } n \text{ is odd} \\ 0, & \text{when } n \text{ is even.} \end{cases} \end{aligned}$$

$$\therefore x(\pi - x) = \frac{8}{\pi} \left[\frac{\sin x}{1^3} + \frac{\sin 3x}{3^3} + \frac{\sin 5x}{5^3} + \dots \right].$$

By Parseval's formula, we get $\frac{2}{\pi} \int_0^\pi x^2 (\pi - x)^2 dx = \sum_{n=1}^\infty b_n^2$

or $\frac{\pi^2}{15} = \frac{64}{\pi^2} \left[\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \dots \right]$ or $\frac{\pi^4}{960} = \frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \dots$

Let $S = \frac{1}{1^6} + \frac{1}{2^6} + \frac{1}{3^6} + \frac{1}{4^6} + \dots = \left(\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \dots \right) + \left(\frac{1}{2^6} + \frac{1}{4^6} + \frac{1}{6^6} + \dots \right)$
 $= \frac{\pi^4}{960} + \left(\frac{1}{2^6} + \frac{1}{4^6} + \frac{1}{6^6} + \dots \right) = \frac{\pi^4}{960} + \frac{1}{2^6} \left[\frac{1}{1^6} + \frac{1}{2^6} + \frac{1}{3^6} + \dots \right] = \frac{\pi^4}{960} + \frac{S}{64}$

$\therefore S - \frac{S}{64} = \frac{\pi^4}{960}$ or $\frac{63S}{64} = \frac{\pi^4}{960}$

or $S = \frac{\pi^4}{960} \times \frac{64}{63} = \frac{\pi^4}{945}$ or $\sum_{n=1}^\infty \frac{1}{n^6} = \frac{\pi^4}{945}$

Example 13: If $f(x) = \begin{cases} \pi x & , 0 < x < 1 \\ \pi(2-x) & , 1 < x < 2; \end{cases}$

using half range cosine series, show that $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots = \frac{\pi^4}{96}$

Solution: The half range cosine series for $f(x)$ in $(0, c)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^\infty a_n \cos \frac{n\pi x}{c}$$

Here $a_0 = \frac{2}{c} \int_0^c f(x) dx = \frac{2}{2} \left[\int_0^1 \pi x dx + \int_1^2 \pi(2-x) dx \right]$
 $= \pi \left[\frac{x^2}{2} \right]_0^1 + \pi \left[2x - \frac{x^2}{2} \right]_1^2 = \frac{\pi}{2} + \pi \left[(4-2) - \left(2 - \frac{1}{2} \right) \right] = \pi;$

$$a_n = \frac{2}{c} \int_0^c f(x) \cos \frac{n\pi x}{c} dx = \frac{2}{2} \left[\int_0^1 \pi x \cos \frac{n\pi x}{2} dx + \int_1^2 \pi(2-x) \cos \frac{n\pi x}{2} dx \right]$$

$$= \pi \left[\frac{x \sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} - \left(-\frac{\cos \frac{n\pi x}{2}}{\frac{n^2 \pi^2}{4}} \right) \right]_0^1 + \pi \left[(2-x) \frac{\sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} - (-1) \left(-\frac{\cos \frac{n\pi x}{2}}{\frac{n^2 \pi^2}{4}} \right) \right]_1^2$$

$$= \pi \left[\frac{2}{n\pi} \sin \frac{n\pi}{2} + \frac{4}{n^2 \pi^2} \cos \frac{n\pi}{2} - \frac{4}{n^2 \pi^2} \right]$$

$$+ \pi \left[0 - \frac{4}{n^2 \pi^2} \cos n\pi - \frac{2}{n\pi} \sin \frac{n\pi}{2} + \frac{4}{n^2 \pi^2} \cos \frac{n\pi}{2} \right]$$

$$= \pi \left[\frac{8}{n^2 \pi^2} \cos \frac{n\pi}{2} - \frac{4}{n^2 \pi^2} - \frac{4}{n^2 \pi^2} \cos n\pi \right] = \frac{4}{n^2 \pi} \left[2 \cos \frac{n\pi}{2} - 1 - \cos n\pi \right].$$

Putting $n = 1, 2, 3, \dots$, we get

$$a_1 = 0, a_2 = \frac{-4}{\pi}, a_3 = 0, a_4 = 0, a_5 = 0, a_6 = \frac{-4}{9\pi} \dots$$

By Parseval's formula, we get

$$\int_0^c [f(x)]^2 dx = \frac{c}{2} \left[\frac{a_0^2}{2} + a_1^2 + a_2^2 + a_3^2 + \dots \right]$$

$$\text{or} \quad \int_0^1 (\pi x)^2 dx + \int_1^2 \pi^2 (2-x)^2 dx = \frac{2}{2} \left[\frac{\pi^2}{2} + \frac{16}{\pi^2} + \frac{16}{81\pi^2} + \dots \right]$$

$$\text{or} \quad \pi^2 \left[\frac{x^3}{3} \right]_0^1 - \pi^2 \left[\frac{(2-x)^3}{3} \right]_1^2 = \frac{\pi^2}{2} + \frac{16}{\pi^2} + \frac{16}{81\pi^2} + \dots$$

$$\text{or} \quad \frac{\pi^2}{3} - \pi^2 \left(0 - \frac{1}{3} \right) = \frac{\pi^2}{2} + \frac{16}{\pi^2} \left[1 + \frac{1}{81} + \dots \right]$$

$$\text{or} \quad \frac{2\pi^2}{3} - \frac{\pi^2}{2} = \frac{16}{\pi^2} \left[1 + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right]$$

$$\text{or} \quad \frac{\pi^2}{6} = \frac{16}{\pi^2} \left[1 + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right] \quad \text{or} \quad \frac{\pi^4}{96} = 1 + \frac{1}{3^4} + \frac{1}{5^4} + \dots$$

7.9 Complex Form of Fourier Series

The Fourier series expansion of a periodic function $f(x)$ of period $2T$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{T} + b_n \frac{n\pi x}{T} \right) \quad \dots(1)$$

We have $\cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$ and $\sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$, therefore from (1) we get

$$\begin{aligned} f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} \left\{ a_n \left(\frac{e^{in\pi x/T} + e^{-in\pi x/T}}{2} \right) + b_n \left(\frac{e^{in\pi x/T} - e^{-in\pi x/T}}{2i} \right) \right\} \\ &= c_0 + \sum_{n=1}^{\infty} \{ c_n e^{in\pi x/T} + c_{-n} e^{-in\pi x/T} \} \end{aligned} \quad \dots(2)$$

where $c_0 = \frac{a_0}{2}$, $c_n = \frac{1}{2}(a_n - ib_n)$, $c_{-n} = \frac{1}{2}(a_n + ib_n)$.

Thus $c_0 = \frac{1}{2T} \int_0^{2T} f(x) dx$,

$$\begin{aligned} c_n &= \frac{1}{2T} \left\{ \int_0^{2T} f(x) \cos \frac{n\pi x}{T} dx - i \int_0^{2T} f(x) \sin \frac{n\pi x}{T} dx \right\} \\ &= \frac{1}{2T} \int_0^{2T} f(x) \left(\cos \frac{n\pi x}{T} - i \sin \frac{n\pi x}{T} \right) dx = \frac{1}{2T} \int_0^{2T} f(x) e^{-in\pi x/T} dx \end{aligned}$$

and $c_{-n} = \frac{1}{2T} \int_0^{2T} f(x) \left(\cos \frac{n\pi x}{T} + i \sin \frac{n\pi x}{T} \right) dx = \frac{1}{2T} \int_0^{2T} f(x) e^{in\pi x/T} dx$.

Combining these, we get $c_n = \frac{1}{2T} \int_0^{2T} f(x) e^{-in\pi x/T} dx$(3)

$$(n = 0, \pm 1, \pm 2, \dots)$$

Hence, the series (2) can be compactly written as $f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/T}$(4)

This is the complex form of Fourier series and its coefficients are given by (3).

If the function is defined in the interval $(-T, T)$, then the coefficients are given by

$$c_n = \frac{1}{2T} \int_{-T}^T f(x) e^{-in\pi x/T} dx. \quad (n = 0, \pm 1, \pm 2, \dots)$$

The complex form of a Fourier series is especially useful in problems on electrical circuits having periodic voltage.

Illustrative Examples

Example 14: Obtain the complex form of the Fourier series of the function

$$f(x) = \begin{cases} 0, & -\pi \leq x \leq 0 \\ 1, & 0 \leq x \leq \pi. \end{cases}$$

Solution: The complex form of Fourier series for $f(x)$ is

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/T}.$$

Here
$$c_n = \frac{1}{2T} \int_{-T}^T f(x) e^{-in\pi x/T} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

$$\begin{aligned} &= \frac{1}{2\pi} \left[\int_{-\pi}^0 0 \cdot e^{-inx} dx + \int_0^{\pi} 1 \cdot e^{-inx} dx \right] \\ &= \frac{1}{2\pi} \int_0^{\pi} e^{-inx} dx = \frac{1}{2\pi} \left[\frac{e^{-inx}}{-in} \right]_0^{\pi} = -\frac{1}{2n\pi i} [e^{-in\pi} - 1] \end{aligned}$$

$$= -\frac{1}{2n\pi i} [\cos n\pi - 1] = -\frac{1}{2n\pi i} [(-1)^n - 1] = \begin{cases} \frac{1}{in\pi}, & n \text{ is odd} \\ 0, & n \text{ is even} \end{cases}$$

and
$$c_0 = \frac{1}{2T} \int_{-T}^T f(x) dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \int_0^{\pi} dx = \frac{1}{2\pi} [x]_0^{\pi} = \frac{1}{2}.$$

$$\begin{aligned} \therefore f(x) &= \frac{1}{2} + \frac{1}{i\pi} \left[\frac{e^{ix}}{1} + \frac{e^{3ix}}{3} + \frac{e^{5ix}}{5} + \dots \right] + \frac{1}{i\pi} \left[\frac{e^{-ix}}{-1} + \frac{e^{-3ix}}{-3} + \frac{e^{-5ix}}{-5} + \dots \right] \\ &= \frac{1}{2} - \frac{1}{i\pi} \left[(e^{ix} - e^{-ix}) + \frac{1}{3} (e^{3ix} - e^{-3ix}) + \frac{1}{5} (e^{5ix} - e^{-5ix}) + \dots \right]. \end{aligned}$$

Comprehensive Exercise 5

- By using the sine series for $f(x) = 1$ in $0 < x < \pi$ show that

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots$$

- Find the Fourier series for $f(x) = x^2$ in $-\pi < x < \pi$. Using the two values of $f(x)$,

$$\text{show that } \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \frac{1}{4^4} + \dots = \frac{\pi^4}{90}.$$

3. Obtain the complex form of the Fourier series of

$$f(x) = e^{-x} \text{ in } -1 \leq x \leq 1.$$

Answers 5

1. $1 = \frac{4}{\pi} \sin x + \frac{4}{3\pi} \sin 3x + \frac{4}{5\pi} \sin 5x + \frac{4}{7\pi} \sin 7x + \dots$
2. $f(x) = \frac{n^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} (-1)^n \cos nx$ 3. $e^{-x} = \sum_{n=-\infty}^{\infty} \frac{(-1)^n (1 - in\pi)}{1 + n^2 \pi^2} \sinh 1 \cdot e^{in\pi x}.$

Objective Type Questions

Multiple Choice Questions

Indicate the correct answer for each question by writing the corresponding letter from (a), (b), (c) and (d).

1. Fourier series for the function $f(x)$ in $(-l, l)$ is
- (a) $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l}; f(x)$ is even in $(-l, l)$
- (b) $f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}; f(x)$ is odd in $(-l, l)$
- (c) $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \frac{\cos n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$
- (d) All of these
2. If $f(x) = x^4$ in $(-1 \leq x \leq 1)$, then the Fourier coefficient b_n is
- (a) 0 (b) 1 (c) 2 (d) None of these
3. The Fourier half range sine series for $f(x) = x$ in $(0, 2)$, then b_n is
- (a) $\frac{4(-1)^n}{n\pi}$ (b) $-\frac{4(-1)^n}{n\pi}$ (c) $\frac{4(-1)^n}{\pi}$ (d) $-\frac{4(-1)^n}{\pi}$
4. The sum of the series $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$ is equal to
- (a) $\frac{\pi^2}{12}$ (b) $\frac{\pi^2}{8}$ (c) $\frac{\pi^2}{6}$ (d) $\frac{\pi^2}{2}$
5. For Fourier expansion of an odd function in $(-\pi, \pi)$
- (a) a_0 is zero (b) a_n is zero
- (c) b_n is zero (d) both a_0 and a_n are zero.

Fill in the Blank(s)

Fill in the blanks "....." so that the following statements are complete and correct.

1. Fourier expansion of an odd function has only terms.
2. If $f(x) = x \sin x$ in $(-\pi, \pi)$ then the value of $b_n = \dots\dots\dots$

3. Fourier expansion of an even function has only terms.
4. The period of a constant function is
5. The period of $\cos 3x$ is
6. In the fourier series for $f(x) = x$ in $-\pi \leq x \leq \pi$, the terms are absent.
7. Fourier expansion of an even function in $(-\pi, \pi)$, b_n is

True or False

Write 'T' for true and 'F' for false statement.

1. A function $f(x)$ is even if $f(-x) = -f(x)$.
2. A function $f(x)$ is odd if $f(-x) = f(x)$.
3. Most functions are neither even nor odd.
4. A function $f(x)$ can always be expressed as an arithmetic mean of an even and odd function as $f(x) = \frac{1}{2}[f(x) + f(-x)] + \frac{1}{2}[f(x) - f(-x)]$.
5. For an even function $f(x)$, the graph of $f(x)$ is symmetric about x -axis.
6. If $f(x) = x^2$ in $(-\pi, \pi)$ then the Fourier series of $f(x)$ contains only sine terms.
7. The period of $f(x) = \sin 2x$ is π .
8. For an odd function $f(x)$, the graph of $f(x)$ is symmetric about the origin.

Answers

Multiple Choice Questions

1. (d) 2. (a) 3. (b) 4. (b) 5. (d)

Fill in the Blank(s)

1. sine 2. 0 3. cosine 4. not defined
5. $\frac{2\pi}{3}$ 6. cosine 7. zero

True or False

1. F 2. F 3. T 4. T 5. F
6. F 7. T 8. T

